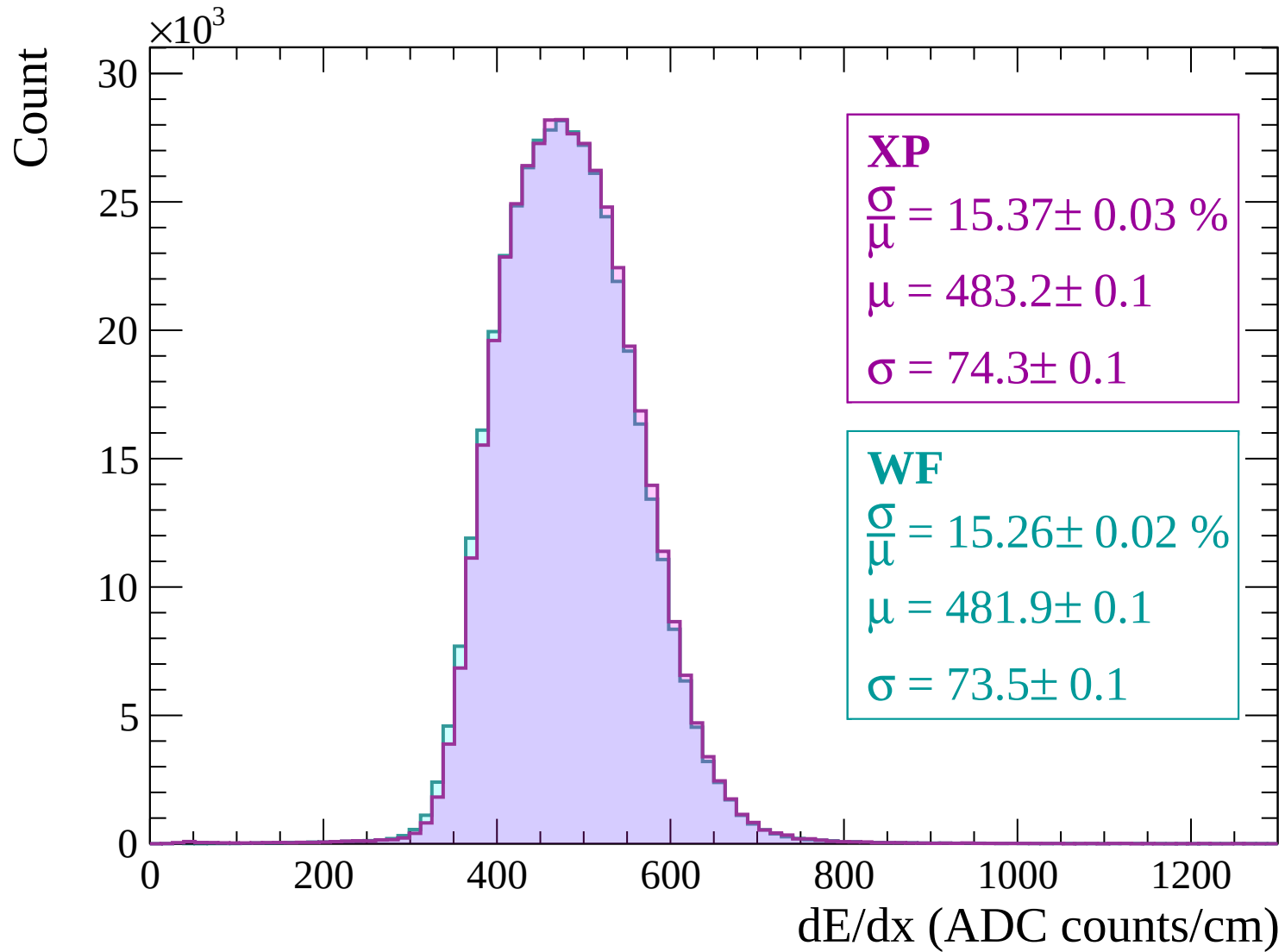
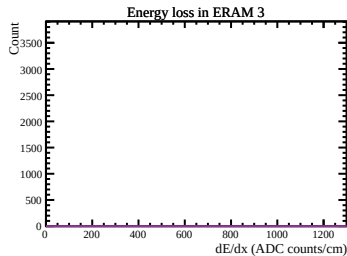
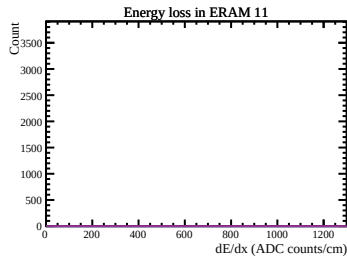
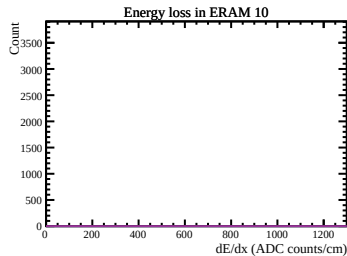
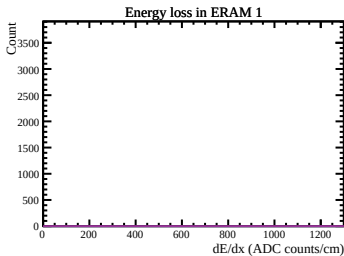
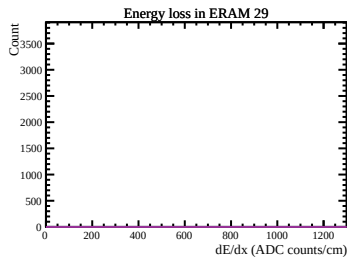
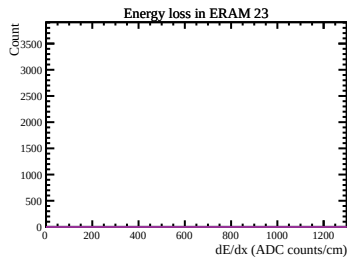
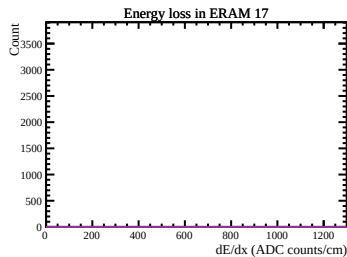
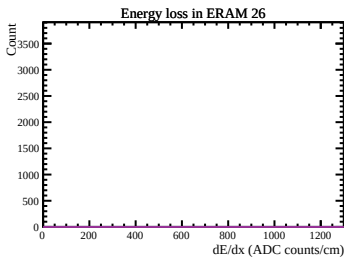
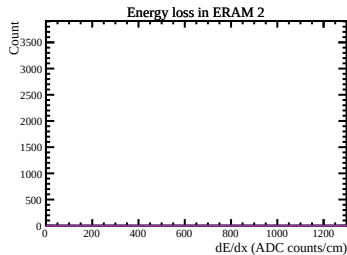
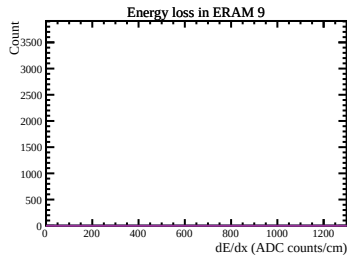
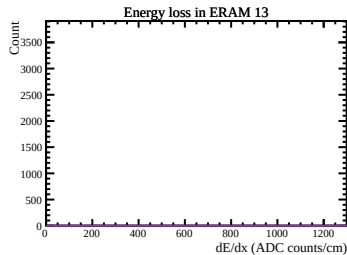
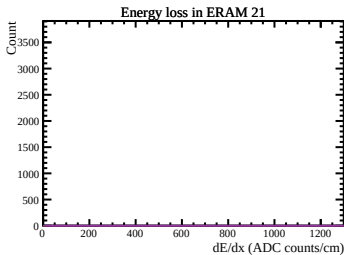
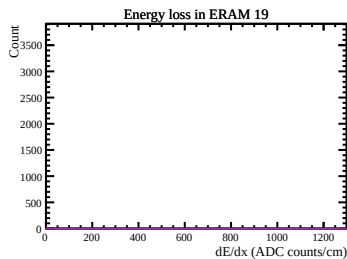
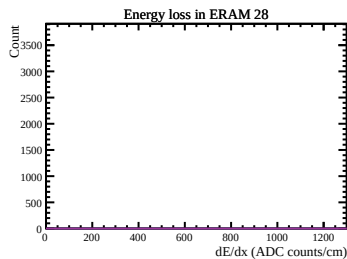
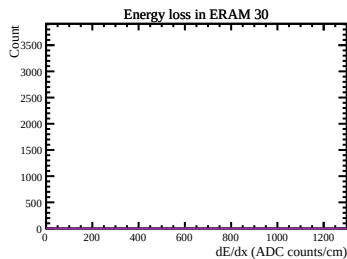
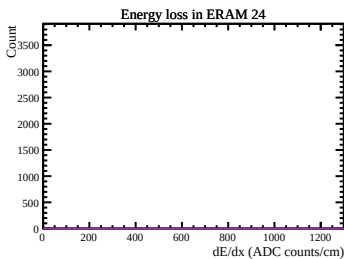
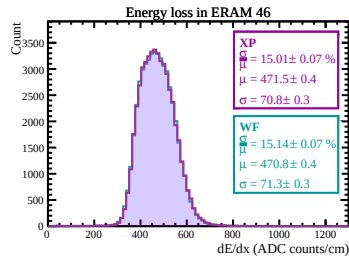
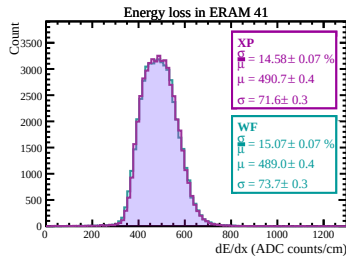
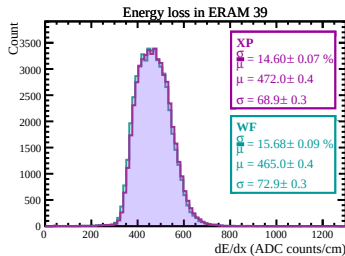
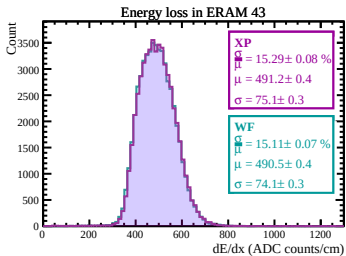
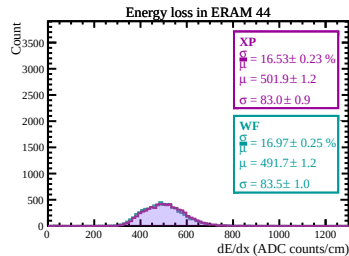
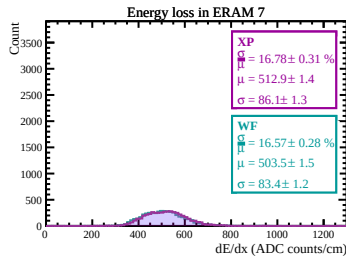
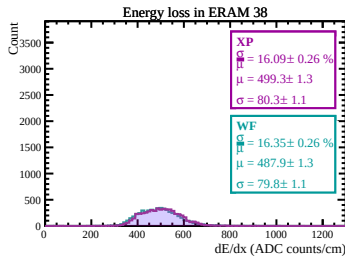
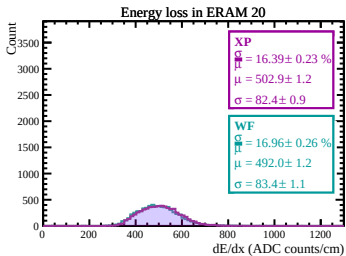
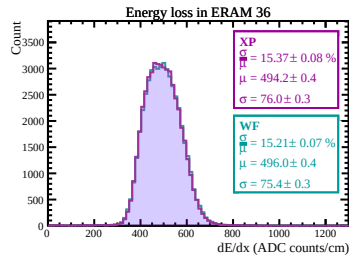
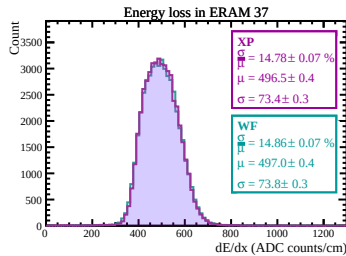
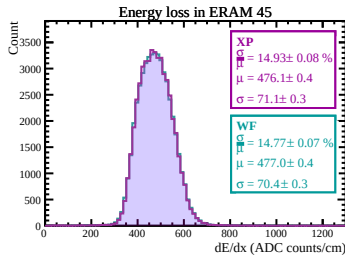
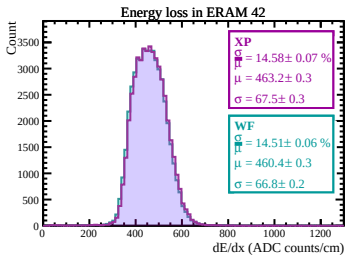
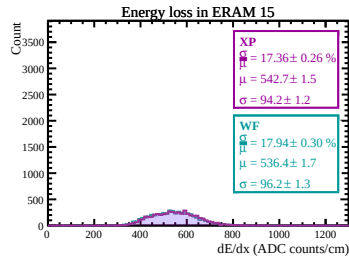
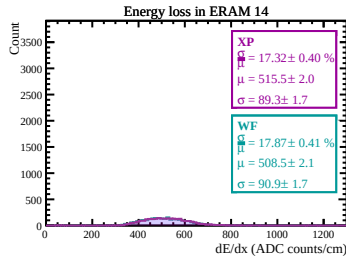
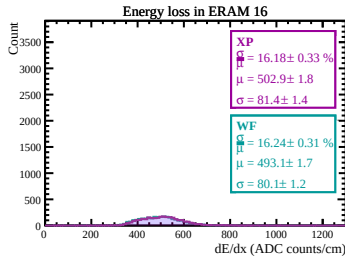
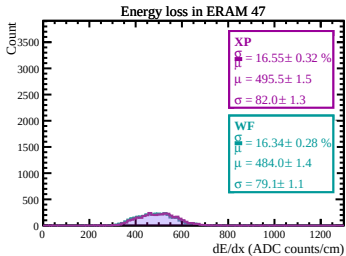
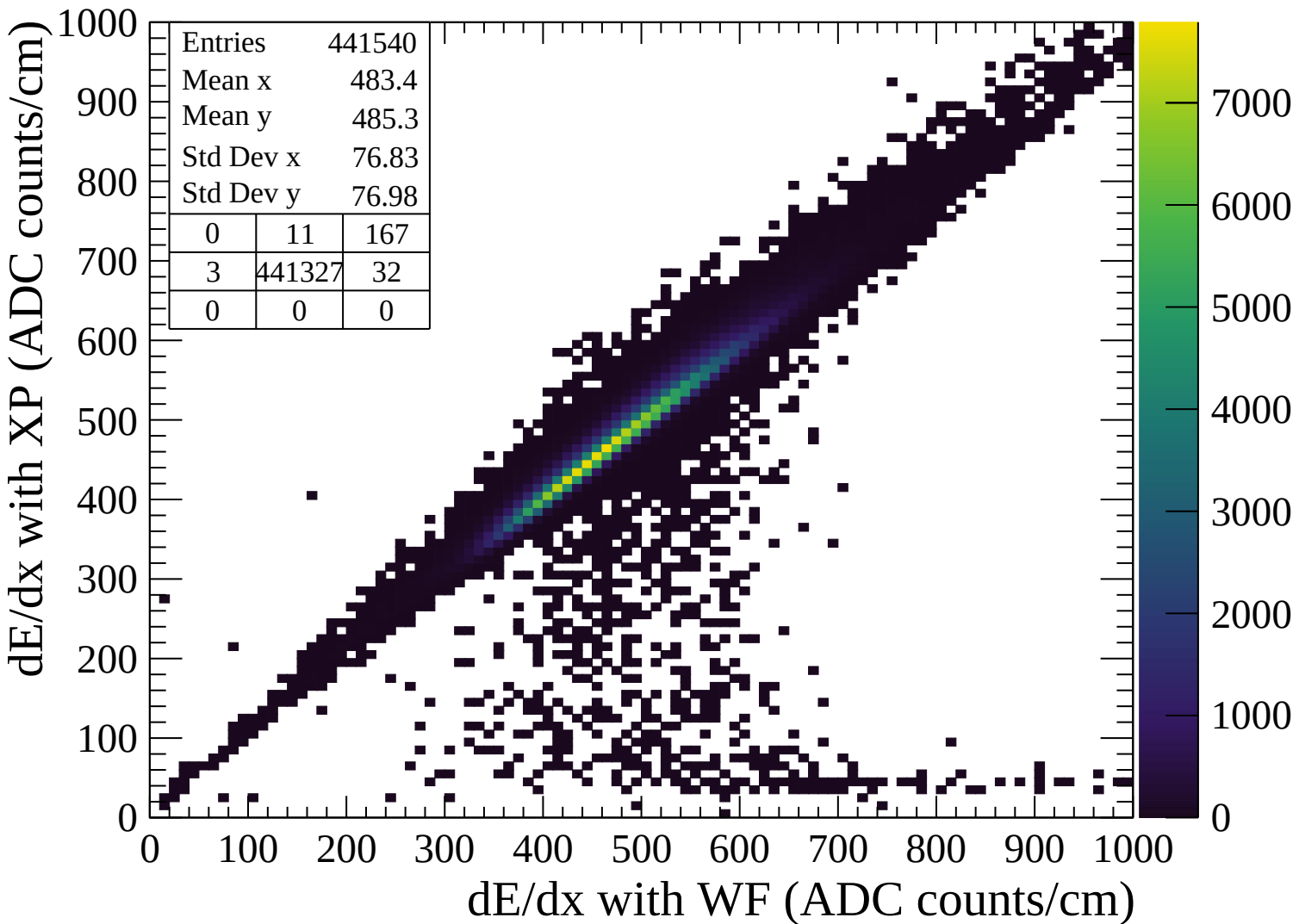
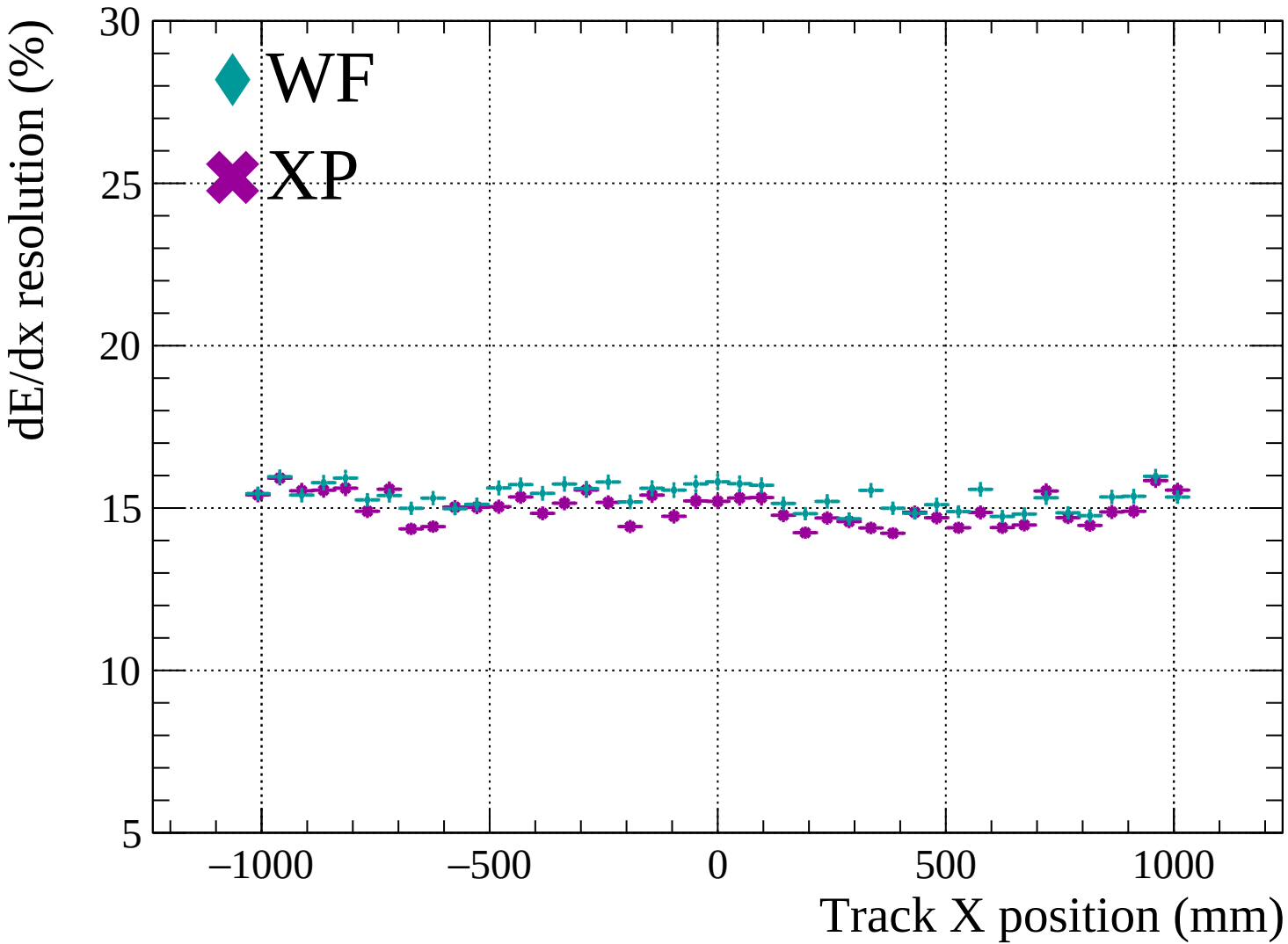


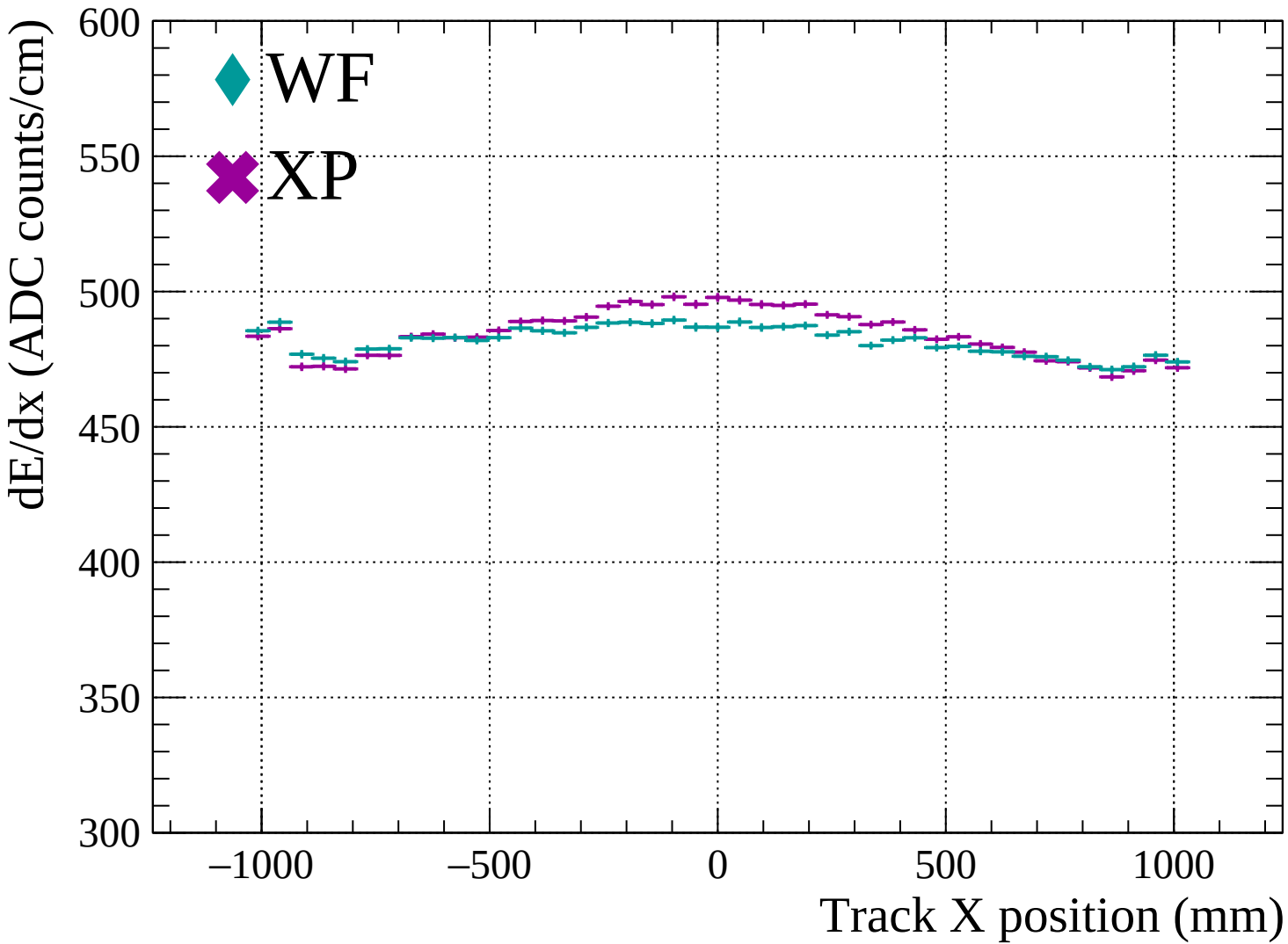


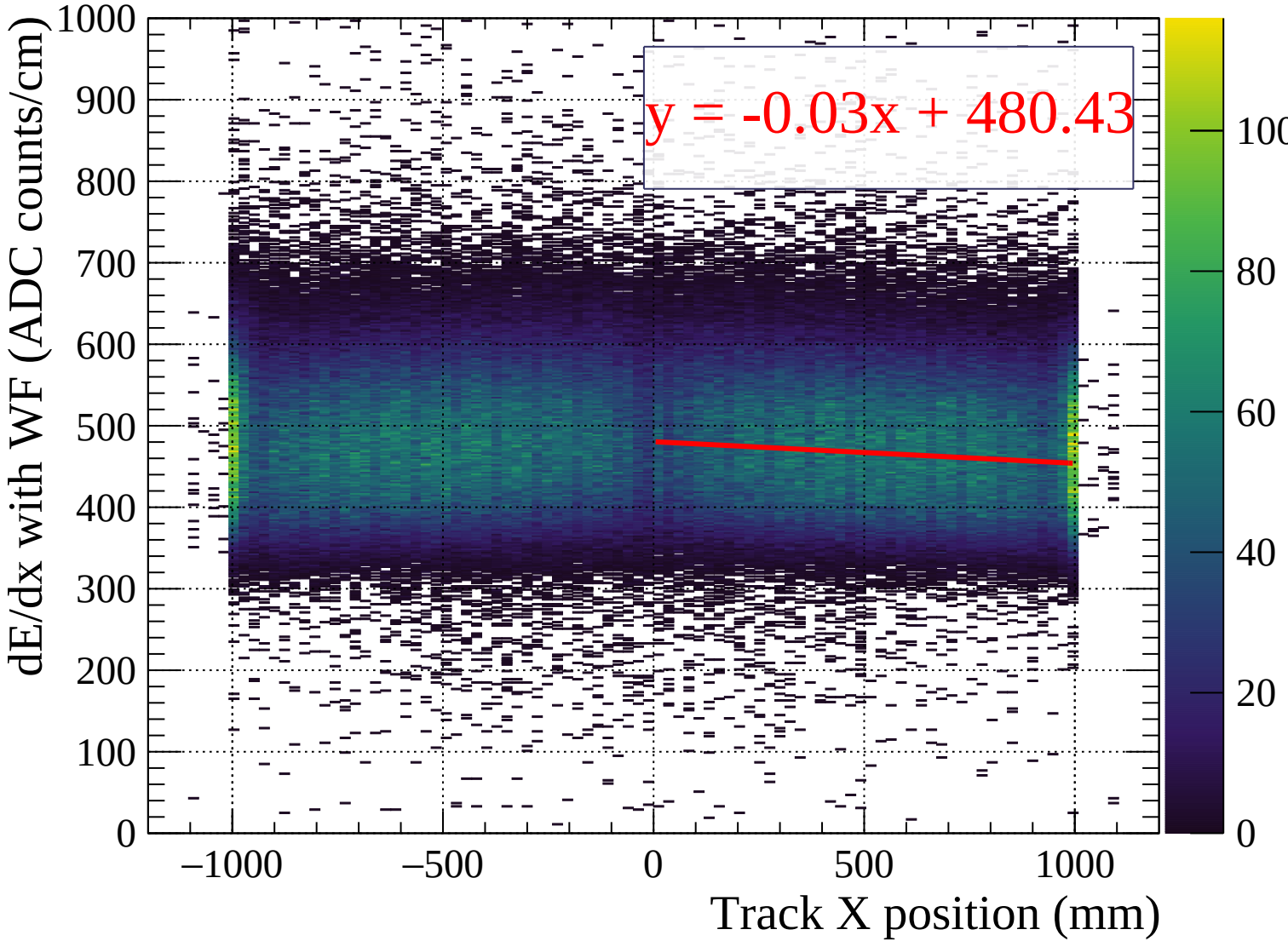


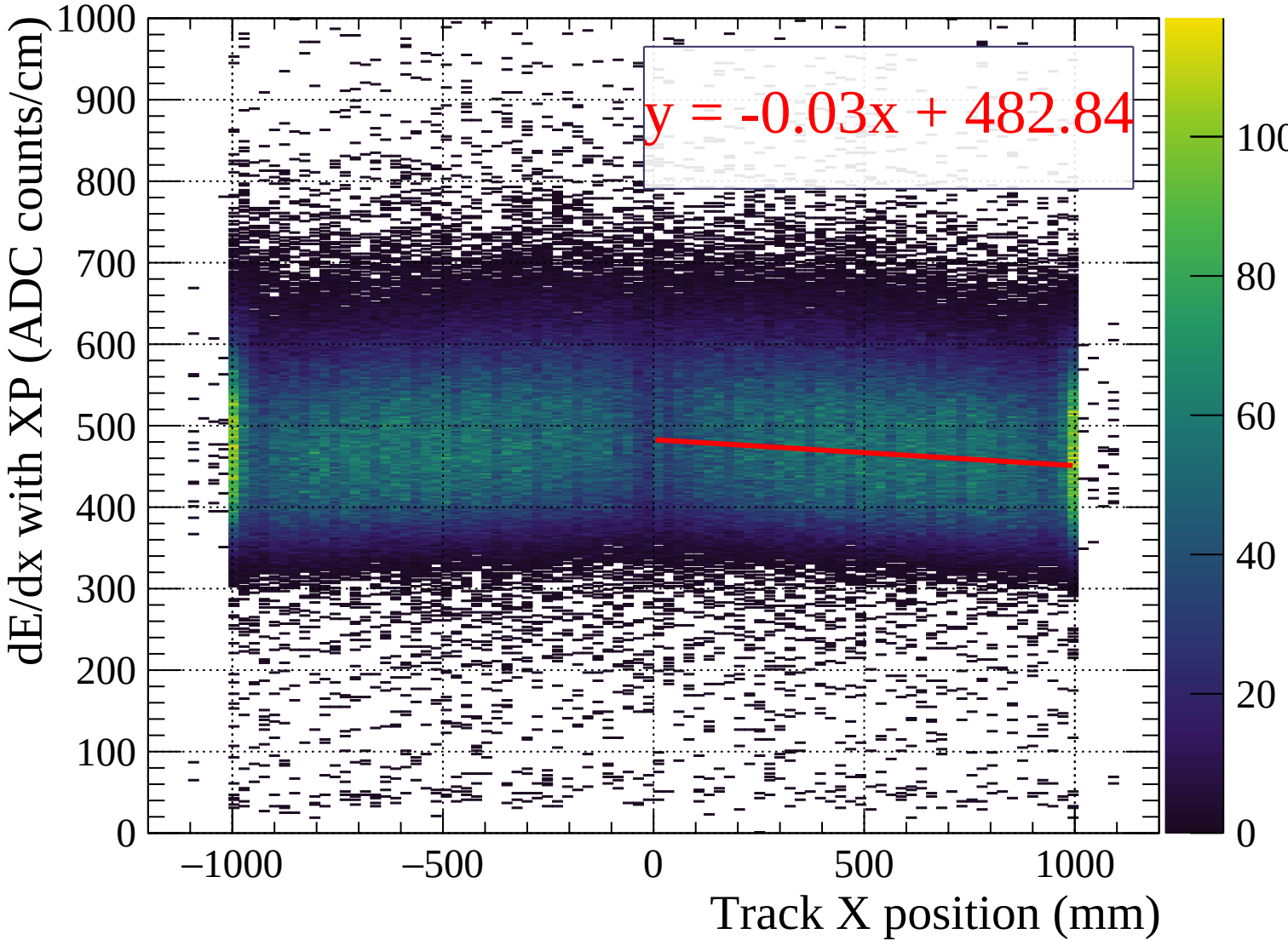


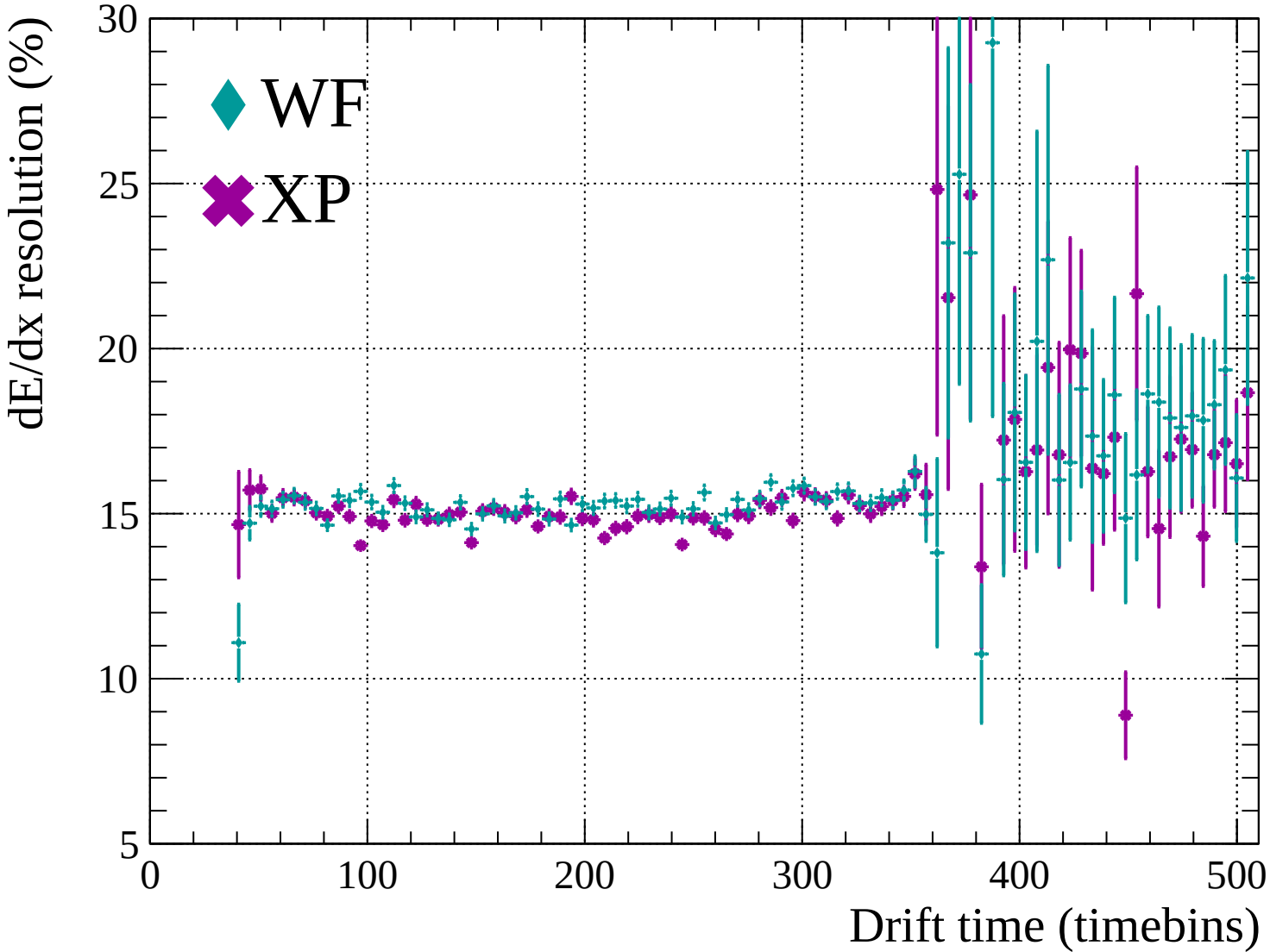


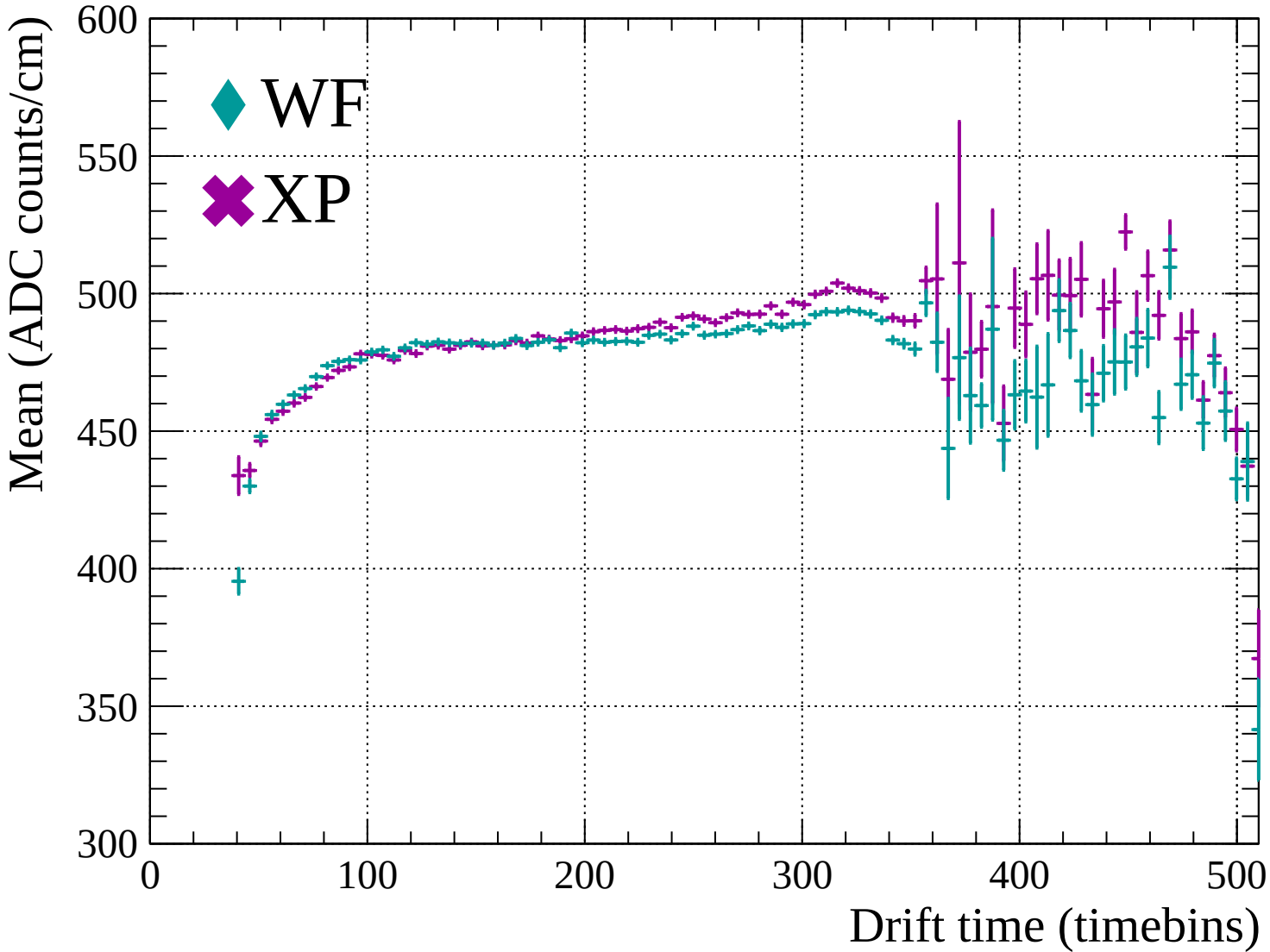


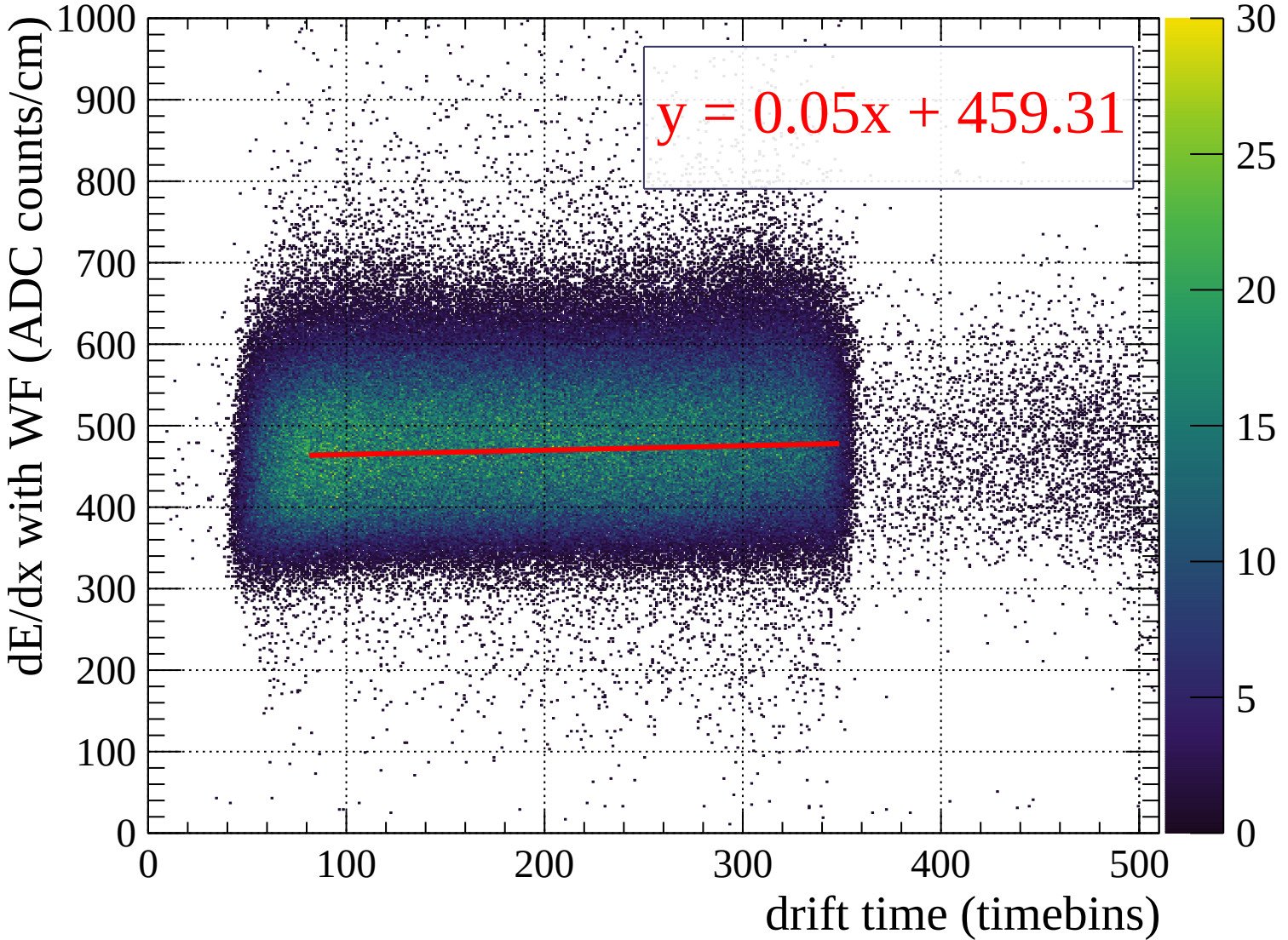


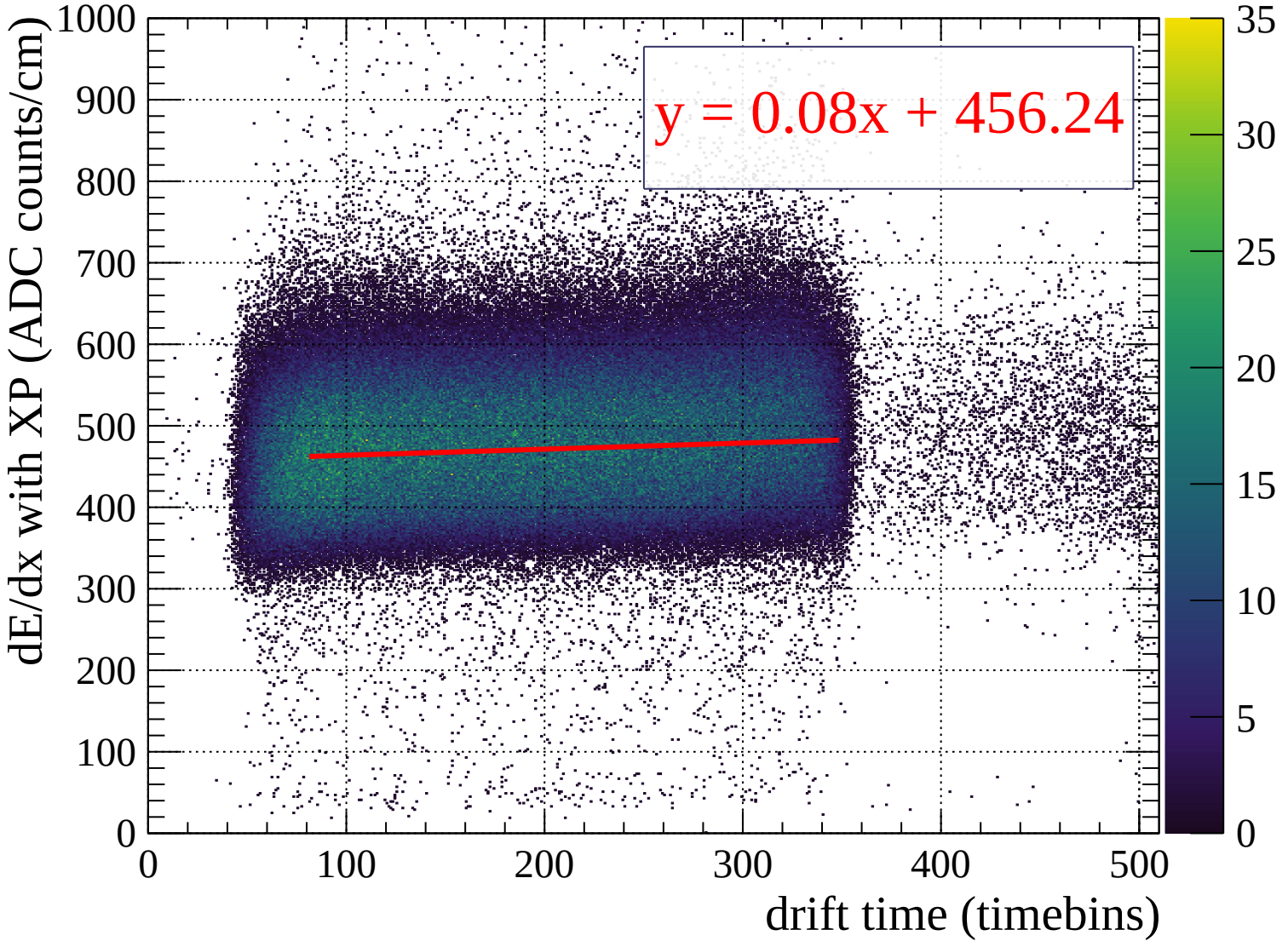


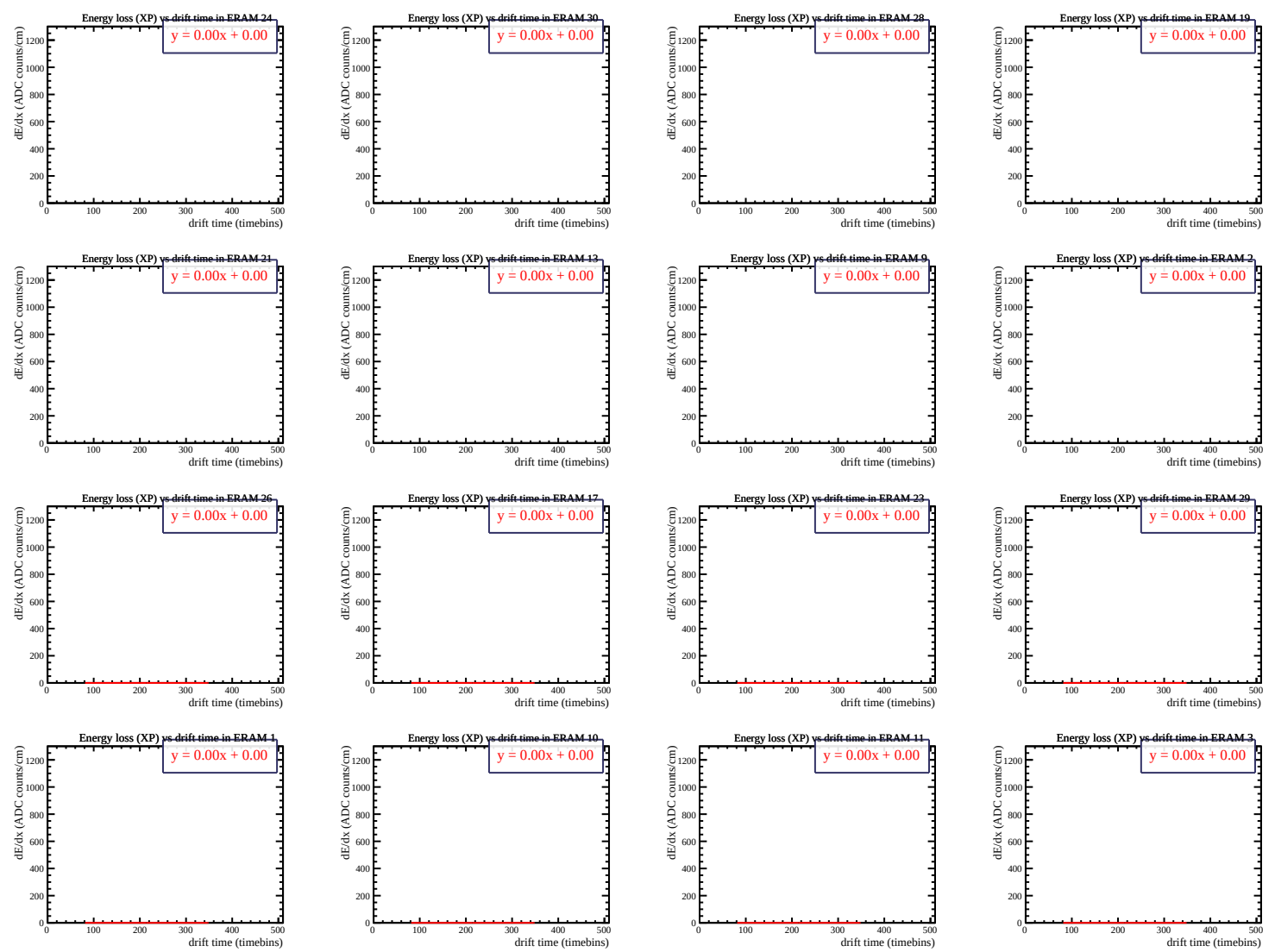


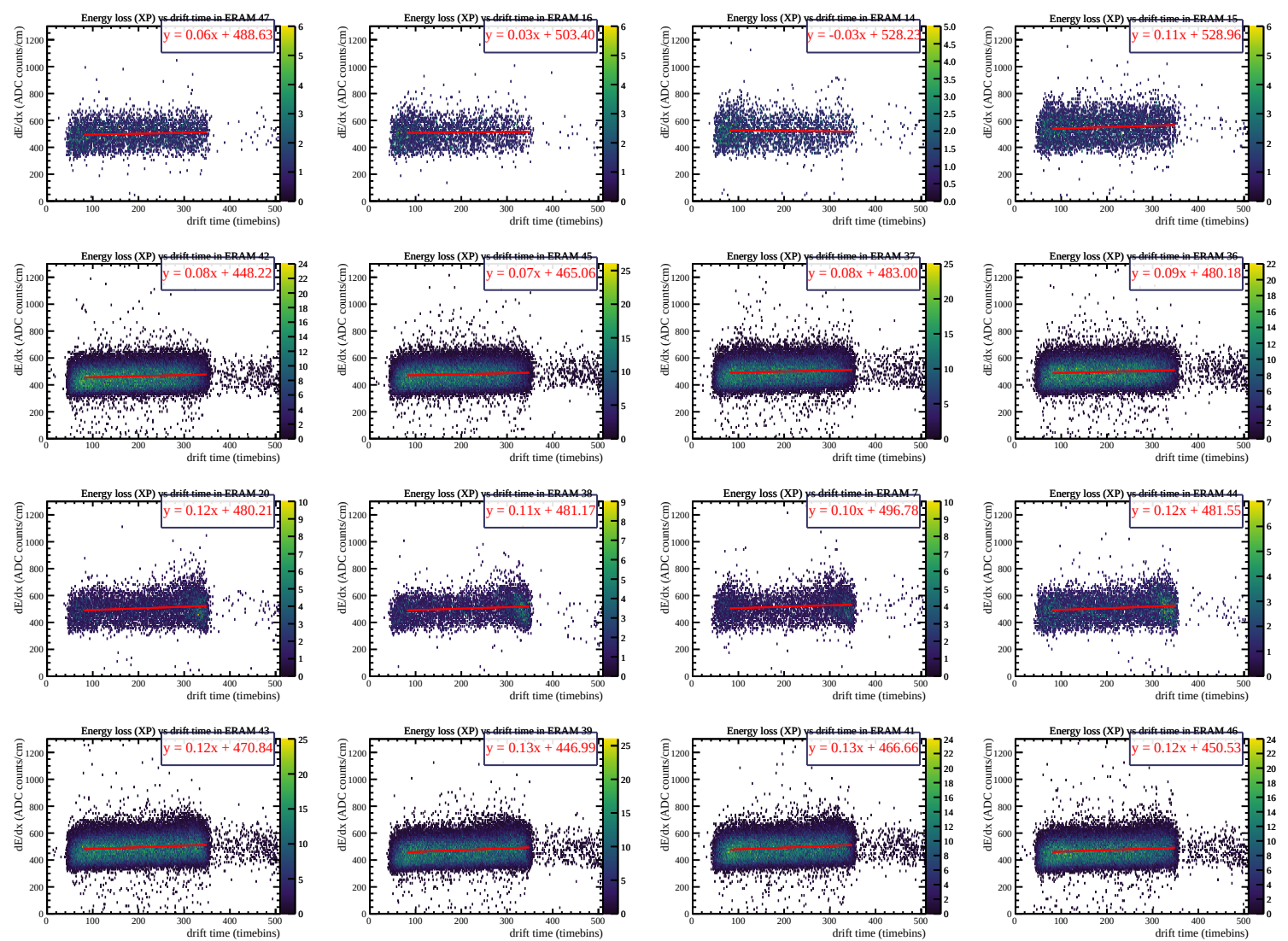


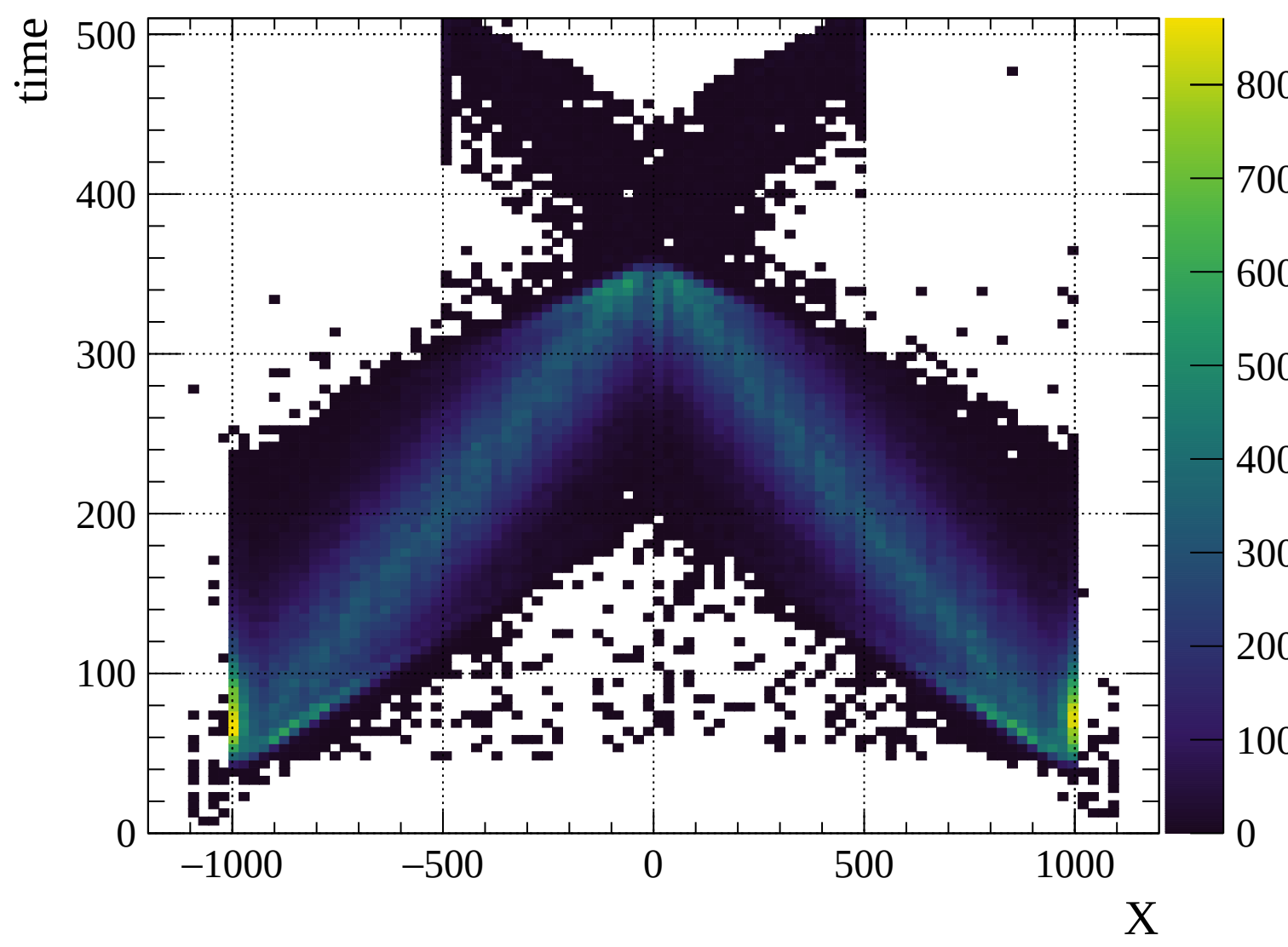


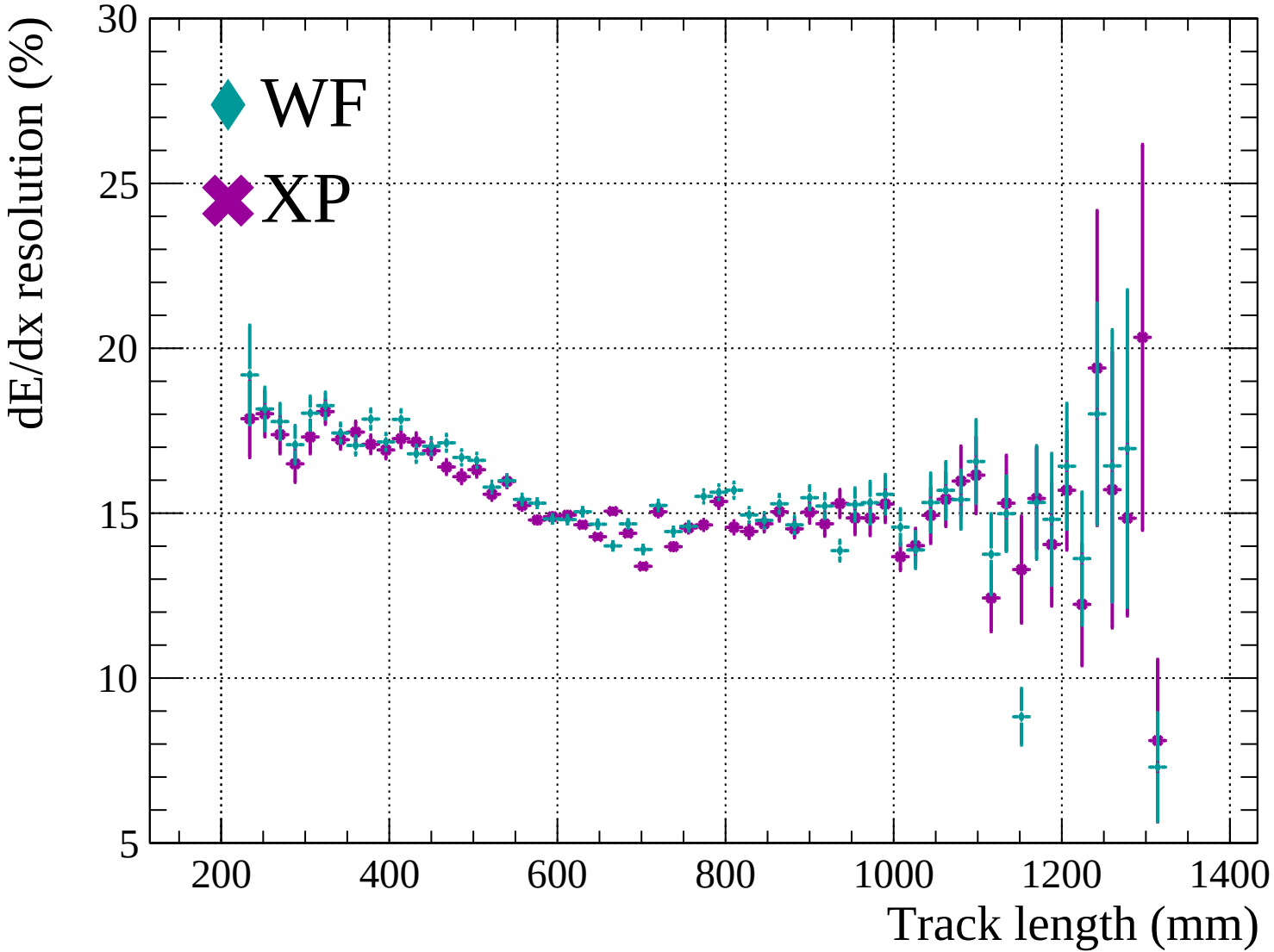


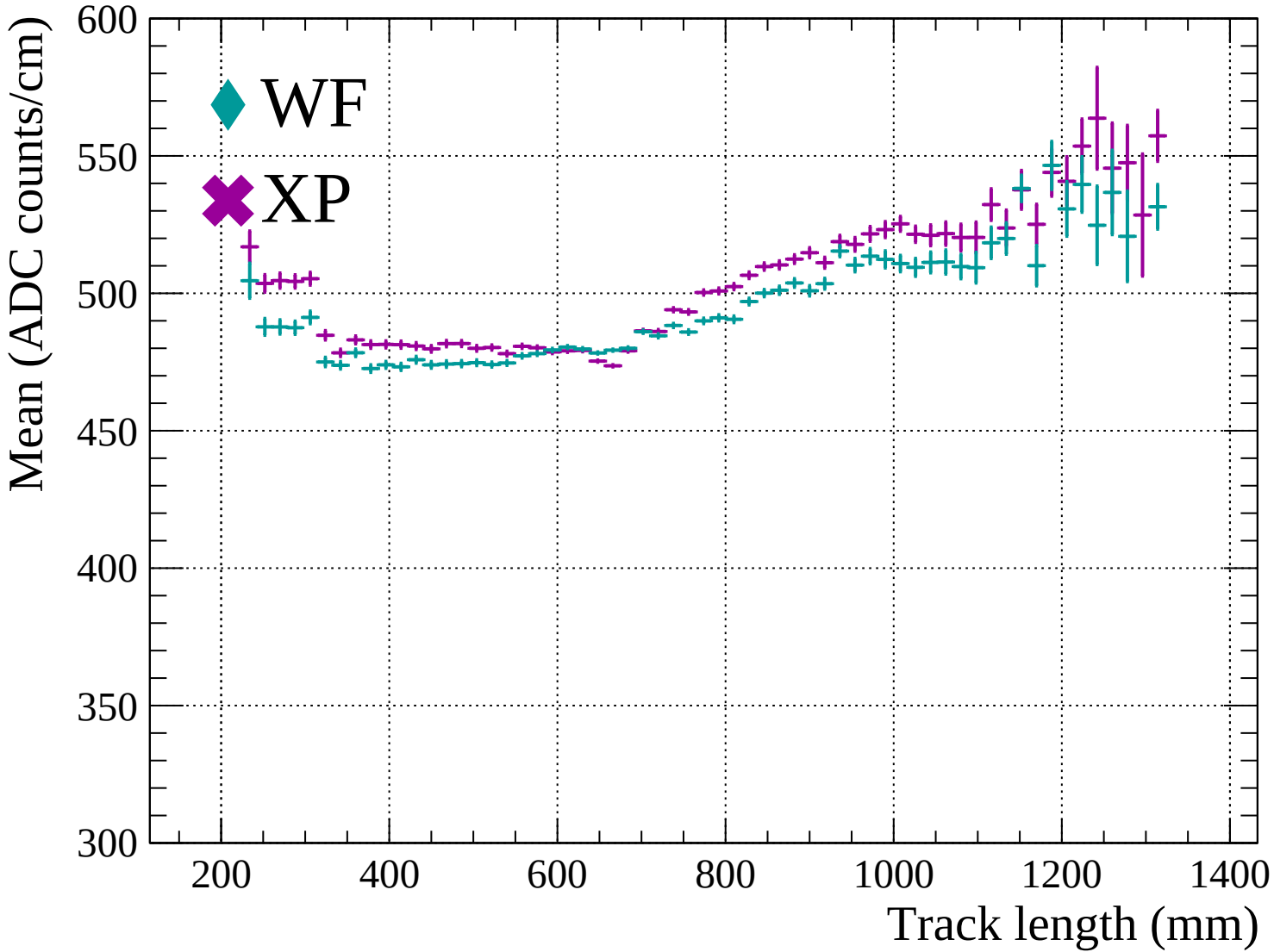


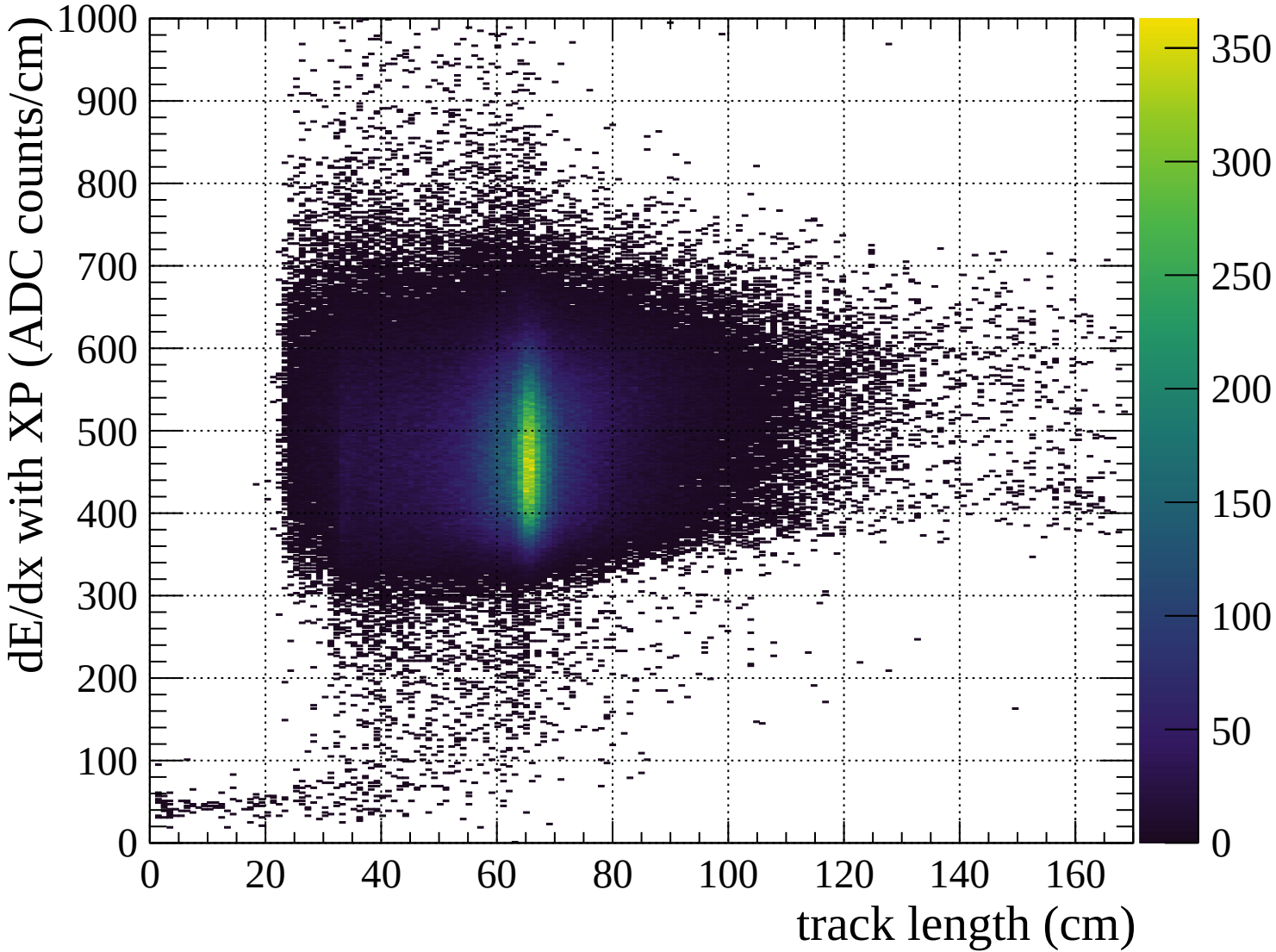


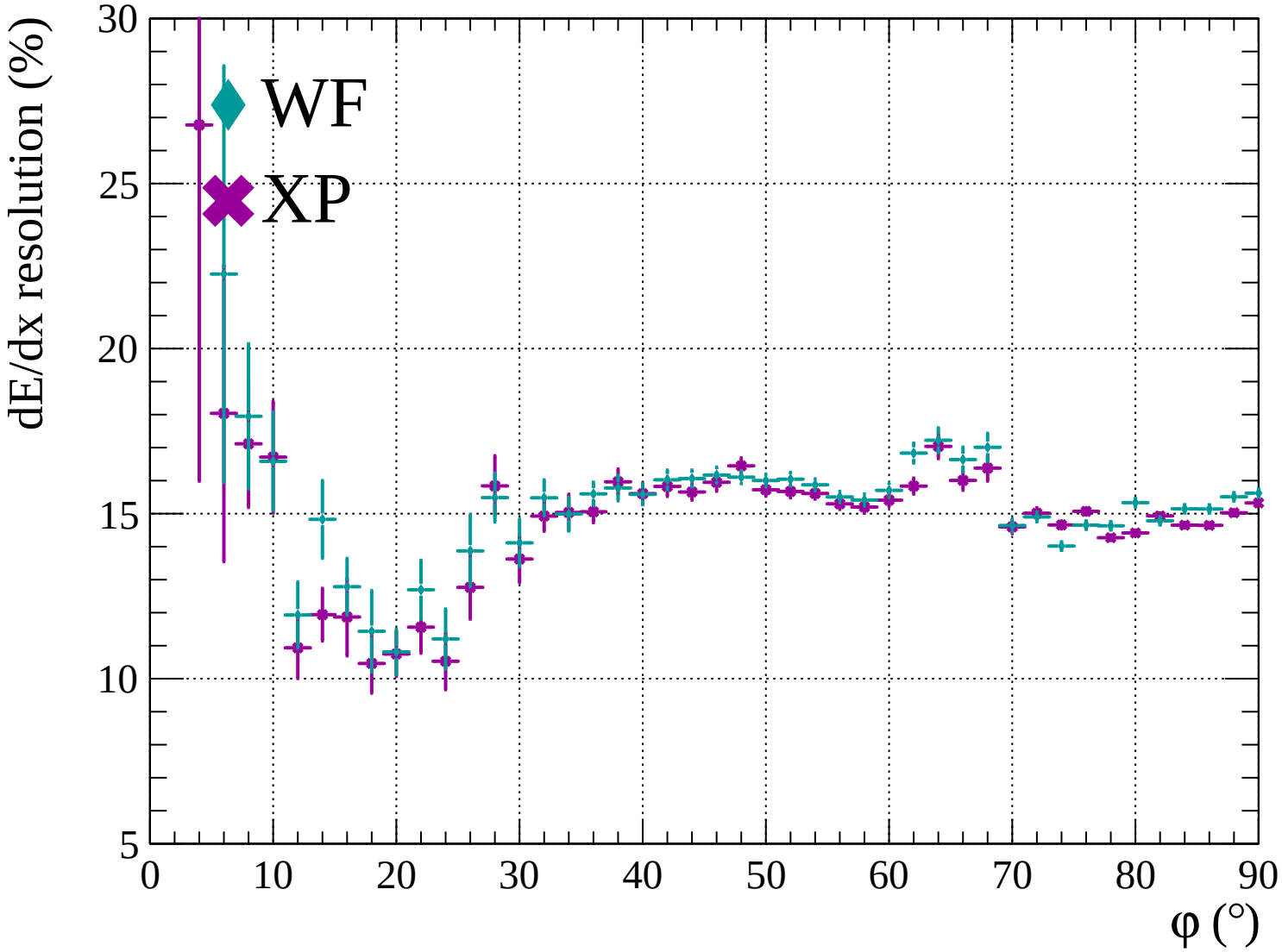


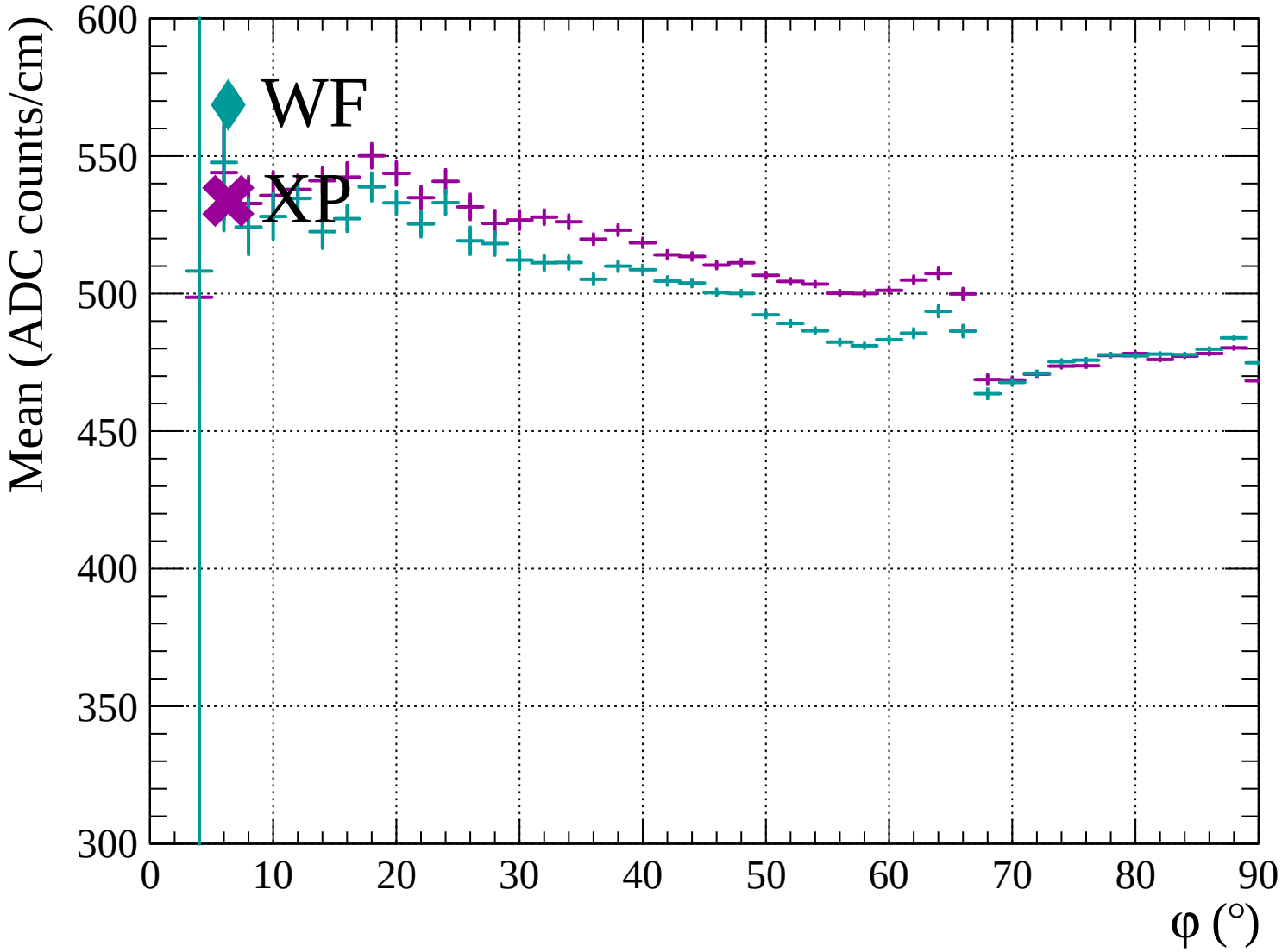


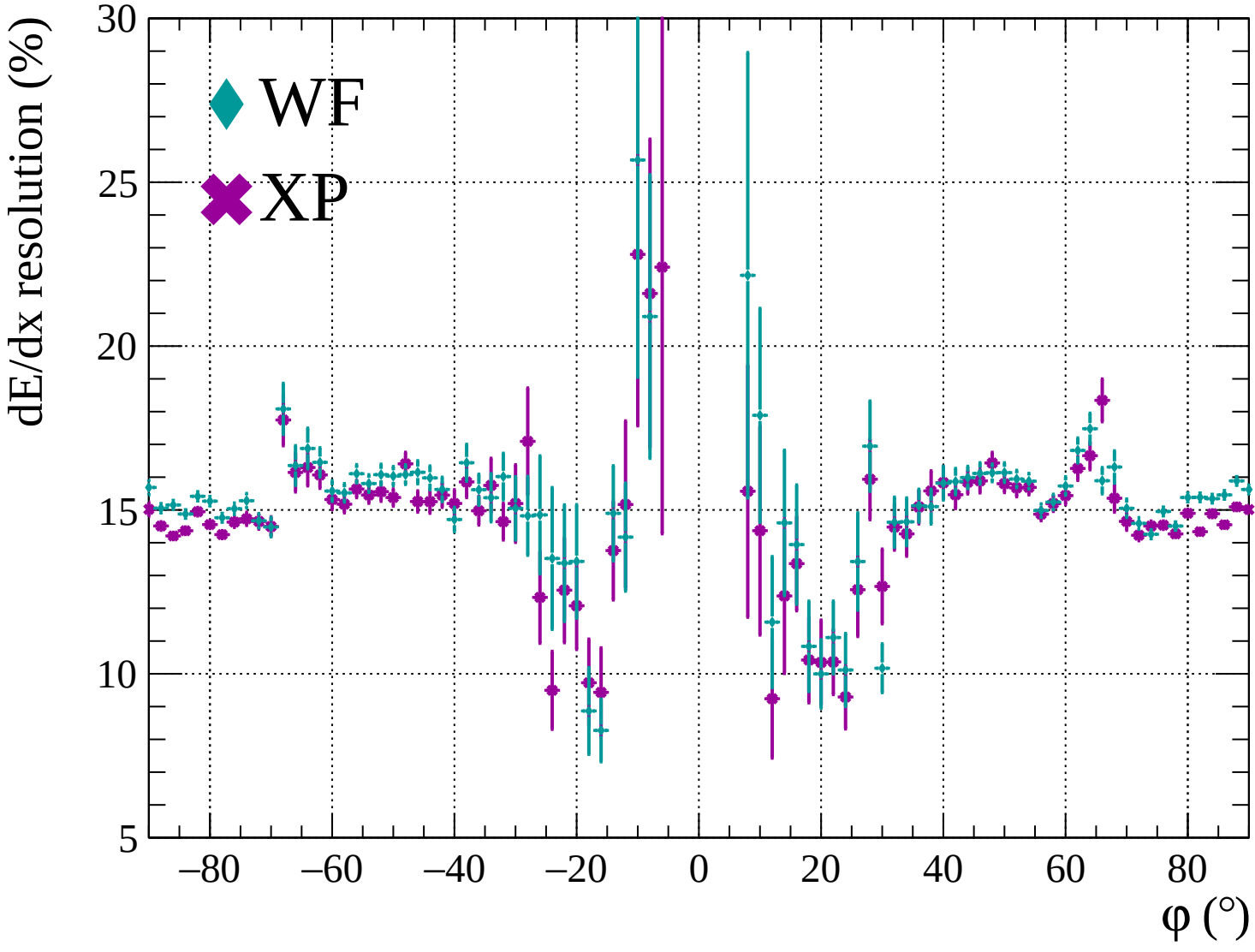


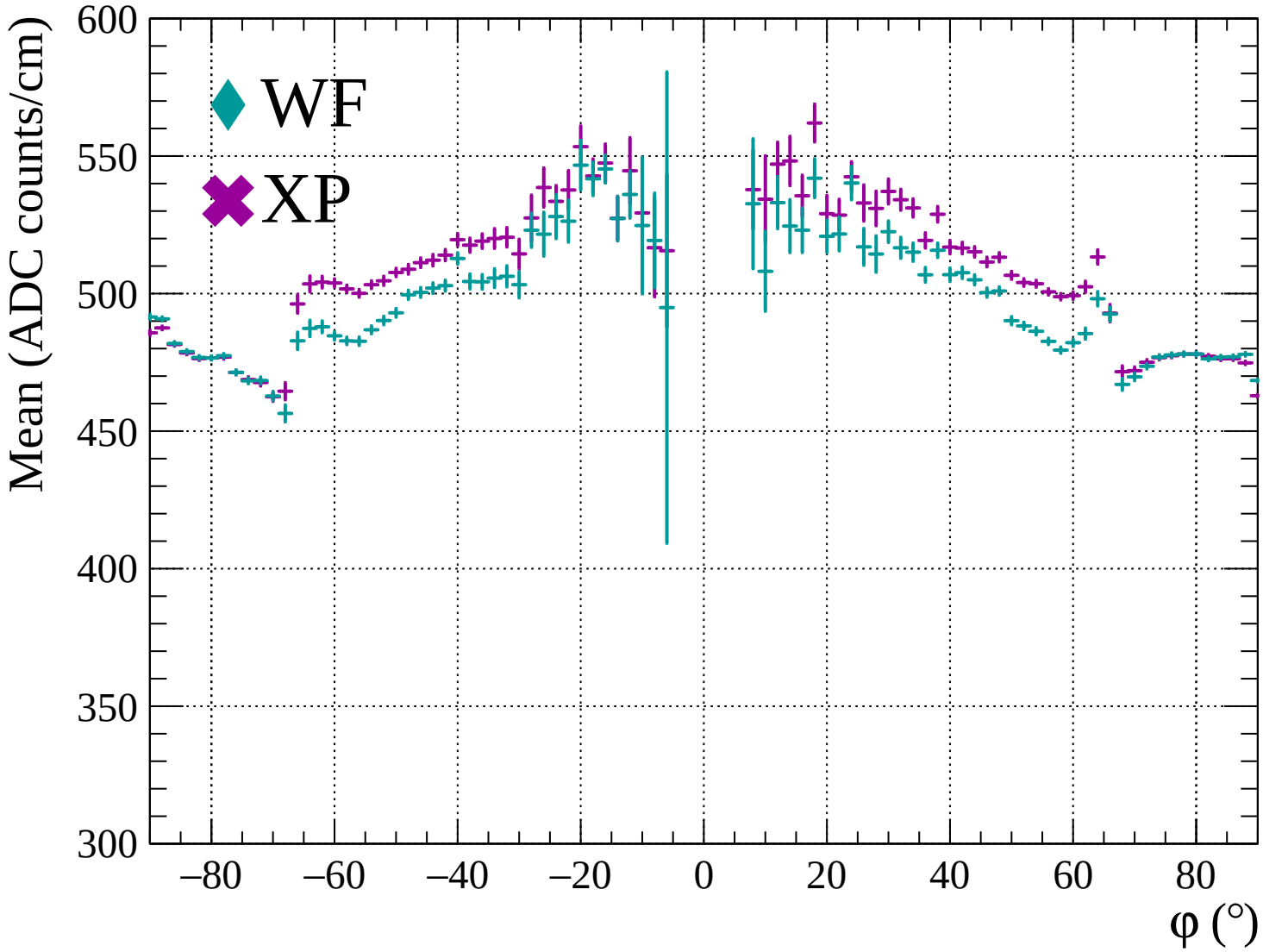


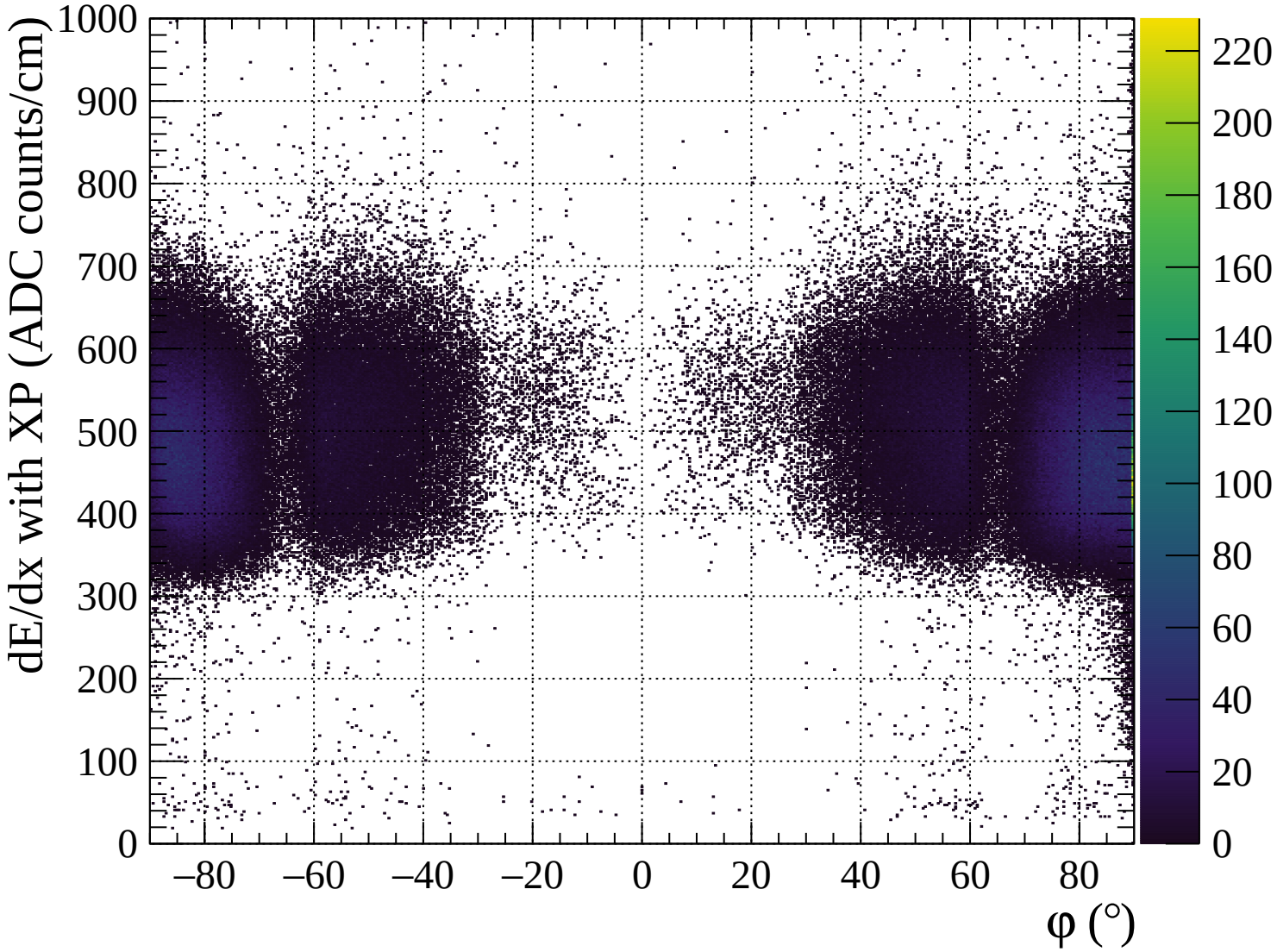


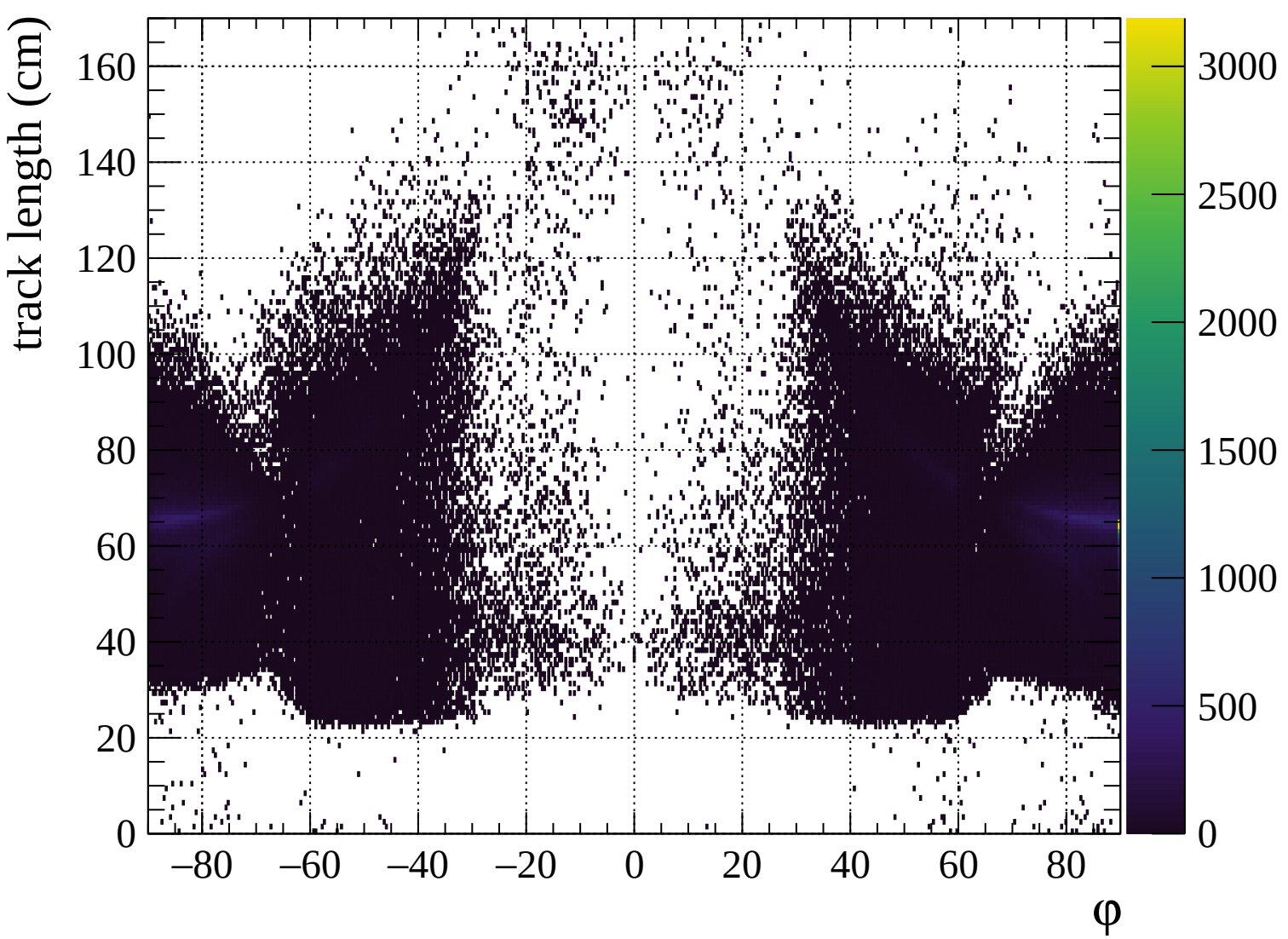


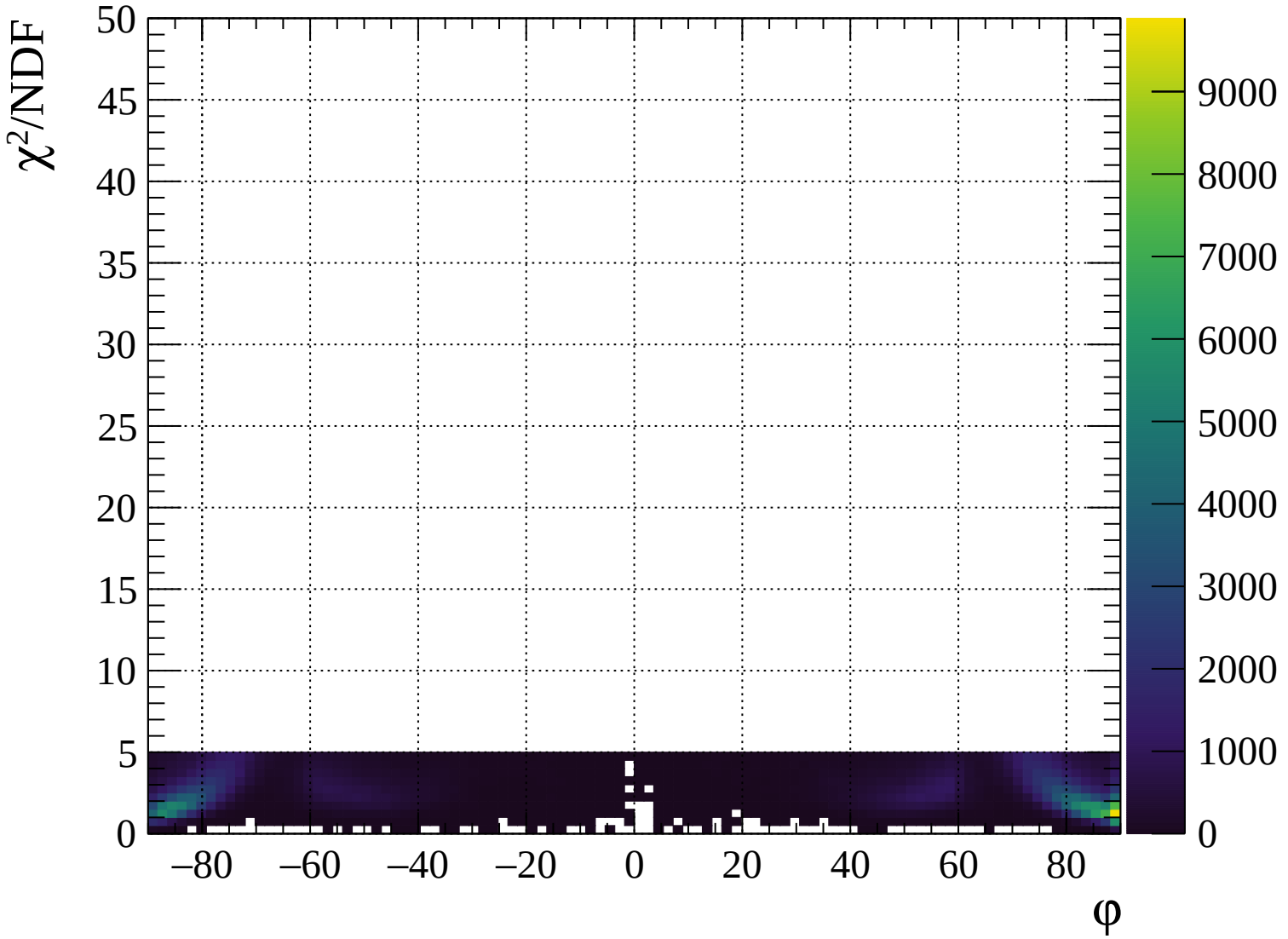


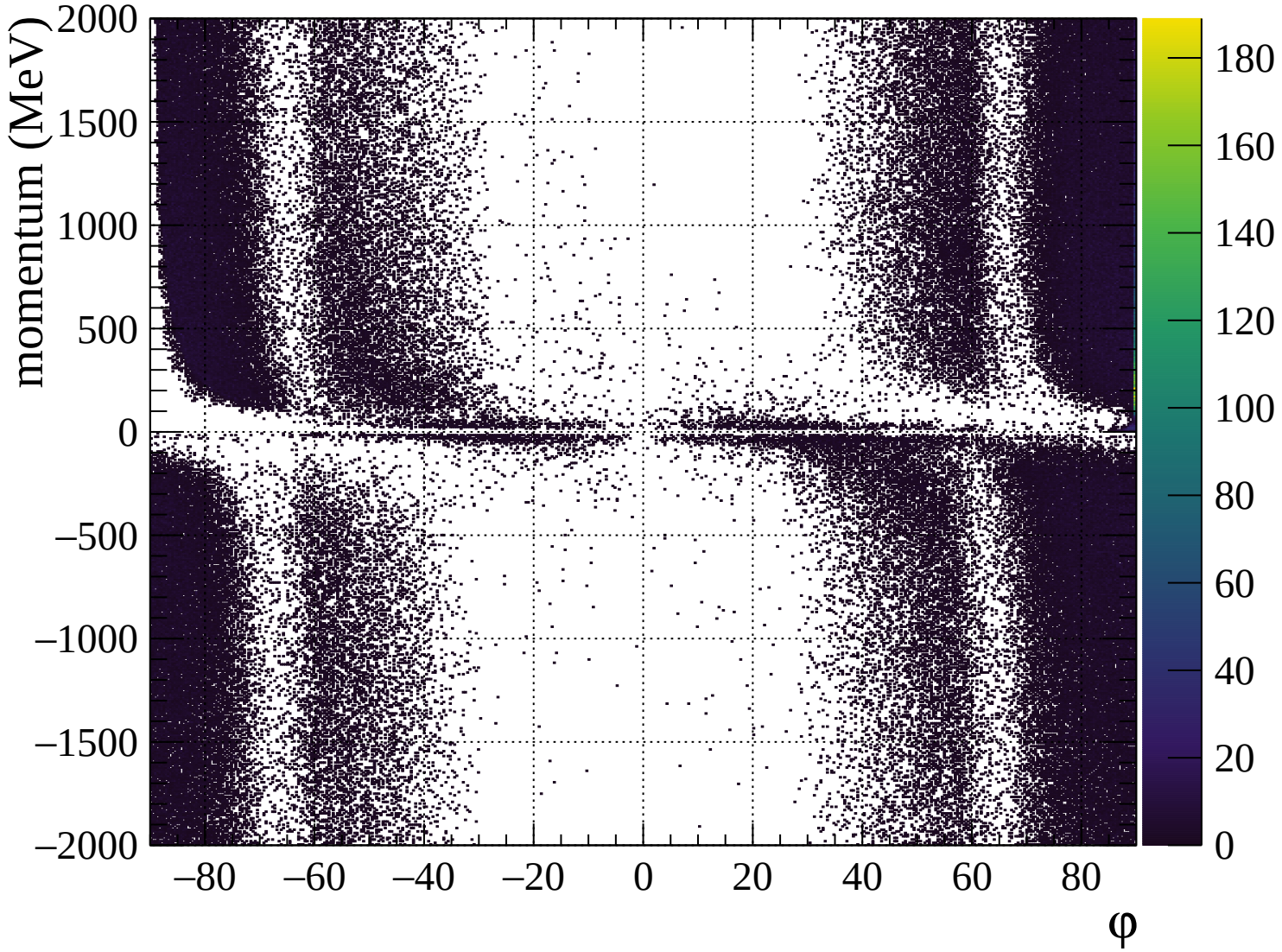


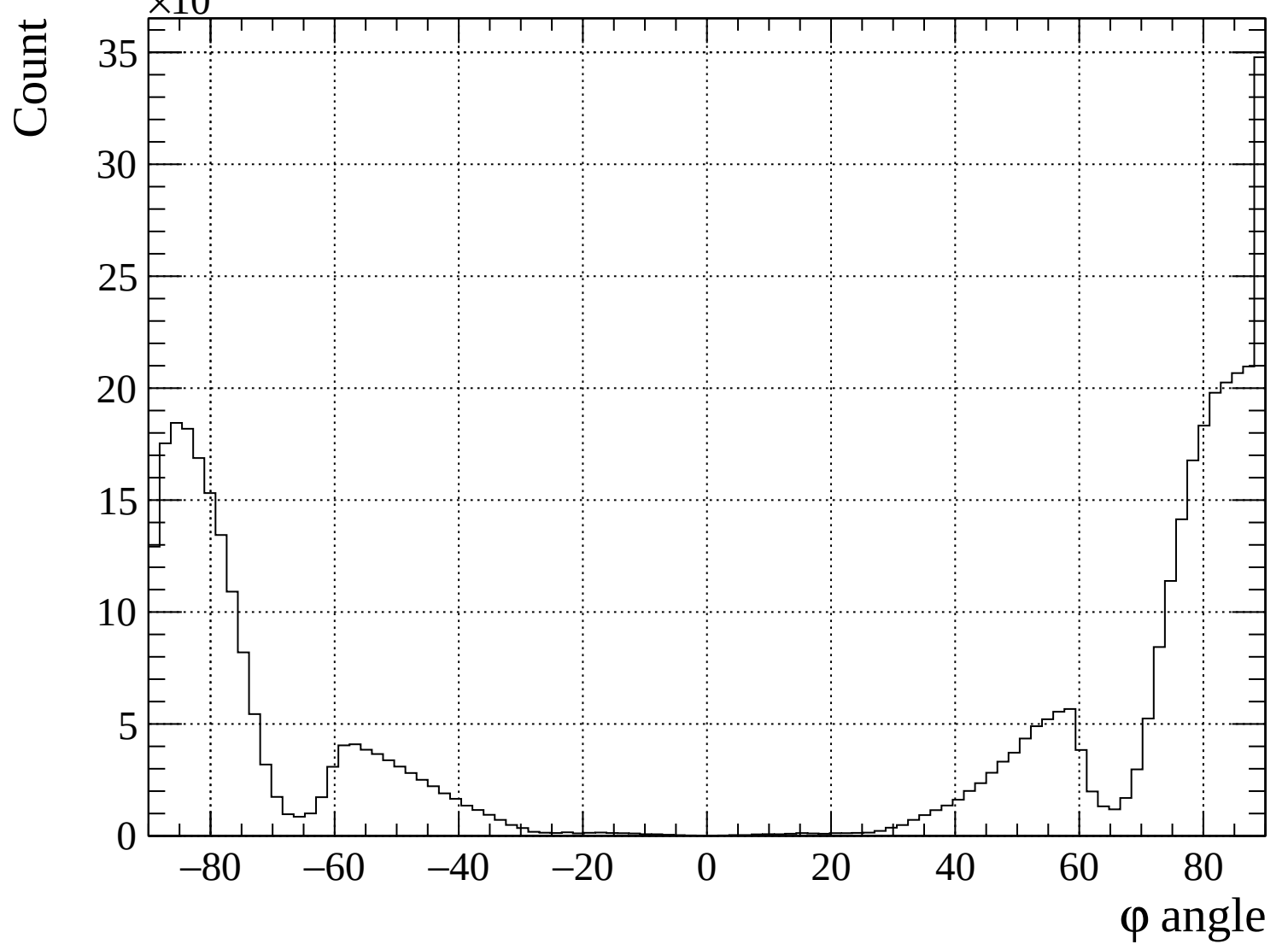


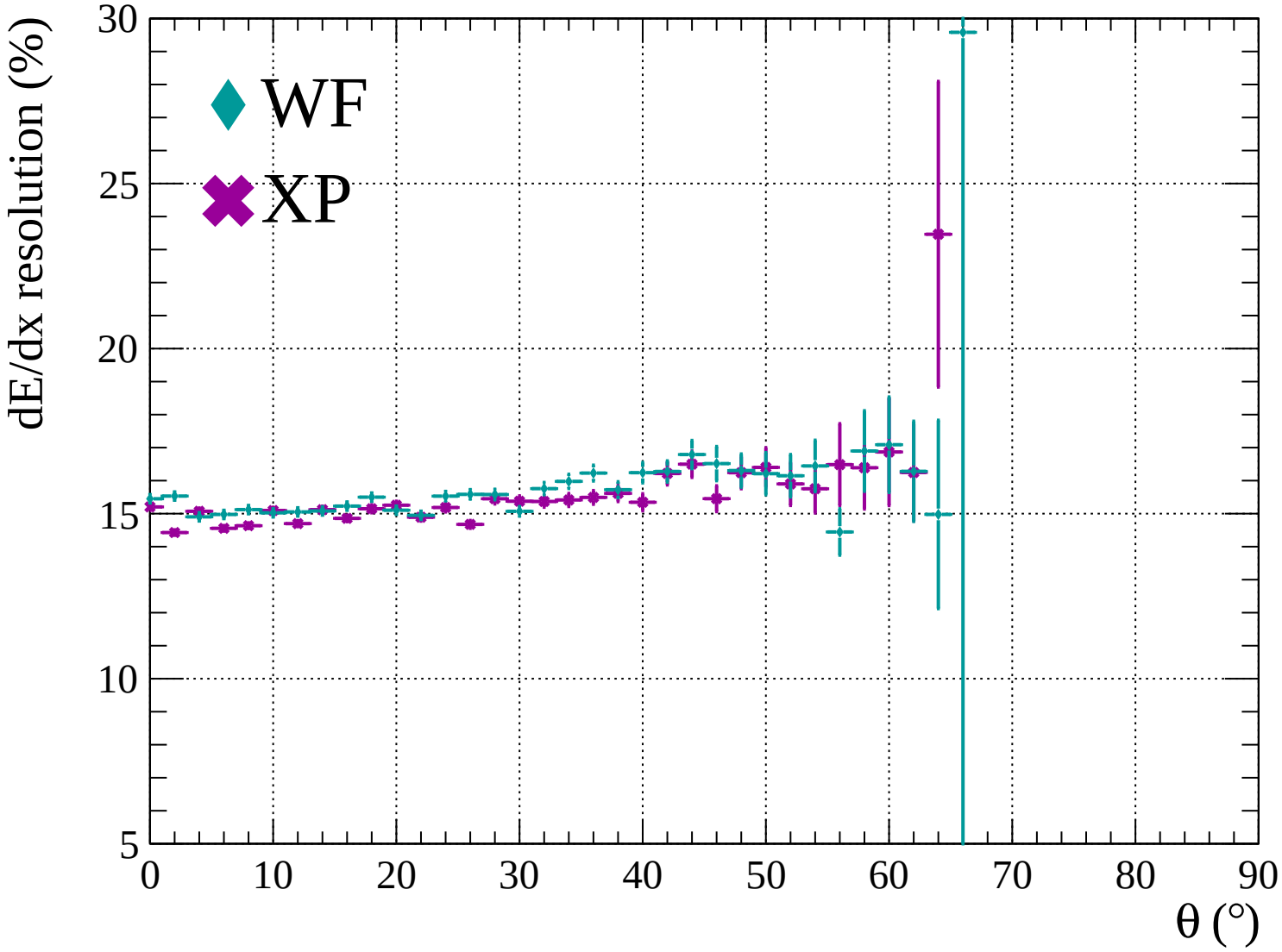


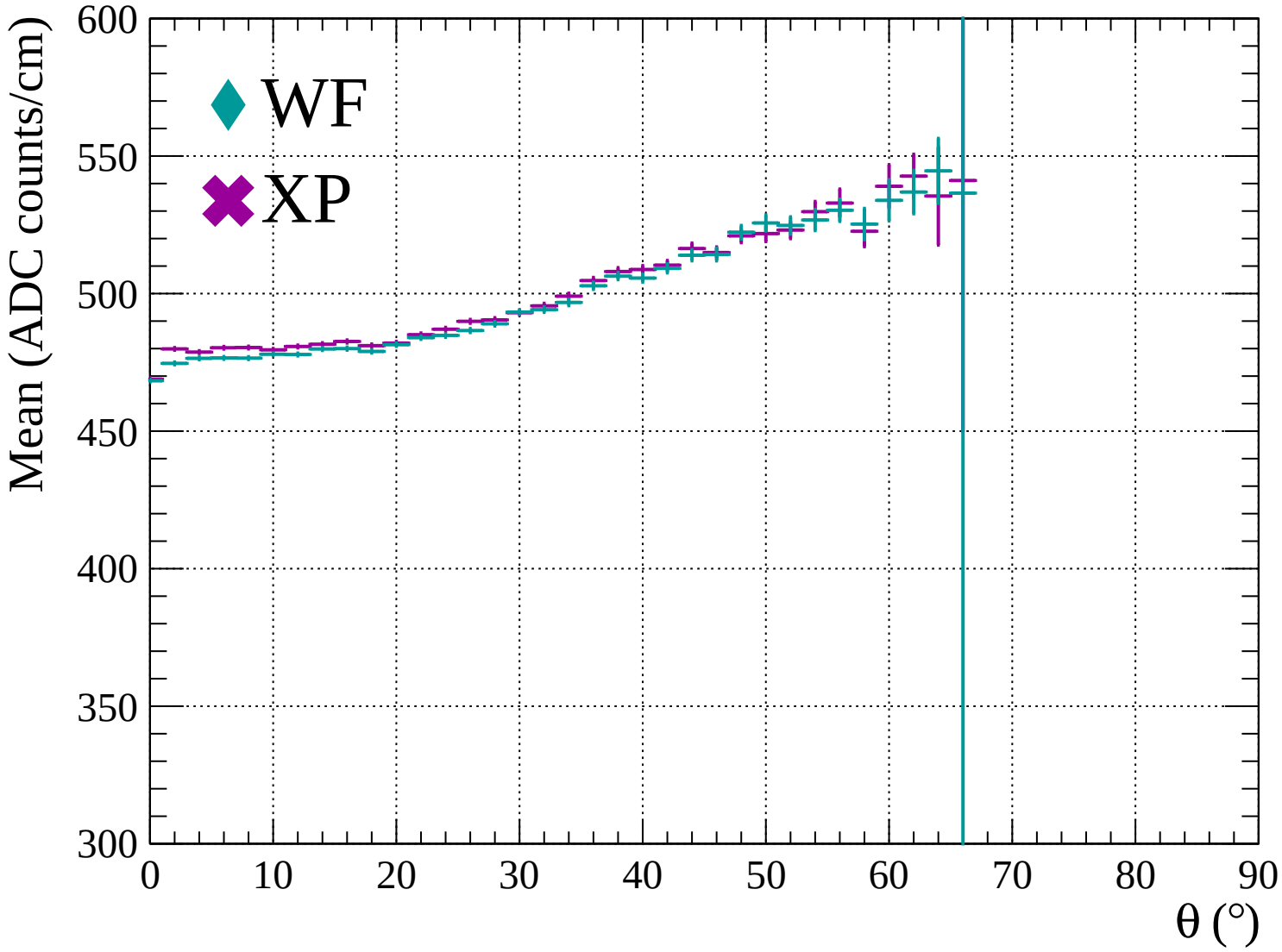


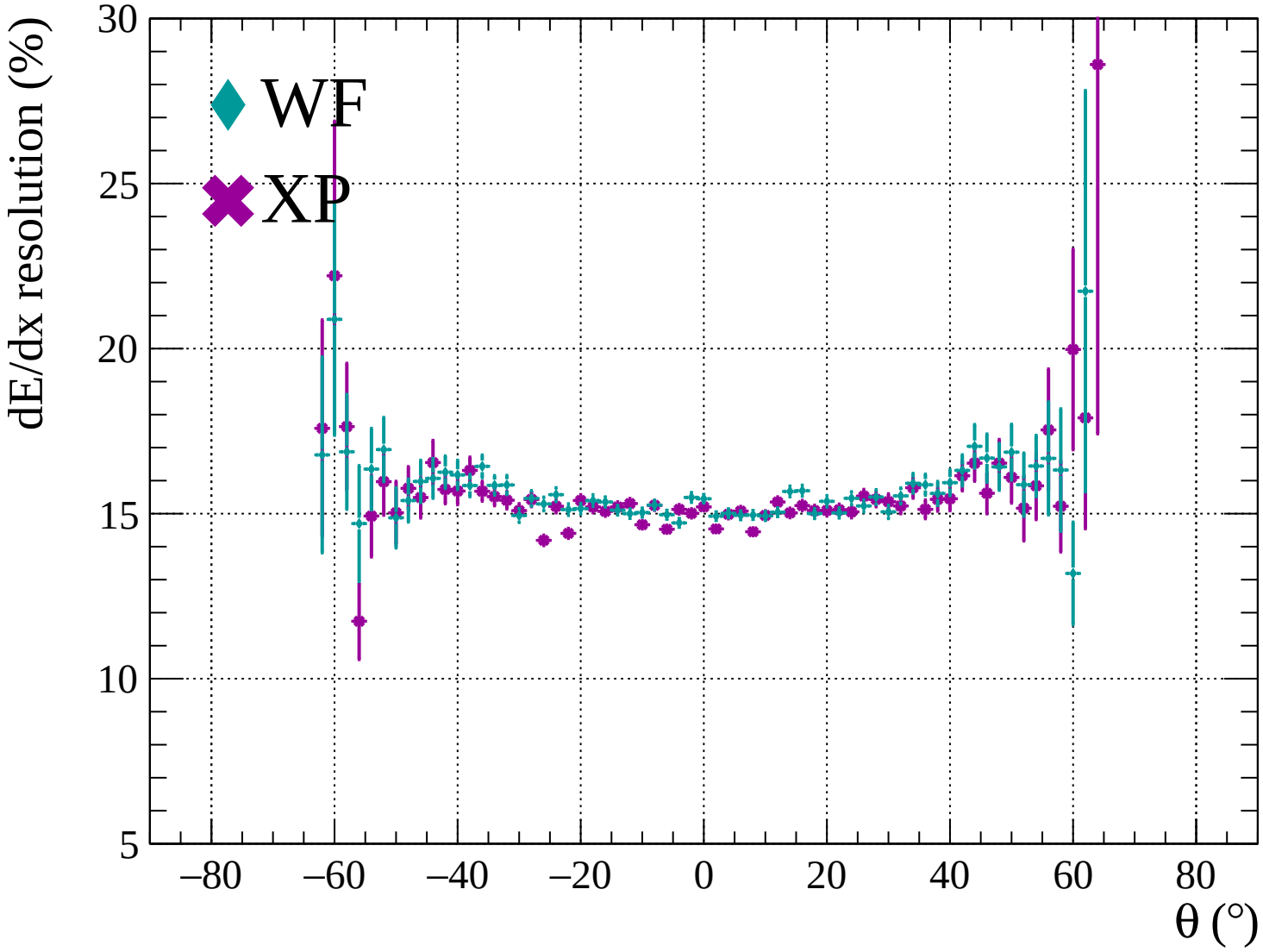


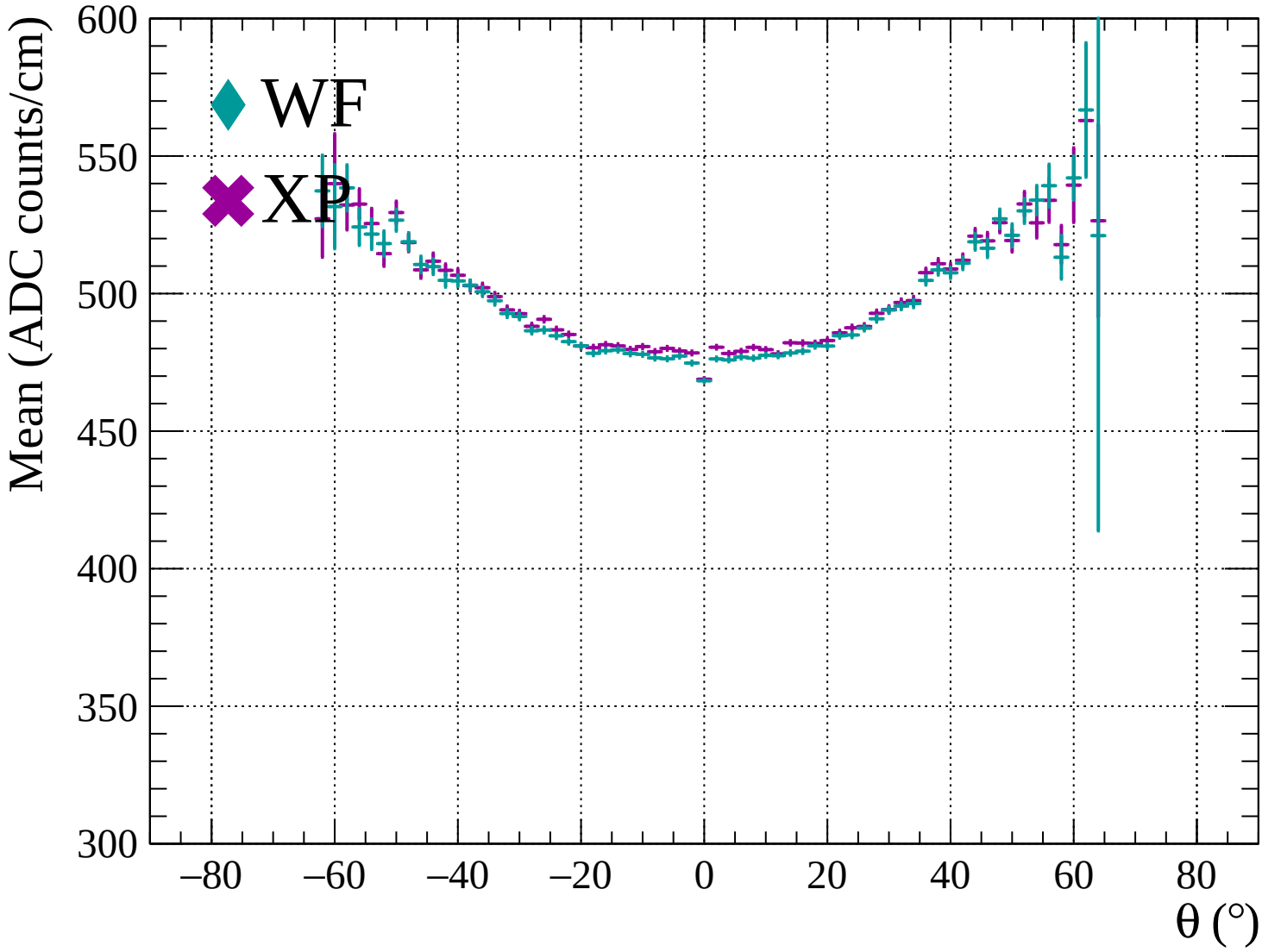


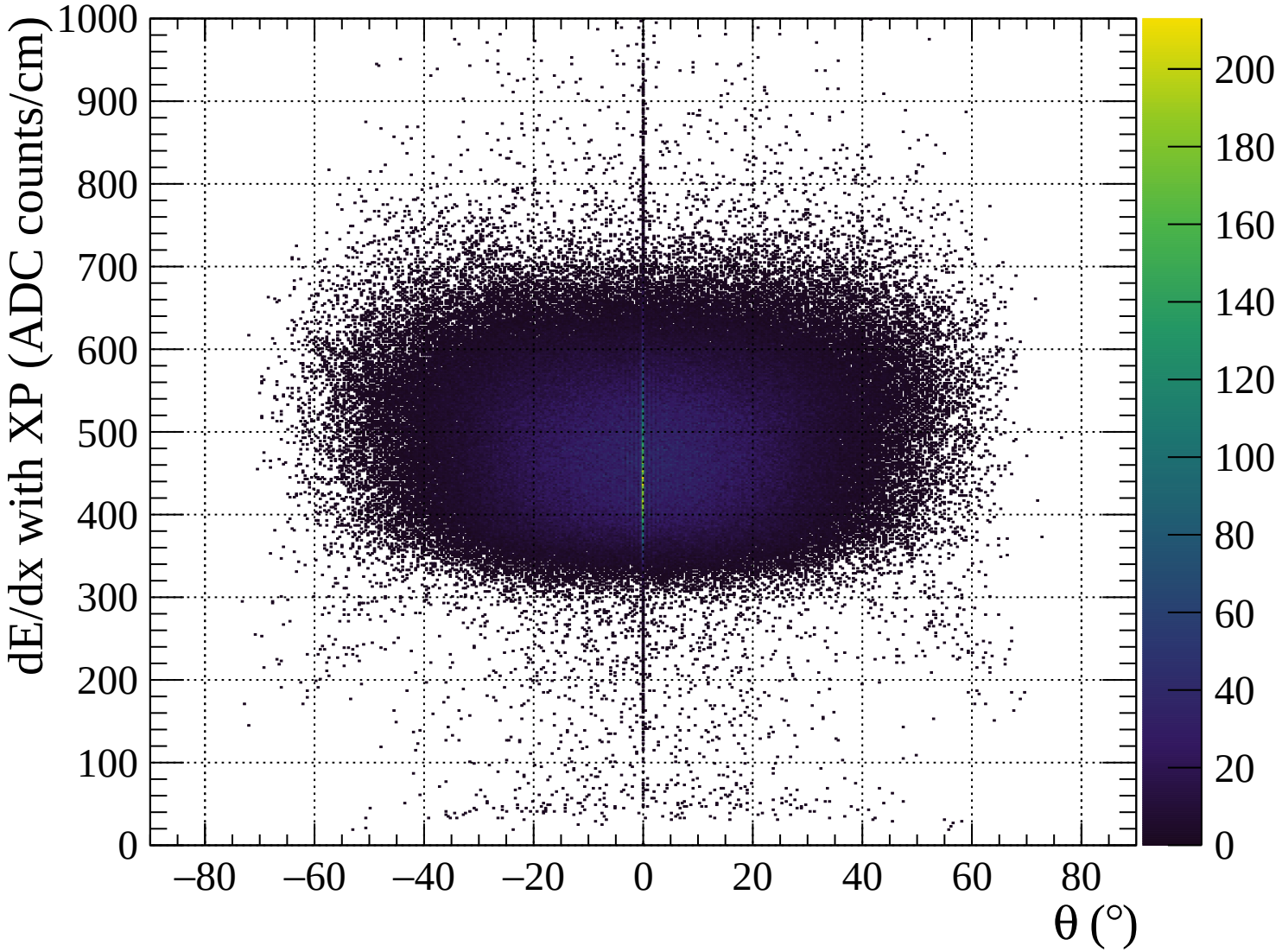


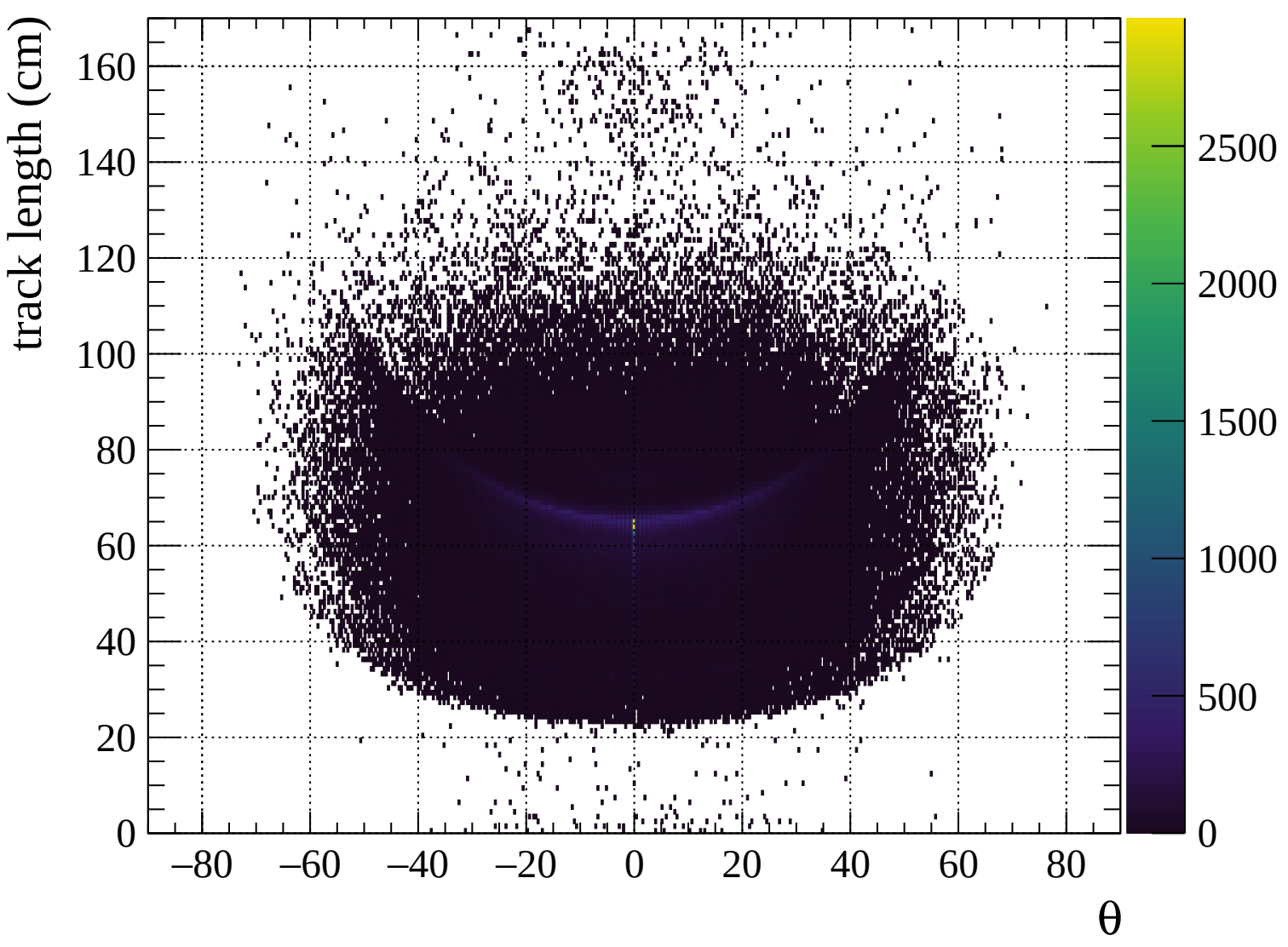


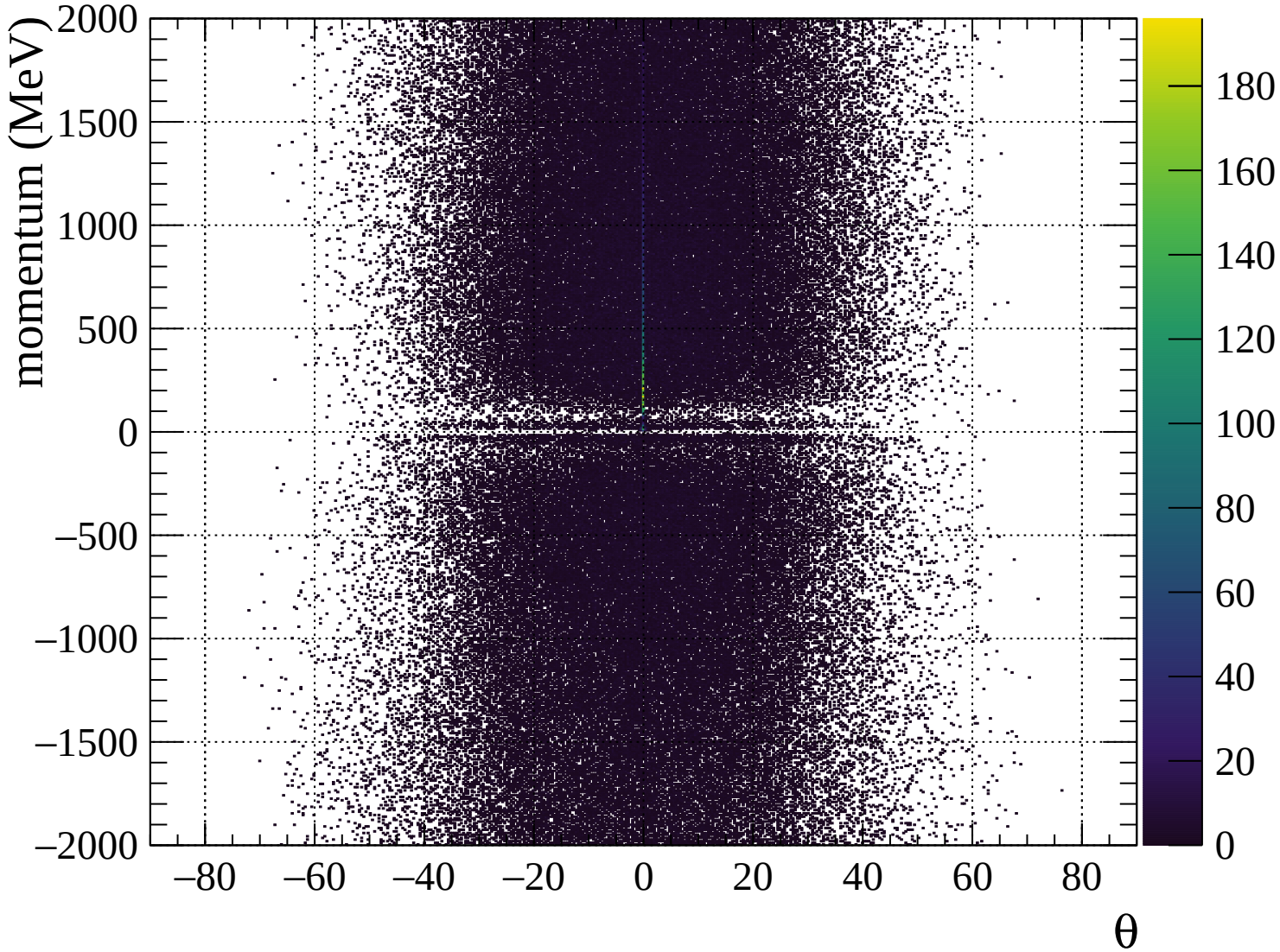


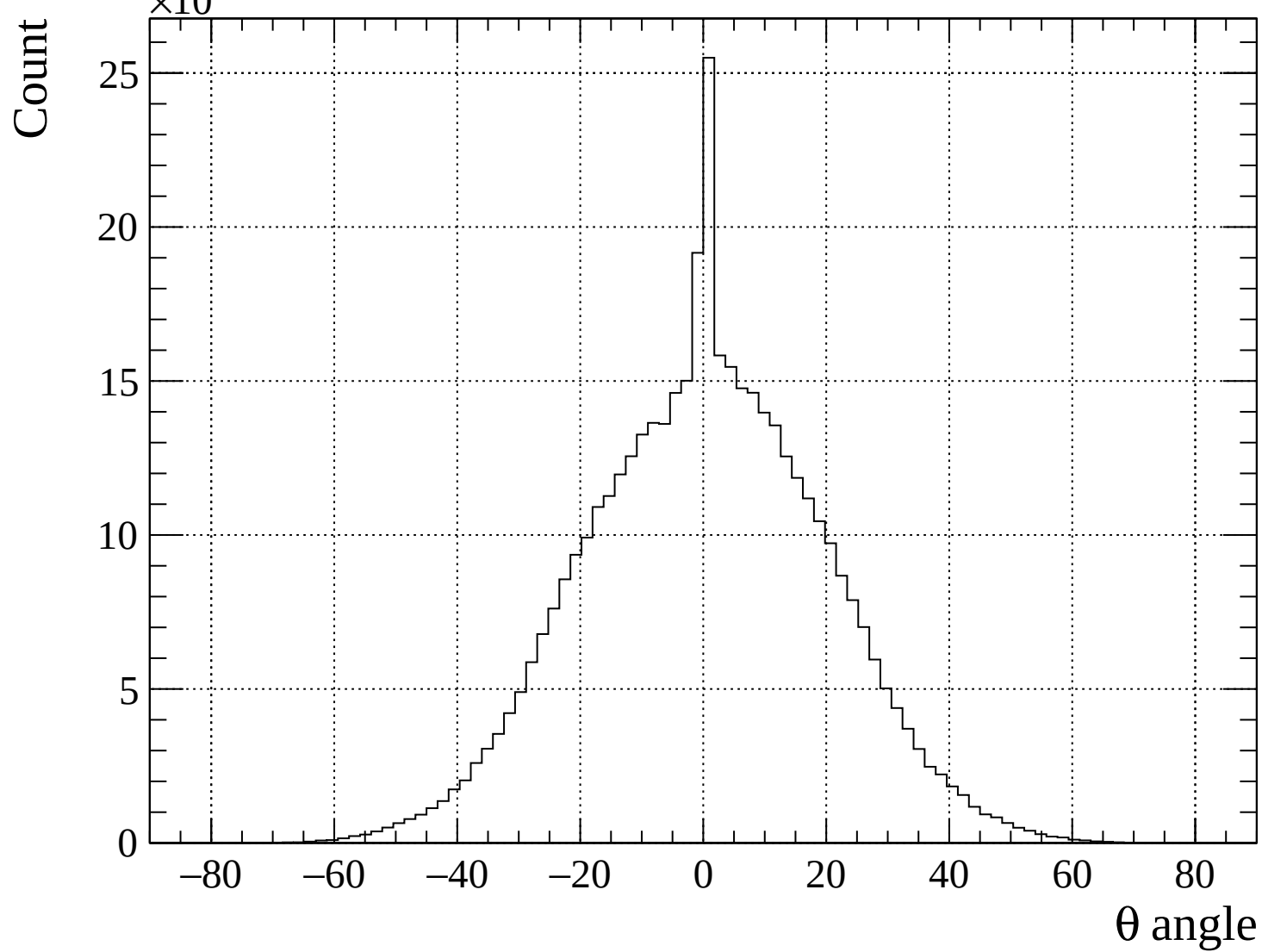


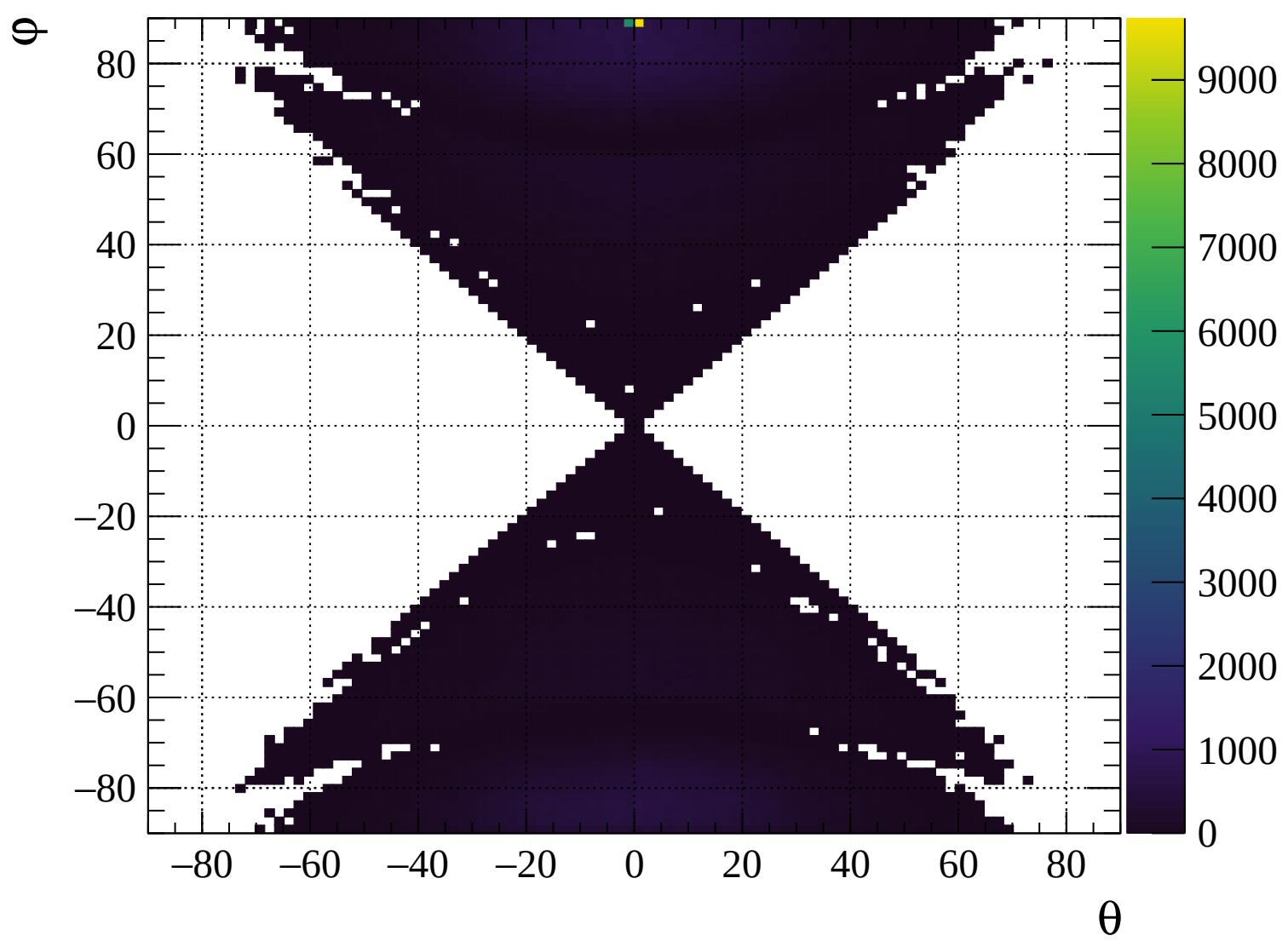


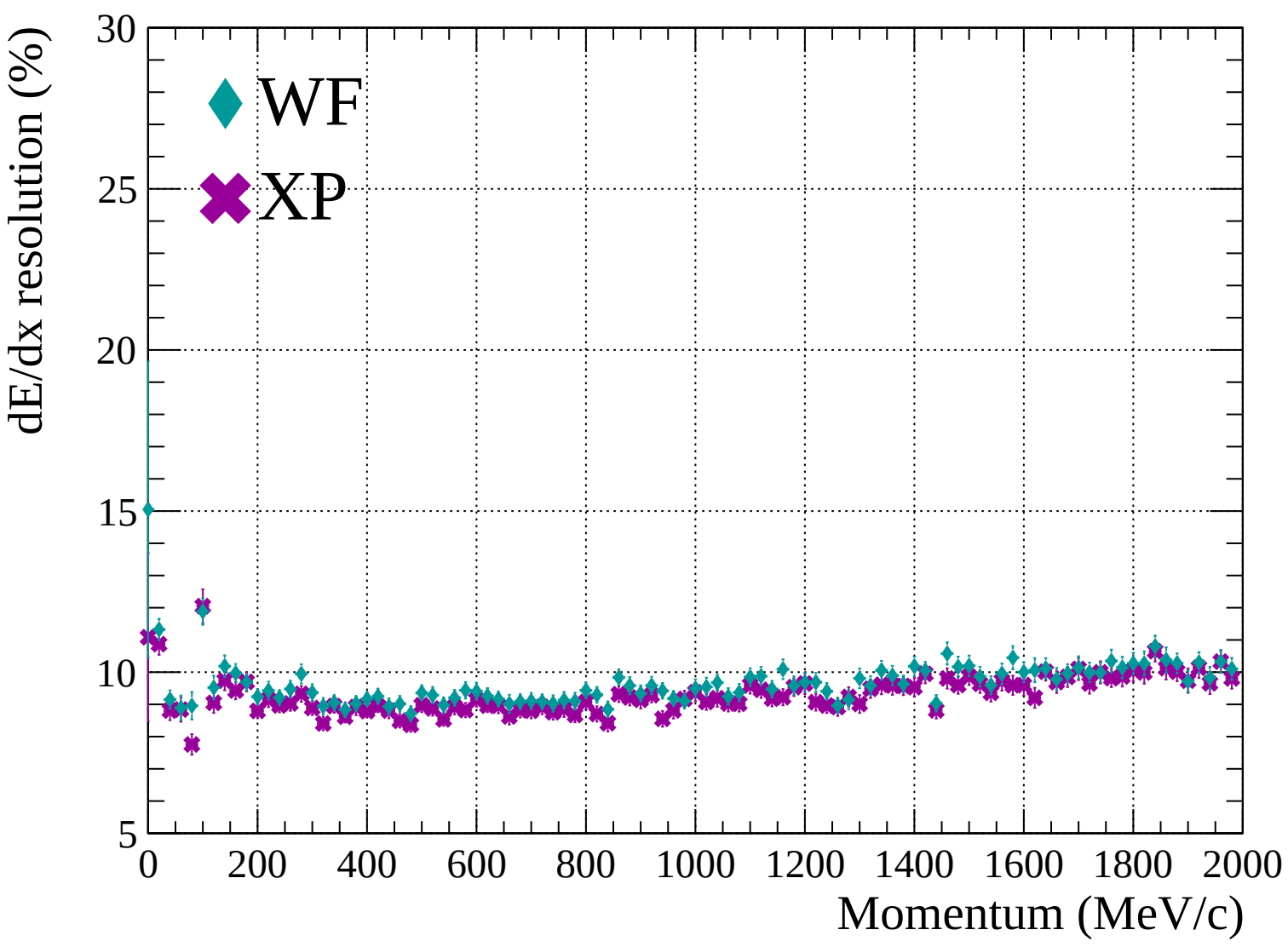


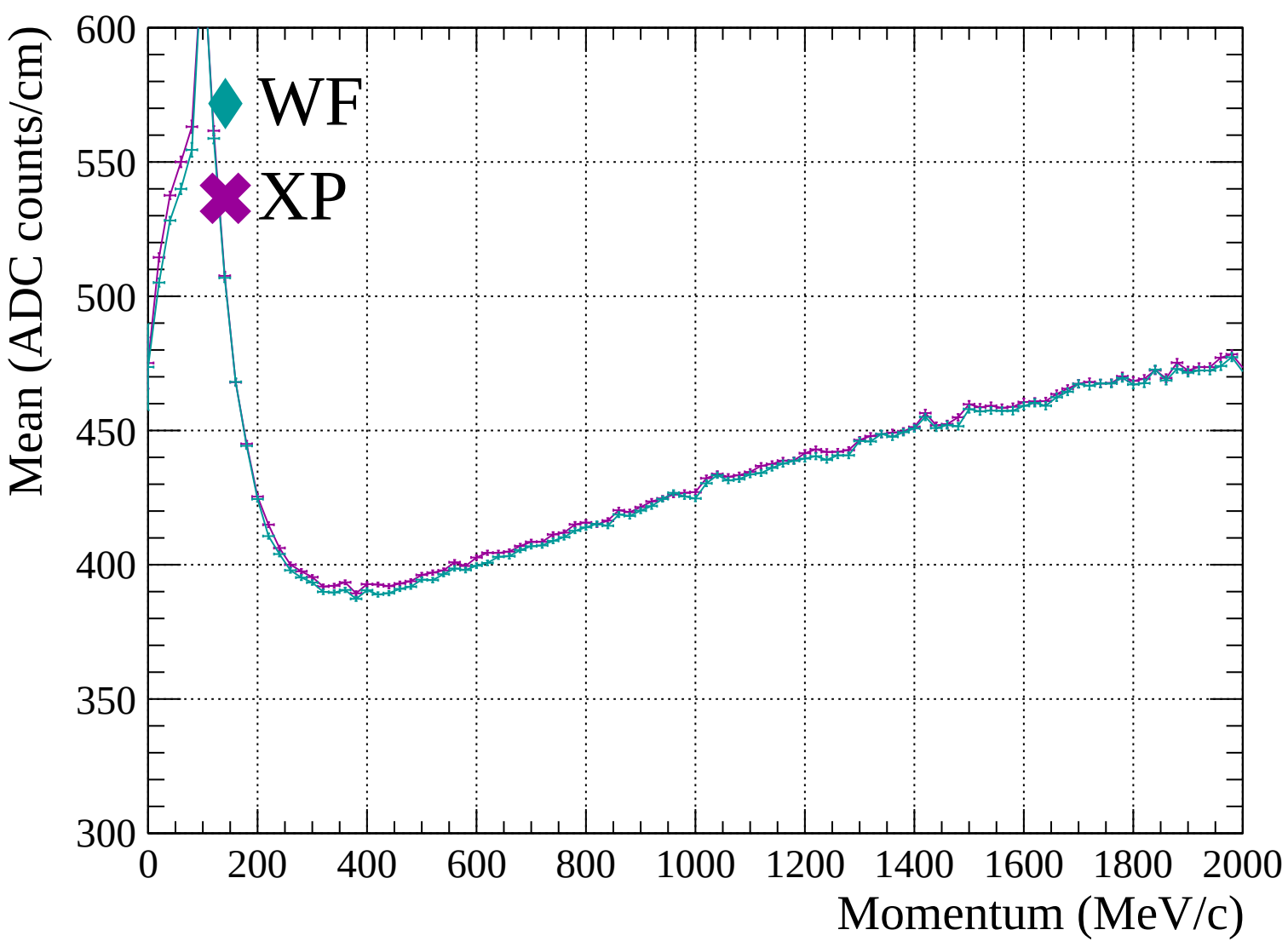


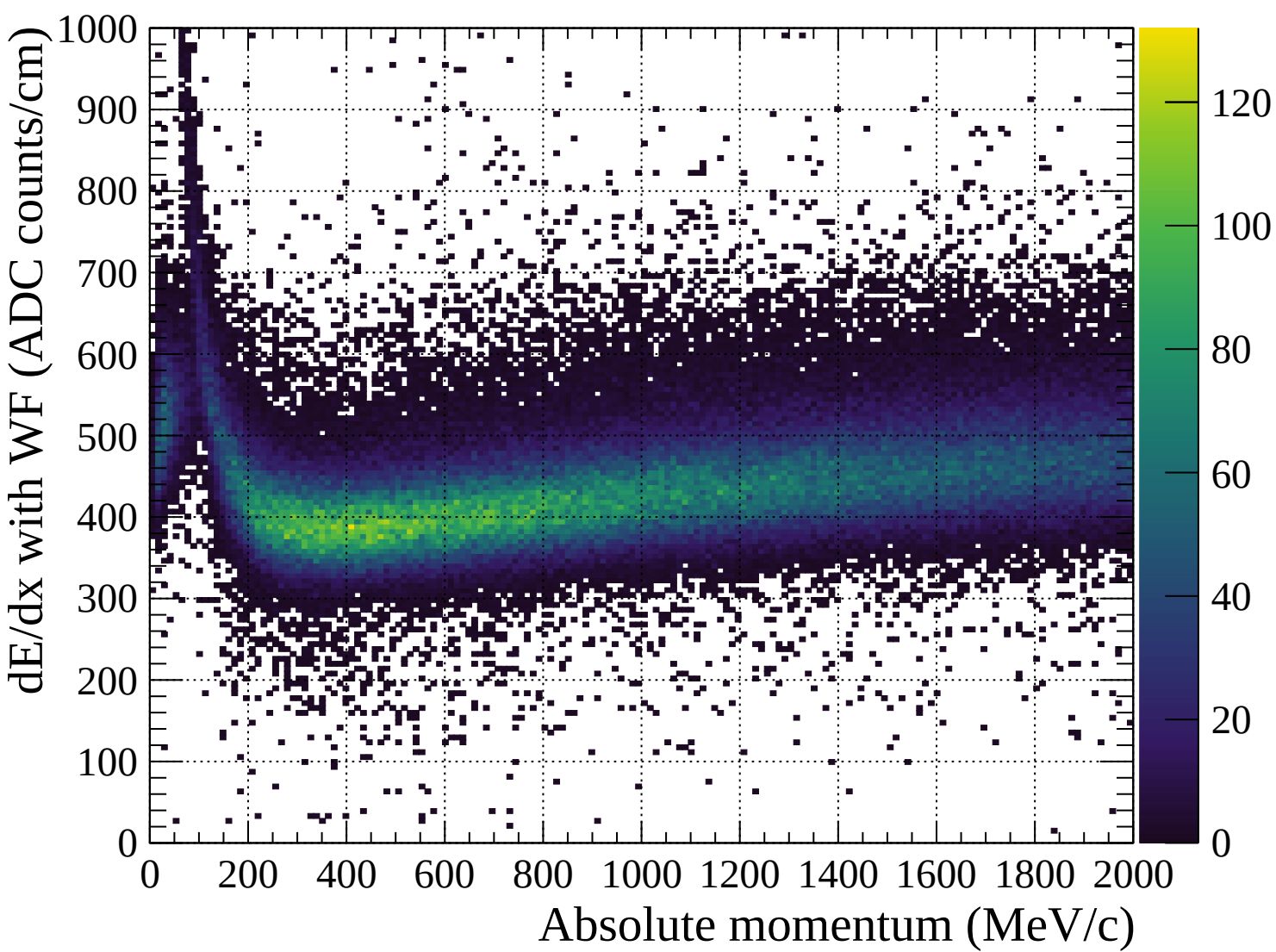


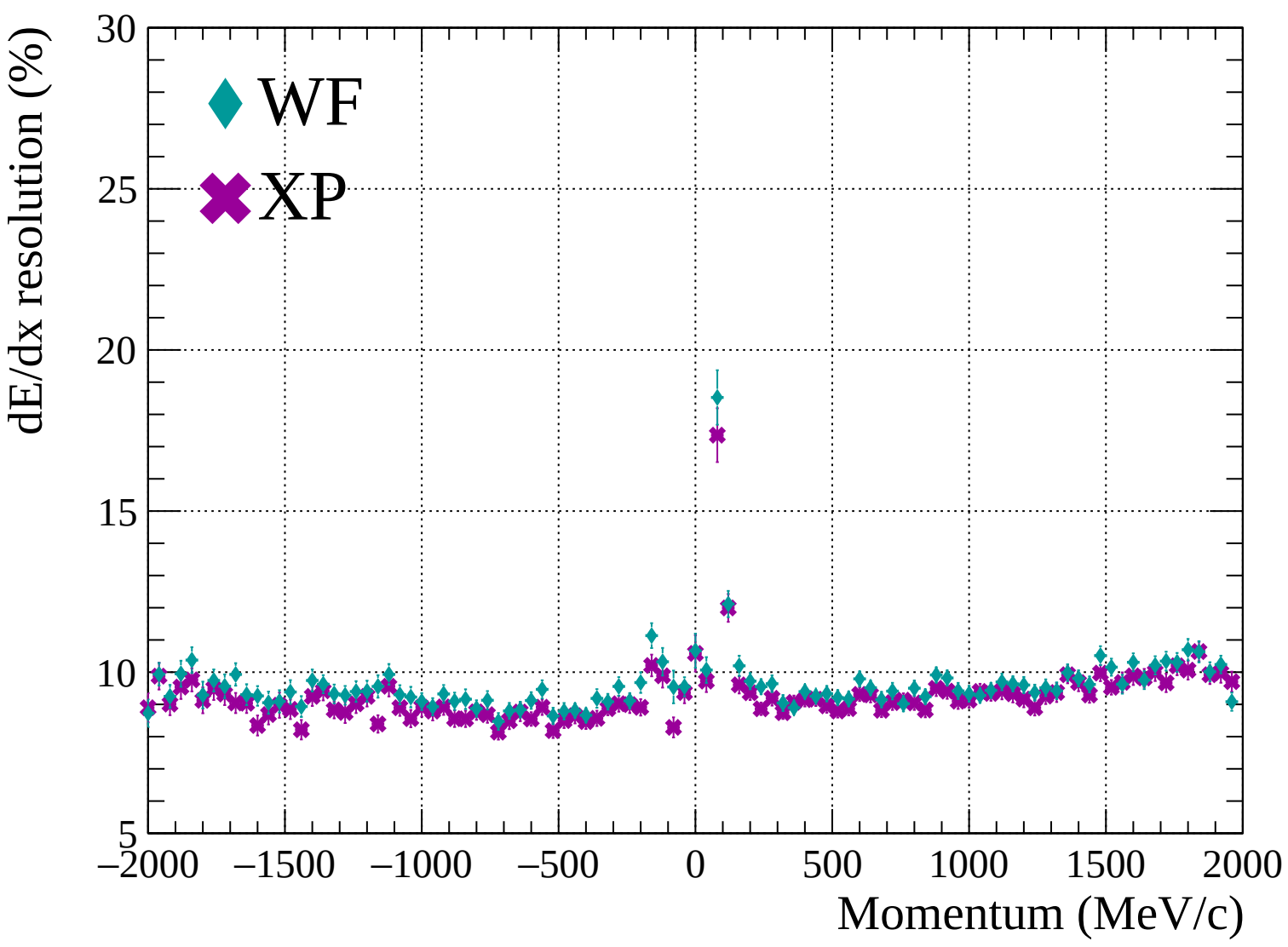


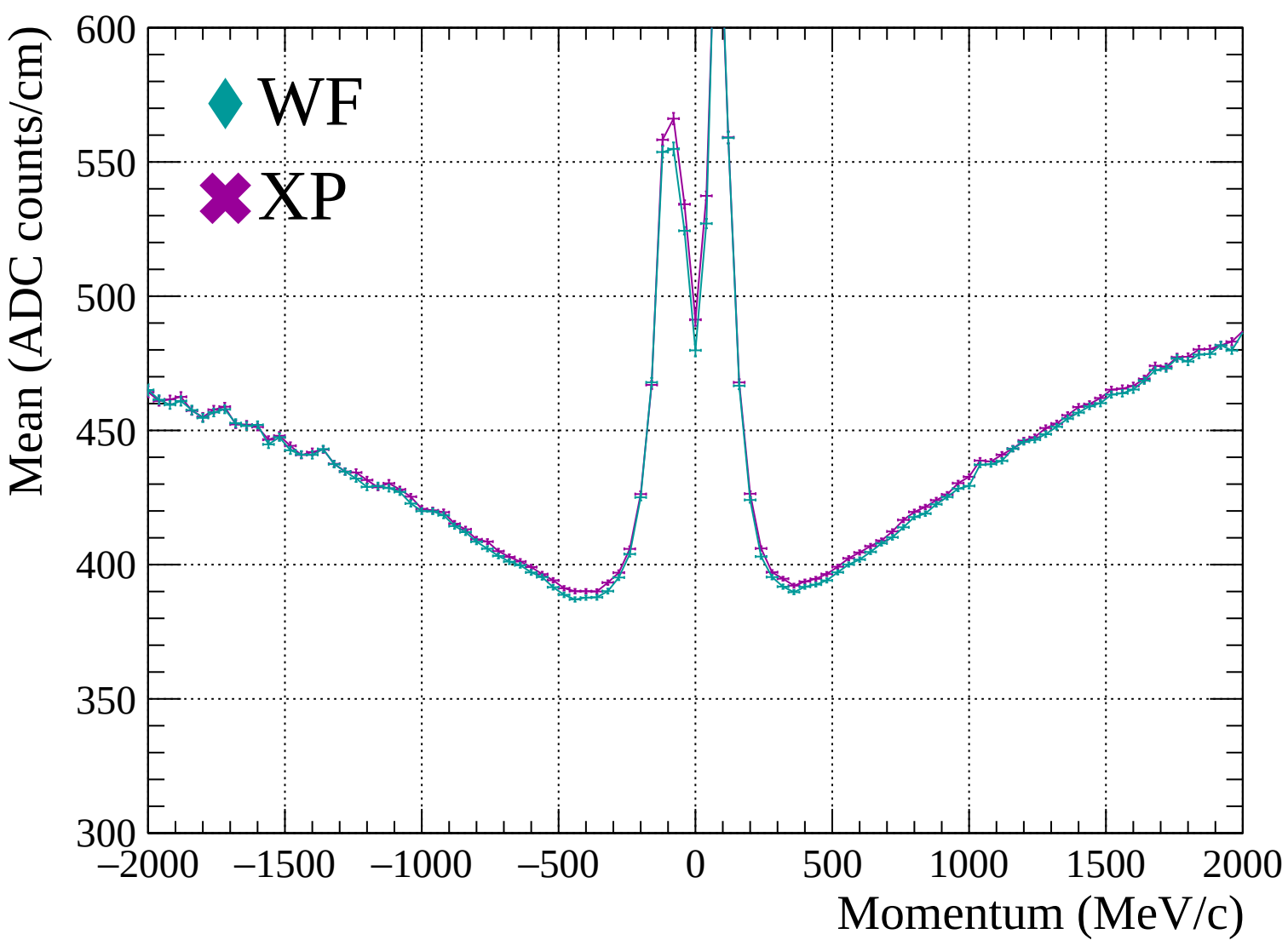


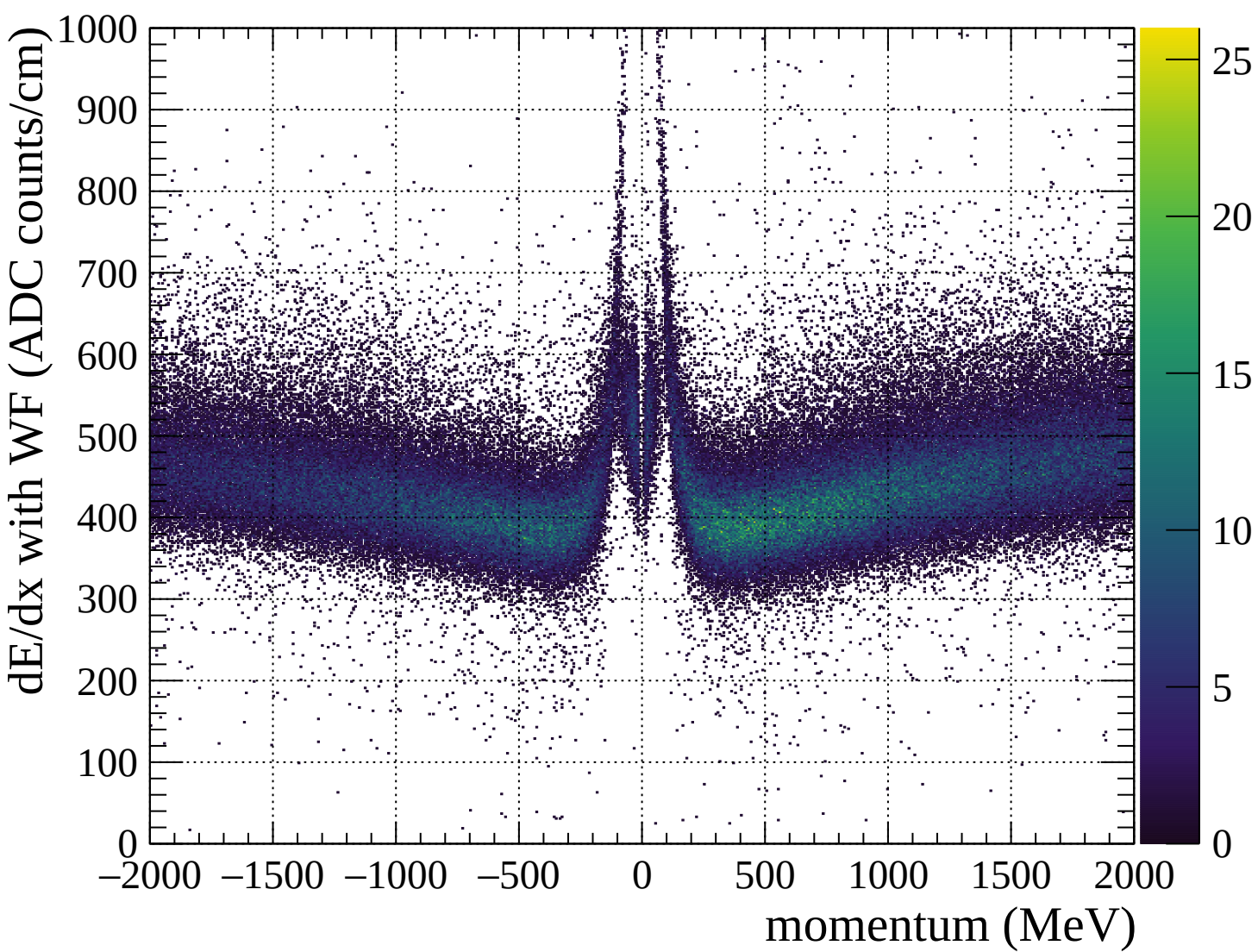


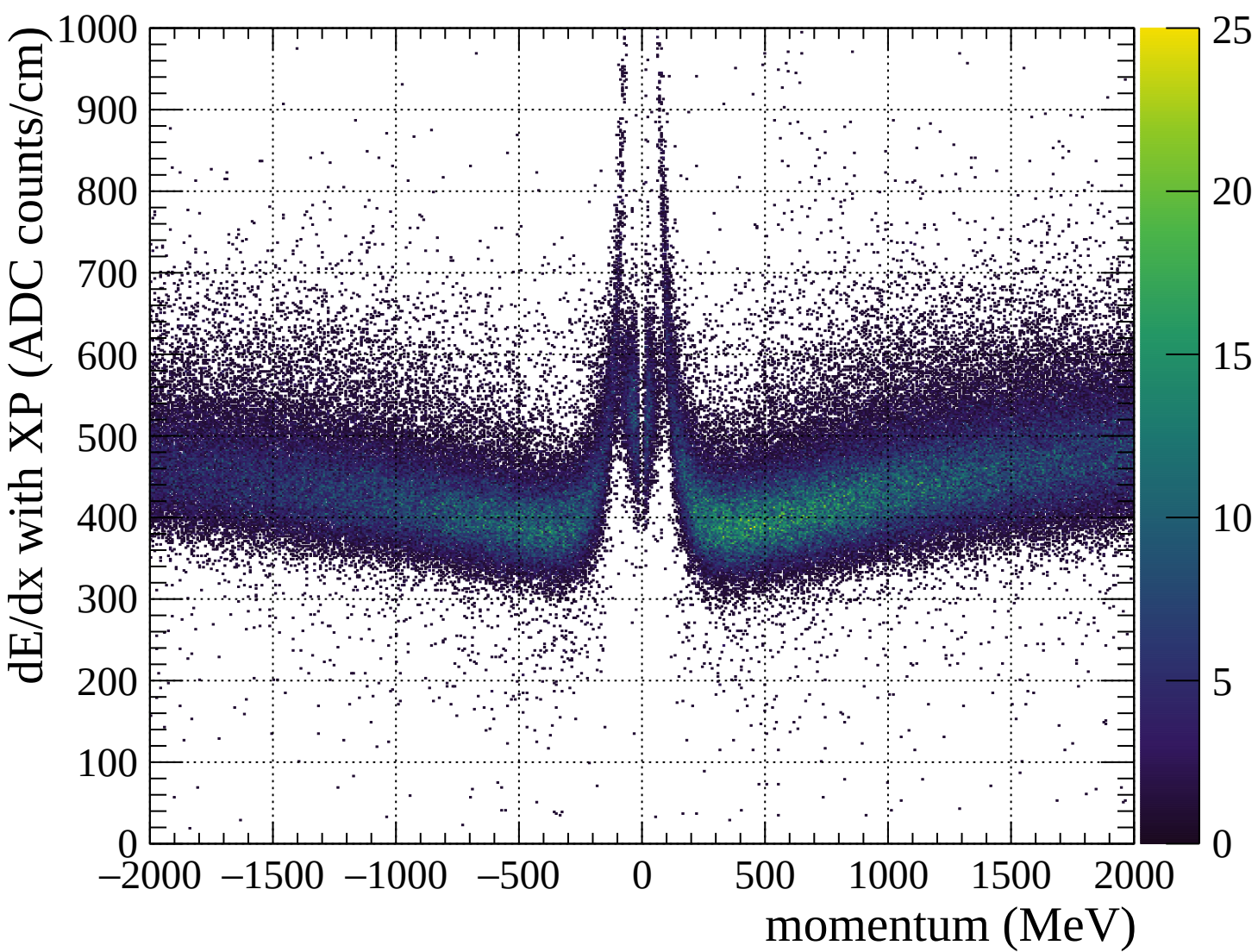


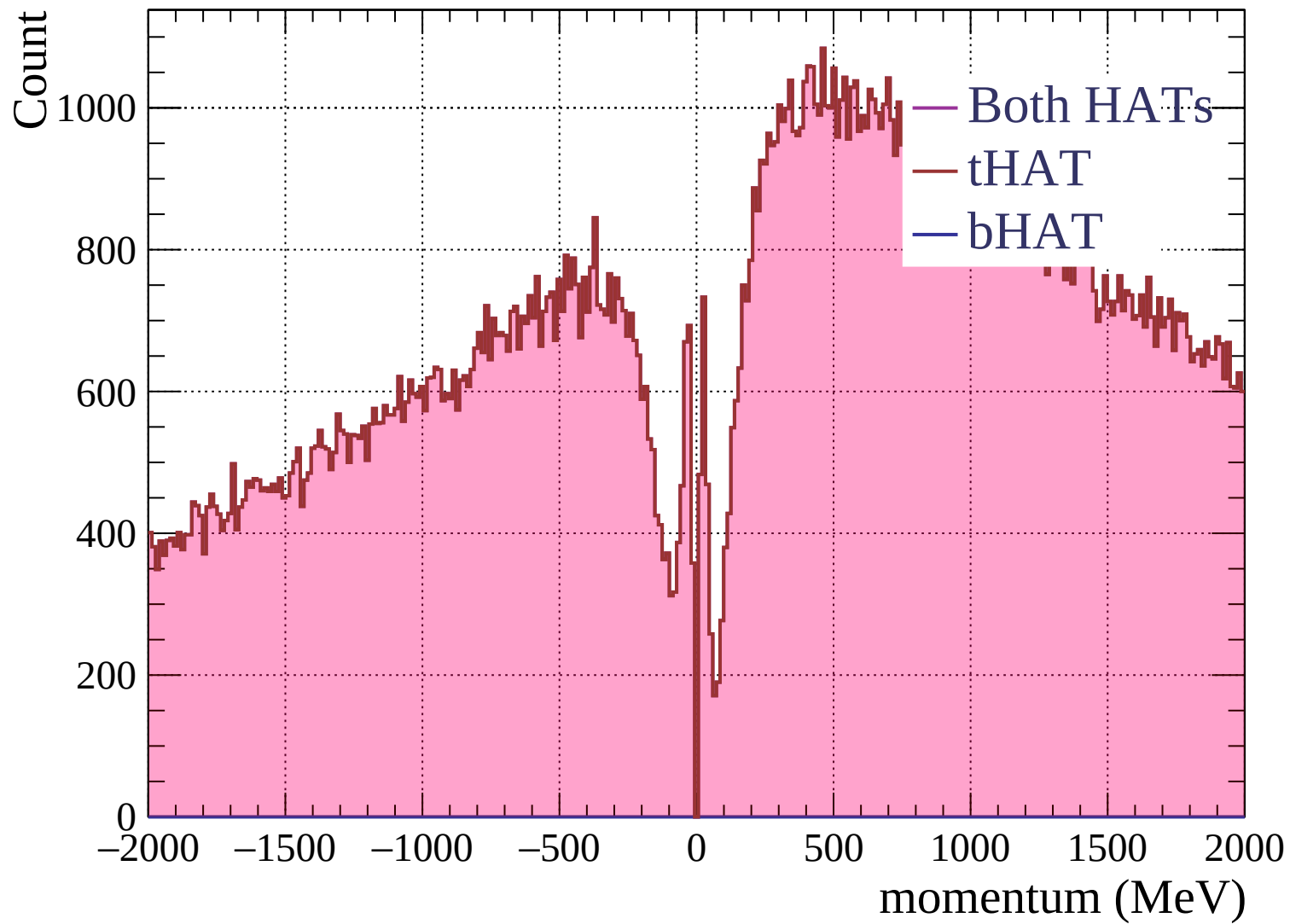


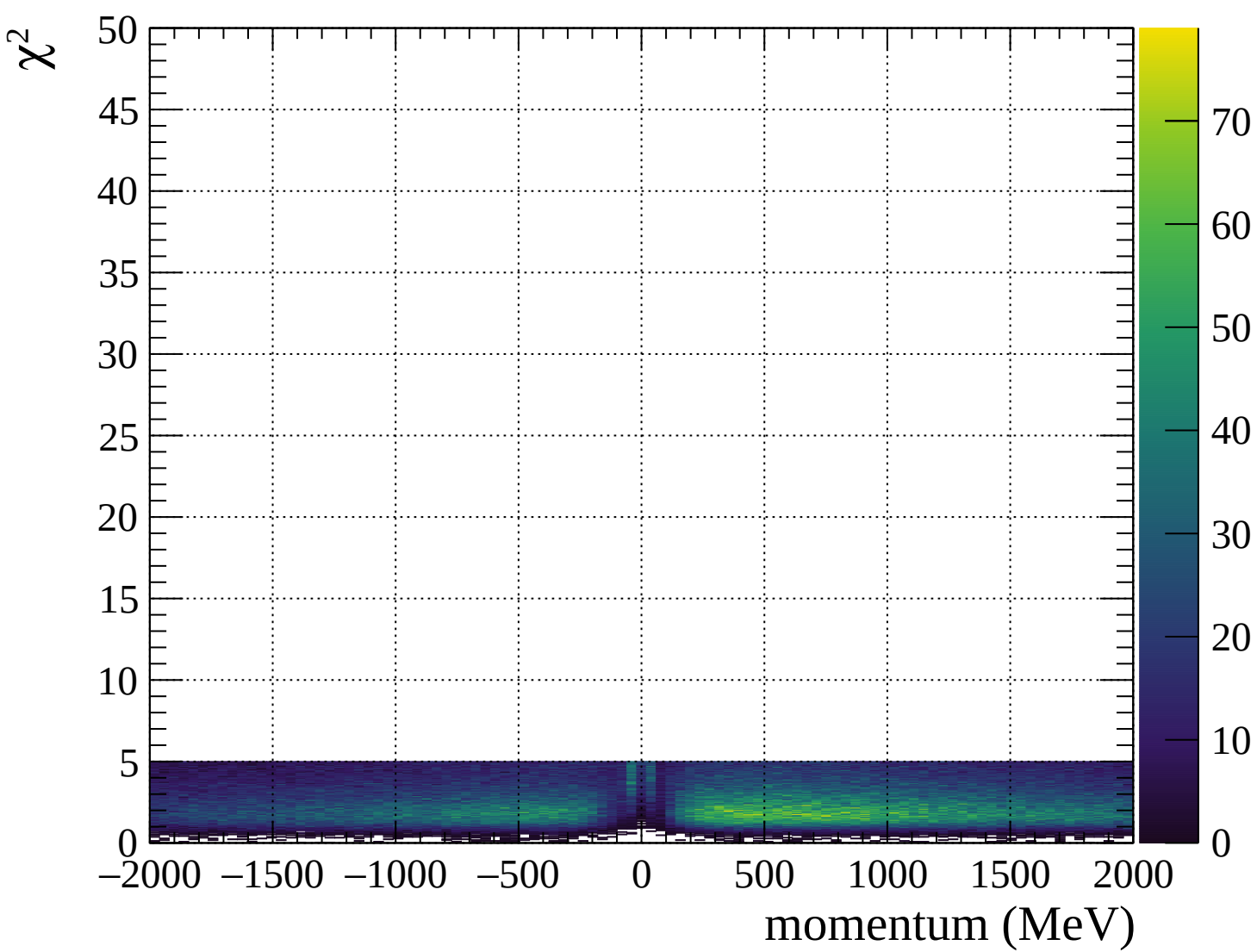




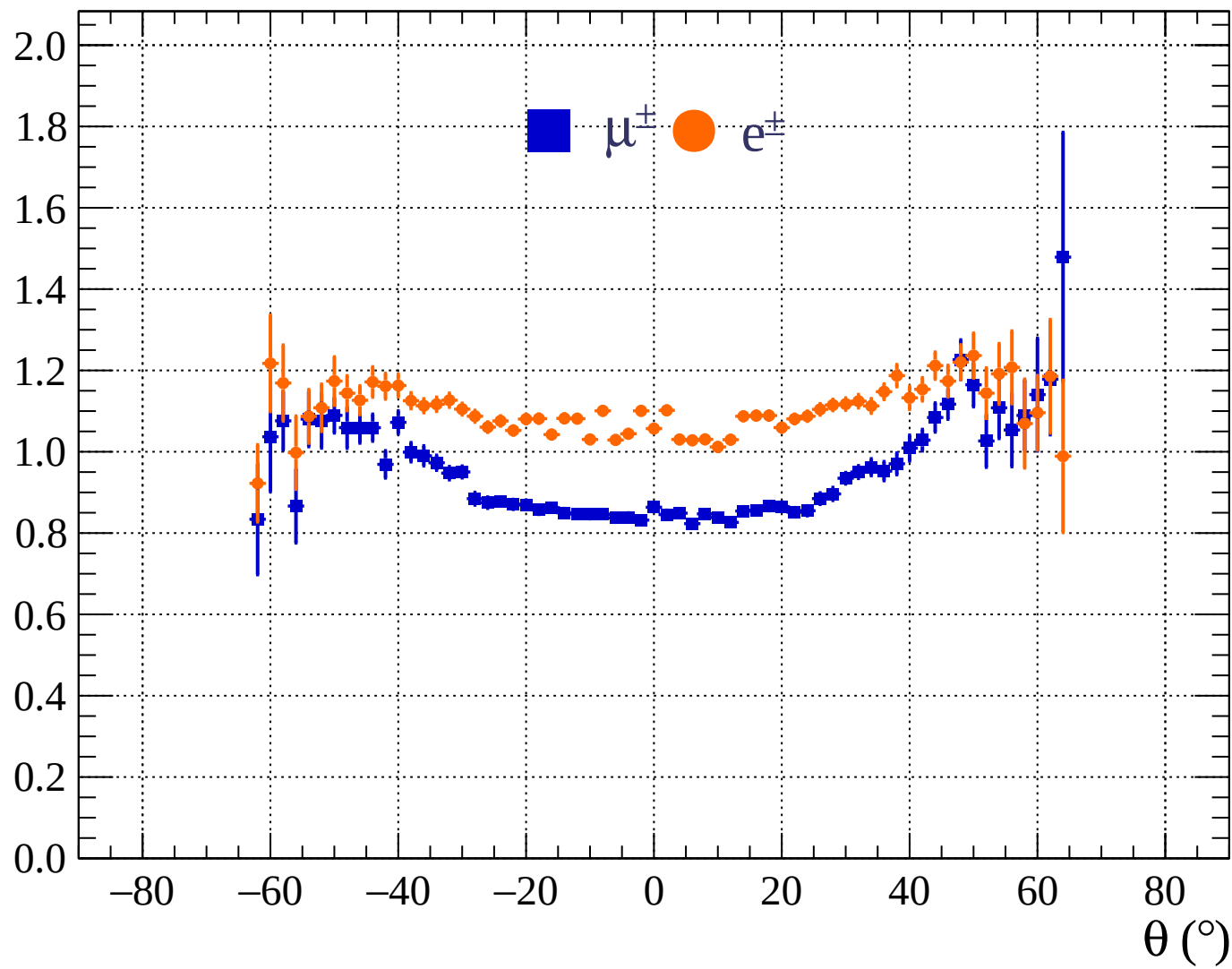


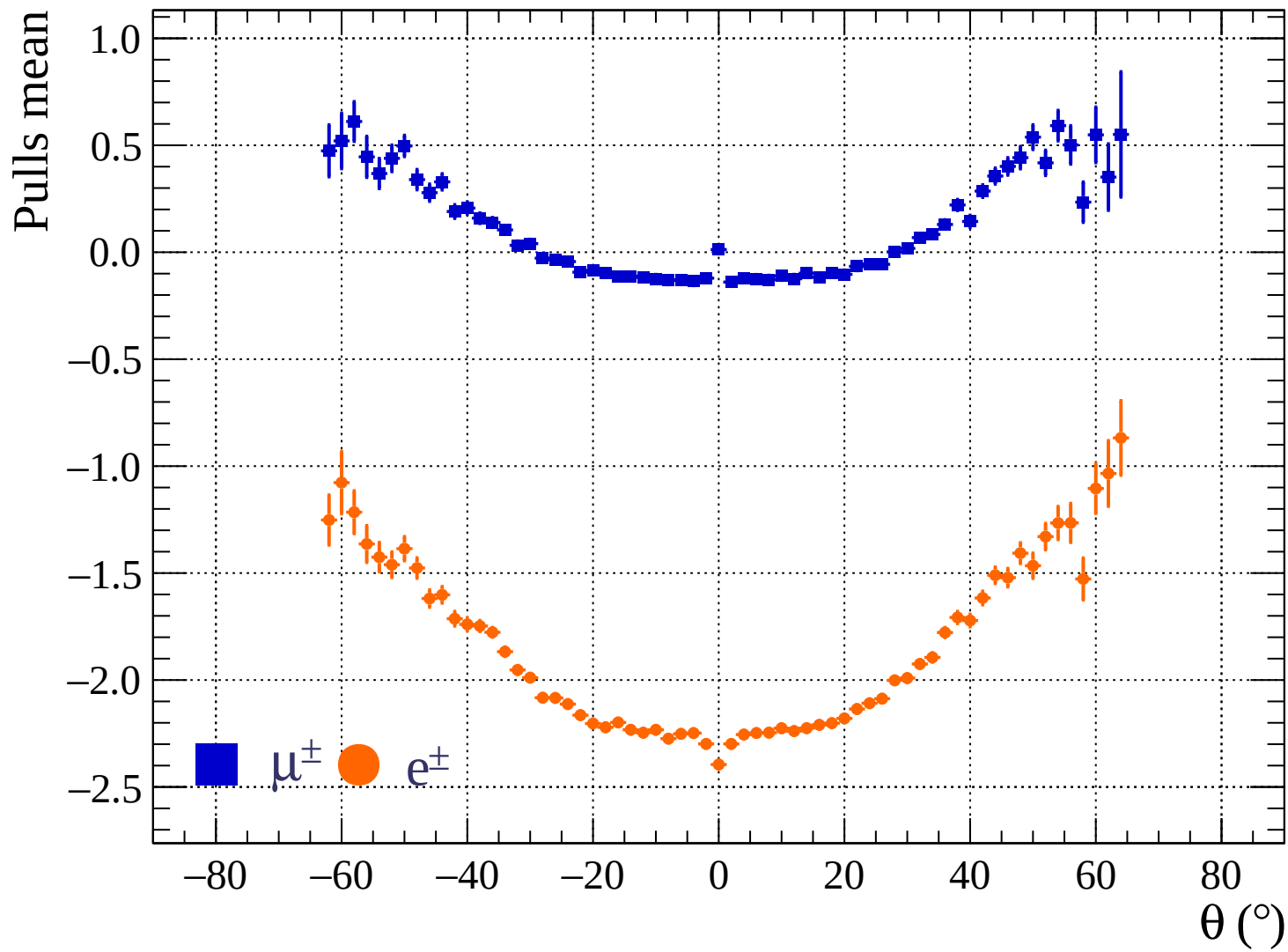




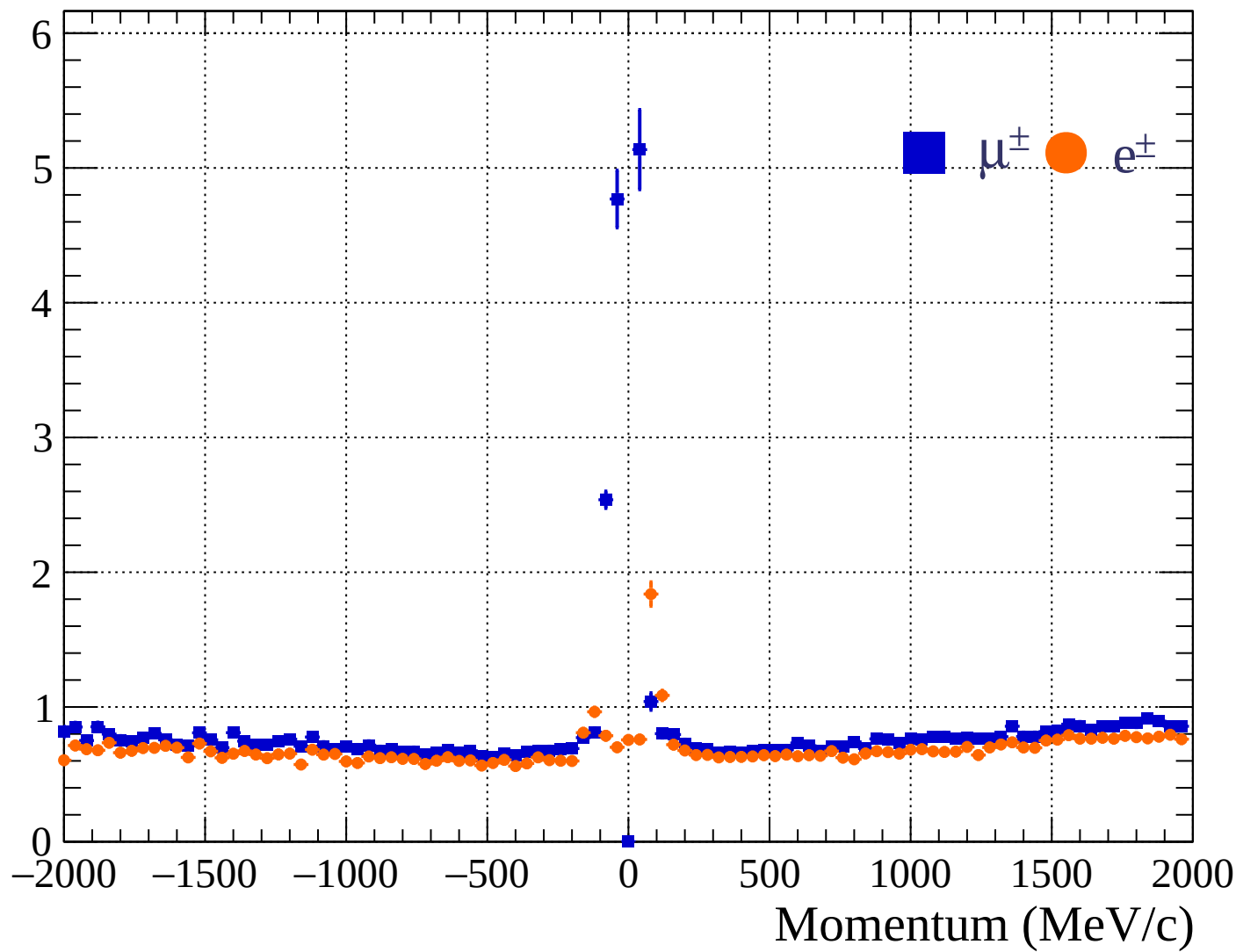


Pulls standard deviation

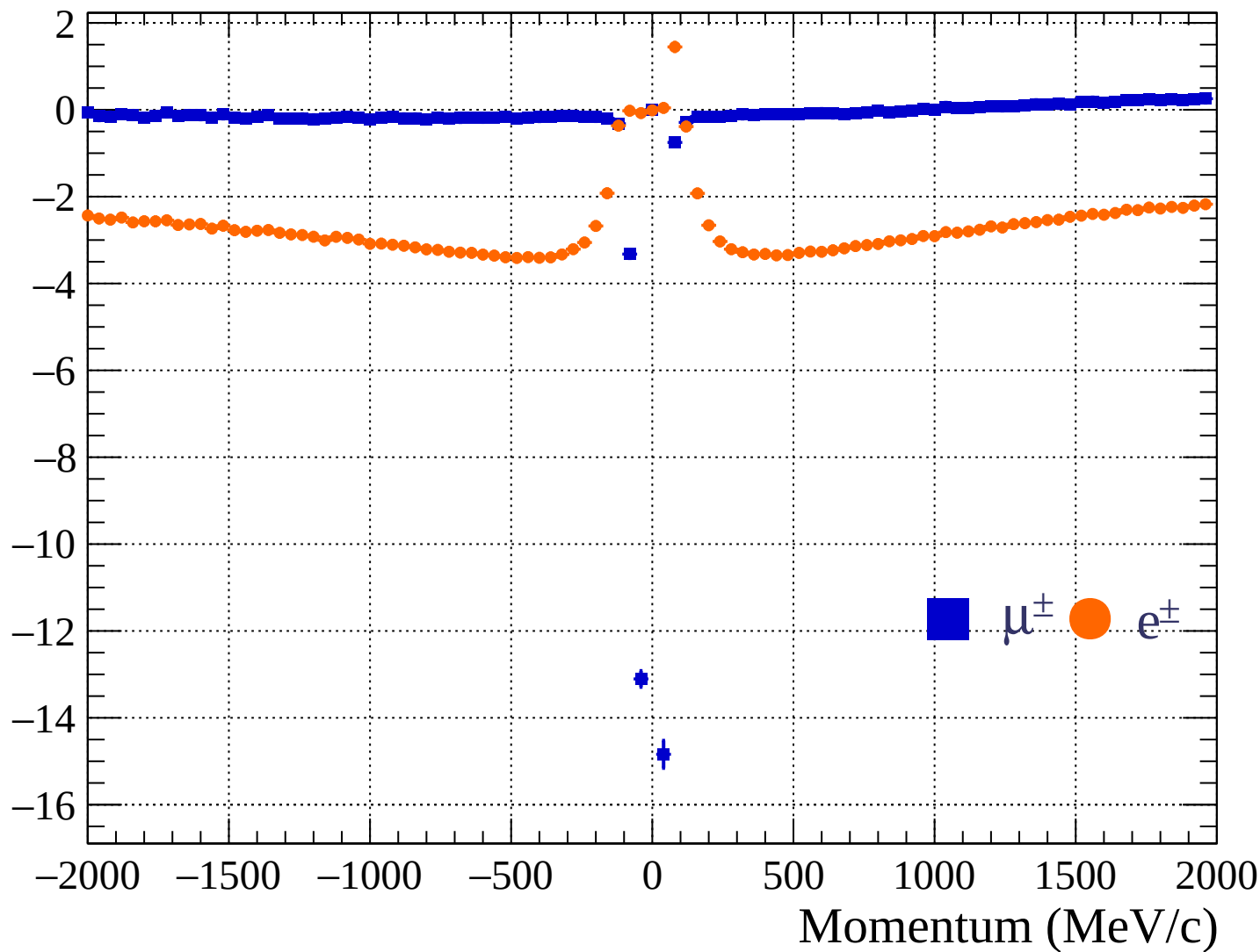


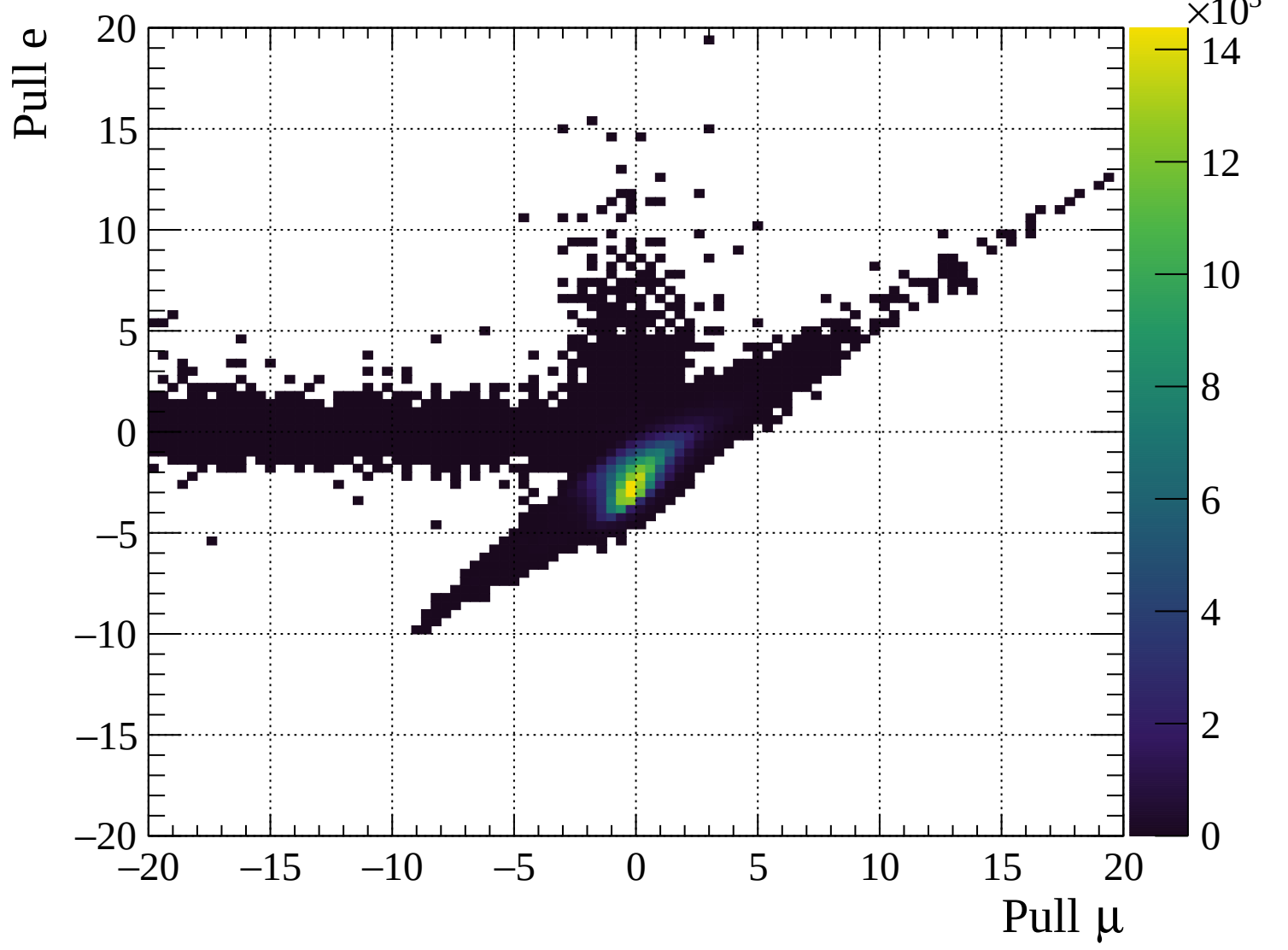


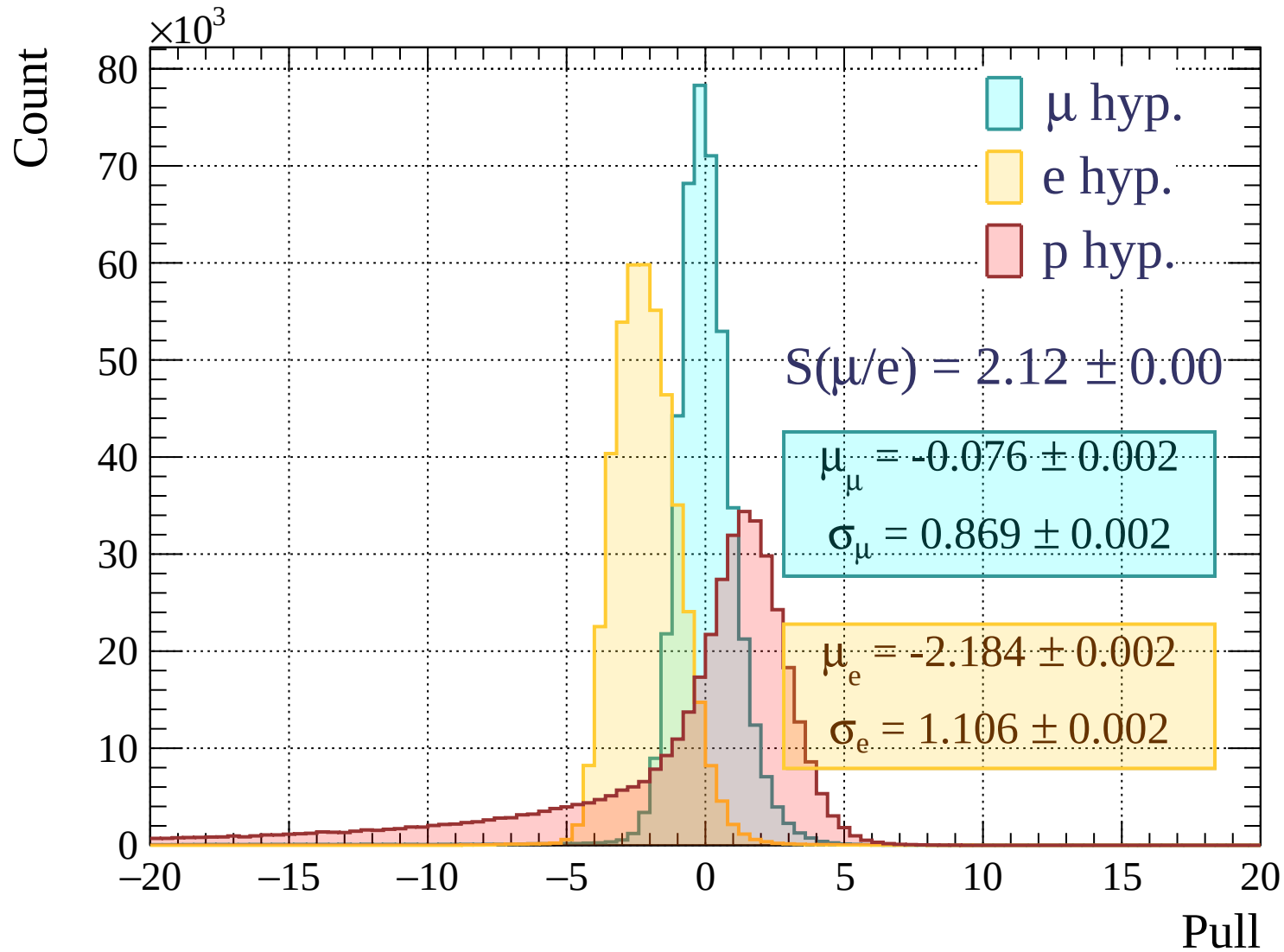
Pulls standard deviation

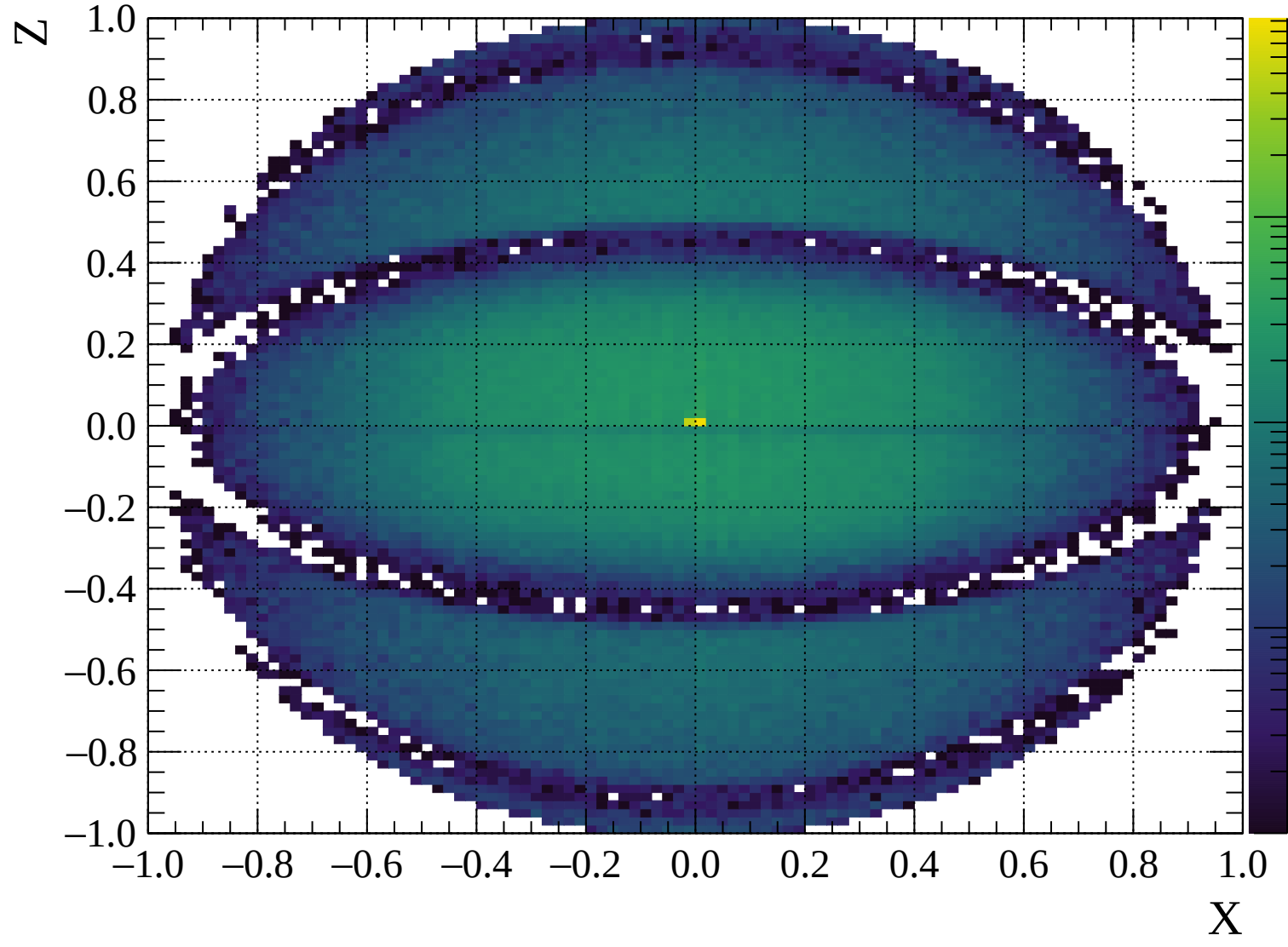


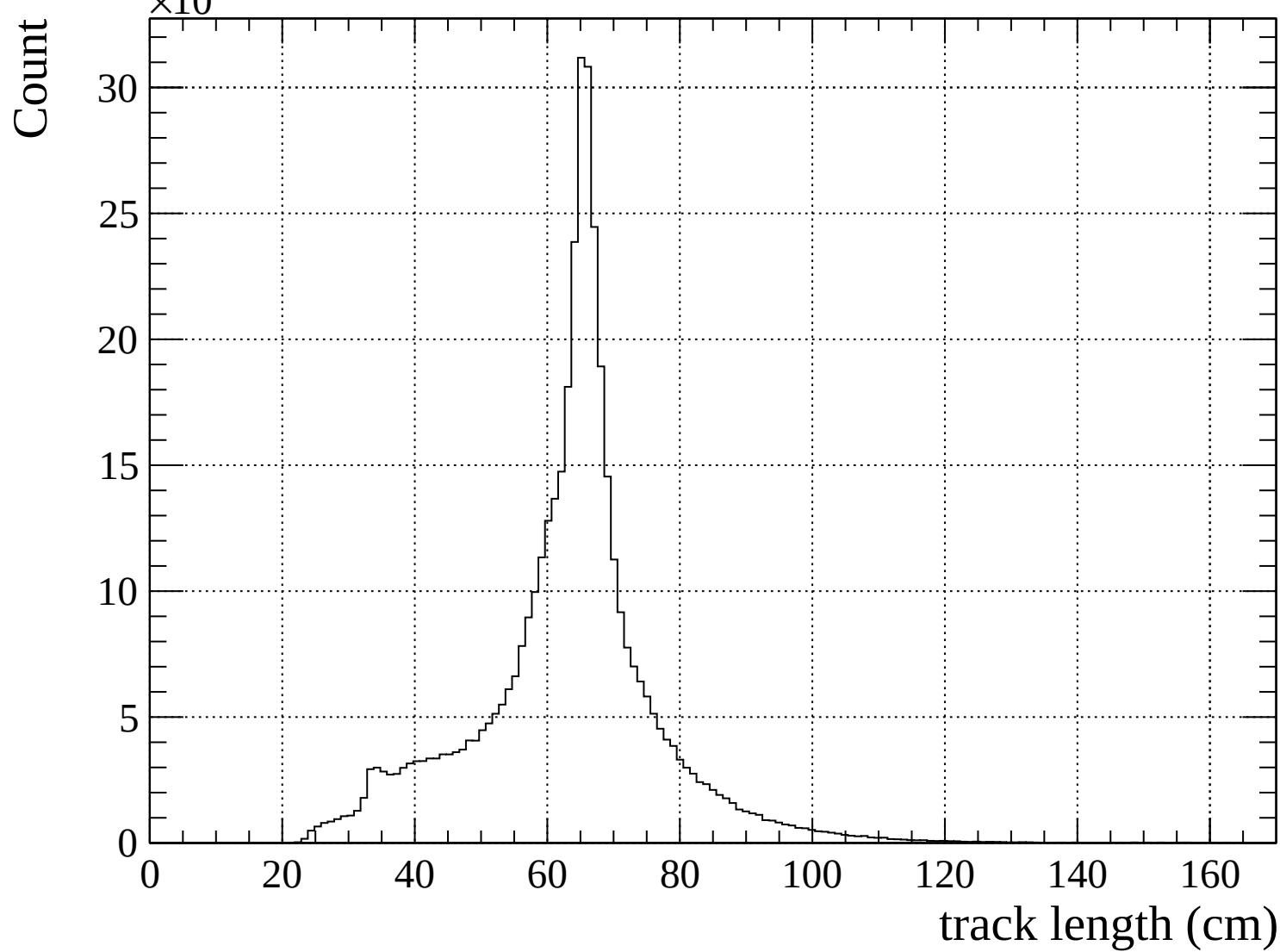
Pulls mean

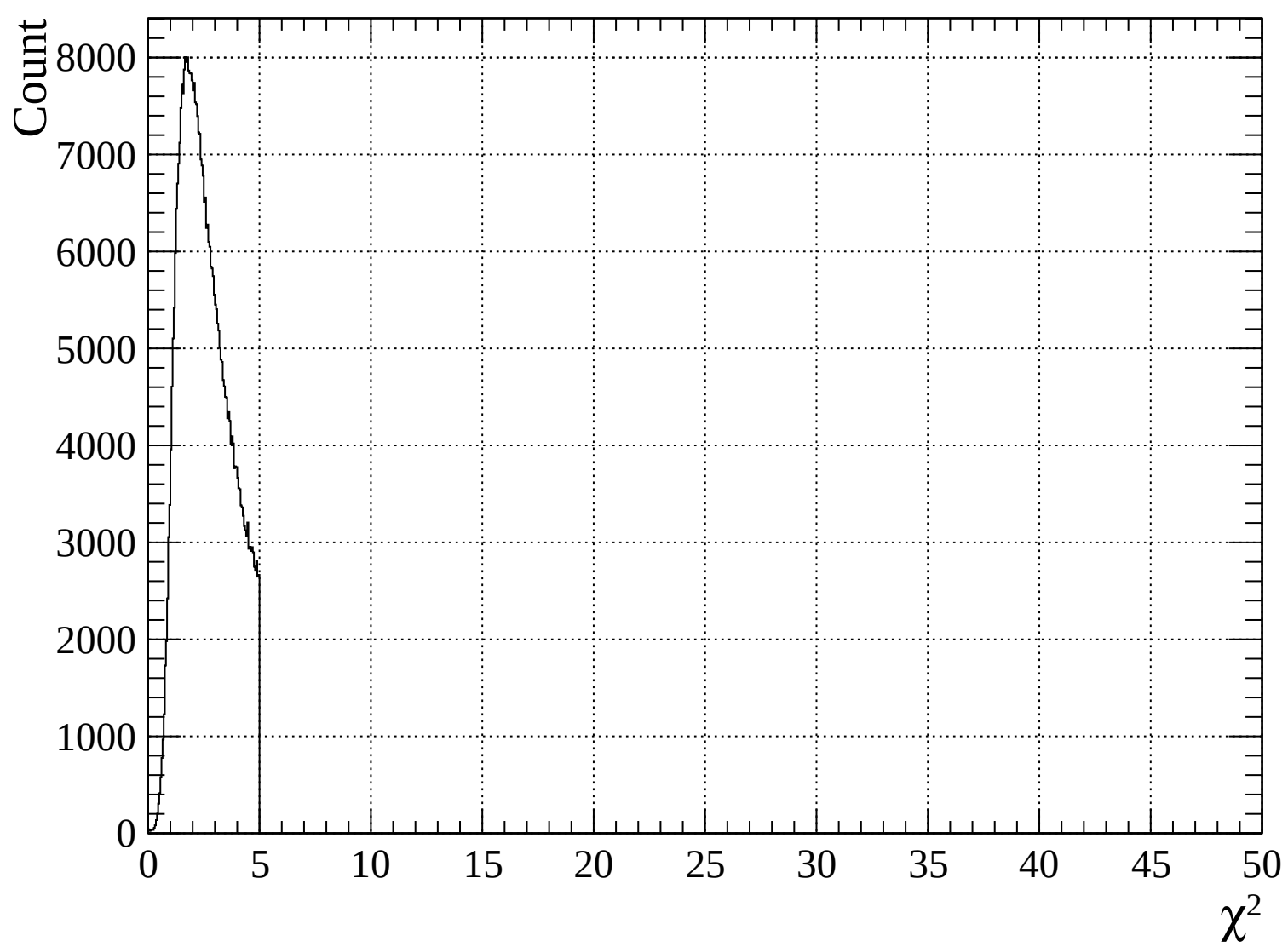


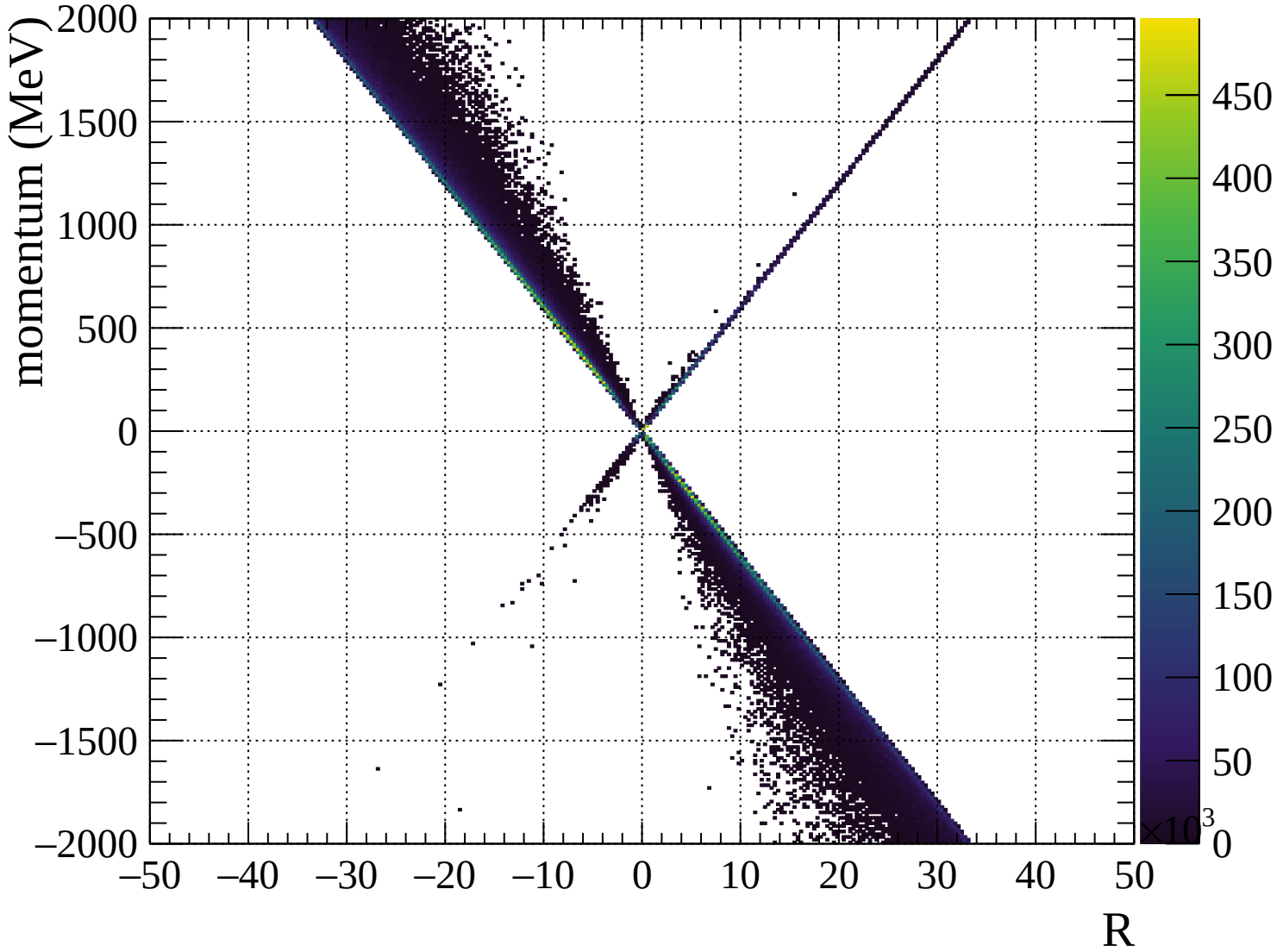


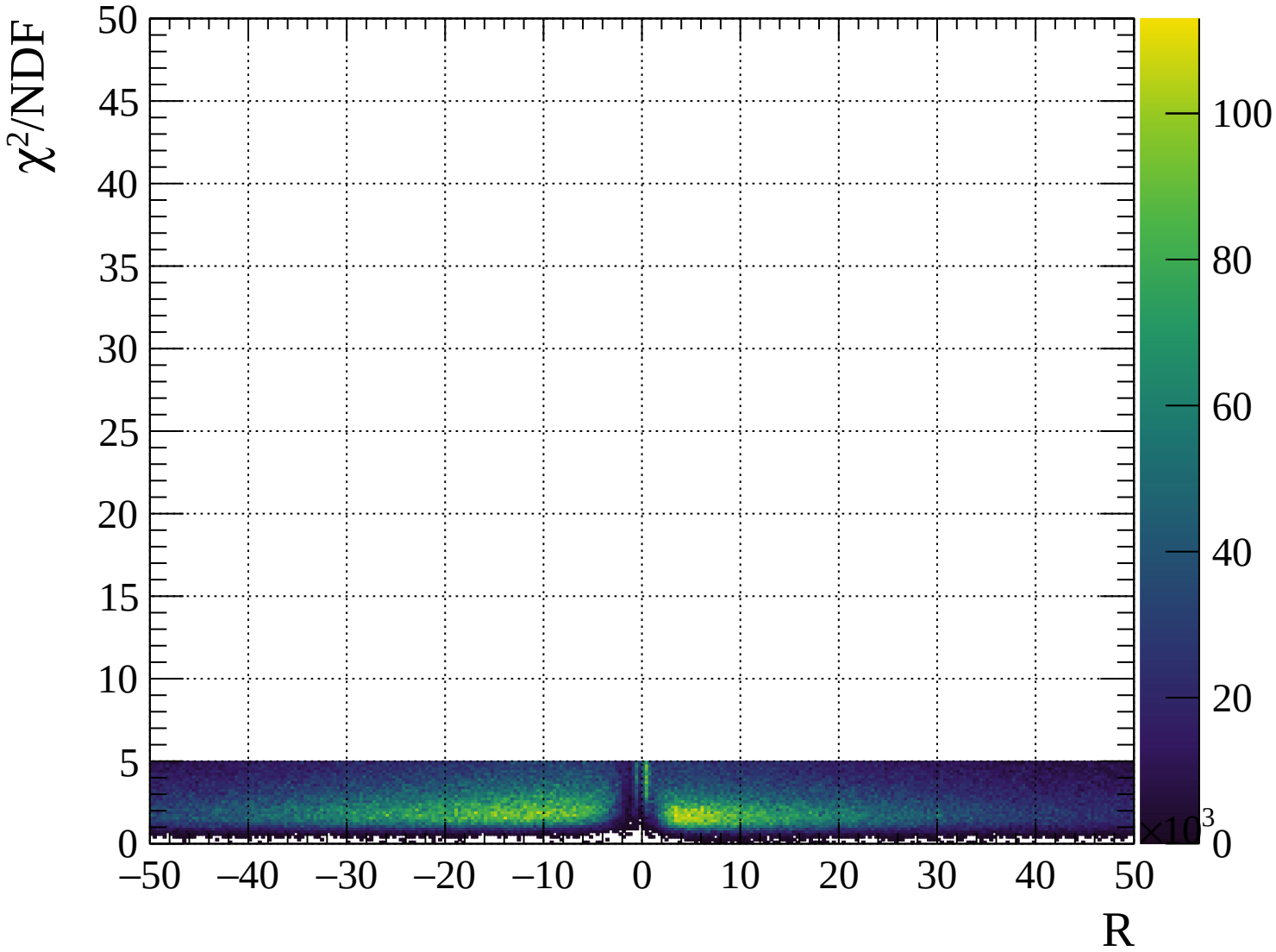






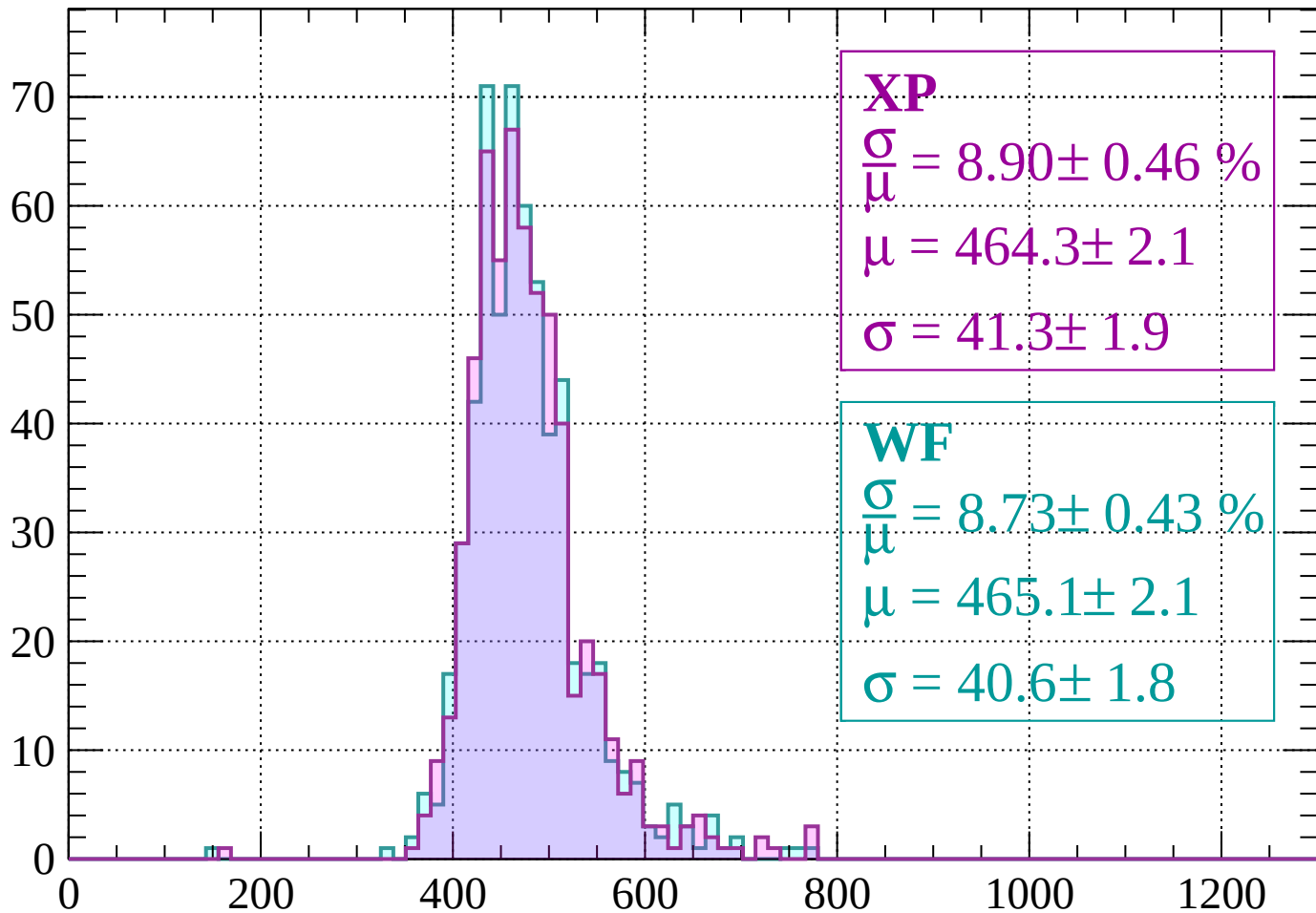






Energy loss | -2000 < p < -1960

Count



XP

$$\frac{\sigma}{\mu} = 8.90 \pm 0.46 \%$$

$$\mu = 464.3 \pm 2.1$$

$$\sigma = 41.3 \pm 1.9$$

WF

$$\frac{\sigma}{\mu} = 8.73 \pm 0.43 \%$$

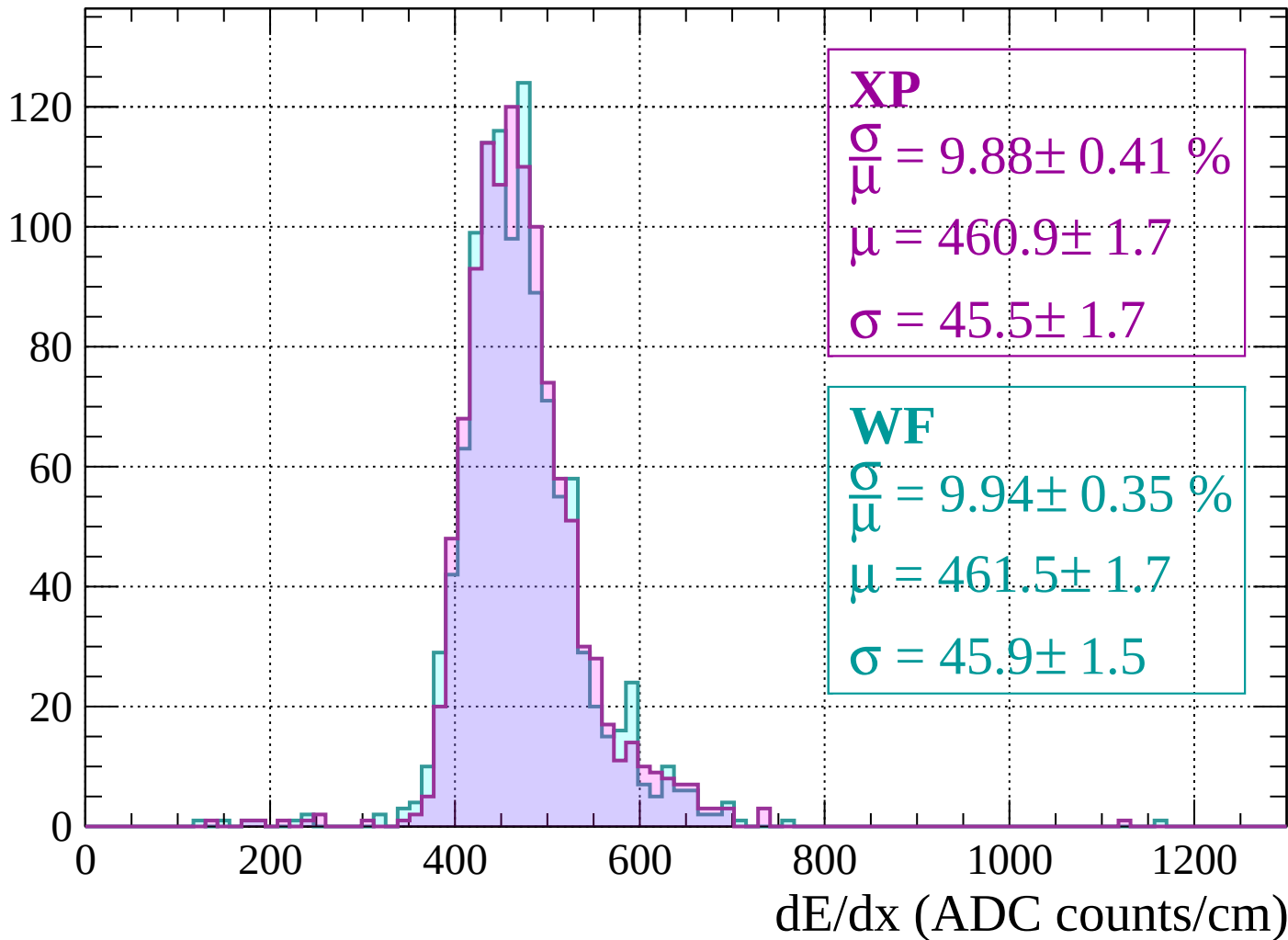
$$\mu = 465.1 \pm 2.1$$

$$\sigma = 40.6 \pm 1.8$$

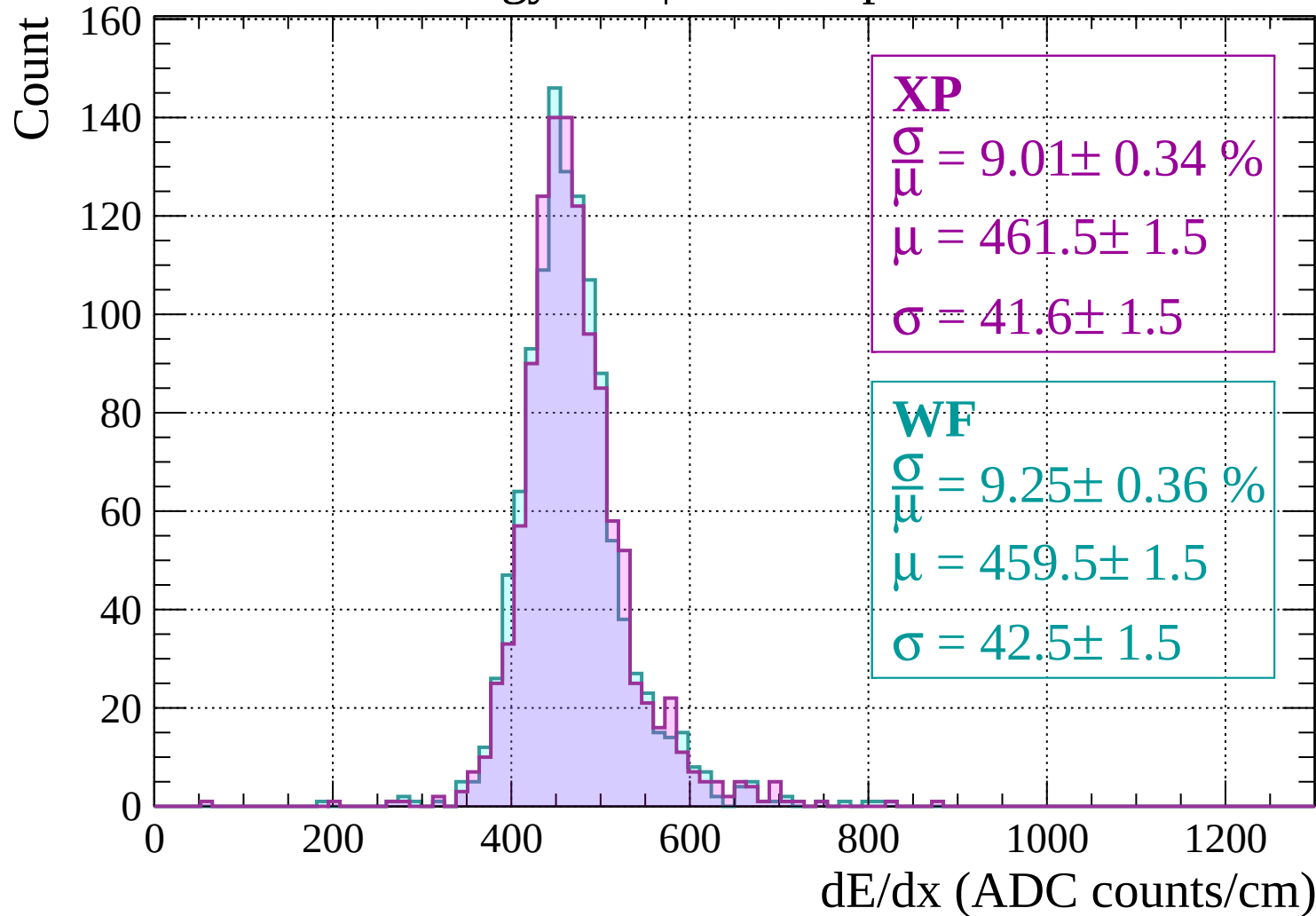
dE/dx (ADC counts/cm)

Energy loss | -1960 < p < -1920

Count

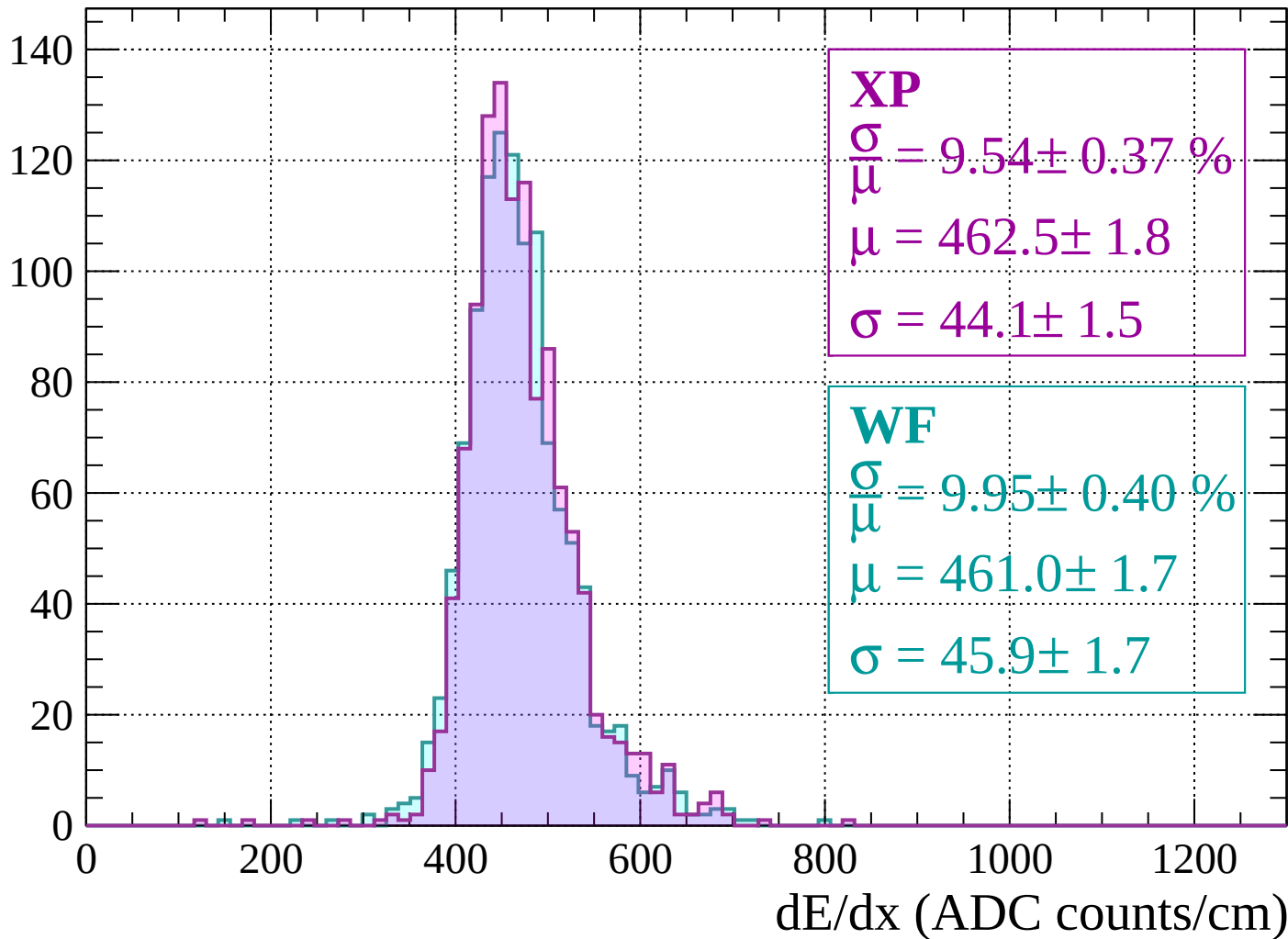


Energy loss | -1920 < p < -1880



Energy loss | $-1880 < p < -1840$

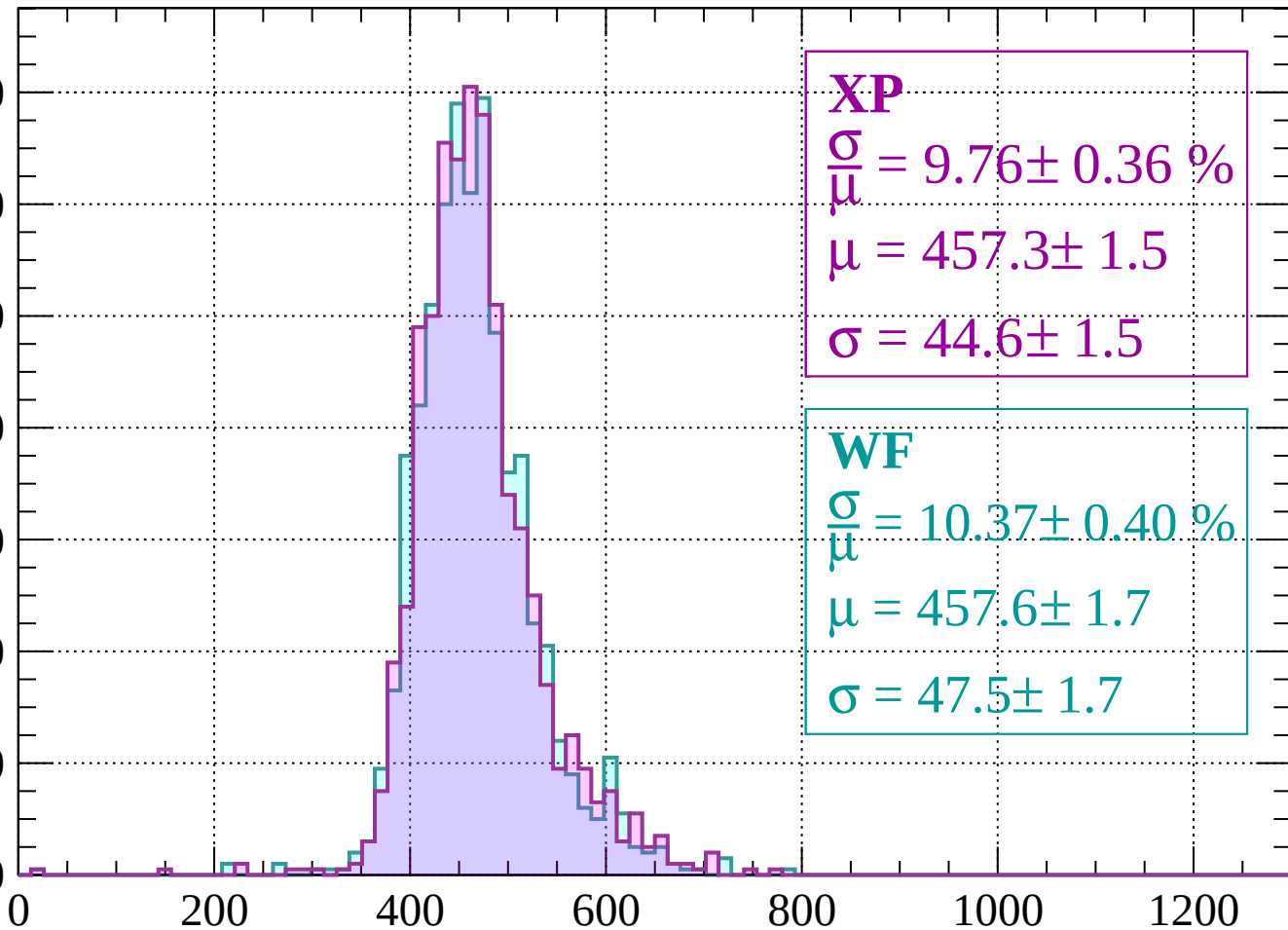
Count



Energy loss | $-1840 < p < -1800$

Count

140
120
100
80
60
40
20
0



XP

$$\frac{\sigma}{\mu} = 9.76 \pm 0.36 \%$$

$$\mu = 457.3 \pm 1.5$$

$$\sigma = 44.6 \pm 1.5$$

WF

$$\frac{\sigma}{\mu} = 10.37 \pm 0.40 \%$$

$$\mu = 457.6 \pm 1.7$$

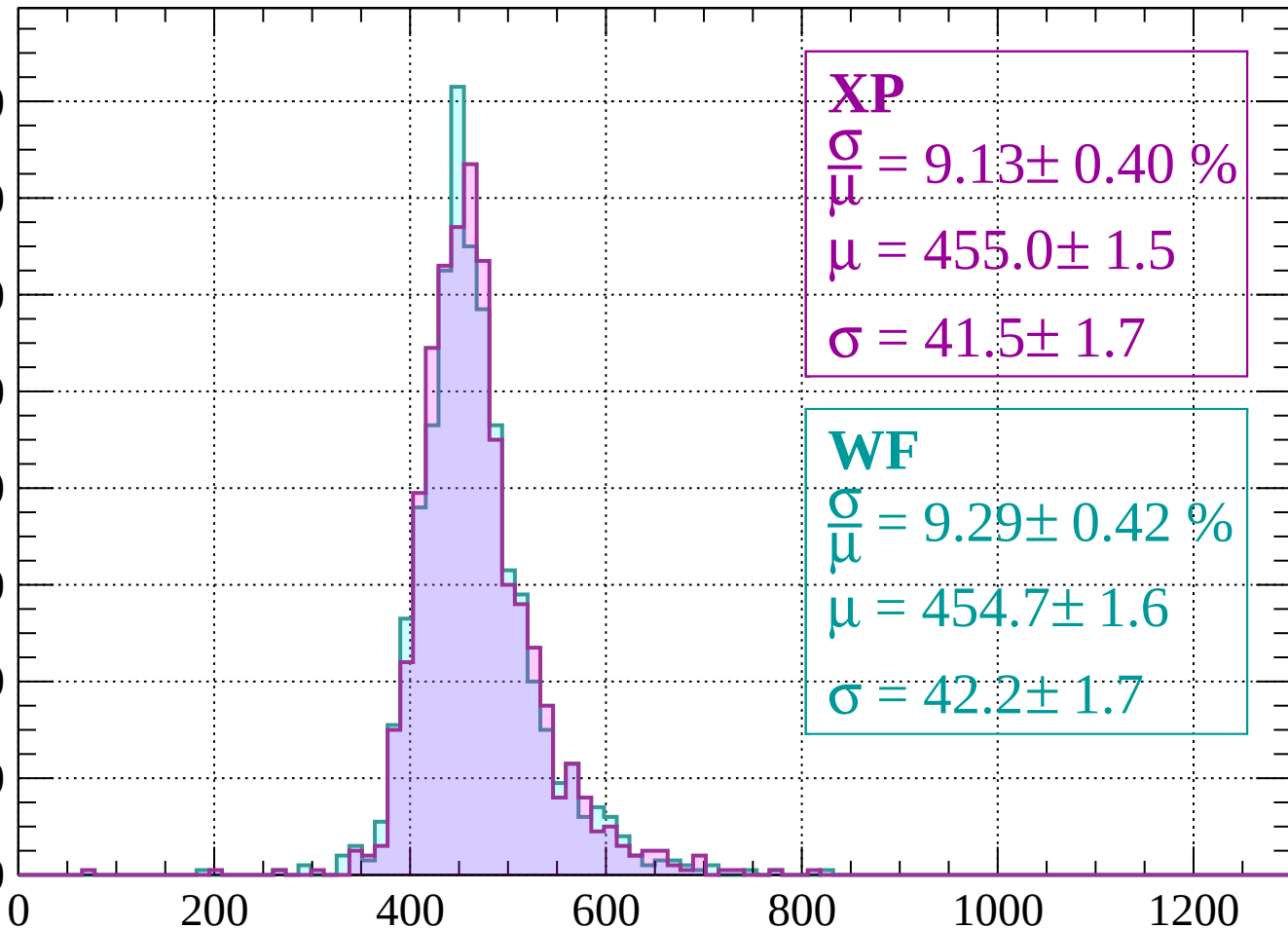
$$\sigma = 47.5 \pm 1.7$$

dE/dx (ADC counts/cm)

Energy loss | $-1800 < p < -1760$

Count

160
140
120
100
80
60
40
20
0



XP

$$\frac{\sigma}{\mu} = 9.13 \pm 0.40 \%$$

$$\mu = 455.0 \pm 1.5$$

$$\sigma = 41.5 \pm 1.7$$

WF

$$\frac{\sigma}{\mu} = 9.29 \pm 0.42 \%$$

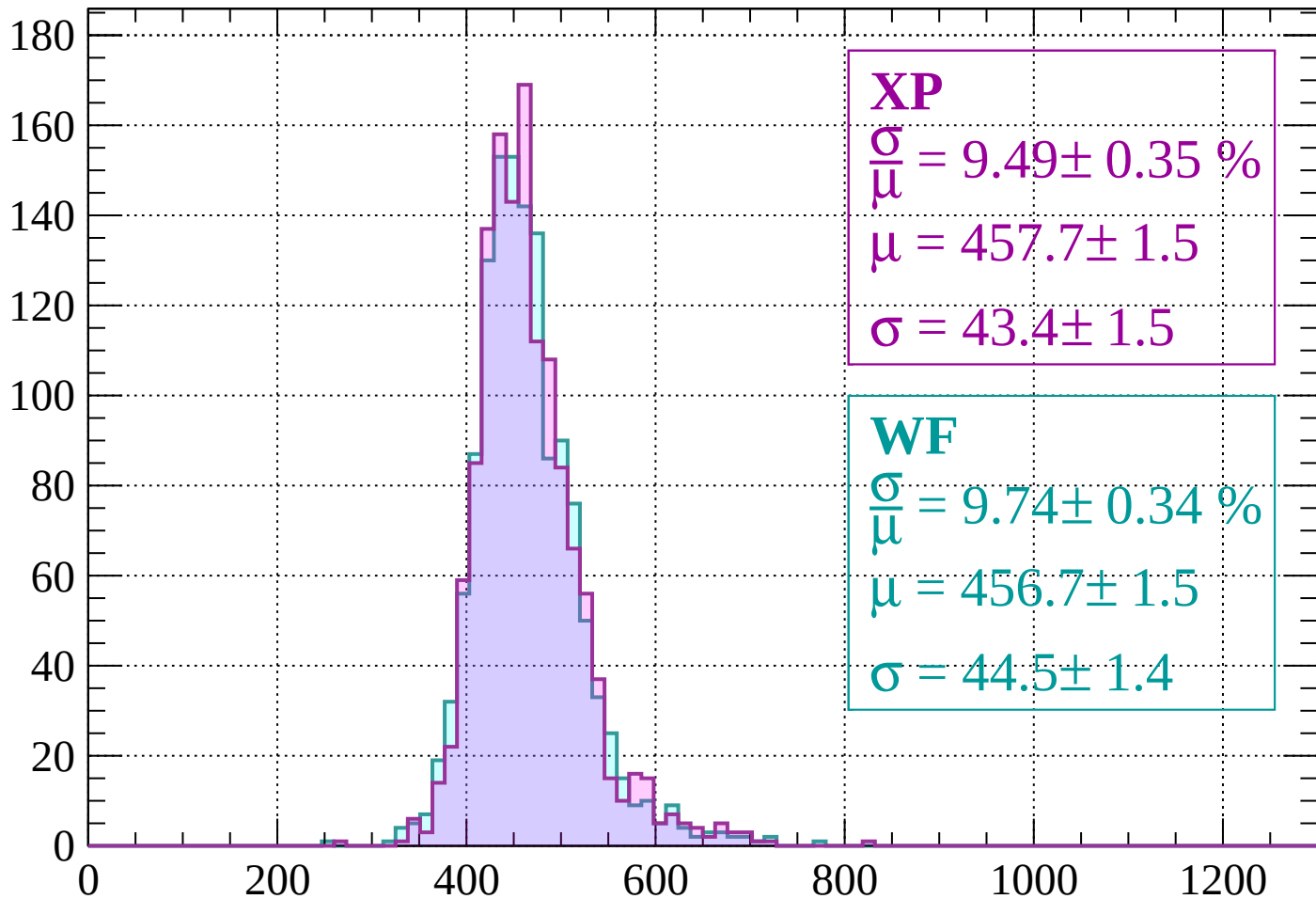
$$\mu = 454.7 \pm 1.6$$

$$\sigma = 42.2 \pm 1.7$$

dE/dx (ADC counts/cm)

Energy loss | $-1760 < p < -1720$

Count



XP

$$\frac{\sigma}{\mu} = 9.49 \pm 0.35 \%$$

$$\mu = 457.7 \pm 1.5$$

$$\sigma = 43.4 \pm 1.5$$

WF

$$\frac{\sigma}{\mu} = 9.74 \pm 0.34 \%$$

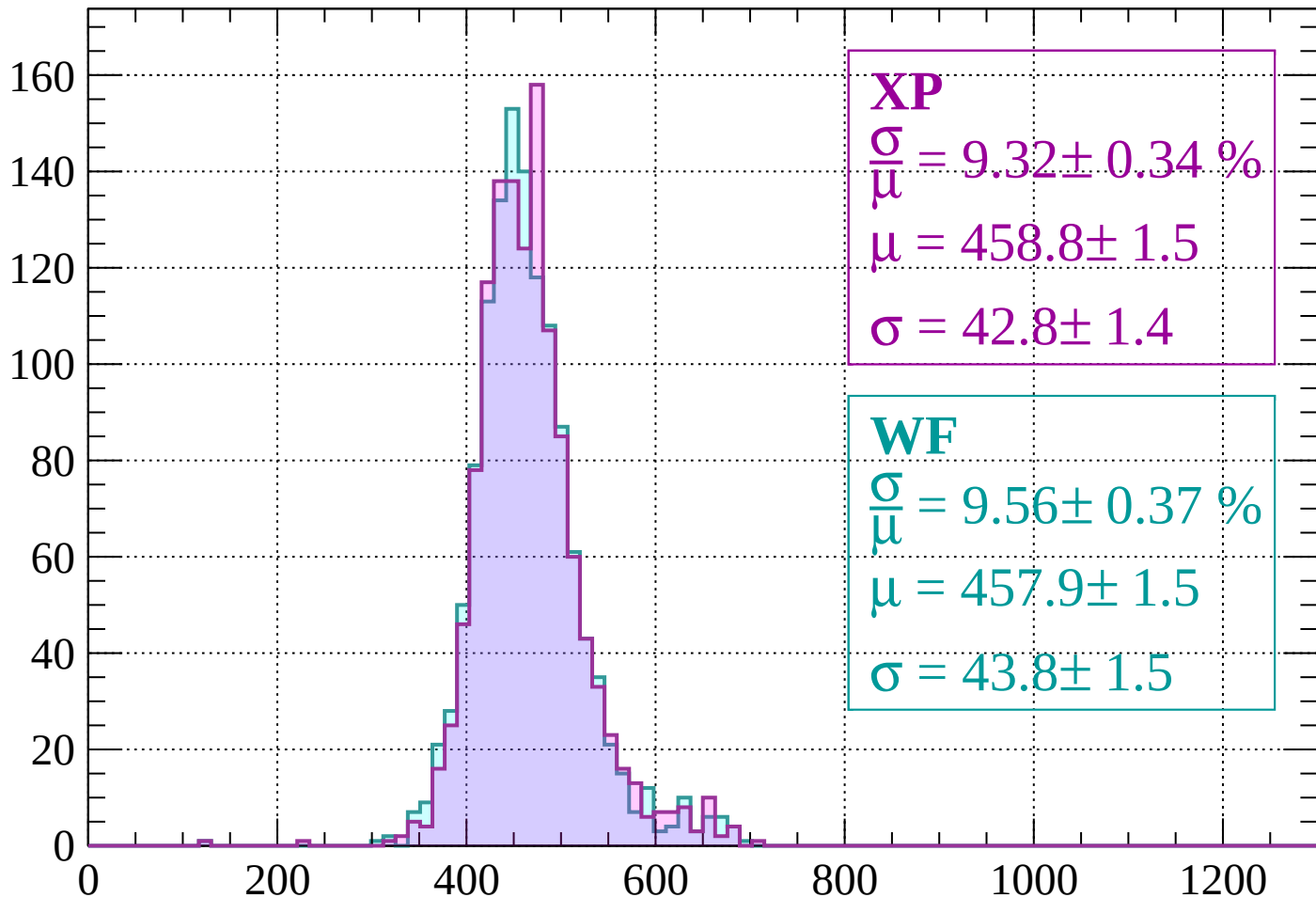
$$\mu = 456.7 \pm 1.5$$

$$\sigma = 44.5 \pm 1.4$$

dE/dx (ADC counts/cm)

Energy loss | $-1720 < p < -1680$

Count



XP

$$\frac{\sigma}{\bar{\mu}} = 9.32 \pm 0.34 \%$$

$$\mu = 458.8 \pm 1.5$$

$$\sigma = 42.8 \pm 1.4$$

WF

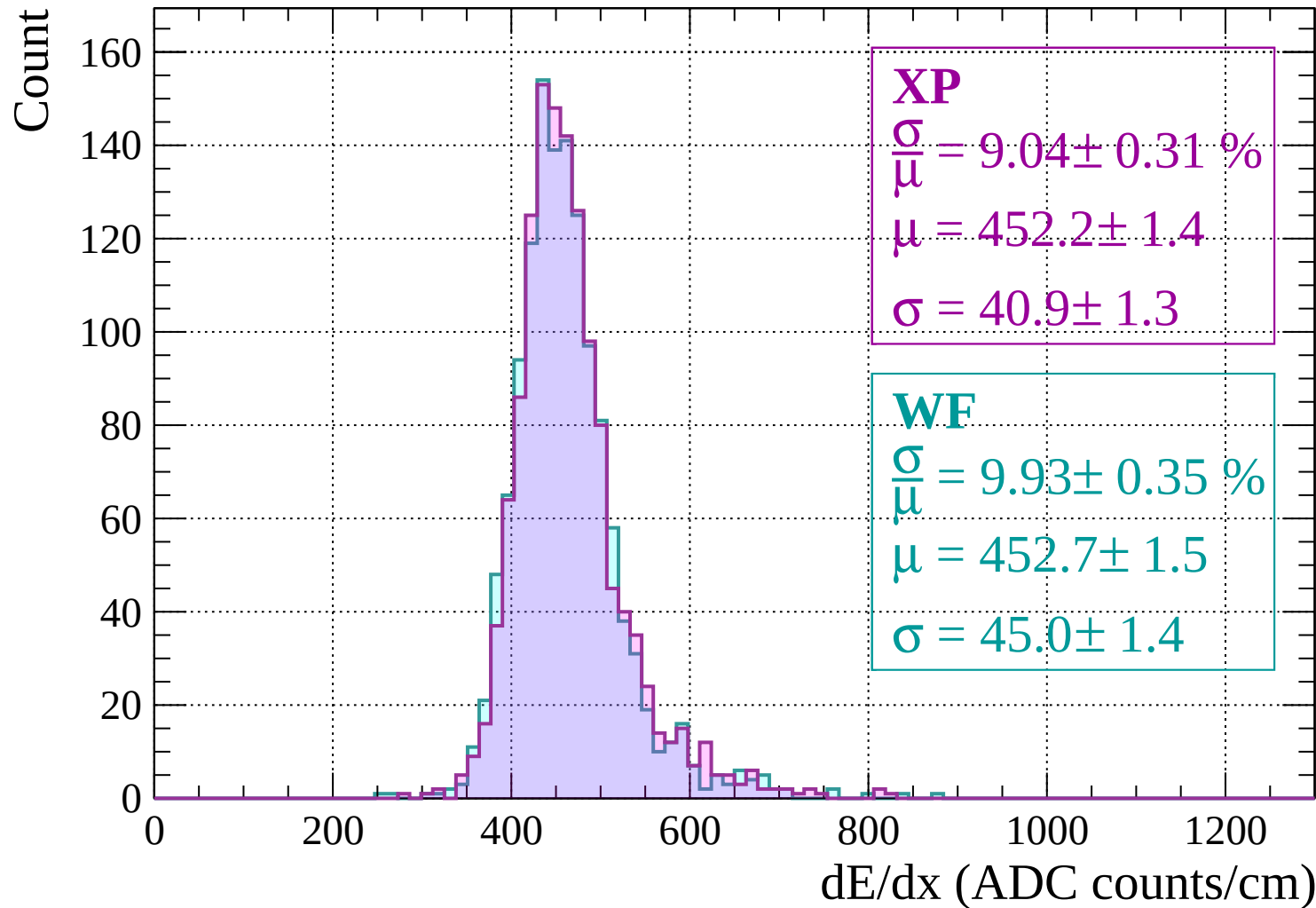
$$\frac{\sigma}{\bar{\mu}} = 9.56 \pm 0.37 \%$$

$$\mu = 457.9 \pm 1.5$$

$$\sigma = 43.8 \pm 1.5$$

dE/dx (ADC counts/cm)

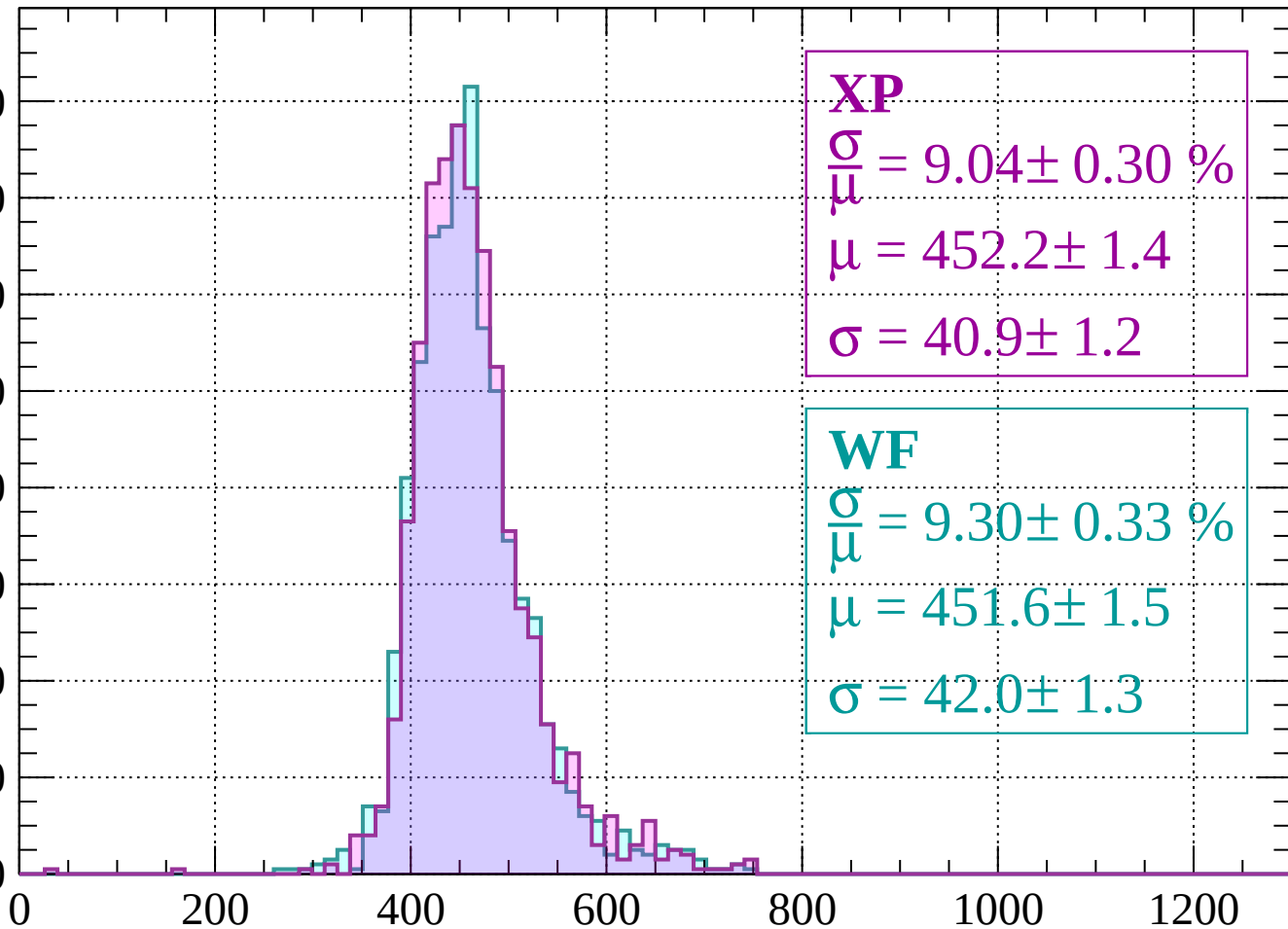
Energy loss | $-1680 < p < -1640$



Energy loss | $-1640 < p < -1600$

Count

160
140
120
100
80
60
40
20
0



XP

$$\frac{\sigma}{\mu} = 9.04 \pm 0.30 \%$$

$$\mu = 452.2 \pm 1.4$$

$$\sigma = 40.9 \pm 1.2$$

WF

$$\frac{\sigma}{\mu} = 9.30 \pm 0.33 \%$$

$$\mu = 451.6 \pm 1.5$$

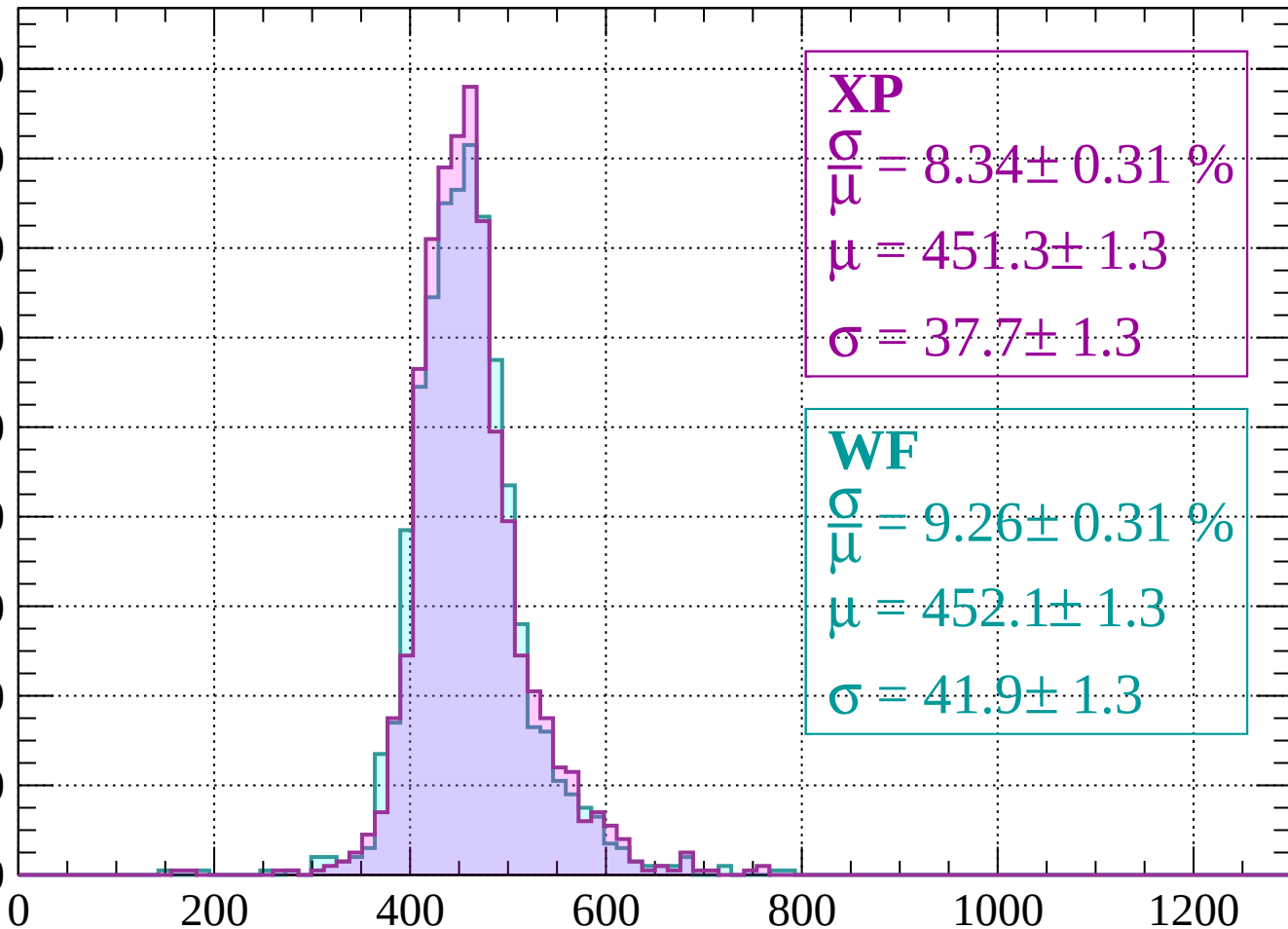
$$\sigma = 42.0 \pm 1.3$$

dE/dx (ADC counts/cm)

Energy loss | -1600 < p < -1560

Count

180
160
140
120
100
80
60
40
20
0



XP

$$\frac{\sigma}{\bar{\mu}} = 8.34 \pm 0.31 \%$$

$$\mu = 451.3 \pm 1.3$$

$$\sigma = 37.7 \pm 1.3$$

WF

$$\frac{\sigma}{\bar{\mu}} = 9.26 \pm 0.31 \%$$

$$\mu = 452.1 \pm 1.3$$

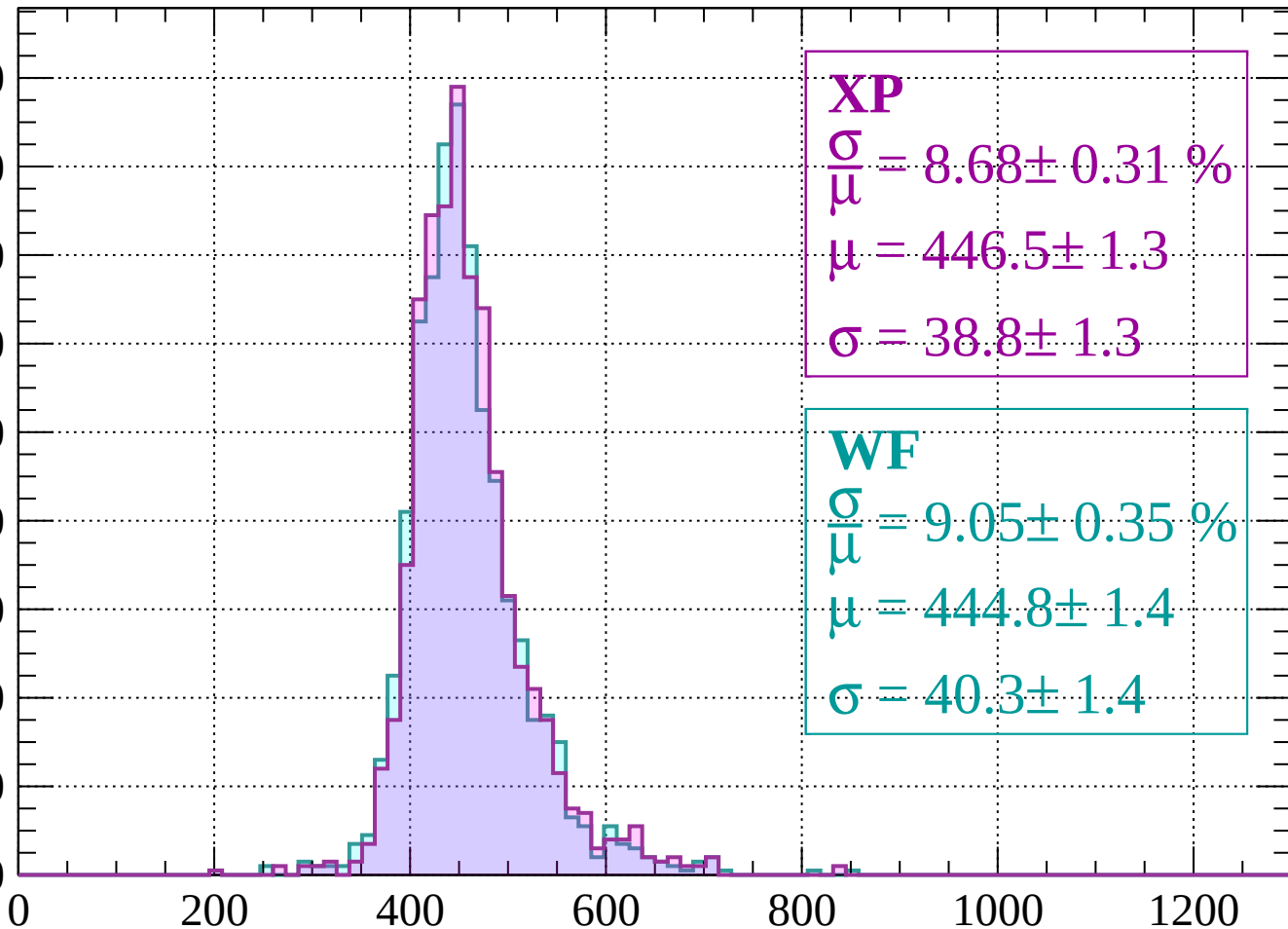
$$\sigma = 41.9 \pm 1.3$$

dE/dx (ADC counts/cm)

Energy loss | -1560 < p < -1520

Count

180
160
140
120
100
80
60
40
20
0



XP

$$\frac{\sigma}{\mu} = 8.68 \pm 0.31 \%$$

$$\mu = 446.5 \pm 1.3$$

$$\sigma = 38.8 \pm 1.3$$

WF

$$\frac{\sigma}{\mu} = 9.05 \pm 0.35 \%$$

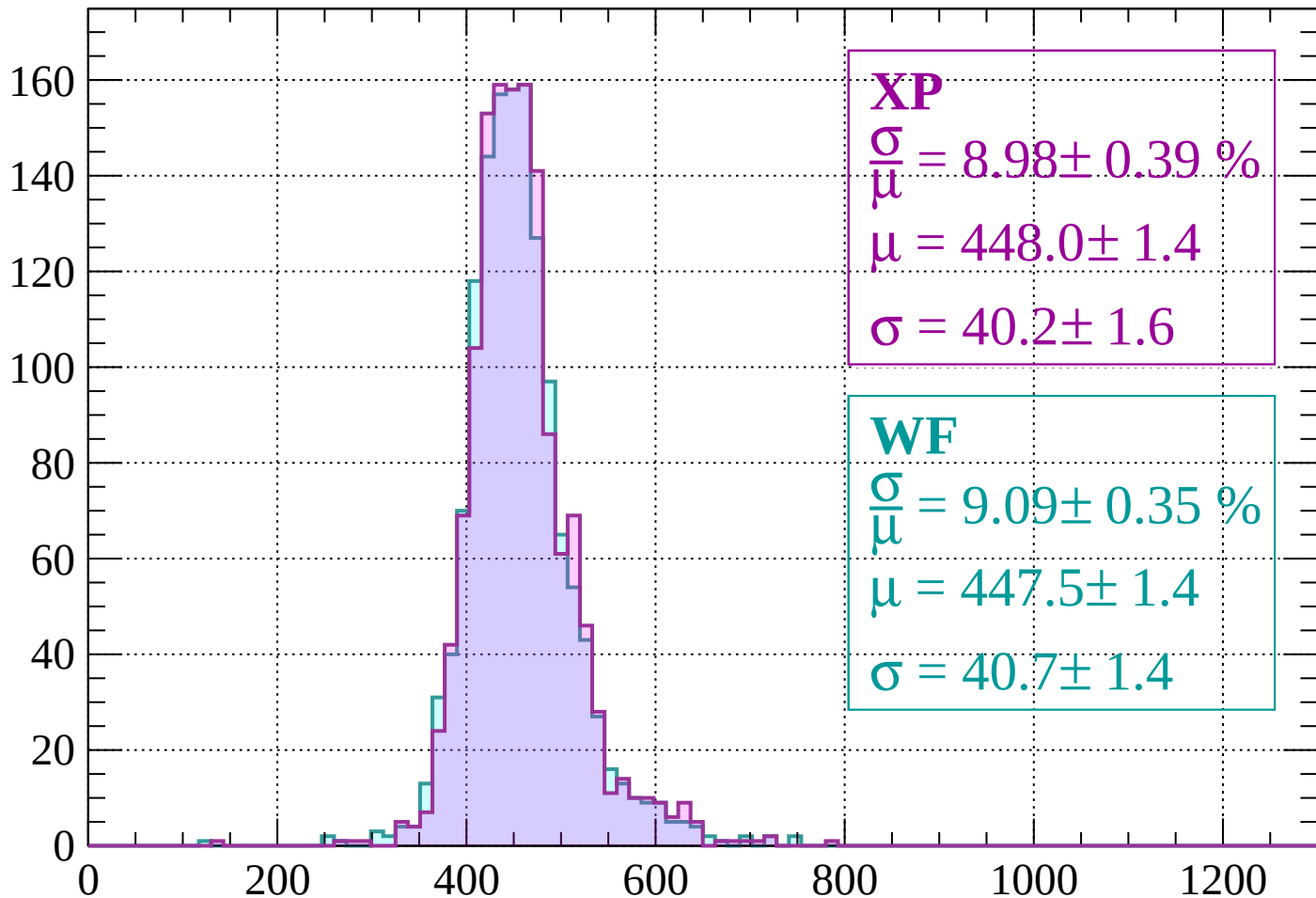
$$\mu = 444.8 \pm 1.4$$

$$\sigma = 40.3 \pm 1.4$$

dE/dx (ADC counts/cm)

Energy loss | $-1520 < p < -1480$

Count



XP

$$\frac{\sigma}{\mu} = 8.98 \pm 0.39 \%$$

$$\mu = 448.0 \pm 1.4$$

$$\sigma = 40.2 \pm 1.6$$

WF

$$\frac{\sigma}{\mu} = 9.09 \pm 0.35 \%$$

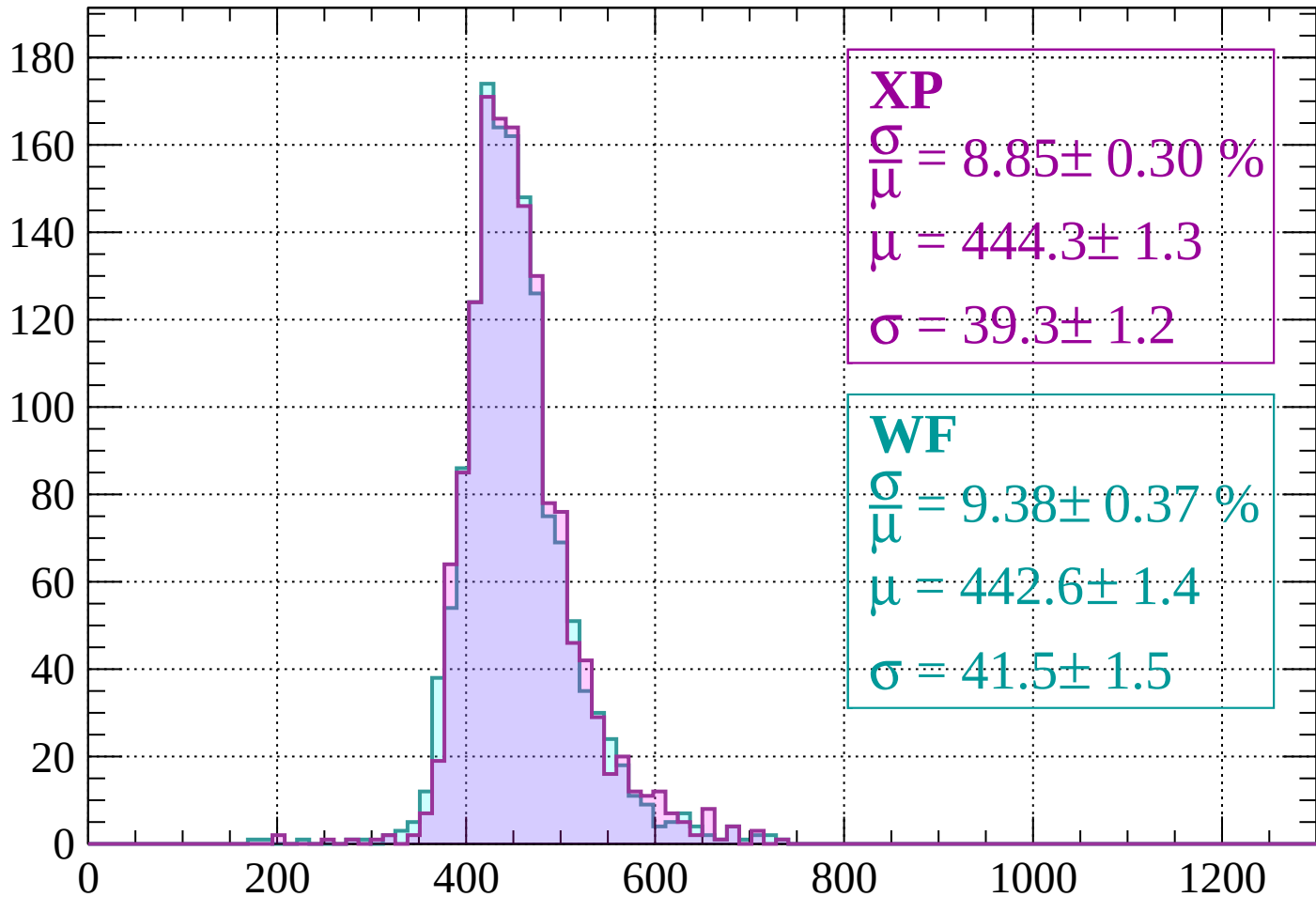
$$\mu = 447.5 \pm 1.4$$

$$\sigma = 40.7 \pm 1.4$$

dE/dx (ADC counts/cm)

Energy loss | $-1480 < p < -1440$

Count



XP

$$\frac{\sigma}{\mu} = 8.85 \pm 0.30 \%$$

$$\mu = 444.3 \pm 1.3$$

$$\sigma = 39.3 \pm 1.2$$

WF

$$\frac{\sigma}{\mu} = 9.38 \pm 0.37 \%$$

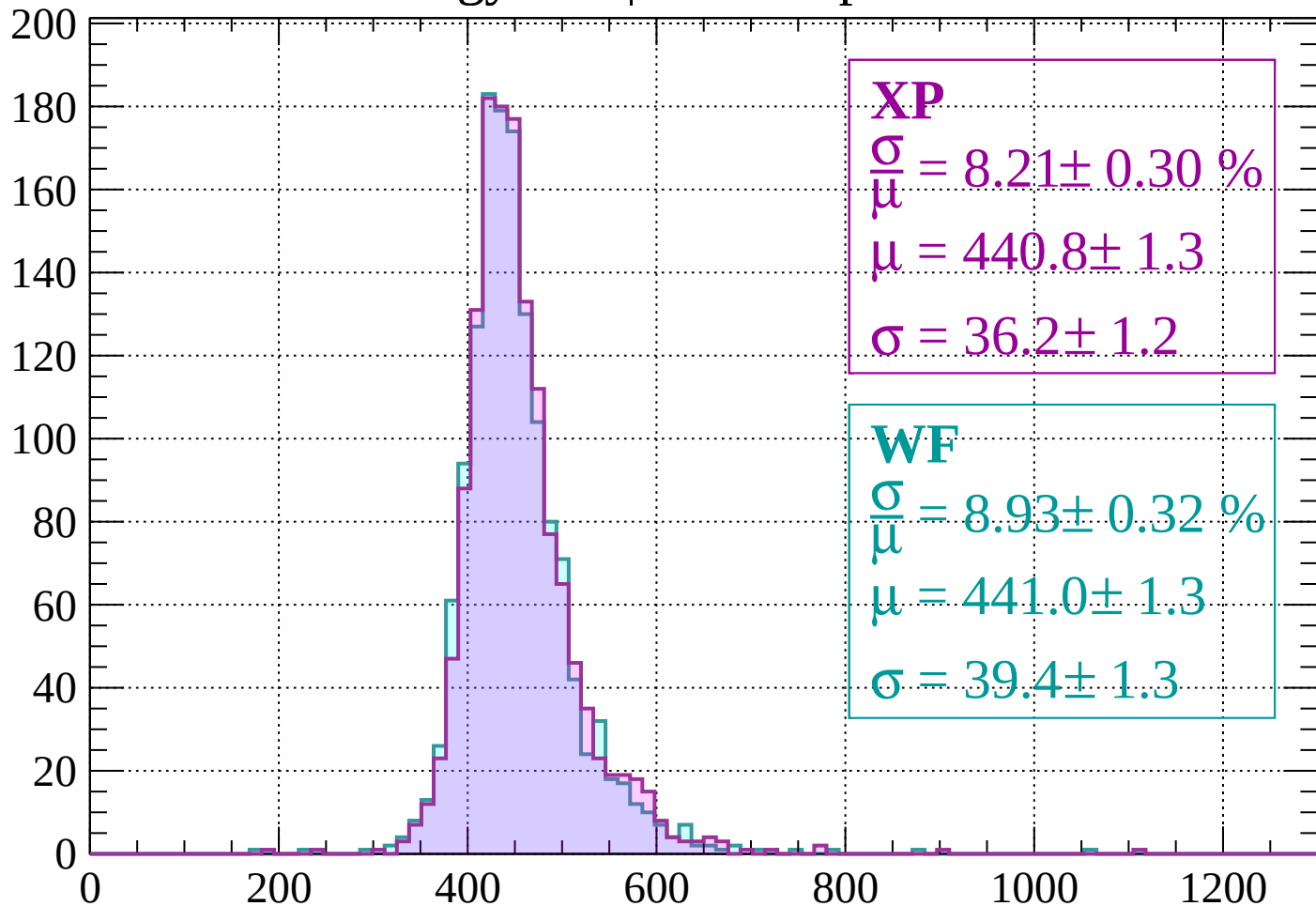
$$\mu = 442.6 \pm 1.4$$

$$\sigma = 41.5 \pm 1.5$$

dE/dx (ADC counts/cm)

Energy loss | $-1440 < p < -1400$

Count



XP

$$\frac{\sigma}{\mu} = 8.21 \pm 0.30 \%$$

$$\mu = 440.8 \pm 1.3$$

$$\sigma = 36.2 \pm 1.2$$

WF

$$\frac{\sigma}{\mu} = 8.93 \pm 0.32 \%$$

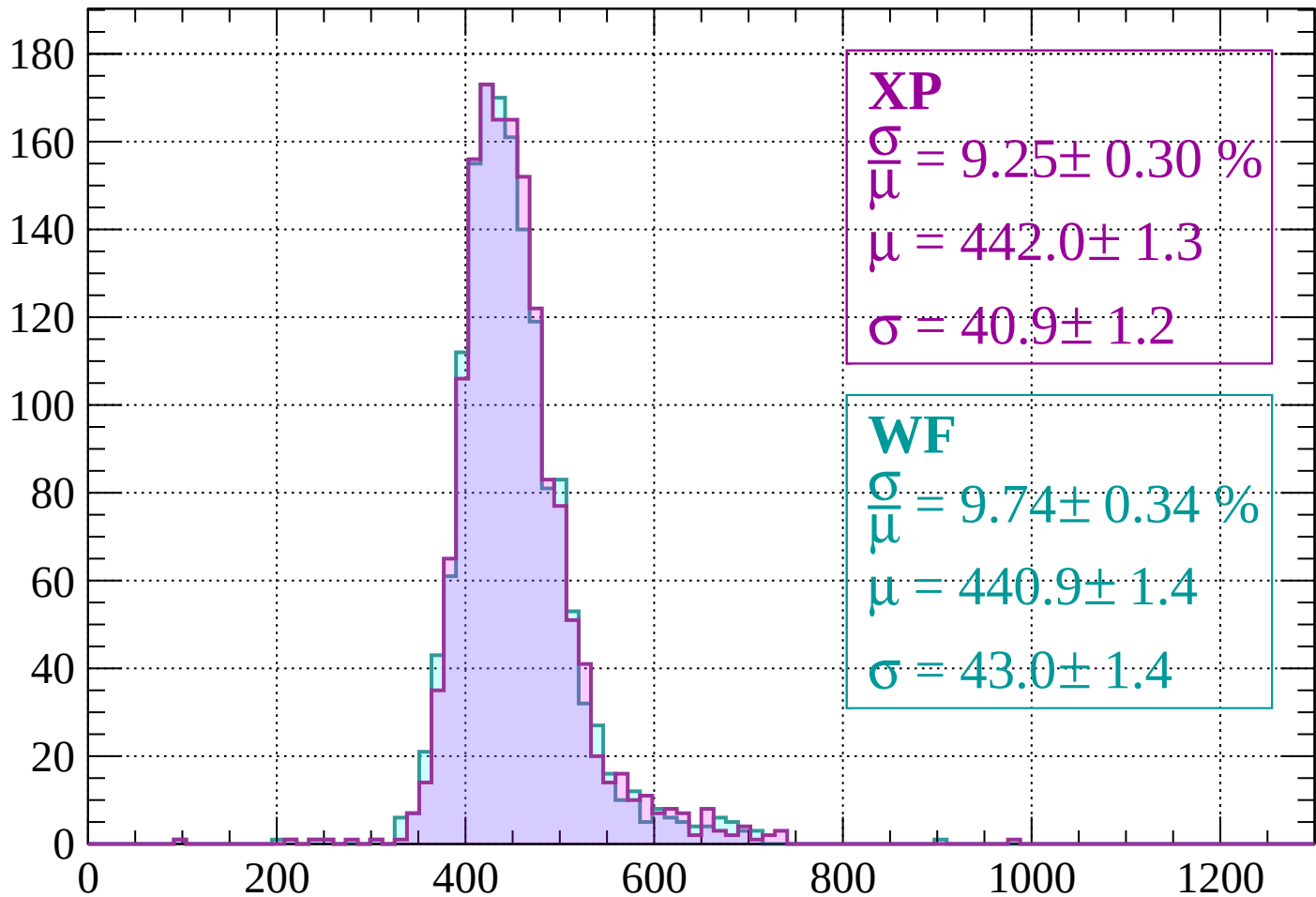
$$\mu = 441.0 \pm 1.3$$

$$\sigma = 39.4 \pm 1.3$$

dE/dx (ADC counts/cm)

Energy loss | $-1400 < p < -1360$

Count



XP

$$\frac{\sigma}{\mu} = 9.25 \pm 0.30 \%$$

$$\mu = 442.0 \pm 1.3$$

$$\sigma = 40.9 \pm 1.2$$

WF

$$\frac{\sigma}{\mu} = 9.74 \pm 0.34 \%$$

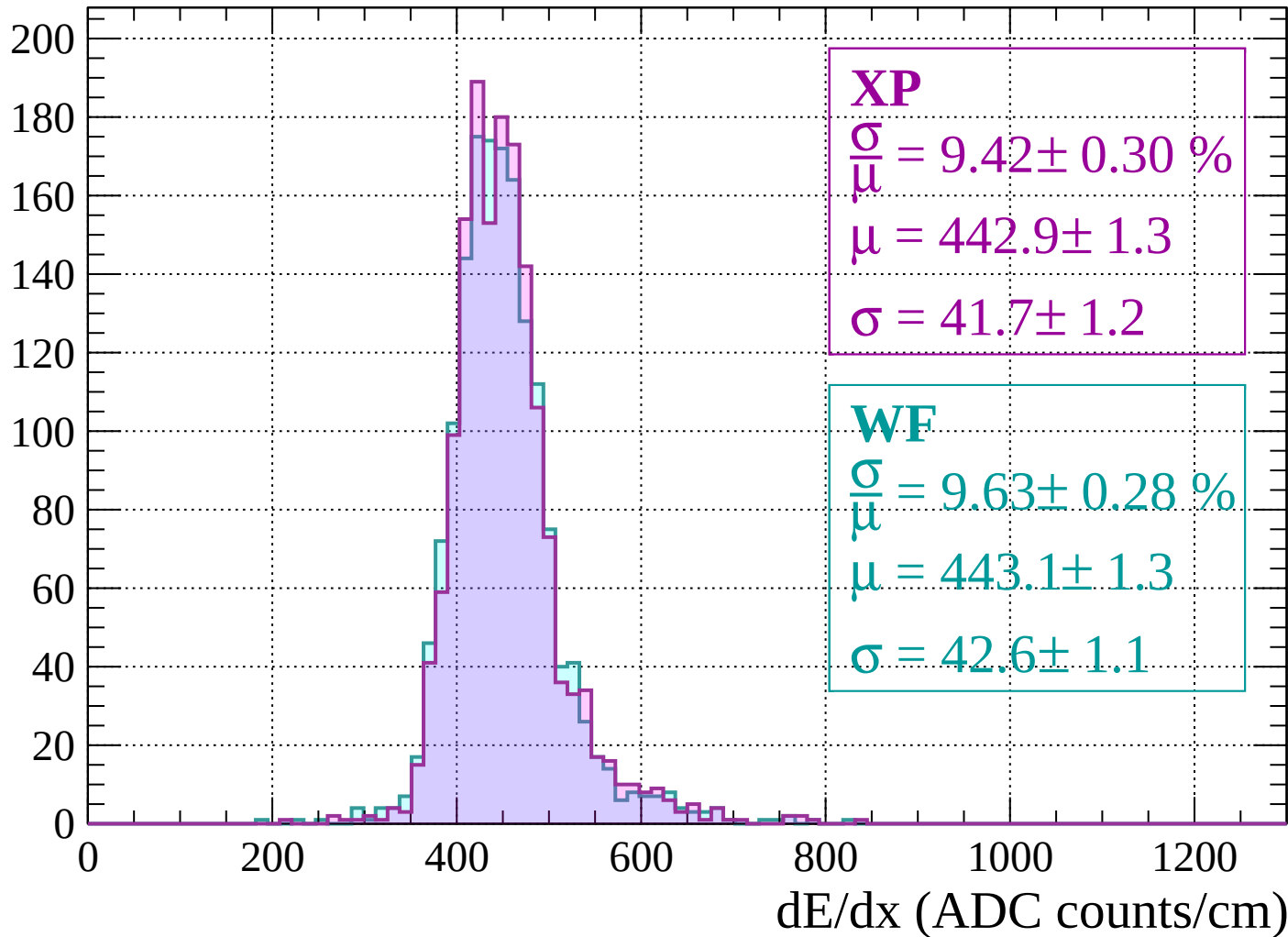
$$\mu = 440.9 \pm 1.4$$

$$\sigma = 43.0 \pm 1.4$$

dE/dx (ADC counts/cm)

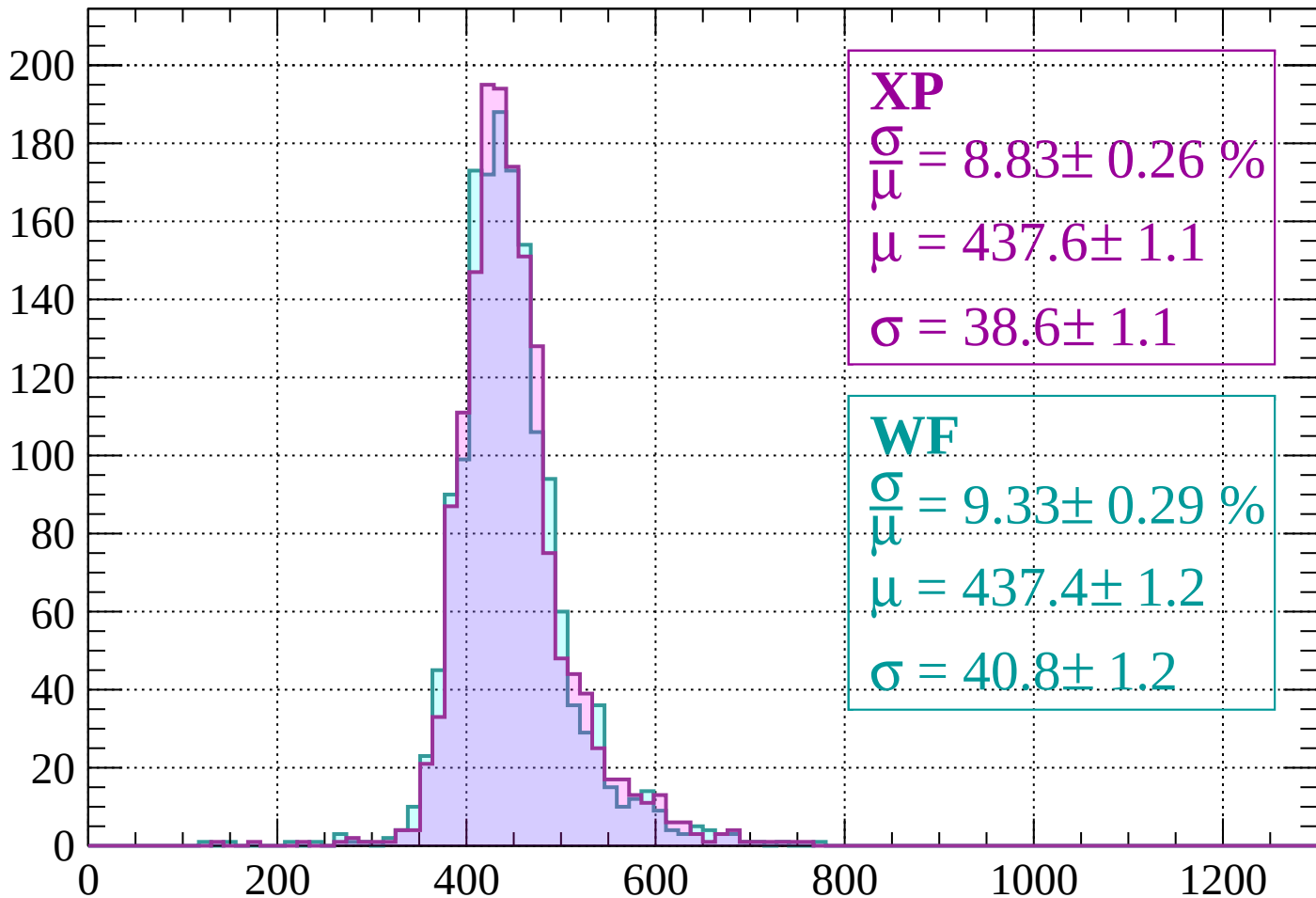
Energy loss | $-1360 < p < -1320$

Count



Energy loss | $-1320 < p < -1280$

Count



XP

$$\frac{\sigma}{\mu} = 8.83 \pm 0.26 \%$$

$$\mu = 437.6 \pm 1.1$$

$$\sigma = 38.6 \pm 1.1$$

WF

$$\frac{\sigma}{\mu} = 9.33 \pm 0.29 \%$$

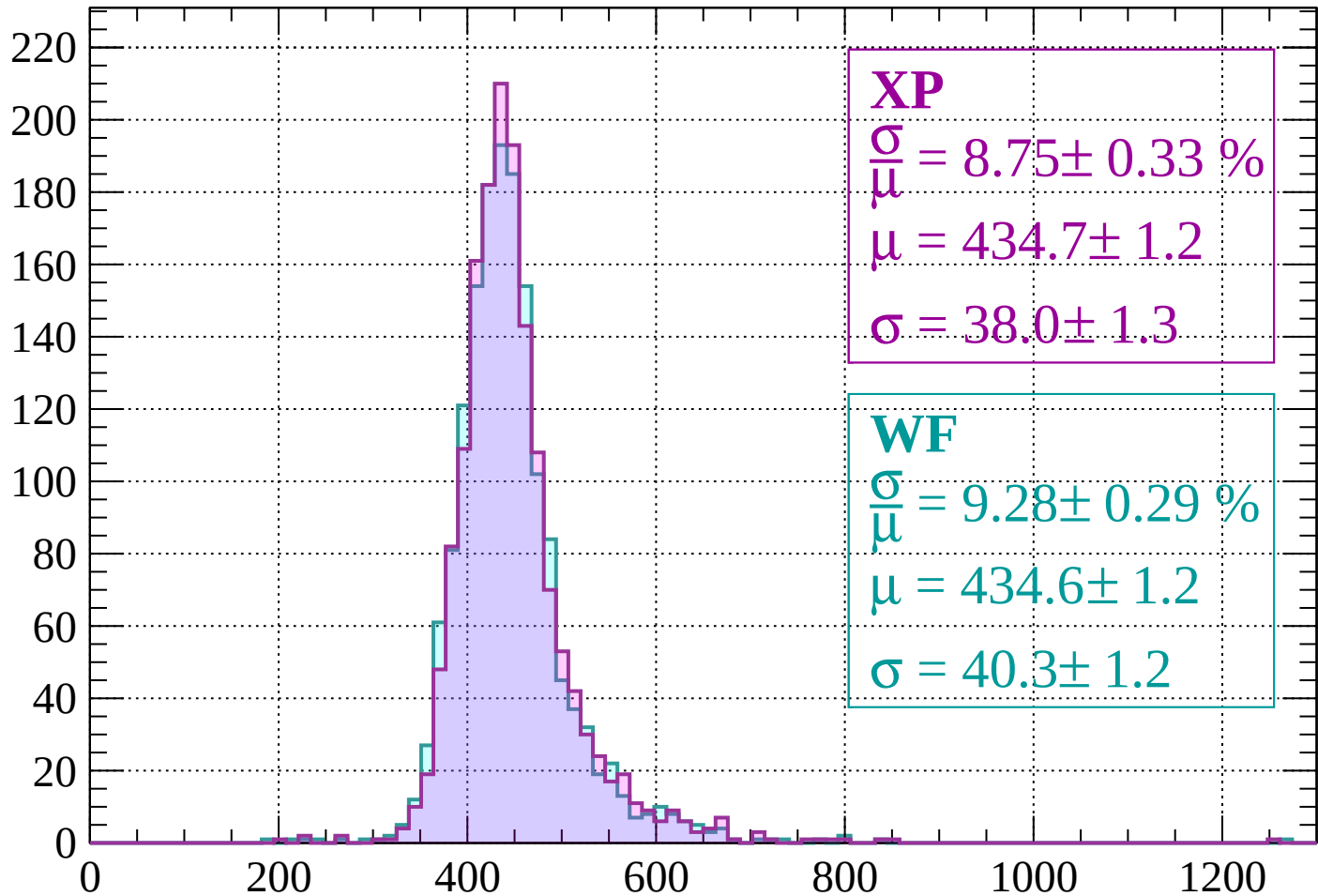
$$\mu = 437.4 \pm 1.2$$

$$\sigma = 40.8 \pm 1.2$$

dE/dx (ADC counts/cm)

Energy loss | $-1280 < p < -1240$

Count



XP

$$\frac{\sigma}{\mu} = 8.75 \pm 0.33 \%$$

$$\mu = 434.7 \pm 1.2$$

$$\sigma = 38.0 \pm 1.3$$

WF

$$\frac{\sigma}{\mu} = 9.28 \pm 0.29 \%$$

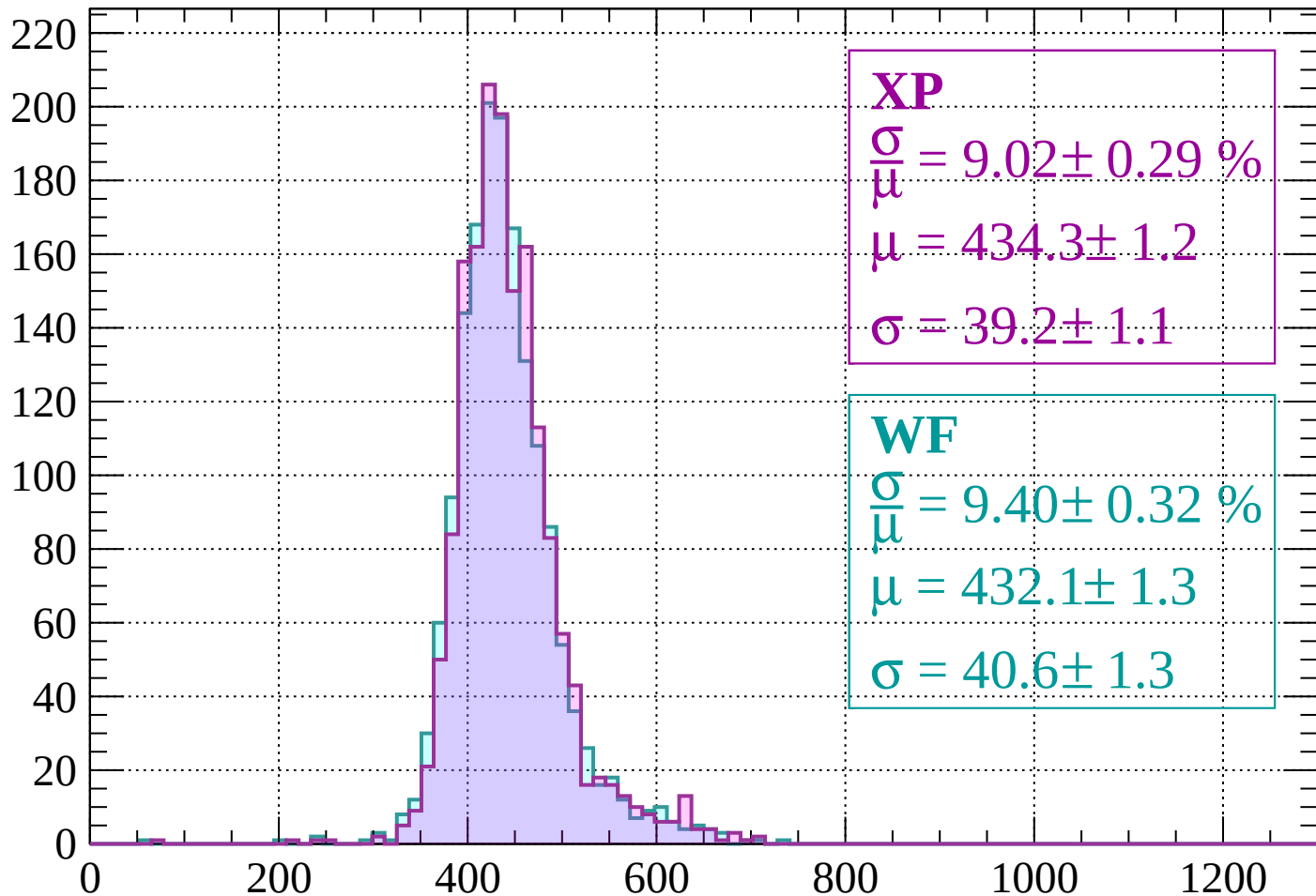
$$\mu = 434.6 \pm 1.2$$

$$\sigma = 40.3 \pm 1.2$$

dE/dx (ADC counts/cm)

Energy loss | -1240 < p < -1200

Count



XP

$$\frac{\sigma}{\mu} = 9.02 \pm 0.29 \%$$

$$\mu = 434.3 \pm 1.2$$

$$\sigma = 39.2 \pm 1.1$$

WF

$$\frac{\sigma}{\mu} = 9.40 \pm 0.32 \%$$

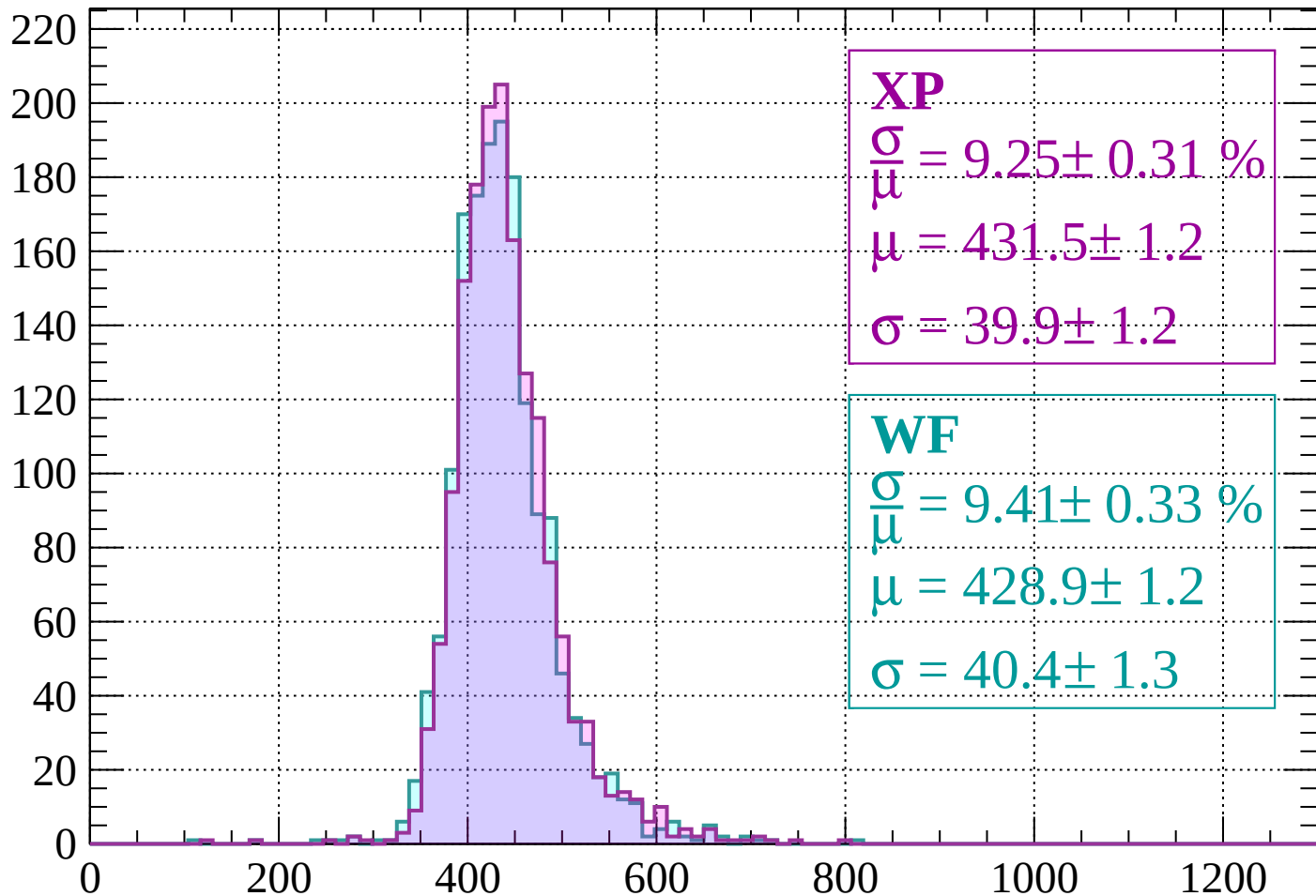
$$\mu = 432.1 \pm 1.3$$

$$\sigma = 40.6 \pm 1.3$$

dE/dx (ADC counts/cm)

Energy loss | -1200 < p < -1160

Count



XP

$$\frac{\sigma}{\mu} = 9.25 \pm 0.31 \%$$

$$\mu = 431.5 \pm 1.2$$

$$\sigma = 39.9 \pm 1.2$$

WF

$$\frac{\sigma}{\mu} = 9.41 \pm 0.33 \%$$

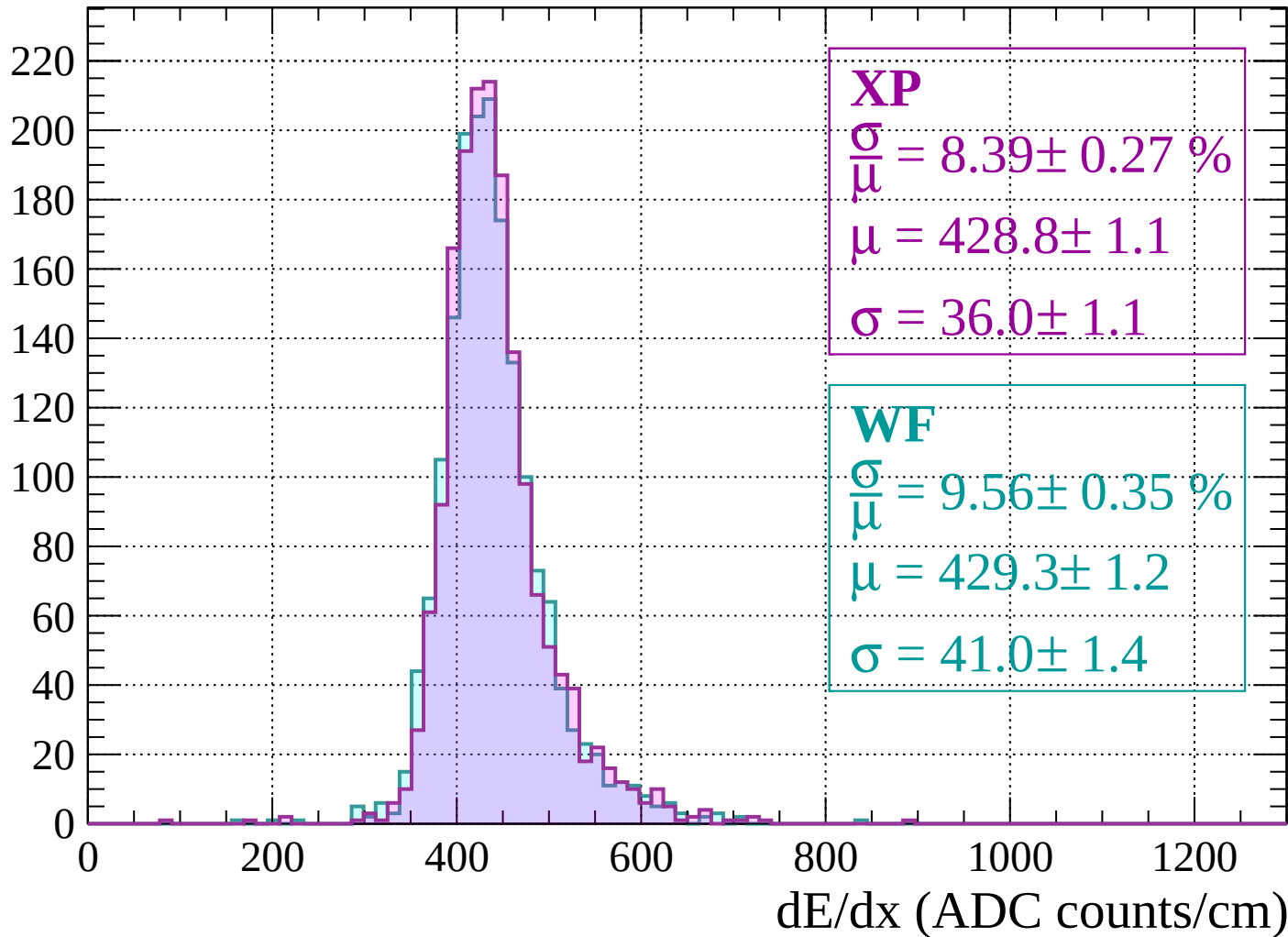
$$\mu = 428.9 \pm 1.2$$

$$\sigma = 40.4 \pm 1.3$$

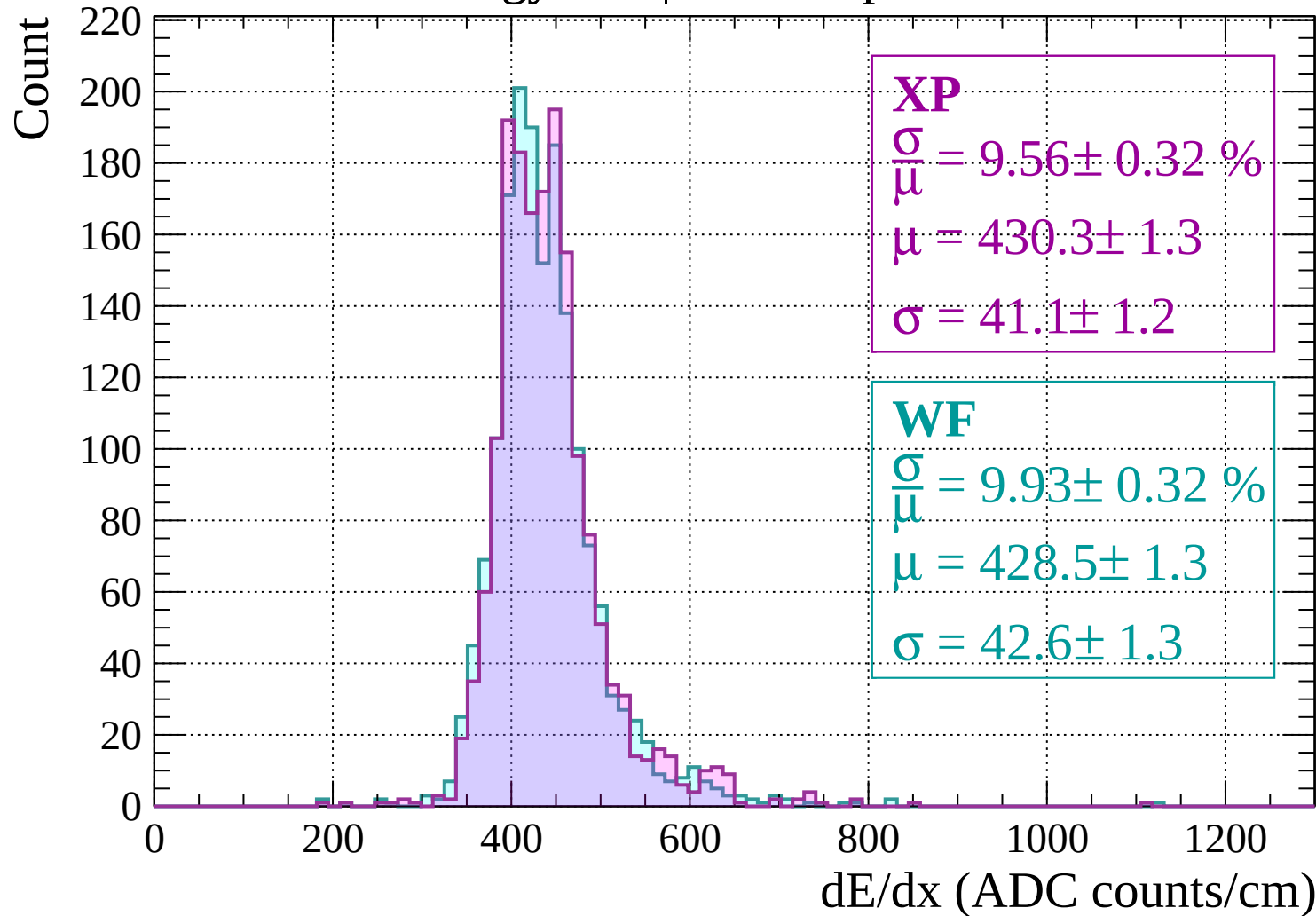
dE/dx (ADC counts/cm)

Energy loss | -1160 < p < -1120

Count

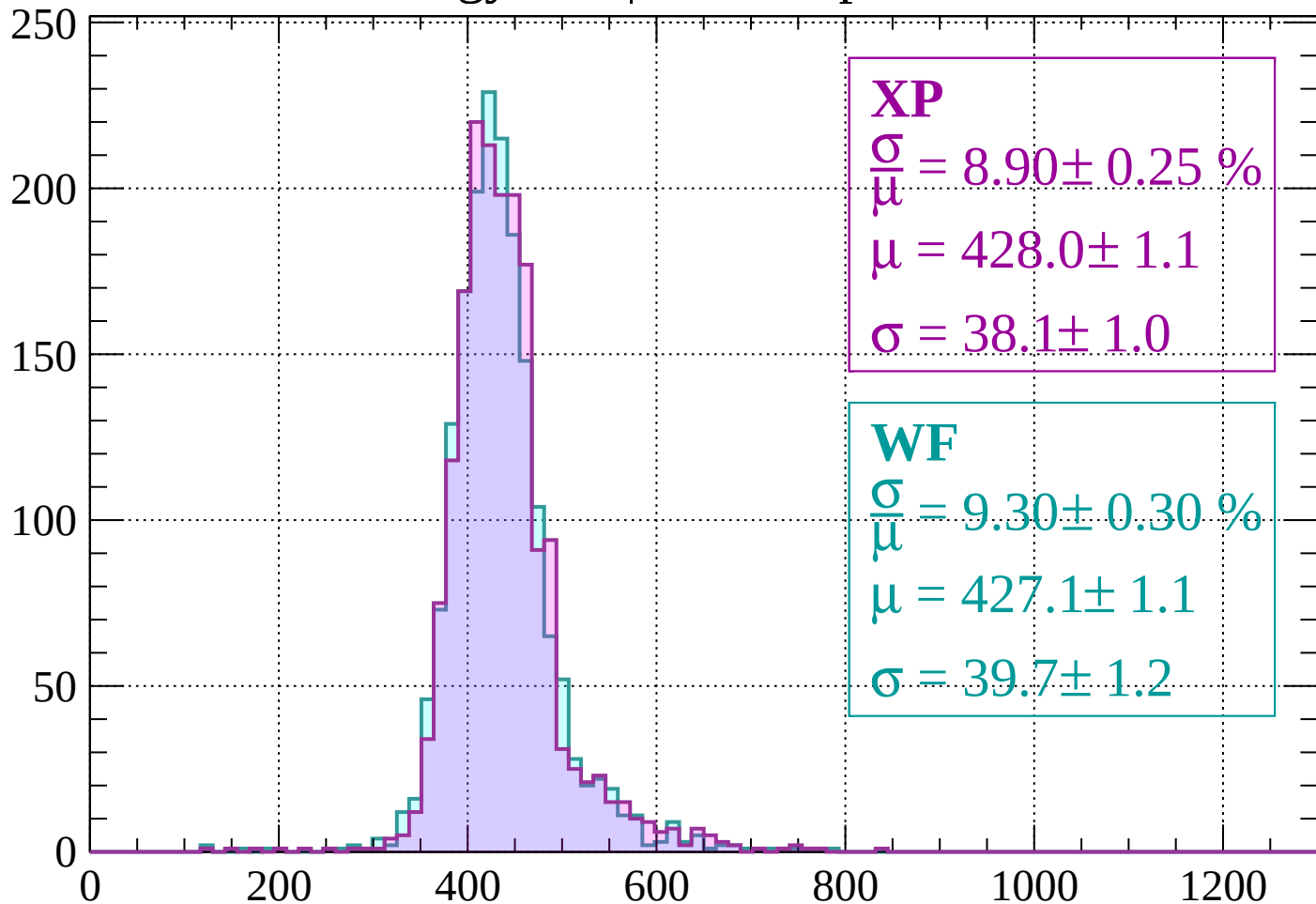


Energy loss | -1120 < p < -1080



Energy loss | -1080 < p < -1040

Count



XP

$$\frac{\sigma}{\mu} = 8.90 \pm 0.25 \%$$

$$\mu = 428.0 \pm 1.1$$

$$\sigma = 38.1 \pm 1.0$$

WF

$$\frac{\sigma}{\mu} = 9.30 \pm 0.30 \%$$

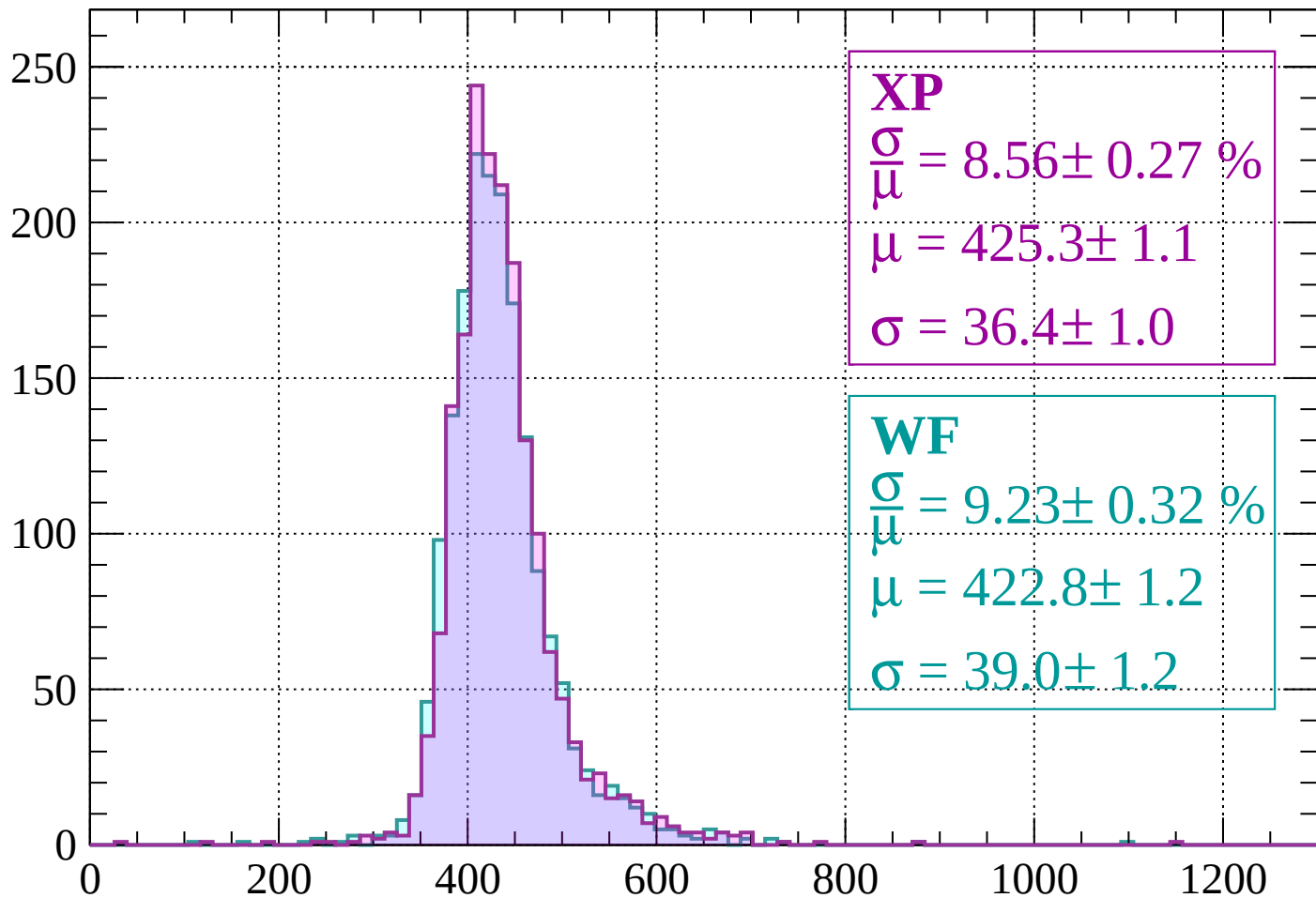
$$\mu = 427.1 \pm 1.1$$

$$\sigma = 39.7 \pm 1.2$$

dE/dx (ADC counts/cm)

Energy loss | $-1040 < p < -1000$

Count



XP

$$\frac{\sigma}{\mu} = 8.56 \pm 0.27 \%$$

$$\mu = 425.3 \pm 1.1$$

$$\sigma = 36.4 \pm 1.0$$

WF

$$\frac{\sigma}{\mu} = 9.23 \pm 0.32 \%$$

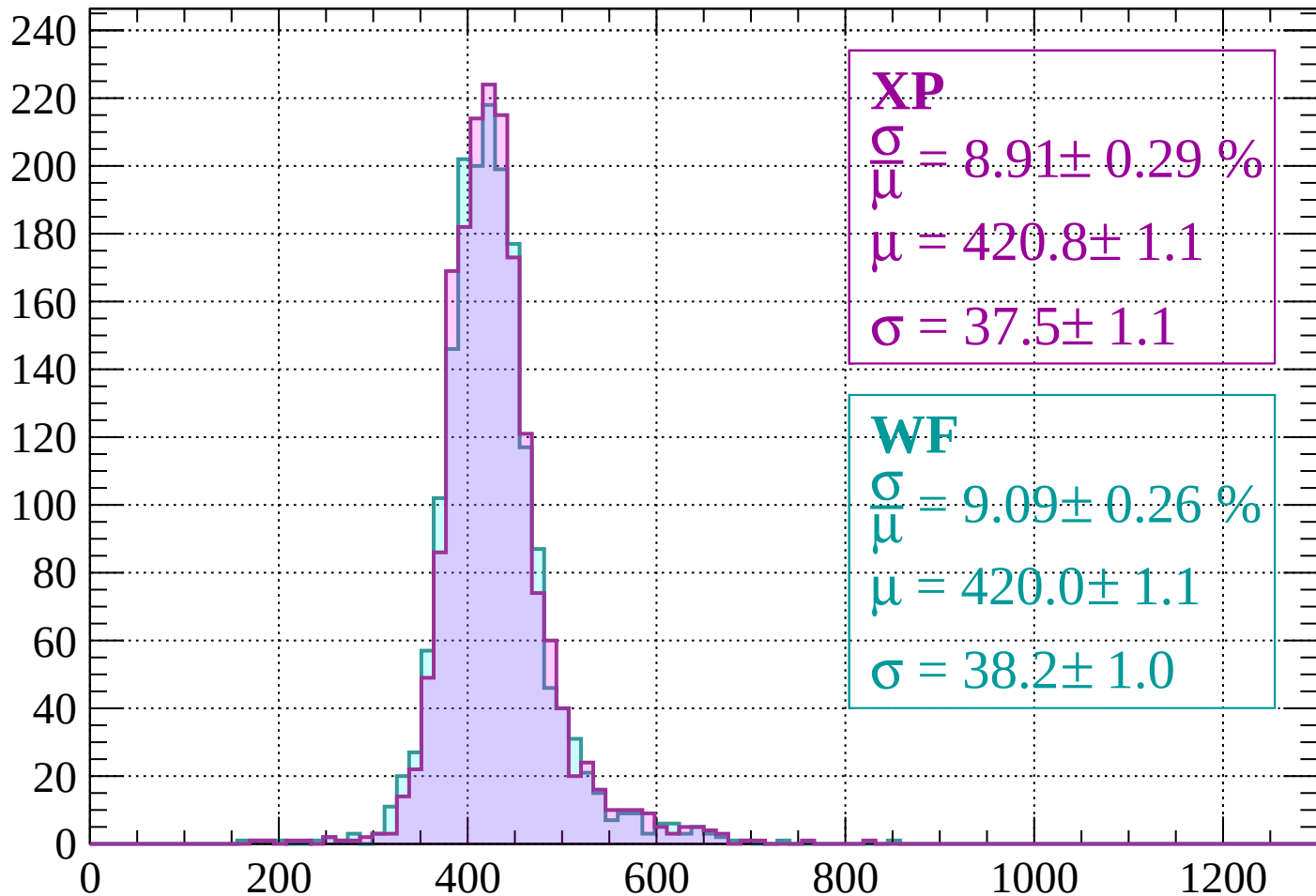
$$\mu = 422.8 \pm 1.2$$

$$\sigma = 39.0 \pm 1.2$$

dE/dx (ADC counts/cm)

Energy loss | $-1000 < p < -960$

Count



XP

$$\frac{\sigma}{\mu} = 8.91 \pm 0.29 \%$$

$$\mu = 420.8 \pm 1.1$$

$$\sigma = 37.5 \pm 1.1$$

WF

$$\frac{\sigma}{\mu} = 9.09 \pm 0.26 \%$$

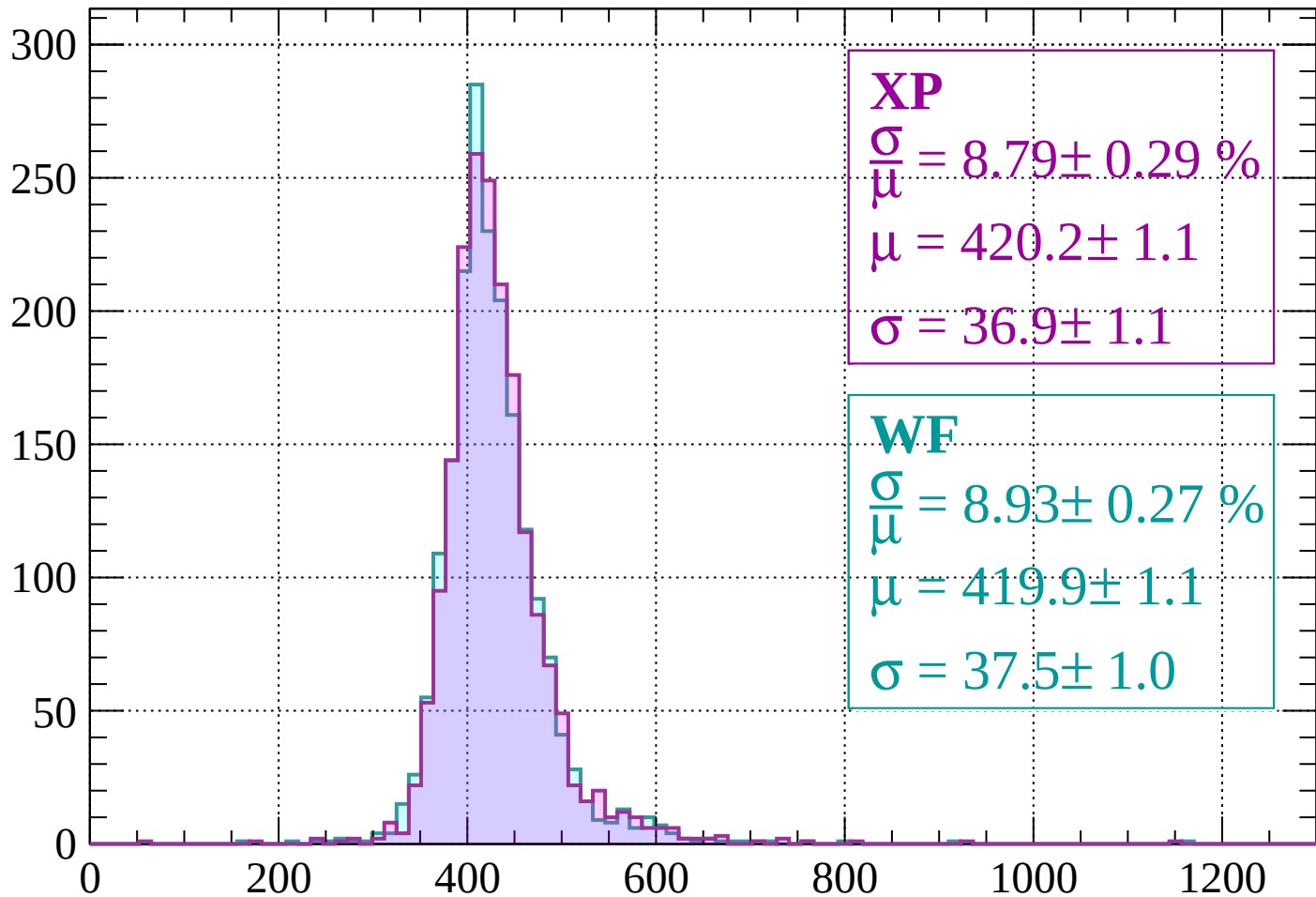
$$\mu = 420.0 \pm 1.1$$

$$\sigma = 38.2 \pm 1.0$$

dE/dx (ADC counts/cm)

Energy loss | $-960 < p < -920$

Count



XP

$$\frac{\sigma}{\mu} = 8.79 \pm 0.29 \%$$

$$\mu = 420.2 \pm 1.1$$

$$\sigma = 36.9 \pm 1.1$$

WF

$$\frac{\sigma}{\mu} = 8.93 \pm 0.27 \%$$

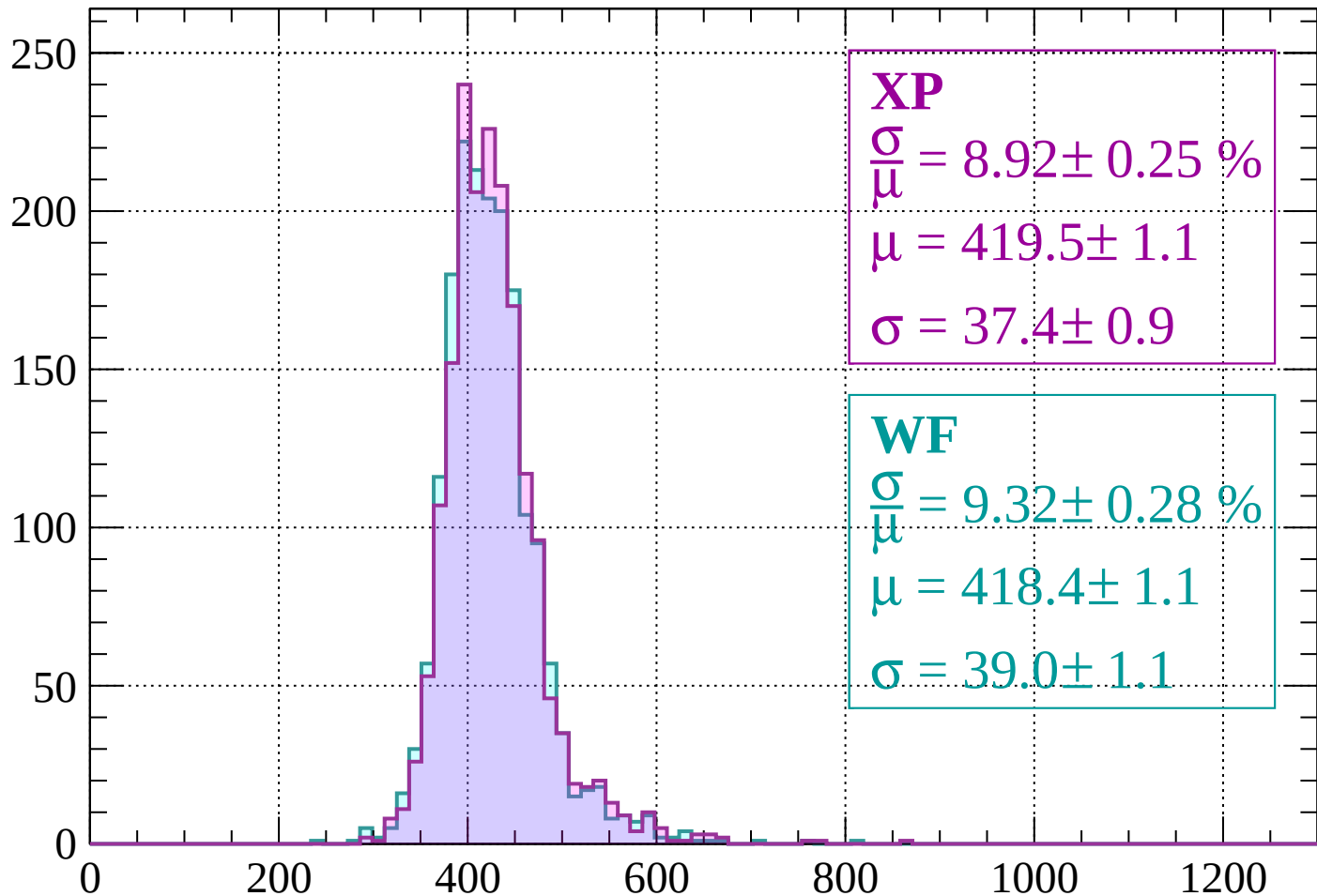
$$\mu = 419.9 \pm 1.1$$

$$\sigma = 37.5 \pm 1.0$$

dE/dx (ADC counts/cm)

Energy loss | $-920 < p < -880$

Count



XP

$$\frac{\sigma}{\mu} = 8.92 \pm 0.25 \%$$

$$\mu = 419.5 \pm 1.1$$

$$\sigma = 37.4 \pm 0.9$$

WF

$$\frac{\sigma}{\mu} = 9.32 \pm 0.28 \%$$

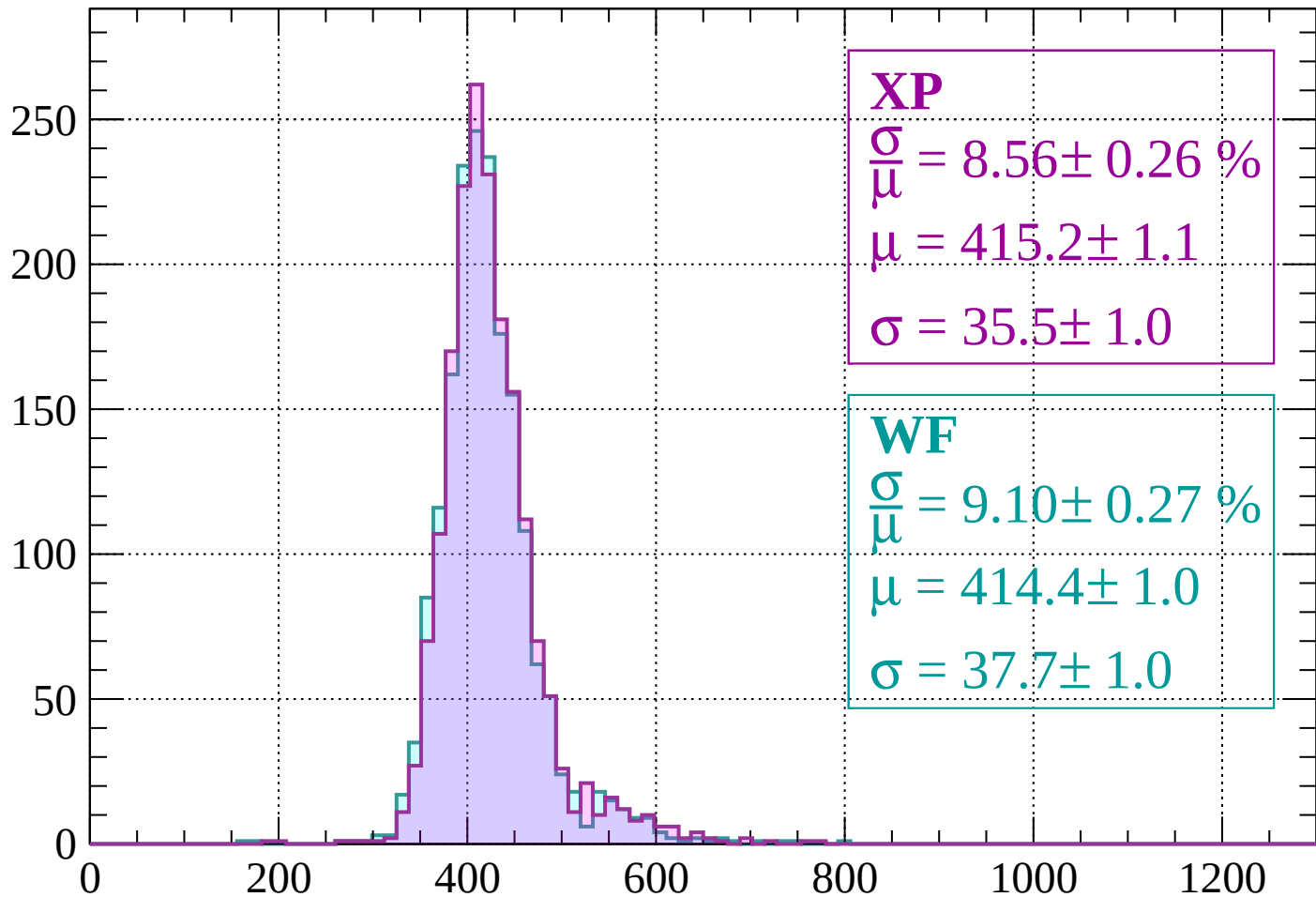
$$\mu = 418.4 \pm 1.1$$

$$\sigma = 39.0 \pm 1.1$$

dE/dx (ADC counts/cm)

Energy loss | $-880 < p < -840$

Count



XP

$$\frac{\sigma}{\mu} = 8.56 \pm 0.26 \%$$

$$\mu = 415.2 \pm 1.1$$

$$\sigma = 35.5 \pm 1.0$$

WF

$$\frac{\sigma}{\mu} = 9.10 \pm 0.27 \%$$

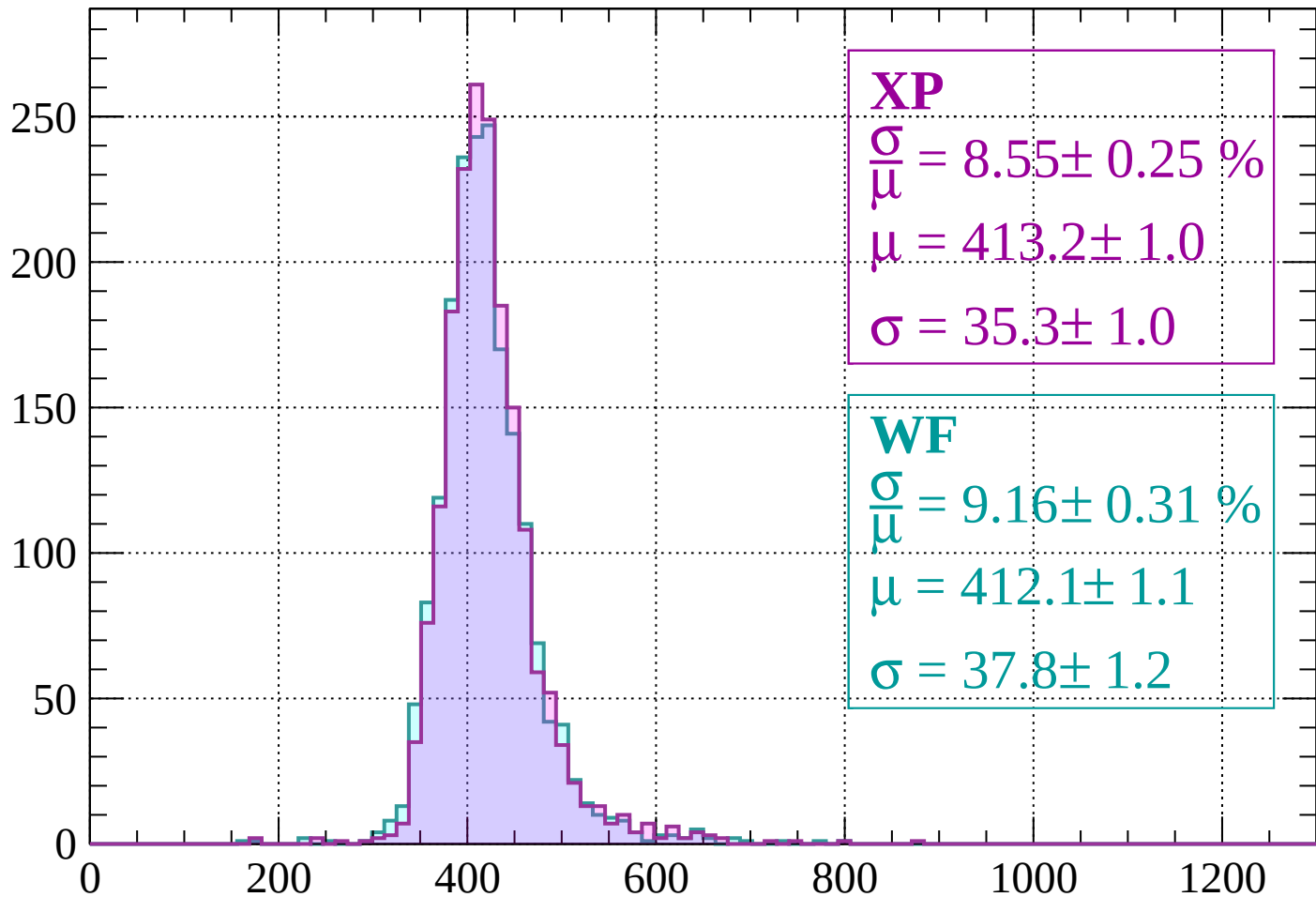
$$\mu = 414.4 \pm 1.0$$

$$\sigma = 37.7 \pm 1.0$$

dE/dx (ADC counts/cm)

Energy loss | $-840 < p < -800$

Count



XP

$$\frac{\sigma}{\bar{\mu}} = 8.55 \pm 0.25 \%$$

$$\mu = 413.2 \pm 1.0$$

$$\sigma = 35.3 \pm 1.0$$

WF

$$\frac{\sigma}{\bar{\mu}} = 9.16 \pm 0.31 \%$$

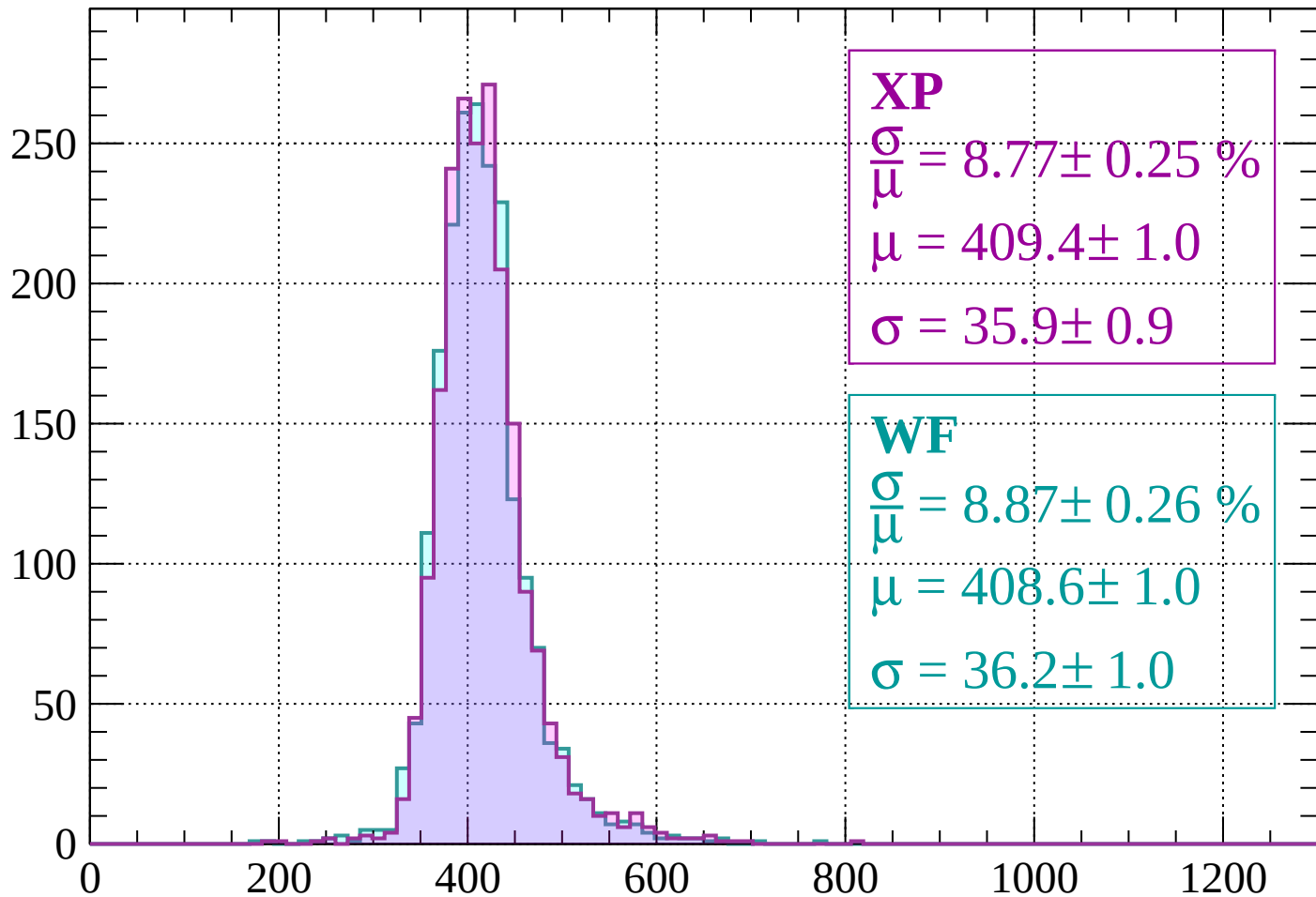
$$\mu = 412.1 \pm 1.1$$

$$\sigma = 37.8 \pm 1.2$$

dE/dx (ADC counts/cm)

Energy loss | $-800 < p < -760$

Count



XP

$$\frac{\sigma}{\mu} = 8.77 \pm 0.25 \%$$

$$\mu = 409.4 \pm 1.0$$

$$\sigma = 35.9 \pm 0.9$$

WF

$$\frac{\sigma}{\mu} = 8.87 \pm 0.26 \%$$

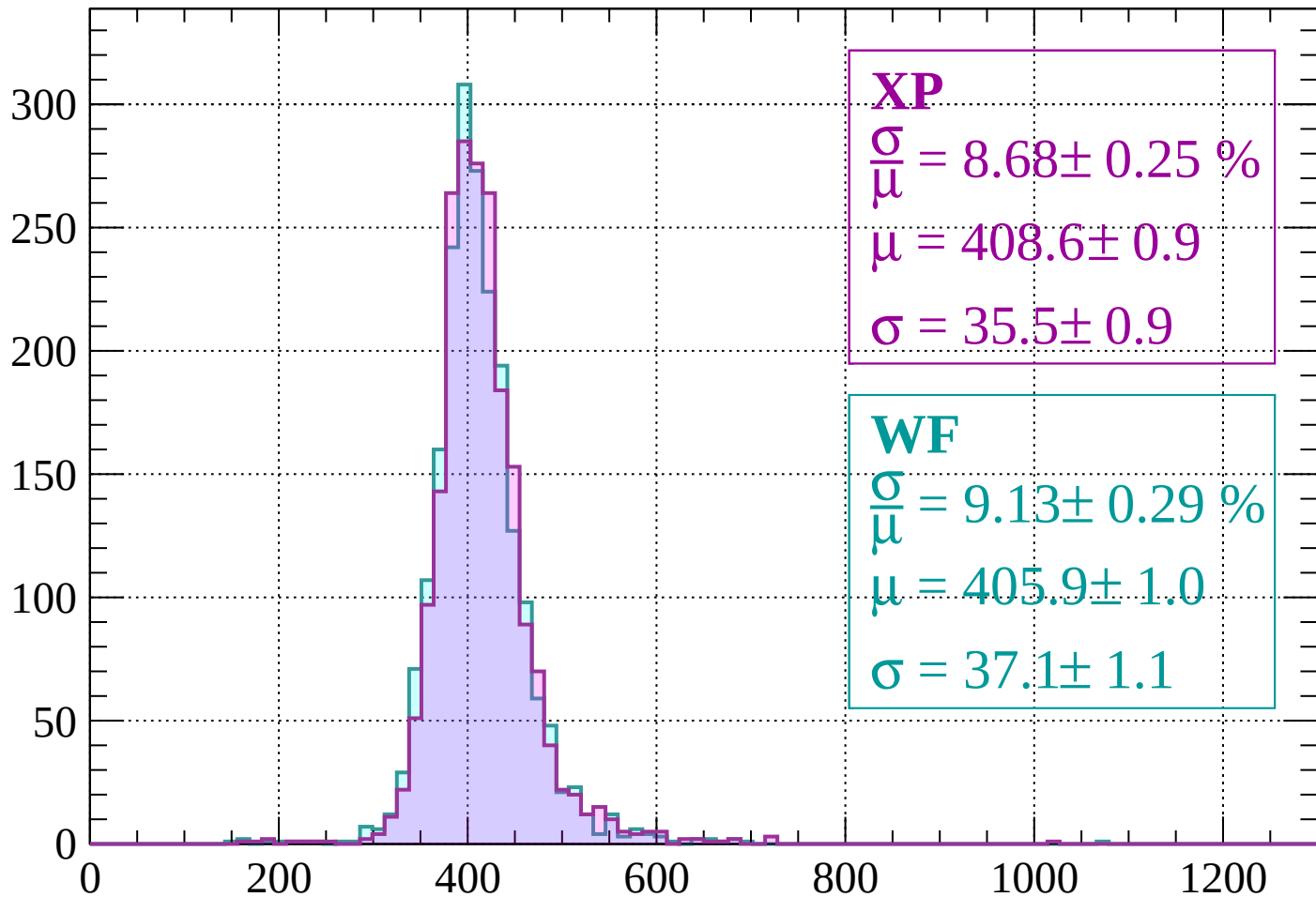
$$\mu = 408.6 \pm 1.0$$

$$\sigma = 36.2 \pm 1.0$$

dE/dx (ADC counts/cm)

Energy loss | $-760 < p < -720$

Count



XP

$$\frac{\sigma}{\mu} = 8.68 \pm 0.25 \%$$

$$\mu = 408.6 \pm 0.9$$

$$\sigma = 35.5 \pm 0.9$$

WF

$$\frac{\sigma}{\mu} = 9.13 \pm 0.29 \%$$

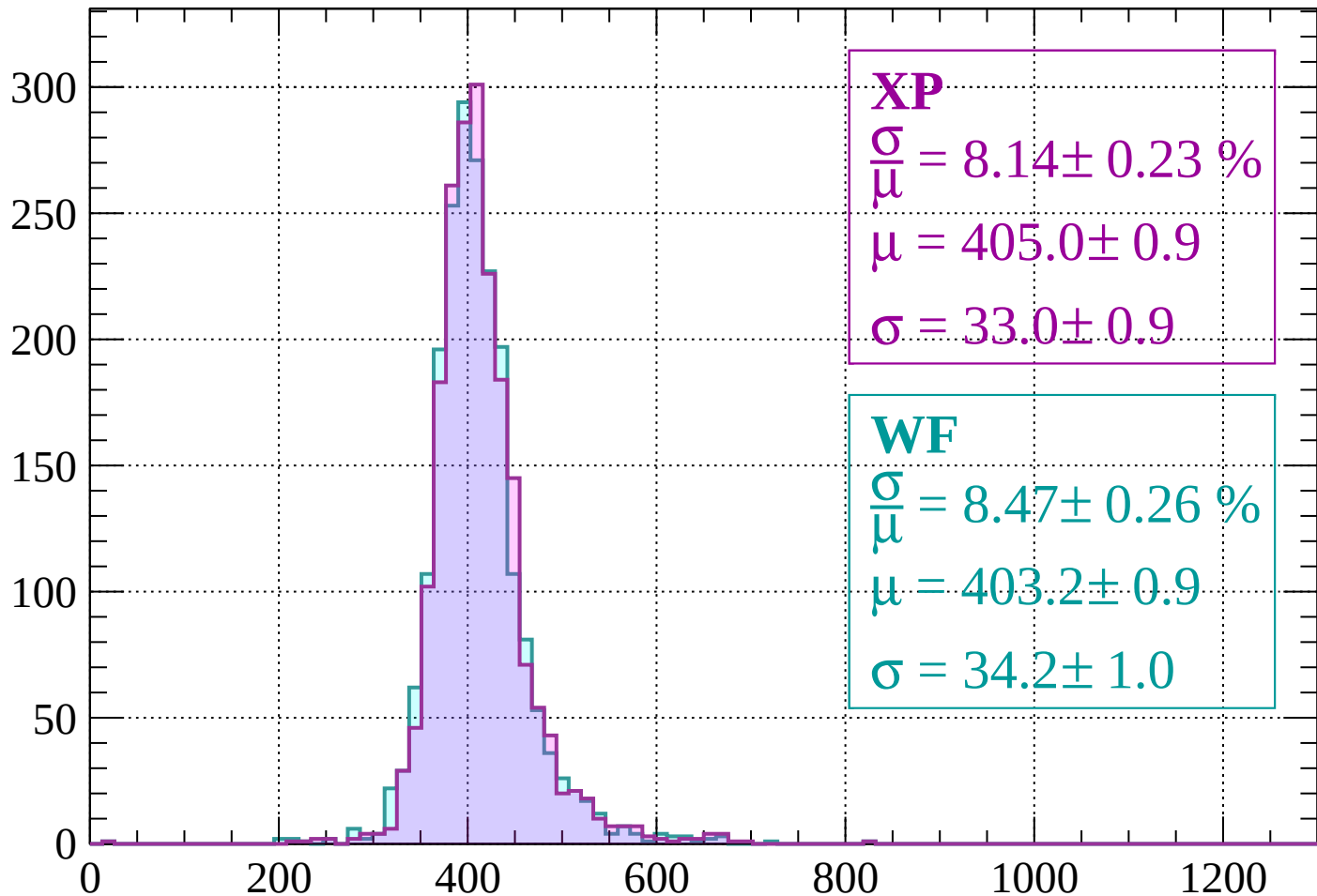
$$\mu = 405.9 \pm 1.0$$

$$\sigma = 37.1 \pm 1.1$$

dE/dx (ADC counts/cm)

Energy loss | $-720 < p < -680$

Count



XP

$$\frac{\sigma}{\mu} = 8.14 \pm 0.23 \%$$

$$\mu = 405.0 \pm 0.9$$

$$\sigma = 33.0 \pm 0.9$$

WF

$$\frac{\sigma}{\mu} = 8.47 \pm 0.26 \%$$

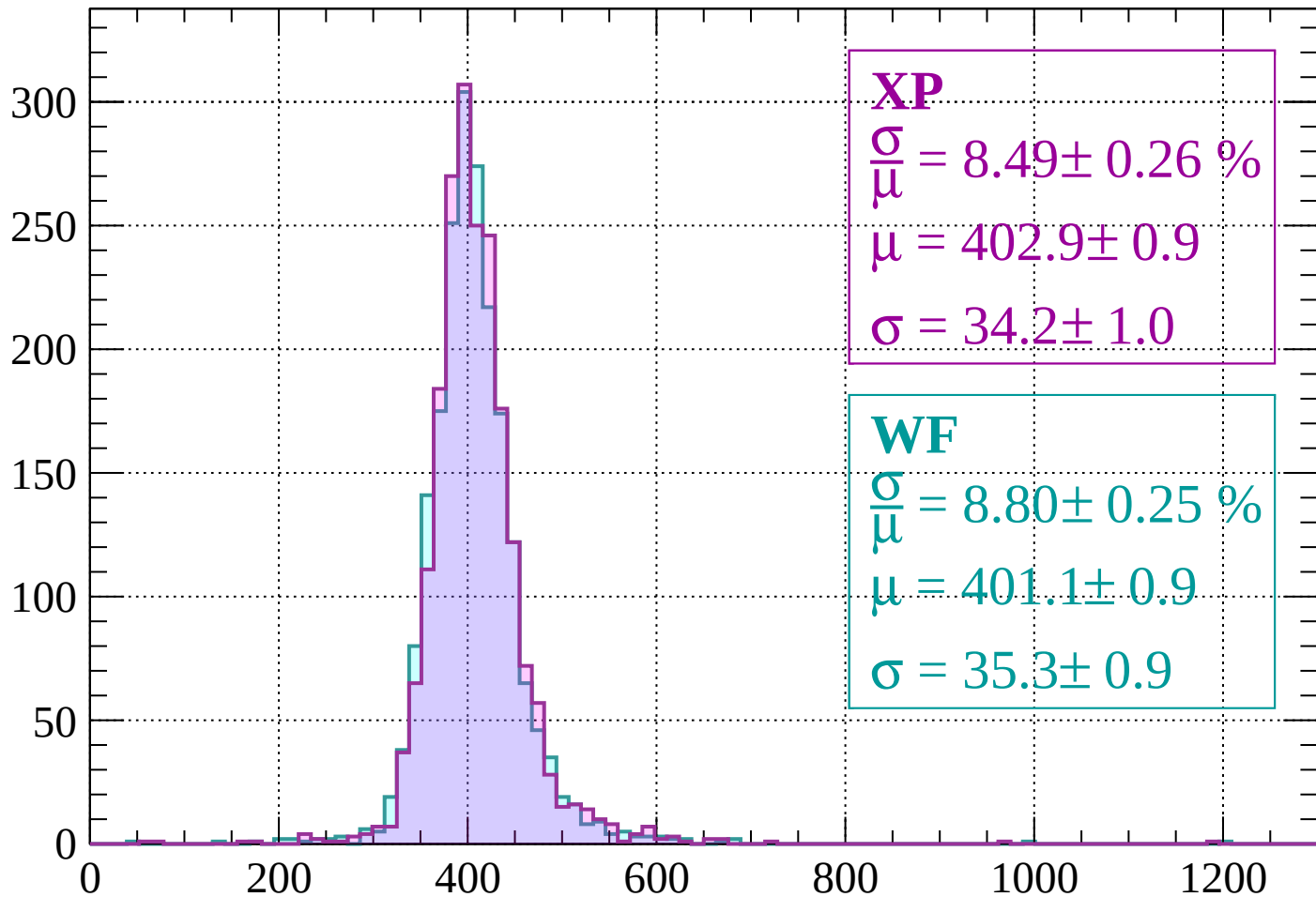
$$\mu = 403.2 \pm 0.9$$

$$\sigma = 34.2 \pm 1.0$$

dE/dx (ADC counts/cm)

Energy loss | $-680 < p < -640$

Count



XP

$$\frac{\sigma}{\mu} = 8.49 \pm 0.26 \%$$

$$\mu = 402.9 \pm 0.9$$

$$\sigma = 34.2 \pm 1.0$$

WF

$$\frac{\sigma}{\mu} = 8.80 \pm 0.25 \%$$

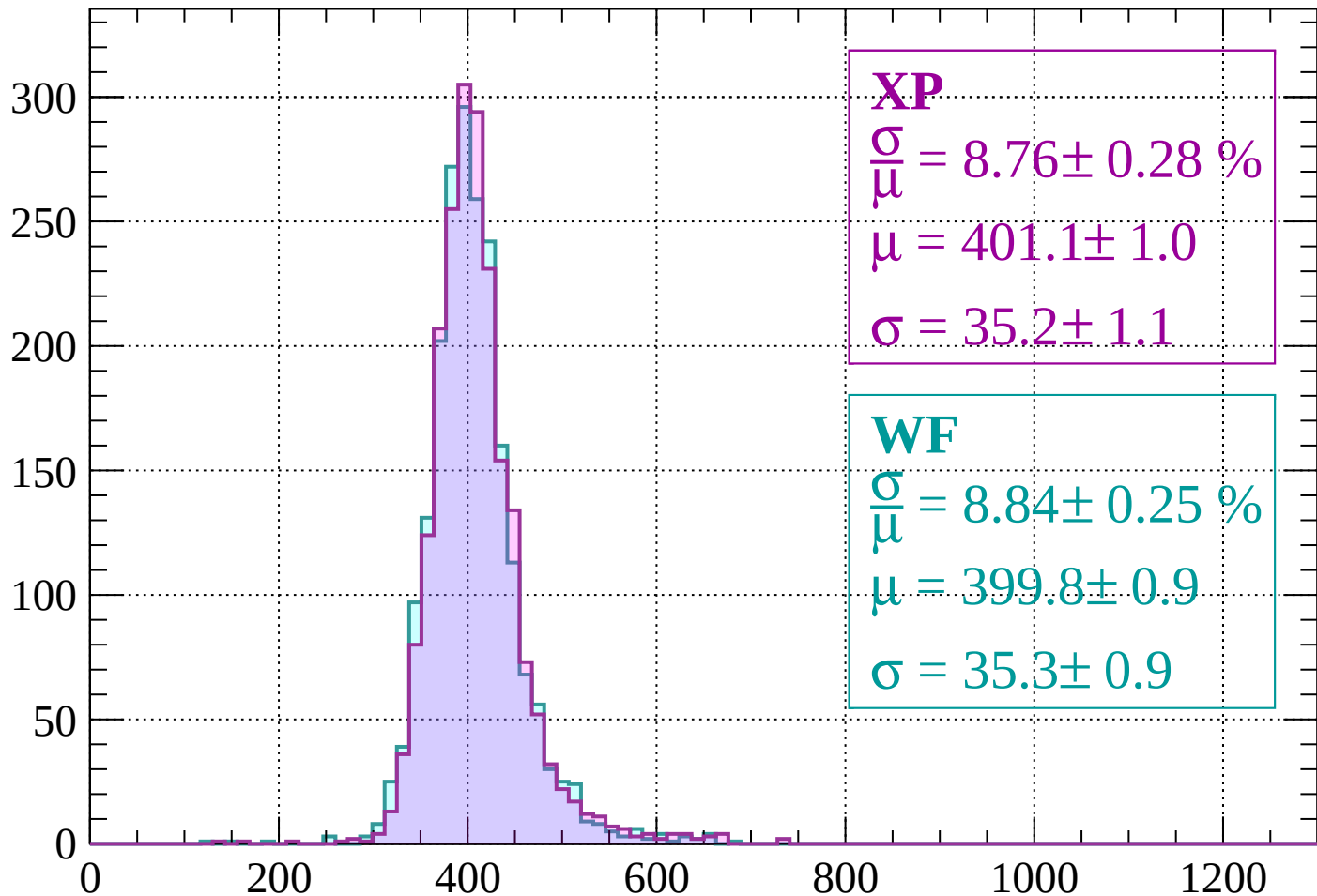
$$\mu = 401.1 \pm 0.9$$

$$\sigma = 35.3 \pm 0.9$$

dE/dx (ADC counts/cm)

Energy loss | $-640 < p < -600$

Count



XP

$$\frac{\sigma}{\mu} = 8.76 \pm 0.28 \%$$

$$\mu = 401.1 \pm 1.0$$

$$\sigma = 35.2 \pm 1.1$$

WF

$$\frac{\sigma}{\mu} = 8.84 \pm 0.25 \%$$

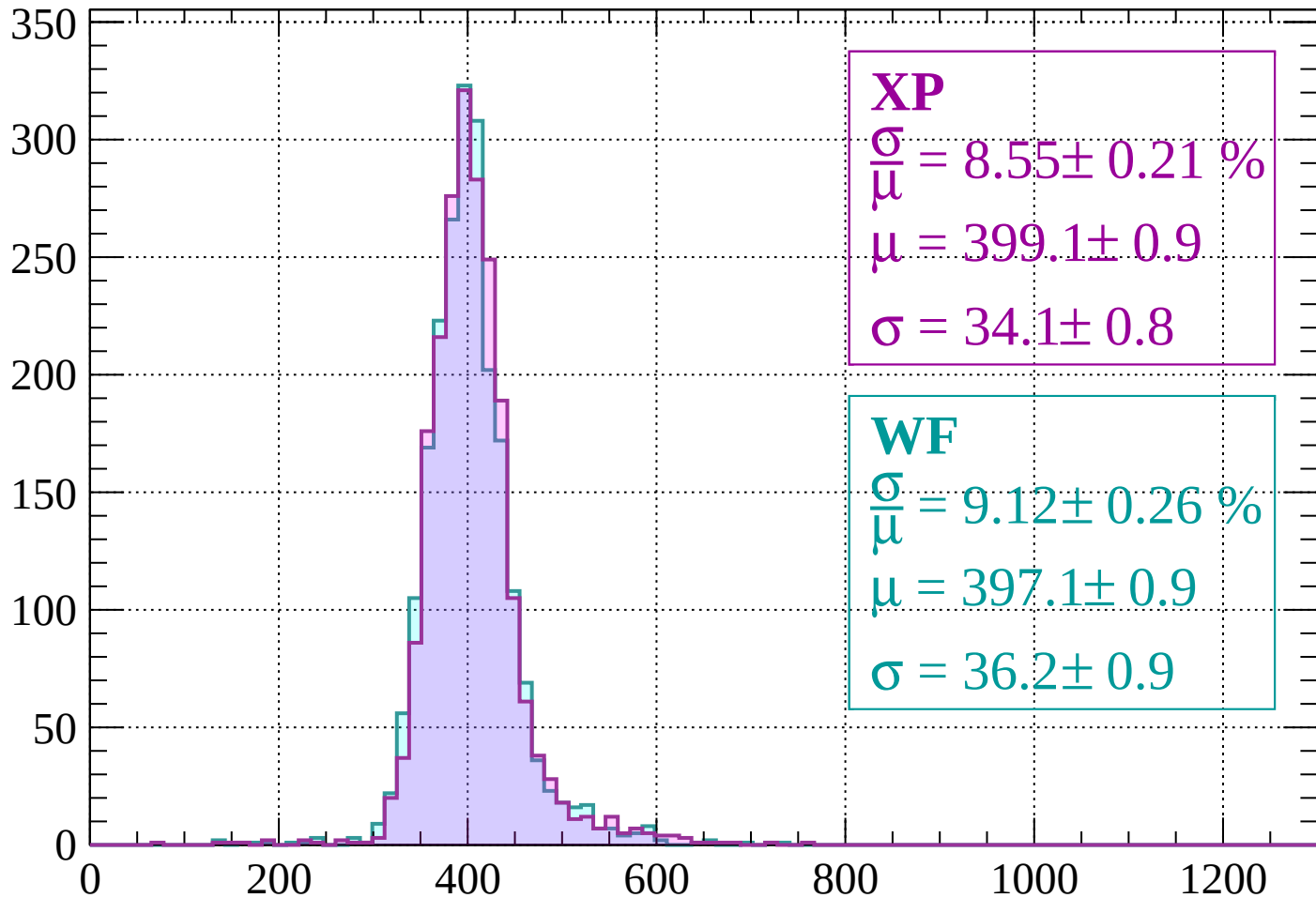
$$\mu = 399.8 \pm 0.9$$

$$\sigma = 35.3 \pm 0.9$$

dE/dx (ADC counts/cm)

Energy loss | $-600 < p < -560$

Count



XP

$$\frac{\sigma}{\mu} = 8.55 \pm 0.21 \%$$

$$\mu = 399.1 \pm 0.9$$

$$\sigma = 34.1 \pm 0.8$$

WF

$$\frac{\sigma}{\mu} = 9.12 \pm 0.26 \%$$

$$\mu = 397.1 \pm 0.9$$

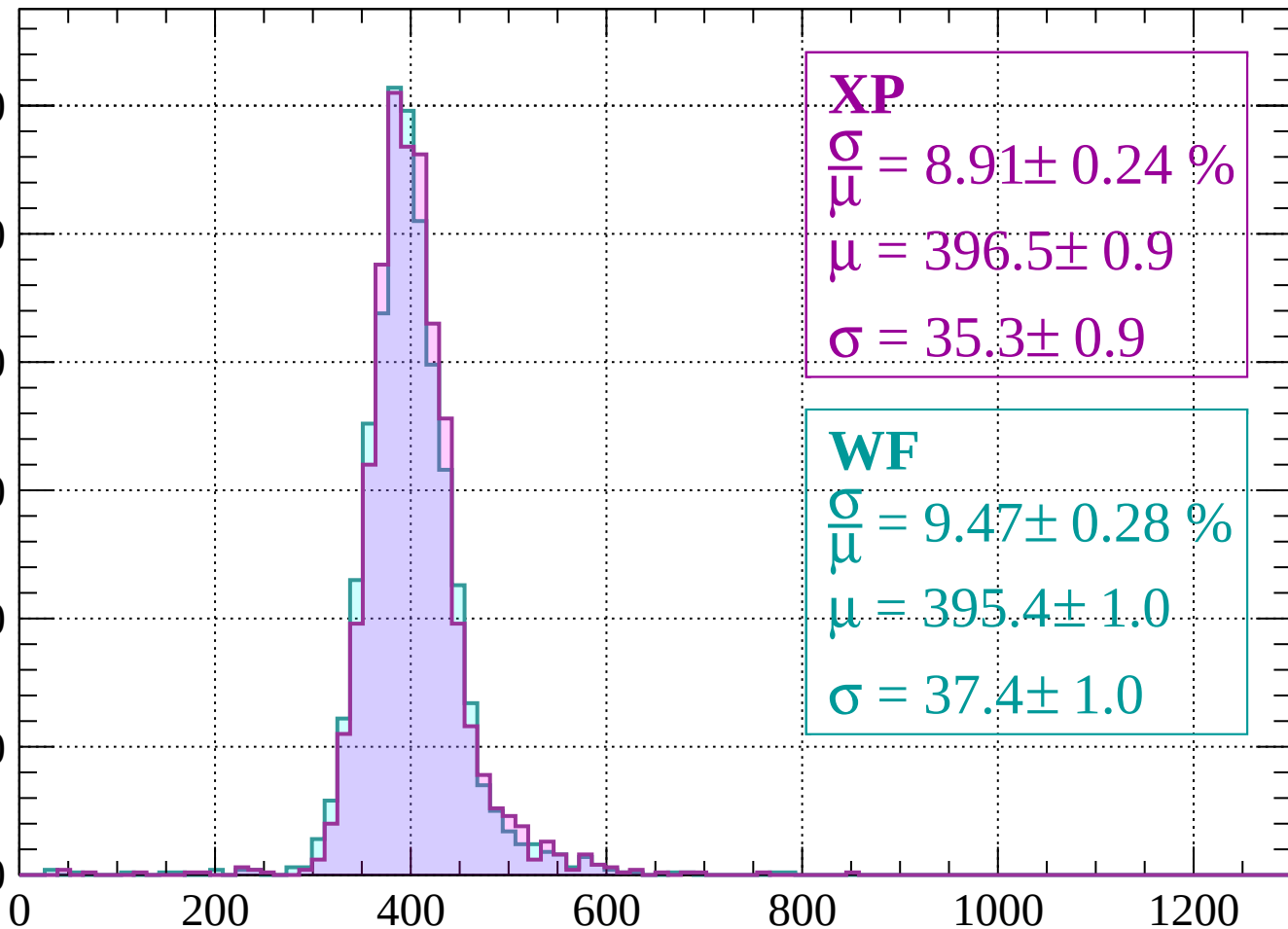
$$\sigma = 36.2 \pm 0.9$$

dE/dx (ADC counts/cm)

Energy loss | $-560 < p < -520$

Count

300
250
200
150
100
50
0



XP

$$\frac{\sigma}{\mu} = 8.91 \pm 0.24 \%$$

$$\mu = 396.5 \pm 0.9$$

$$\sigma = 35.3 \pm 0.9$$

WF

$$\frac{\sigma}{\mu} = 9.47 \pm 0.28 \%$$

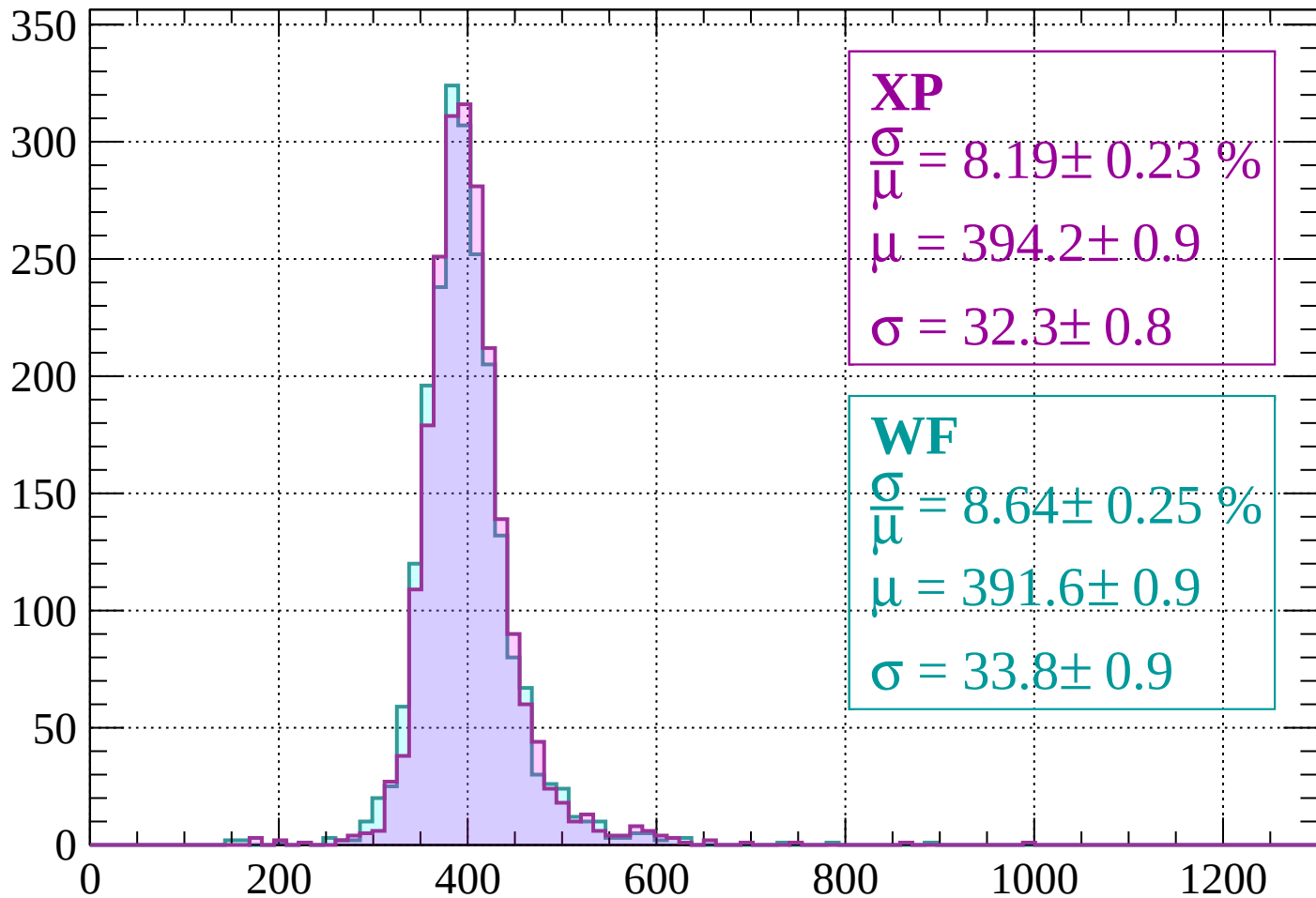
$$\mu = 395.4 \pm 1.0$$

$$\sigma = 37.4 \pm 1.0$$

dE/dx (ADC counts/cm)

Energy loss | $-520 < p < -480$

Count



XP

$$\frac{\sigma}{\mu} = 8.19 \pm 0.23 \%$$

$$\mu = 394.2 \pm 0.9$$

$$\sigma = 32.3 \pm 0.8$$

WF

$$\frac{\sigma}{\mu} = 8.64 \pm 0.25 \%$$

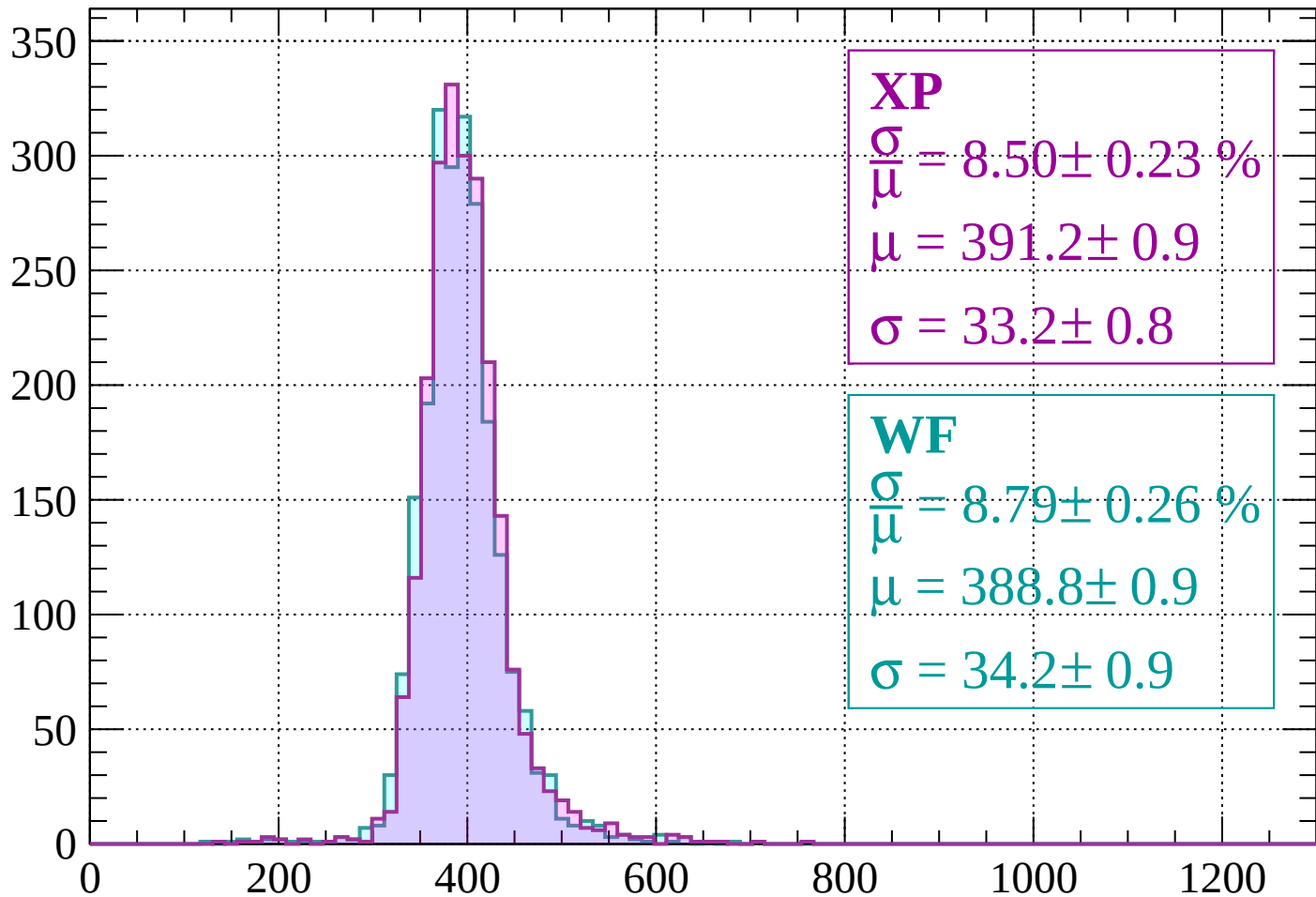
$$\mu = 391.6 \pm 0.9$$

$$\sigma = 33.8 \pm 0.9$$

dE/dx (ADC counts/cm)

Energy loss | $-480 < p < -440$

Count



XP

$$\frac{\sigma}{\bar{\mu}} = 8.50 \pm 0.23 \%$$

$$\mu = 391.2 \pm 0.9$$

$$\sigma = 33.2 \pm 0.8$$

WF

$$\frac{\sigma}{\bar{\mu}} = 8.79 \pm 0.26 \%$$

$$\mu = 388.8 \pm 0.9$$

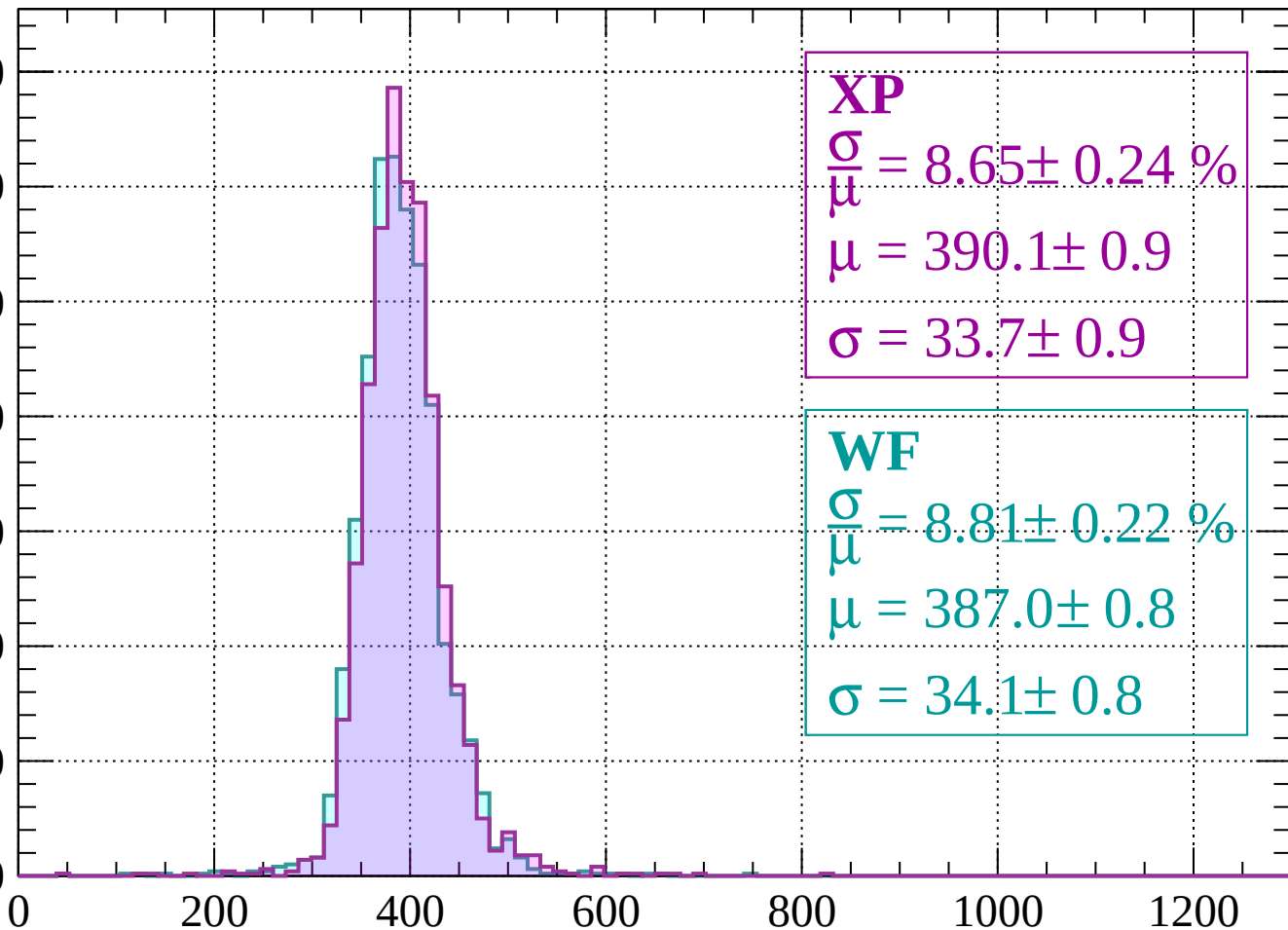
$$\sigma = 34.2 \pm 0.9$$

dE/dx (ADC counts/cm)

Energy loss | $-440 < p < -400$

Count

350
300
250
200
150
100
50
0



XP

$$\frac{\sigma}{\mu} = 8.65 \pm 0.24 \%$$

$$\mu = 390.1 \pm 0.9$$

$$\sigma = 33.7 \pm 0.9$$

WF

$$\frac{\sigma}{\mu} = 8.81 \pm 0.22 \%$$

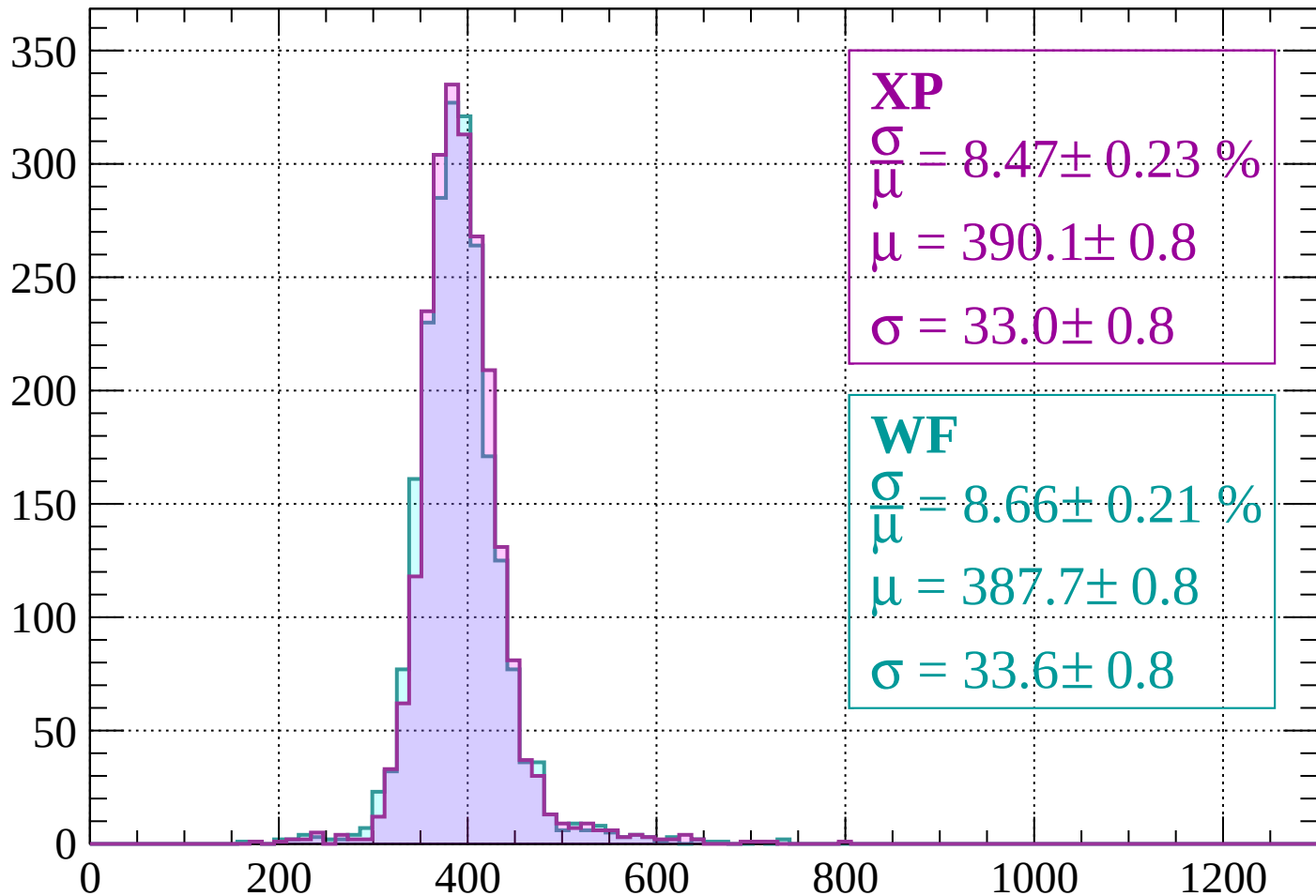
$$\mu = 387.0 \pm 0.8$$

$$\sigma = 34.1 \pm 0.8$$

dE/dx (ADC counts/cm)

Energy loss | $-400 < p < -360$

Count



XP

$$\frac{\sigma}{\mu} = 8.47 \pm 0.23 \%$$

$$\mu = 390.1 \pm 0.8$$

$$\sigma = 33.0 \pm 0.8$$

WF

$$\frac{\sigma}{\mu} = 8.66 \pm 0.21 \%$$

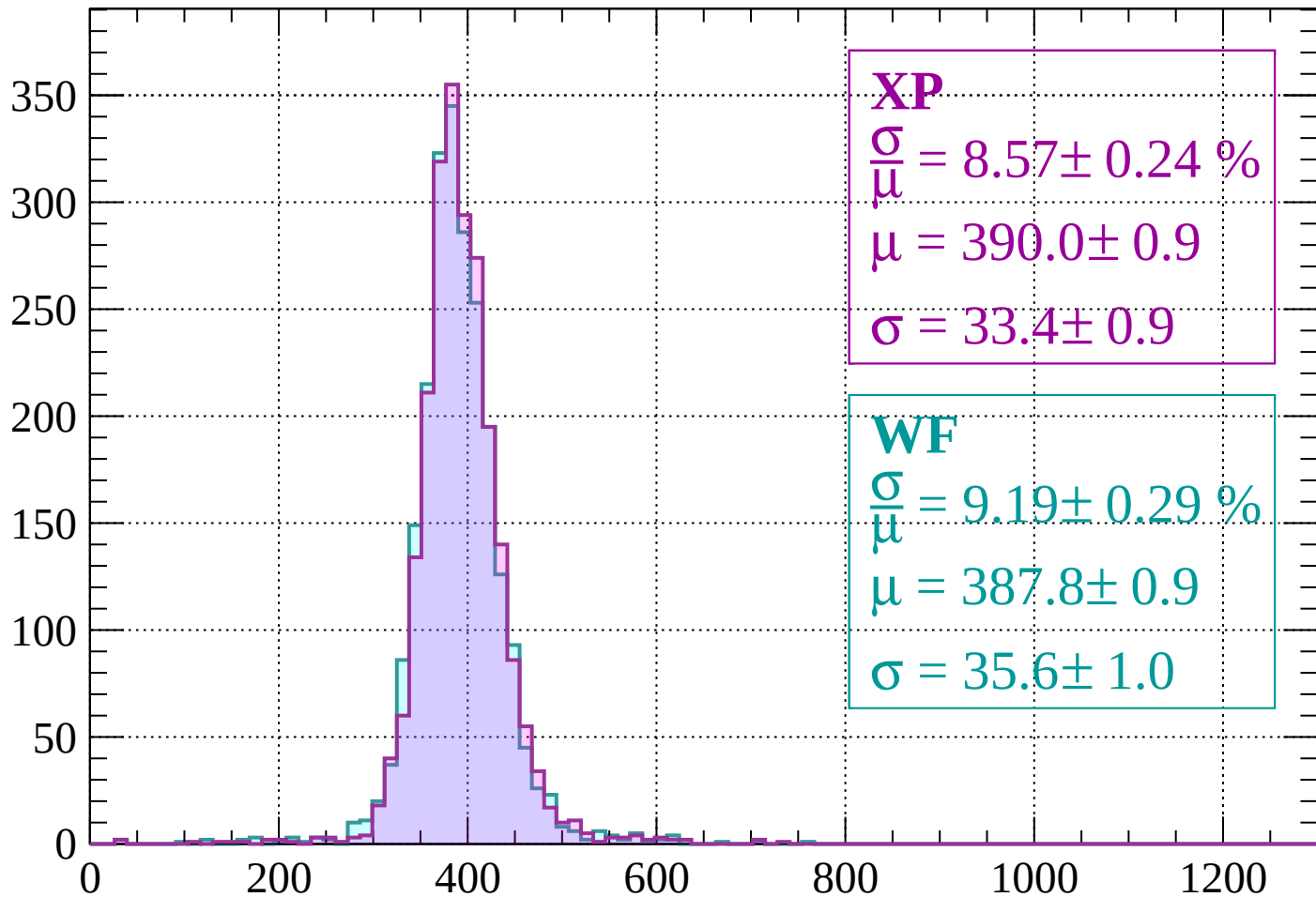
$$\mu = 387.7 \pm 0.8$$

$$\sigma = 33.6 \pm 0.8$$

dE/dx (ADC counts/cm)

Energy loss | $-360 < p < -320$

Count



XP

$$\frac{\sigma}{\mu} = 8.57 \pm 0.24 \%$$

$$\mu = 390.0 \pm 0.9$$

$$\sigma = 33.4 \pm 0.9$$

WF

$$\frac{\sigma}{\mu} = 9.19 \pm 0.29 \%$$

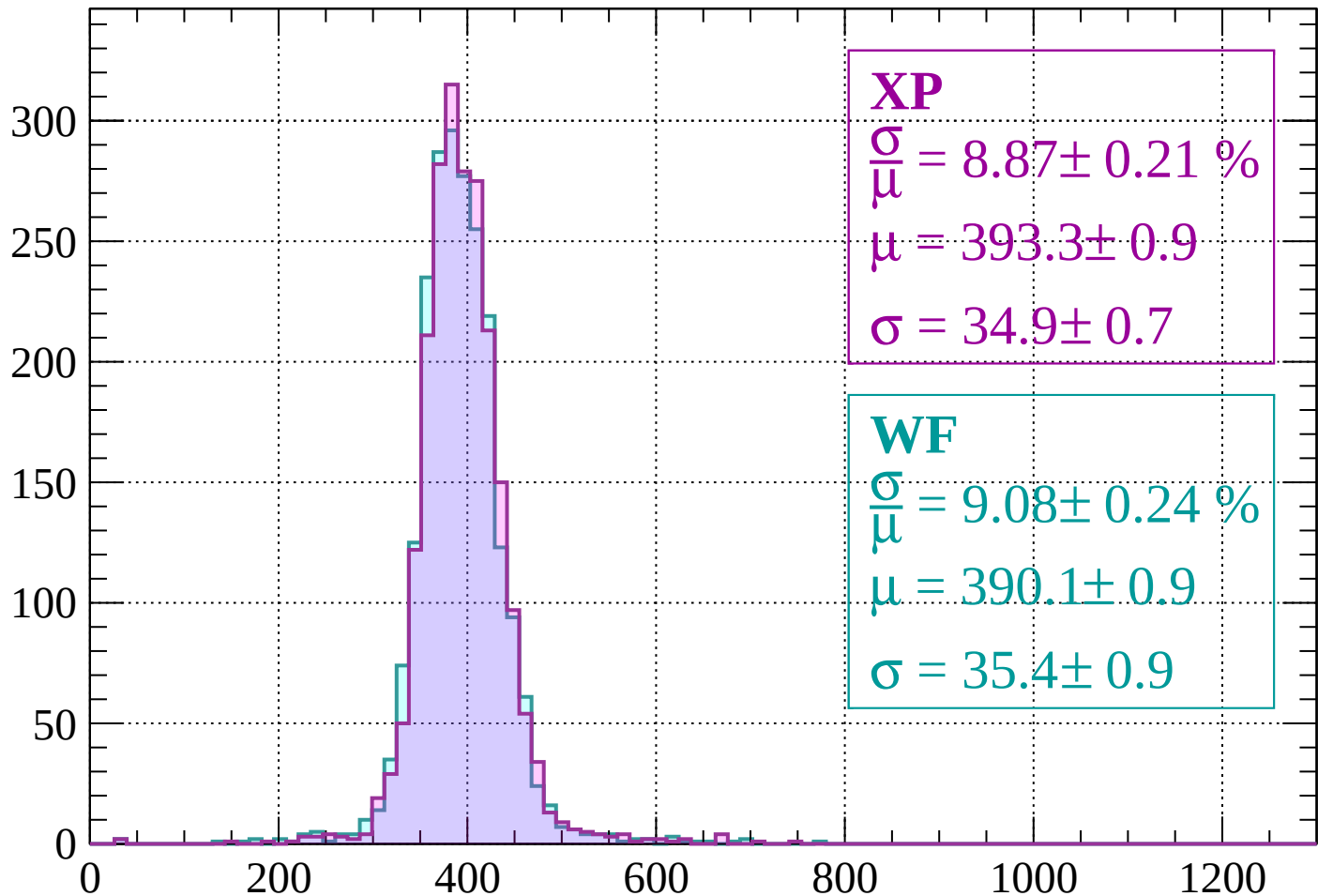
$$\mu = 387.8 \pm 0.9$$

$$\sigma = 35.6 \pm 1.0$$

dE/dx (ADC counts/cm)

Energy loss | $-320 < p < -280$

Count



XP

$$\frac{\sigma}{\mu} = 8.87 \pm 0.21 \%$$

$$\mu = 393.3 \pm 0.9$$

$$\sigma = 34.9 \pm 0.7$$

WF

$$\frac{\sigma}{\mu} = 9.08 \pm 0.24 \%$$

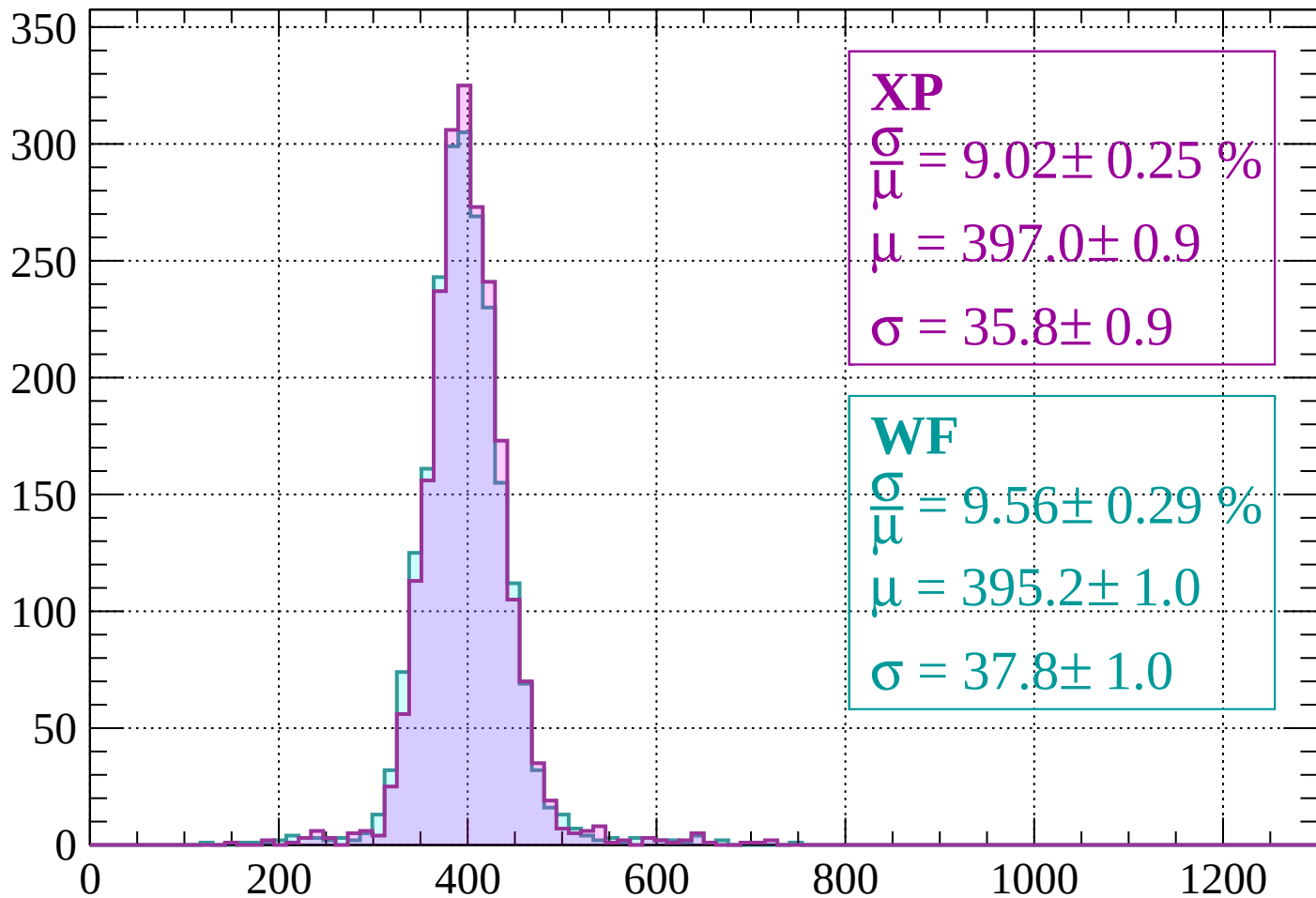
$$\mu = 390.1 \pm 0.9$$

$$\sigma = 35.4 \pm 0.9$$

dE/dx (ADC counts/cm)

Energy loss | $-280 < p < -240$

Count



XP

$$\frac{\sigma}{\mu} = 9.02 \pm 0.25 \%$$

$$\mu = 397.0 \pm 0.9$$

$$\sigma = 35.8 \pm 0.9$$

WF

$$\frac{\sigma}{\mu} = 9.56 \pm 0.29 \%$$

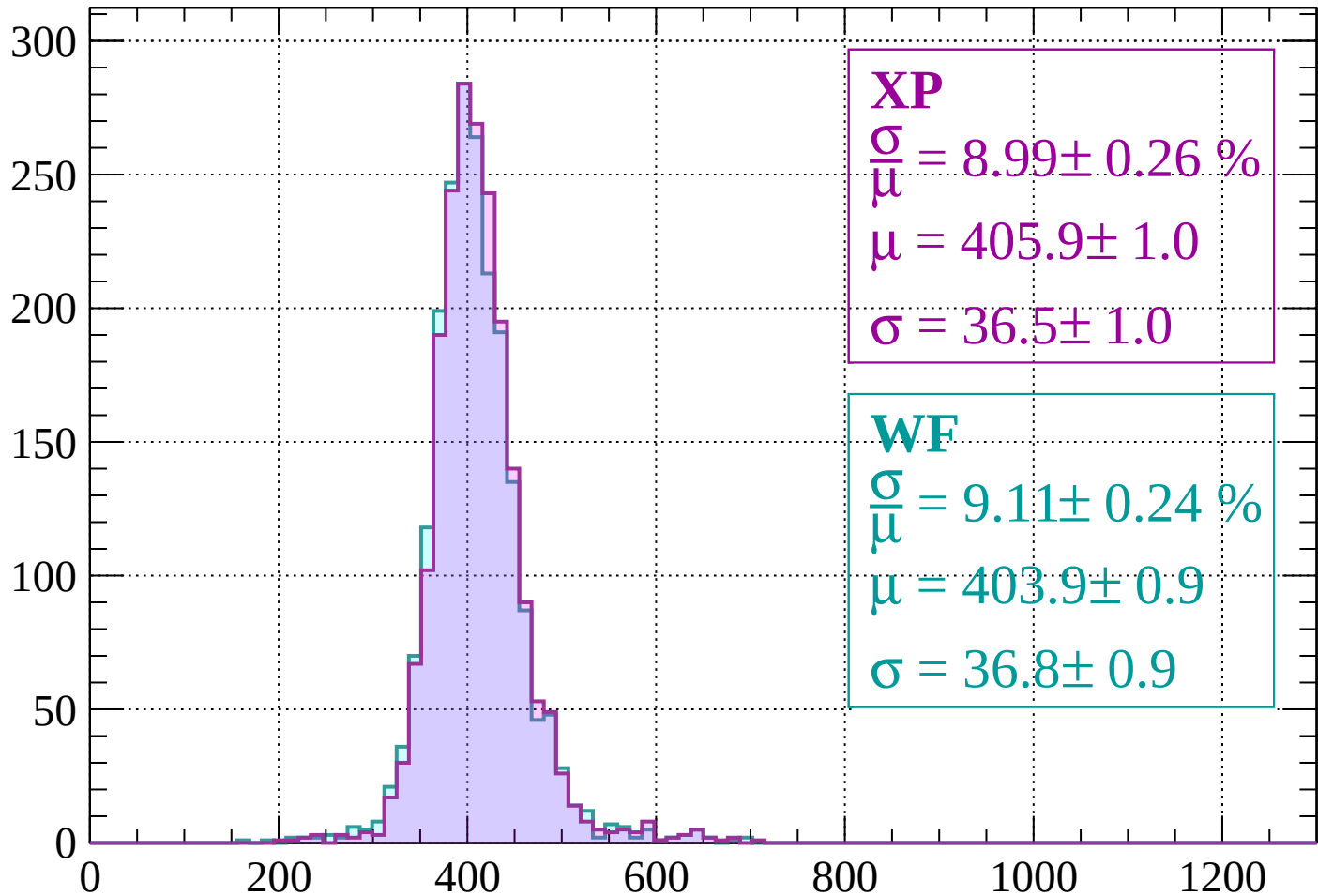
$$\mu = 395.2 \pm 1.0$$

$$\sigma = 37.8 \pm 1.0$$

dE/dx (ADC counts/cm)

Energy loss | $-240 < p < -200$

Count



XP

$$\frac{\sigma}{\mu} = 8.99 \pm 0.26 \%$$

$$\mu = 405.9 \pm 1.0$$

$$\sigma = 36.5 \pm 1.0$$

WF

$$\frac{\sigma}{\mu} = 9.11 \pm 0.24 \%$$

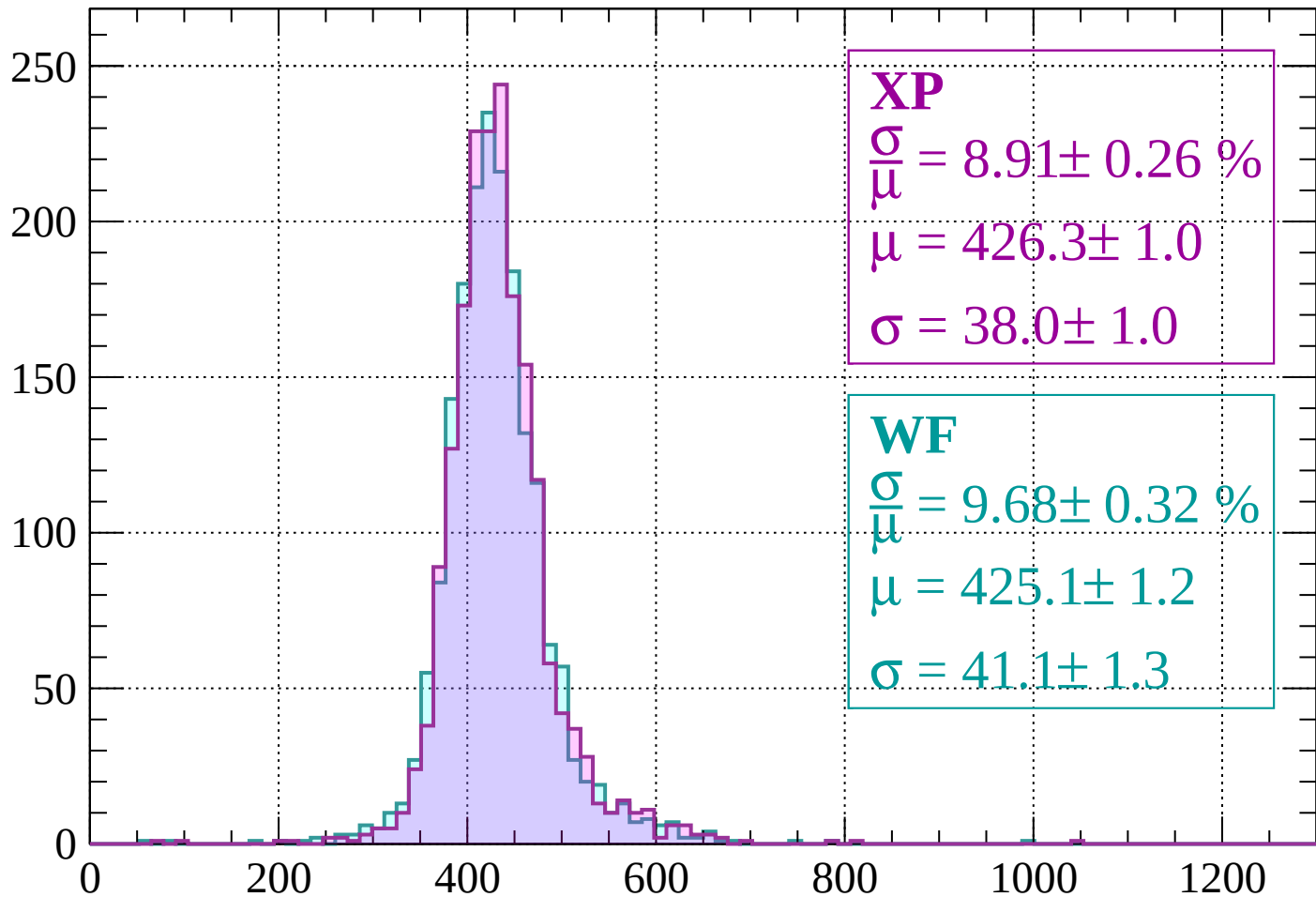
$$\mu = 403.9 \pm 0.9$$

$$\sigma = 36.8 \pm 0.9$$

dE/dx (ADC counts/cm)

Energy loss | $-200 < p < -160$

Count



XP

$$\frac{\sigma}{\mu} = 8.91 \pm 0.26 \%$$

$$\mu = 426.3 \pm 1.0$$

$$\sigma = 38.0 \pm 1.0$$

WF

$$\frac{\sigma}{\mu} = 9.68 \pm 0.32 \%$$

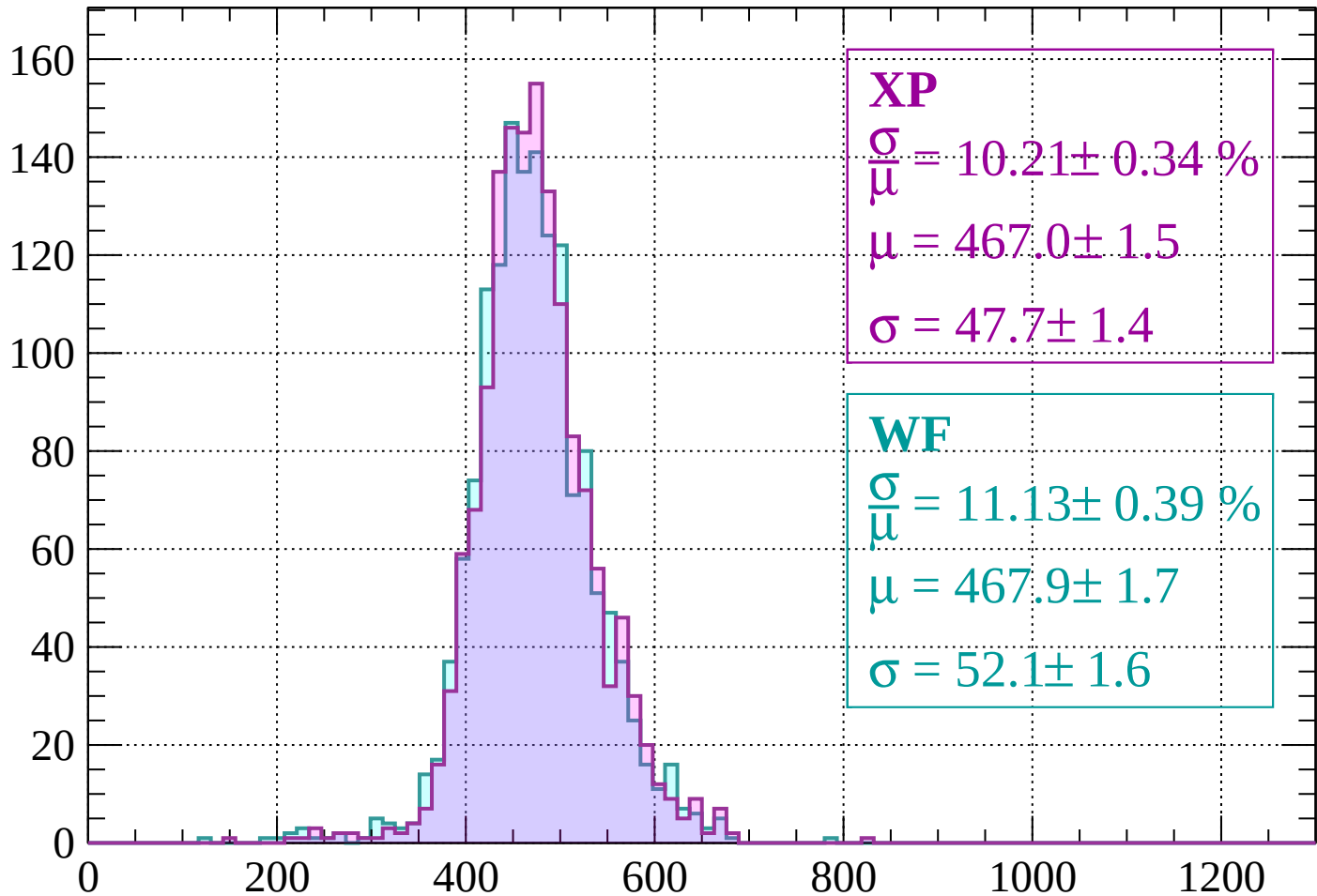
$$\mu = 425.1 \pm 1.2$$

$$\sigma = 41.1 \pm 1.3$$

dE/dx (ADC counts/cm)

Energy loss | $-160 < p < -120$

Count



XP

$$\frac{\sigma}{\mu} = 10.21 \pm 0.34 \%$$

$$\mu = 467.0 \pm 1.5$$

$$\sigma = 47.7 \pm 1.4$$

WF

$$\frac{\sigma}{\mu} = 11.13 \pm 0.39 \%$$

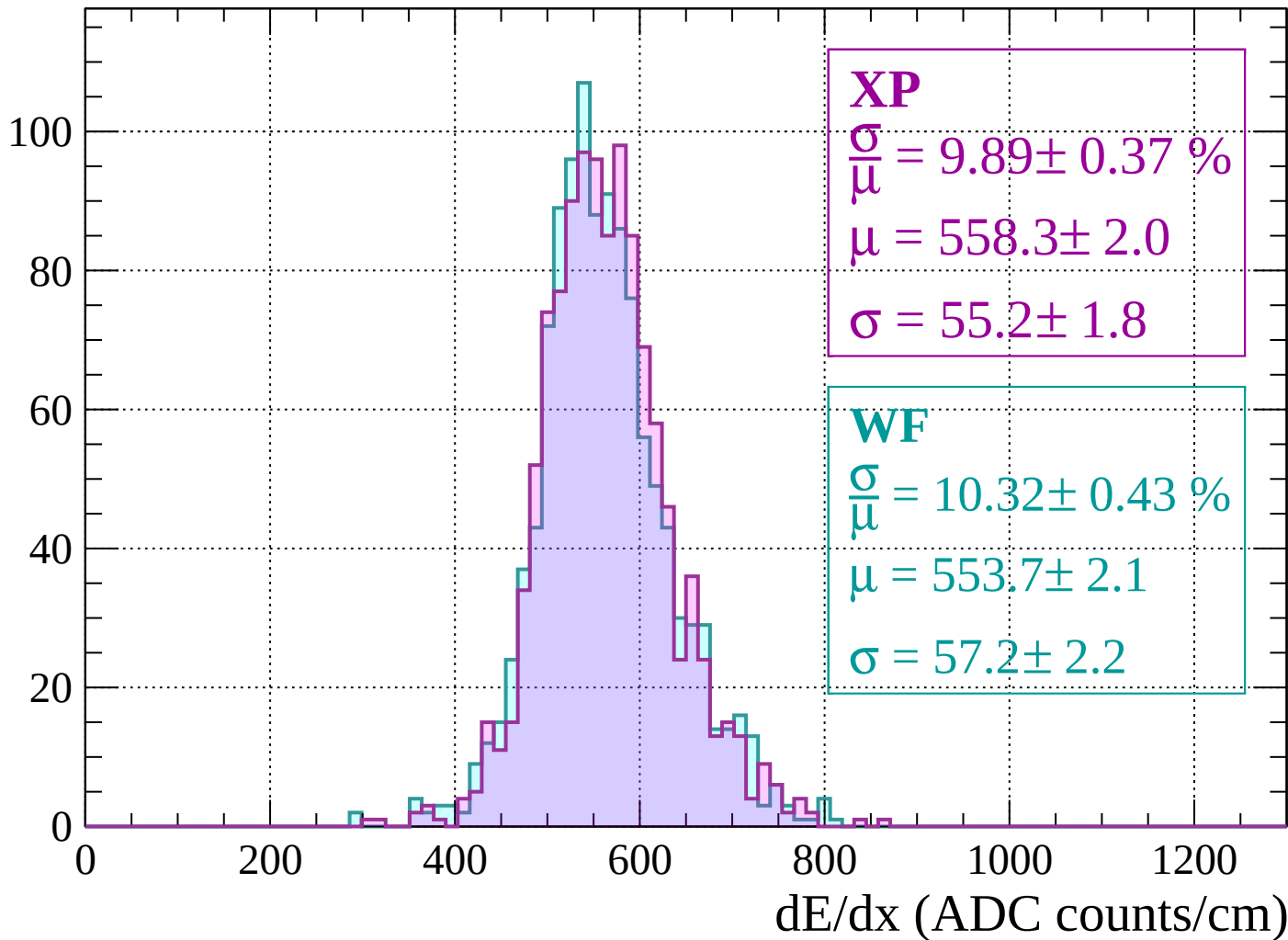
$$\mu = 467.9 \pm 1.7$$

$$\sigma = 52.1 \pm 1.6$$

dE/dx (ADC counts/cm)

Energy loss | $-120 < p < -80$

Count



Energy loss | $-80 < p < -40$

Count

XP

$$\frac{\sigma}{\mu} = 8.29 \pm 0.31 \%$$

$$\mu = 566.1 \pm 2.2$$

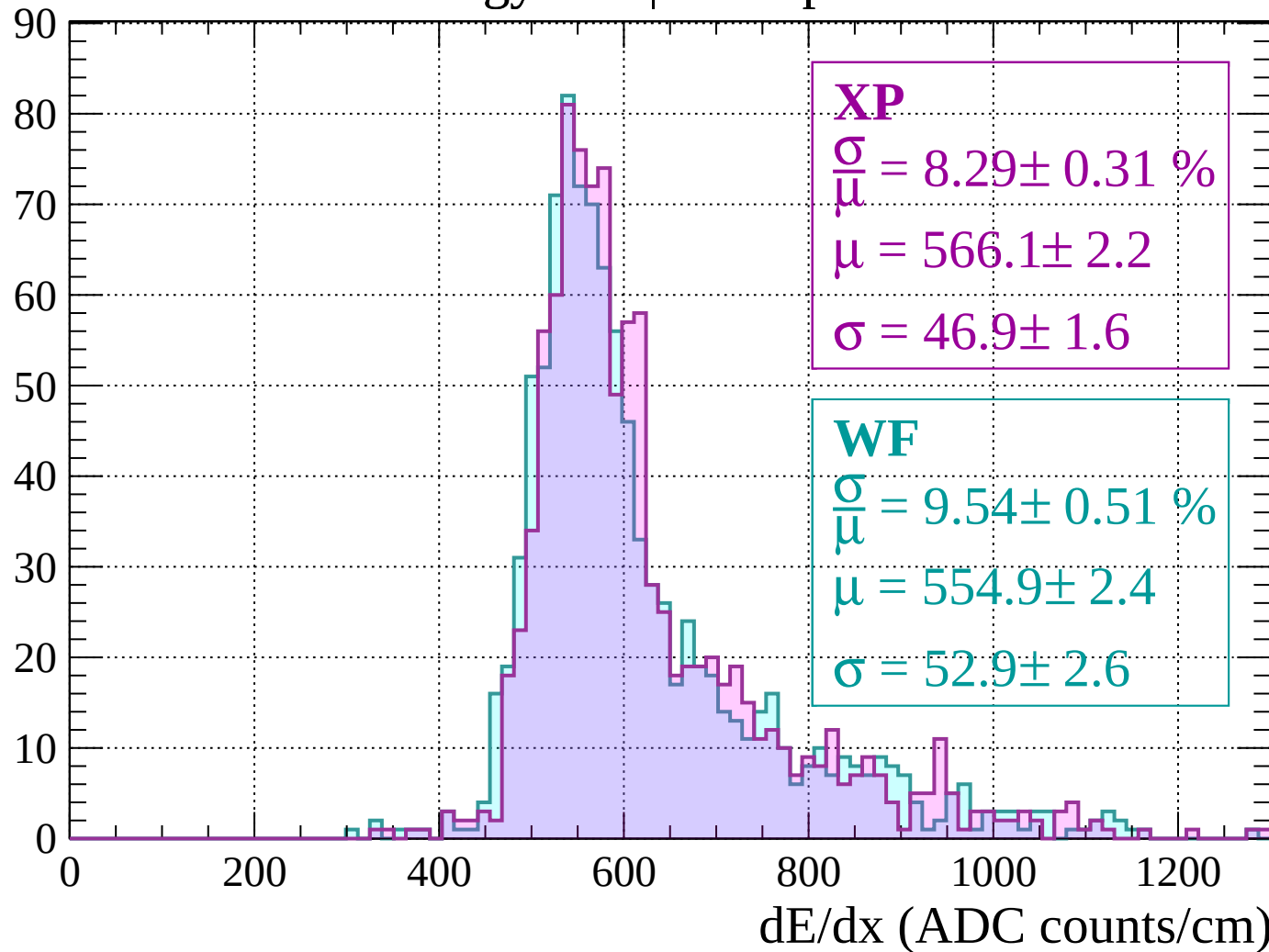
$$\sigma = 46.9 \pm 1.6$$

WF

$$\frac{\sigma}{\mu} = 9.54 \pm 0.51 \%$$

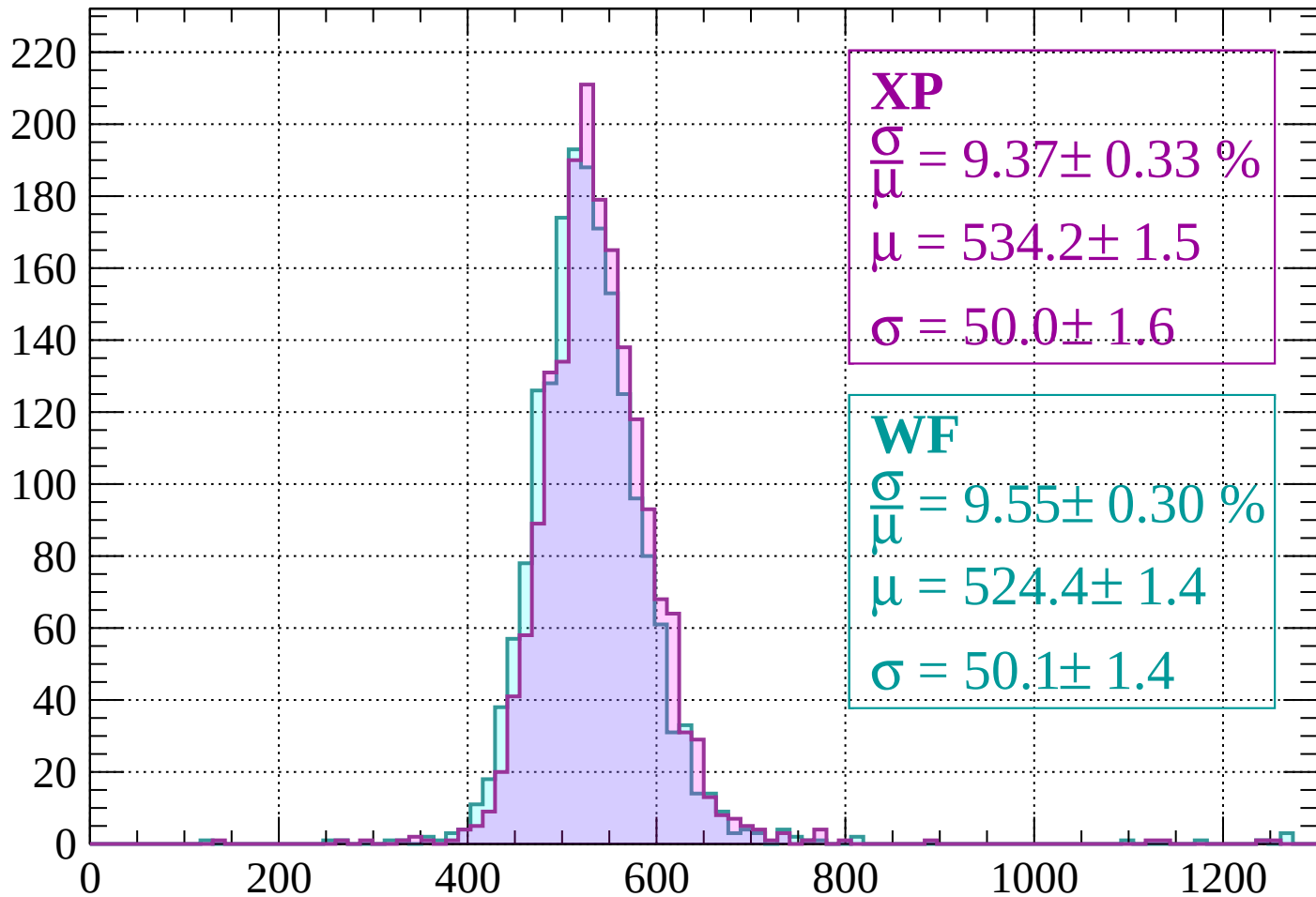
$$\mu = 554.9 \pm 2.4$$

$$\sigma = 52.9 \pm 2.6$$



Energy loss | $-40 < p < 0$

Count



XP

$$\frac{\sigma}{\mu} = 9.37 \pm 0.33 \%$$

$$\mu = 534.2 \pm 1.5$$

$$\sigma = 50.0 \pm 1.6$$

WF

$$\frac{\sigma}{\mu} = 9.55 \pm 0.30 \%$$

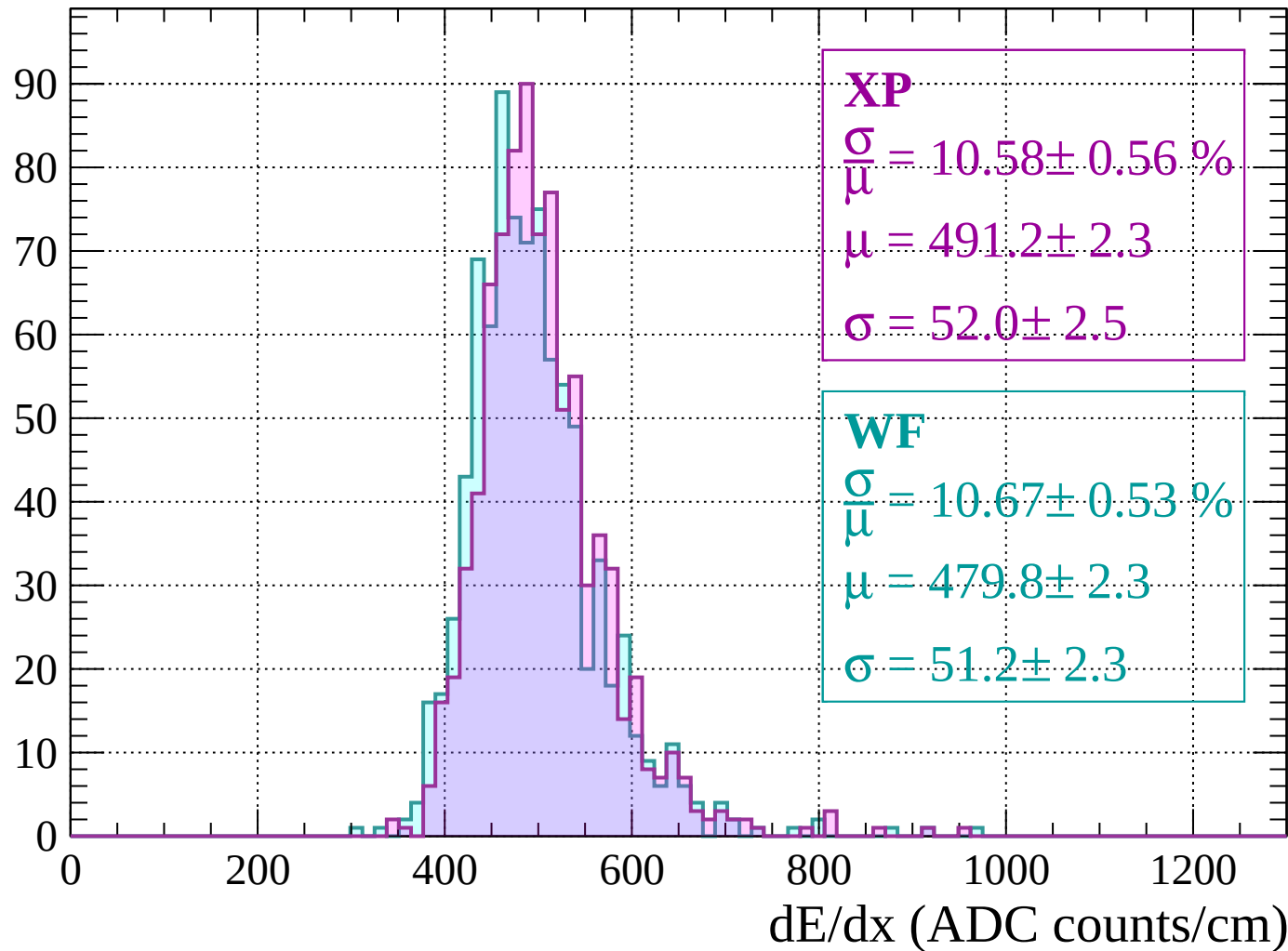
$$\mu = 524.4 \pm 1.4$$

$$\sigma = 50.1 \pm 1.4$$

dE/dx (ADC counts/cm)

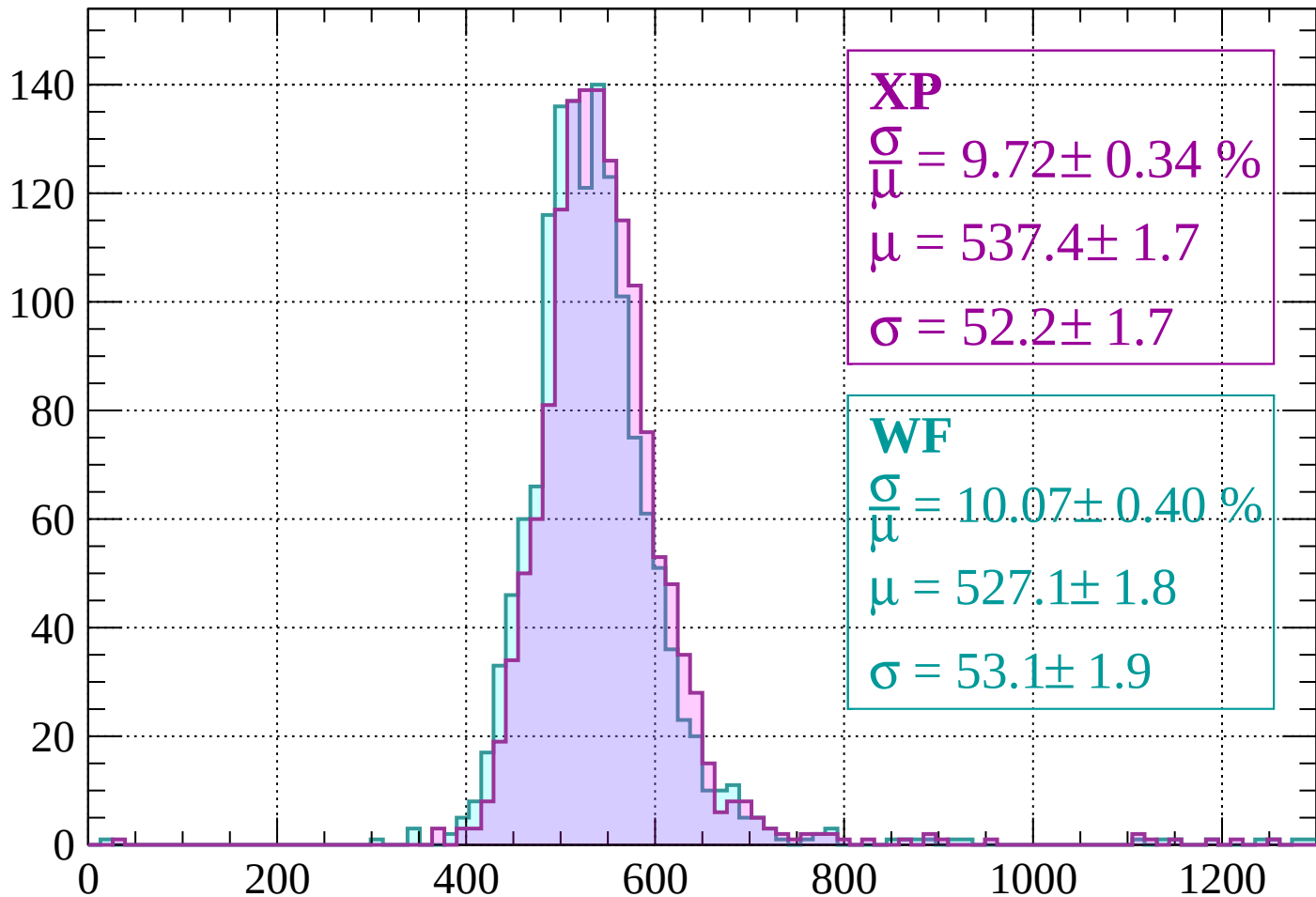
Energy loss | $0 < p < 40$

Count



Energy loss | $40 < p < 80$

Count



XP

$$\frac{\sigma}{\mu} = 9.72 \pm 0.34 \%$$

$$\mu = 537.4 \pm 1.7$$

$$\sigma = 52.2 \pm 1.7$$

WF

$$\frac{\sigma}{\mu} = 10.07 \pm 0.40 \%$$

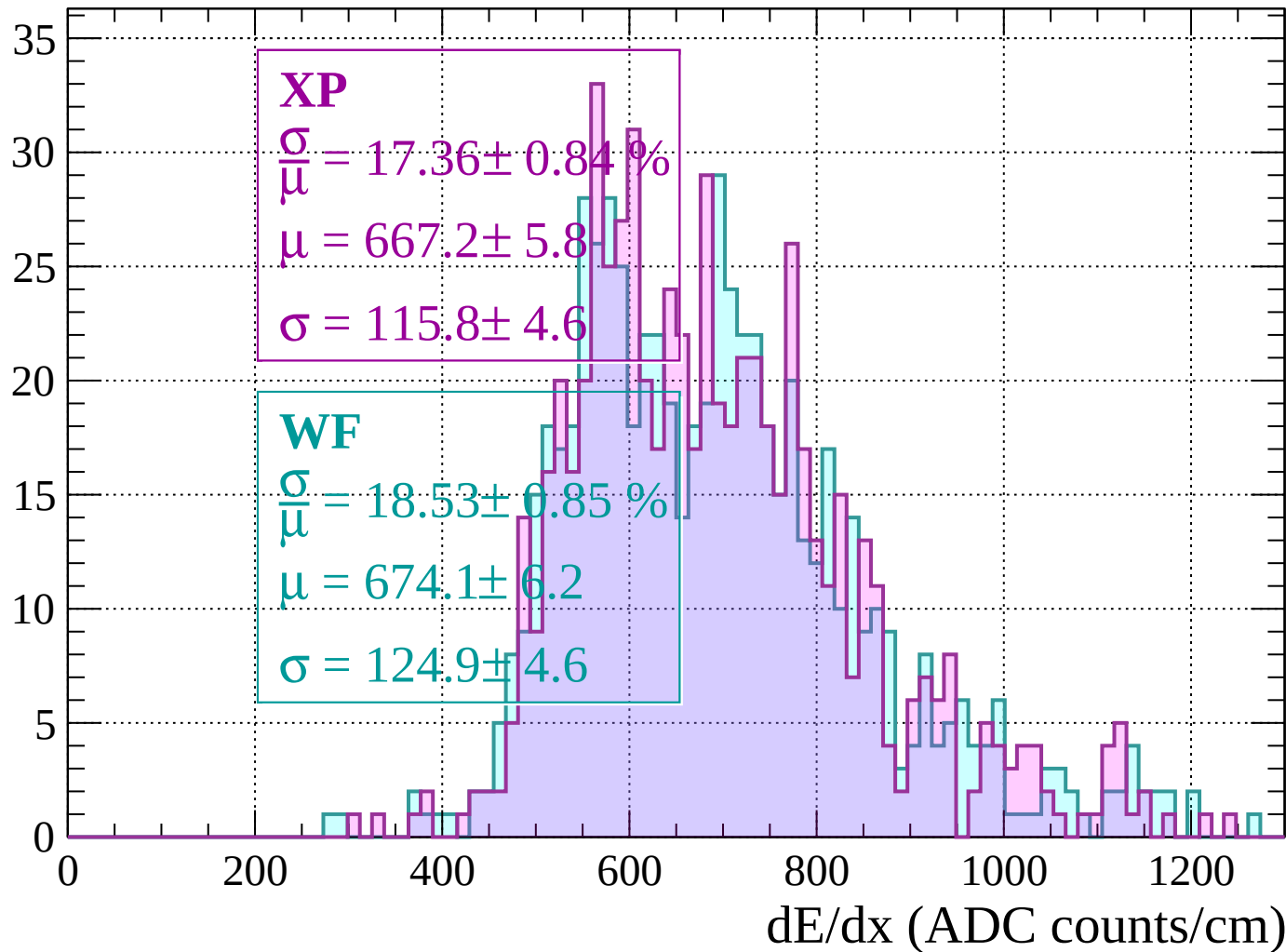
$$\mu = 527.1 \pm 1.8$$

$$\sigma = 53.1 \pm 1.9$$

dE/dx (ADC counts/cm)

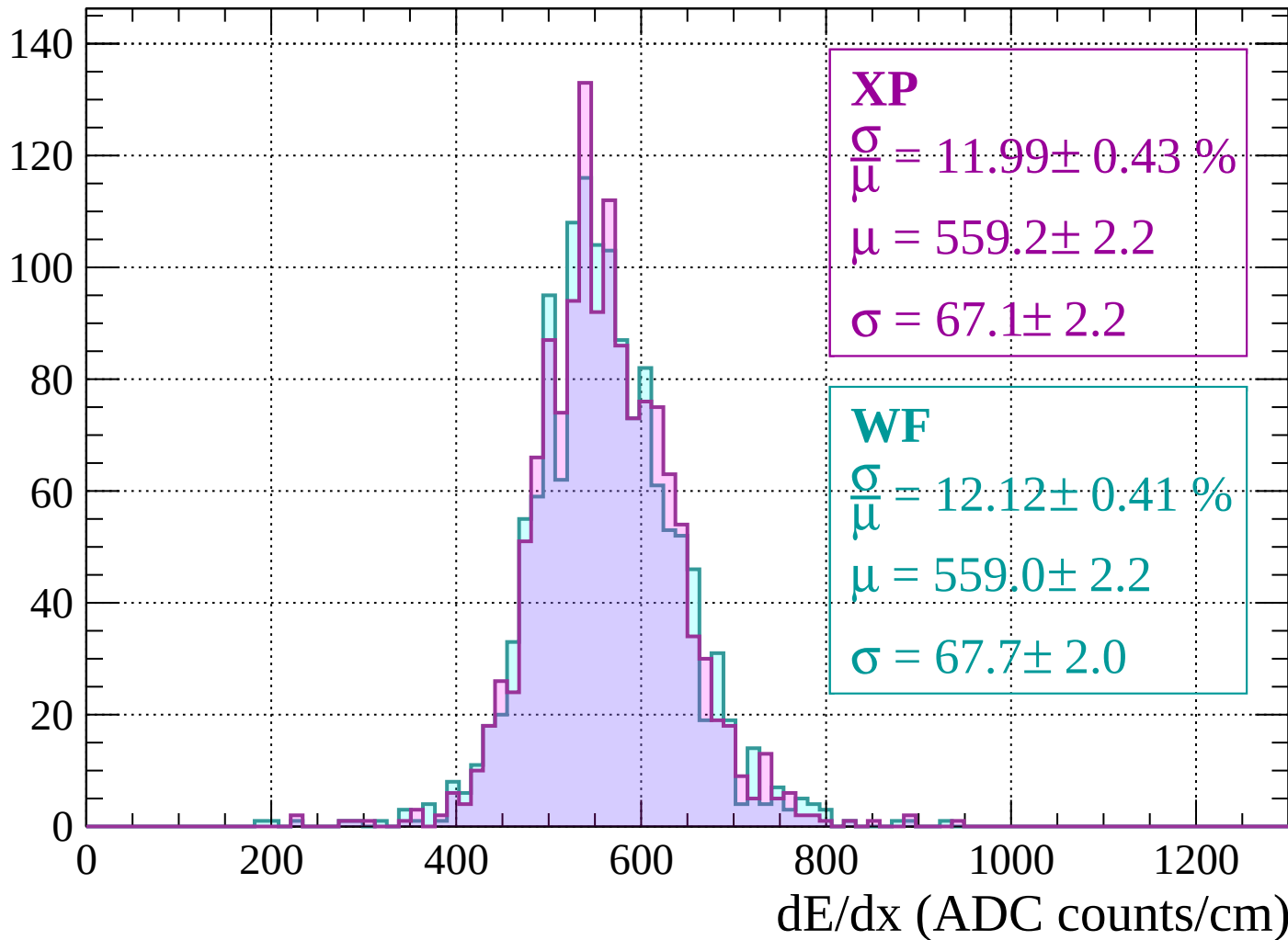
Energy loss | $80 < p < 120$

Count



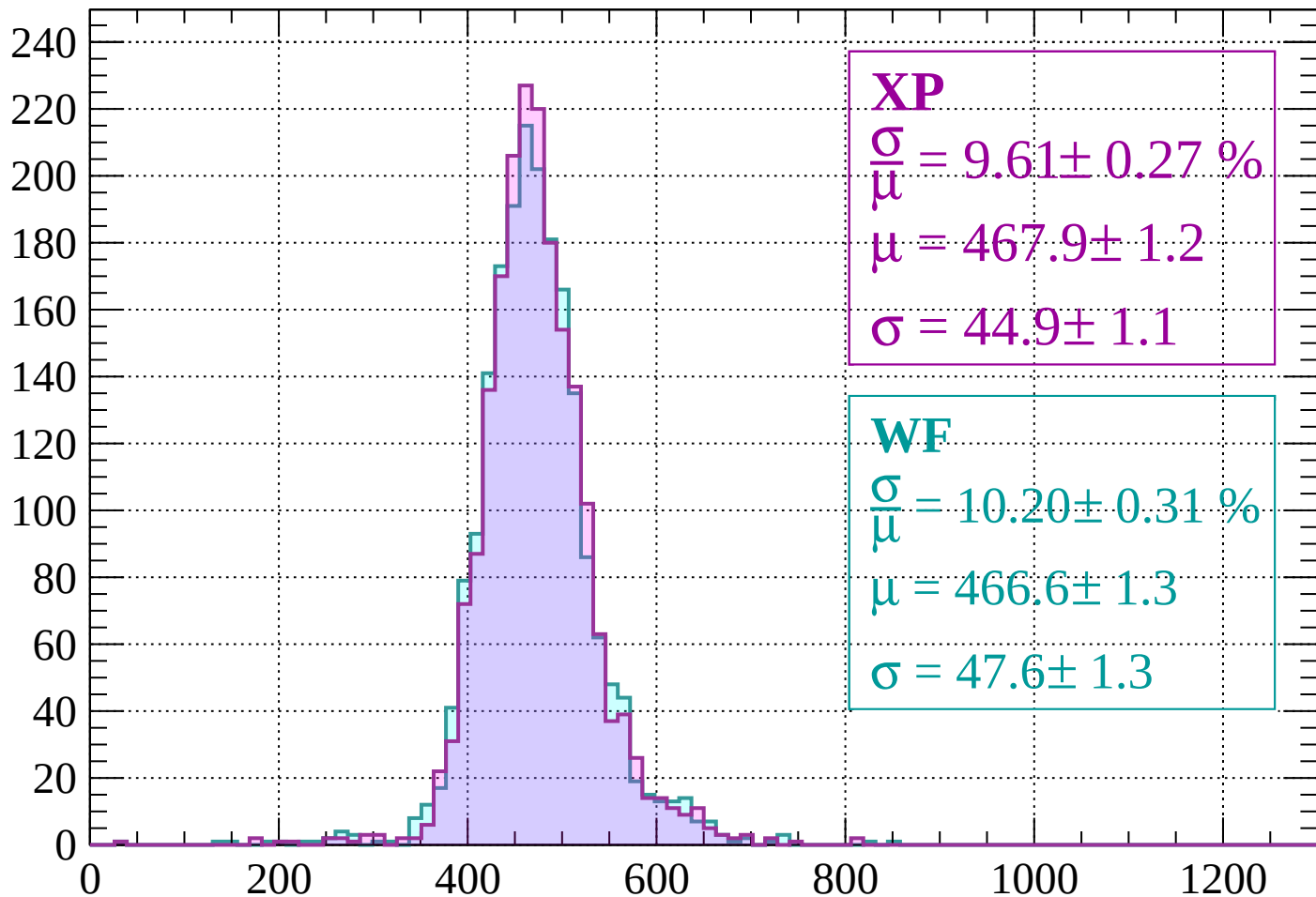
Energy loss | $120 < p < 160$

Count



Energy loss | $160 < p < 200$

Count



XP

$$\frac{\sigma}{\mu} = 9.61 \pm 0.27 \%$$

$$\mu = 467.9 \pm 1.2$$

$$\sigma = 44.9 \pm 1.1$$

WF

$$\frac{\sigma}{\mu} = 10.20 \pm 0.31 \%$$

$$\mu = 466.6 \pm 1.3$$

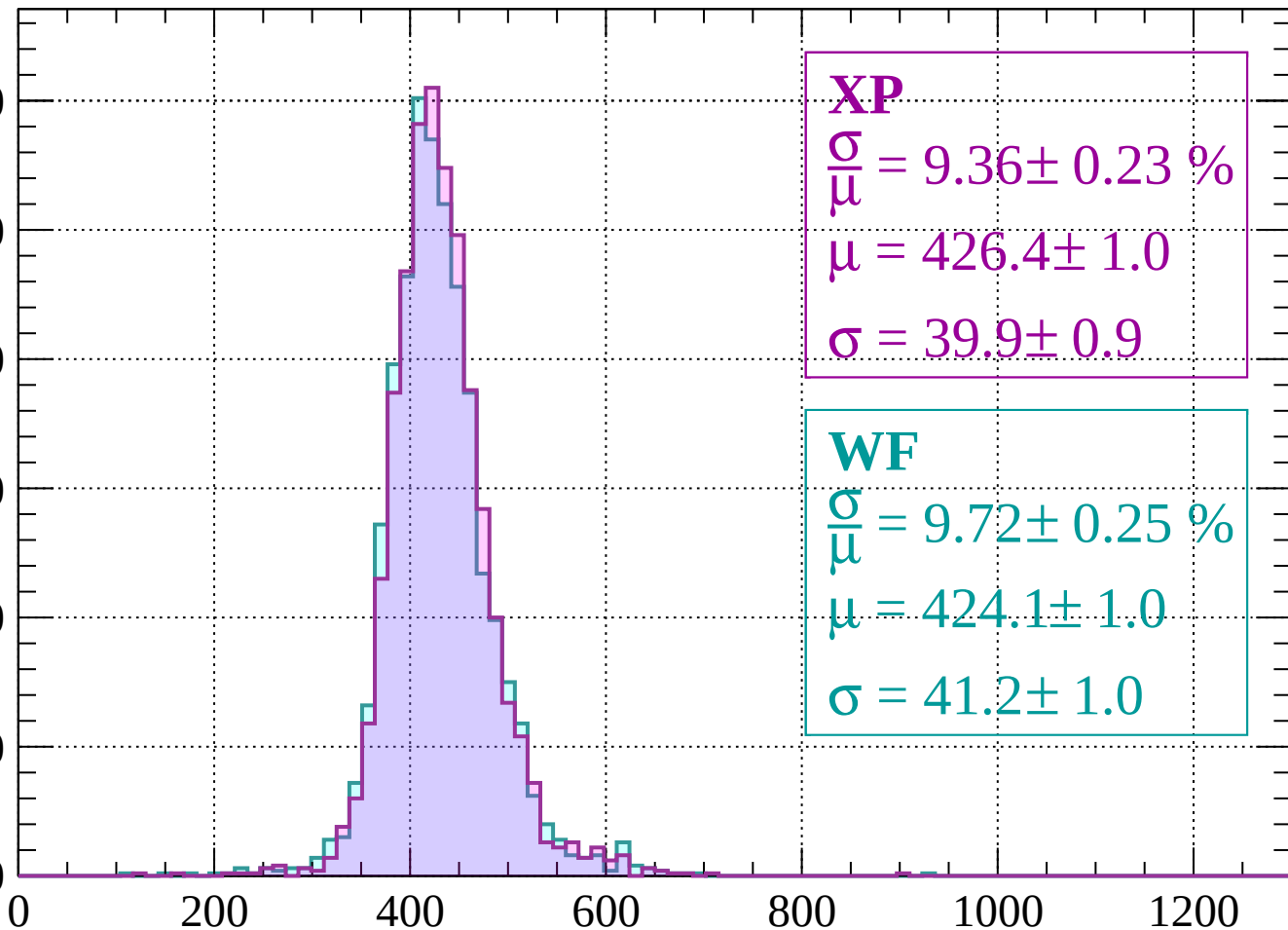
$$\sigma = 47.6 \pm 1.3$$

dE/dx (ADC counts/cm)

Energy loss | $200 < p < 240$

Count

300
250
200
150
100
50
0



XP

$$\frac{\sigma}{\mu} = 9.36 \pm 0.23 \%$$

$$\mu = 426.4 \pm 1.0$$

$$\sigma = 39.9 \pm 0.9$$

WF

$$\frac{\sigma}{\mu} = 9.72 \pm 0.25 \%$$

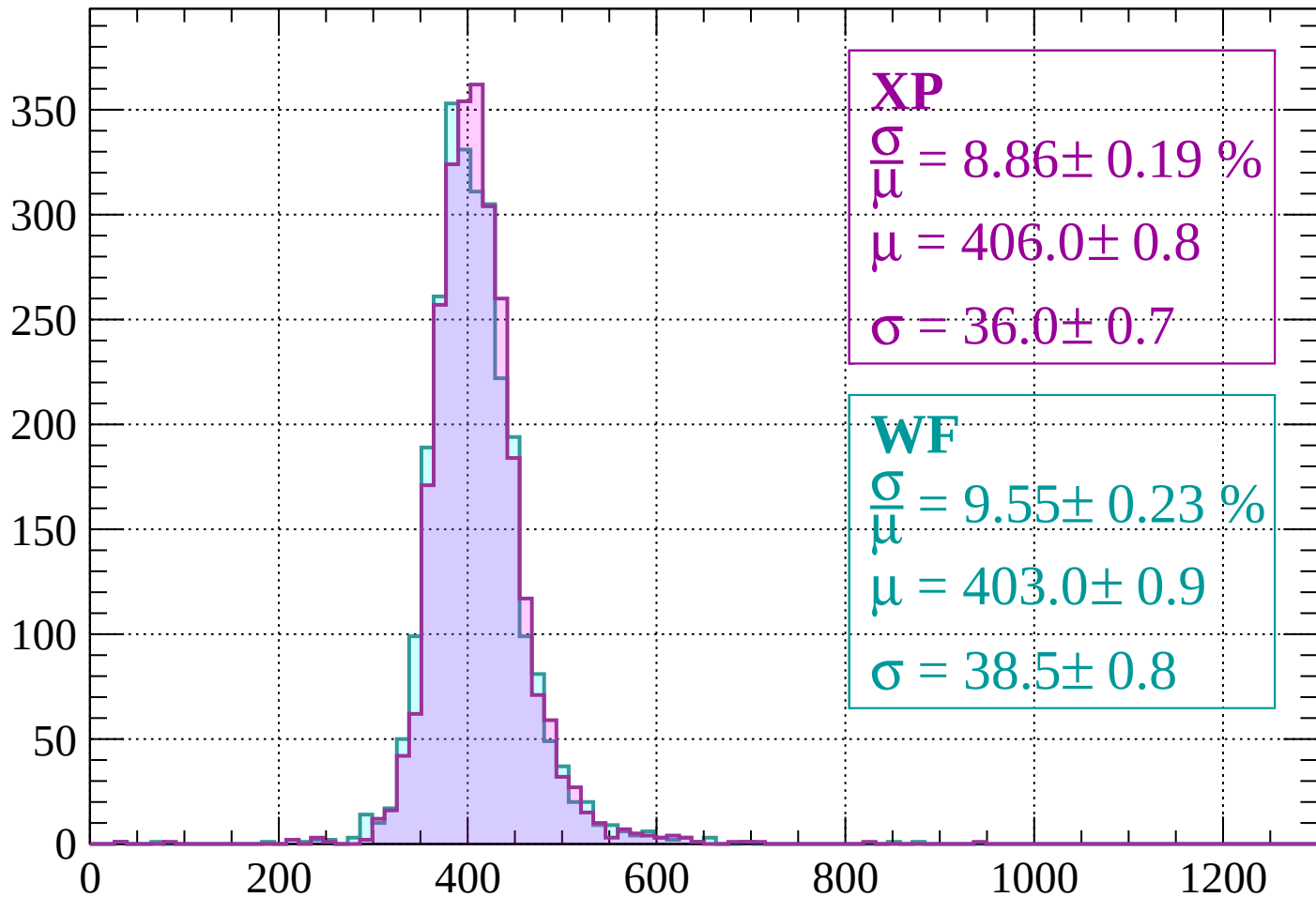
$$\mu = 424.1 \pm 1.0$$

$$\sigma = 41.2 \pm 1.0$$

dE/dx (ADC counts/cm)

Energy loss | $240 < p < 280$

Count



XP

$$\frac{\sigma}{\mu} = 8.86 \pm 0.19 \%$$

$$\mu = 406.0 \pm 0.8$$

$$\sigma = 36.0 \pm 0.7$$

WF

$$\frac{\sigma}{\mu} = 9.55 \pm 0.23 \%$$

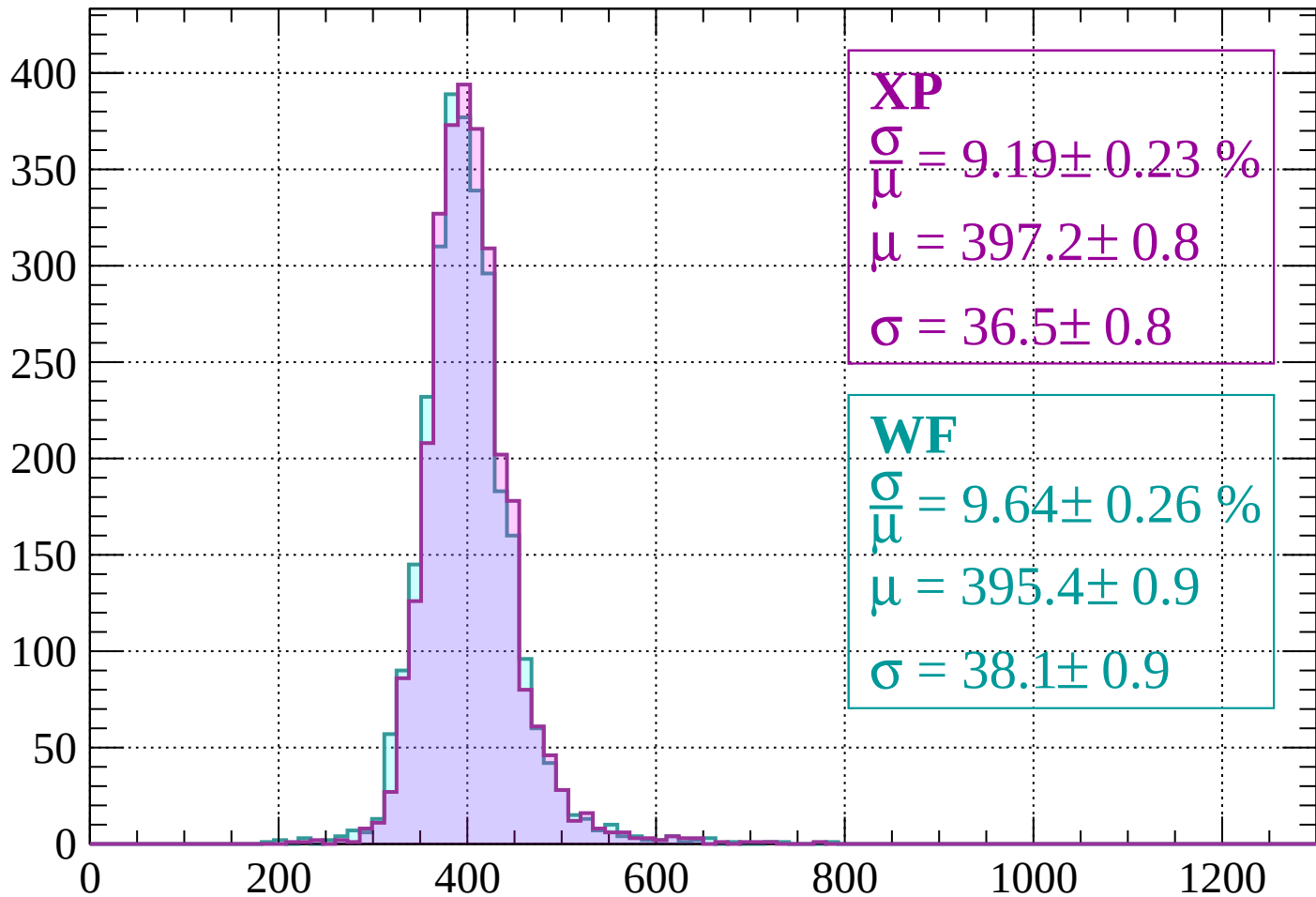
$$\mu = 403.0 \pm 0.9$$

$$\sigma = 38.5 \pm 0.8$$

dE/dx (ADC counts/cm)

Energy loss | $280 < p < 320$

Count



XP

$$\frac{\sigma}{\mu} = 9.19 \pm 0.23 \%$$

$$\mu = 397.2 \pm 0.8$$

$$\sigma = 36.5 \pm 0.8$$

WF

$$\frac{\sigma}{\mu} = 9.64 \pm 0.26 \%$$

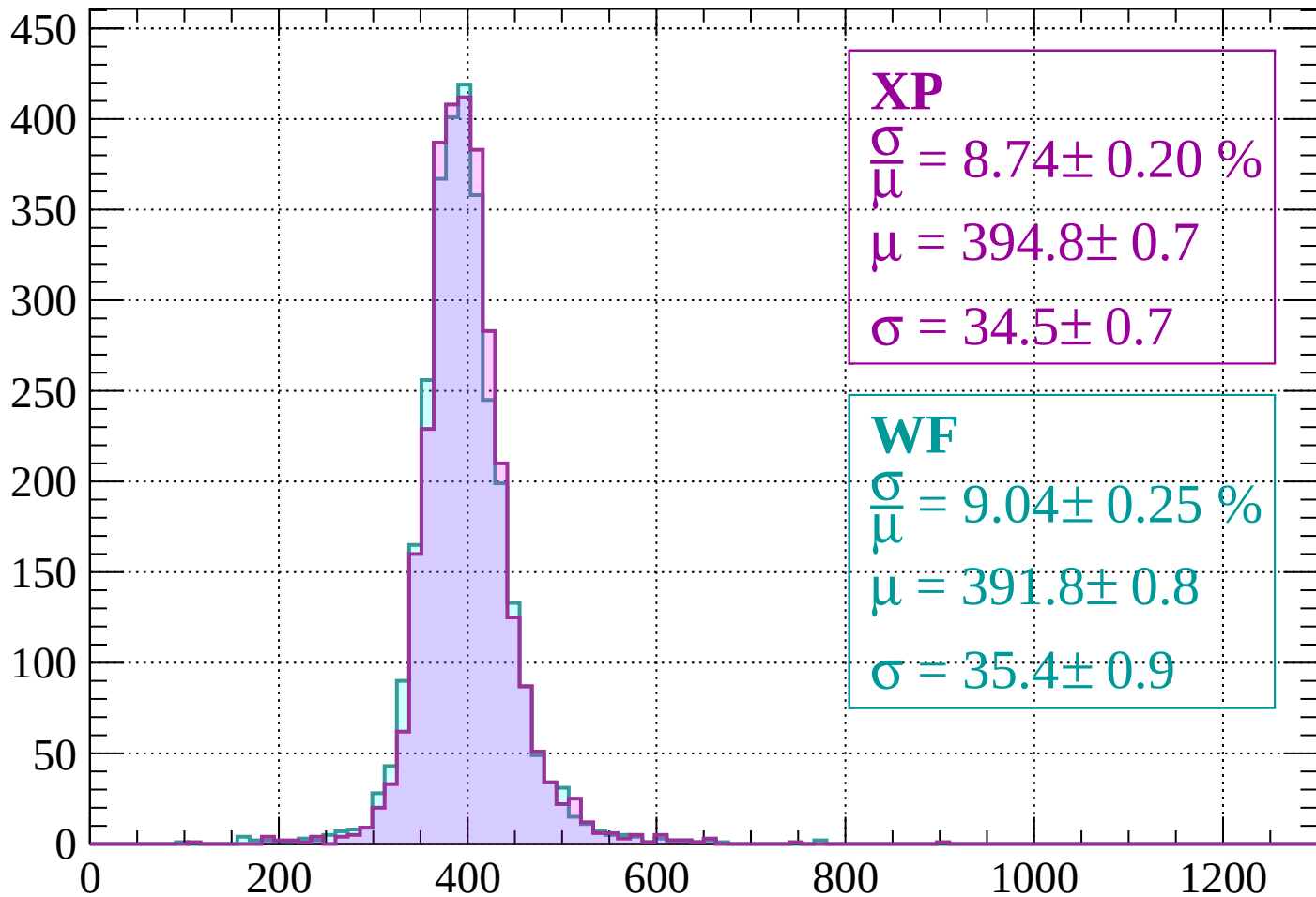
$$\mu = 395.4 \pm 0.9$$

$$\sigma = 38.1 \pm 0.9$$

dE/dx (ADC counts/cm)

Energy loss | $320 < p < 360$

Count



XP

$$\frac{\sigma}{\mu} = 8.74 \pm 0.20 \%$$

$$\mu = 394.8 \pm 0.7$$

$$\sigma = 34.5 \pm 0.7$$

WF

$$\frac{\sigma}{\mu} = 9.04 \pm 0.25 \%$$

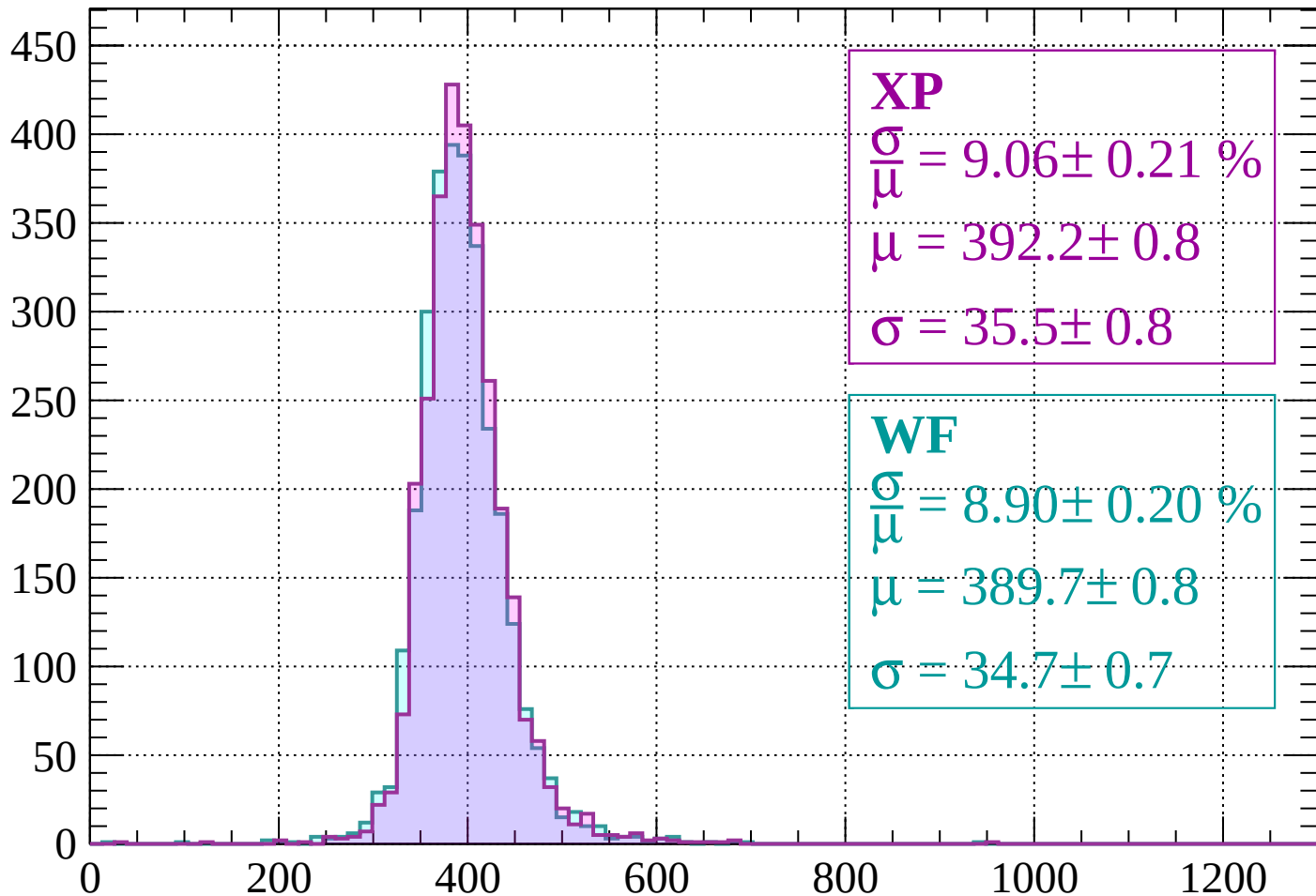
$$\mu = 391.8 \pm 0.8$$

$$\sigma = 35.4 \pm 0.9$$

dE/dx (ADC counts/cm)

Energy loss | $360 < p < 400$

Count



XP

$$\frac{\sigma}{\mu} = 9.06 \pm 0.21 \%$$

$$\mu = 392.2 \pm 0.8$$

$$\sigma = 35.5 \pm 0.8$$

WF

$$\frac{\sigma}{\mu} = 8.90 \pm 0.20 \%$$

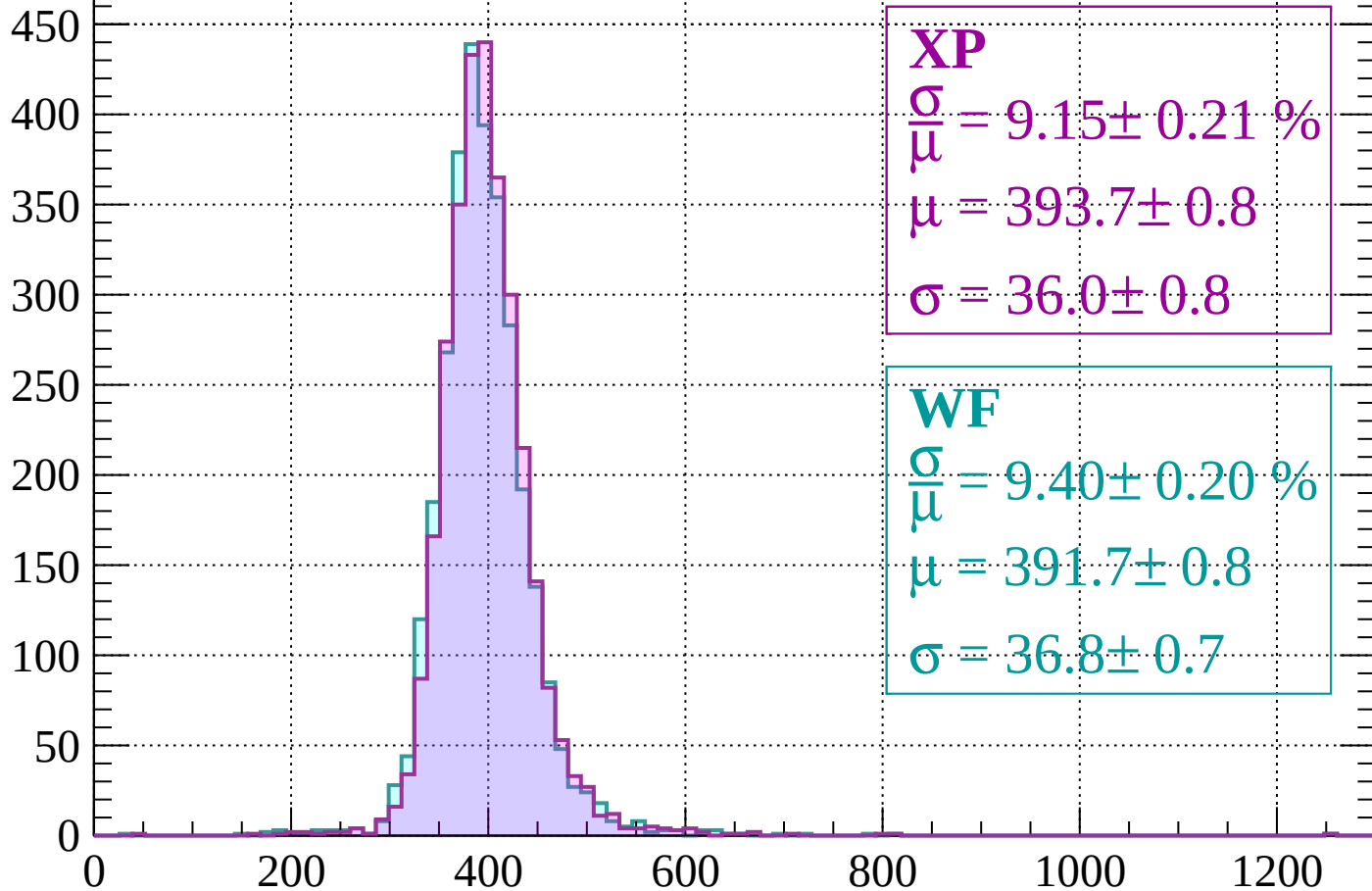
$$\mu = 389.7 \pm 0.8$$

$$\sigma = 34.7 \pm 0.7$$

dE/dx (ADC counts/cm)

Energy loss | $400 < p < 440$

Count



XP

$$\frac{\sigma}{\mu} = 9.15 \pm 0.21 \%$$

$$\mu = 393.7 \pm 0.8$$

$$\sigma = 36.0 \pm 0.8$$

WF

$$\frac{\sigma}{\mu} = 9.40 \pm 0.20 \%$$

$$\mu = 391.7 \pm 0.8$$

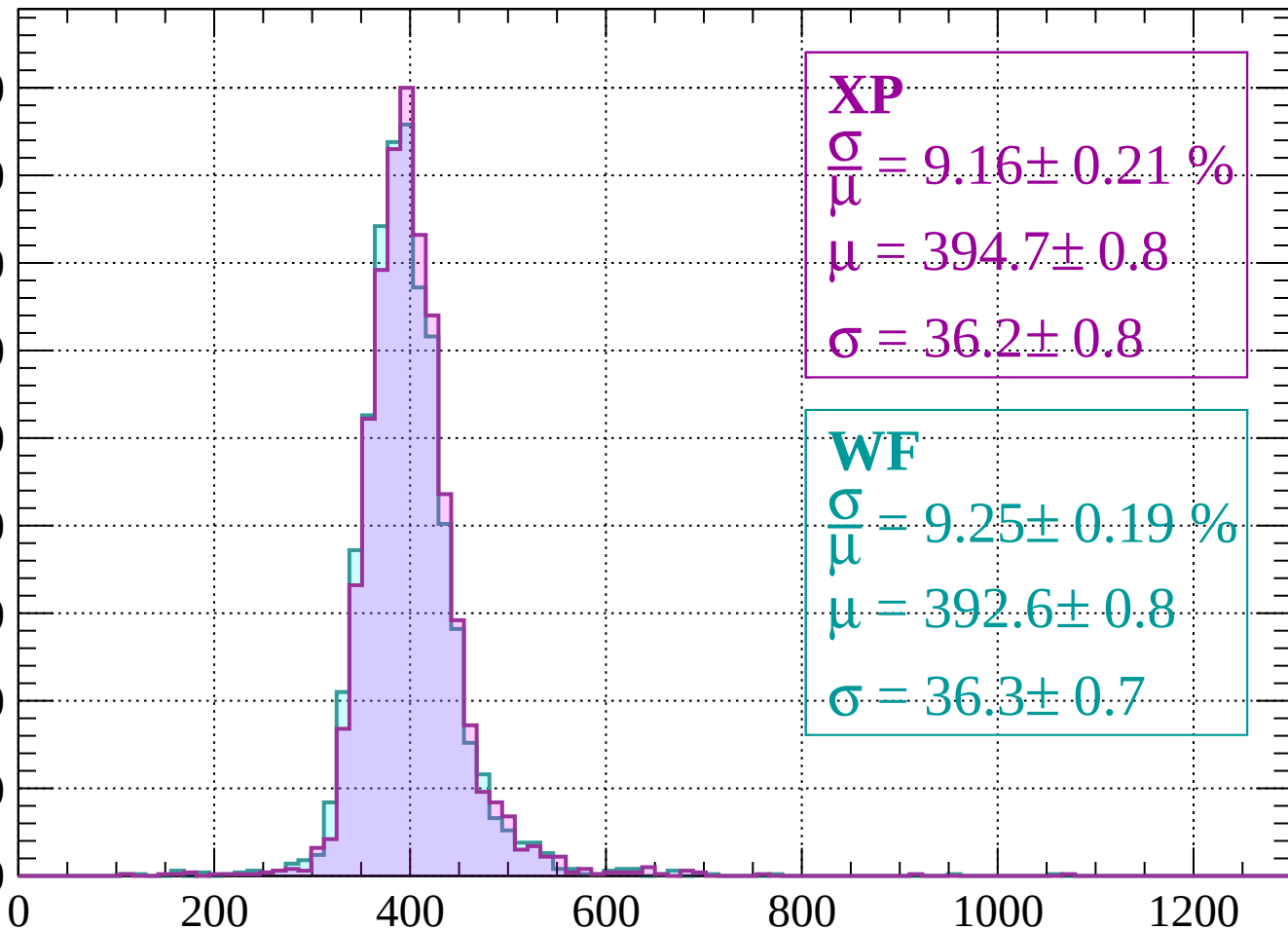
$$\sigma = 36.8 \pm 0.7$$

dE/dx (ADC counts/cm)

Energy loss | $440 < p < 480$

Count

450
400
350
300
250
200
150
100
50
0



XP

$$\frac{\sigma}{\mu} = 9.16 \pm 0.21 \%$$

$$\mu = 394.7 \pm 0.8$$

$$\sigma = 36.2 \pm 0.8$$

WF

$$\frac{\sigma}{\mu} = 9.25 \pm 0.19 \%$$

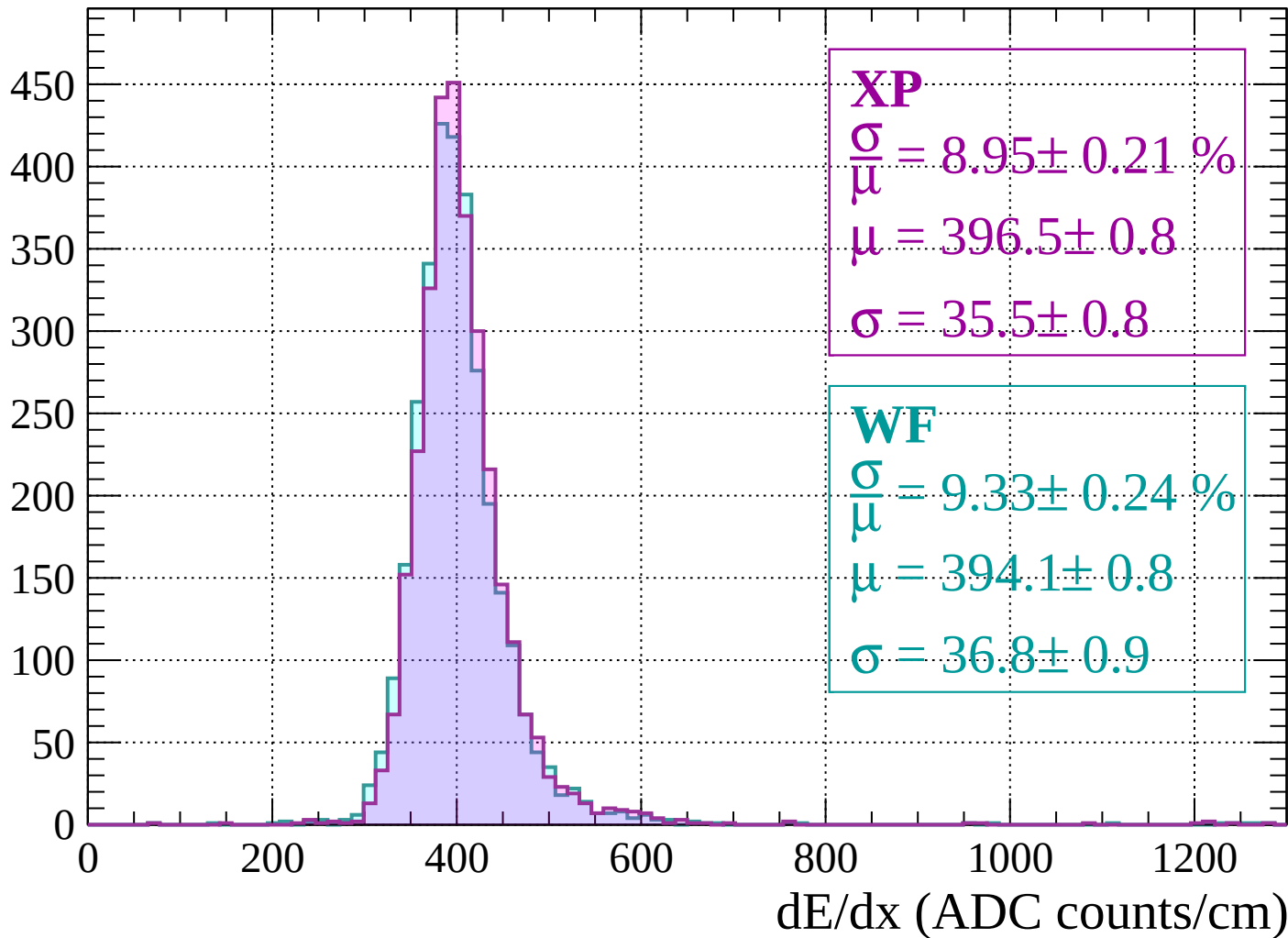
$$\mu = 392.6 \pm 0.8$$

$$\sigma = 36.3 \pm 0.7$$

dE/dx (ADC counts/cm)

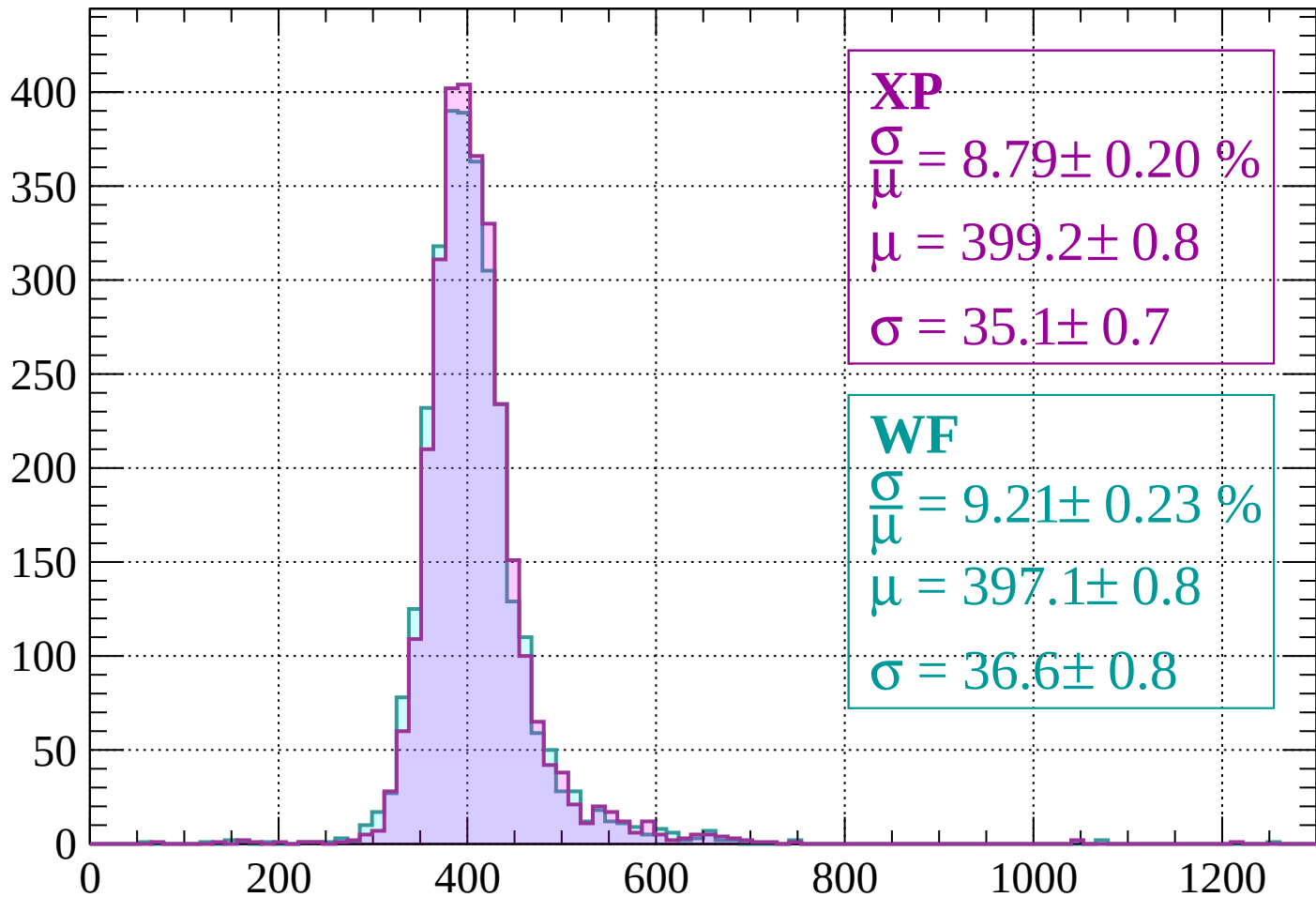
Energy loss | $480 < p < 520$

Count



Energy loss | $520 < p < 560$

Count



XP

$$\frac{\sigma}{\mu} = 8.79 \pm 0.20 \%$$

$$\mu = 399.2 \pm 0.8$$

$$\sigma = 35.1 \pm 0.7$$

WF

$$\frac{\sigma}{\mu} = 9.21 \pm 0.23 \%$$

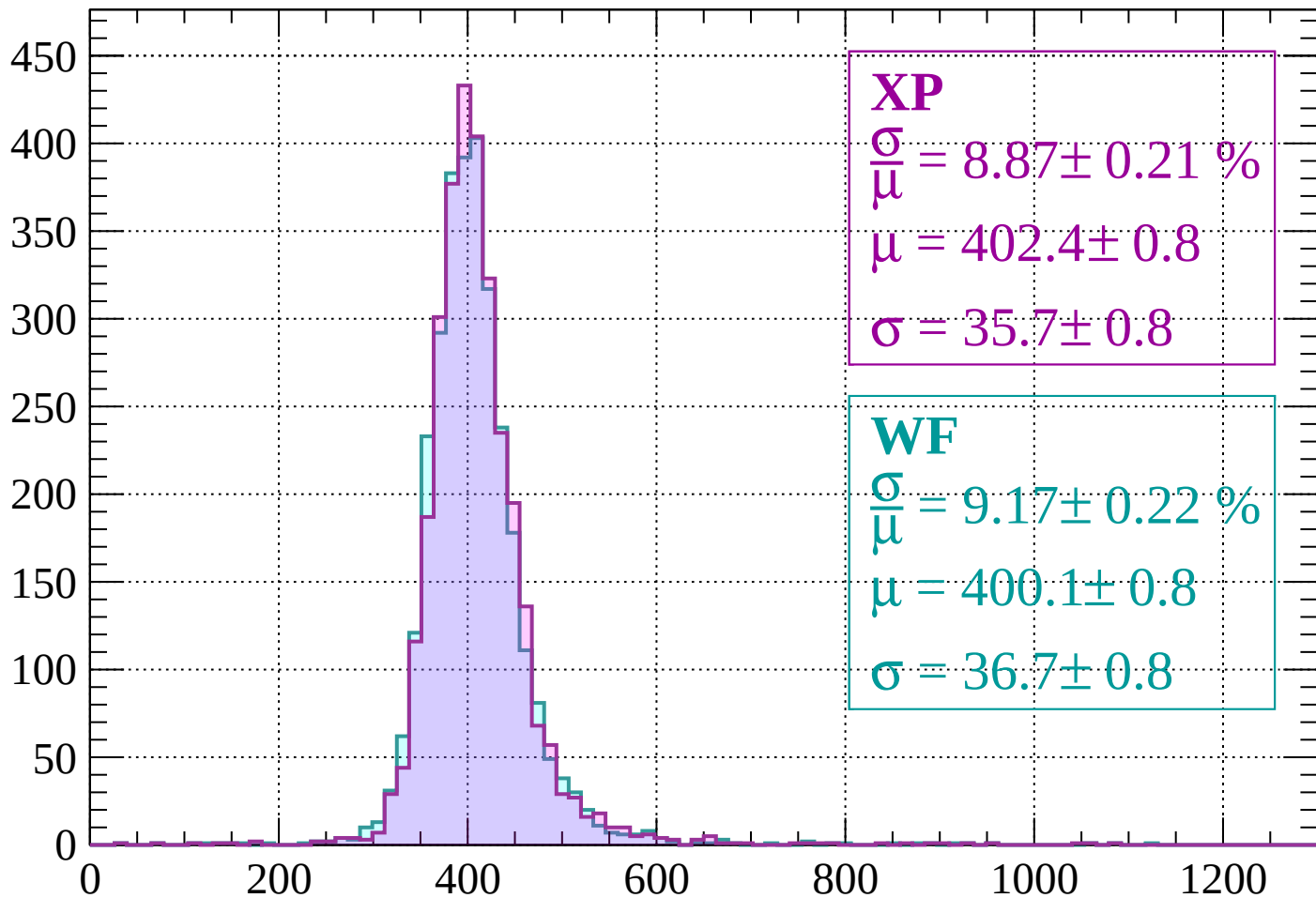
$$\mu = 397.1 \pm 0.8$$

$$\sigma = 36.6 \pm 0.8$$

dE/dx (ADC counts/cm)

Energy loss | $560 < p < 600$

Count



XP

$$\frac{\sigma}{\mu} = 8.87 \pm 0.21 \%$$

$$\mu = 402.4 \pm 0.8$$

$$\sigma = 35.7 \pm 0.8$$

WF

$$\frac{\sigma}{\mu} = 9.17 \pm 0.22 \%$$

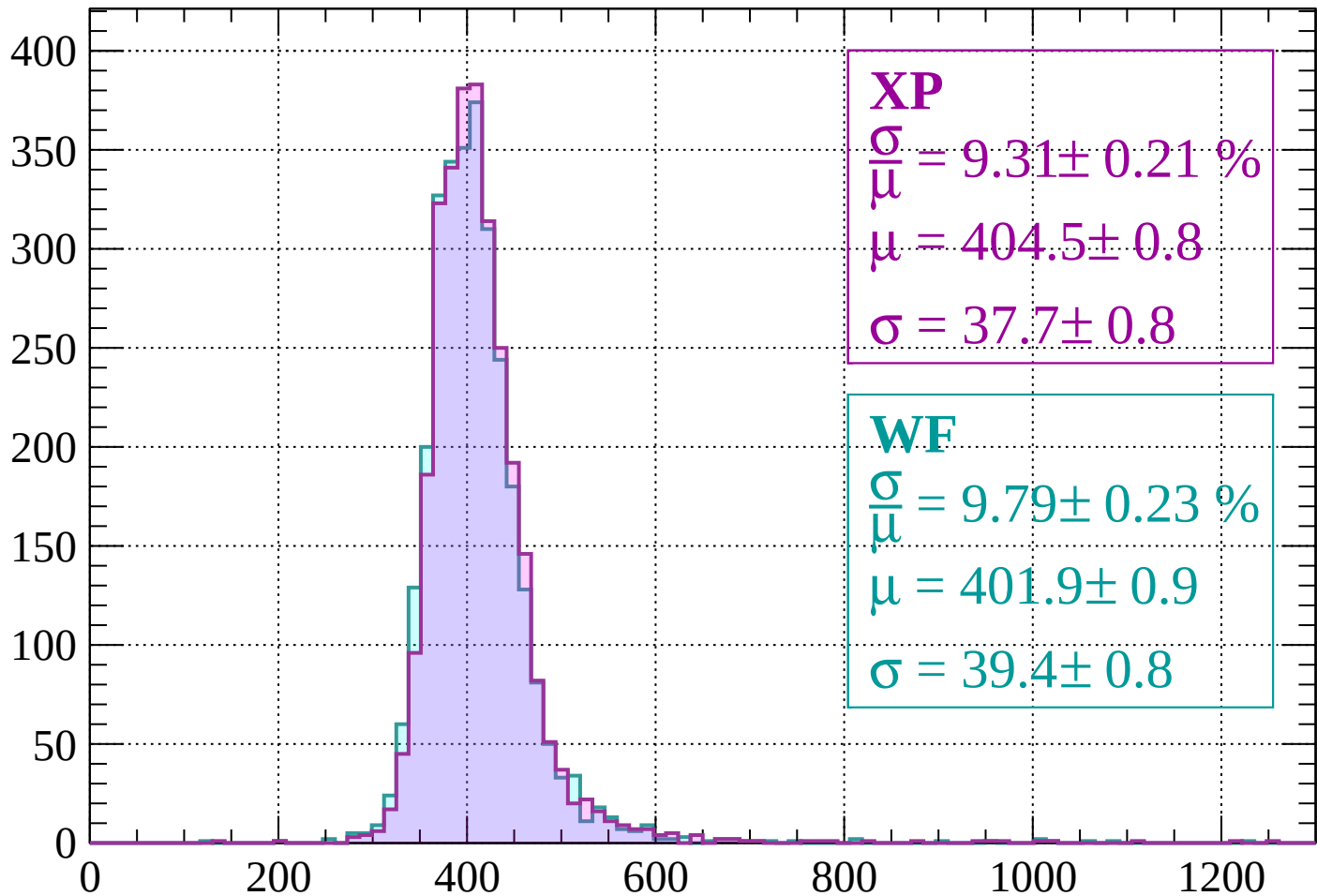
$$\mu = 400.1 \pm 0.8$$

$$\sigma = 36.7 \pm 0.8$$

dE/dx (ADC counts/cm)

Energy loss | $600 < p < 640$

Count



XP

$$\frac{\sigma}{\mu} = 9.31 \pm 0.21 \%$$

$$\mu = 404.5 \pm 0.8$$

$$\sigma = 37.7 \pm 0.8$$

WF

$$\frac{\sigma}{\mu} = 9.79 \pm 0.23 \%$$

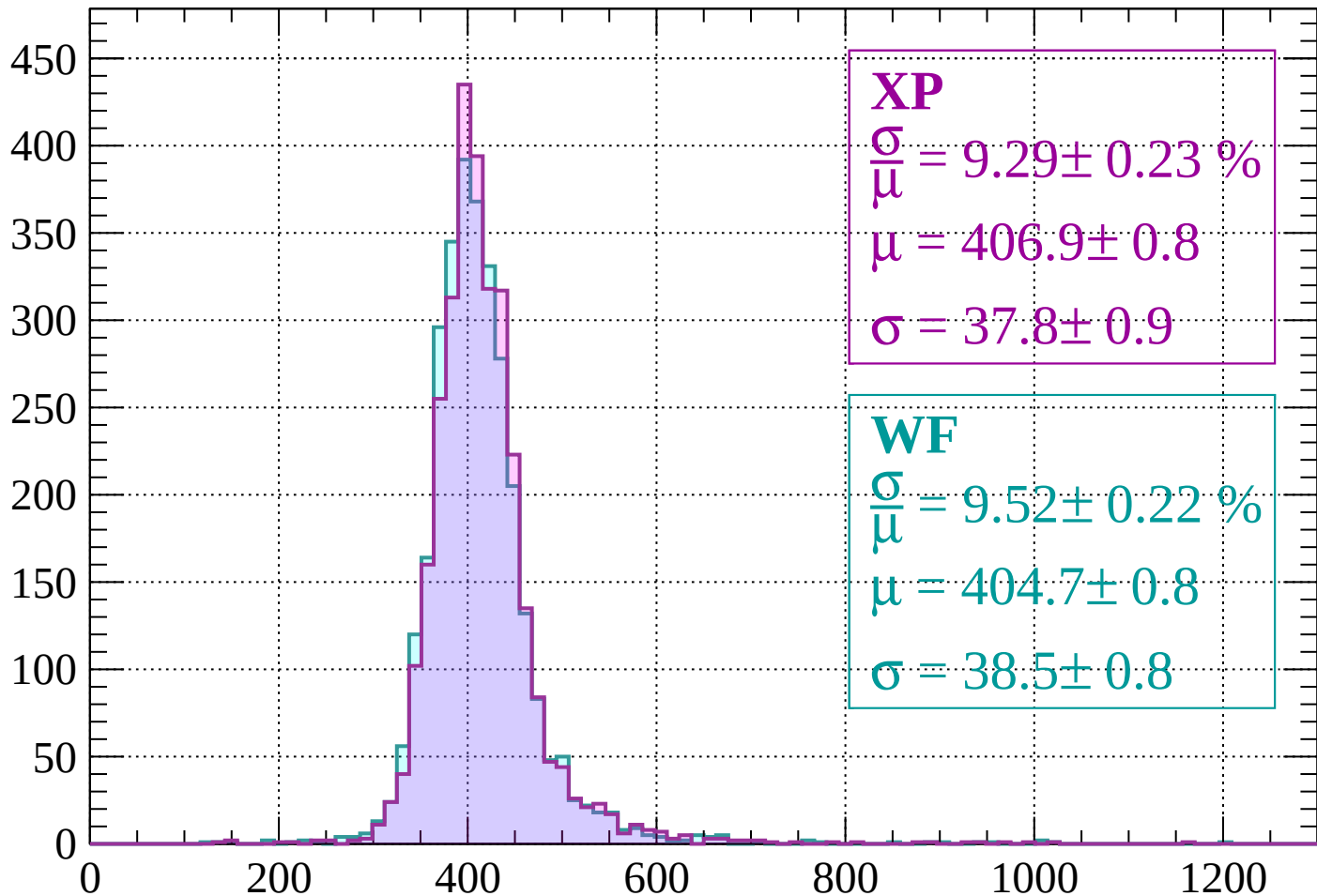
$$\mu = 401.9 \pm 0.9$$

$$\sigma = 39.4 \pm 0.8$$

dE/dx (ADC counts/cm)

Energy loss | $640 < p < 680$

Count



XP

$$\frac{\sigma}{\mu} = 9.29 \pm 0.23 \%$$

$$\mu = 406.9 \pm 0.8$$

$$\sigma = 37.8 \pm 0.9$$

WF

$$\frac{\sigma}{\mu} = 9.52 \pm 0.22 \%$$

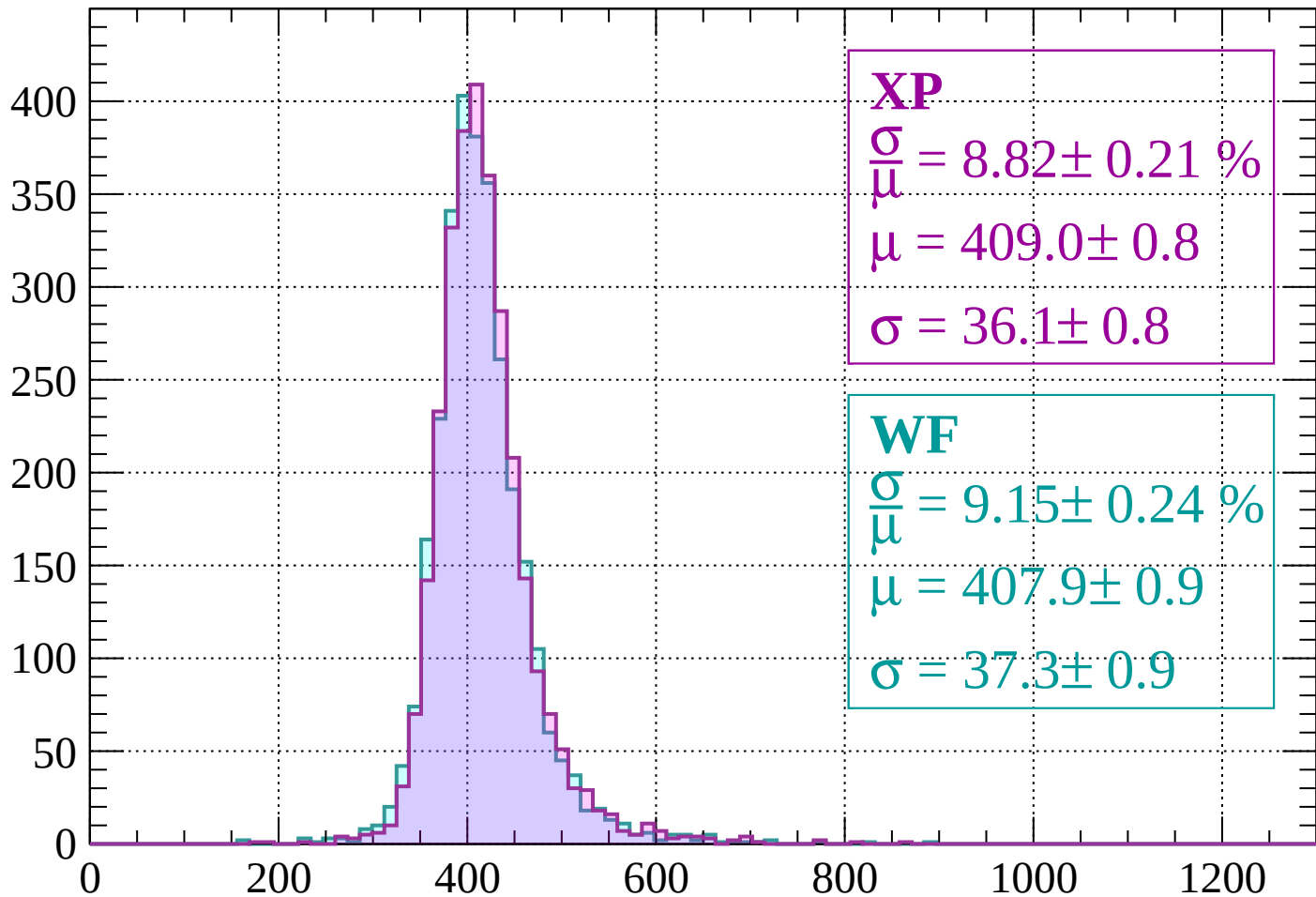
$$\mu = 404.7 \pm 0.8$$

$$\sigma = 38.5 \pm 0.8$$

dE/dx (ADC counts/cm)

Energy loss | $680 < p < 720$

Count



XP

$$\frac{\sigma}{\mu} = 8.82 \pm 0.21 \%$$

$$\mu = 409.0 \pm 0.8$$

$$\sigma = 36.1 \pm 0.8$$

WF

$$\frac{\sigma}{\mu} = 9.15 \pm 0.24 \%$$

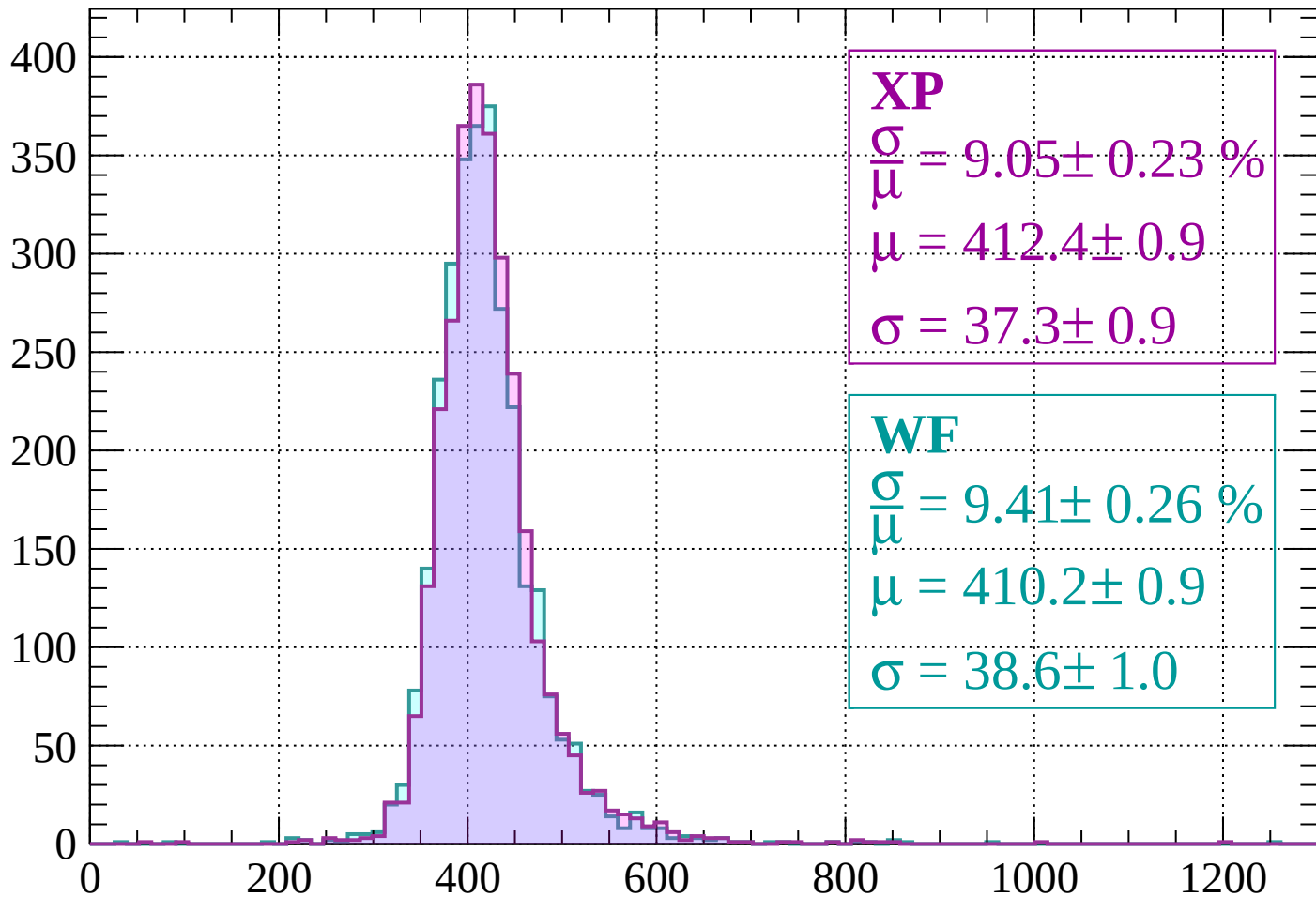
$$\mu = 407.9 \pm 0.9$$

$$\sigma = 37.3 \pm 0.9$$

dE/dx (ADC counts/cm)

Energy loss | $720 < p < 760$

Count



XP

$$\frac{\sigma}{\mu} = 9.05 \pm 0.23 \%$$

$$\mu = 412.4 \pm 0.9$$

$$\sigma = 37.3 \pm 0.9$$

WF

$$\frac{\sigma}{\mu} = 9.41 \pm 0.26 \%$$

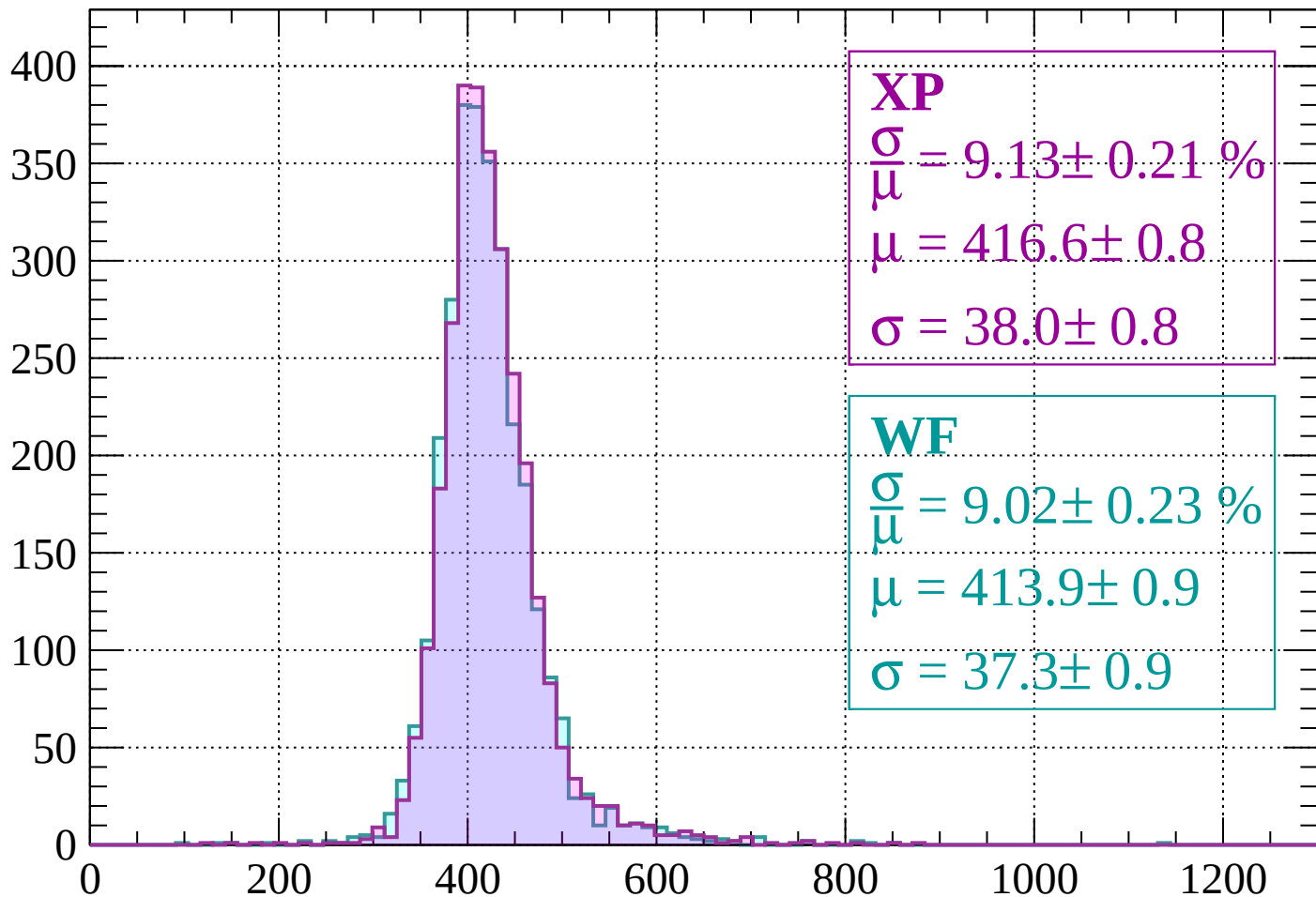
$$\mu = 410.2 \pm 0.9$$

$$\sigma = 38.6 \pm 1.0$$

dE/dx (ADC counts/cm)

Energy loss | $760 < p < 800$

Count



XP

$$\frac{\sigma}{\mu} = 9.13 \pm 0.21 \%$$

$$\mu = 416.6 \pm 0.8$$

$$\sigma = 38.0 \pm 0.8$$

WF

$$\frac{\sigma}{\mu} = 9.02 \pm 0.23 \%$$

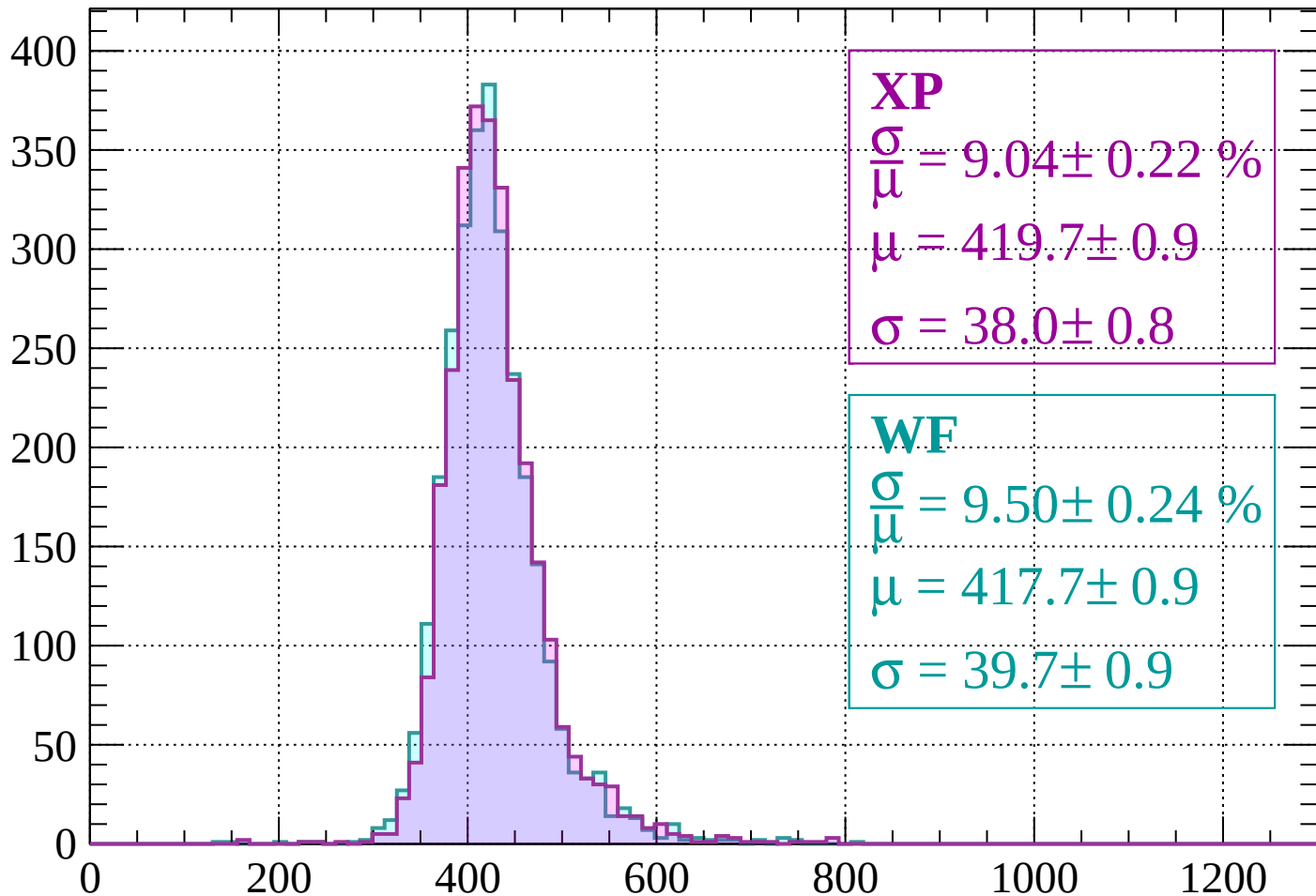
$$\mu = 413.9 \pm 0.9$$

$$\sigma = 37.3 \pm 0.9$$

dE/dx (ADC counts/cm)

Energy loss | $800 < p < 840$

Count



XP

$$\frac{\sigma}{\mu} = 9.04 \pm 0.22 \%$$

$$\mu = 419.7 \pm 0.9$$

$$\sigma = 38.0 \pm 0.8$$

WF

$$\frac{\sigma}{\mu} = 9.50 \pm 0.24 \%$$

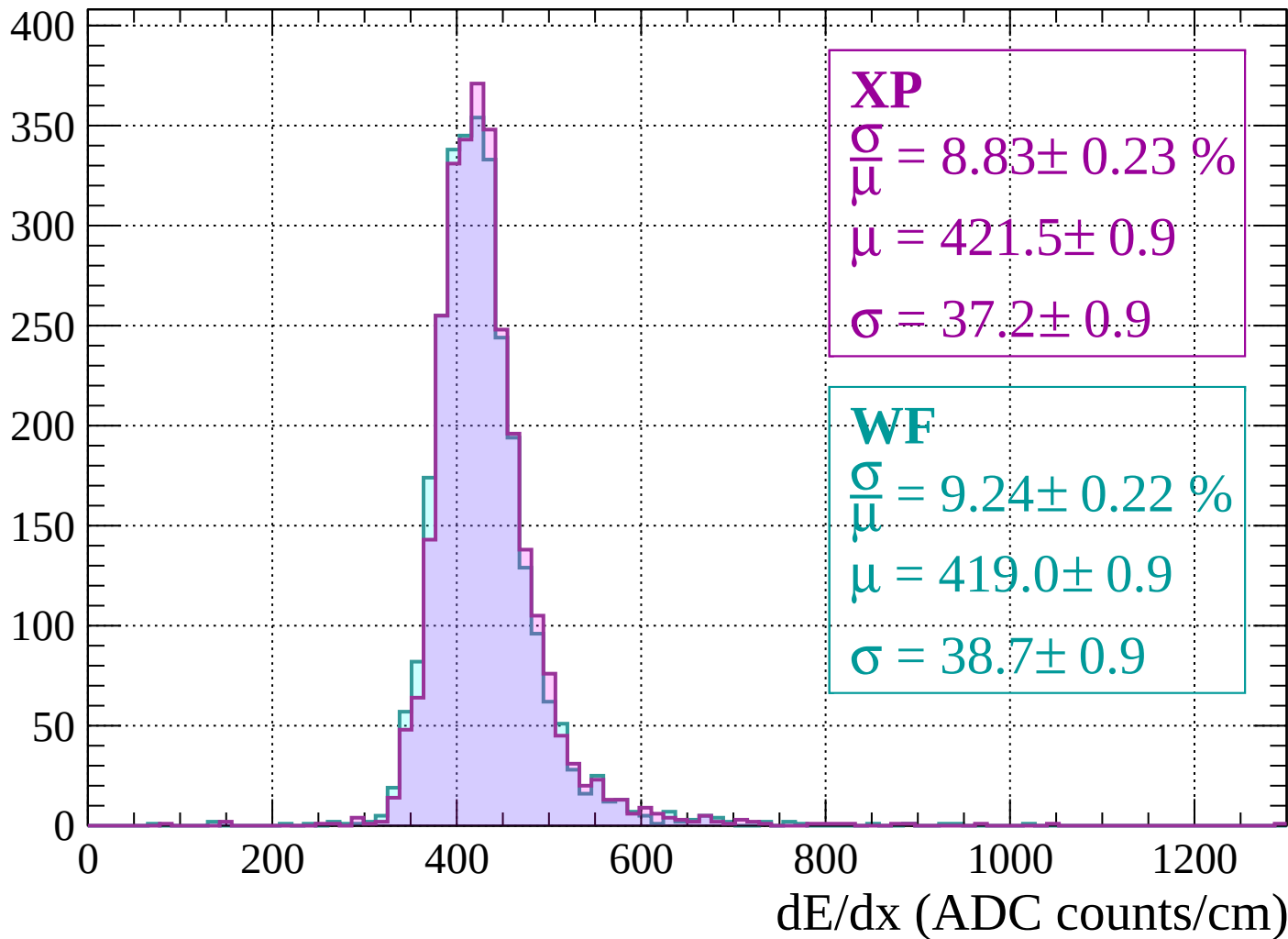
$$\mu = 417.7 \pm 0.9$$

$$\sigma = 39.7 \pm 0.9$$

dE/dx (ADC counts/cm)

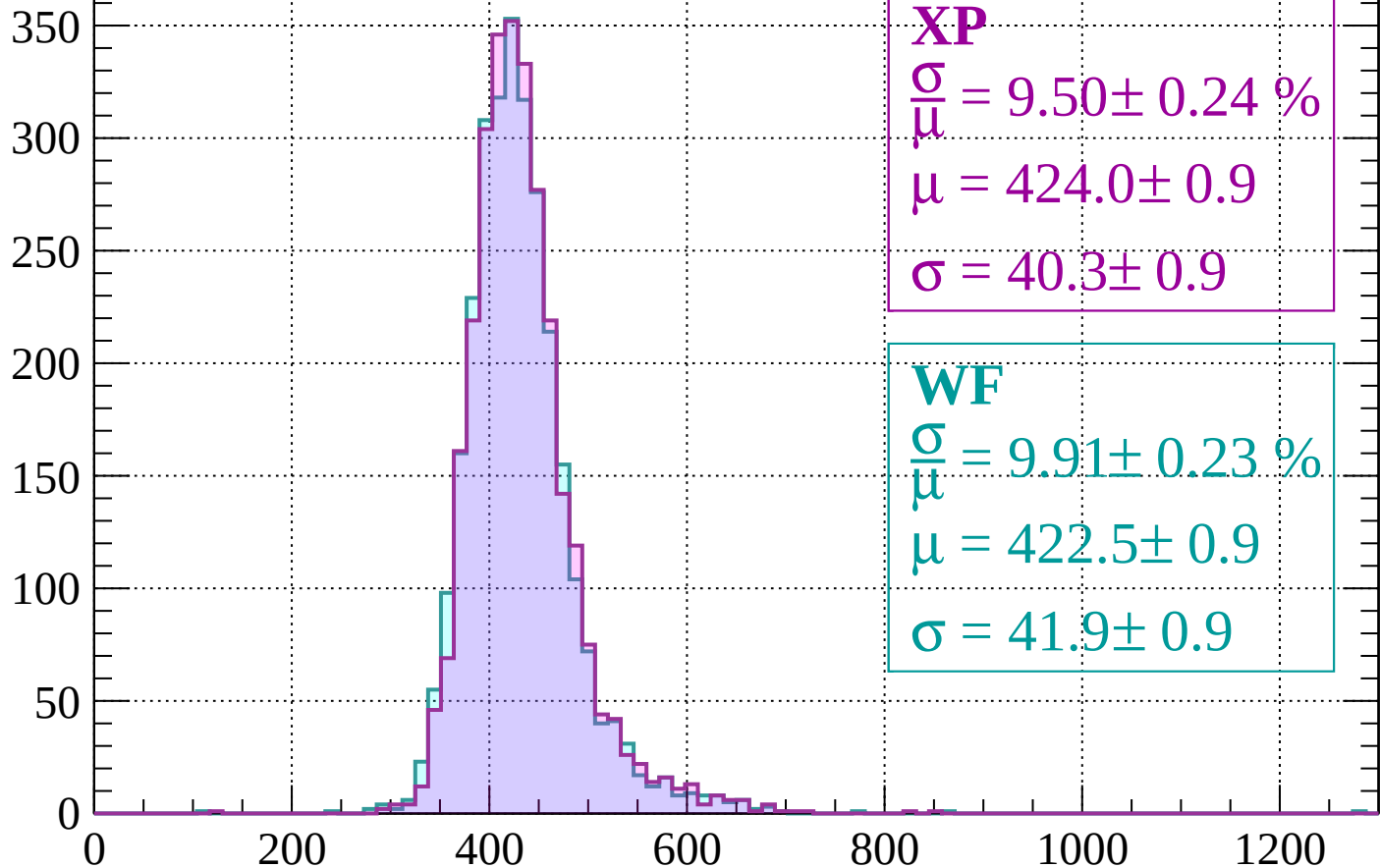
Energy loss | $840 < p < 880$

Count



Energy loss | $880 < p < 920$

Count



XP

$$\frac{\sigma}{\mu} = 9.50 \pm 0.24 \%$$

$$\mu = 424.0 \pm 0.9$$

$$\sigma = 40.3 \pm 0.9$$

WF

$$\frac{\sigma}{\mu} = 9.91 \pm 0.23 \%$$

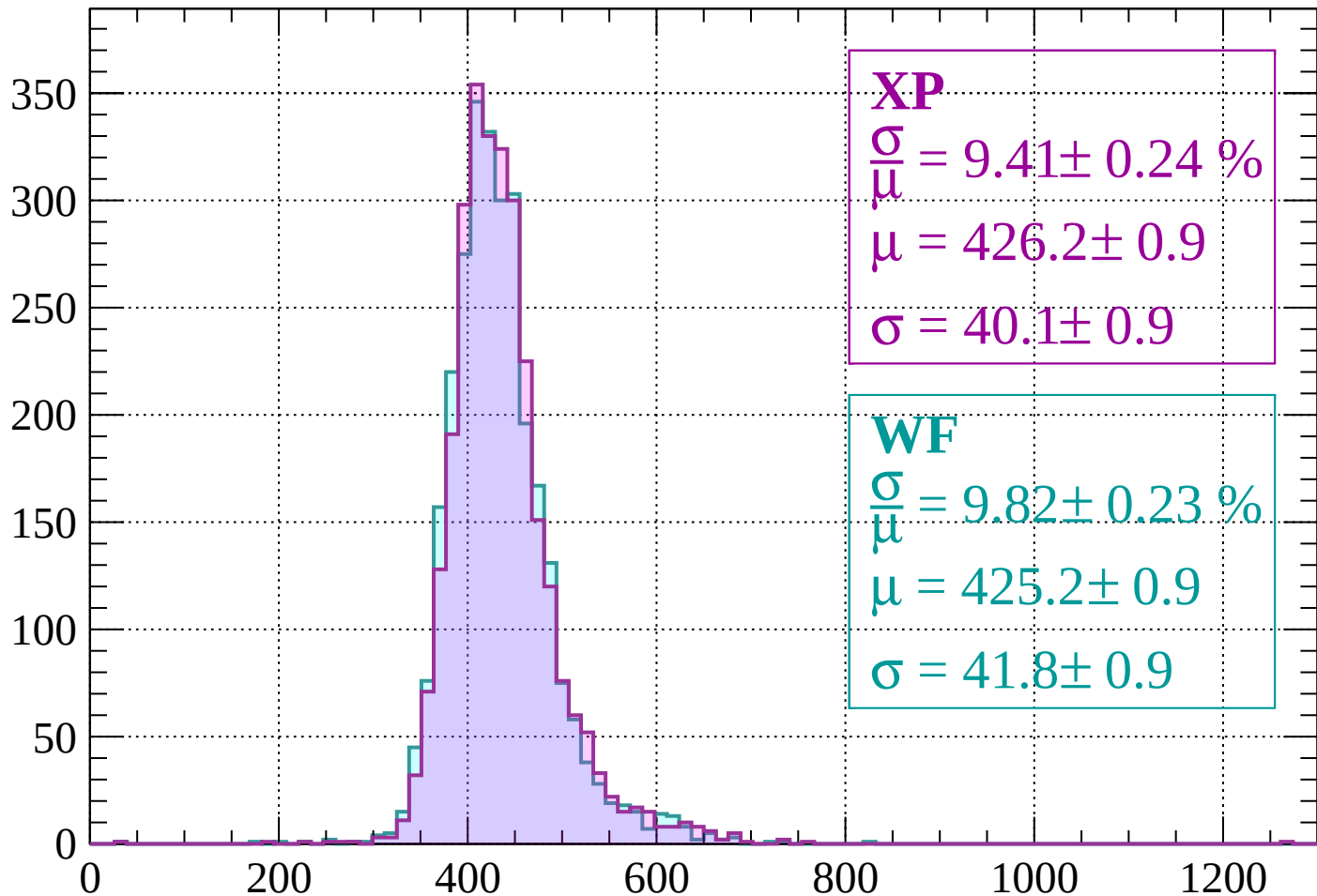
$$\mu = 422.5 \pm 0.9$$

$$\sigma = 41.9 \pm 0.9$$

dE/dx (ADC counts/cm)

Energy loss | $920 < p < 960$

Count



XP

$$\frac{\sigma}{\mu} = 9.41 \pm 0.24 \%$$

$$\mu = 426.2 \pm 0.9$$

$$\sigma = 40.1 \pm 0.9$$

WF

$$\frac{\sigma}{\mu} = 9.82 \pm 0.23 \%$$

$$\mu = 425.2 \pm 0.9$$

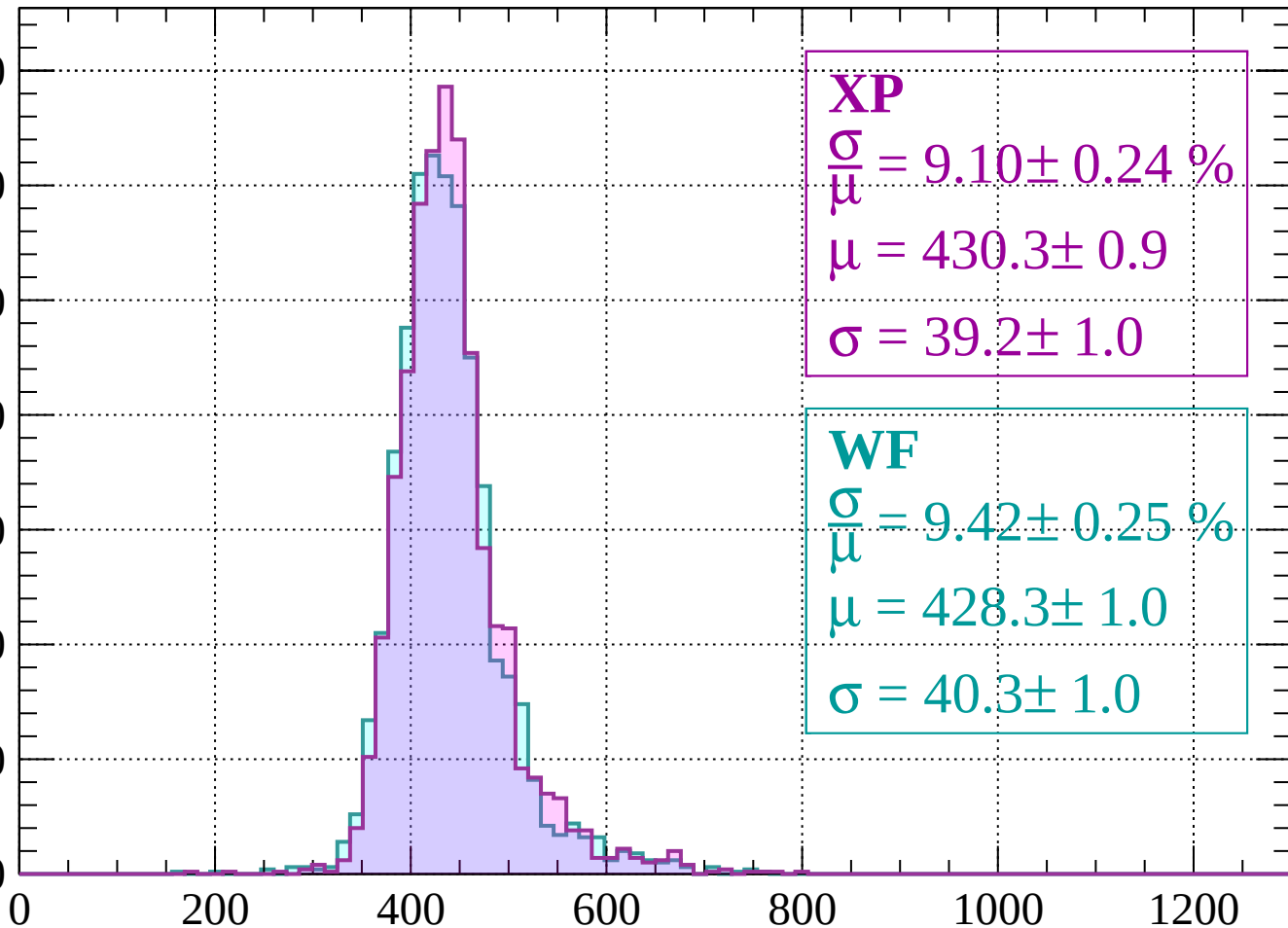
$$\sigma = 41.8 \pm 0.9$$

dE/dx (ADC counts/cm)

Energy loss | $960 < p < 1000$

Count

350
300
250
200
150
100
50
0



XP

$$\frac{\sigma}{\mu} = 9.10 \pm 0.24 \%$$

$$\mu = 430.3 \pm 0.9$$

$$\sigma = 39.2 \pm 1.0$$

WF

$$\frac{\sigma}{\mu} = 9.42 \pm 0.25 \%$$

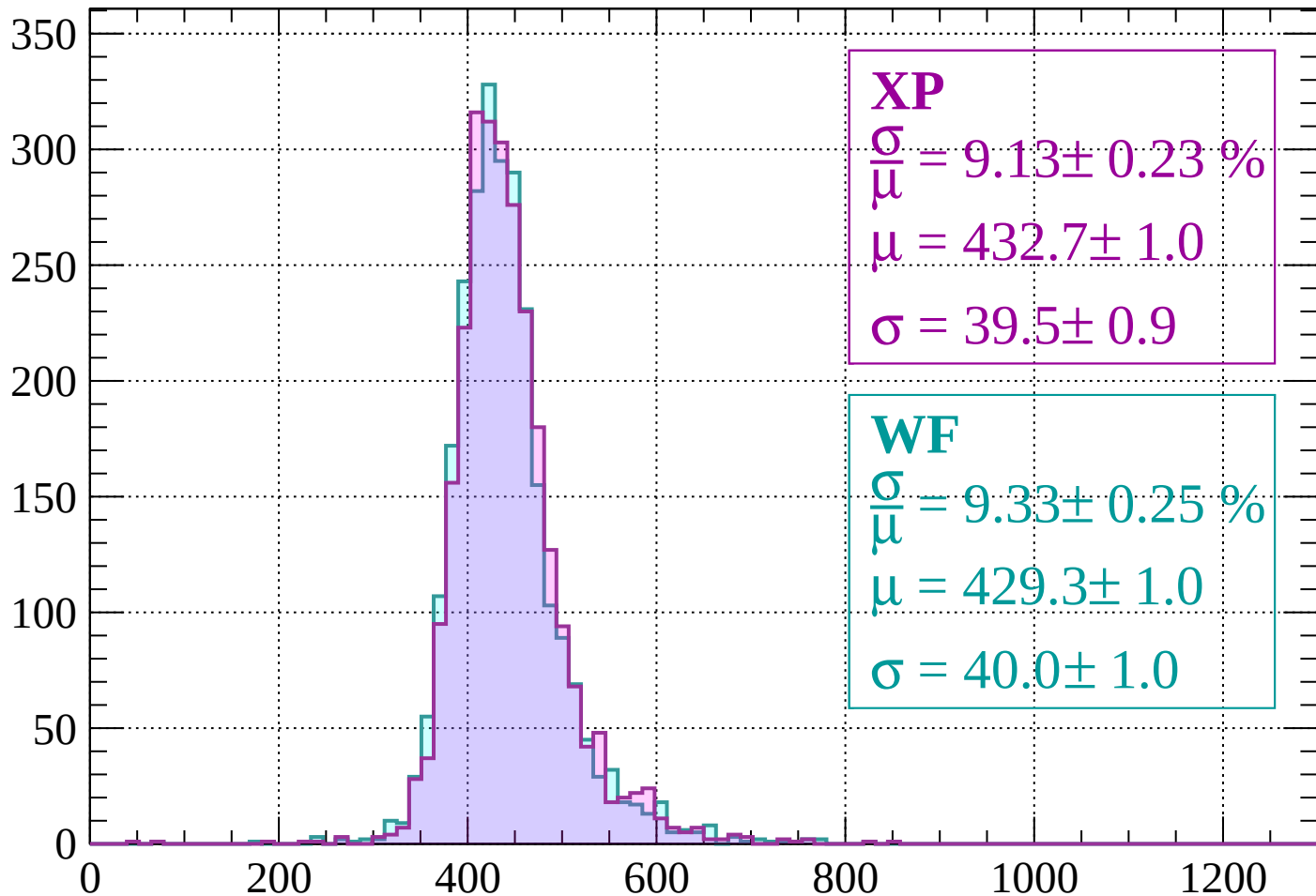
$$\mu = 428.3 \pm 1.0$$

$$\sigma = 40.3 \pm 1.0$$

dE/dx (ADC counts/cm)

Energy loss | $1000 < p < 1040$

Count



XP

$$\frac{\sigma}{\mu} = 9.13 \pm 0.23 \%$$

$$\mu = 432.7 \pm 1.0$$

$$\sigma = 39.5 \pm 0.9$$

WF

$$\frac{\sigma}{\mu} = 9.33 \pm 0.25 \%$$

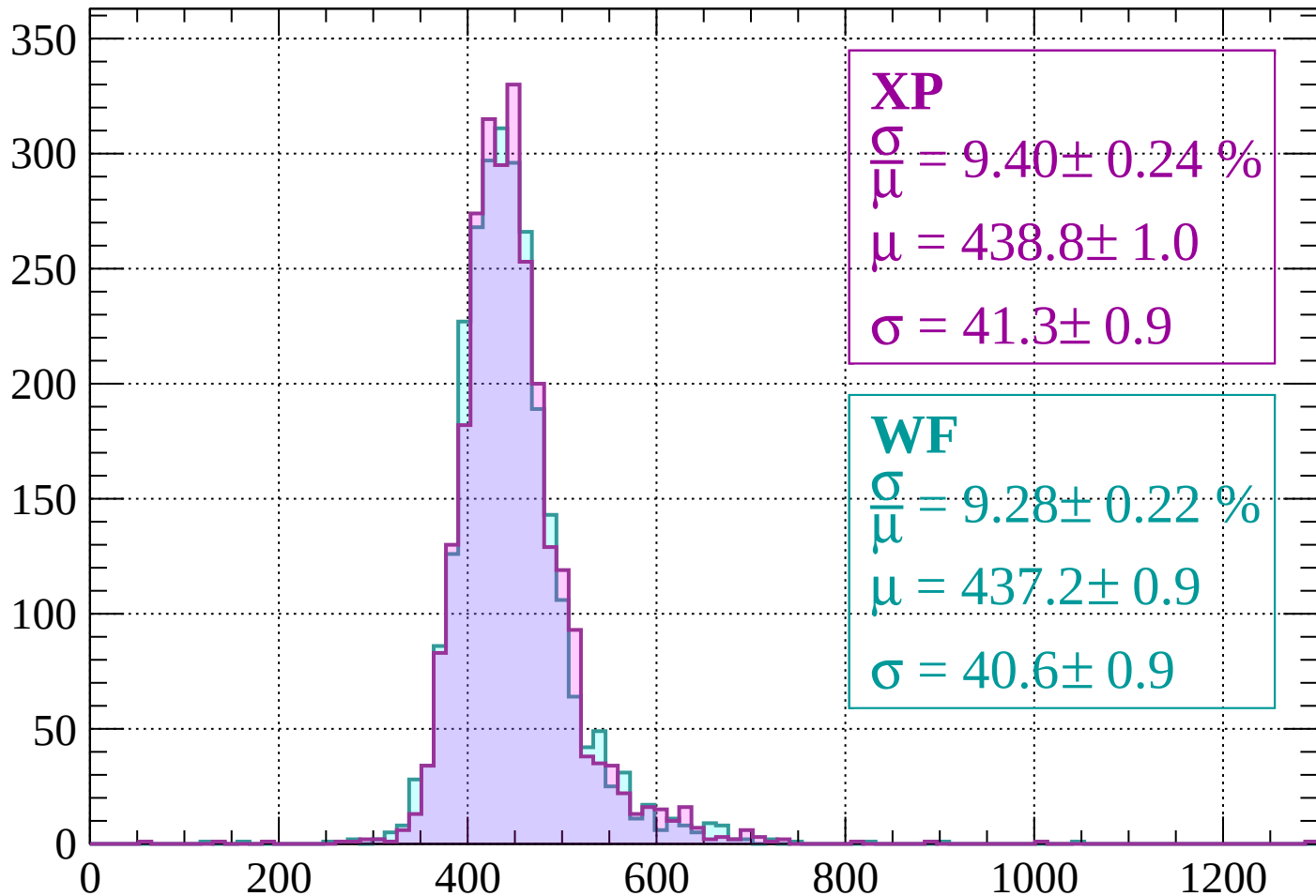
$$\mu = 429.3 \pm 1.0$$

$$\sigma = 40.0 \pm 1.0$$

dE/dx (ADC counts/cm)

Energy loss | $1040 < p < 1080$

Count



XP

$$\frac{\sigma}{\mu} = 9.40 \pm 0.24 \%$$

$$\mu = 438.8 \pm 1.0$$

$$\sigma = 41.3 \pm 0.9$$

WF

$$\frac{\sigma}{\mu} = 9.28 \pm 0.22 \%$$

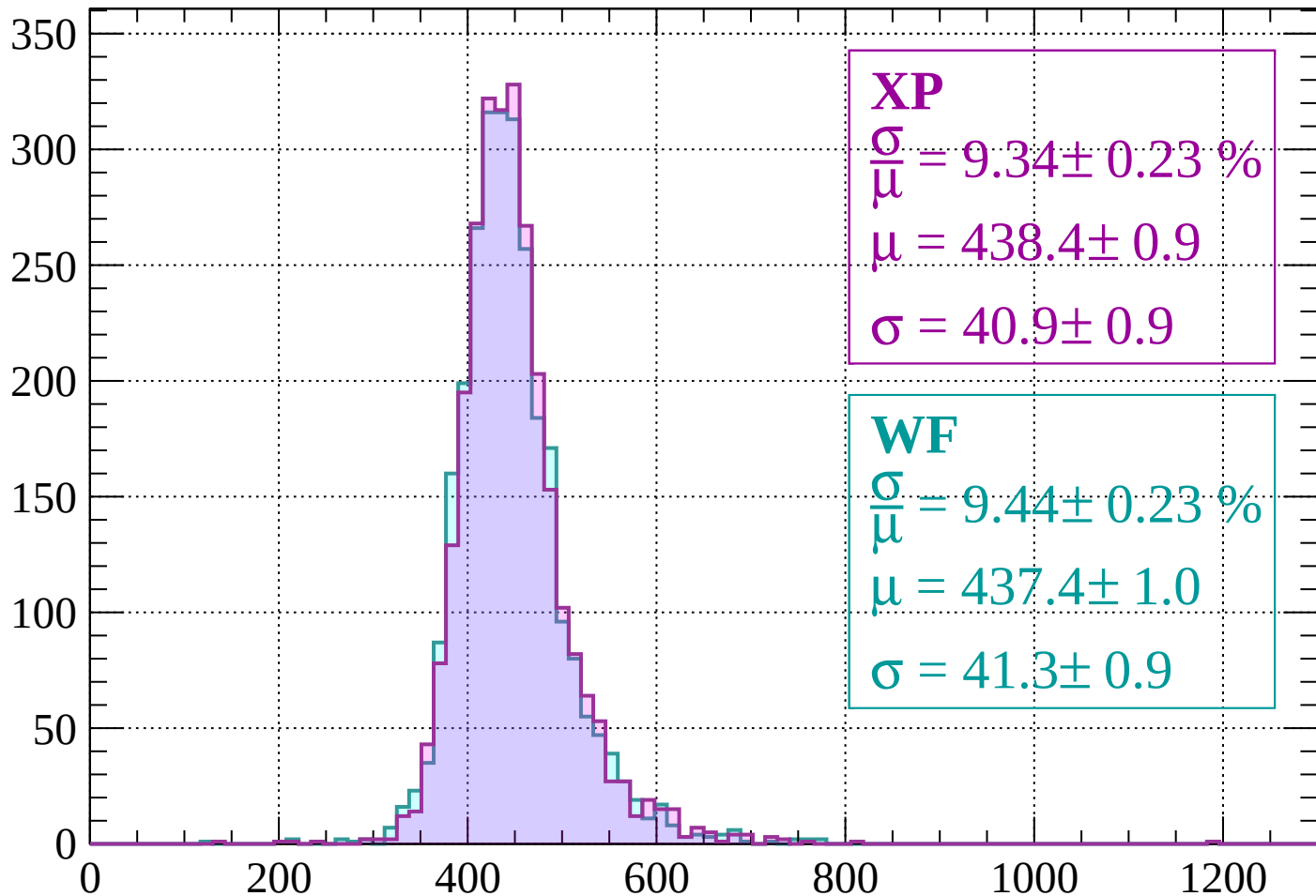
$$\mu = 437.2 \pm 0.9$$

$$\sigma = 40.6 \pm 0.9$$

dE/dx (ADC counts/cm)

Energy loss | $1080 < p < 1120$

Count



XP

$$\frac{\sigma}{\mu} = 9.34 \pm 0.23 \%$$

$$\mu = 438.4 \pm 0.9$$

$$\sigma = 40.9 \pm 0.9$$

WF

$$\frac{\sigma}{\mu} = 9.44 \pm 0.23 \%$$

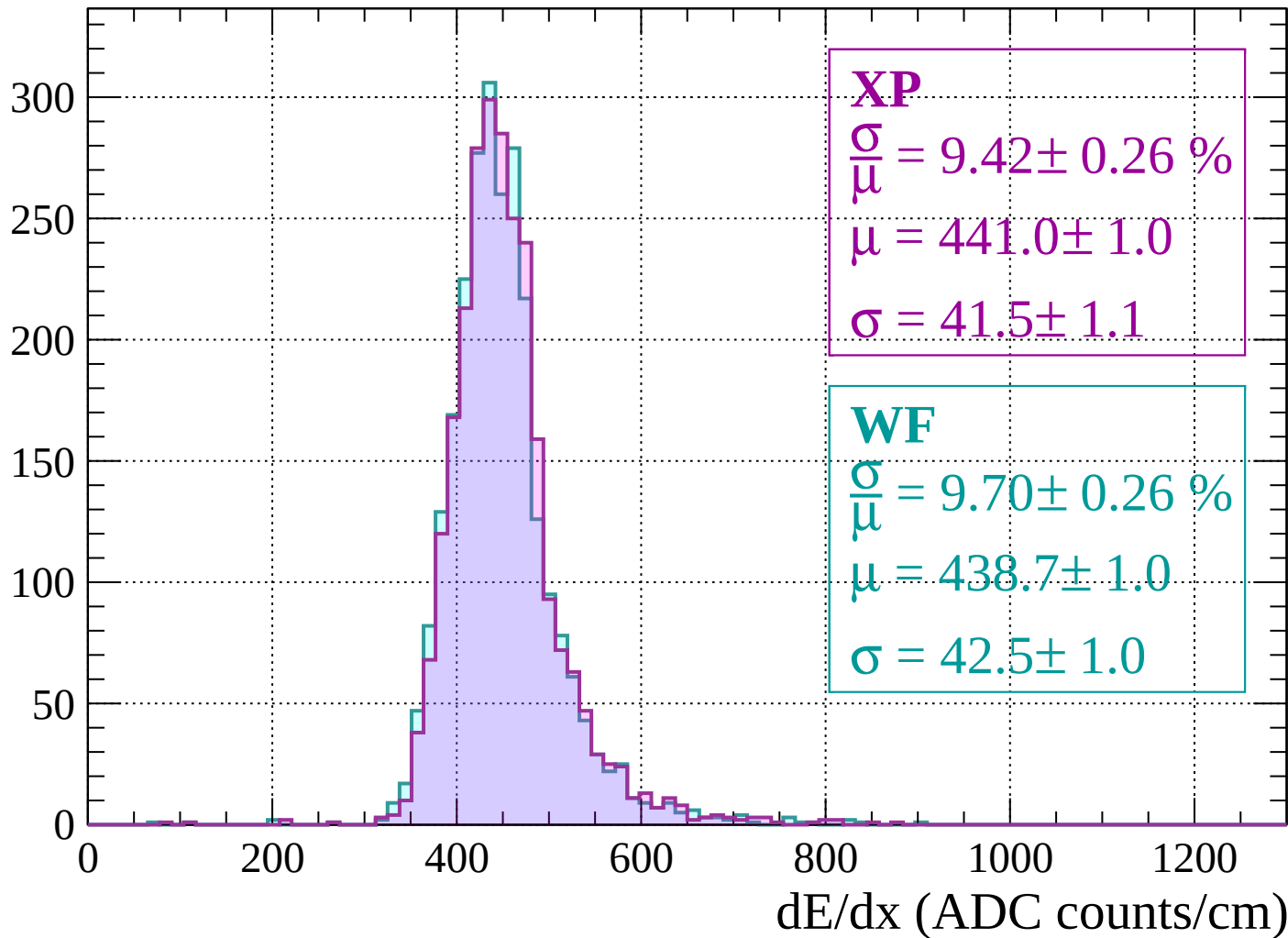
$$\mu = 437.4 \pm 1.0$$

$$\sigma = 41.3 \pm 0.9$$

dE/dx (ADC counts/cm)

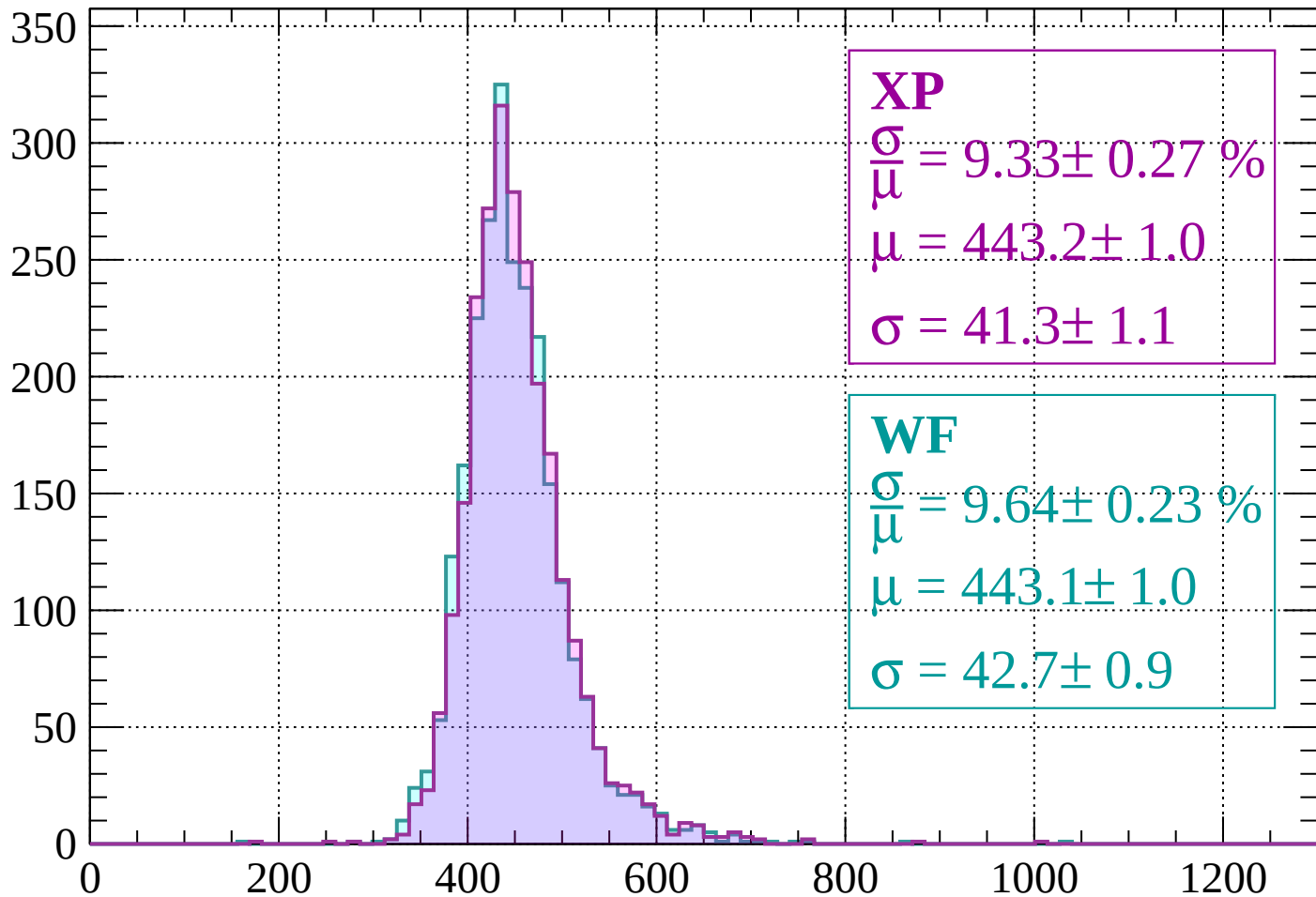
Energy loss | $1120 < p < 1160$

Count



Energy loss | $1160 < p < 1200$

Count



XP

$$\frac{\sigma}{\mu} = 9.33 \pm 0.27 \%$$

$$\mu = 443.2 \pm 1.0$$

$$\sigma = 41.3 \pm 1.1$$

WF

$$\frac{\sigma}{\mu} = 9.64 \pm 0.23 \%$$

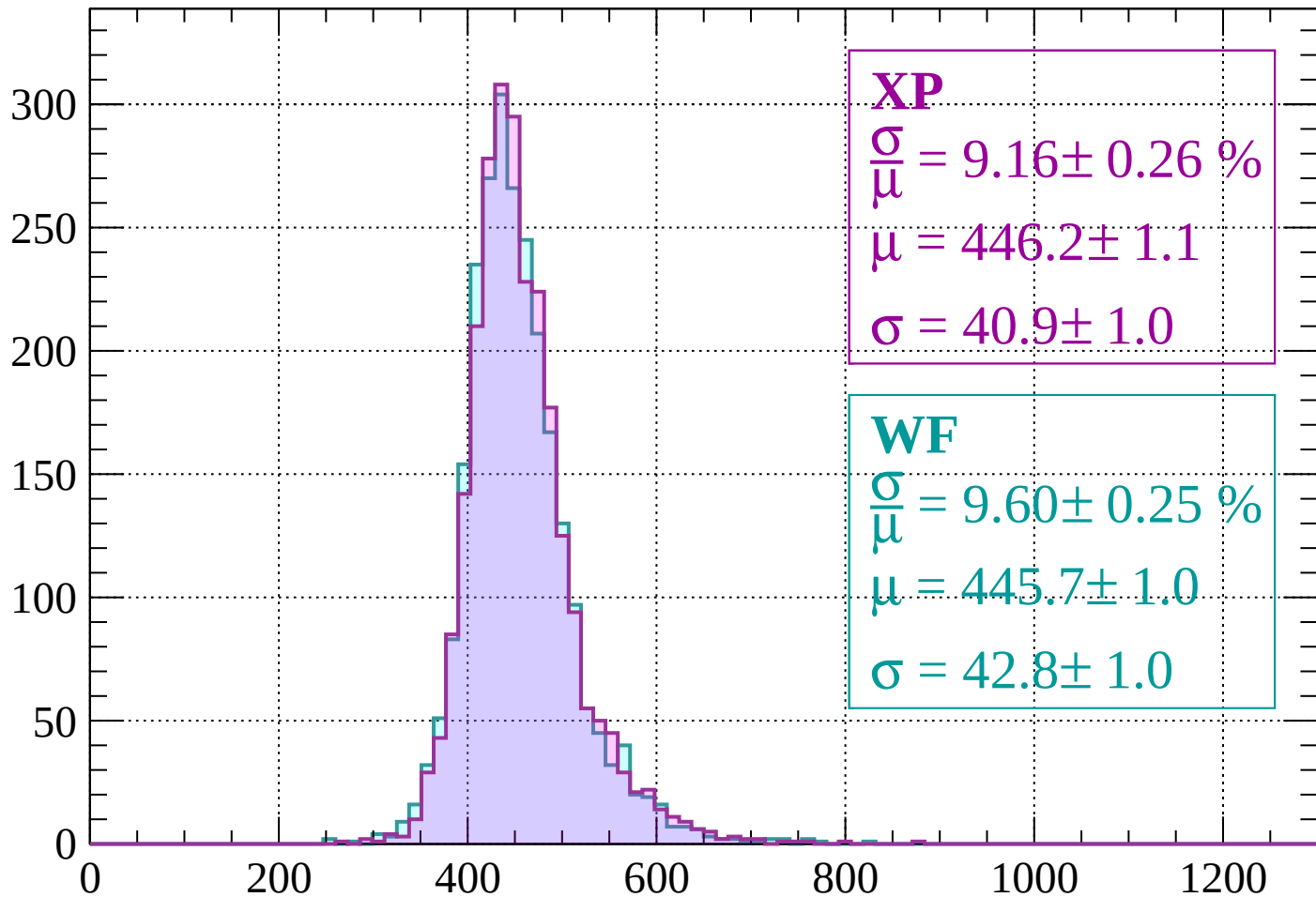
$$\mu = 443.1 \pm 1.0$$

$$\sigma = 42.7 \pm 0.9$$

dE/dx (ADC counts/cm)

Energy loss | $1200 < p < 1240$

Count



XP

$$\frac{\sigma}{\mu} = 9.16 \pm 0.26 \%$$

$$\mu = 446.2 \pm 1.1$$

$$\sigma = 40.9 \pm 1.0$$

WF

$$\frac{\sigma}{\mu} = 9.60 \pm 0.25 \%$$

$$\mu = 445.7 \pm 1.0$$

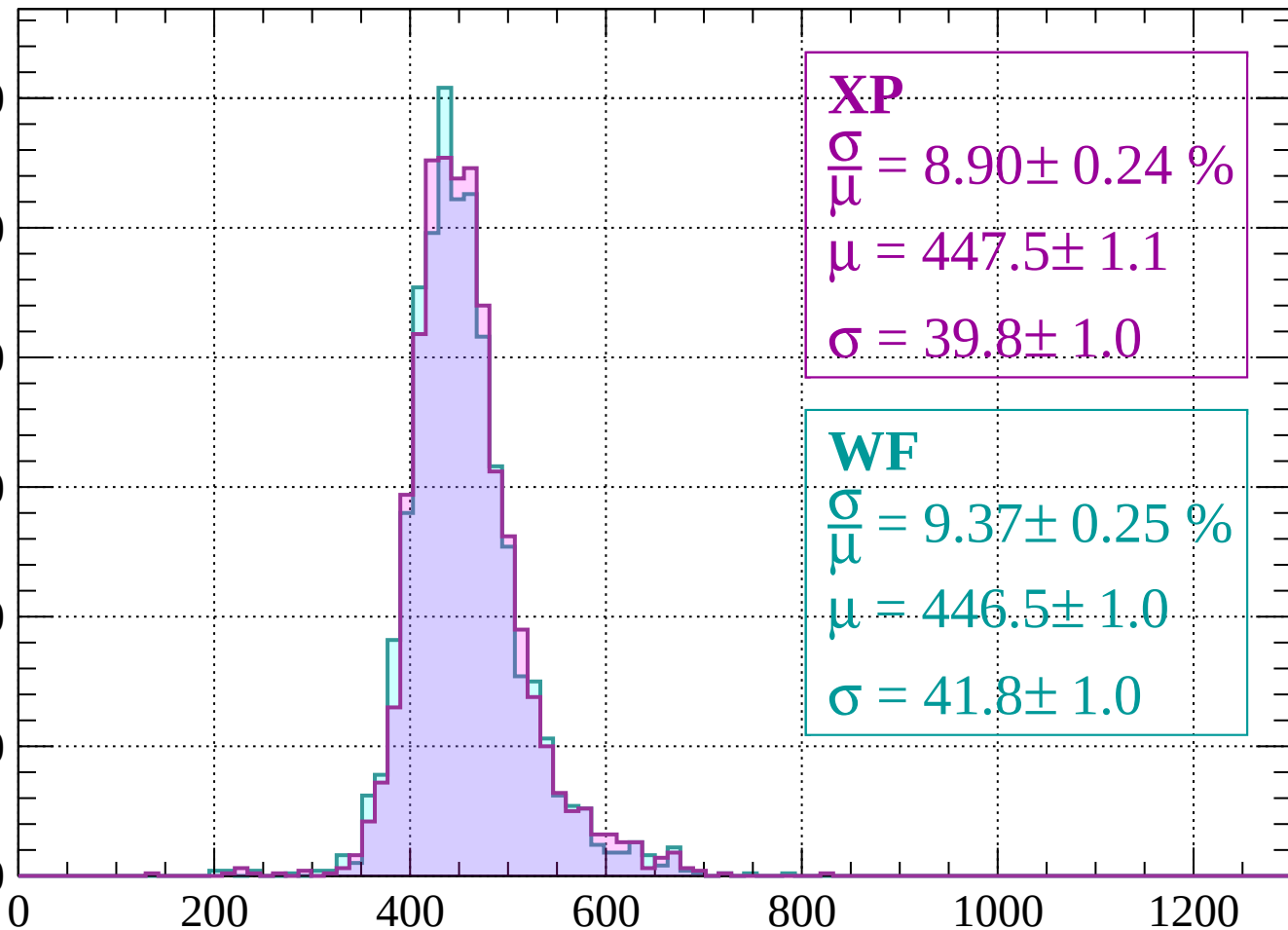
$$\sigma = 42.8 \pm 1.0$$

dE/dx (ADC counts/cm)

Energy loss | $1240 < p < 1280$

Count

300
250
200
150
100
50
0



XP

$$\frac{\sigma}{\mu} = 8.90 \pm 0.24 \%$$

$$\mu = 447.5 \pm 1.1$$

$$\sigma = 39.8 \pm 1.0$$

WF

$$\frac{\sigma}{\mu} = 9.37 \pm 0.25 \%$$

$$\mu = 446.5 \pm 1.0$$

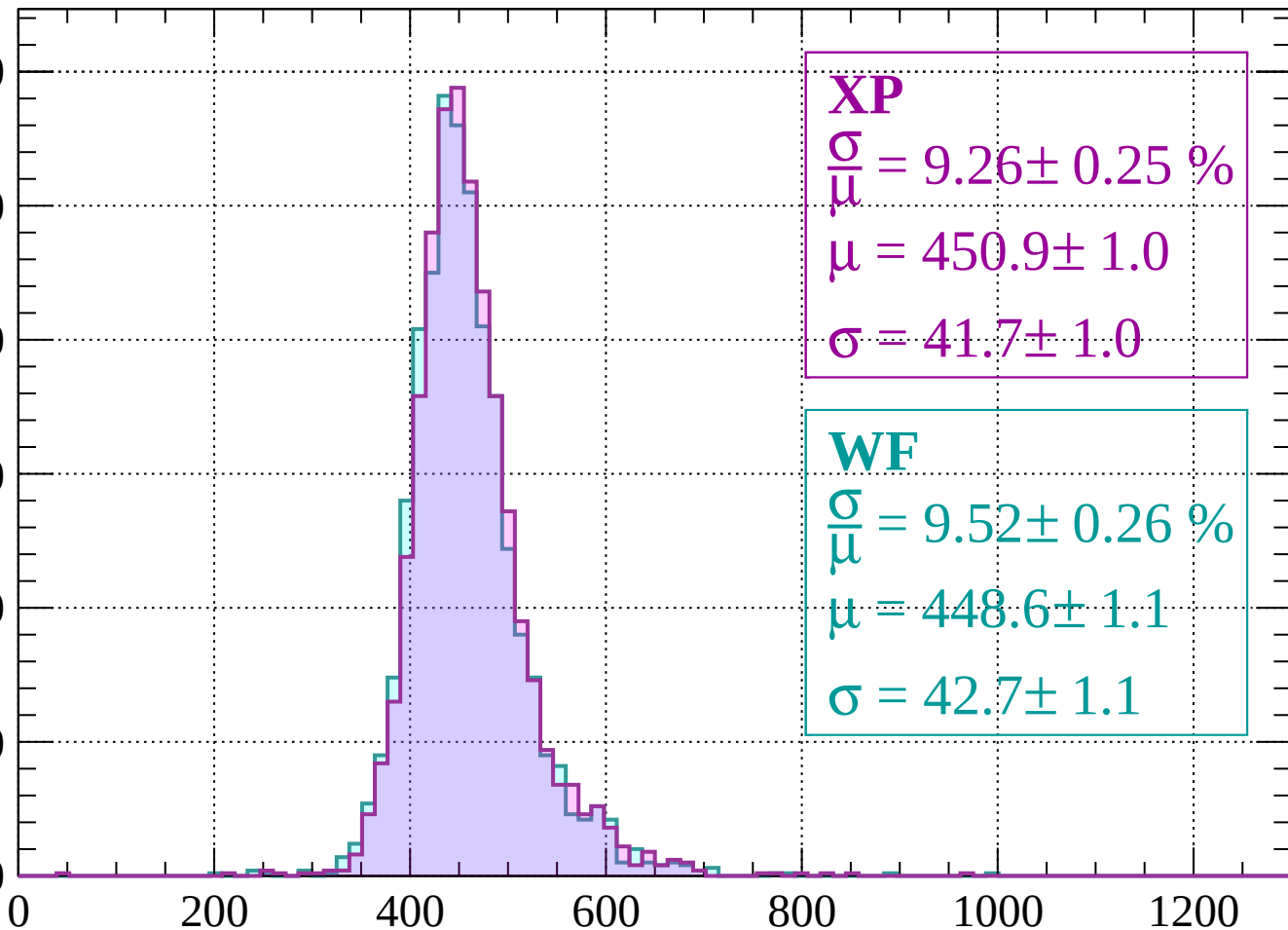
$$\sigma = 41.8 \pm 1.0$$

dE/dx (ADC counts/cm)

Energy loss | $1280 < p < 1320$

Count

300
250
200
150
100
50
0



XP

$$\frac{\sigma}{\mu} = 9.26 \pm 0.25 \%$$

$$\mu = 450.9 \pm 1.0$$

$$\sigma = 41.7 \pm 1.0$$

WF

$$\frac{\sigma}{\mu} = 9.52 \pm 0.26 \%$$

$$\mu = 448.6 \pm 1.1$$

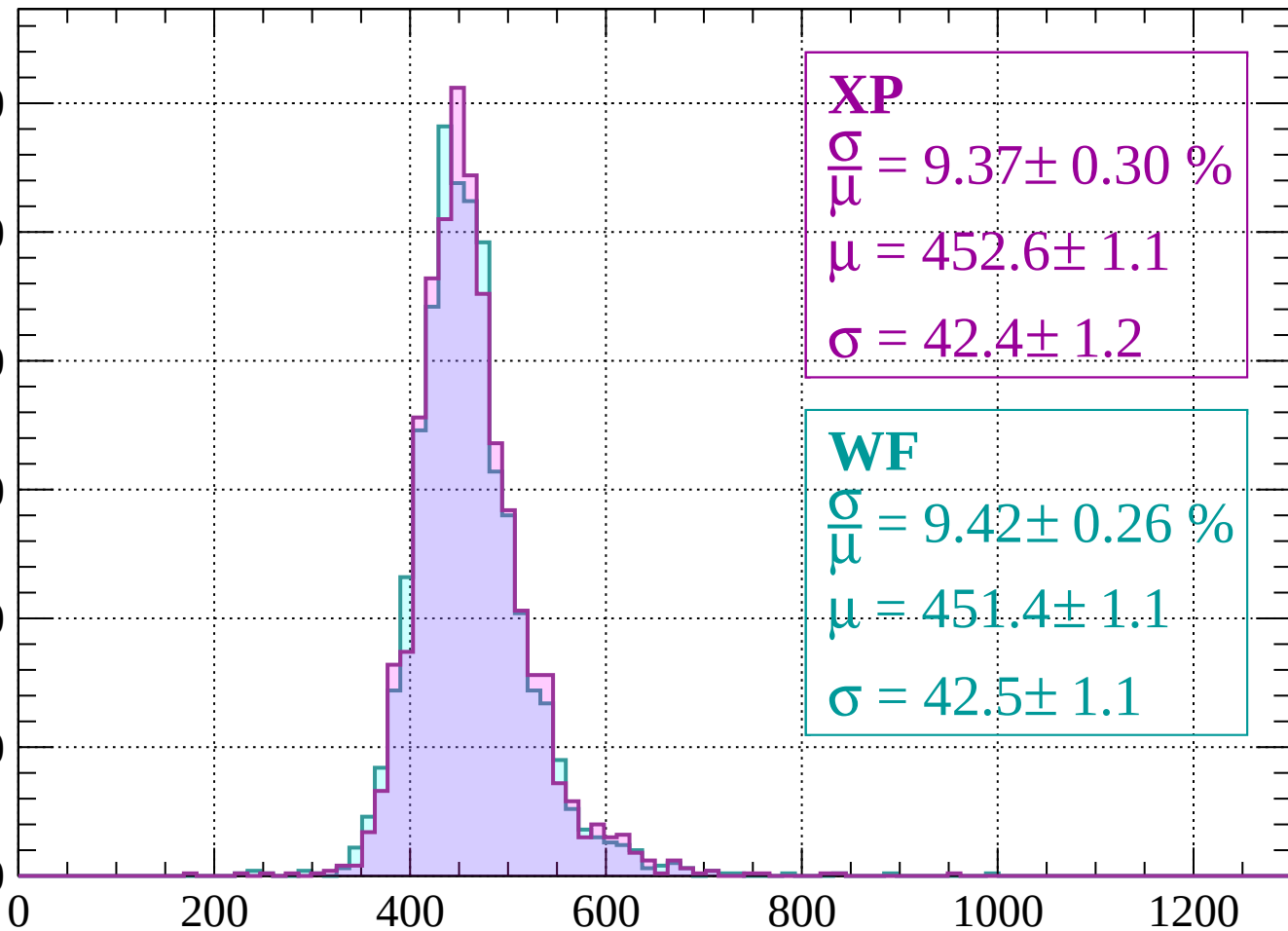
$$\sigma = 42.7 \pm 1.1$$

dE/dx (ADC counts/cm)

Energy loss | $1320 < p < 1360$

Count

300
250
200
150
100
50
0



XP

$$\frac{\sigma}{\mu} = 9.37 \pm 0.30 \%$$

$$\mu = 452.6 \pm 1.1$$

$$\sigma = 42.4 \pm 1.2$$

WF

$$\frac{\sigma}{\mu} = 9.42 \pm 0.26 \%$$

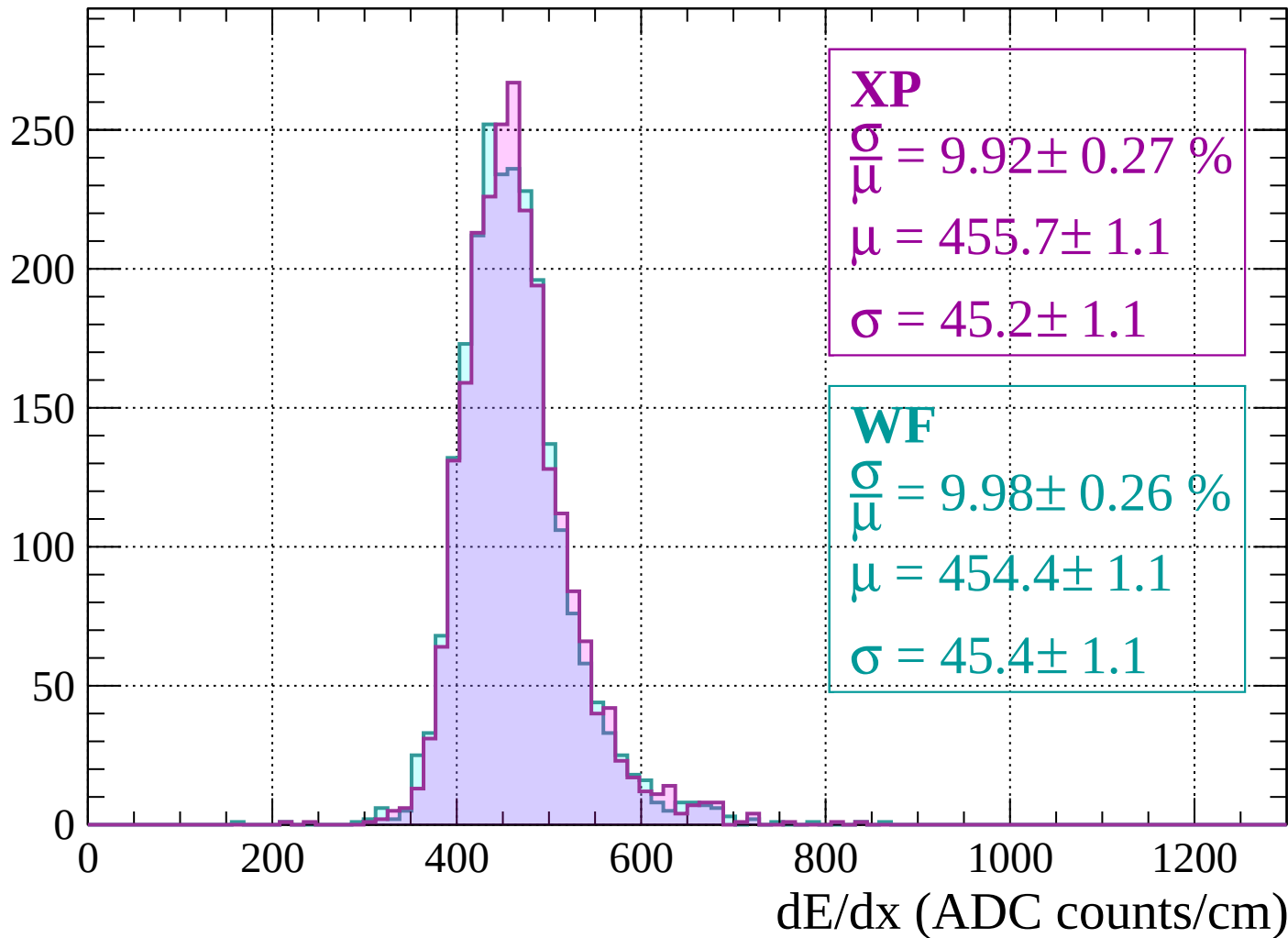
$$\mu = 451.4 \pm 1.1$$

$$\sigma = 42.5 \pm 1.1$$

dE/dx (ADC counts/cm)

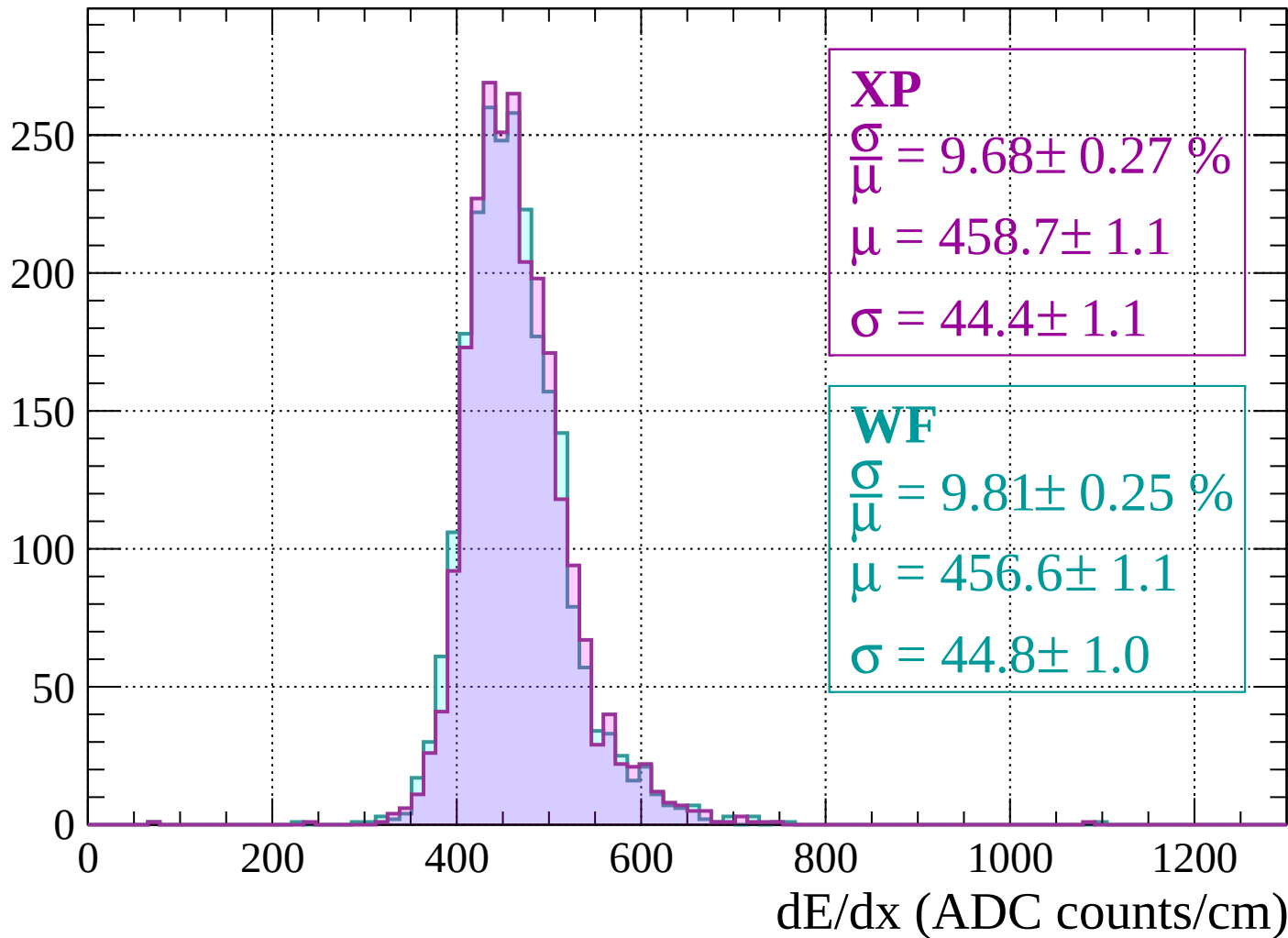
Energy loss | $1360 < p < 1400$

Count



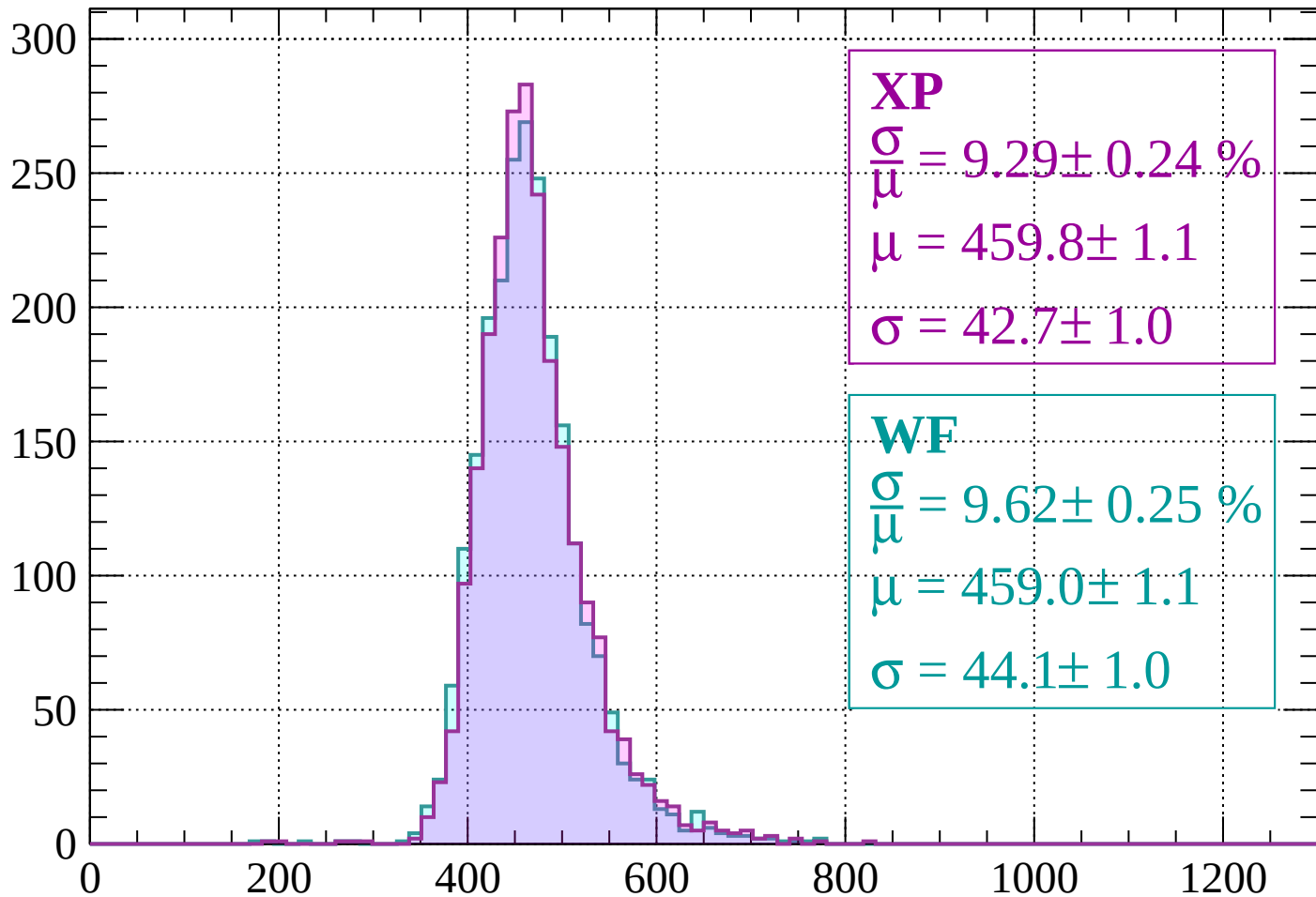
Energy loss | $1400 < p < 1440$

Count



Energy loss | $1440 < p < 1480$

Count



XP

$$\frac{\sigma}{\mu} = 9.29 \pm 0.24 \%$$

$$\mu = 459.8 \pm 1.1$$

$$\sigma = 42.7 \pm 1.0$$

WF

$$\frac{\sigma}{\mu} = 9.62 \pm 0.25 \%$$

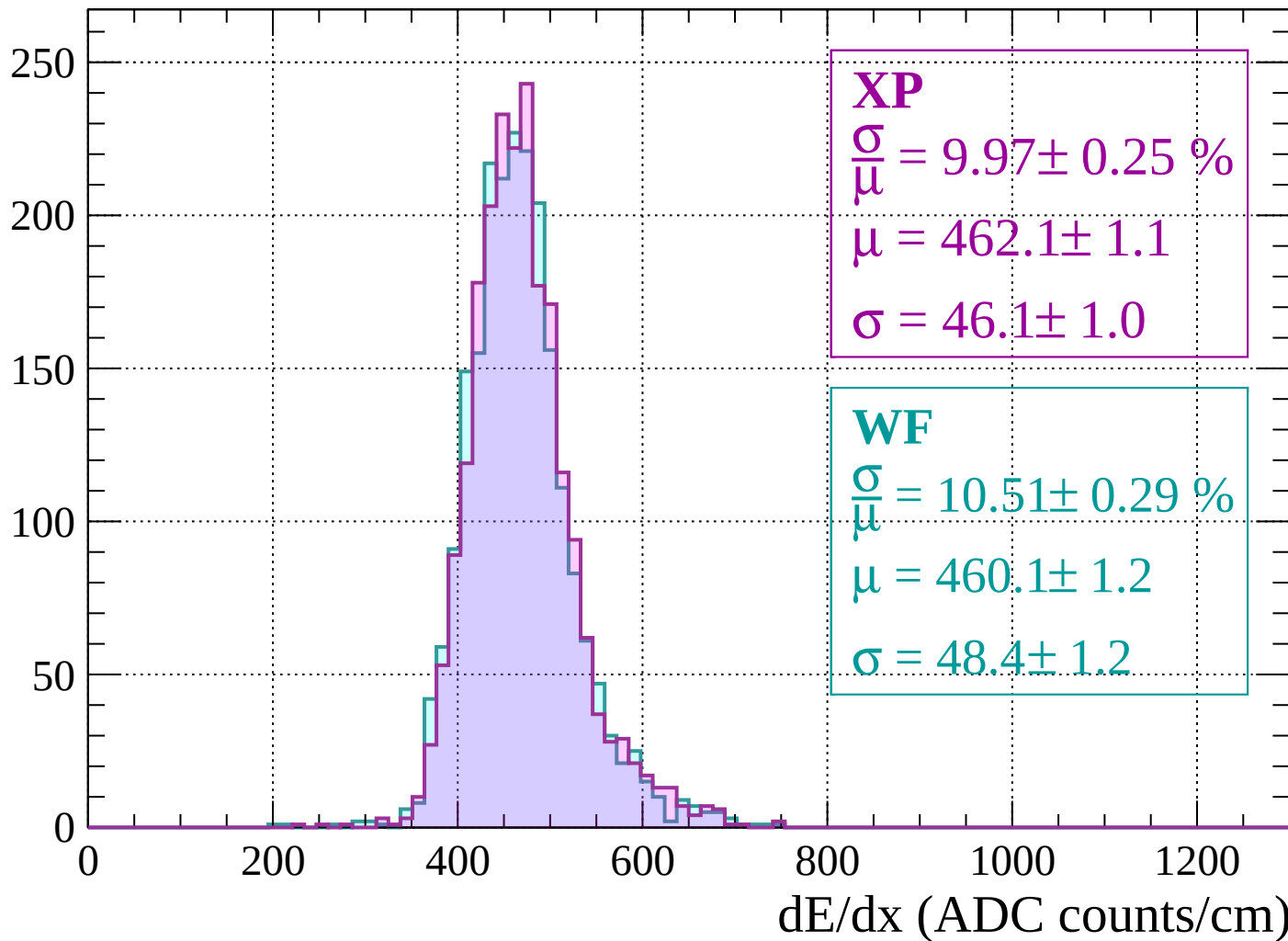
$$\mu = 459.0 \pm 1.1$$

$$\sigma = 44.1 \pm 1.0$$

dE/dx (ADC counts/cm)

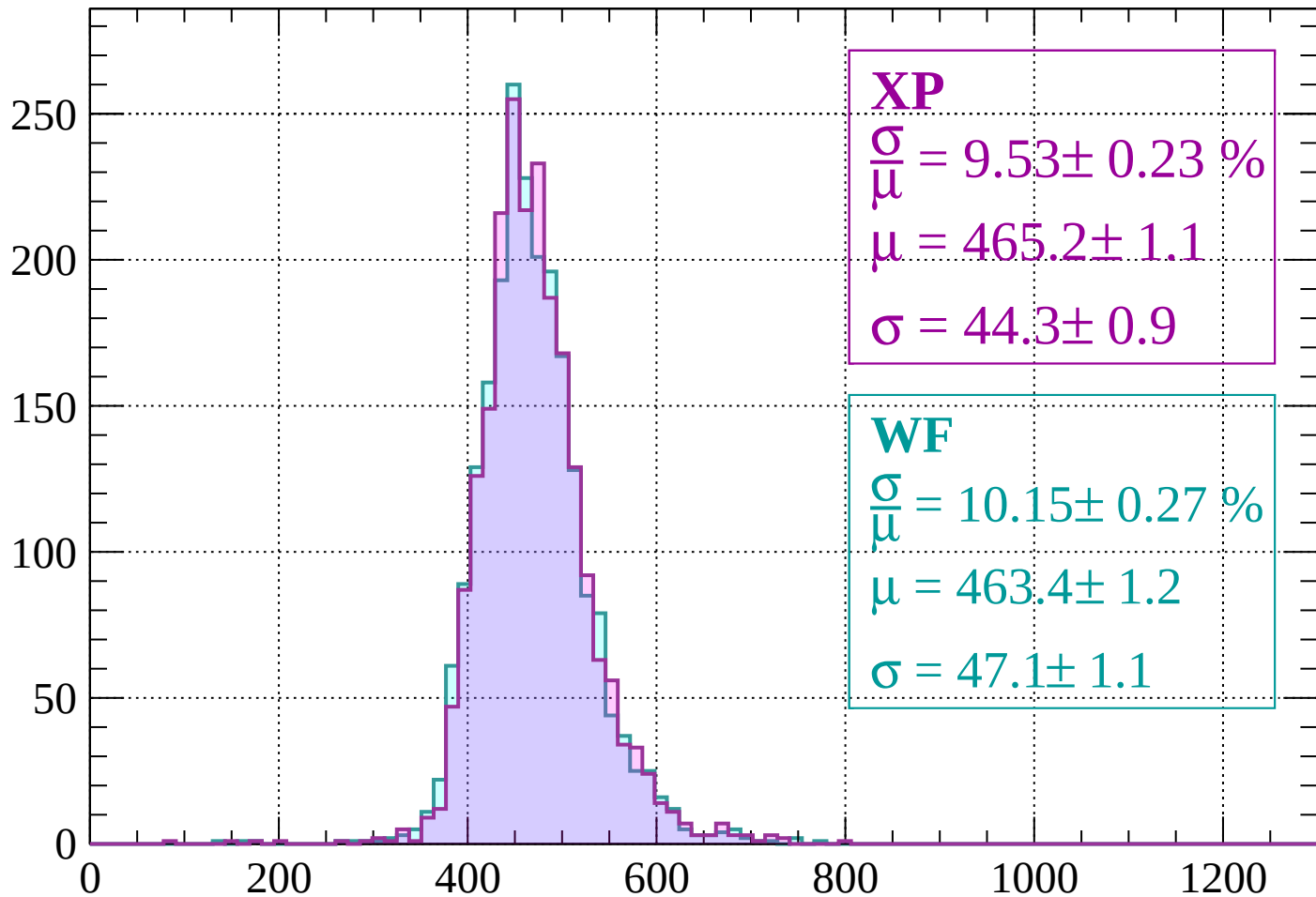
Energy loss | $1480 < p < 1520$

Count



Energy loss | $1520 < p < 1560$

Count



XP

$$\frac{\sigma}{\mu} = 9.53 \pm 0.23 \%$$

$$\mu = 465.2 \pm 1.1$$

$$\sigma = 44.3 \pm 0.9$$

WF

$$\frac{\sigma}{\mu} = 10.15 \pm 0.27 \%$$

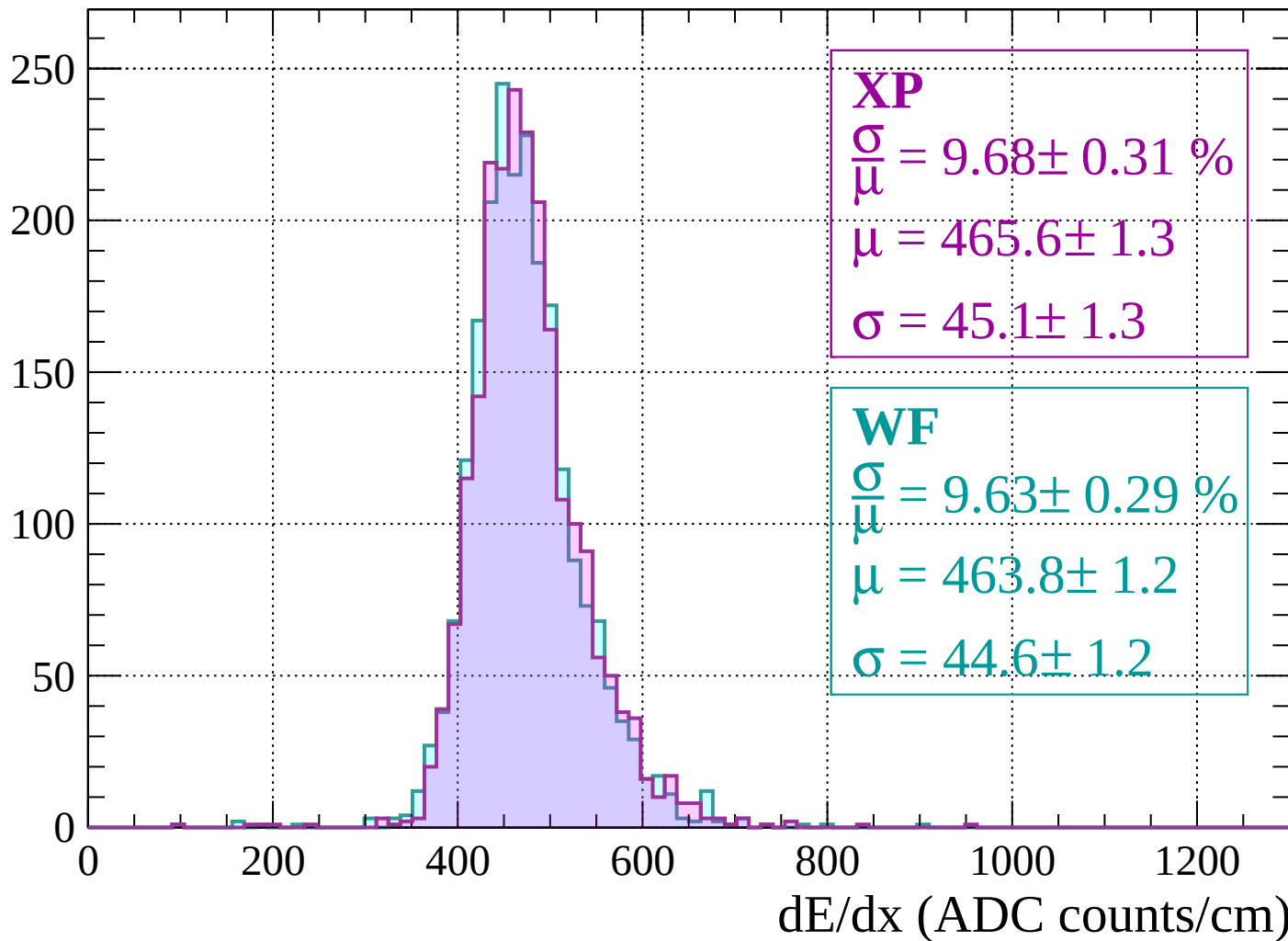
$$\mu = 463.4 \pm 1.2$$

$$\sigma = 47.1 \pm 1.1$$

dE/dx (ADC counts/cm)

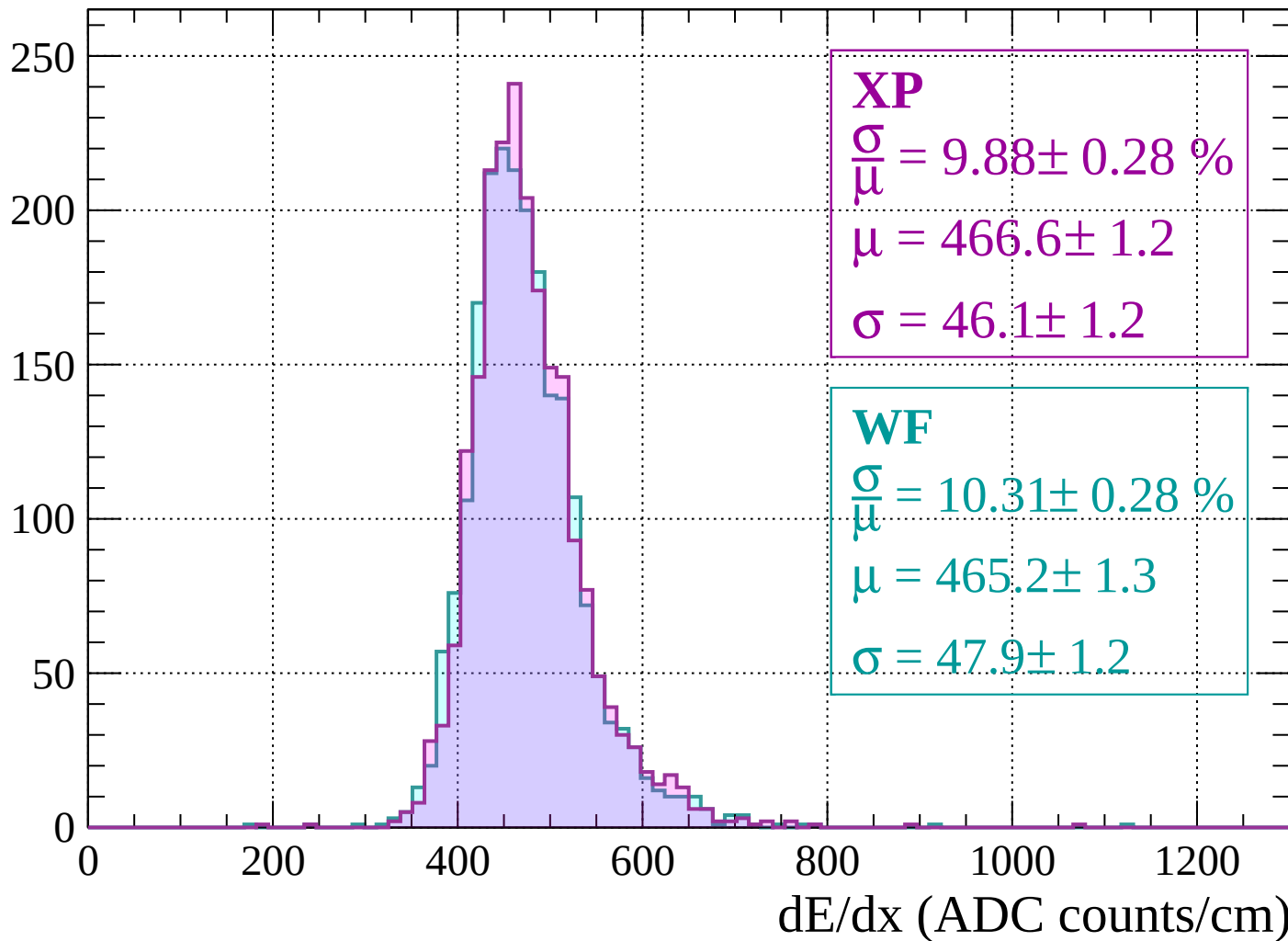
Energy loss | $1560 < p < 1600$

Count



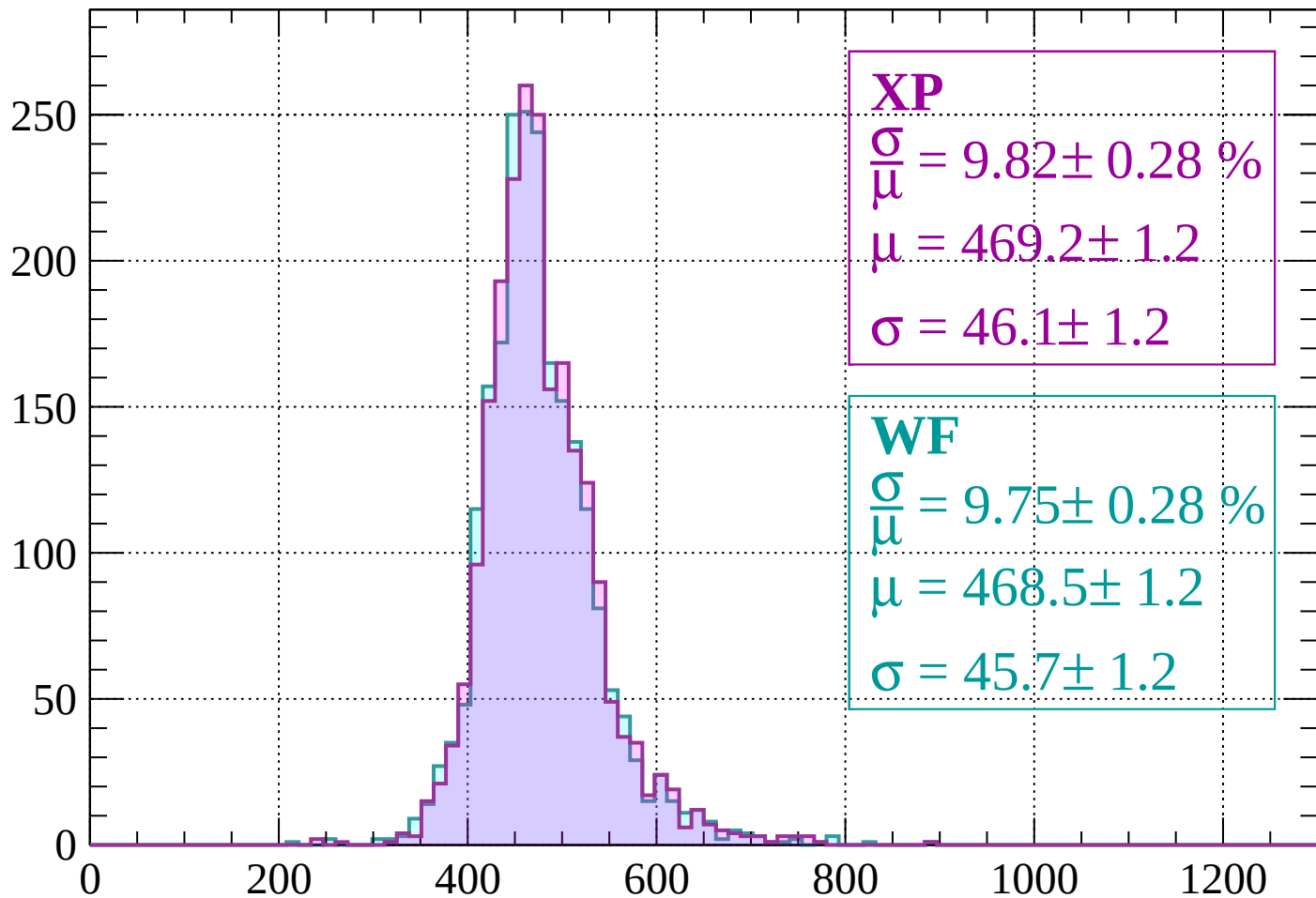
Energy loss | $1600 < p < 1640$

Count



Energy loss | $1640 < p < 1680$

Count



XP

$$\frac{\sigma}{\mu} = 9.82 \pm 0.28 \%$$

$$\mu = 469.2 \pm 1.2$$

$$\sigma = 46.1 \pm 1.2$$

WF

$$\frac{\sigma}{\mu} = 9.75 \pm 0.28 \%$$

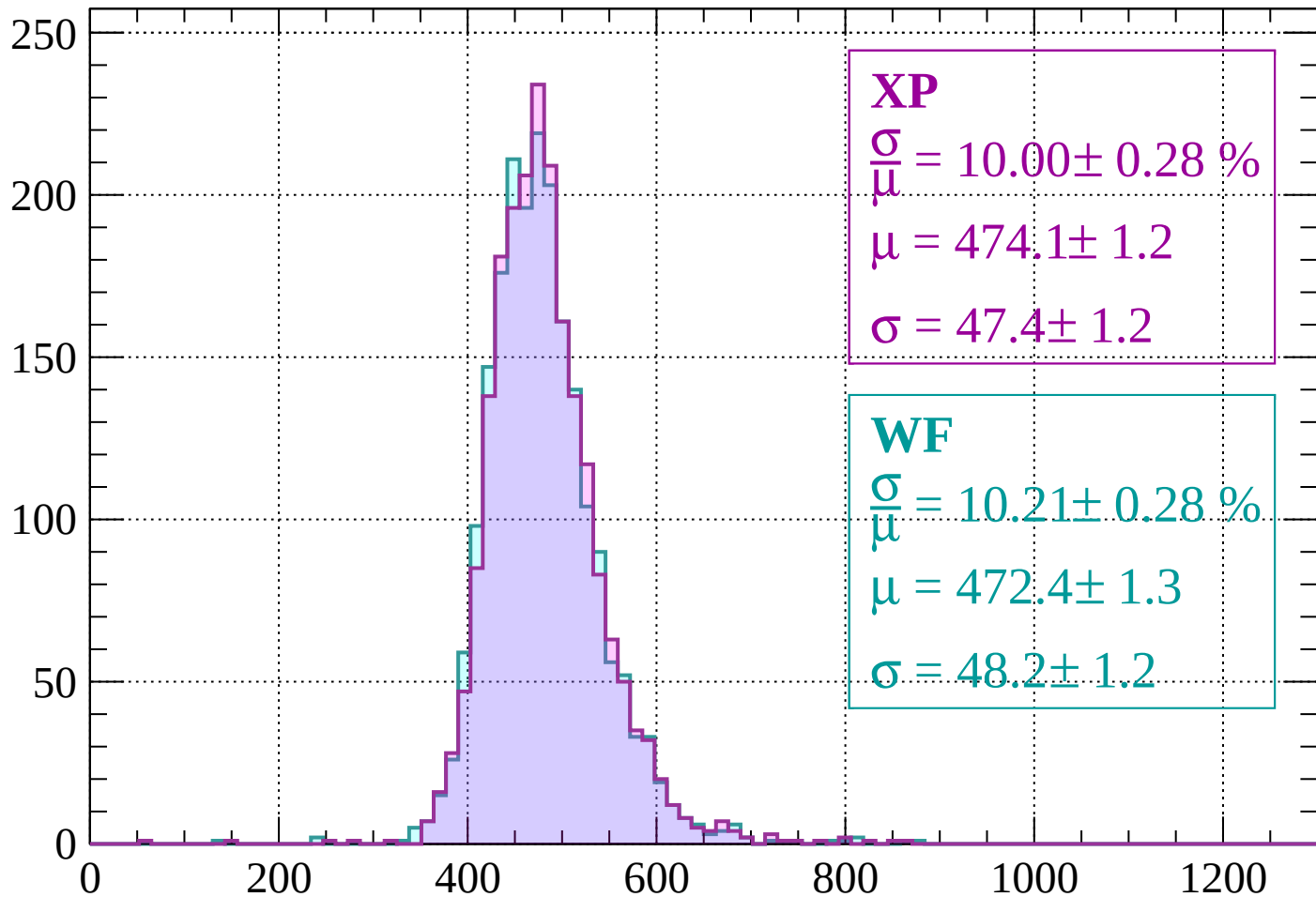
$$\mu = 468.5 \pm 1.2$$

$$\sigma = 45.7 \pm 1.2$$

dE/dx (ADC counts/cm)

Energy loss | $1680 < p < 1720$

Count



XP

$$\frac{\sigma}{\mu} = 10.00 \pm 0.28 \%$$

$$\mu = 474.1 \pm 1.2$$

$$\sigma = 47.4 \pm 1.2$$

WF

$$\frac{\sigma}{\mu} = 10.21 \pm 0.28 \%$$

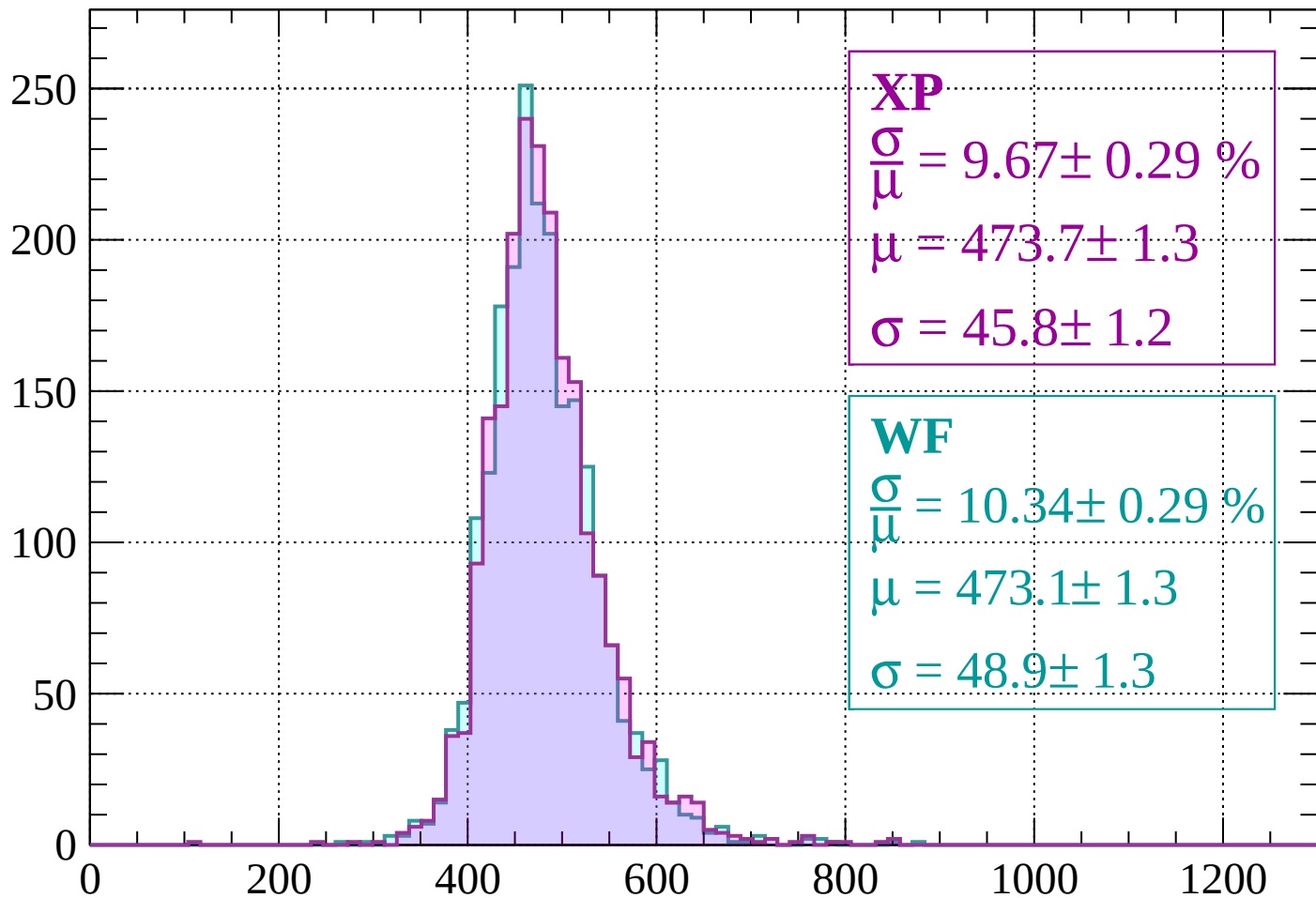
$$\mu = 472.4 \pm 1.3$$

$$\sigma = 48.2 \pm 1.2$$

dE/dx (ADC counts/cm)

Energy loss | $1720 < p < 1760$

Count



XP

$$\frac{\sigma}{\mu} = 9.67 \pm 0.29 \%$$

$$\mu = 473.7 \pm 1.3$$

$$\sigma = 45.8 \pm 1.2$$

WF

$$\frac{\sigma}{\mu} = 10.34 \pm 0.29 \%$$

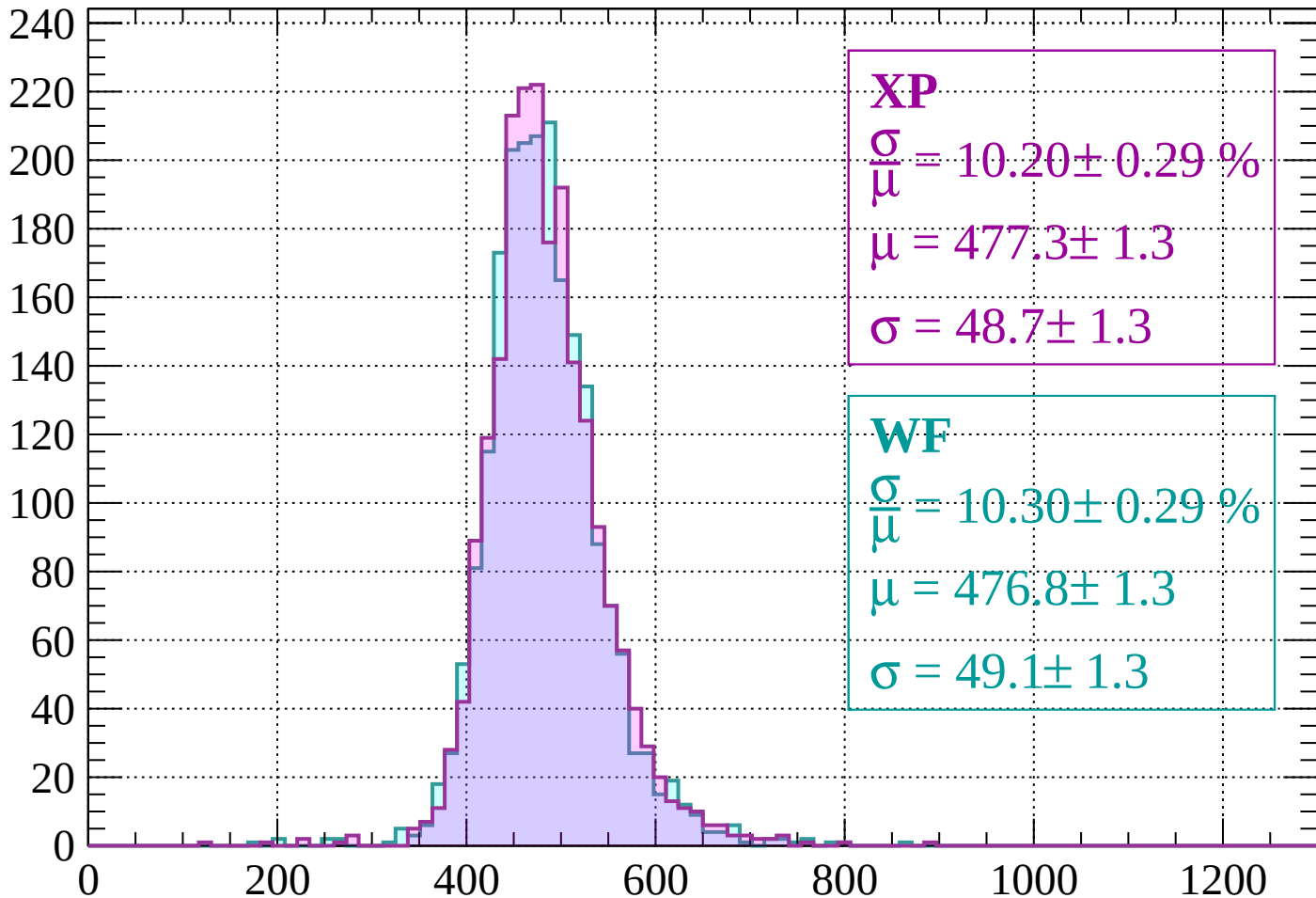
$$\mu = 473.1 \pm 1.3$$

$$\sigma = 48.9 \pm 1.3$$

dE/dx (ADC counts/cm)

Energy loss | $1760 < p < 1800$

Count



XP

$$\frac{\sigma}{\mu} = 10.20 \pm 0.29 \%$$

$$\mu = 477.3 \pm 1.3$$

$$\sigma = 48.7 \pm 1.3$$

WF

$$\frac{\sigma}{\mu} = 10.30 \pm 0.29 \%$$

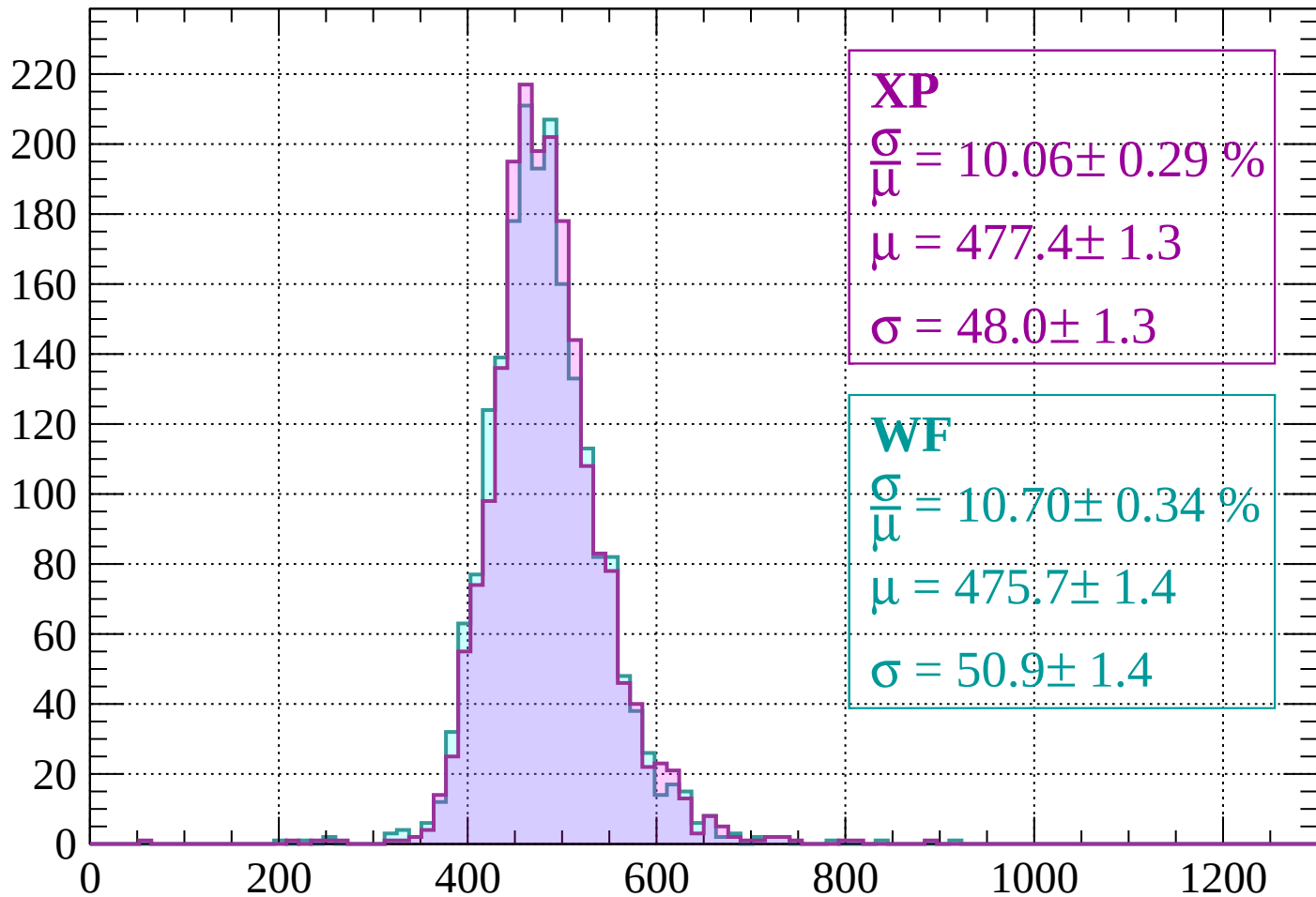
$$\mu = 476.8 \pm 1.3$$

$$\sigma = 49.1 \pm 1.3$$

dE/dx (ADC counts/cm)

Energy loss | $1800 < p < 1840$

Count



XP

$$\frac{\sigma}{\mu} = 10.06 \pm 0.29 \%$$

$$\mu = 477.4 \pm 1.3$$

$$\sigma = 48.0 \pm 1.3$$

WF

$$\frac{\sigma}{\mu} = 10.70 \pm 0.34 \%$$

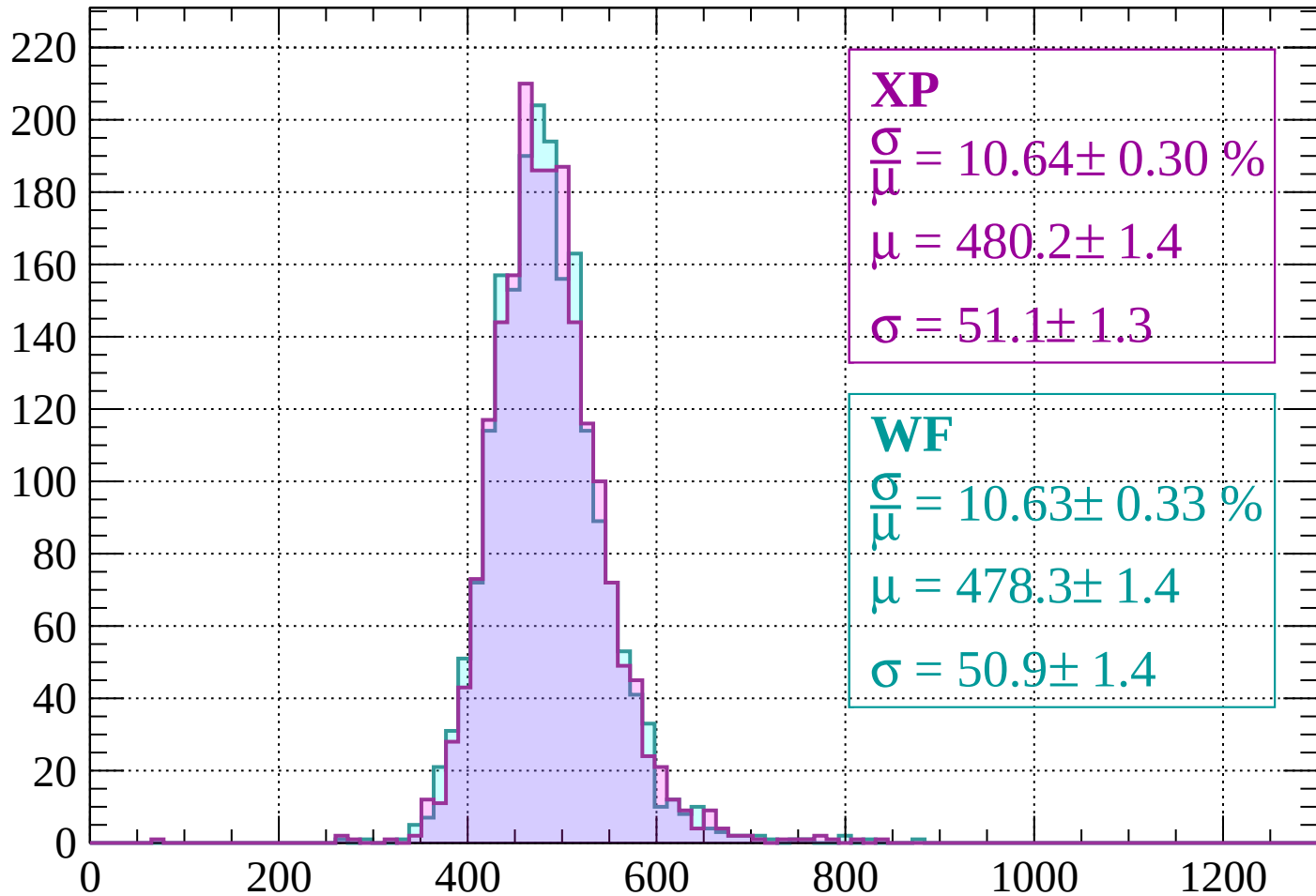
$$\mu = 475.7 \pm 1.4$$

$$\sigma = 50.9 \pm 1.4$$

dE/dx (ADC counts/cm)

Energy loss | $1840 < p < 1880$

Count



XP

$$\frac{\sigma}{\mu} = 10.64 \pm 0.30 \%$$

$$\mu = 480.2 \pm 1.4$$

$$\sigma = 51.1 \pm 1.3$$

WF

$$\frac{\sigma}{\mu} = 10.63 \pm 0.33 \%$$

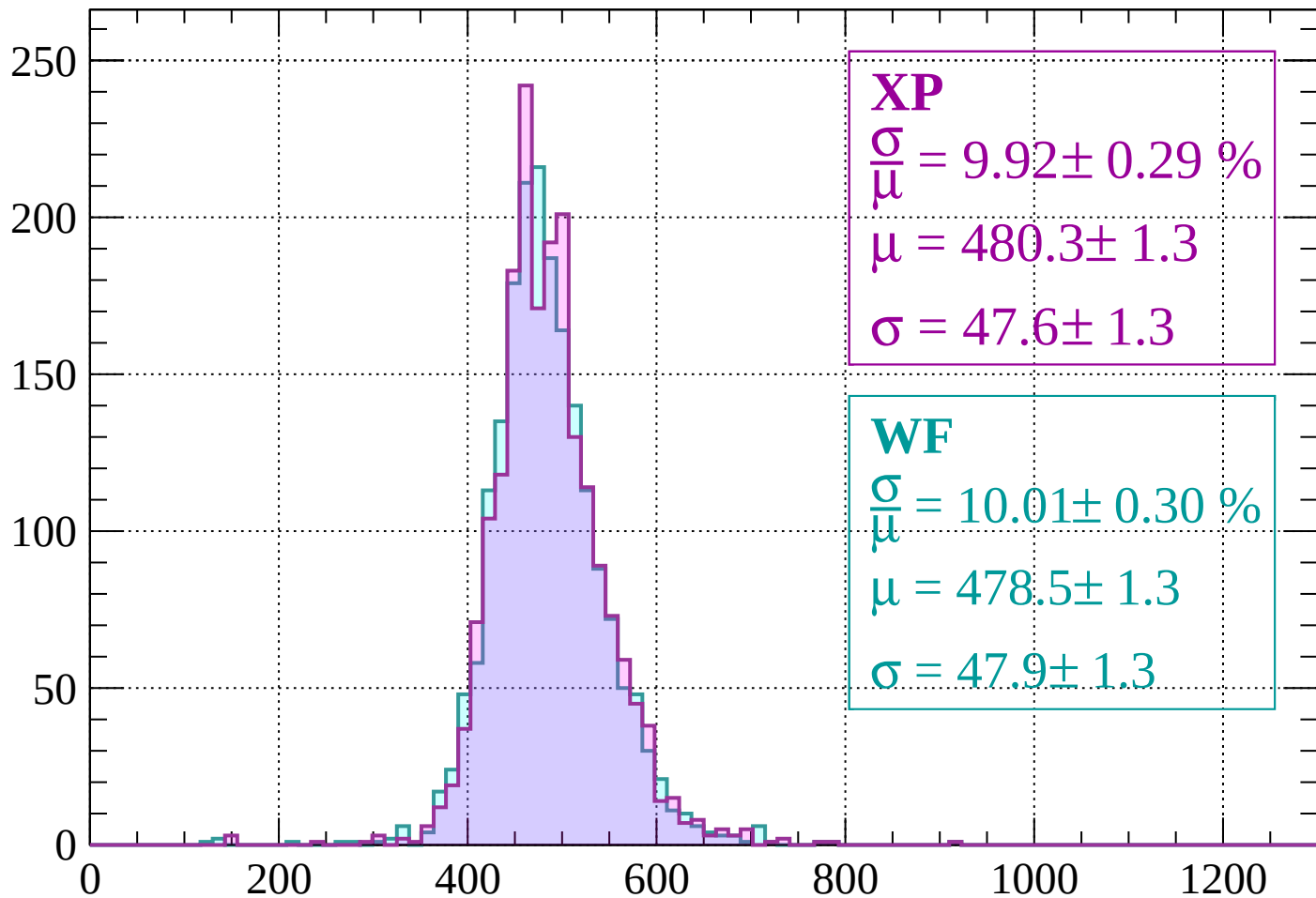
$$\mu = 478.3 \pm 1.4$$

$$\sigma = 50.9 \pm 1.4$$

dE/dx (ADC counts/cm)

Energy loss | $1880 < p < 1920$

Count



XP

$$\frac{\sigma}{\mu} = 9.92 \pm 0.29 \%$$

$$\mu = 480.3 \pm 1.3$$

$$\sigma = 47.6 \pm 1.3$$

WF

$$\frac{\sigma}{\mu} = 10.01 \pm 0.30 \%$$

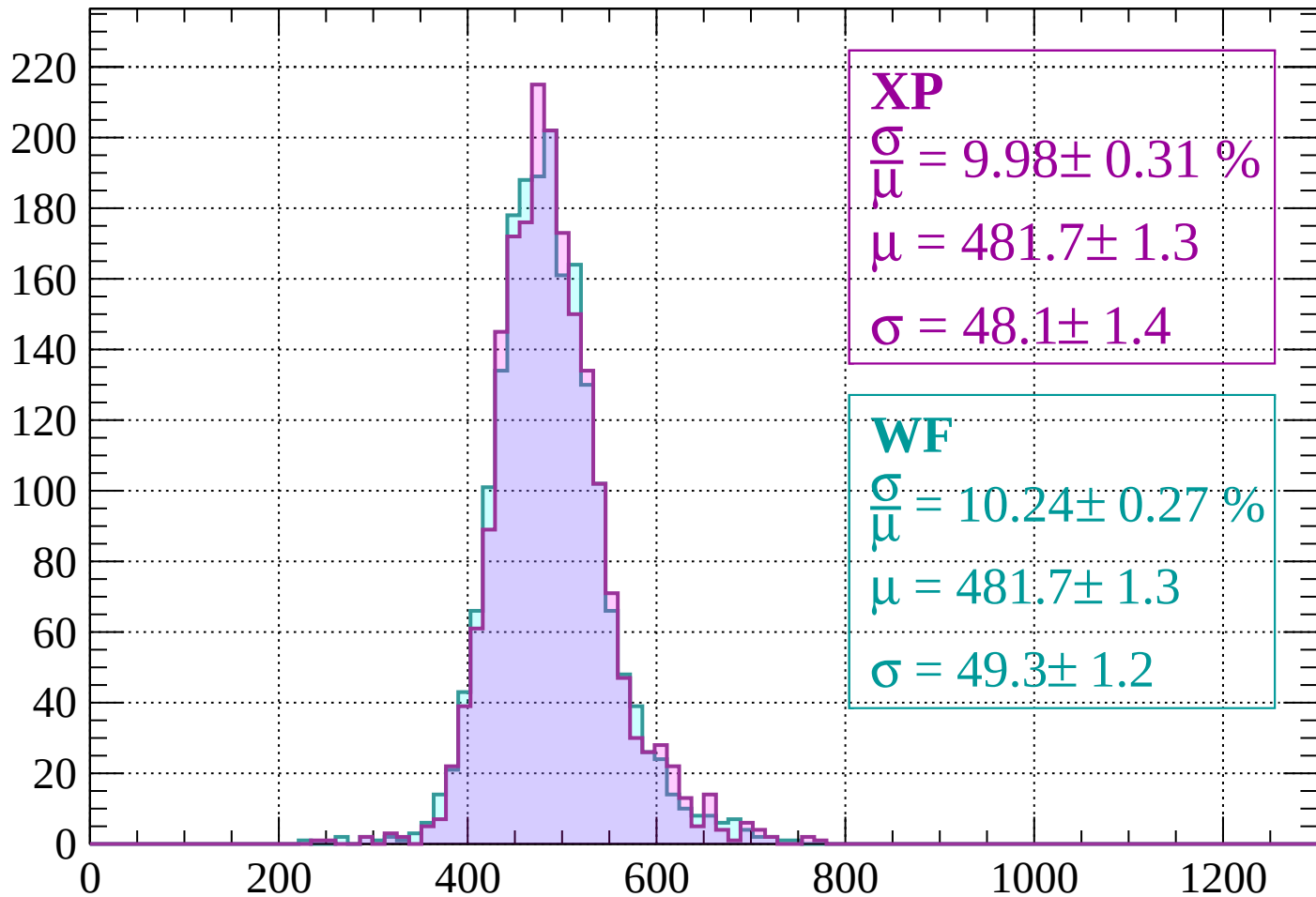
$$\mu = 478.5 \pm 1.3$$

$$\sigma = 47.9 \pm 1.3$$

dE/dx (ADC counts/cm)

Energy loss | $1920 < p < 1960$

Count



XP

$$\frac{\sigma}{\mu} = 9.98 \pm 0.31 \%$$

$$\mu = 481.7 \pm 1.3$$

$$\sigma = 48.1 \pm 1.4$$

WF

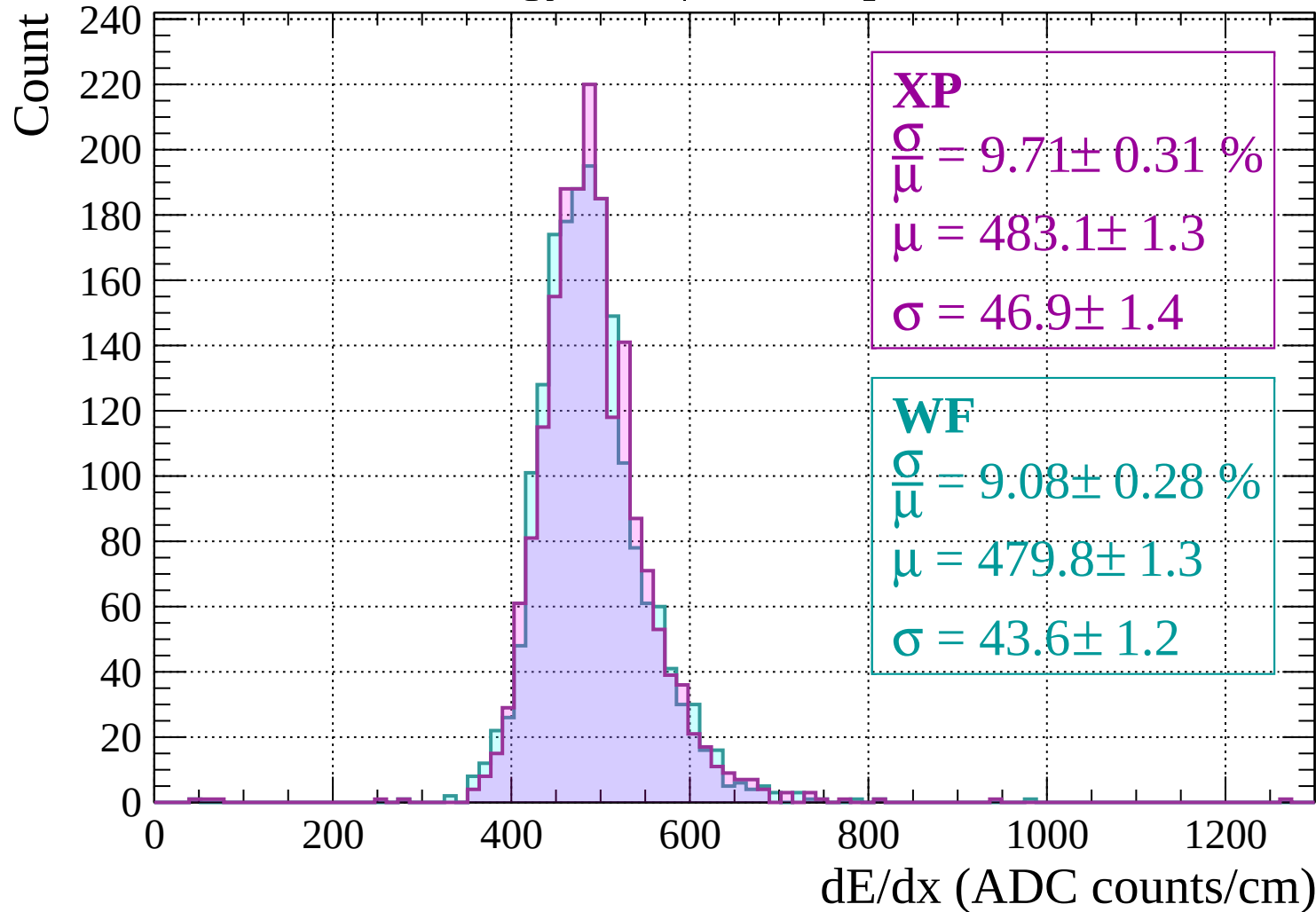
$$\frac{\sigma}{\mu} = 10.24 \pm 0.27 \%$$

$$\mu = 481.7 \pm 1.3$$

$$\sigma = 49.3 \pm 1.2$$

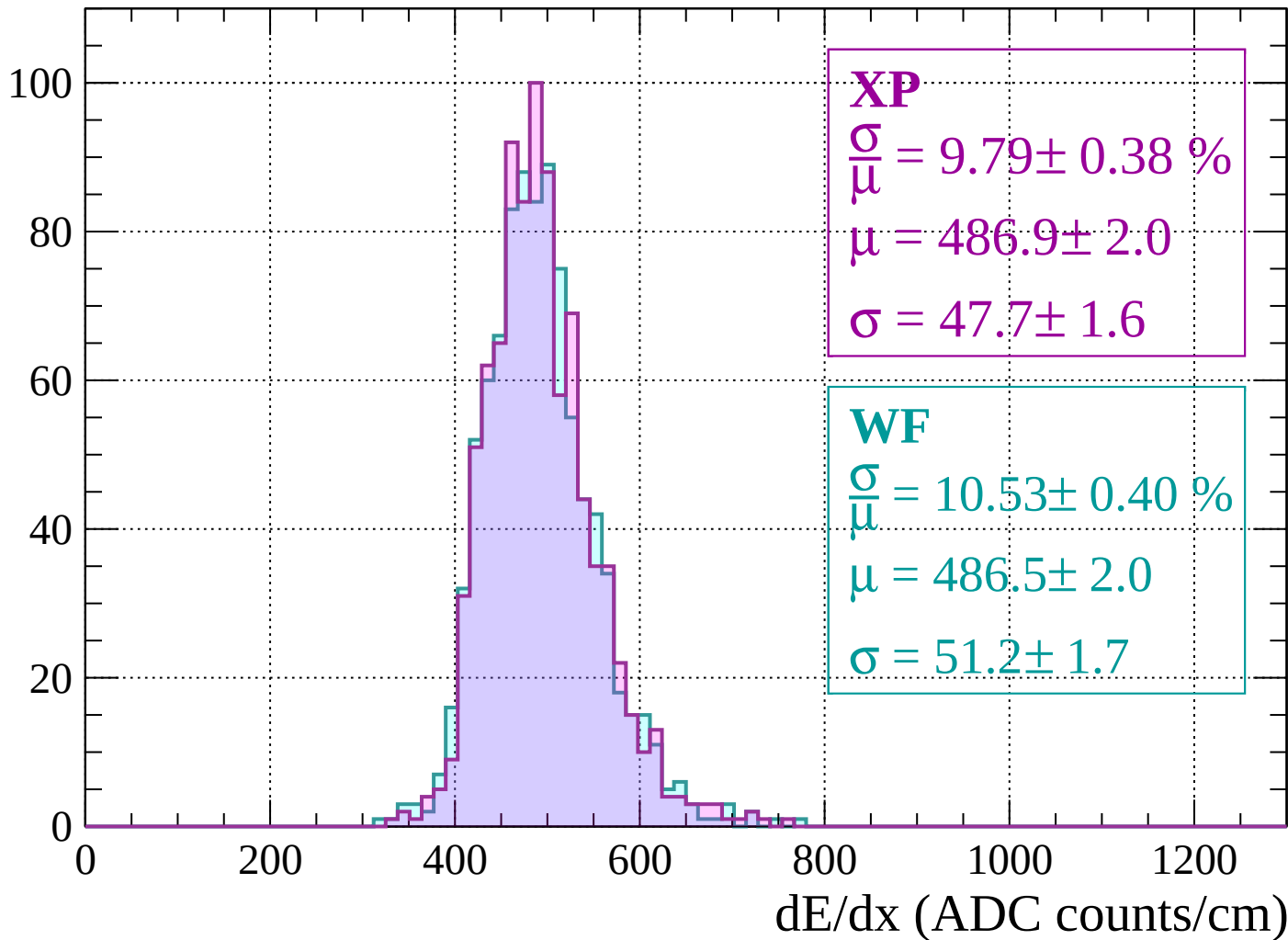
dE/dx (ADC counts/cm)

Energy loss | $1960 < p < 2000$



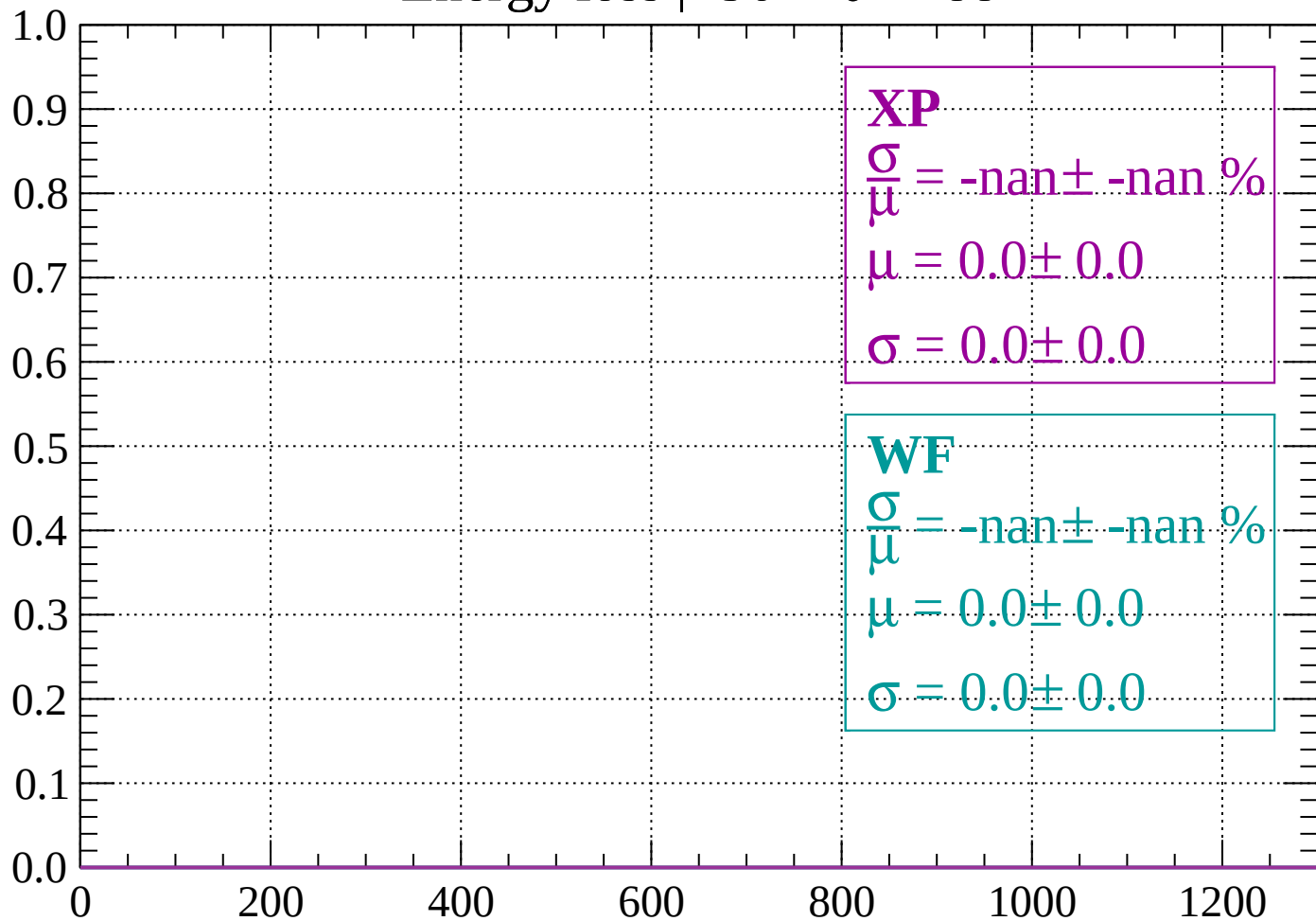
Energy loss | $2000 < p < 2040$

Count



Energy loss | $-90 < \theta < -88$

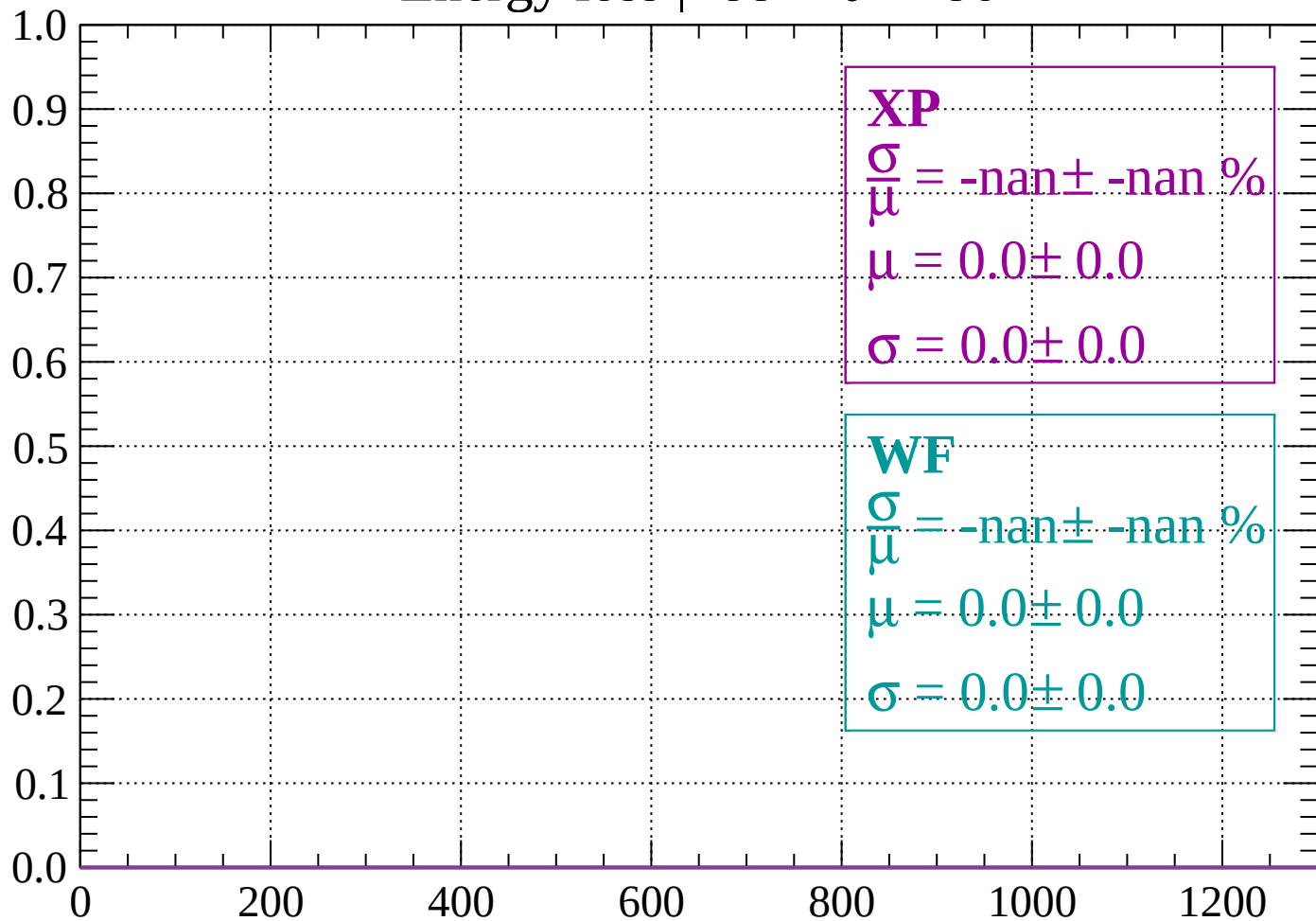
Count



dE/dx (ADC counts/cm)

Energy loss | $-88 < \theta < -86$

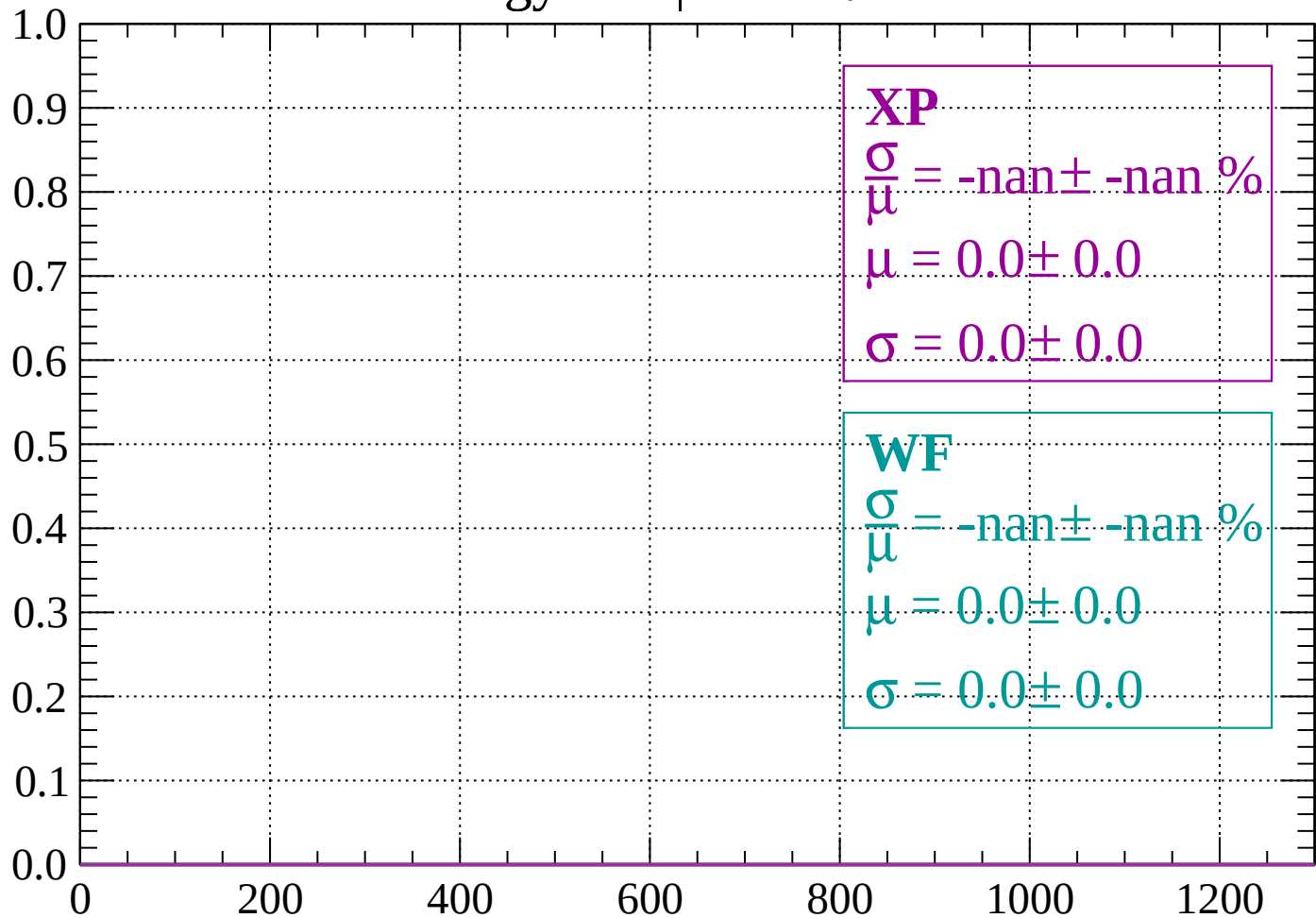
Count



dE/dx (ADC counts/cm)

Energy loss | $-86 < \theta < -84$

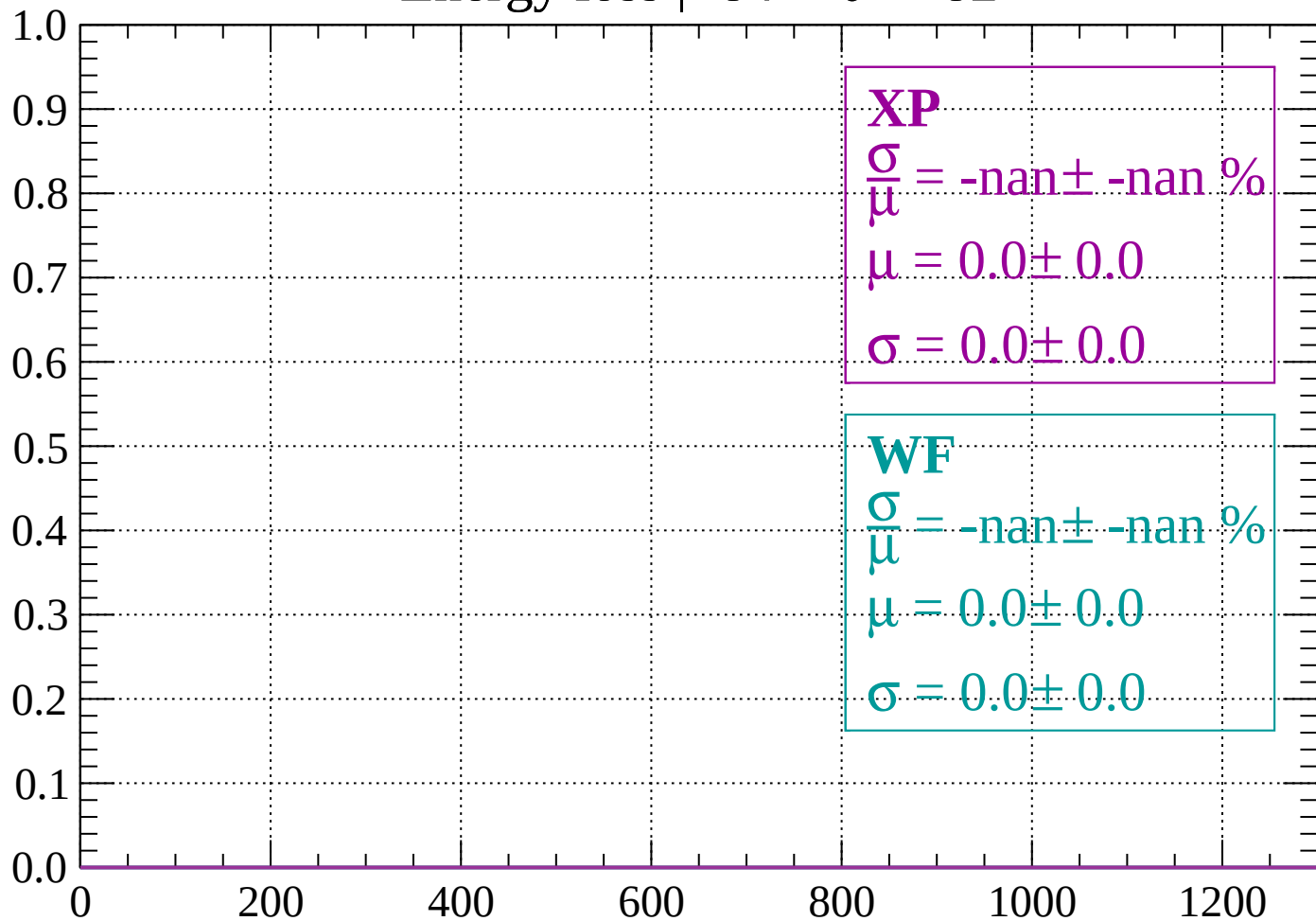
Count



dE/dx (ADC counts/cm)

Energy loss | $-84 < \theta < -82$

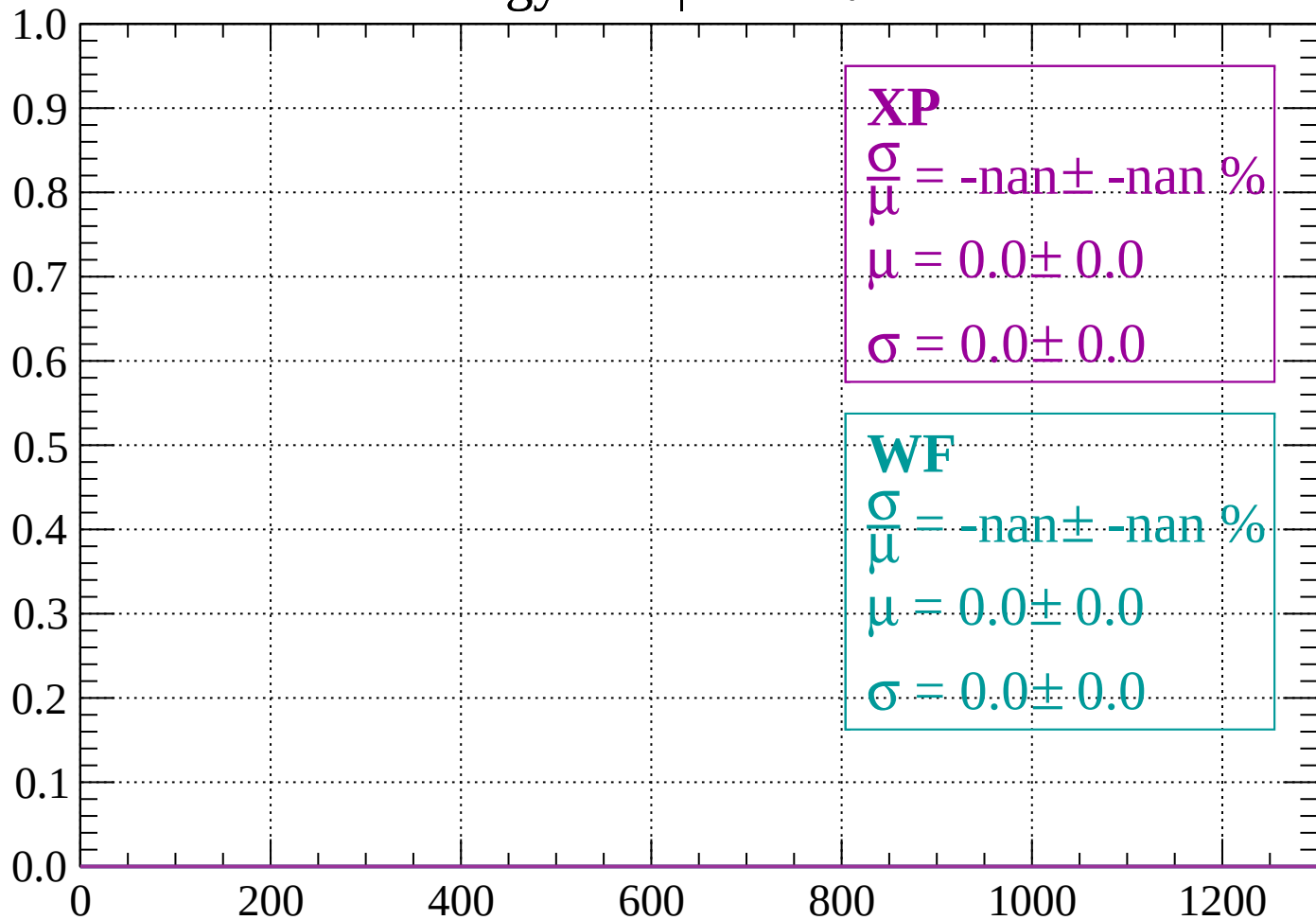
Count



dE/dx (ADC counts/cm)

Energy loss | $-82 < \theta < -80$

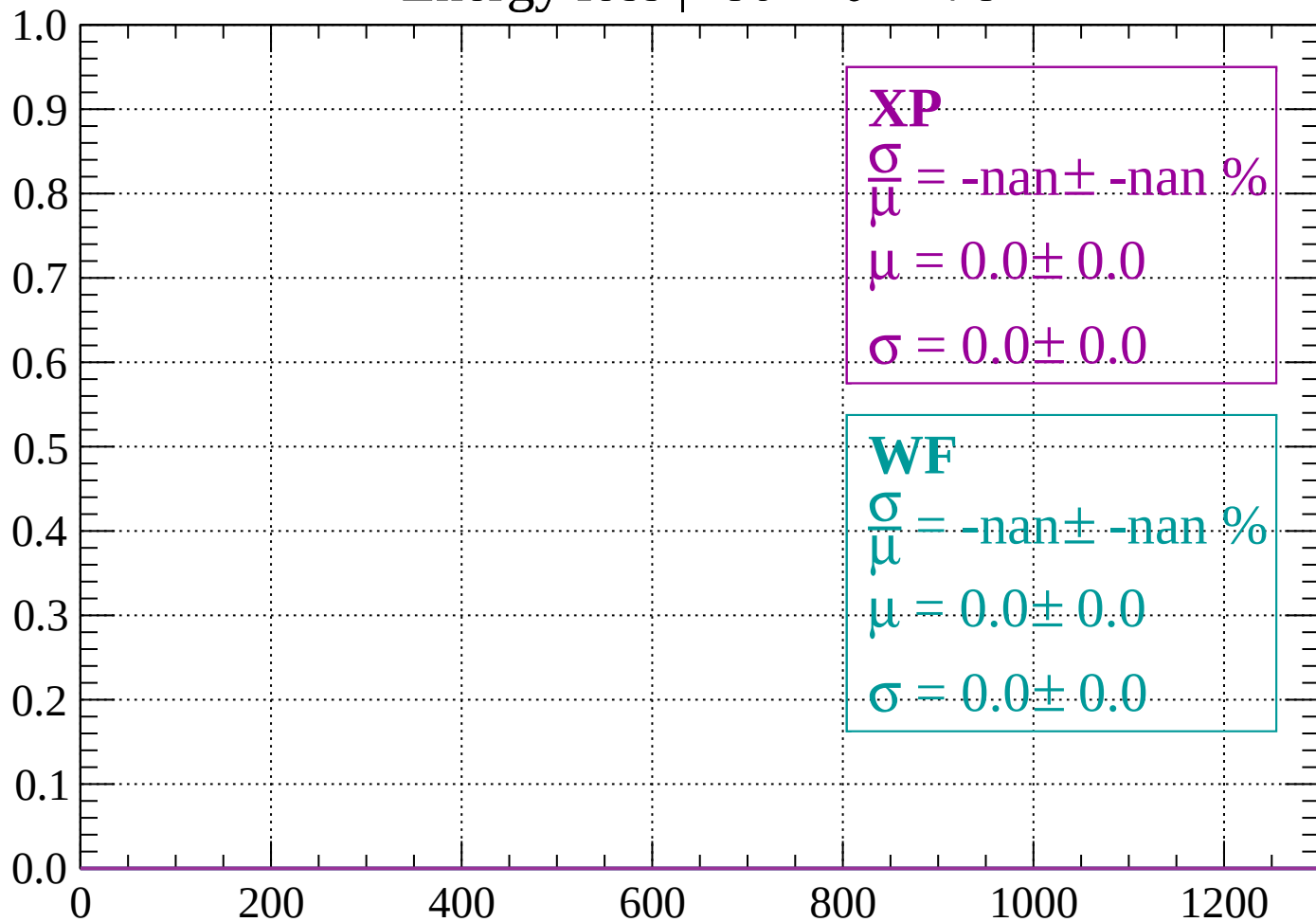
Count



dE/dx (ADC counts/cm)

Energy loss | $-80 < \theta < -78$

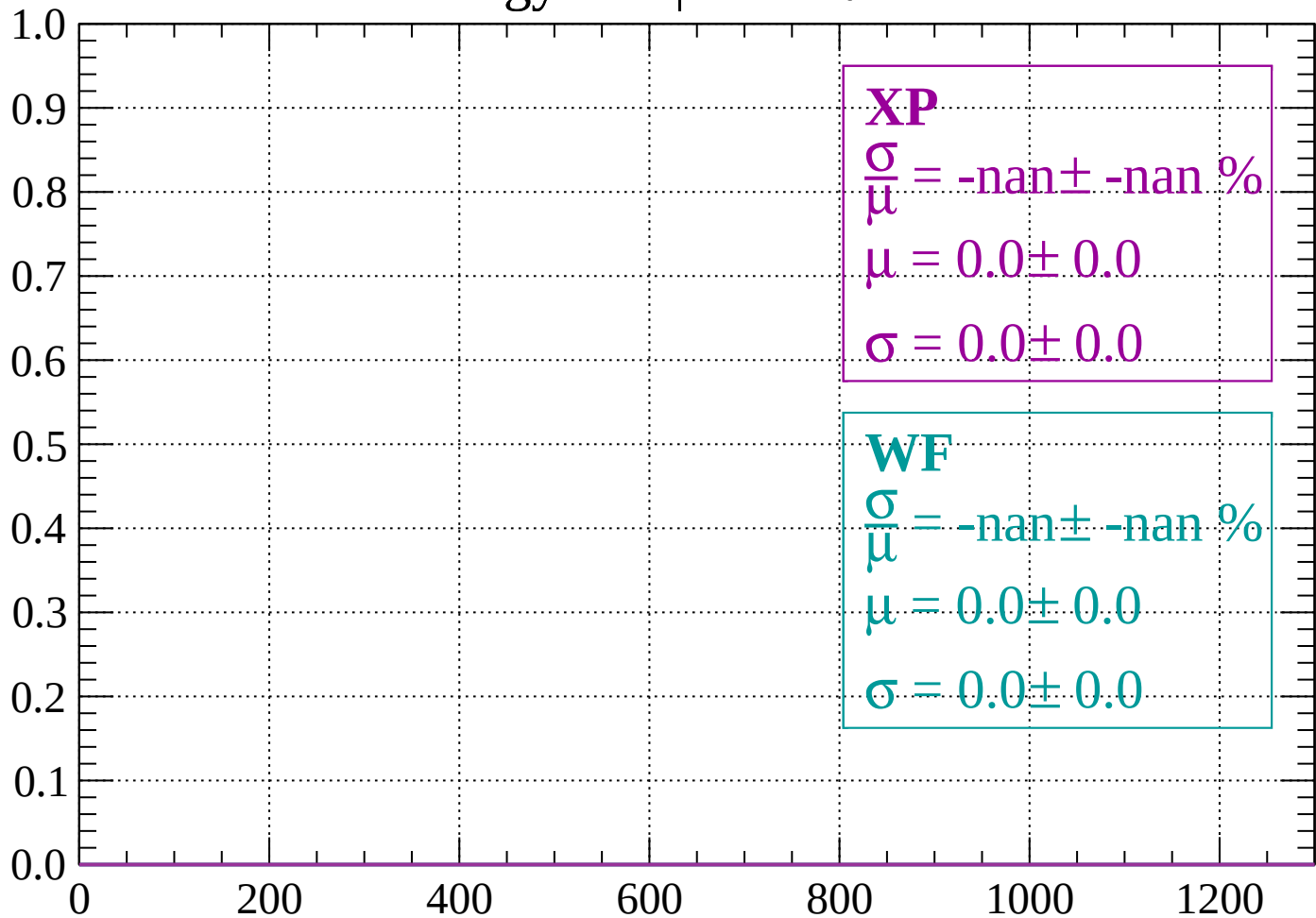
Count



dE/dx (ADC counts/cm)

Energy loss | $-78 < \theta < -76$

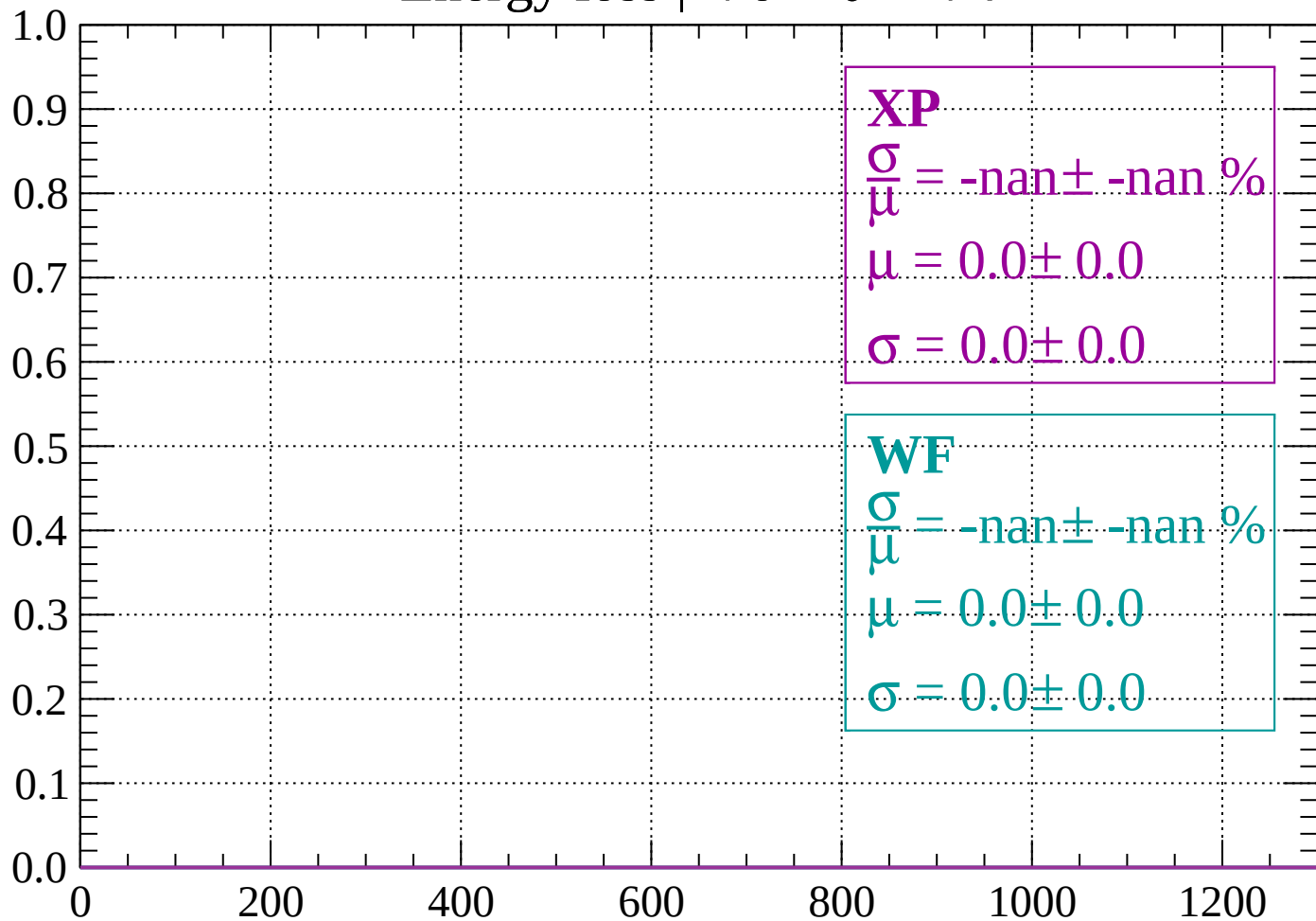
Count



dE/dx (ADC counts/cm)

Energy loss | $-76 < \theta < -74$

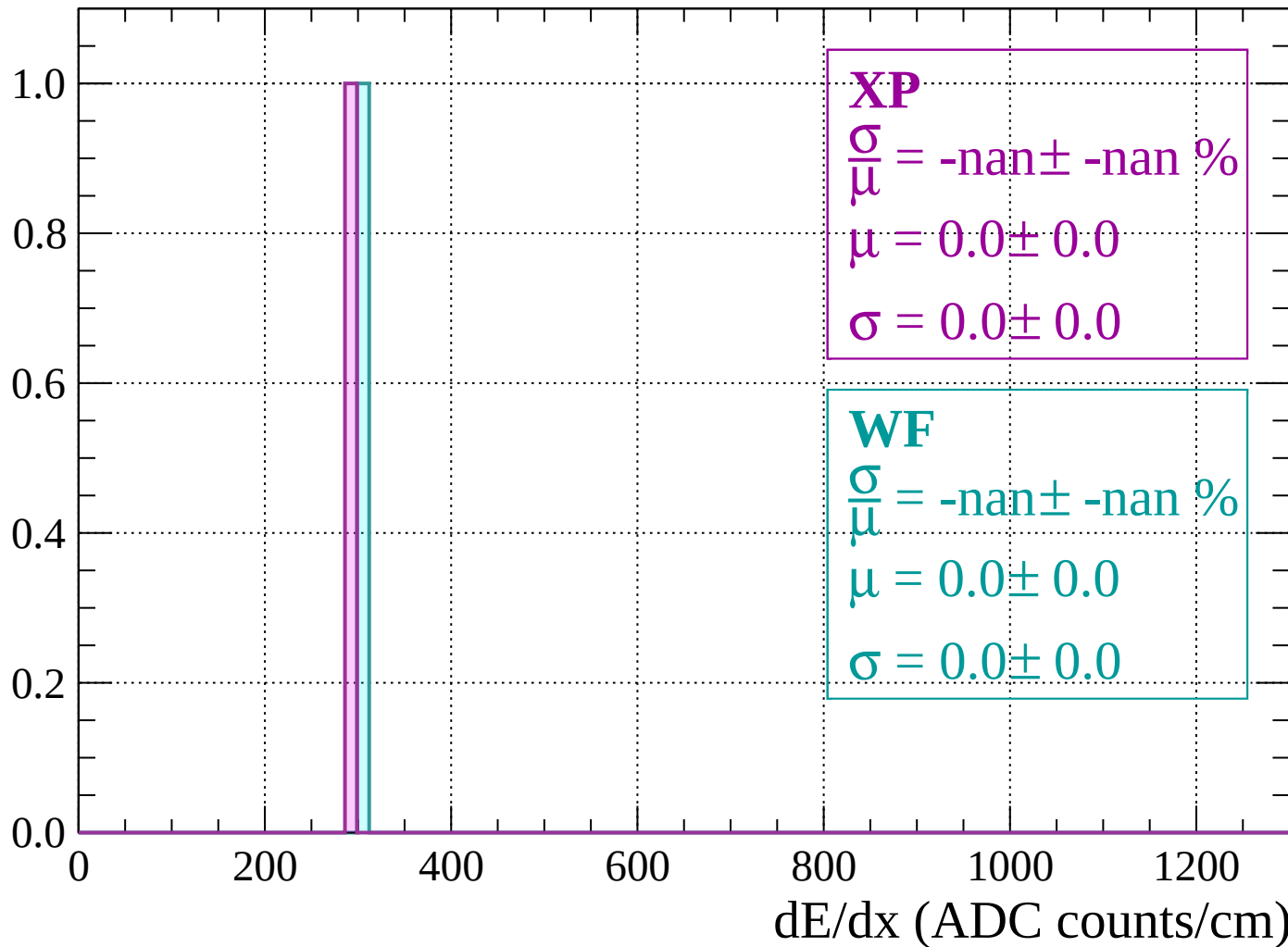
Count



dE/dx (ADC counts/cm)

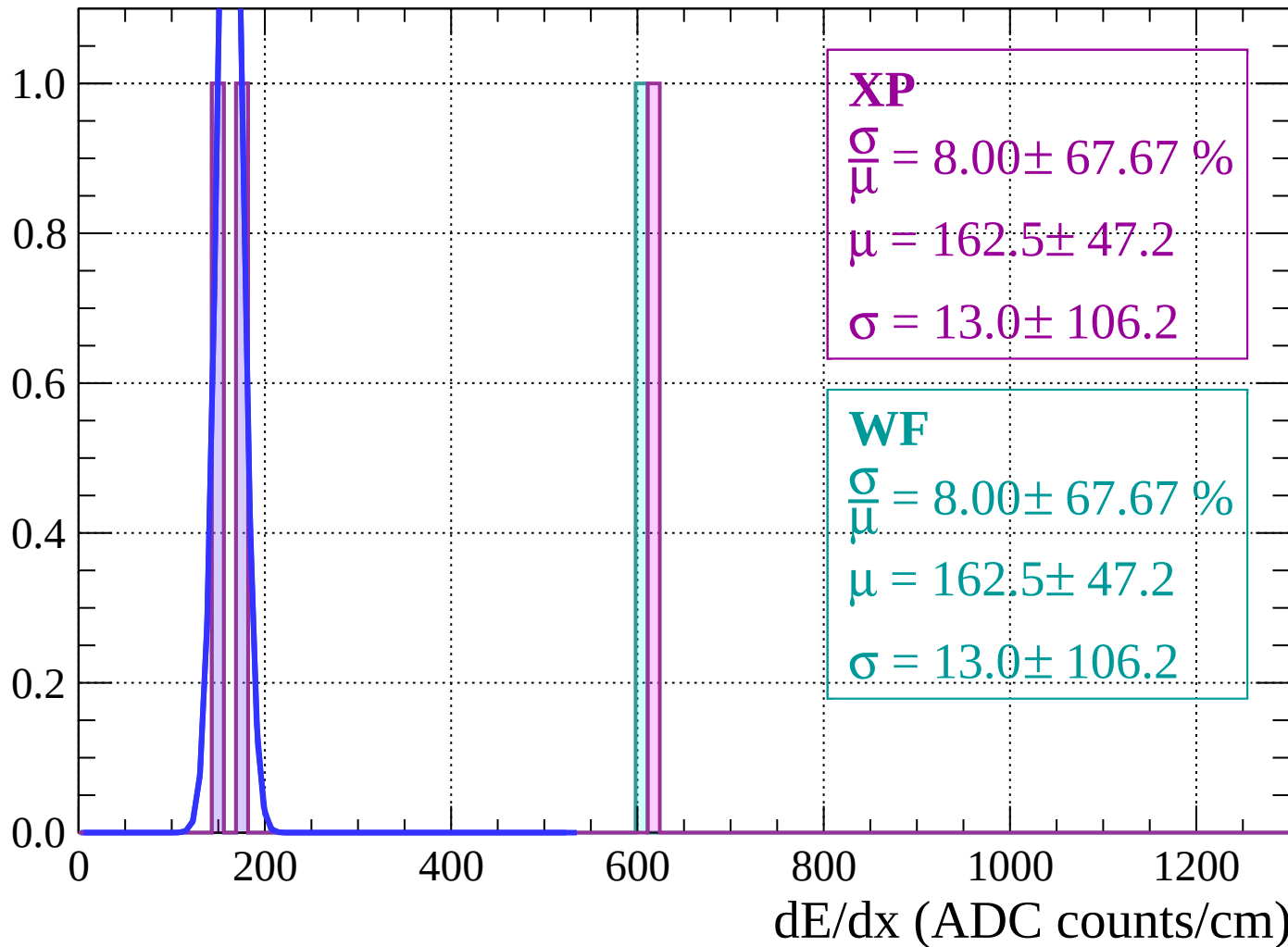
Energy loss | $-74 < \theta < -72$

Count



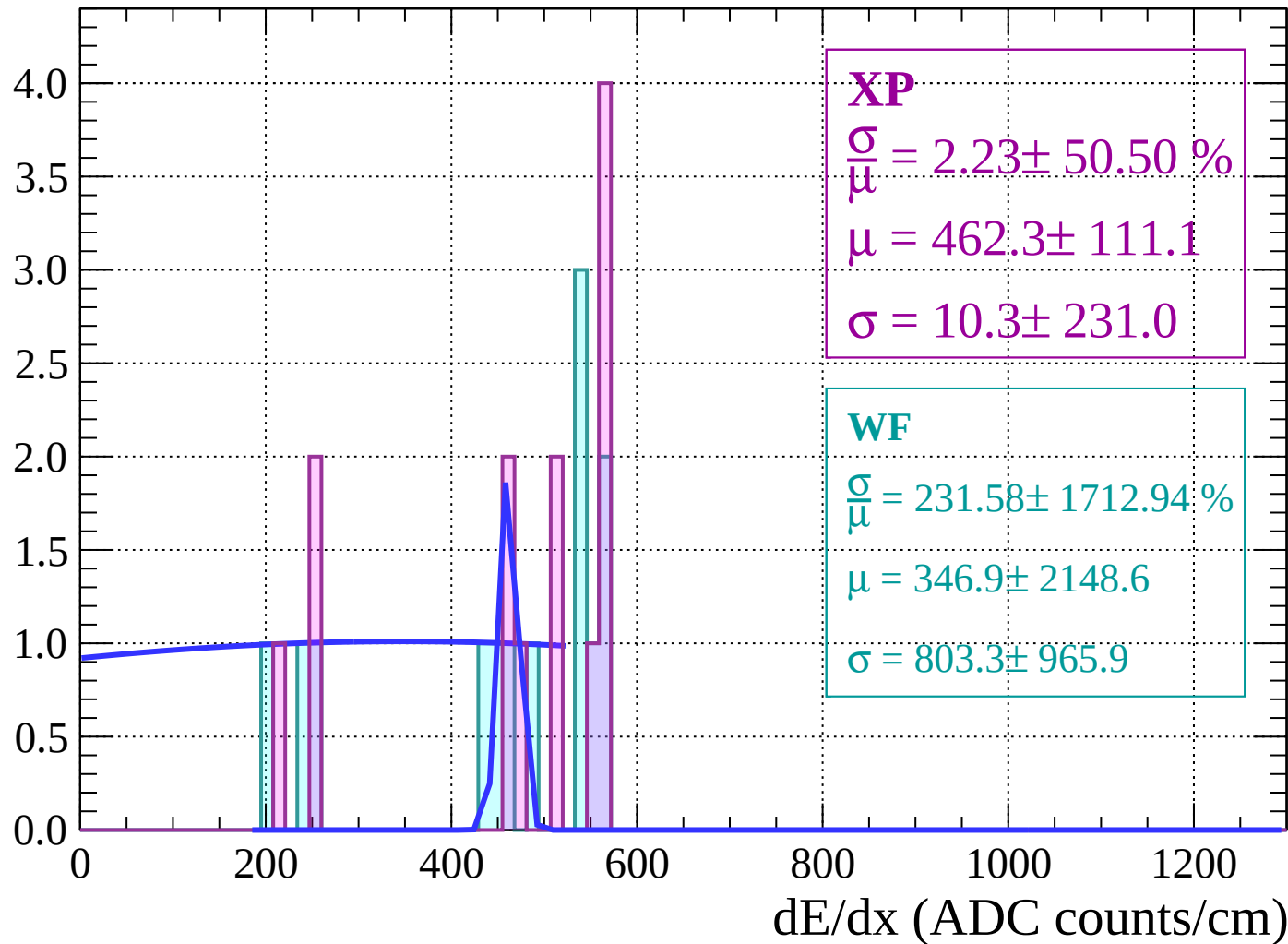
Energy loss $|-72 < \theta < -70$

Count



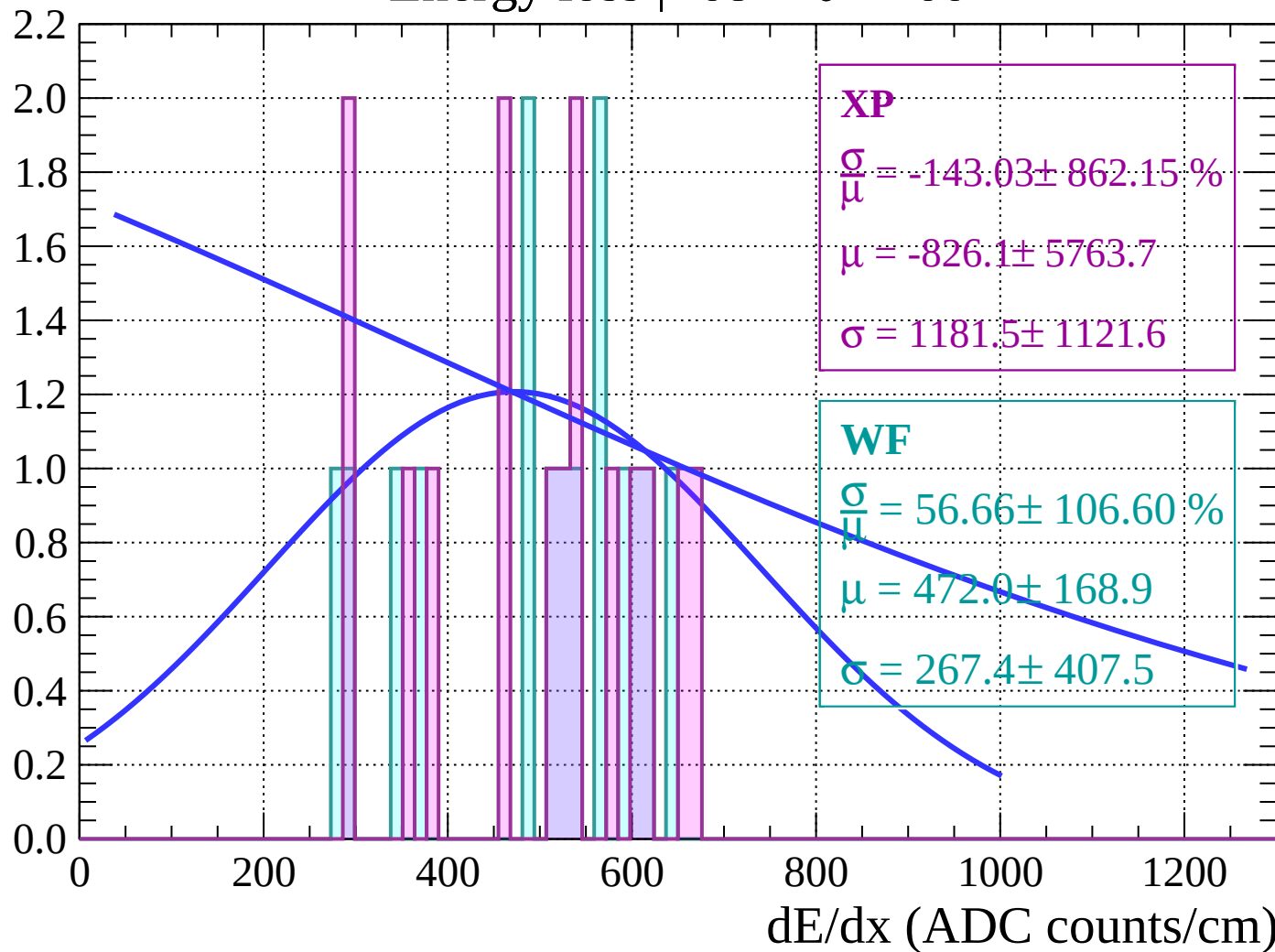
Energy loss | $-70 < \theta < -68$

Count



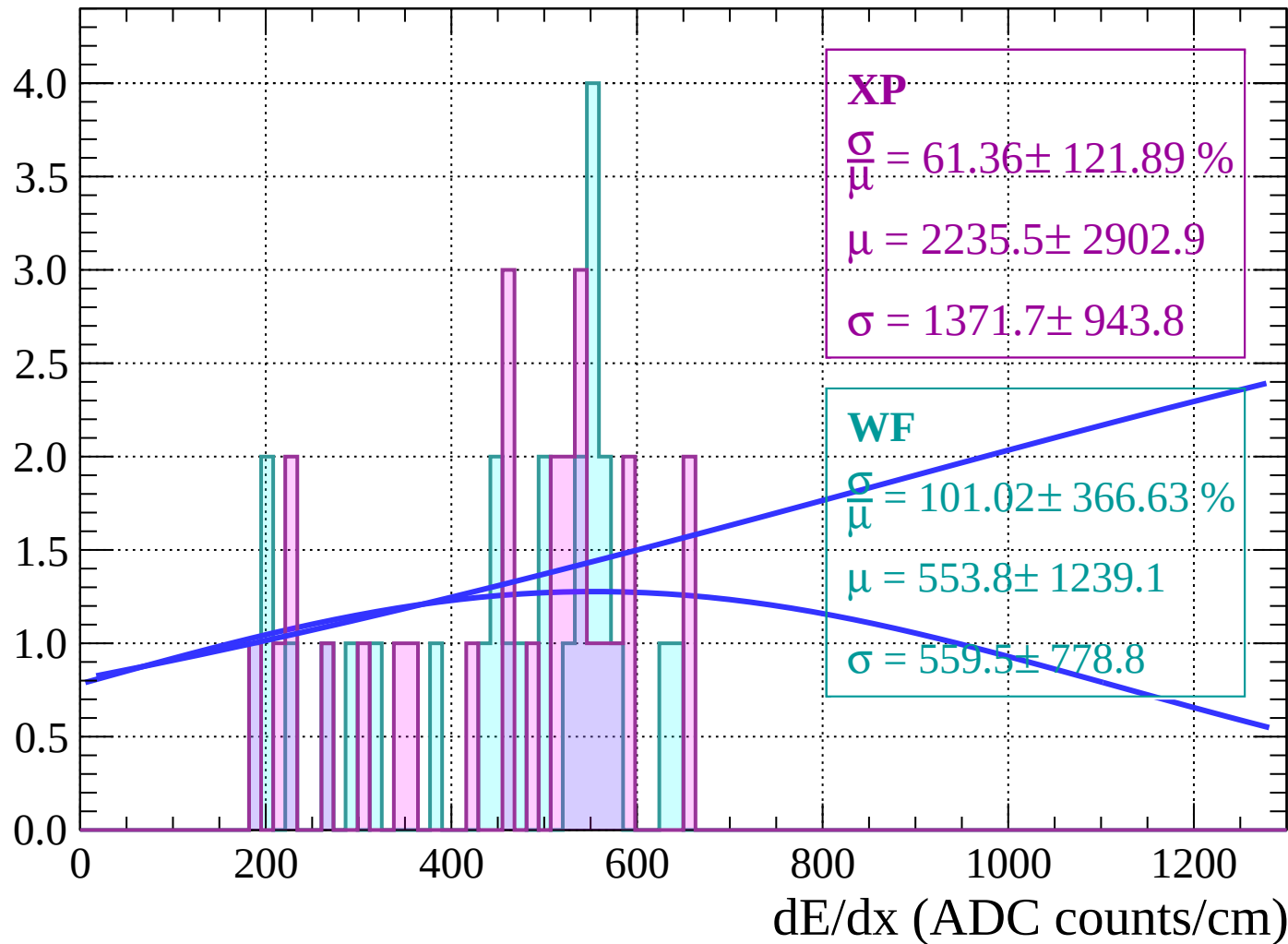
Energy loss | $-68 < \theta < -66$

Count



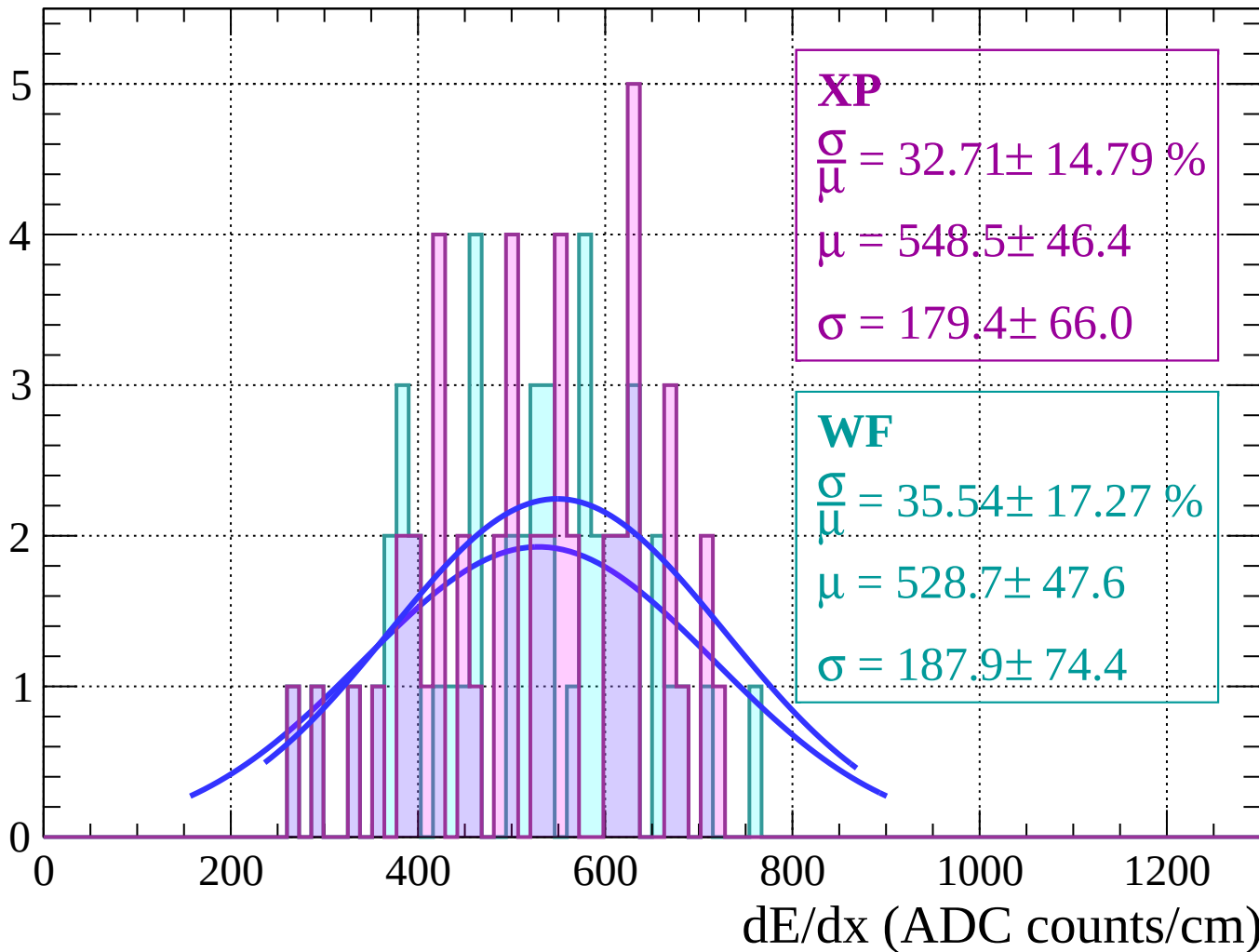
Energy loss | $-66 < \theta < -64$

Count



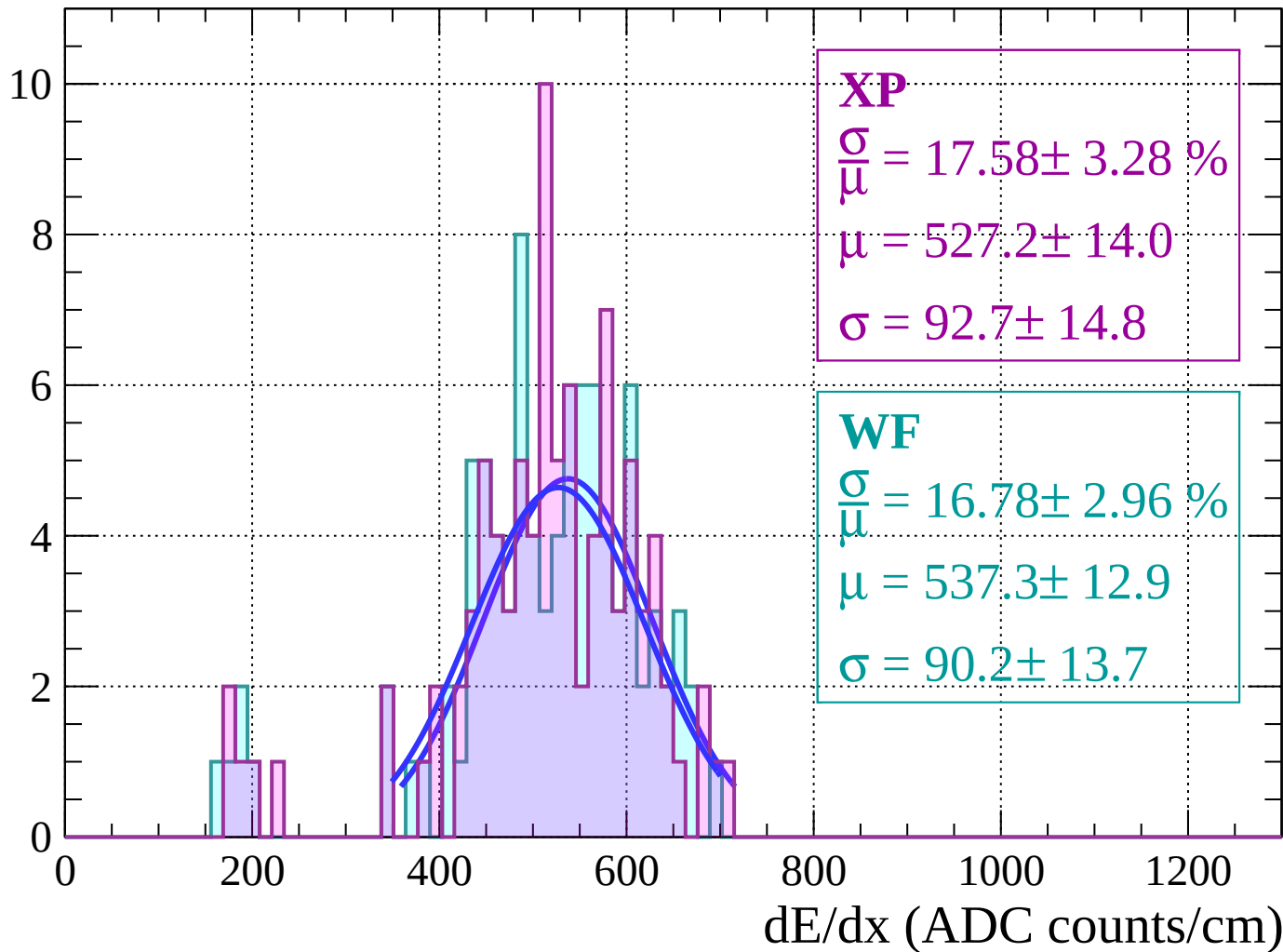
Energy loss | $-64 < \theta < -62$

Count



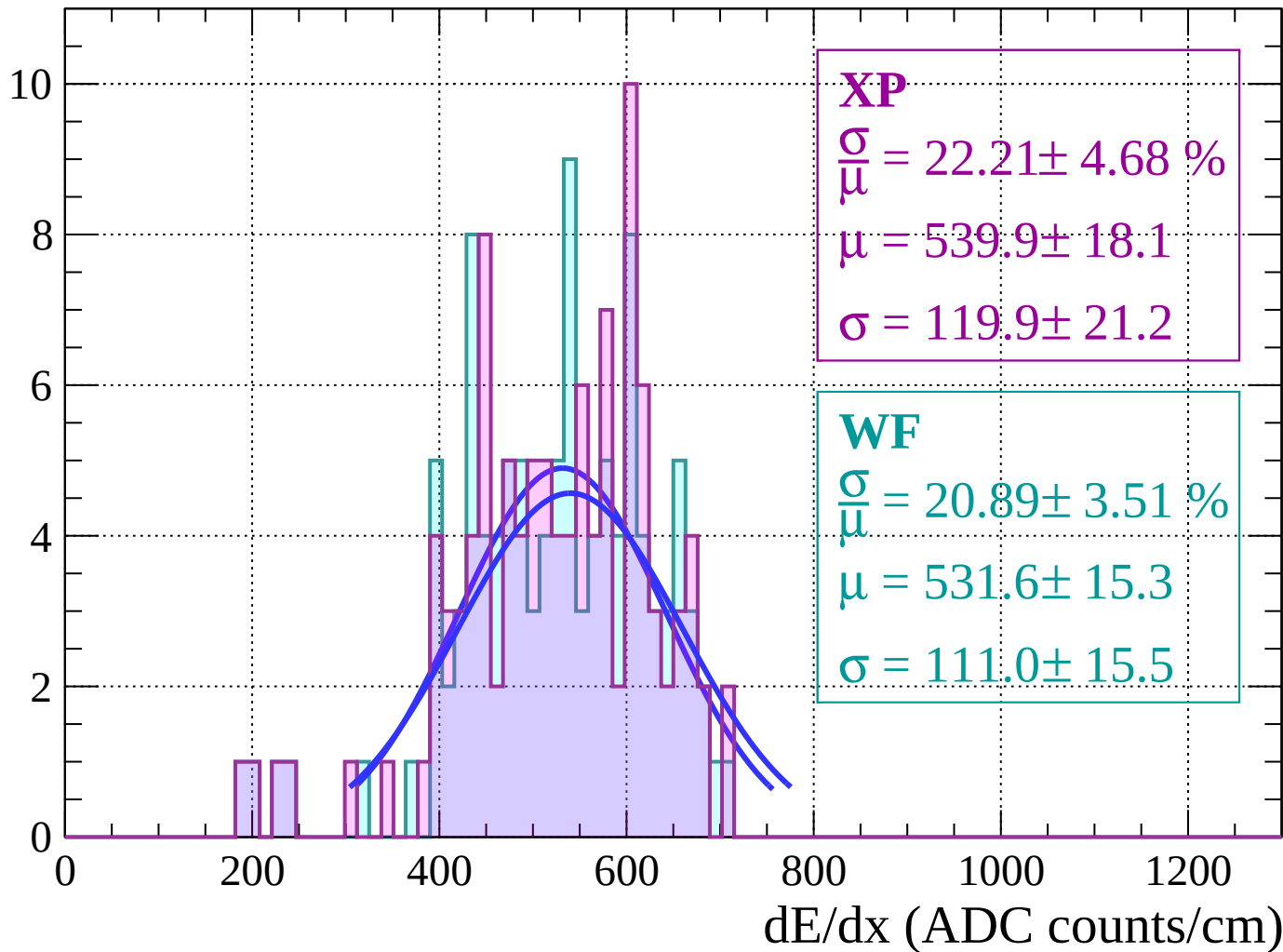
Energy loss | $-62 < \theta < -60$

Count



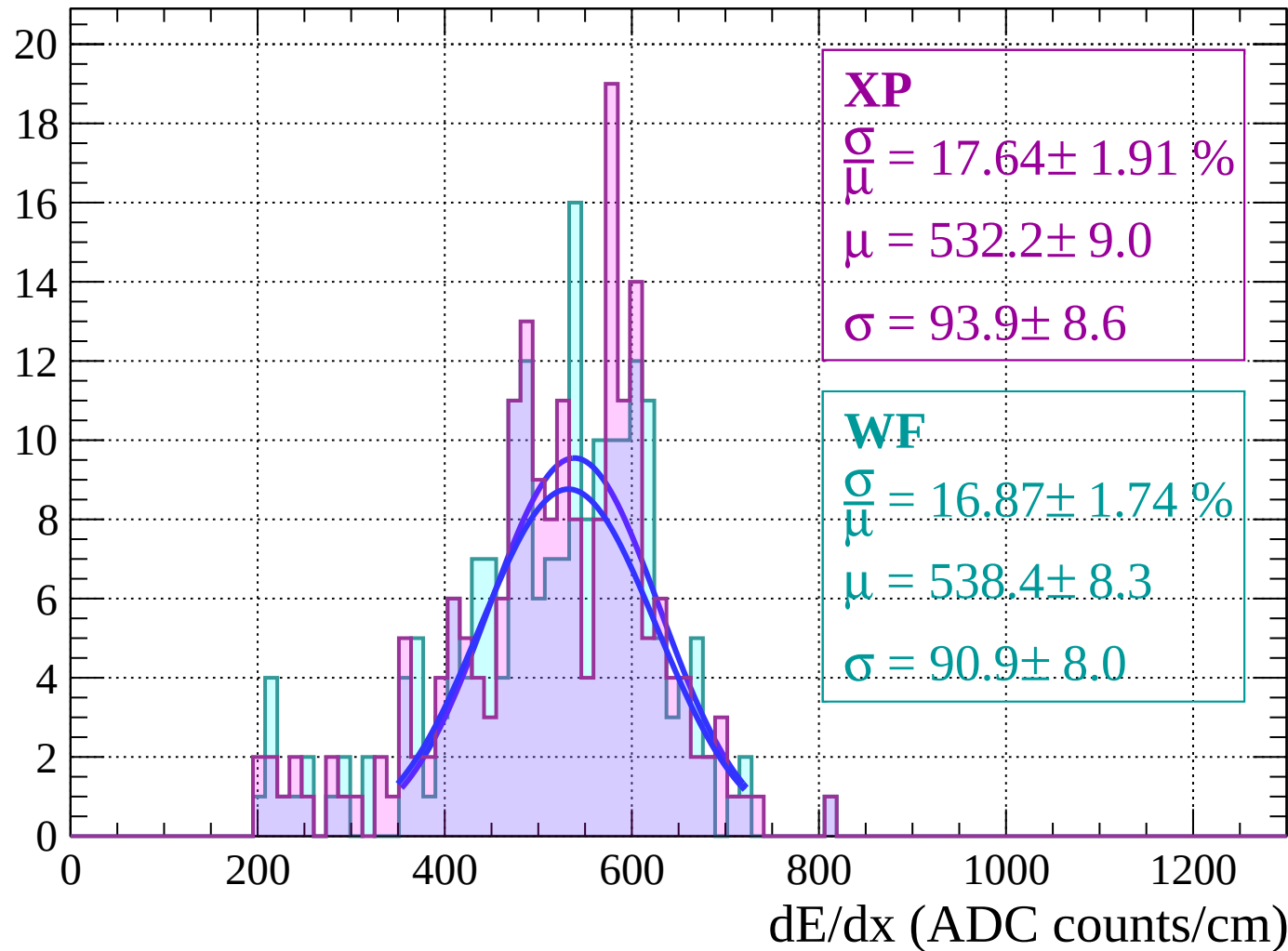
Energy loss | $-60 < \theta < -58$

Count



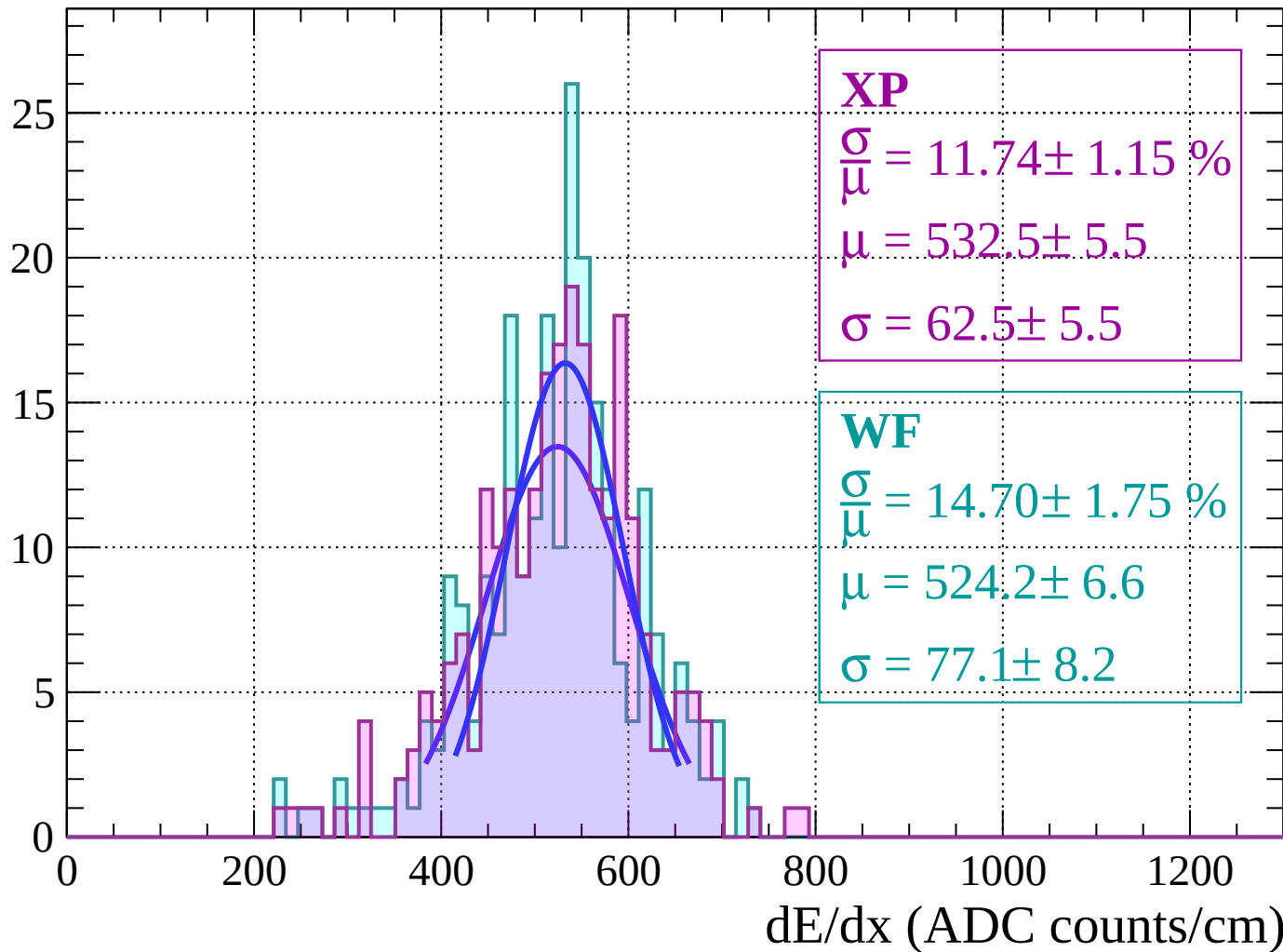
Energy loss | $-58 < \theta < -56$

Count



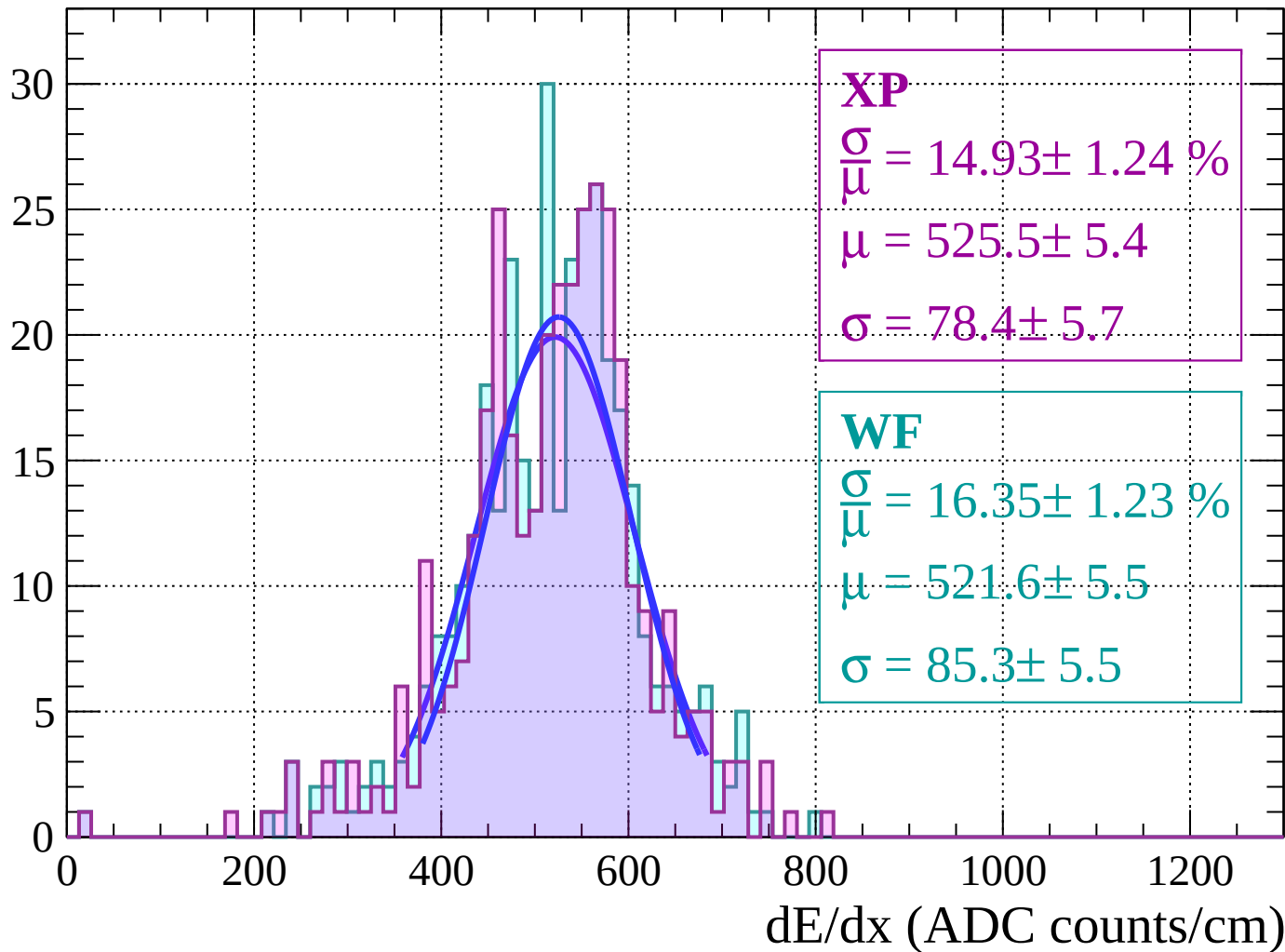
Energy loss | $-56 < \theta < -54$

Count



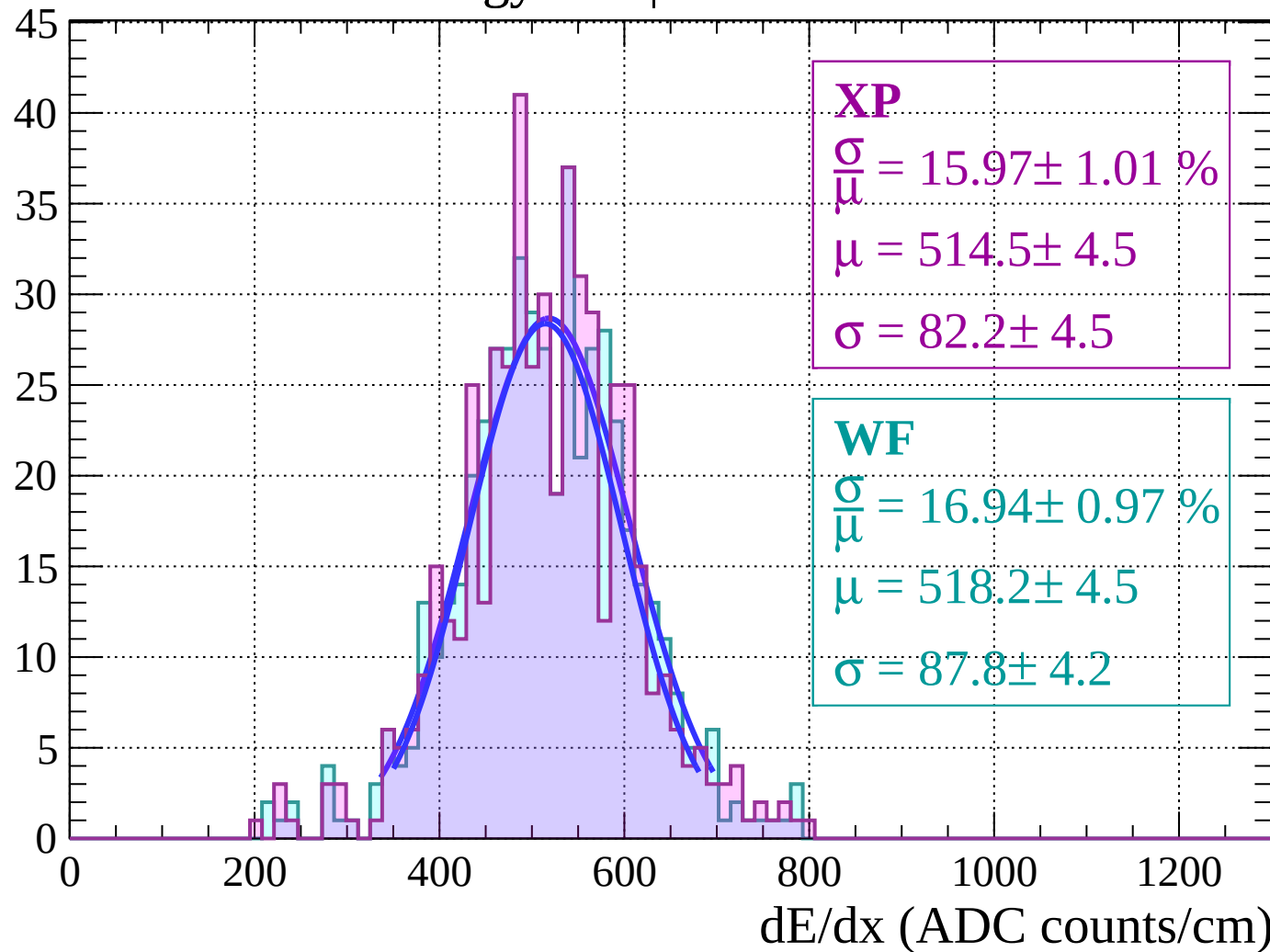
Energy loss | $-54 < \theta < -52$

Count



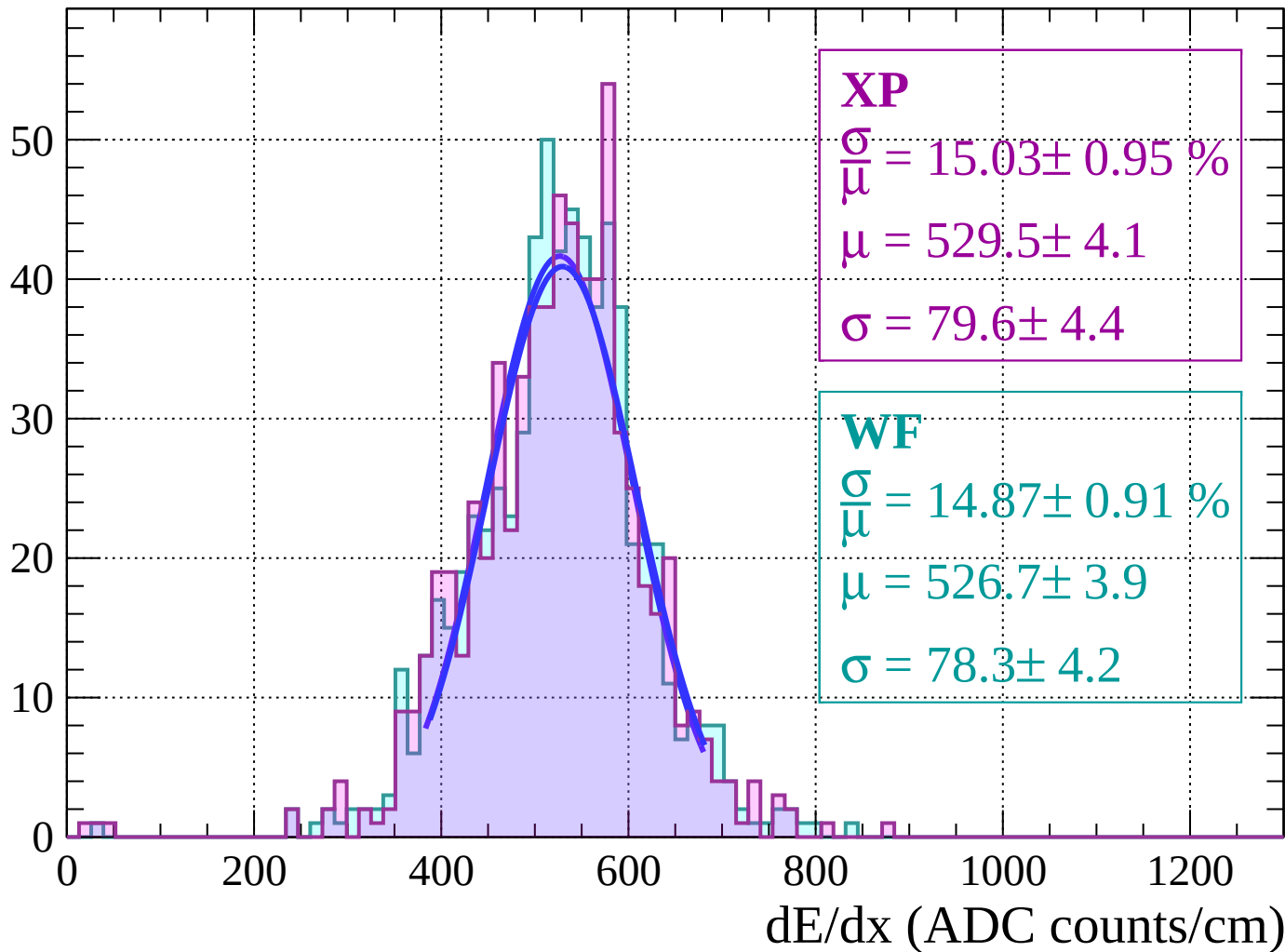
Energy loss | $-52 < \theta < -50$

Count



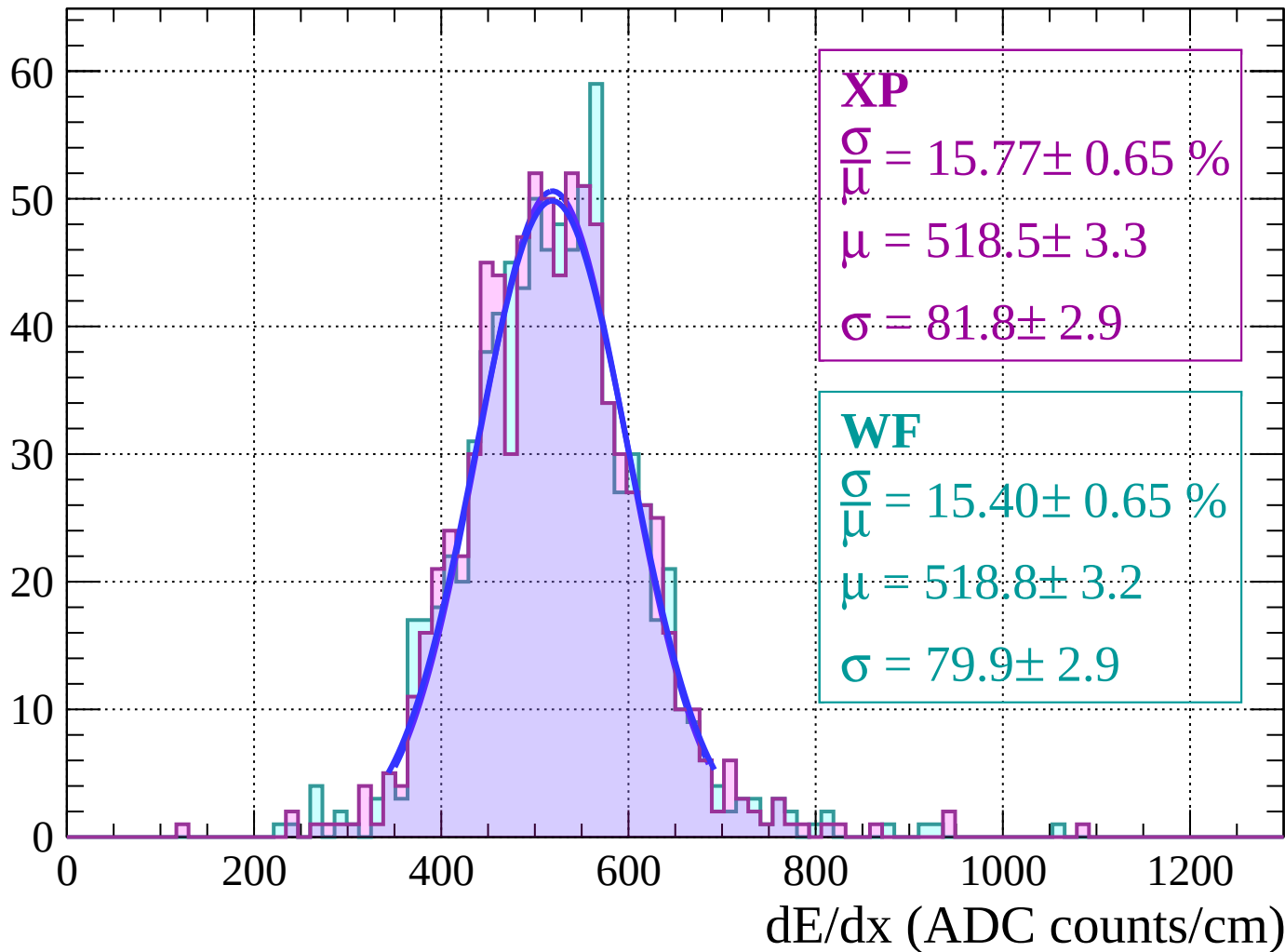
Energy loss $|-50 < \theta < -48$

Count



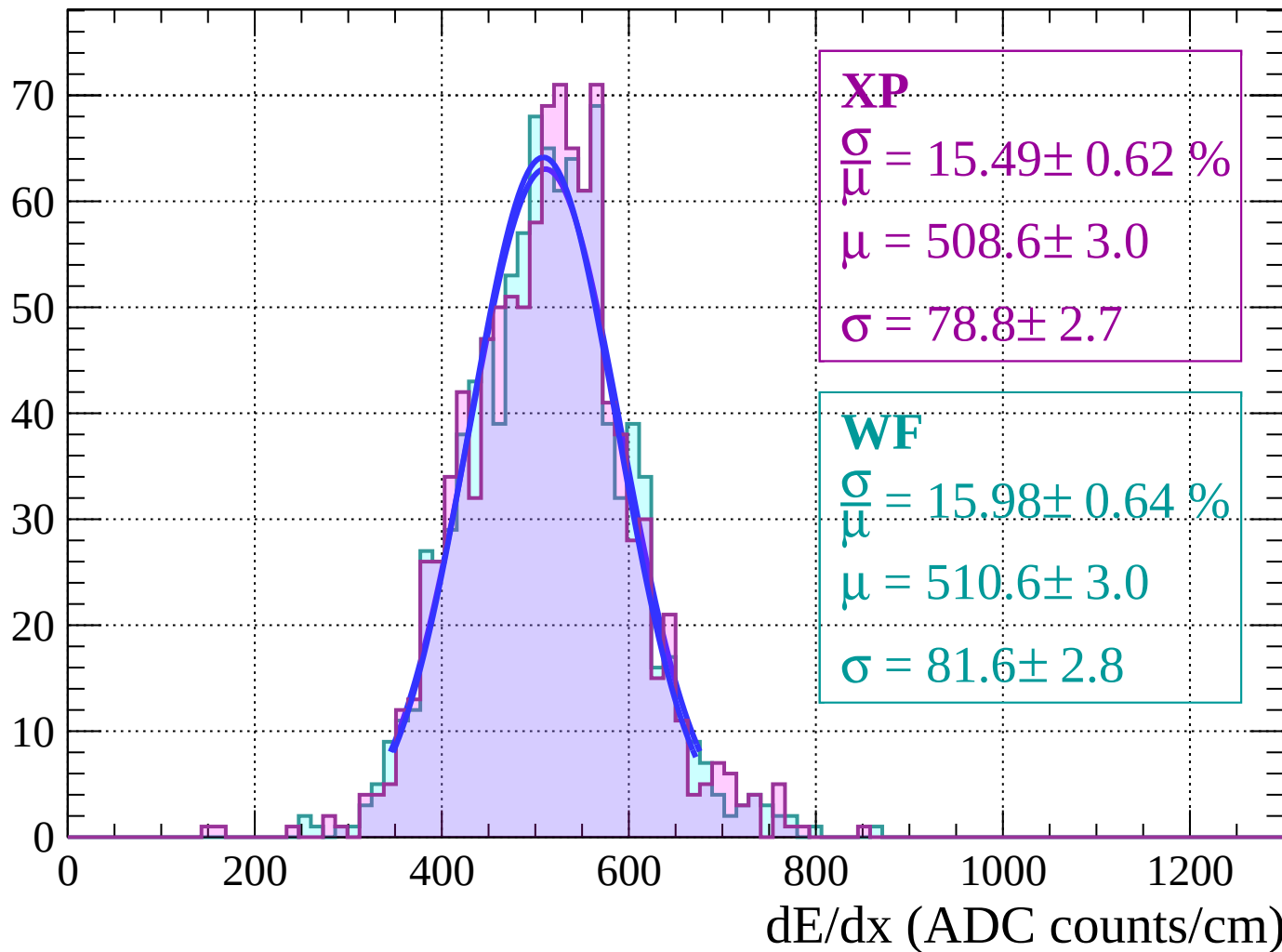
Energy loss $|-48 < \theta < -46$

Count



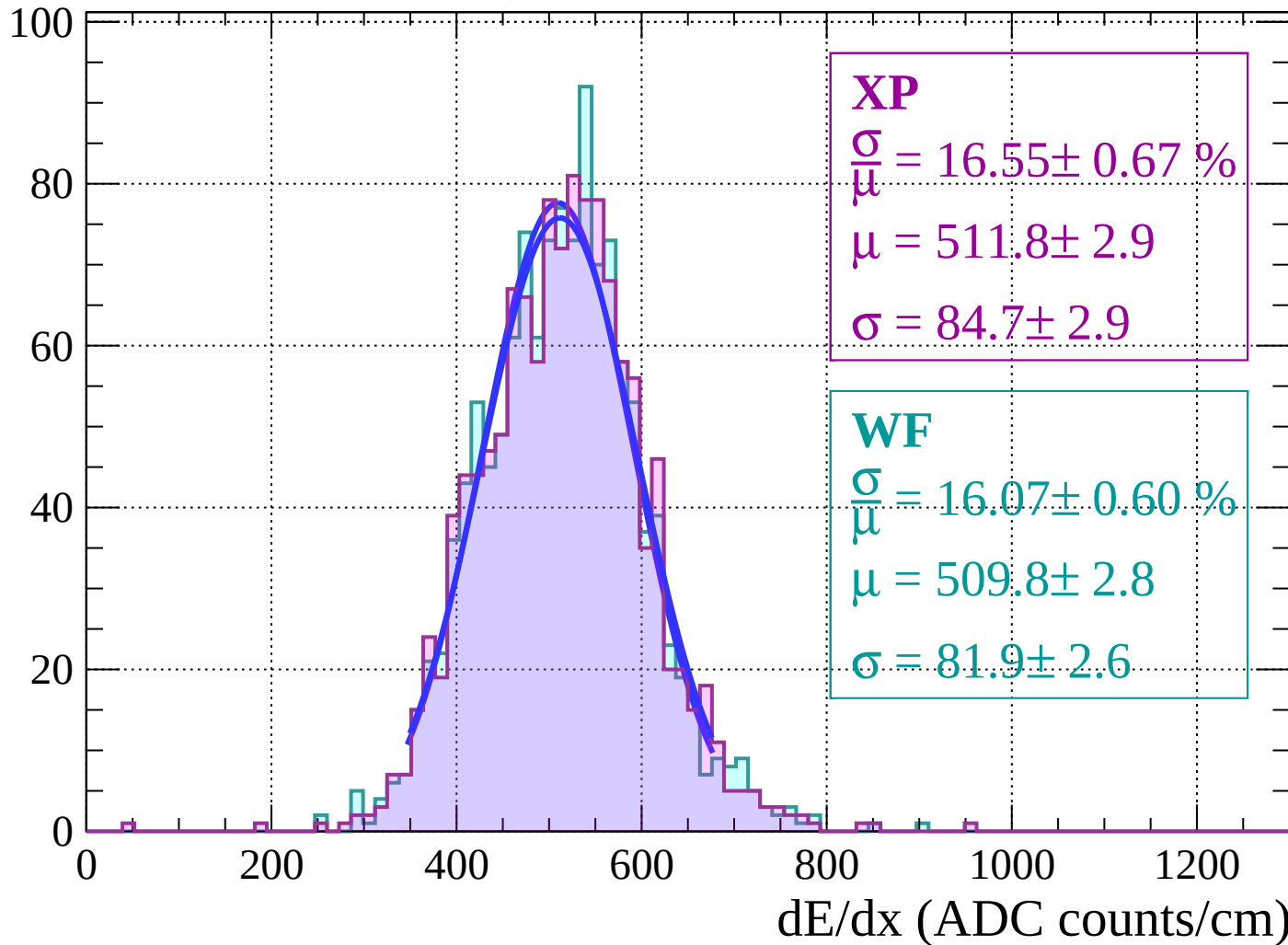
Energy loss $|-46 < \theta < -44$

Count



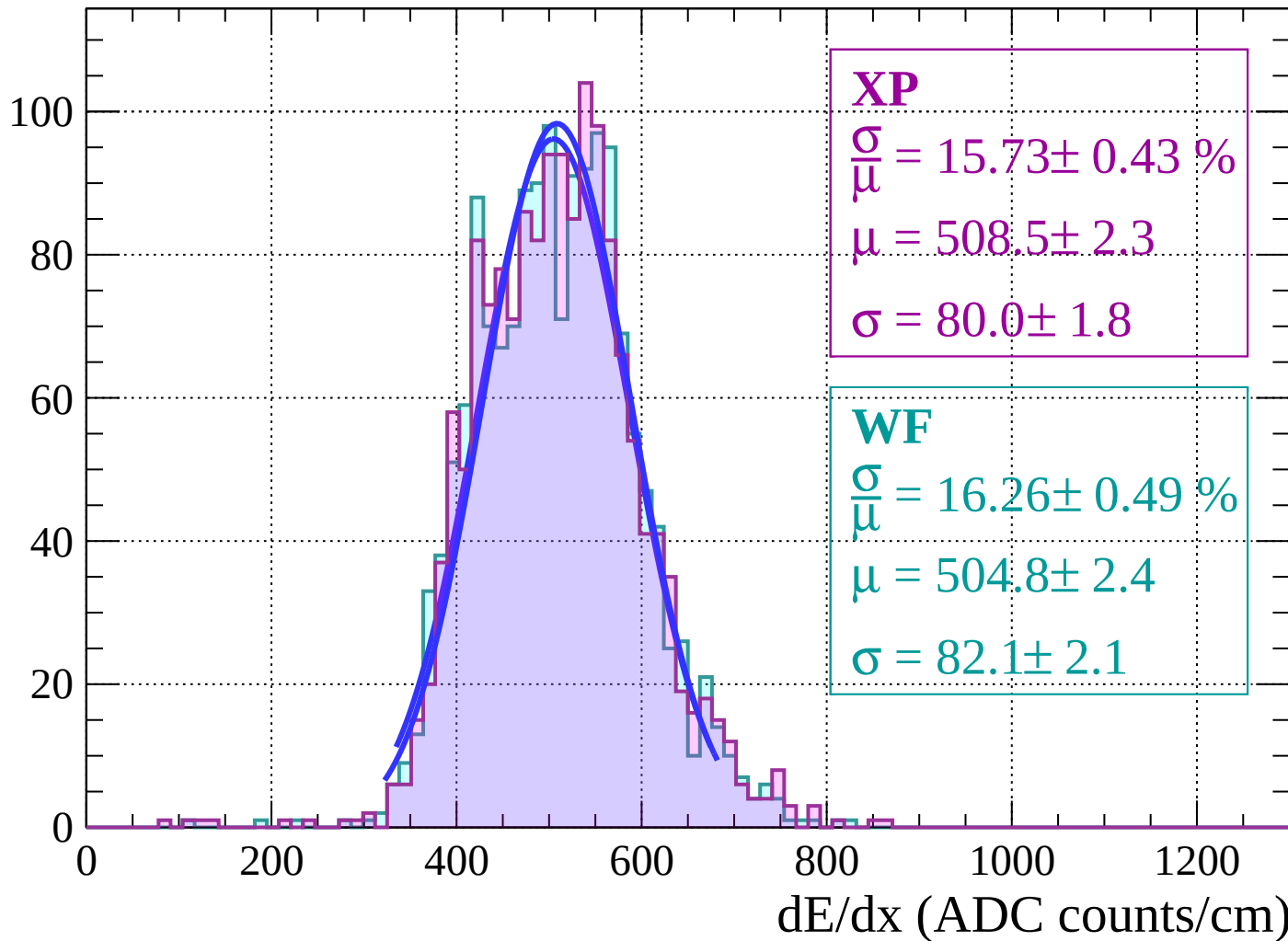
Energy loss | $-44 < \theta < -42$

Count

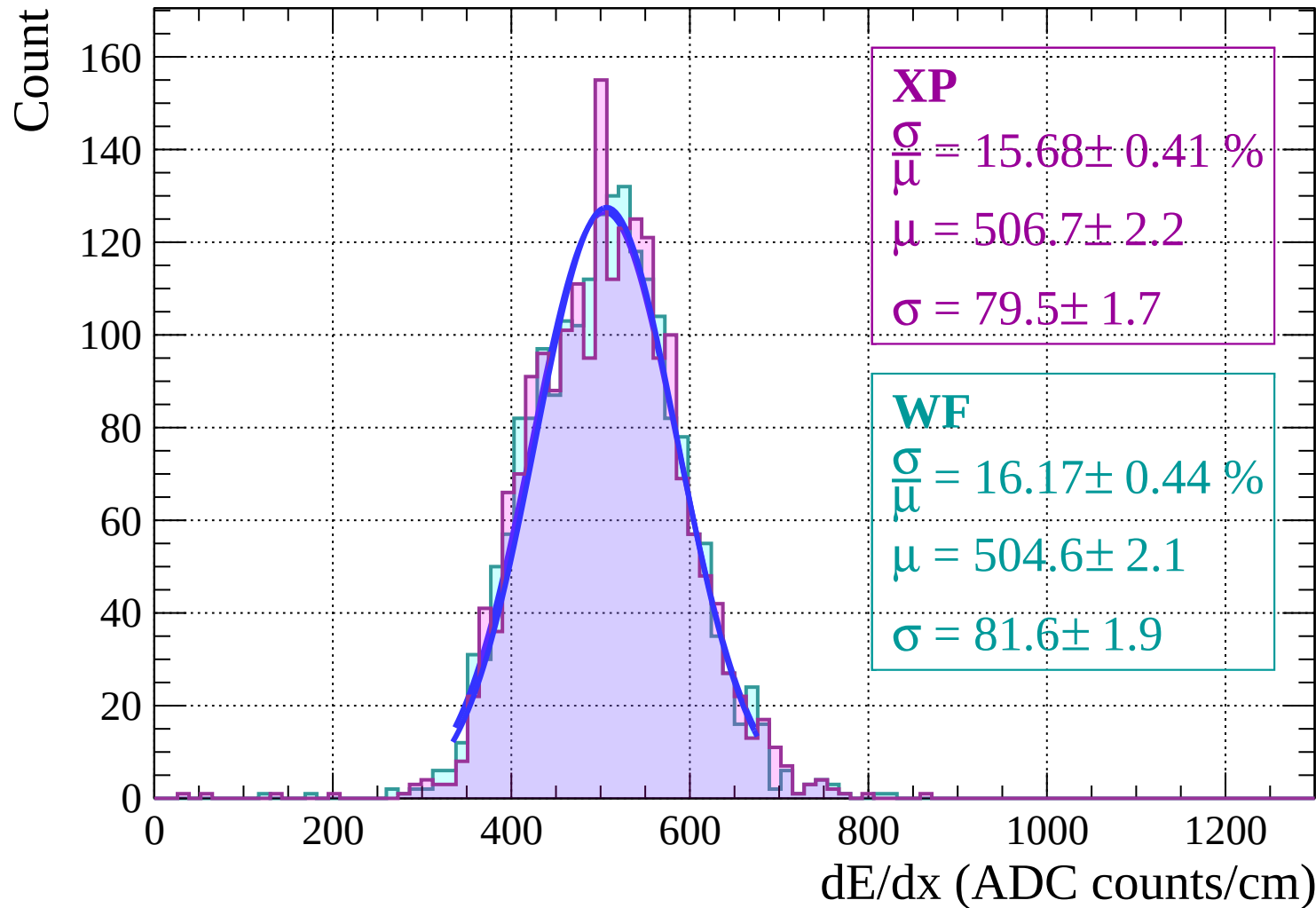


Energy loss | $-42 < \theta < -40$

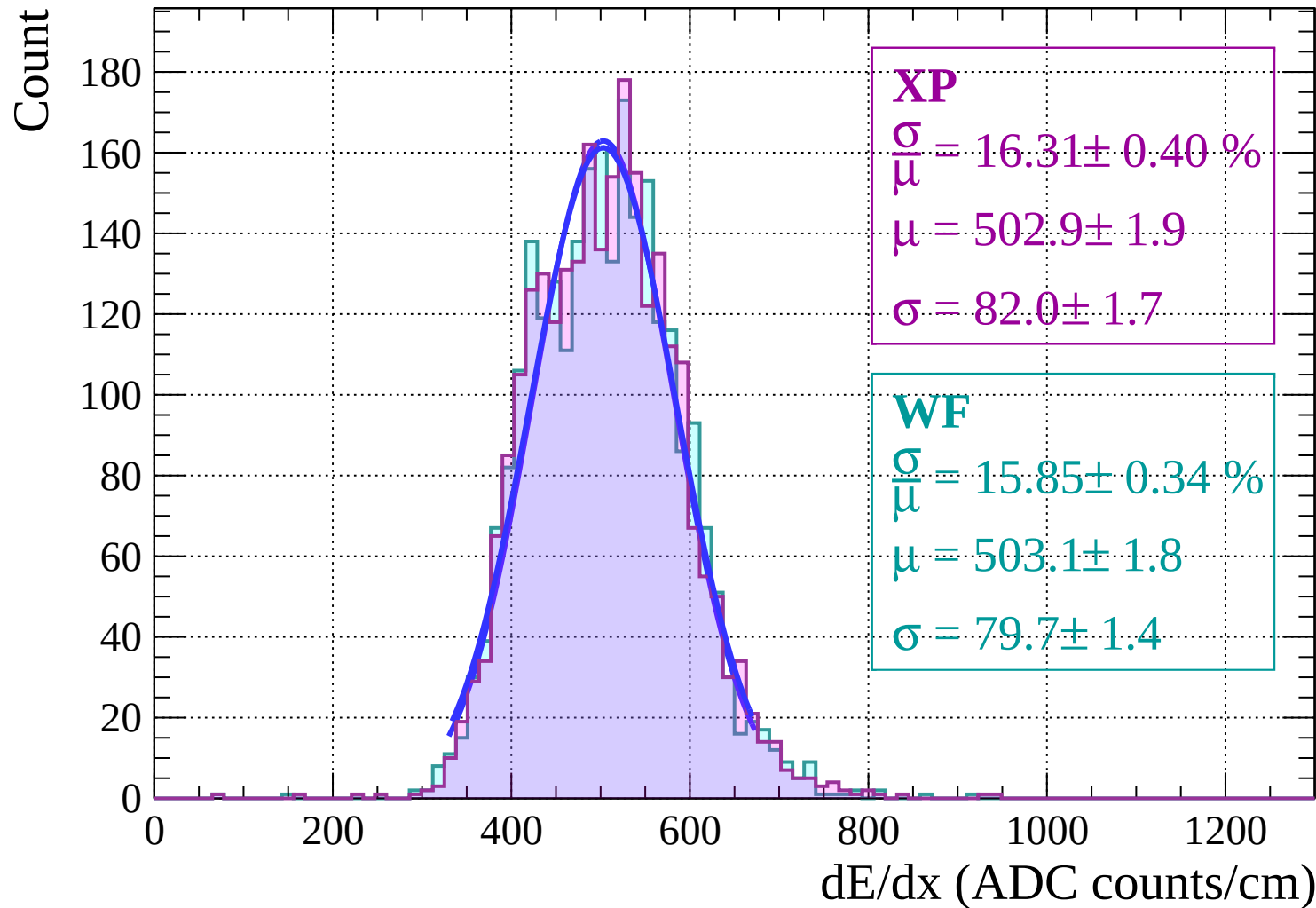
Count



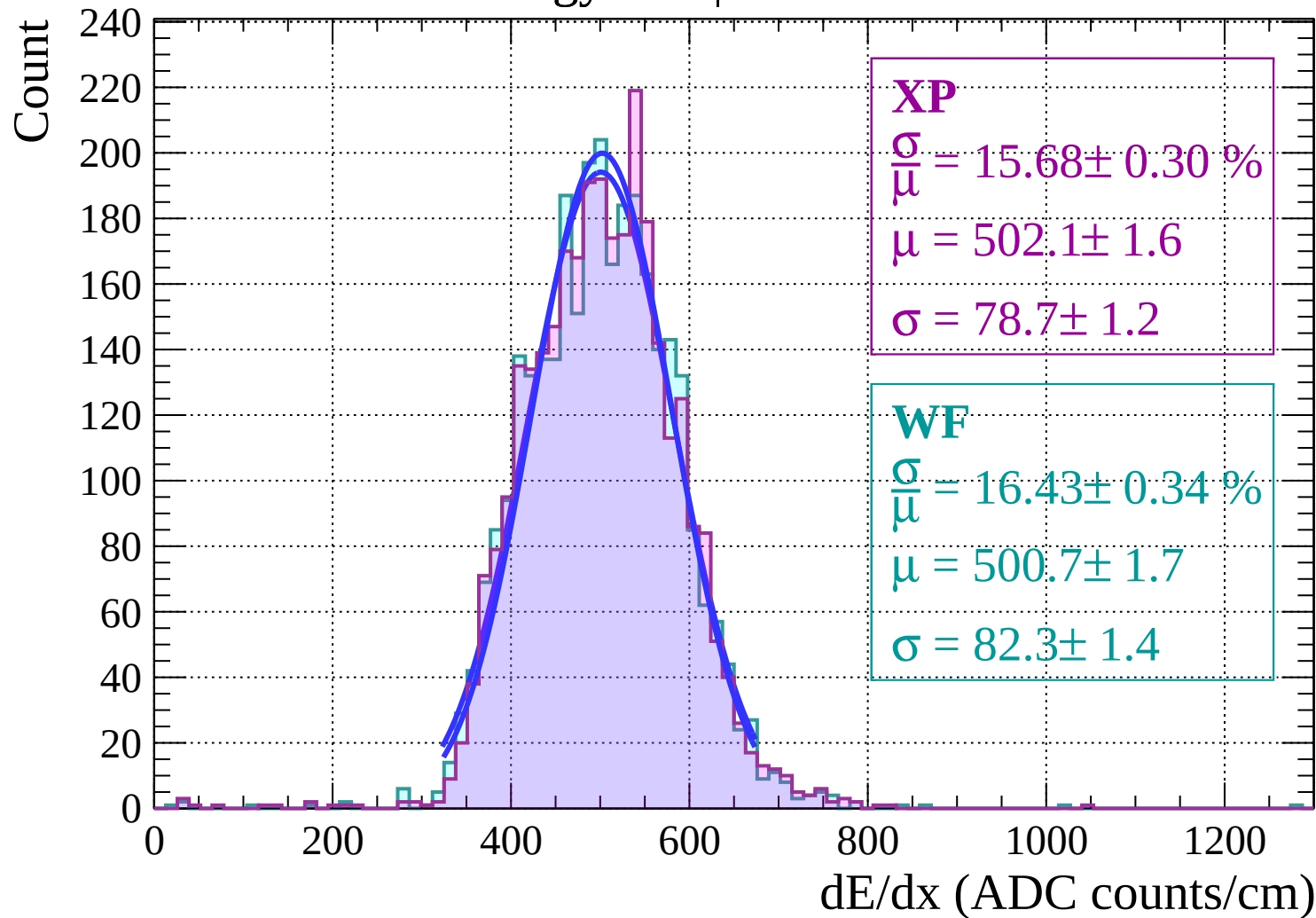
Energy loss | $-40 < \theta < -38$



Energy loss | $-38 < \theta < -36$

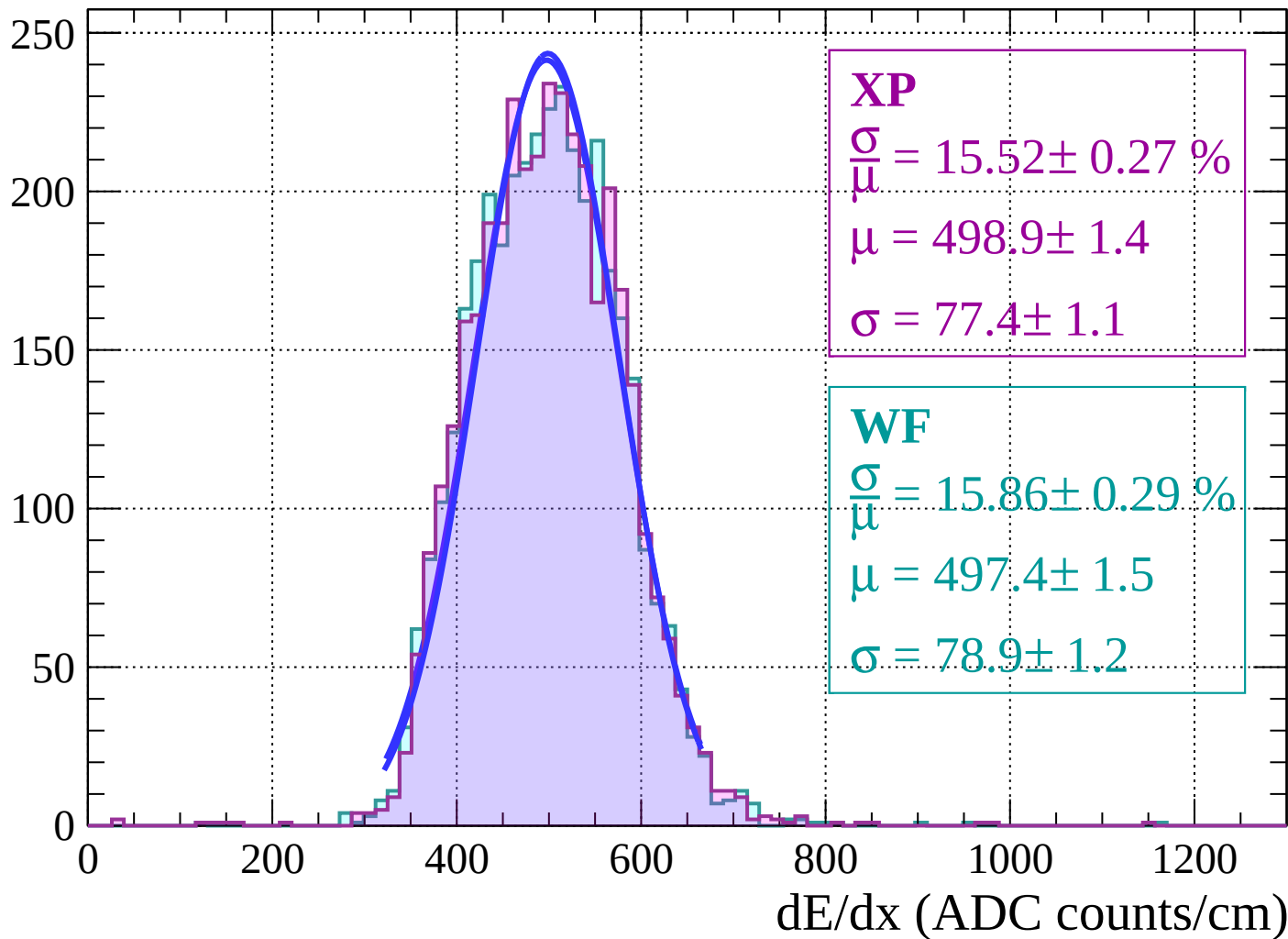


Energy loss $|-36 < \theta < -34$



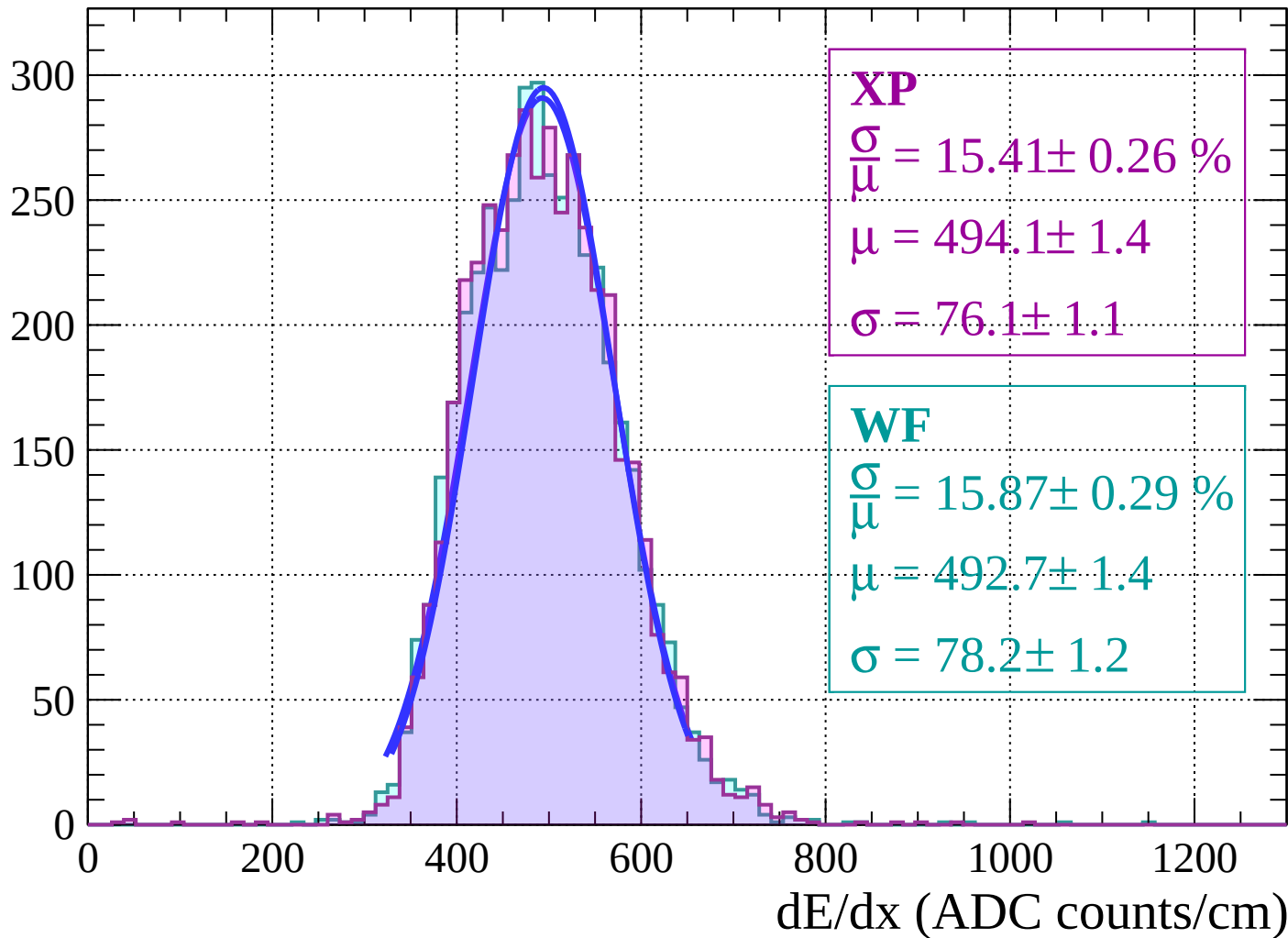
Energy loss $|-34 < \theta < -32$

Count



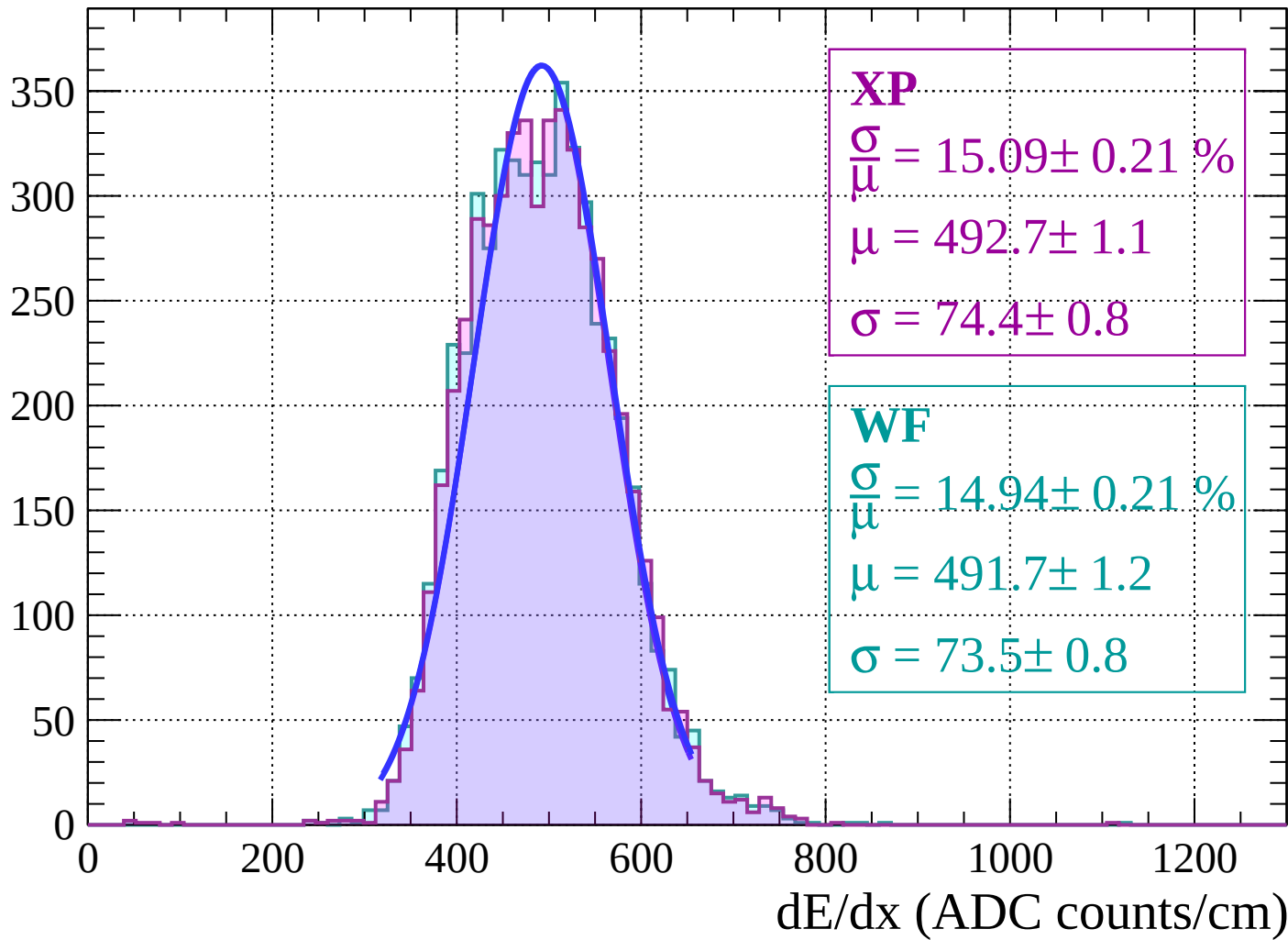
Energy loss | $-32 < \theta < -30$

Count

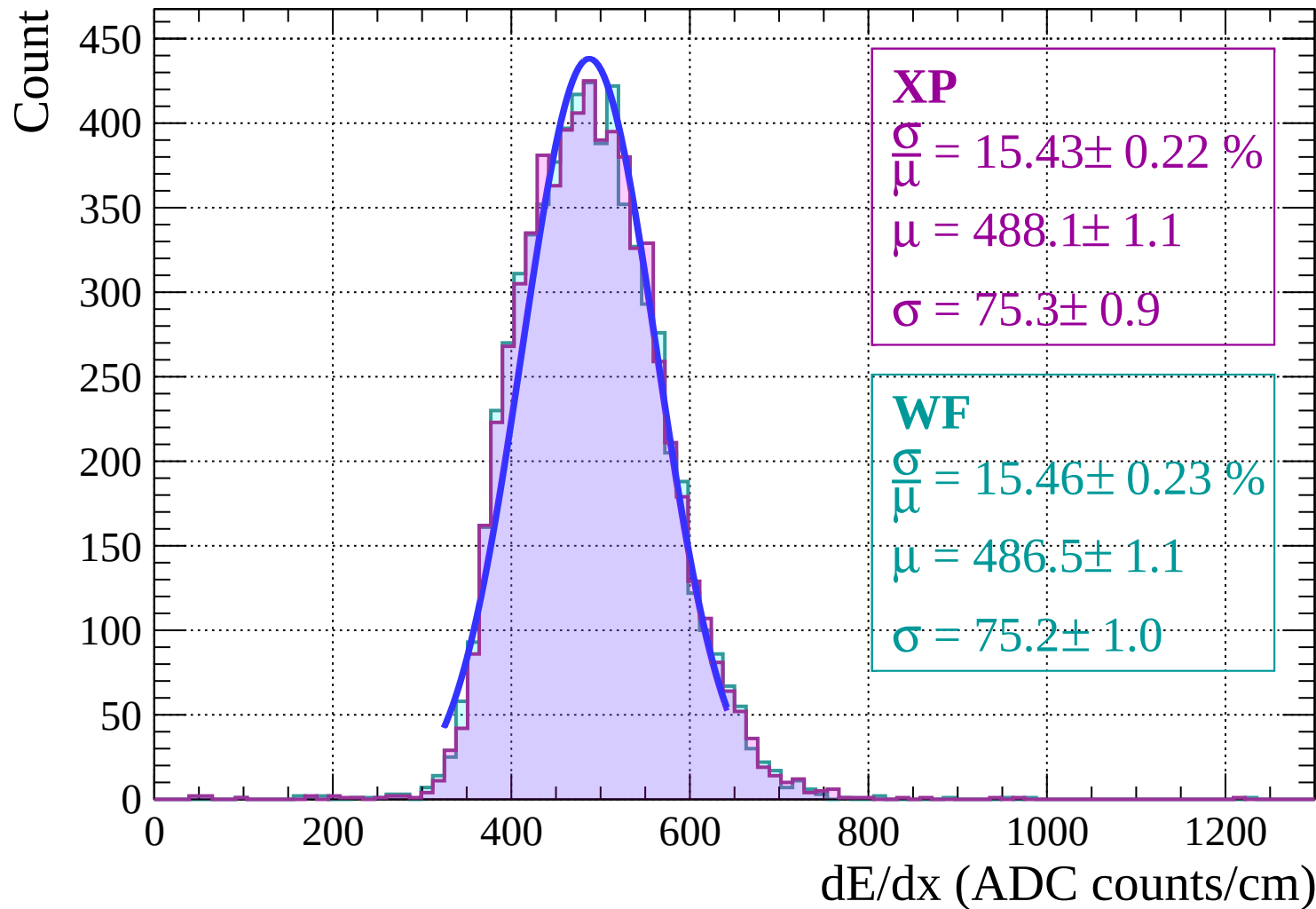


Energy loss | $-30 < \theta < -28$

Count

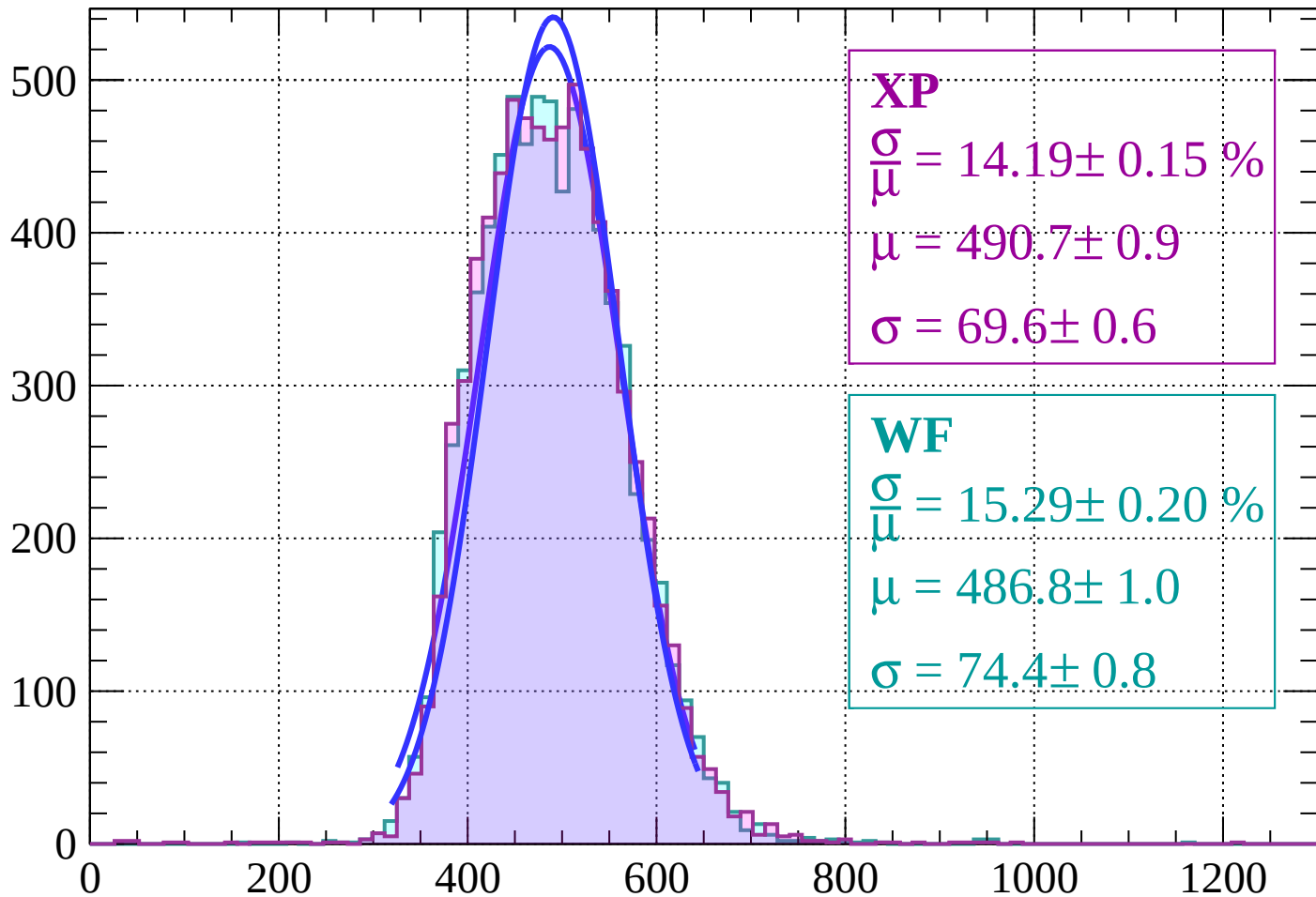


Energy loss | $-28 < \theta < -26$



Energy loss | $-26 < \theta < -24$

Count



XP

$$\frac{\sigma}{\mu} = 14.19 \pm 0.15 \%$$

$$\mu = 490.7 \pm 0.9$$

$$\sigma = 69.6 \pm 0.6$$

WF

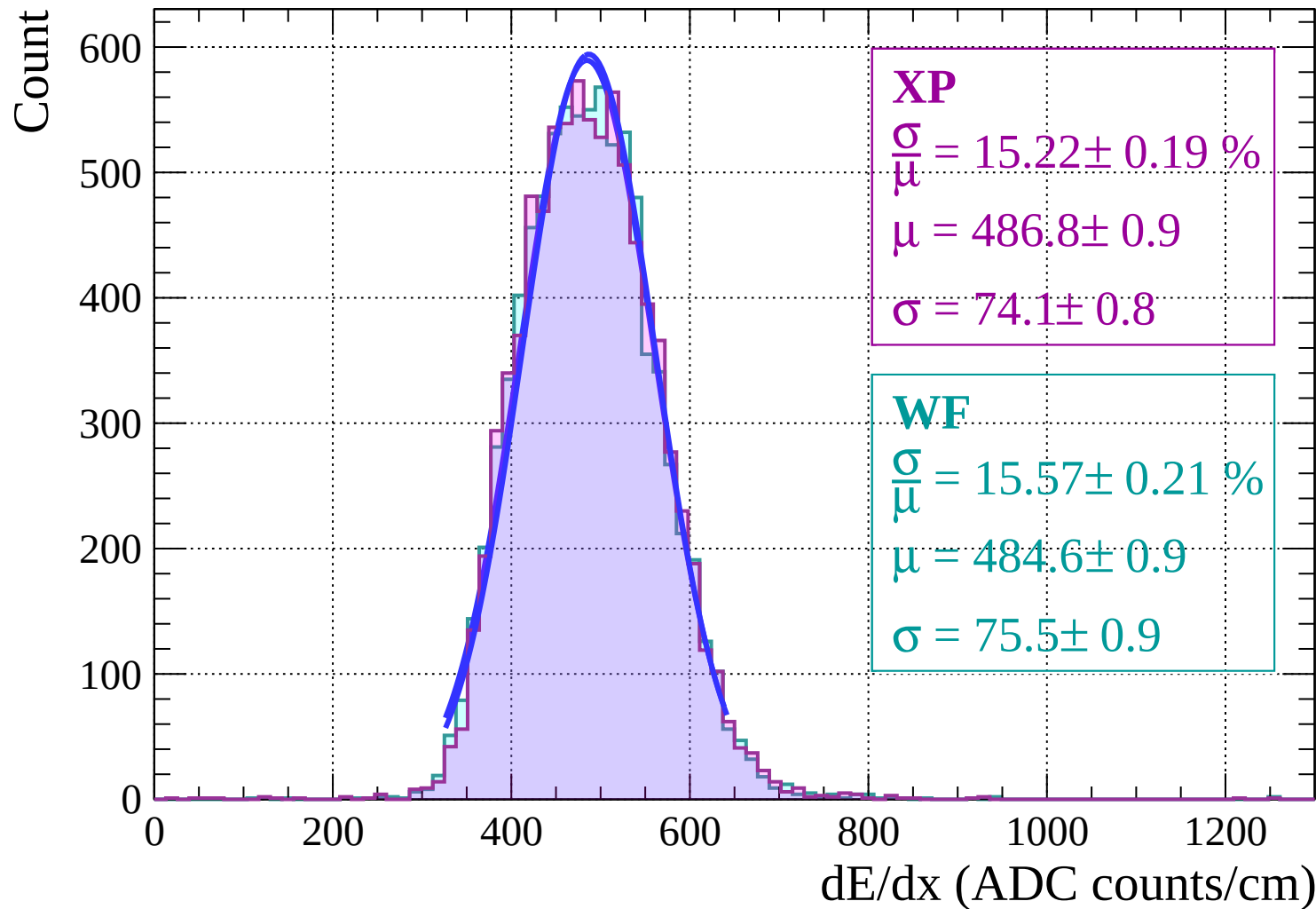
$$\frac{\sigma}{\mu} = 15.29 \pm 0.20 \%$$

$$\mu = 486.8 \pm 1.0$$

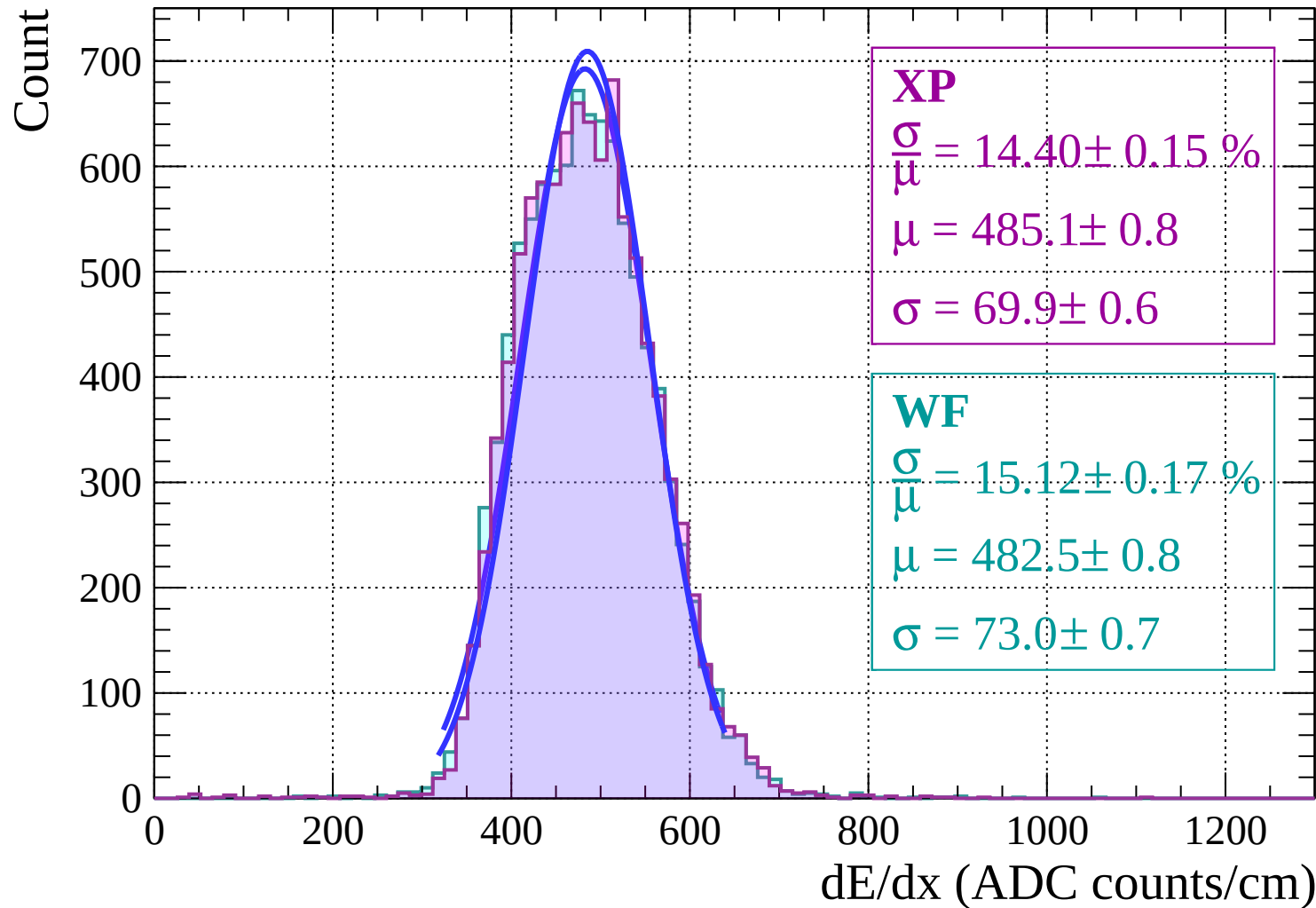
$$\sigma = 74.4 \pm 0.8$$

dE/dx (ADC counts/cm)

Energy loss | $-24 < \theta < -22$

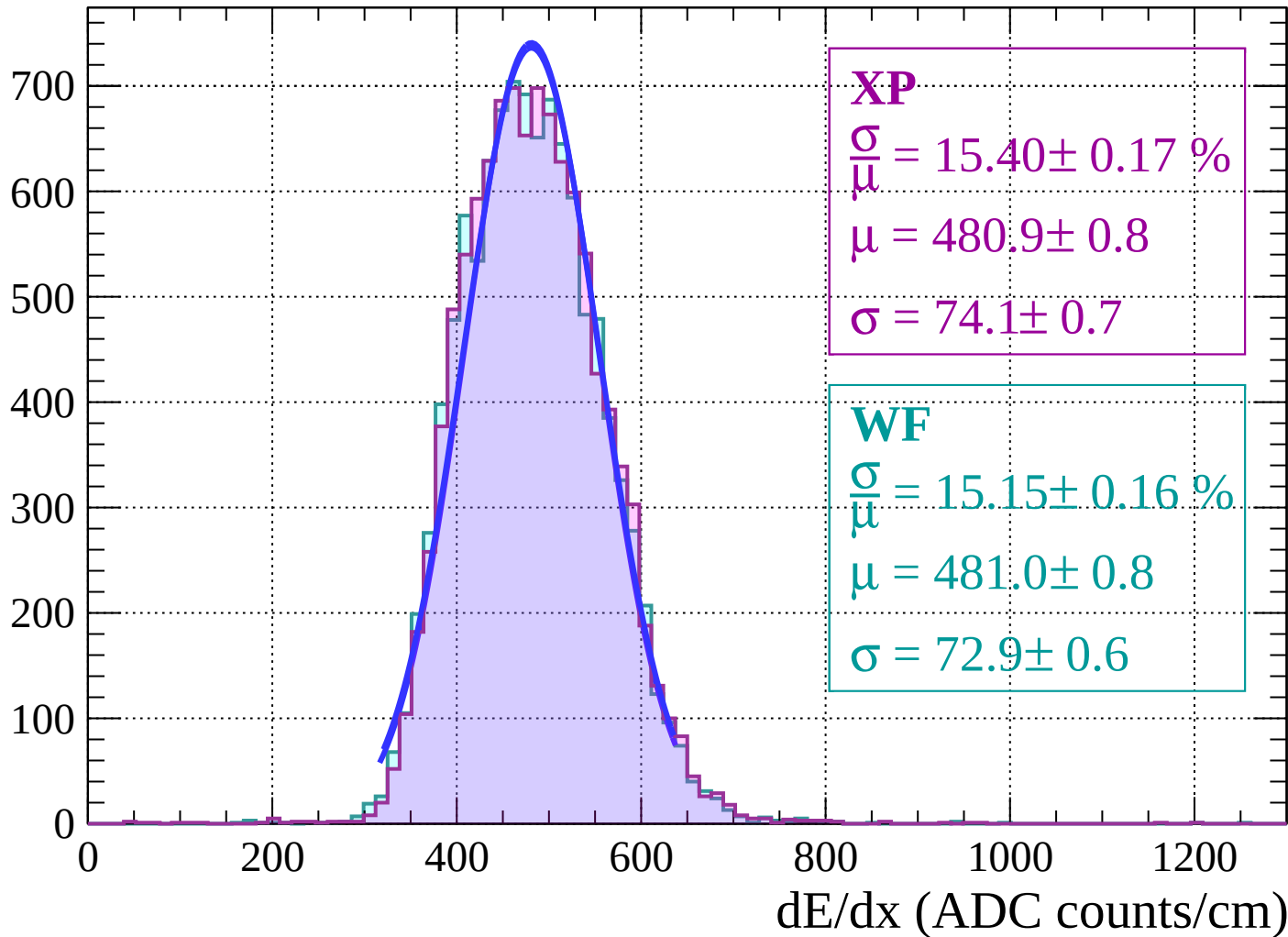


Energy loss | $-22 < \theta < -20$



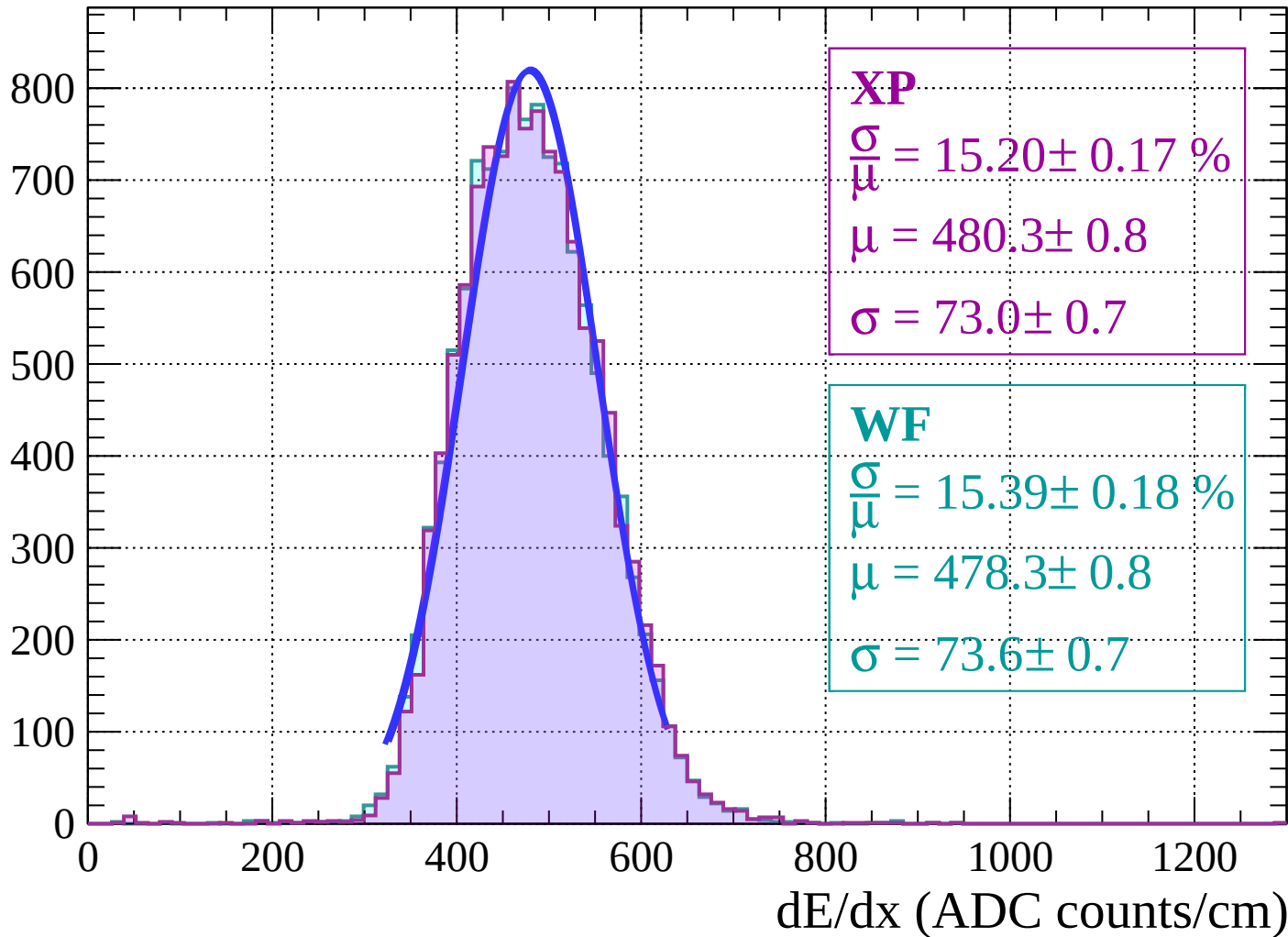
Energy loss | $-20 < \theta < -18$

Count

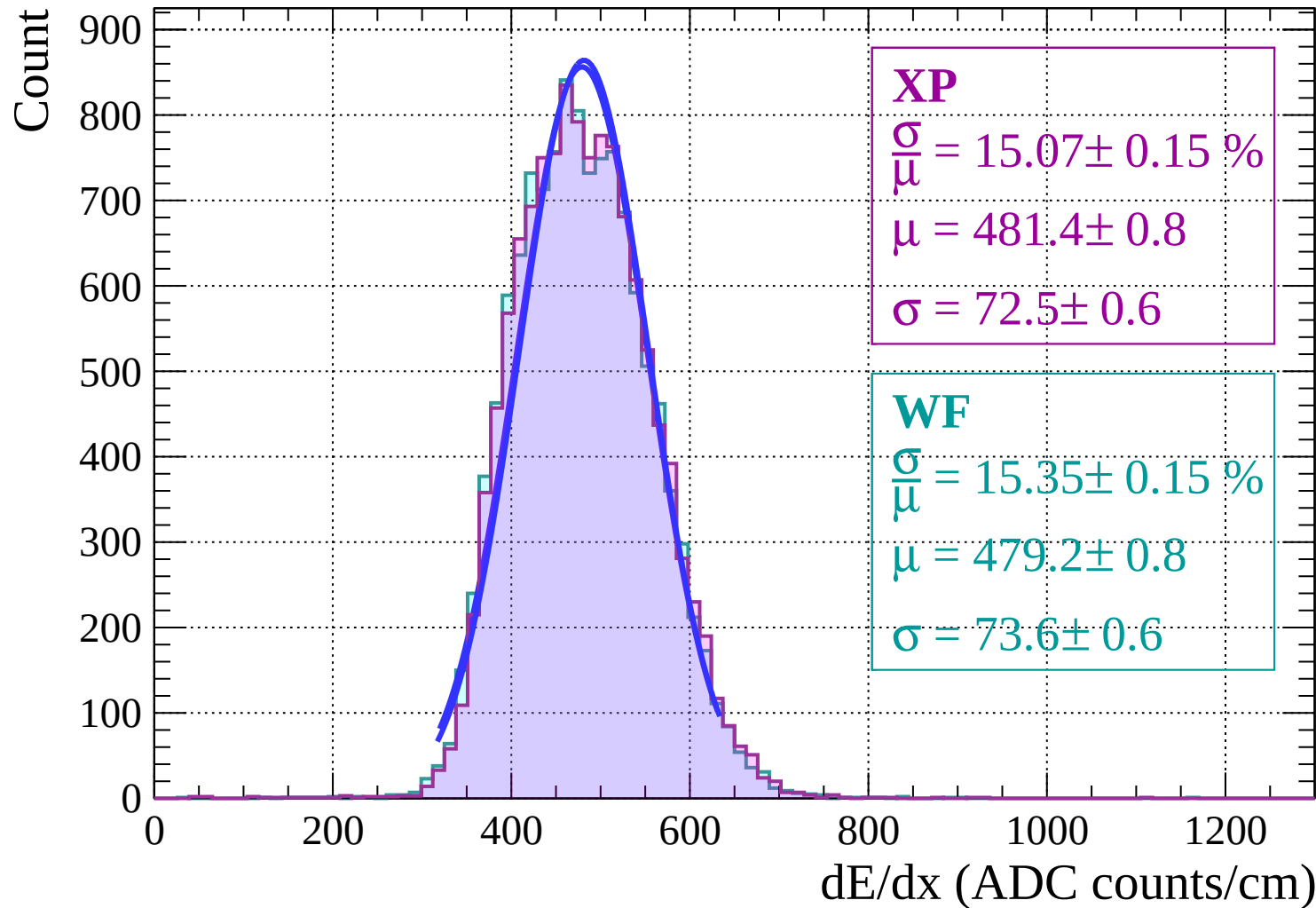


Energy loss | $-18 < \theta < -16$

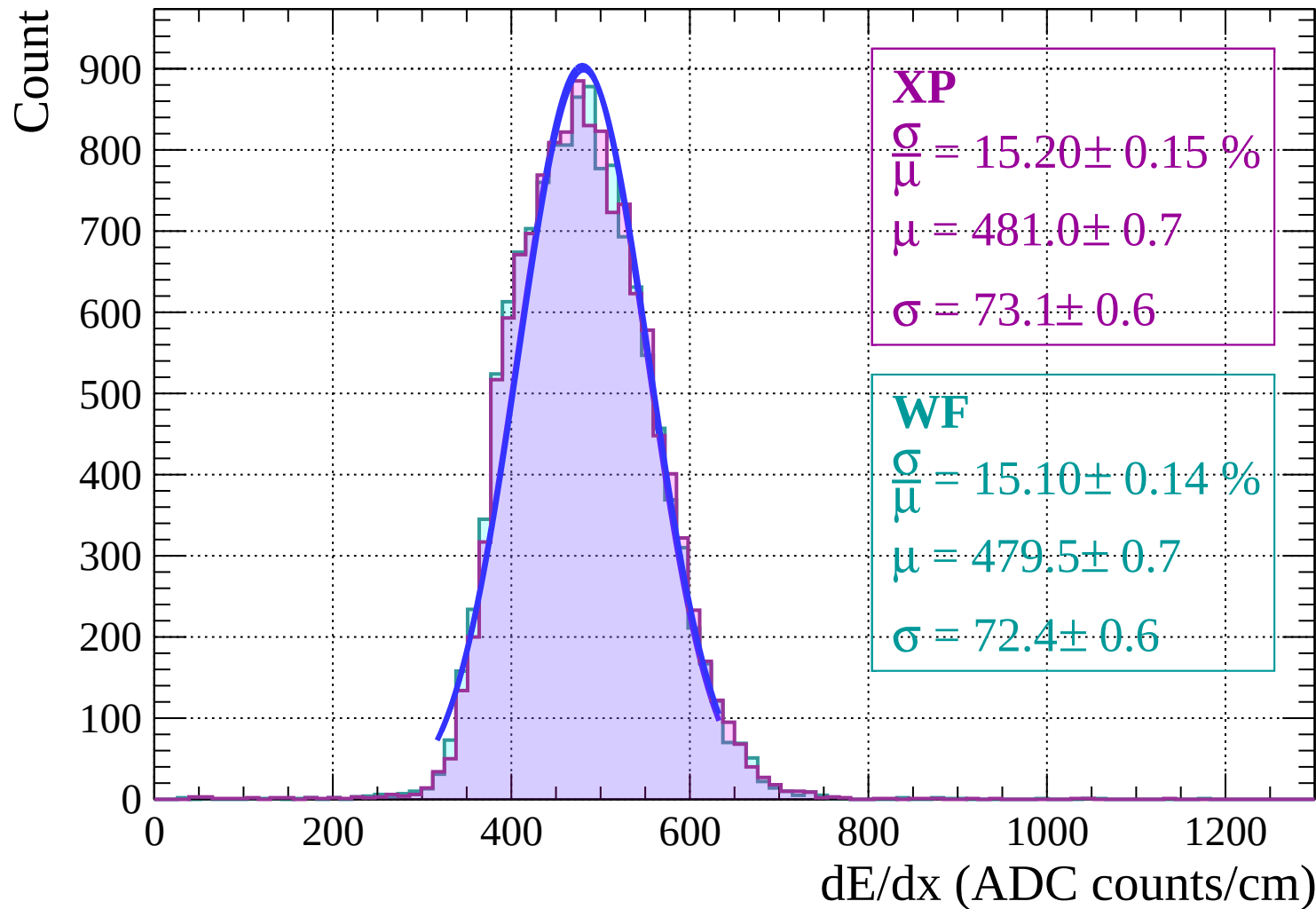
Count



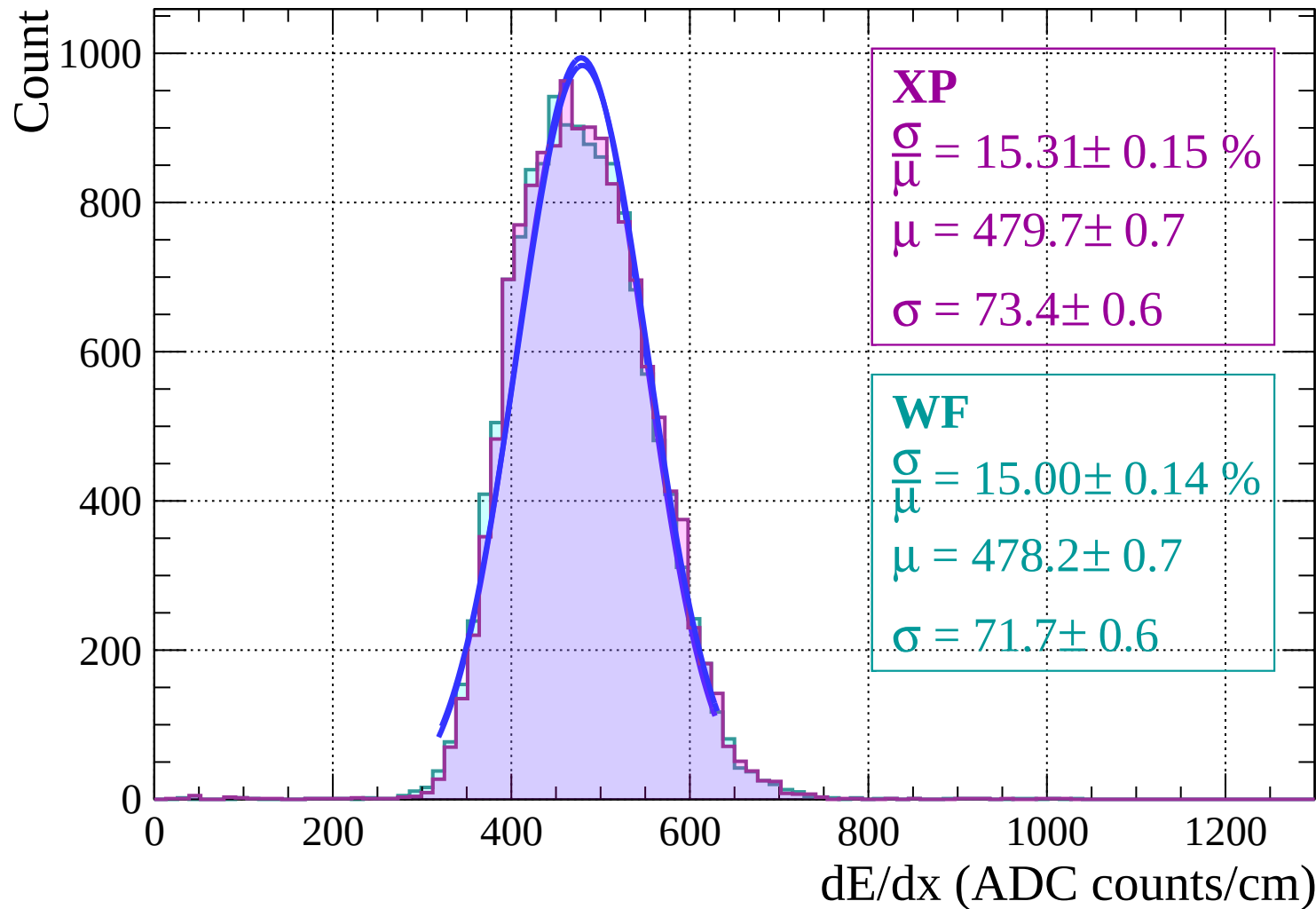
Energy loss $|-16 < \theta < -14$



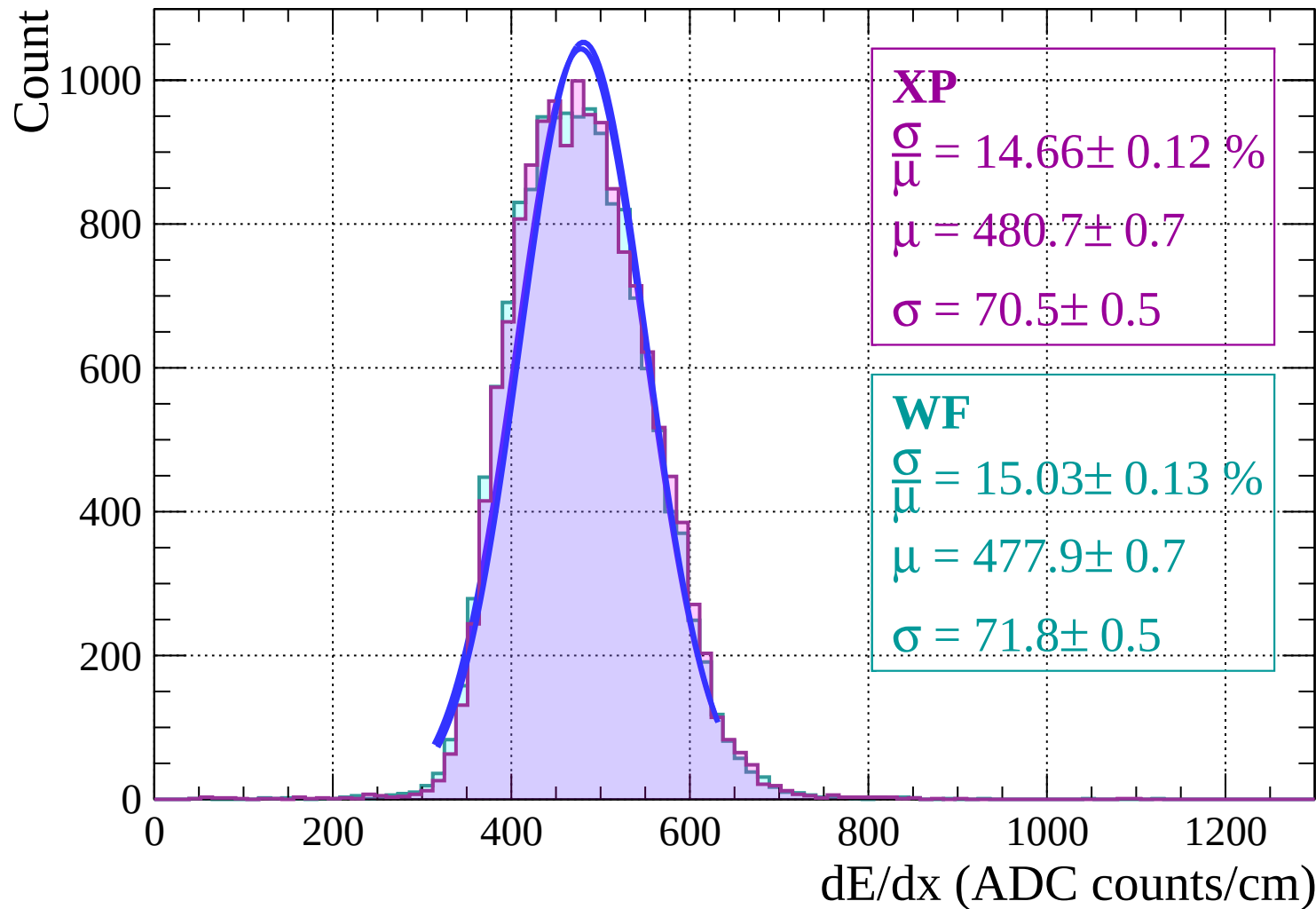
Energy loss | $-14 < \theta < -12$



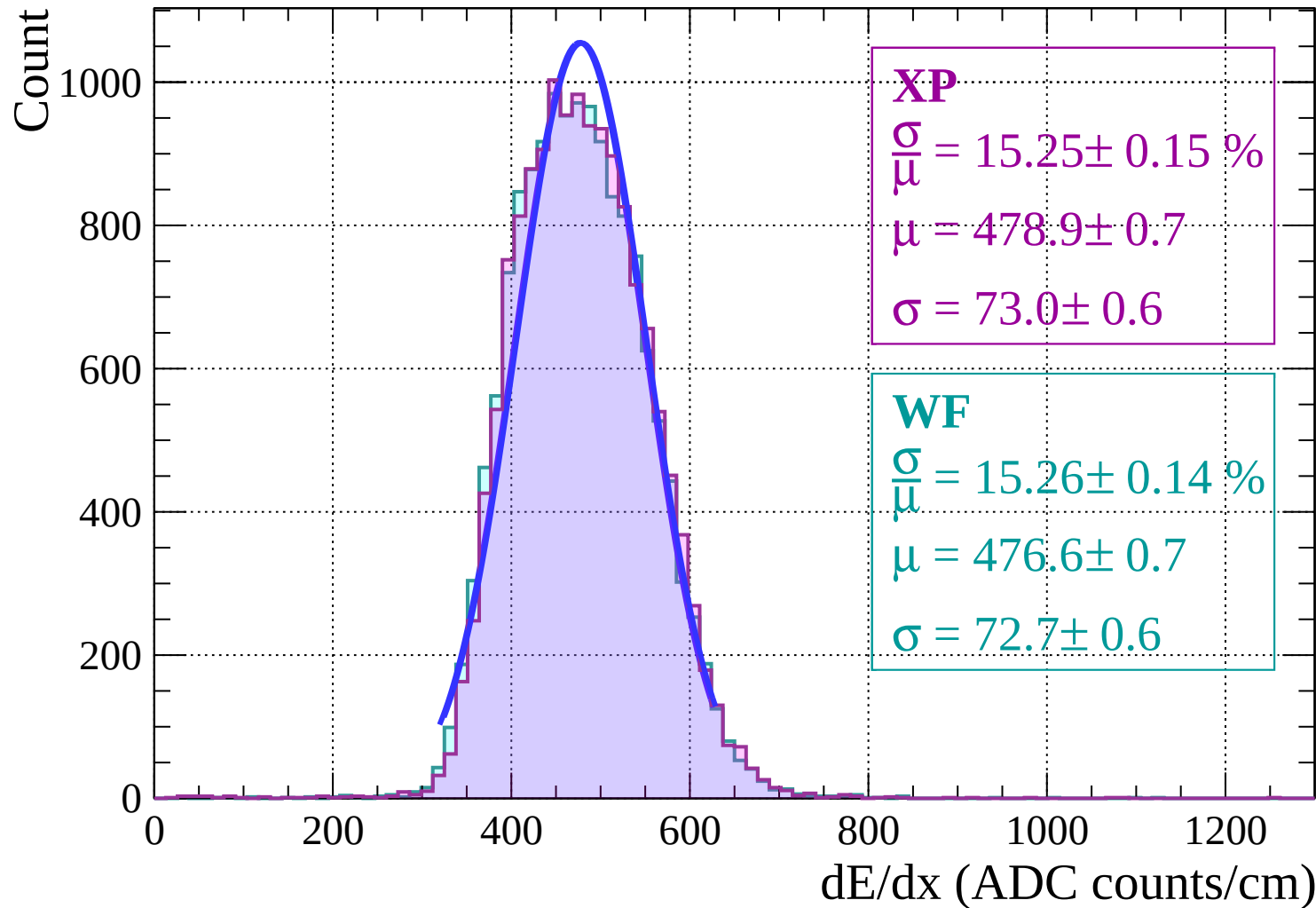
Energy loss | $-12 < \theta < -10$



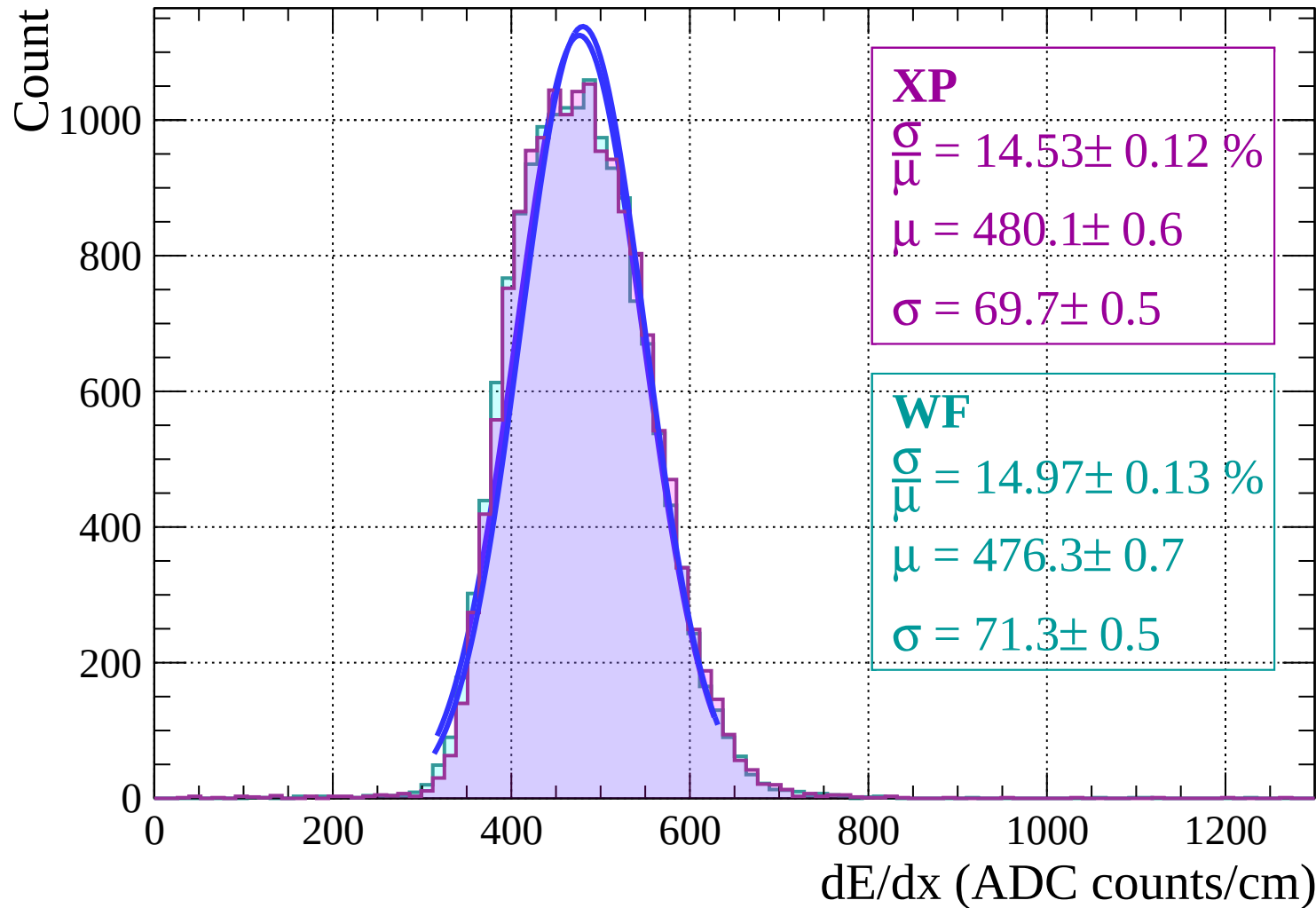
Energy loss | $-10 < \theta < -8$



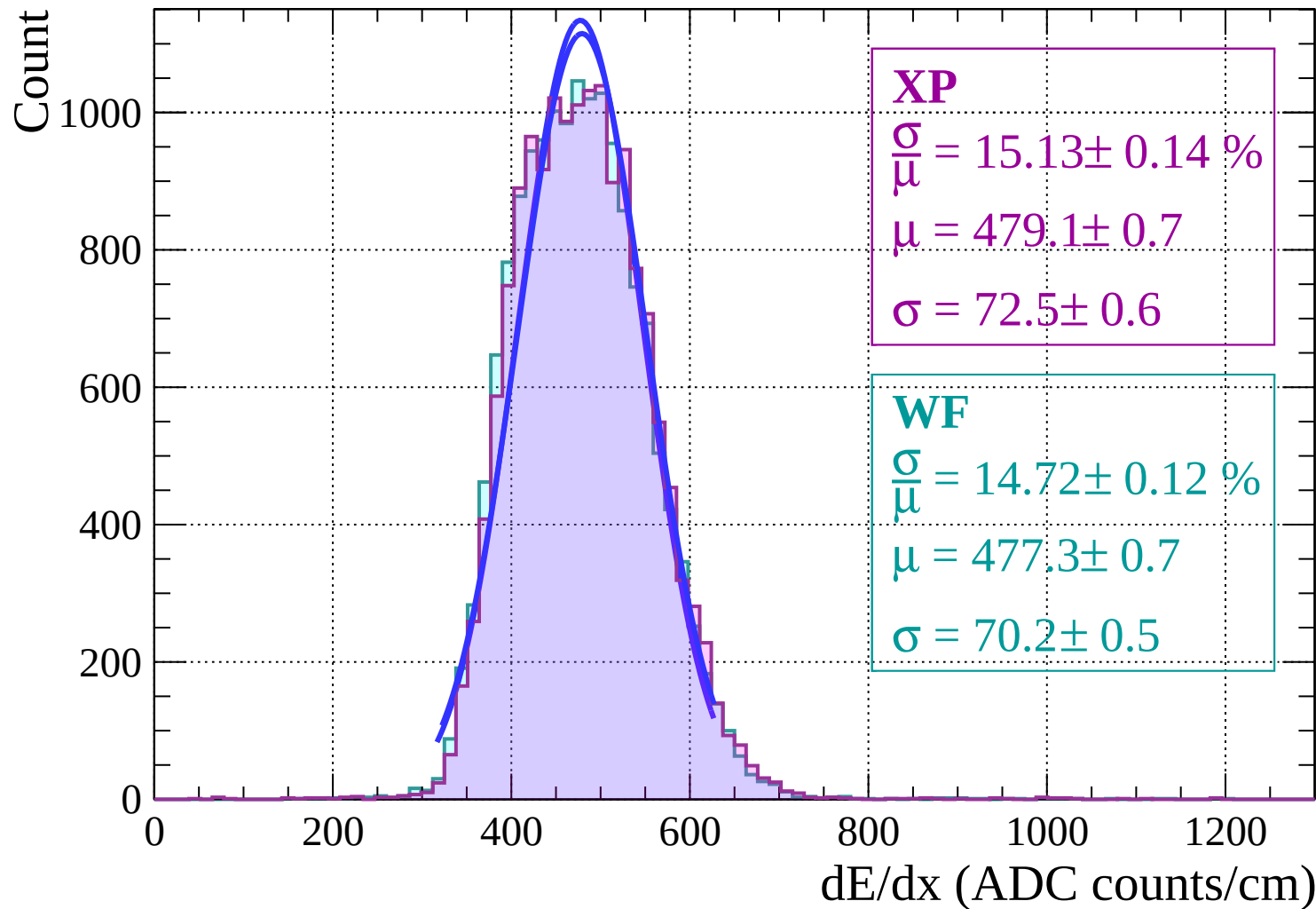
Energy loss $|-8 < \theta < -6$



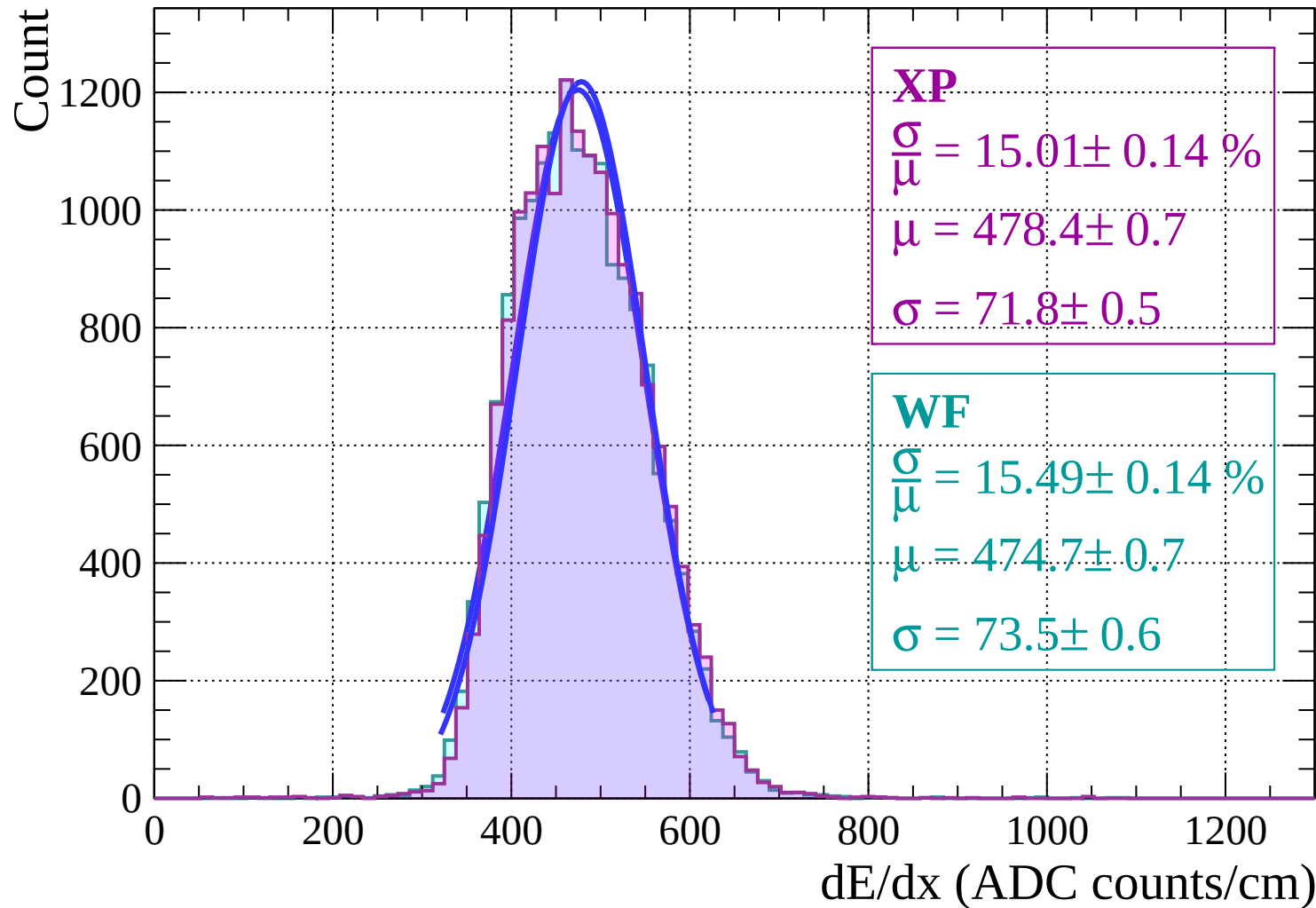
Energy loss | $-6 < \theta < -4$



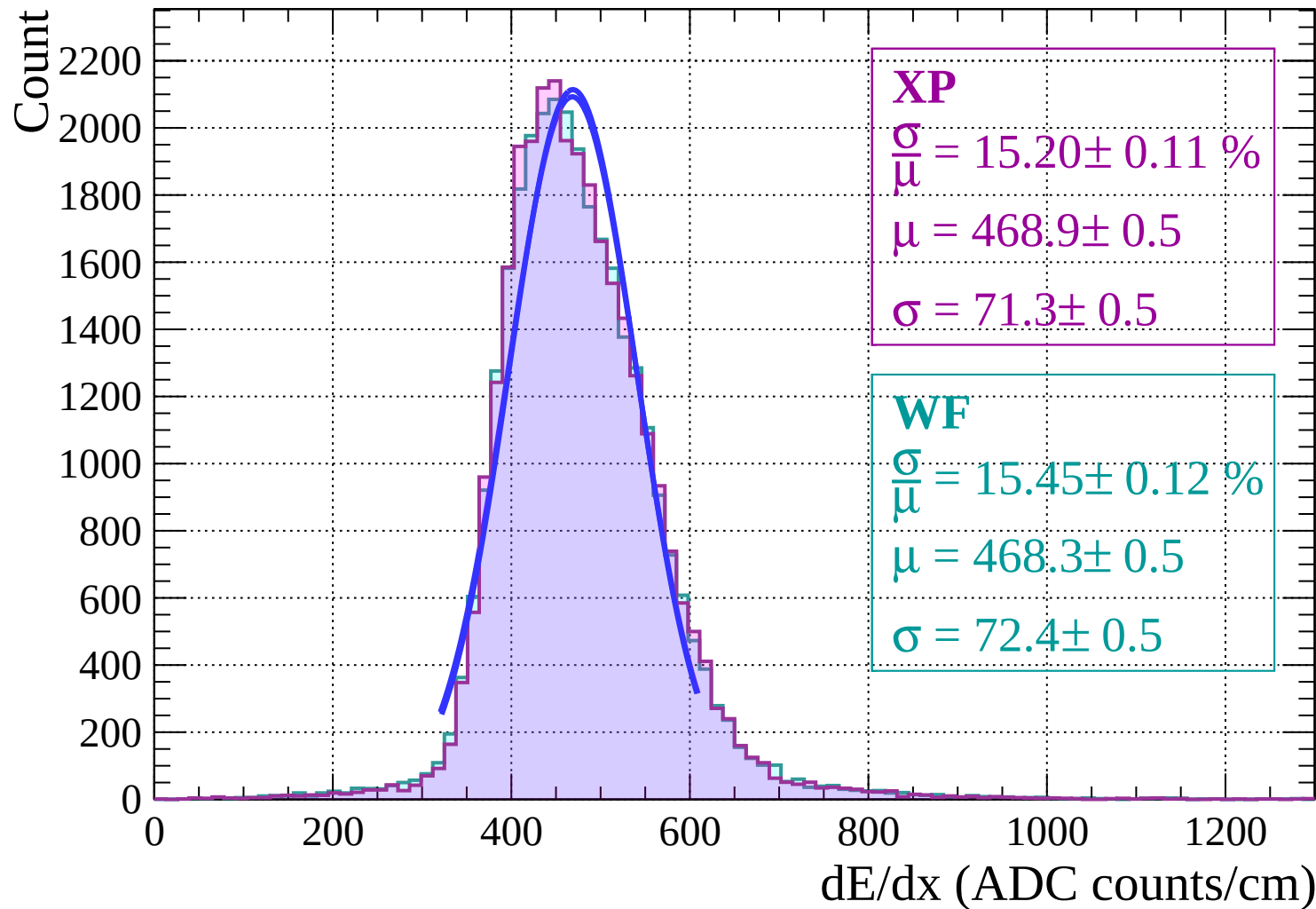
Energy loss | $-4 < \theta < -2$



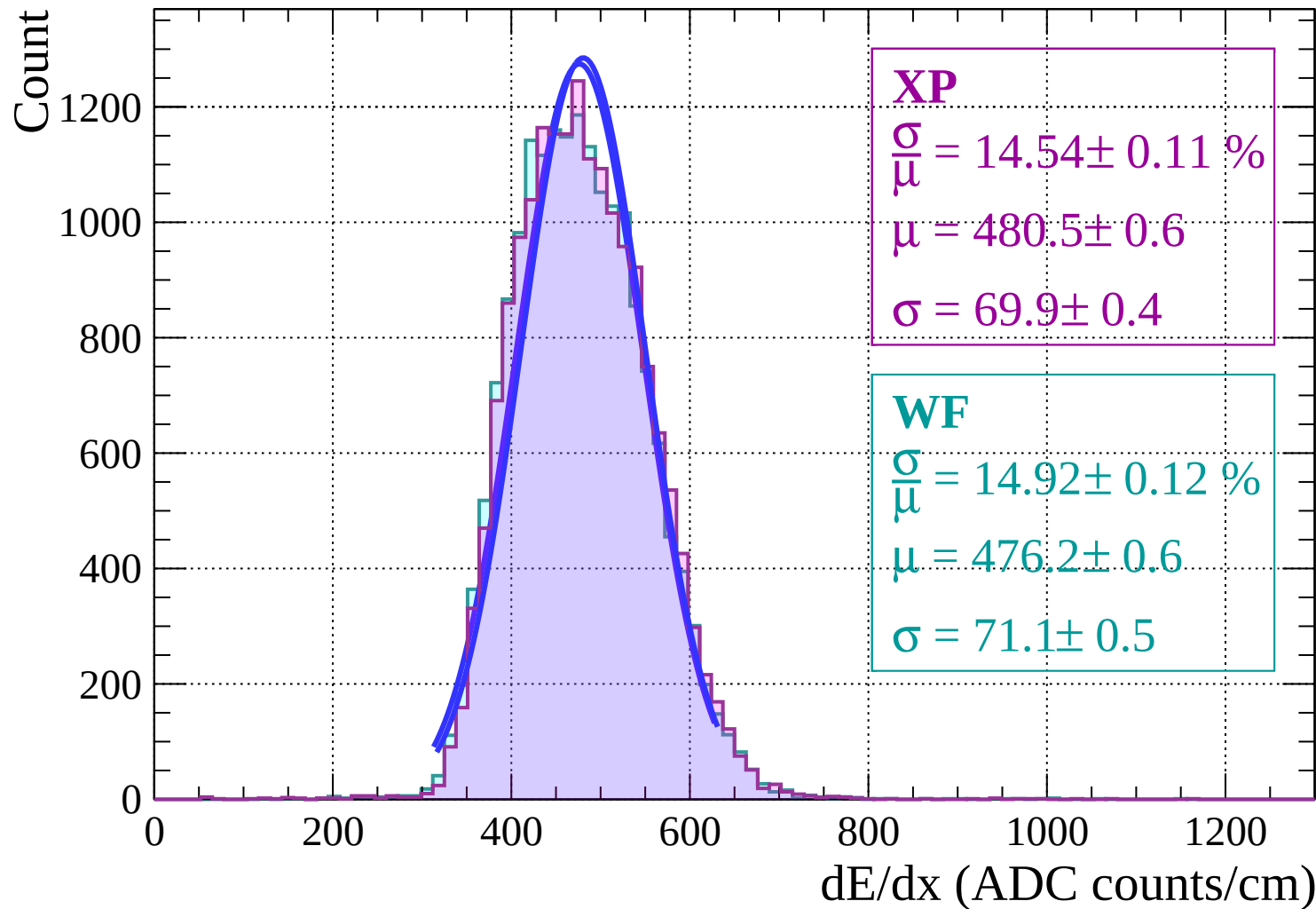
Energy loss | $-2 < \theta < 0$



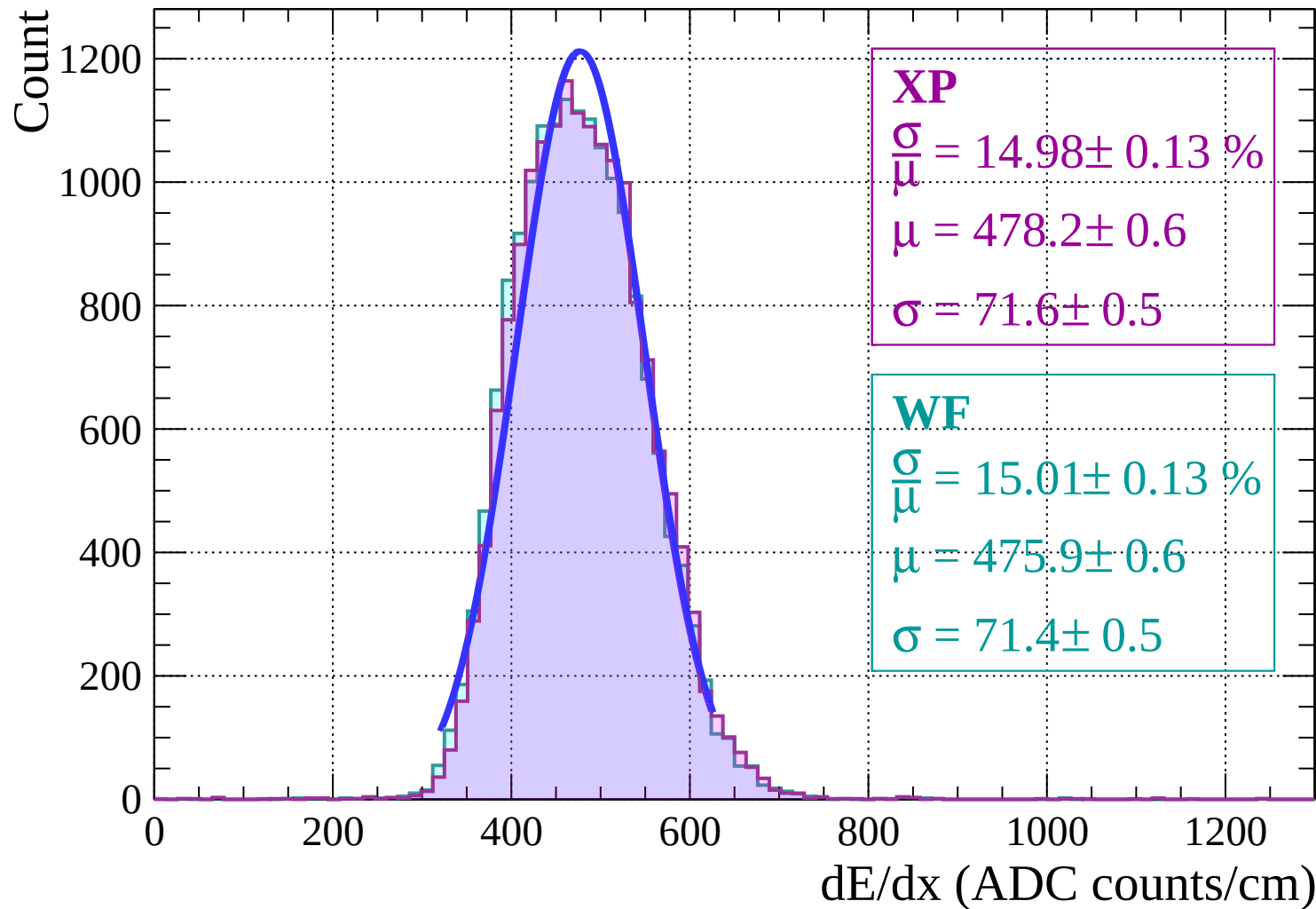
Energy loss | $0 < \theta < 2$



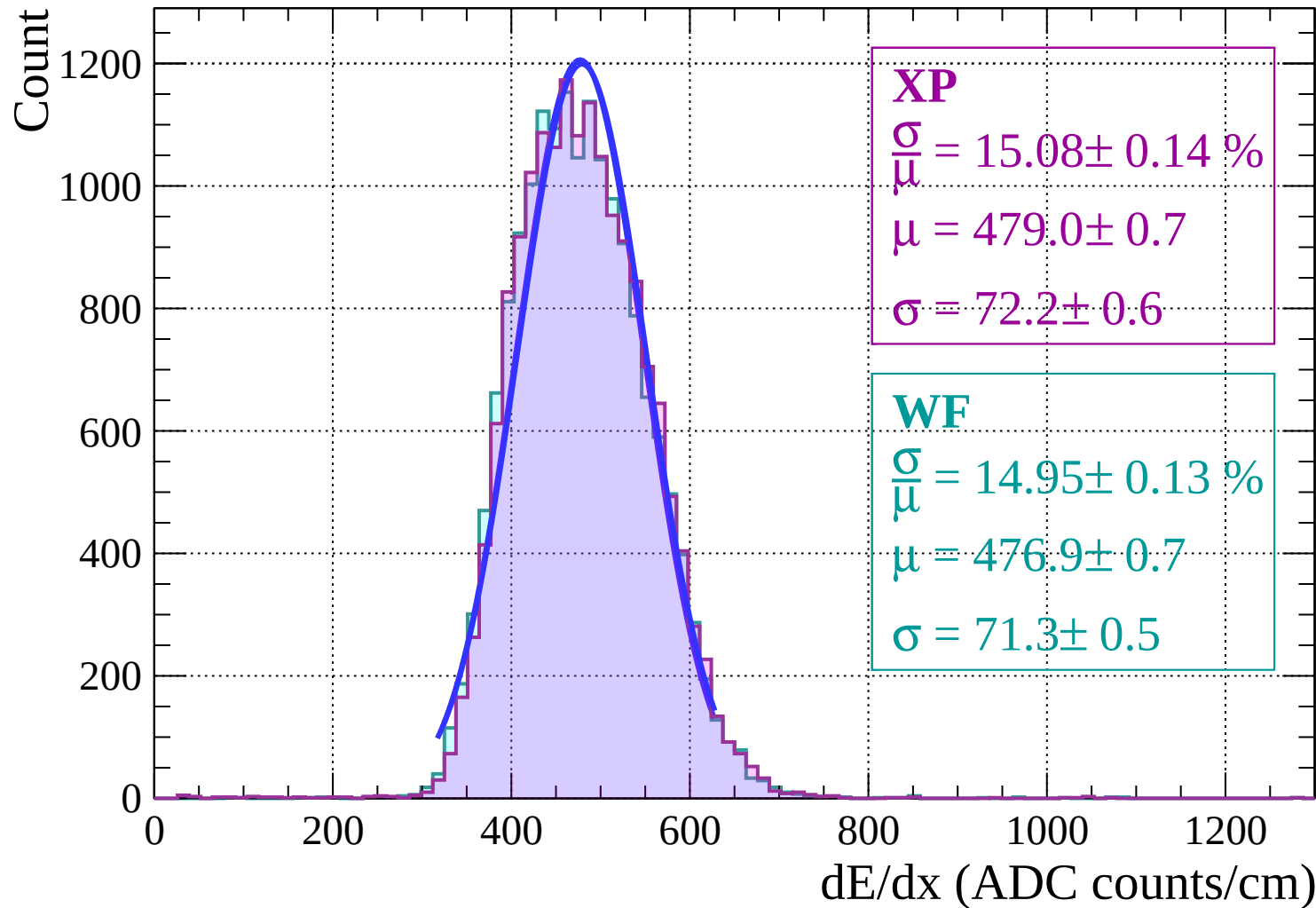
Energy loss | $2 < \theta < 4$



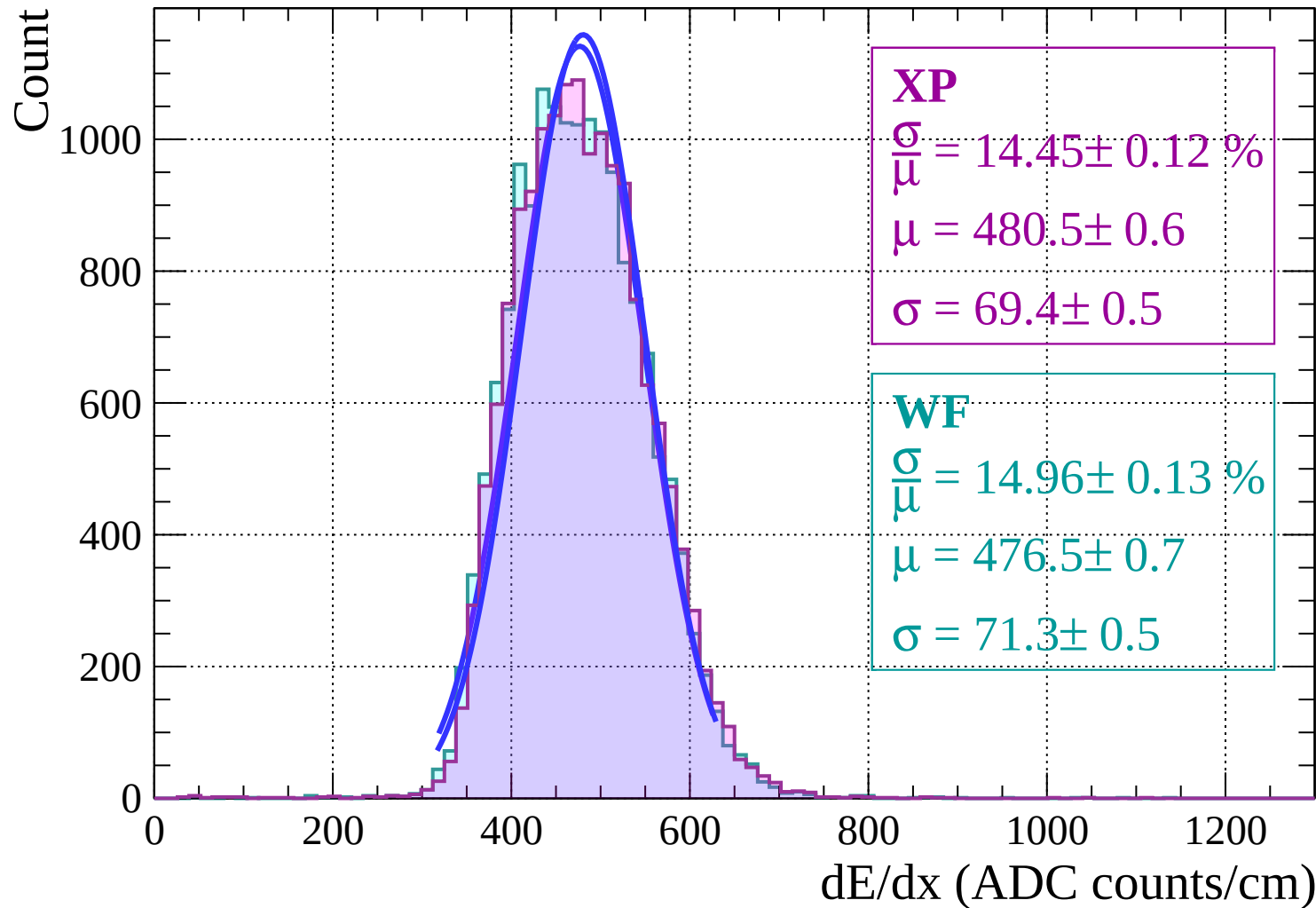
Energy loss | $4 < \theta < 6$



Energy loss | $6 < \theta < 8$



Energy loss | $8 < \theta < 10$



Energy loss | $10 < \theta < 12$

Count

1000

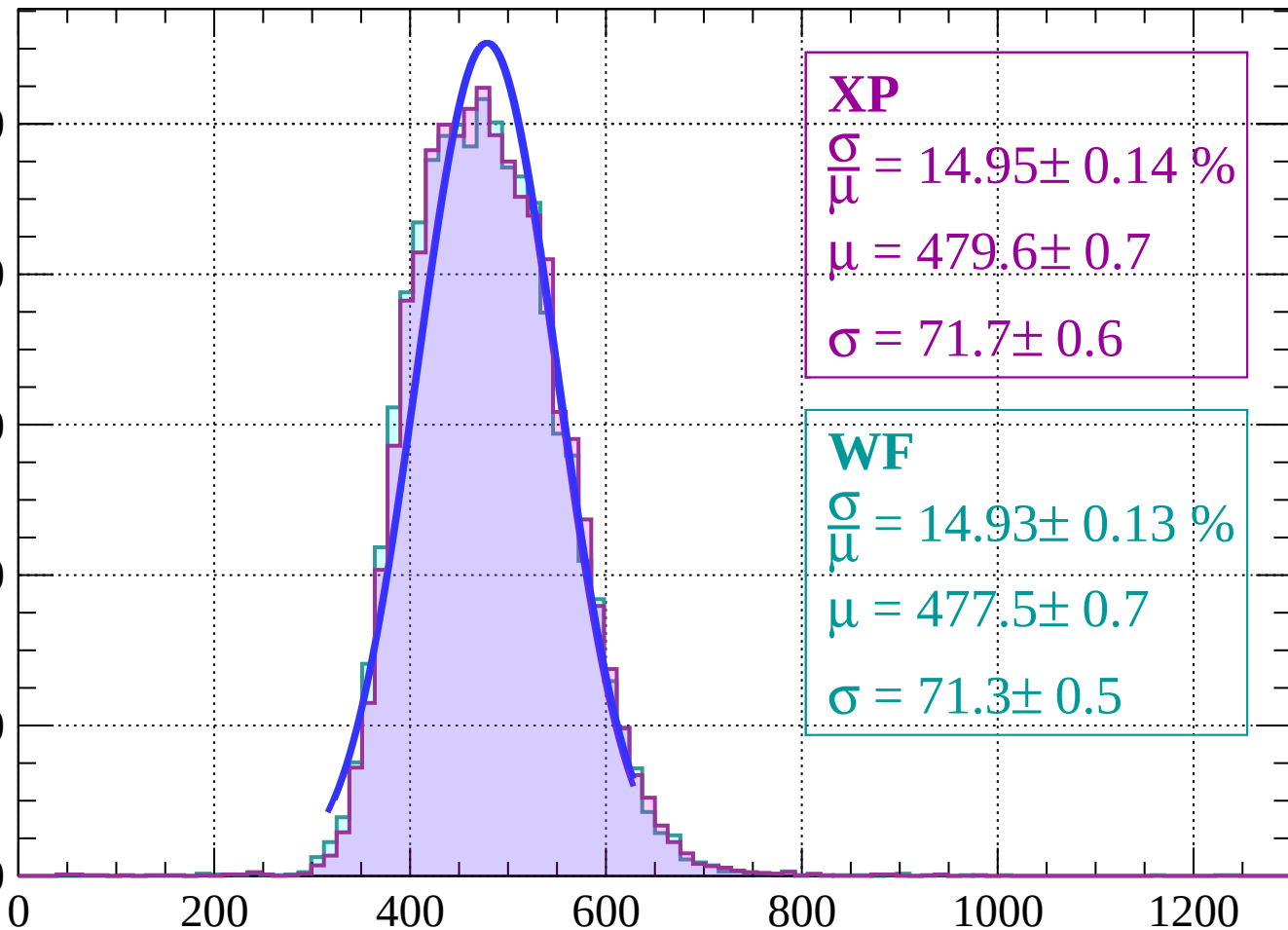
800

600

400

200

0



XP

$$\frac{\sigma}{\mu} = 14.95 \pm 0.14 \%$$

$$\mu = 479.6 \pm 0.7$$

$$\sigma = 71.7 \pm 0.6$$

WF

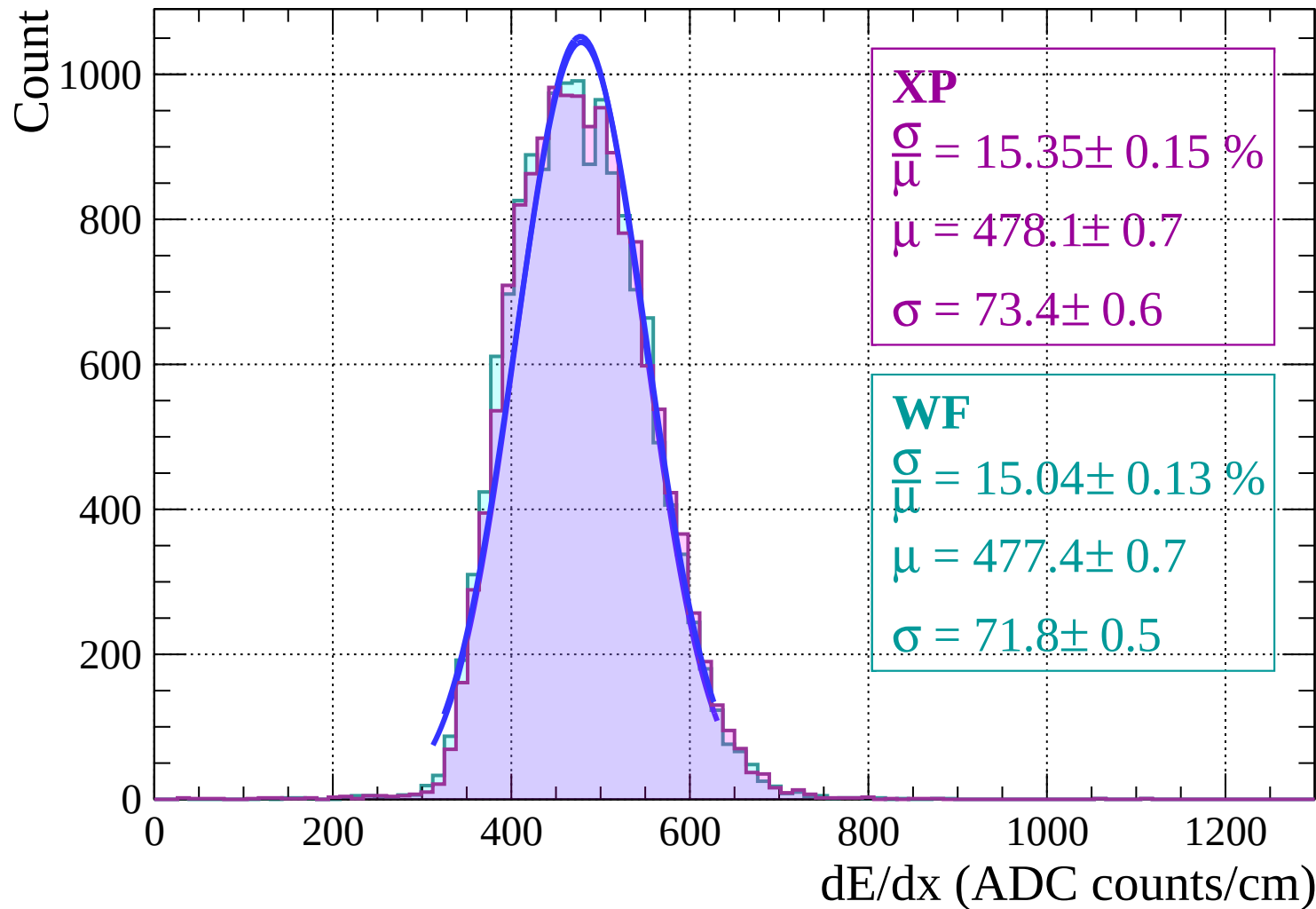
$$\frac{\sigma}{\mu} = 14.93 \pm 0.13 \%$$

$$\mu = 477.5 \pm 0.7$$

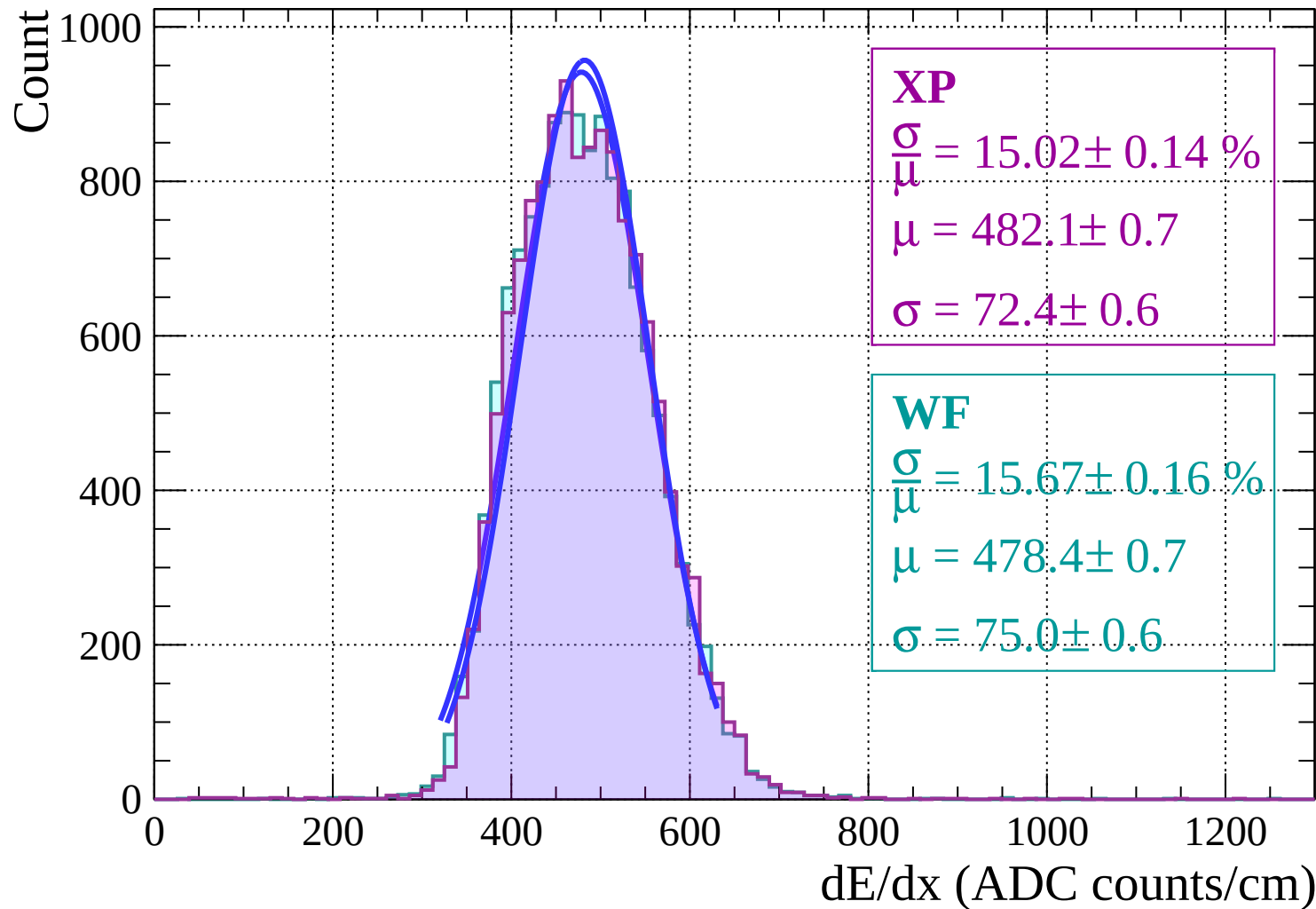
$$\sigma = 71.3 \pm 0.5$$

dE/dx (ADC counts/cm)

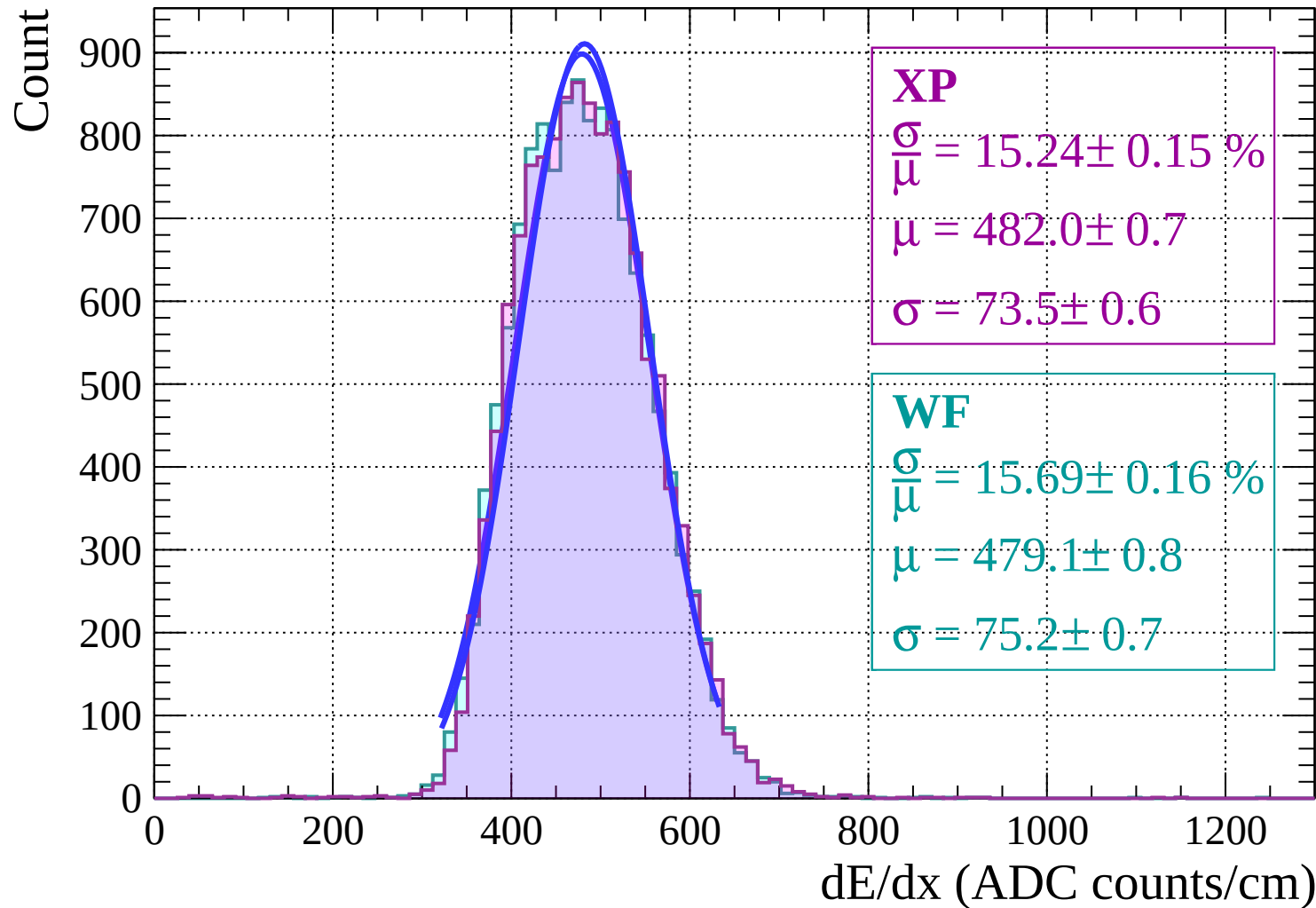
Energy loss | $12 < \theta < 14$



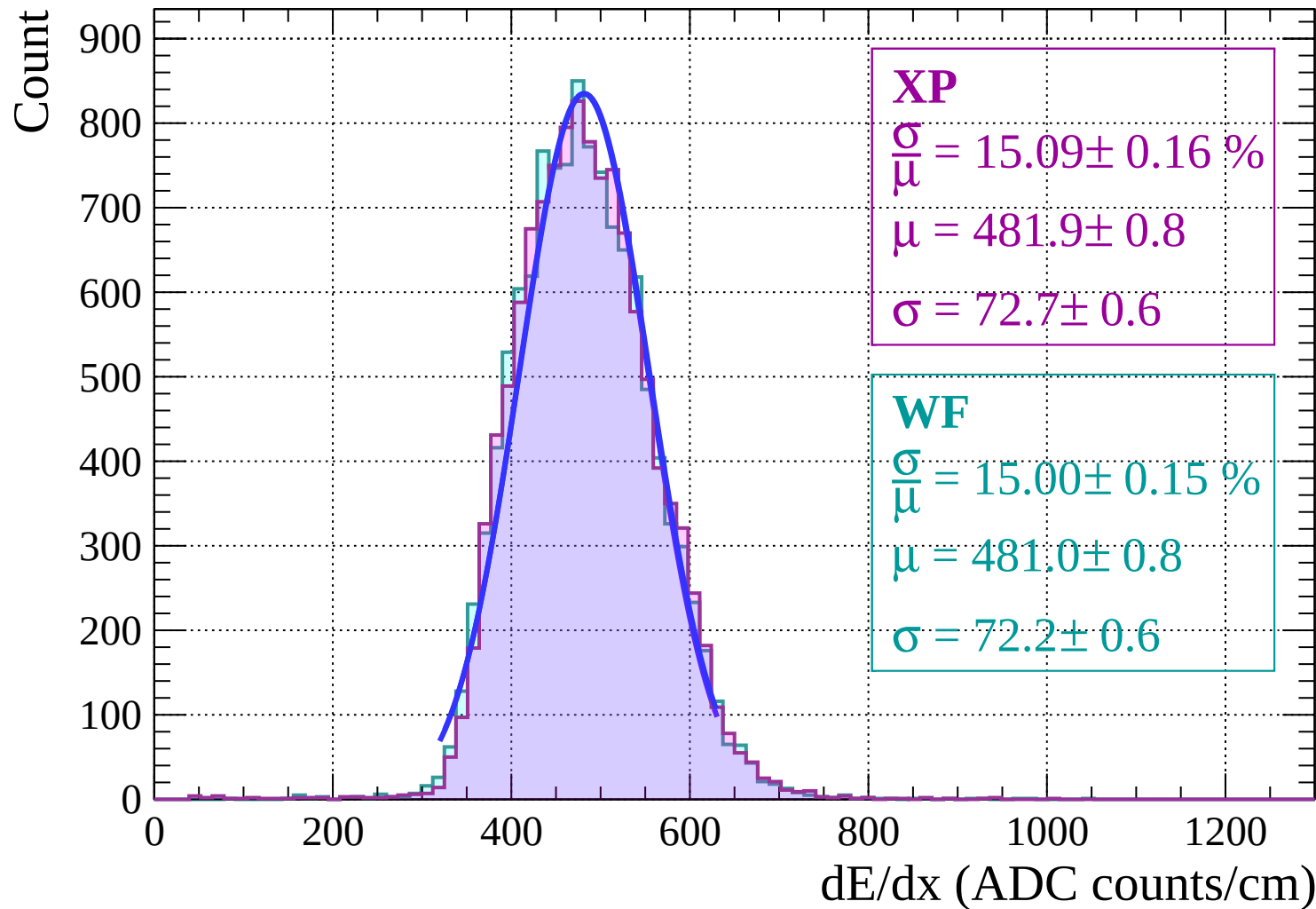
Energy loss | $14 < \theta < 16$



Energy loss | $16 < \theta < 18$

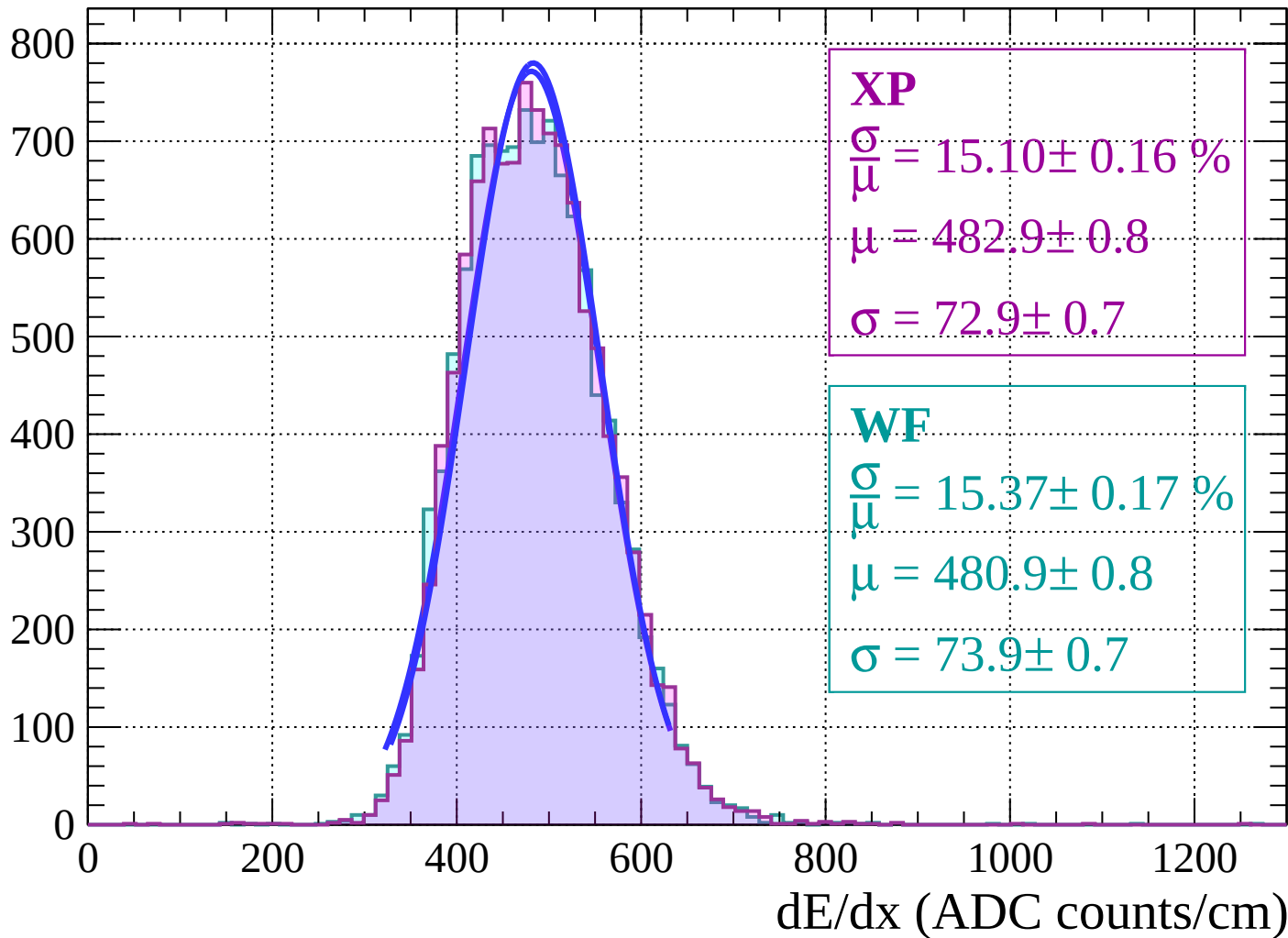


Energy loss | $18 < \theta < 20$



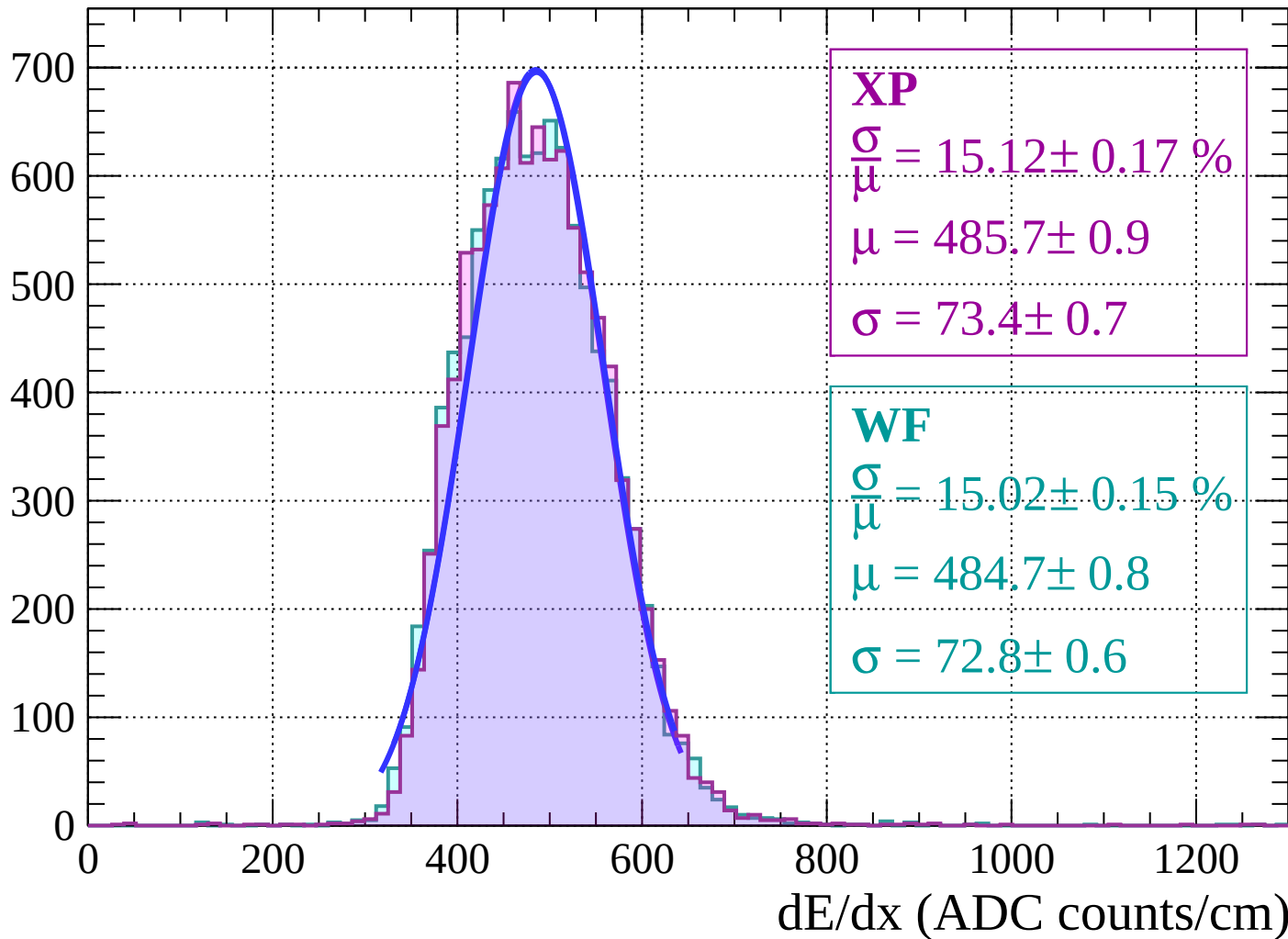
Energy loss | $20 < \theta < 22$

Count



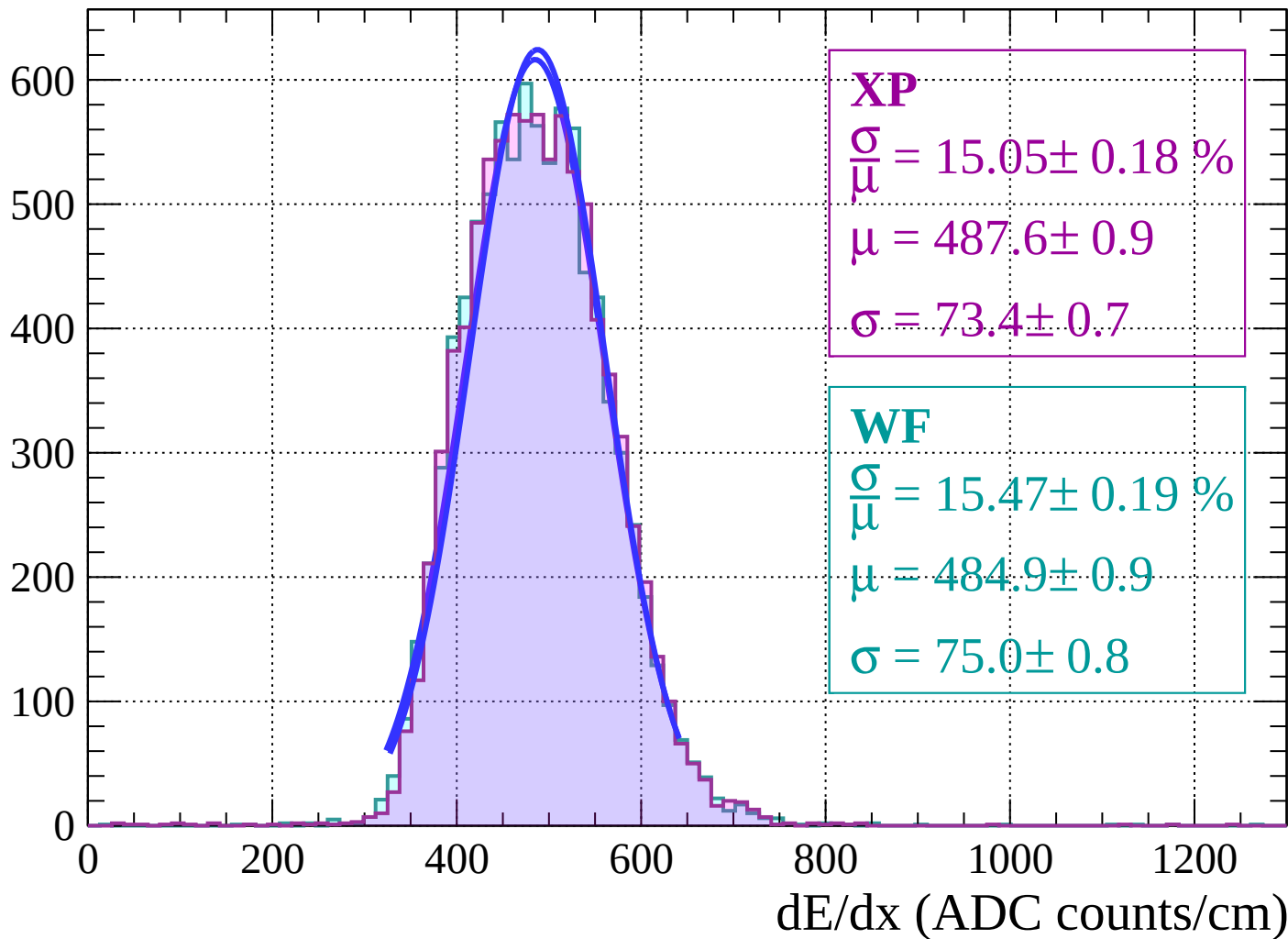
Energy loss | $22 < \theta < 24$

Count



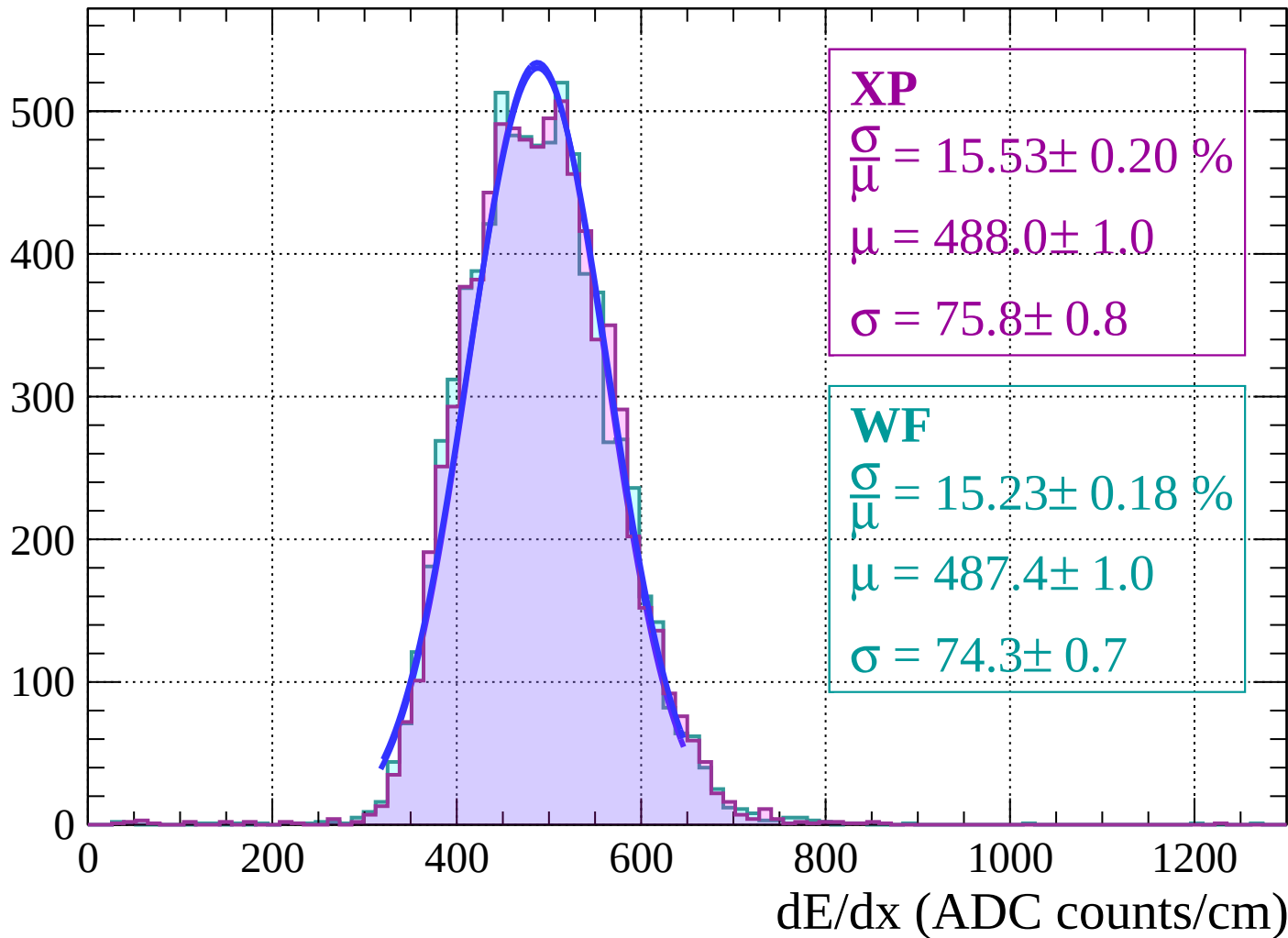
Energy loss | $24 < \theta < 26$

Count

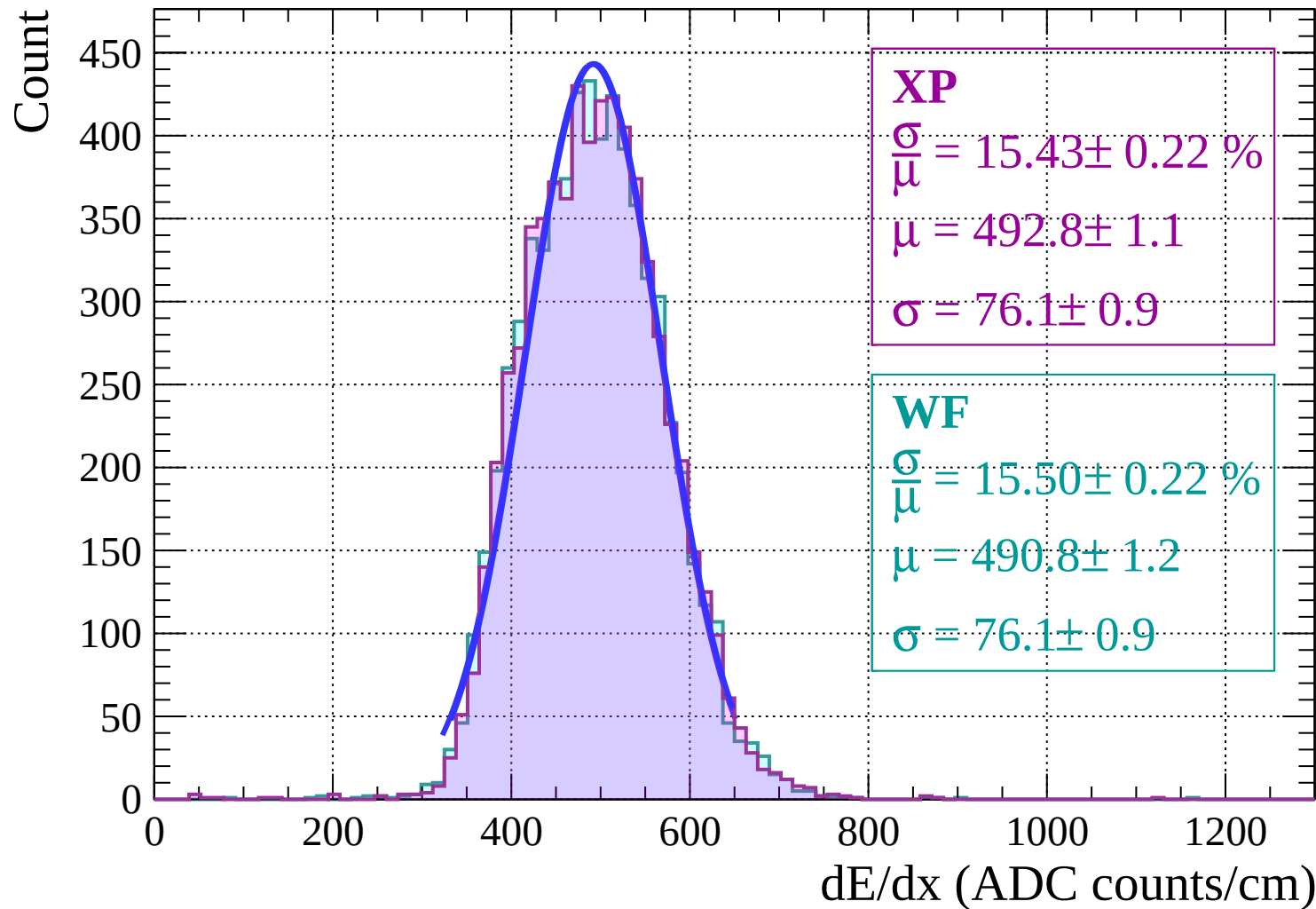


Energy loss | $26 < \theta < 28$

Count

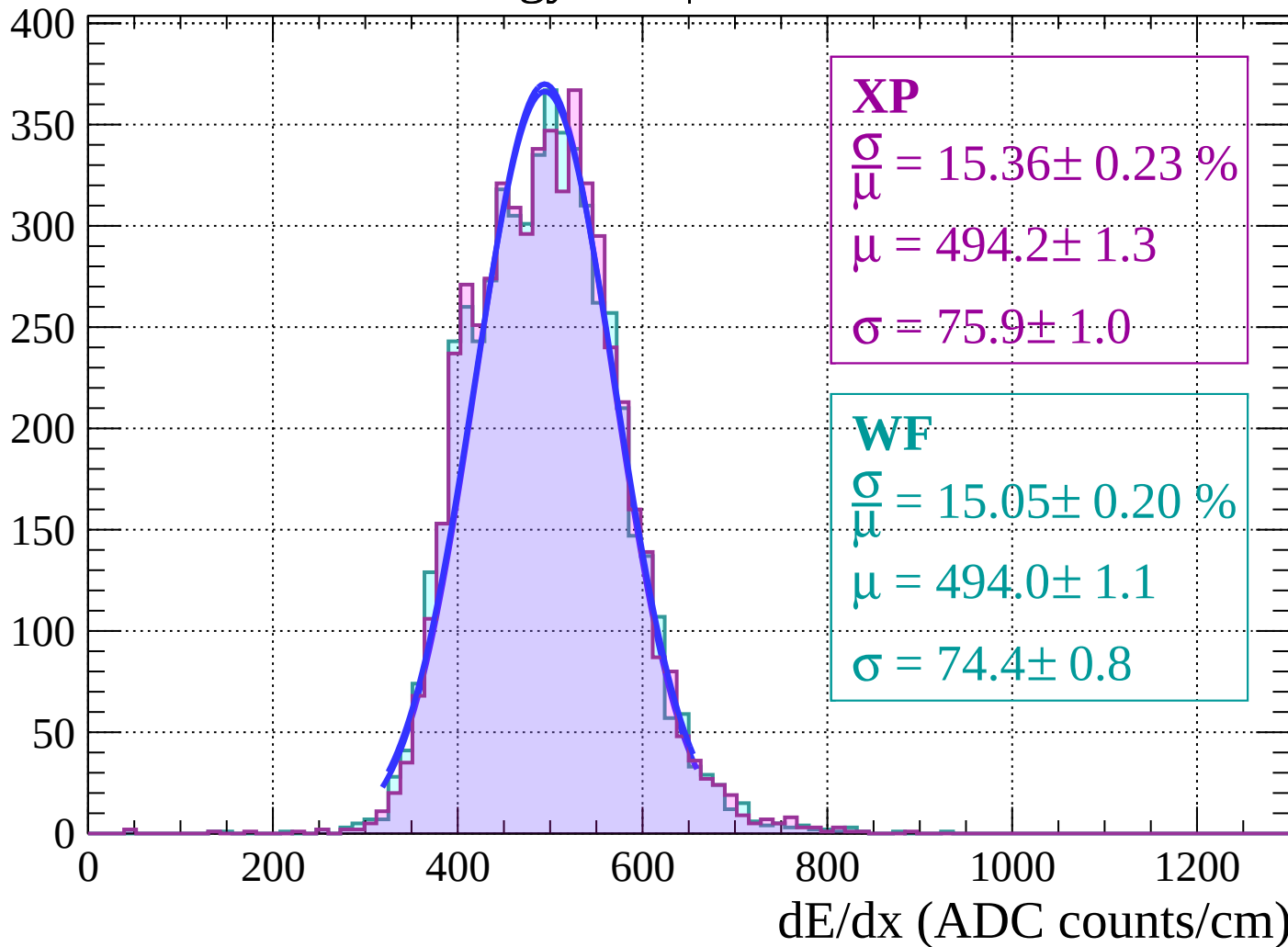


Energy loss | $28 < \theta < 30$



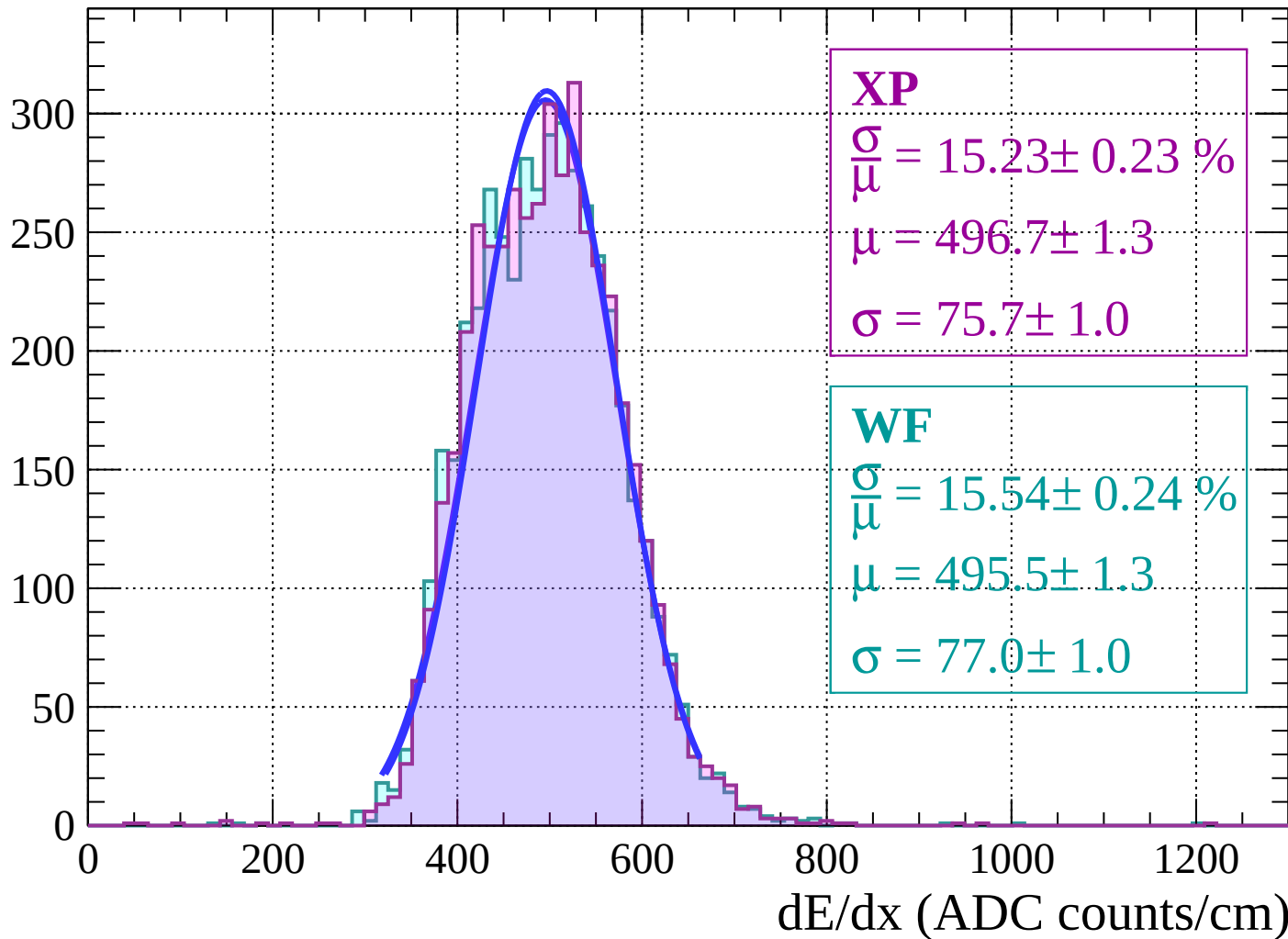
Energy loss | $30 < \theta < 32$

Count



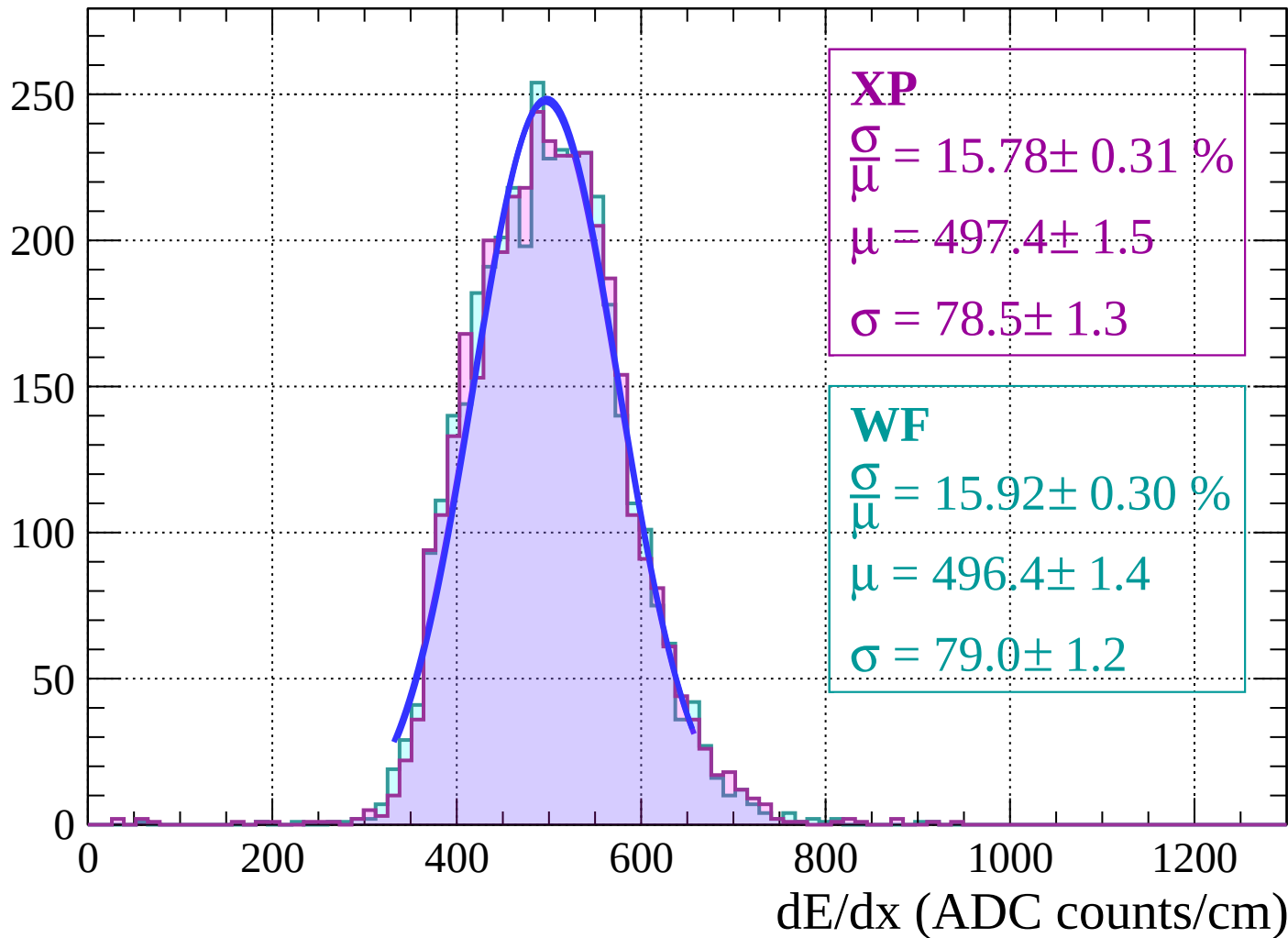
Energy loss | $32 < \theta < 34$

Count

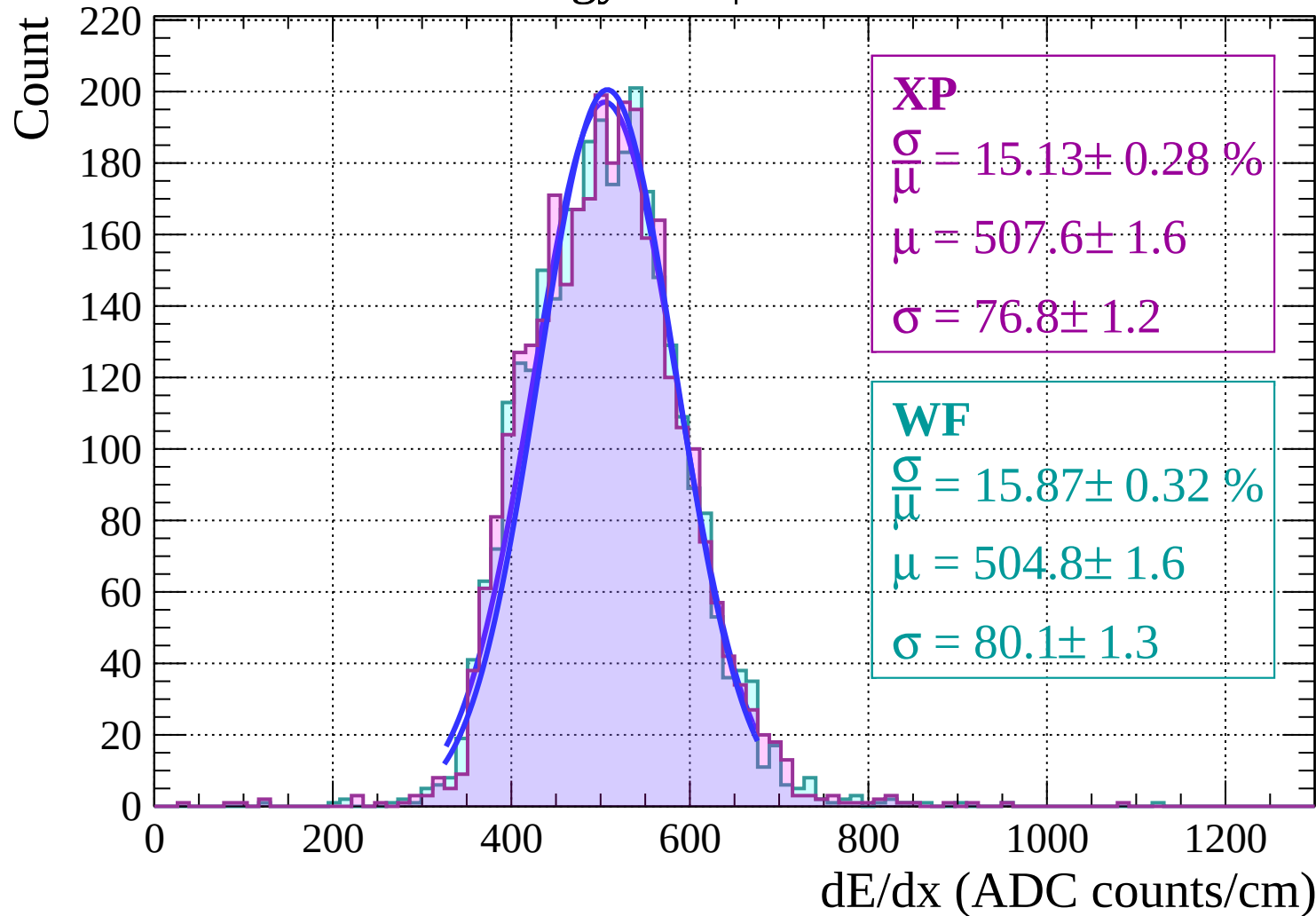


Energy loss | $34 < \theta < 36$

Count

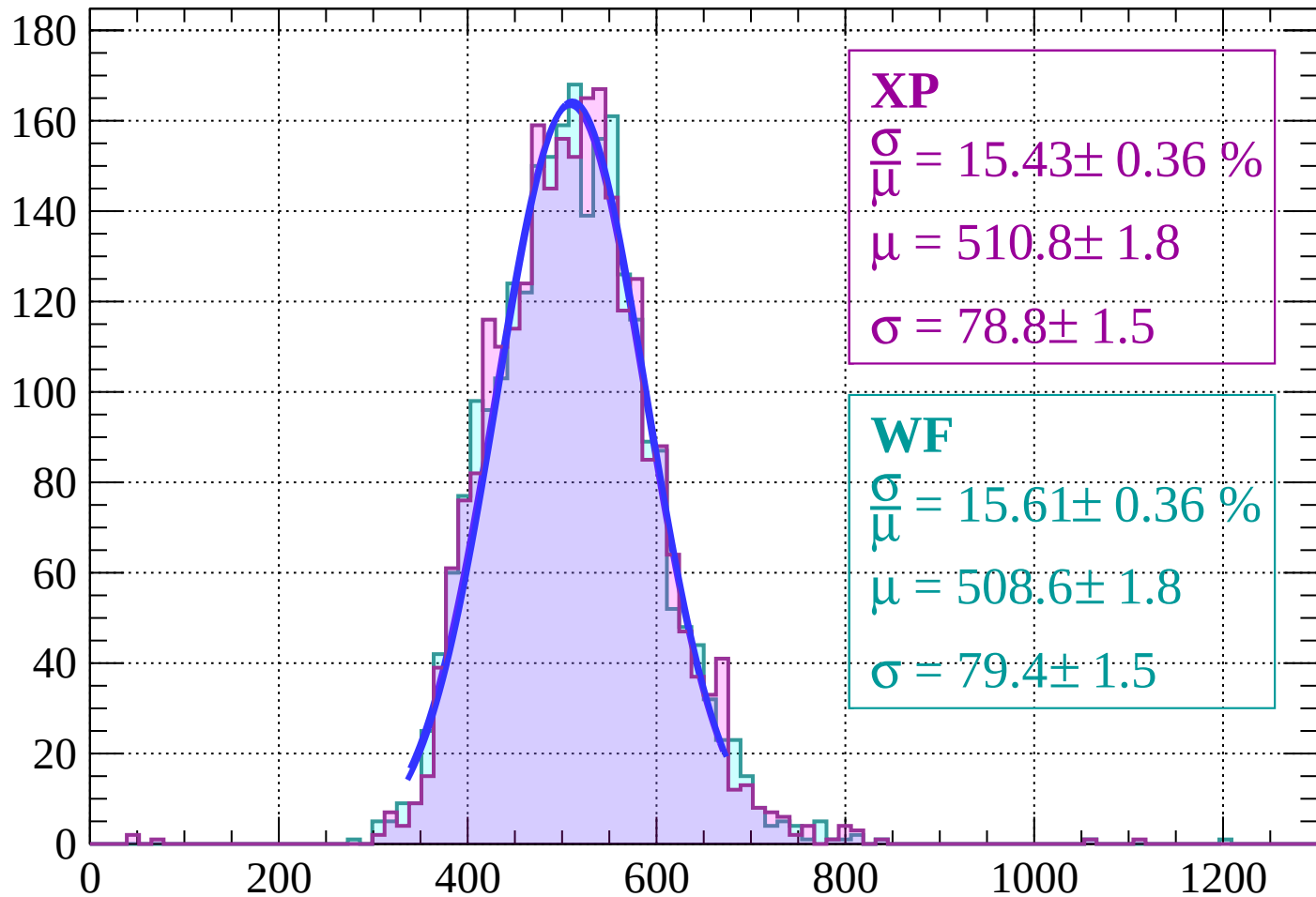


Energy loss | $36 < \theta < 38$



Energy loss | $38 < \theta < 40$

Count



XP

$$\frac{\sigma}{\mu} = 15.43 \pm 0.36 \%$$

$$\mu = 510.8 \pm 1.8$$

$$\sigma = 78.8 \pm 1.5$$

WF

$$\frac{\sigma}{\mu} = 15.61 \pm 0.36 \%$$

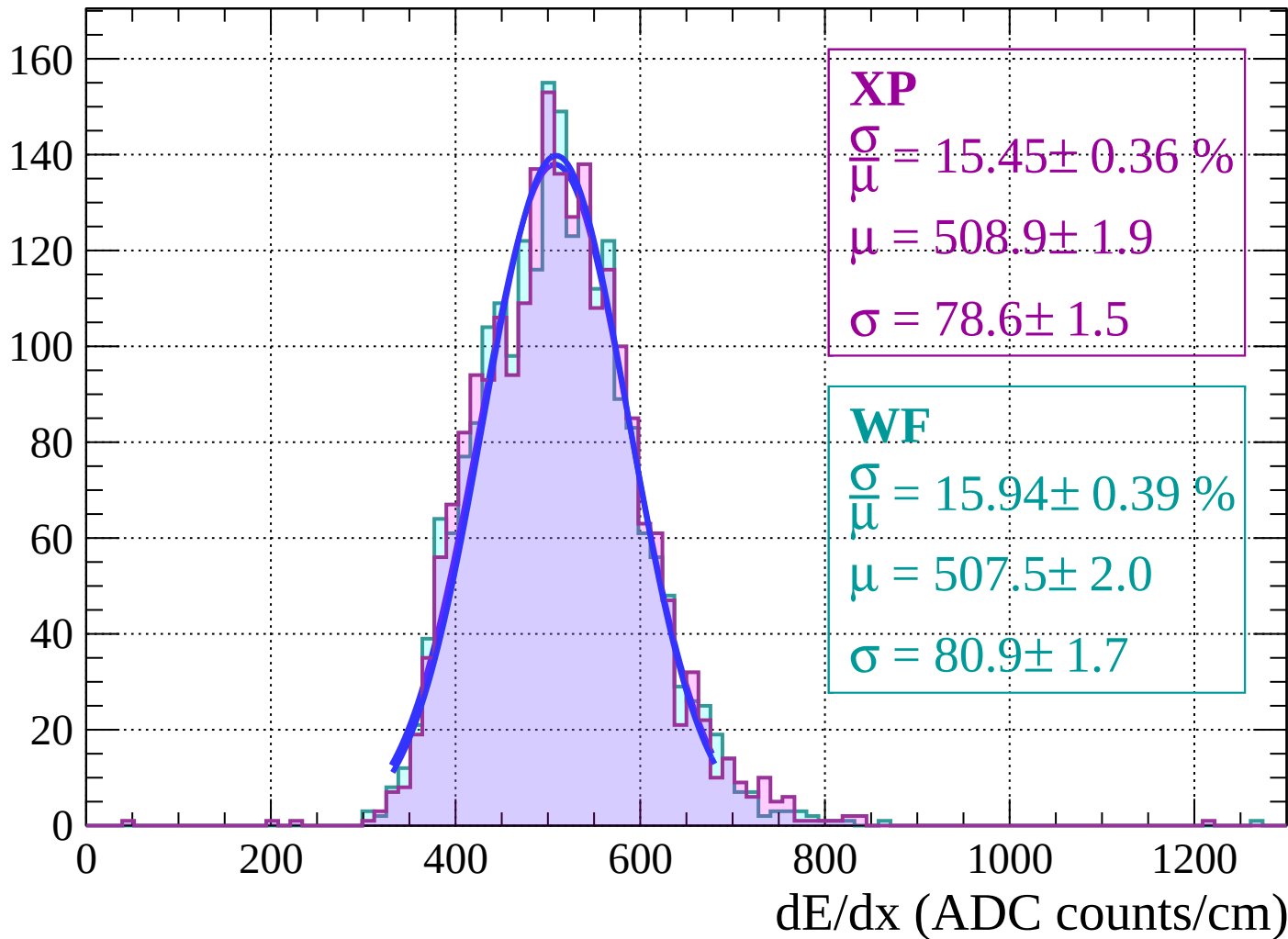
$$\mu = 508.6 \pm 1.8$$

$$\sigma = 79.4 \pm 1.5$$

dE/dx (ADC counts/cm)

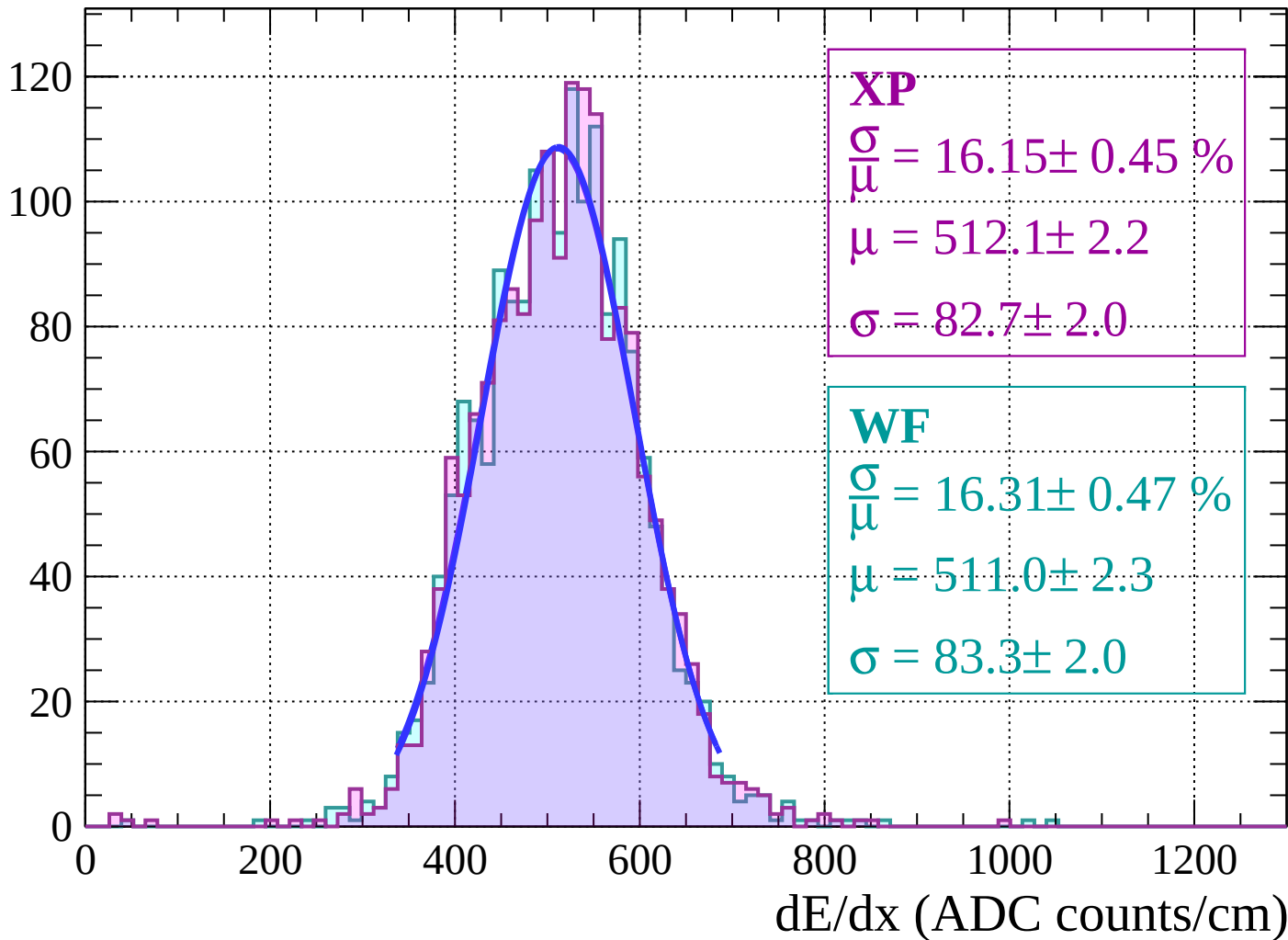
Energy loss | $40 < \theta < 42$

Count



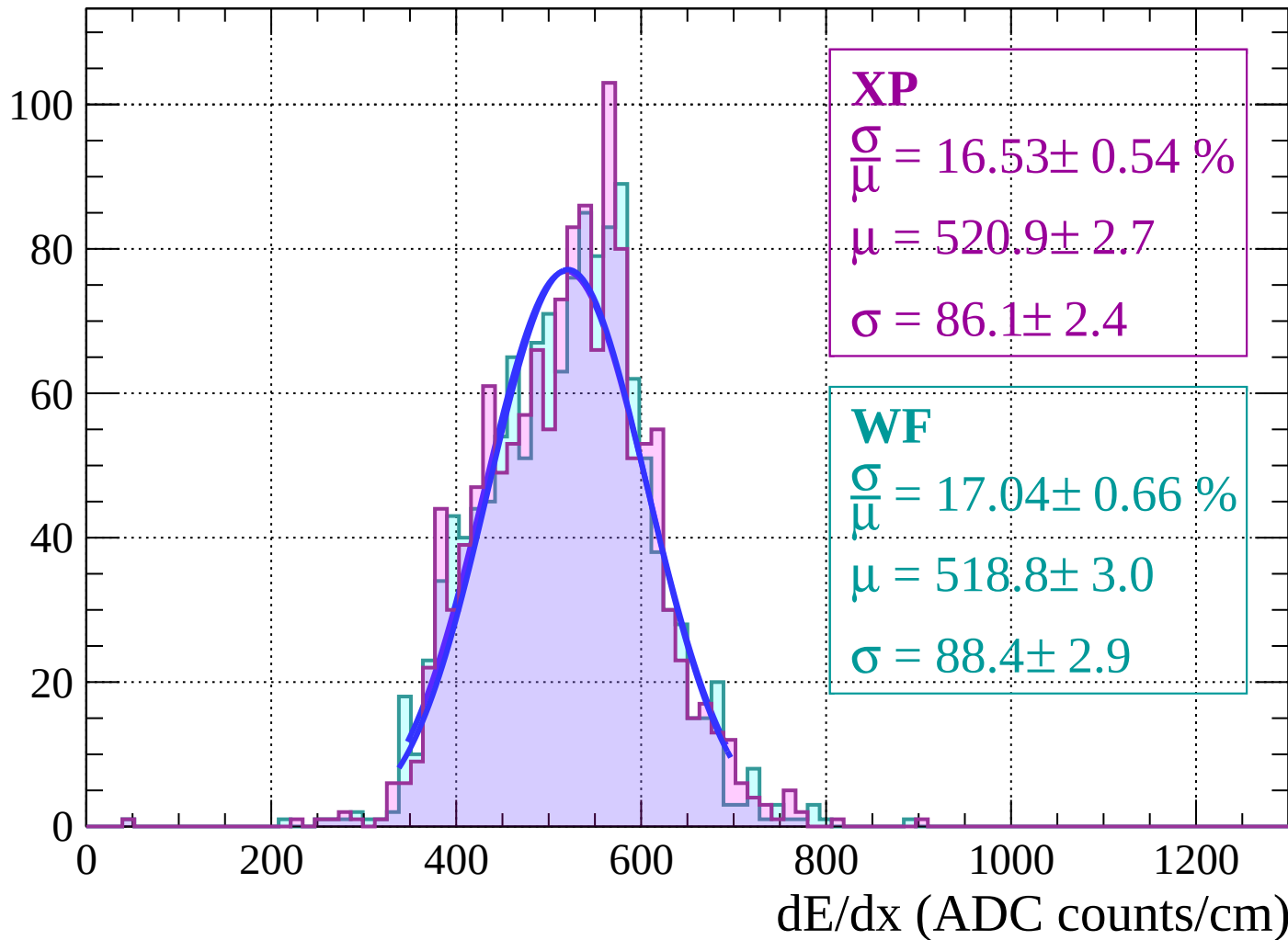
Energy loss | $42 < \theta < 44$

Count



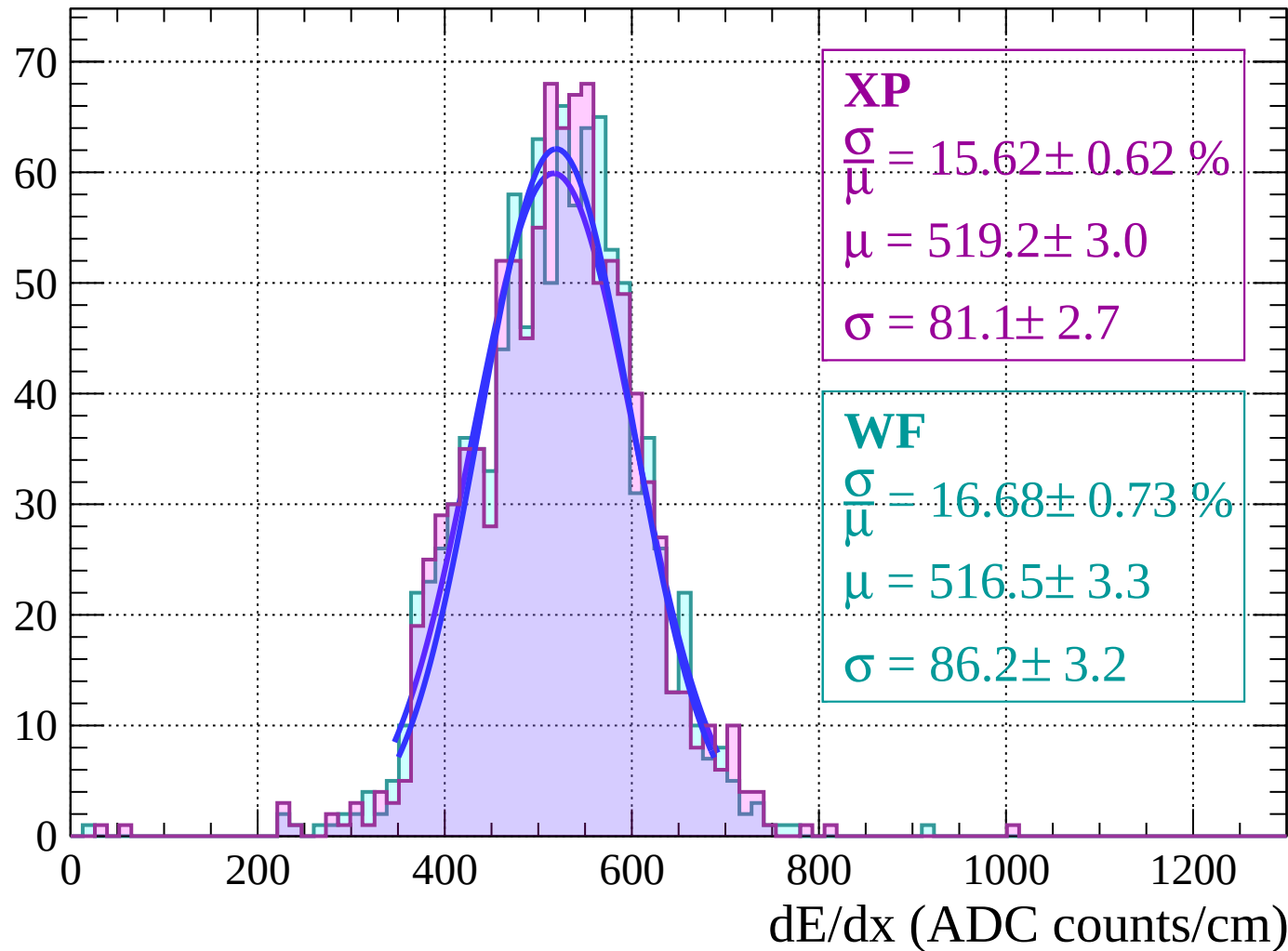
Energy loss | $44 < \theta < 46$

Count



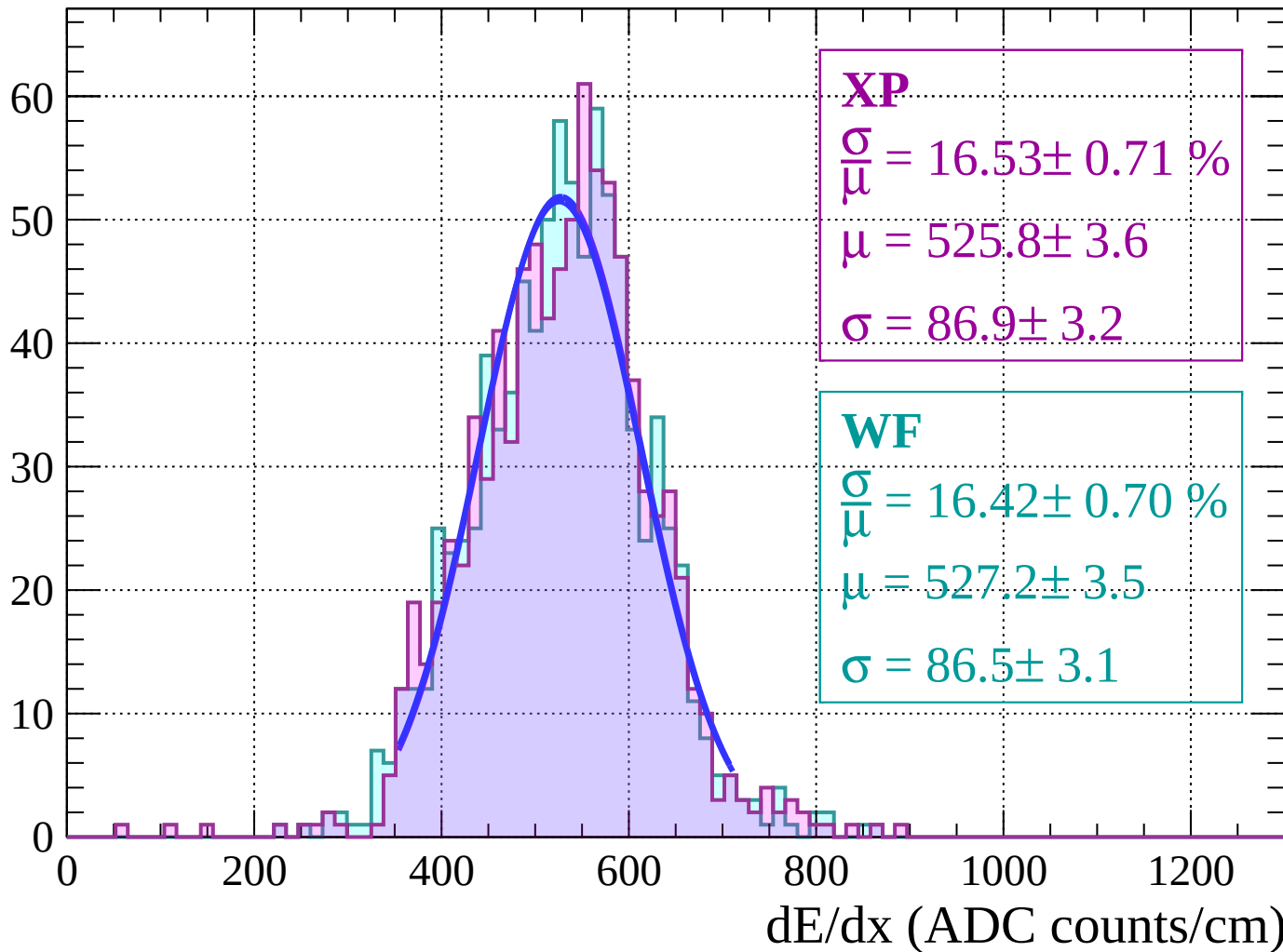
Energy loss | $46 < \theta < 48$

Count



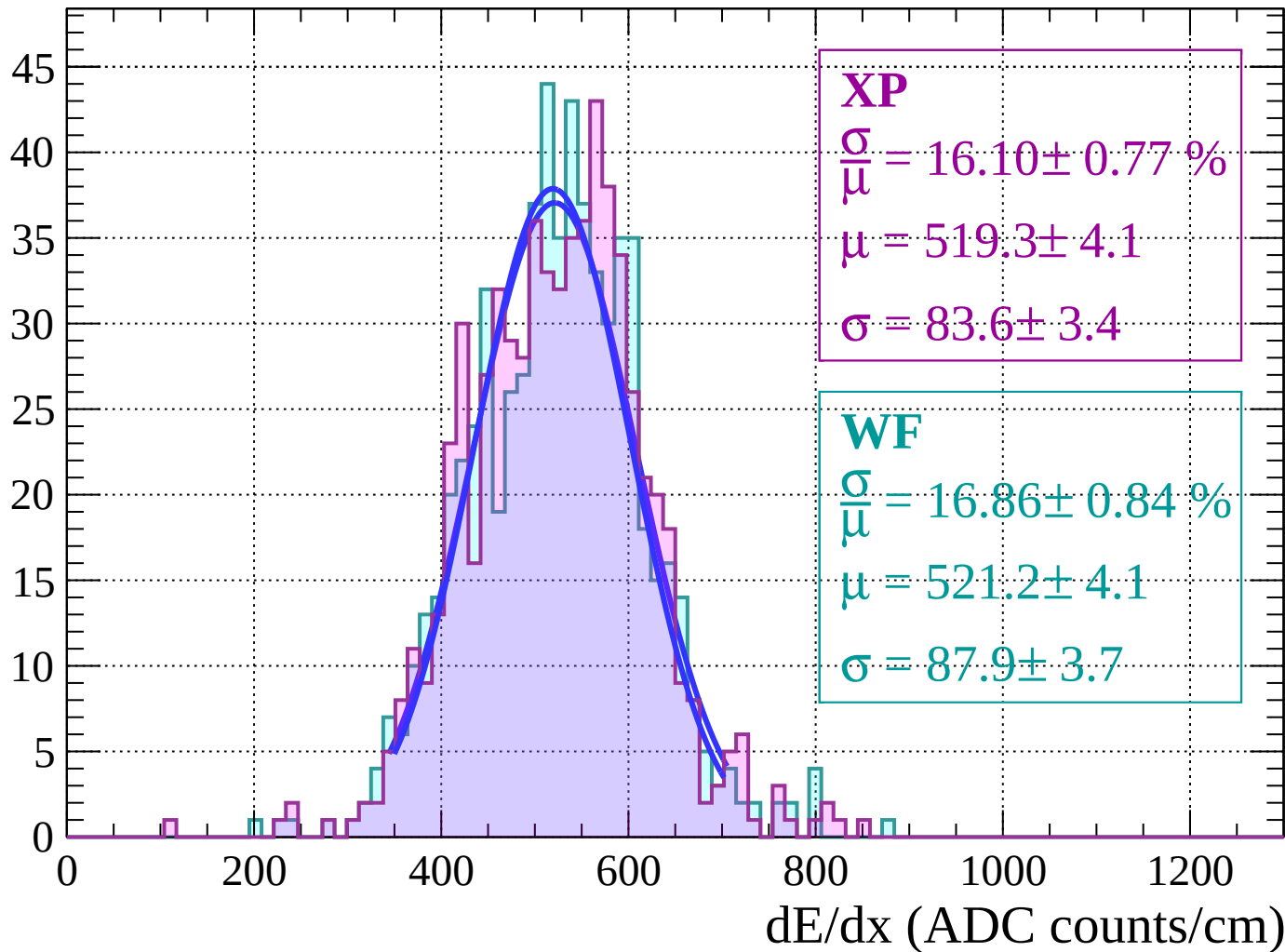
Energy loss | $48 < \theta < 50$

Count



Energy loss | $50 < \theta < 52$

Count



Energy loss | $52 < \theta < 54$

Count

XP

$$\frac{\sigma}{\mu} = 15.16 \pm 0.99 \%$$

$$\mu = 532.6 \pm 4.5$$

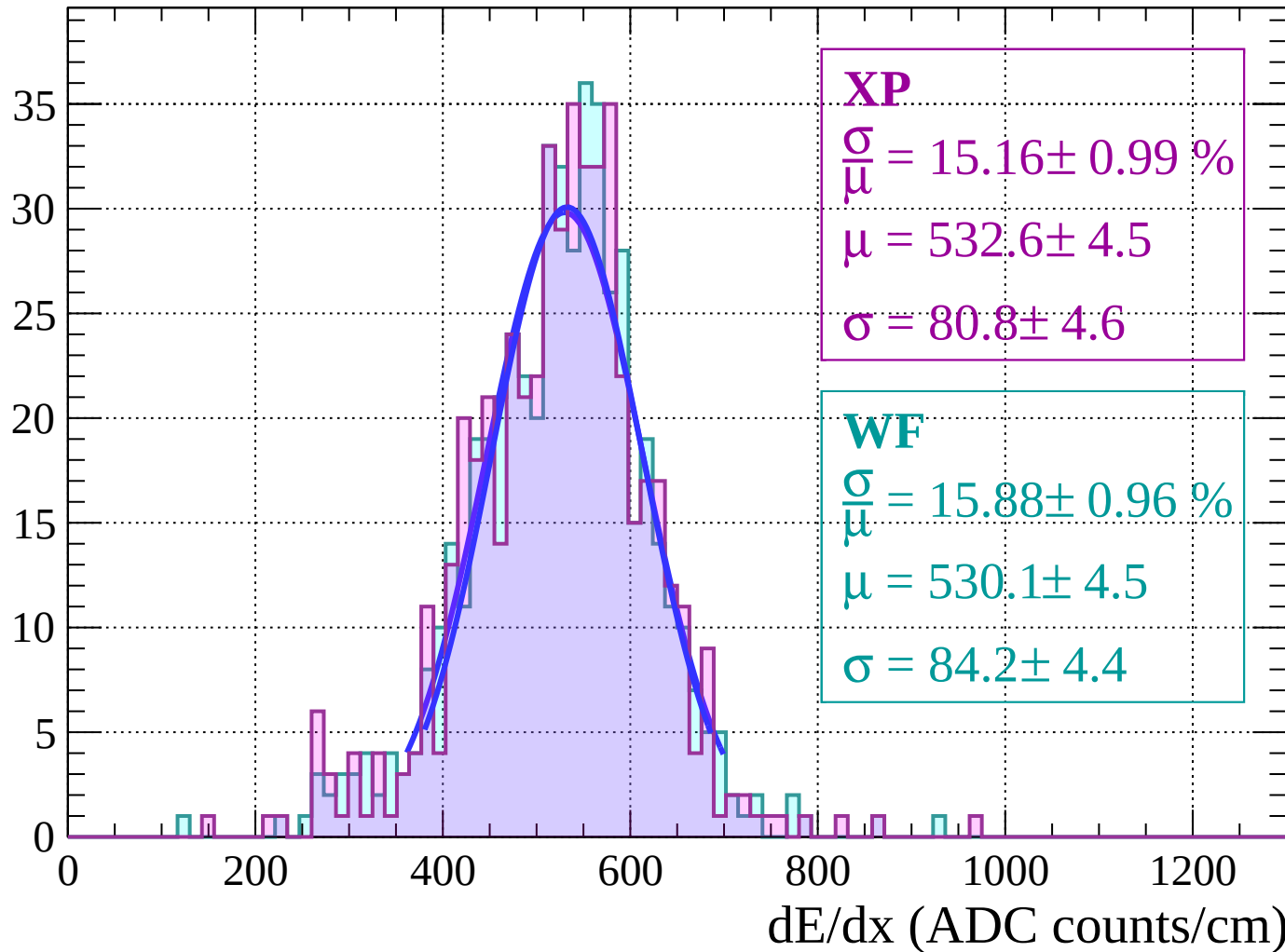
$$\sigma = 80.8 \pm 4.6$$

WF

$$\frac{\sigma}{\mu} = 15.88 \pm 0.96 \%$$

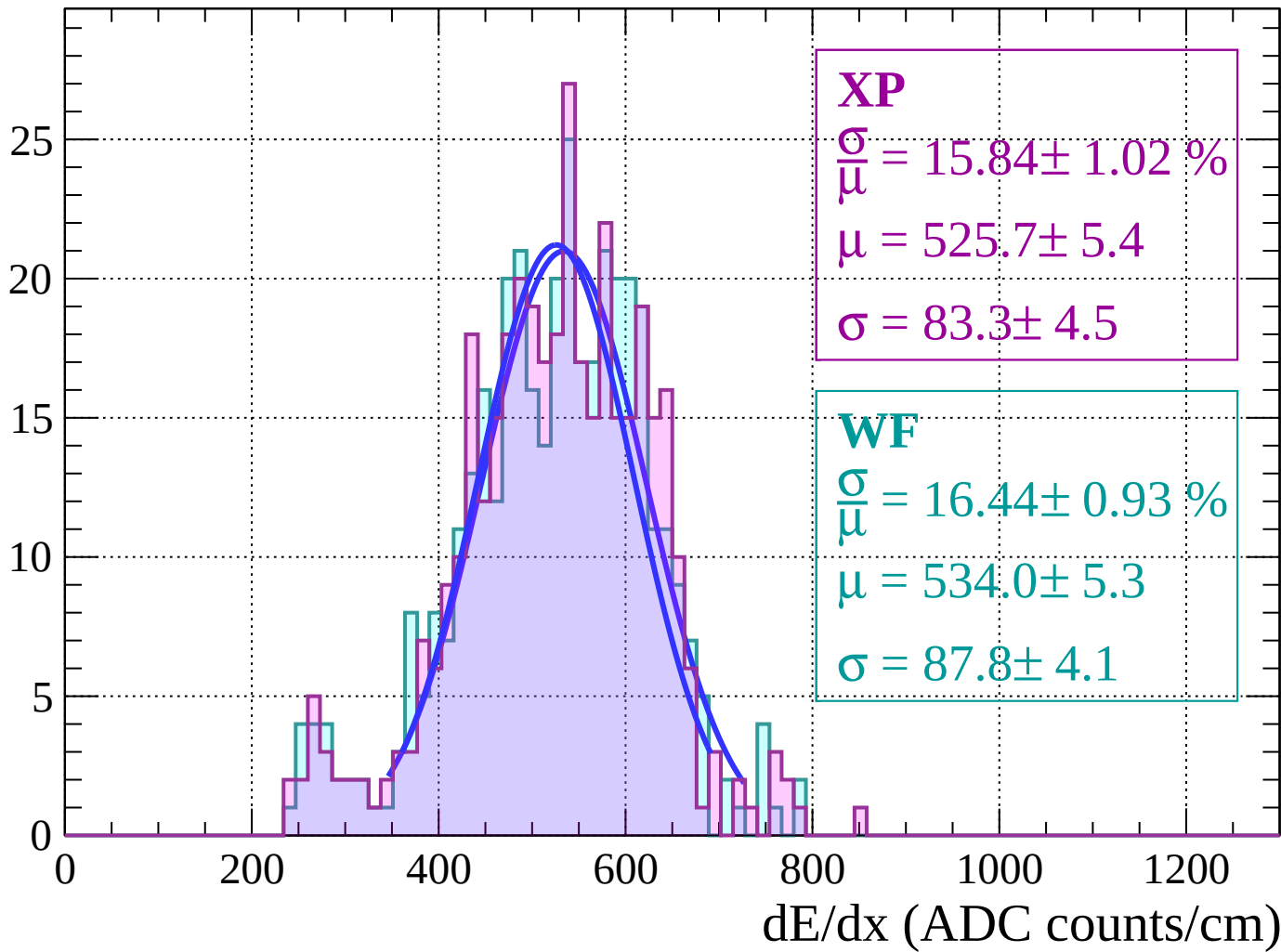
$$\mu = 530.1 \pm 4.5$$

$$\sigma = 84.2 \pm 4.4$$



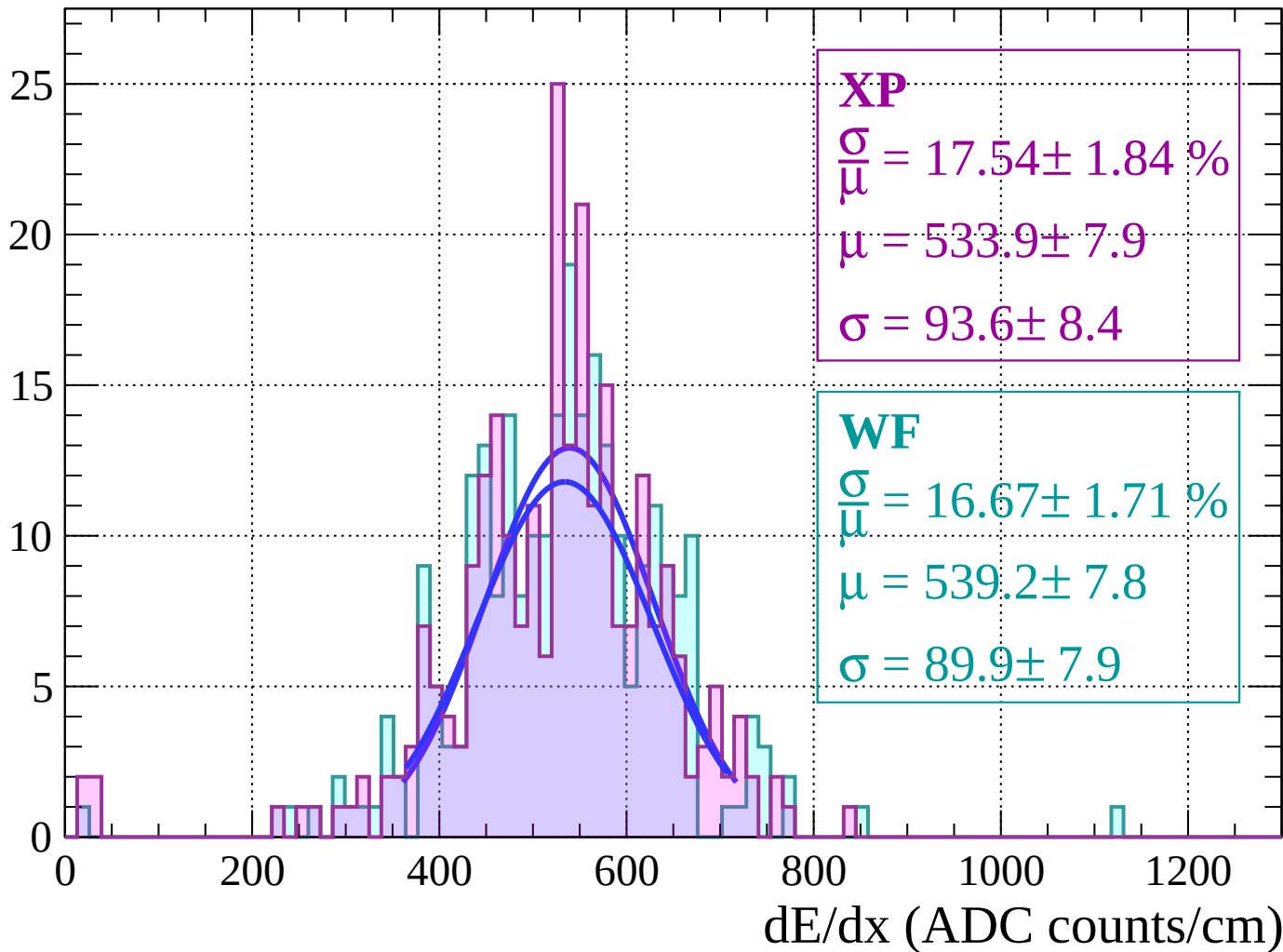
Energy loss | $54 < \theta < 56$

Count



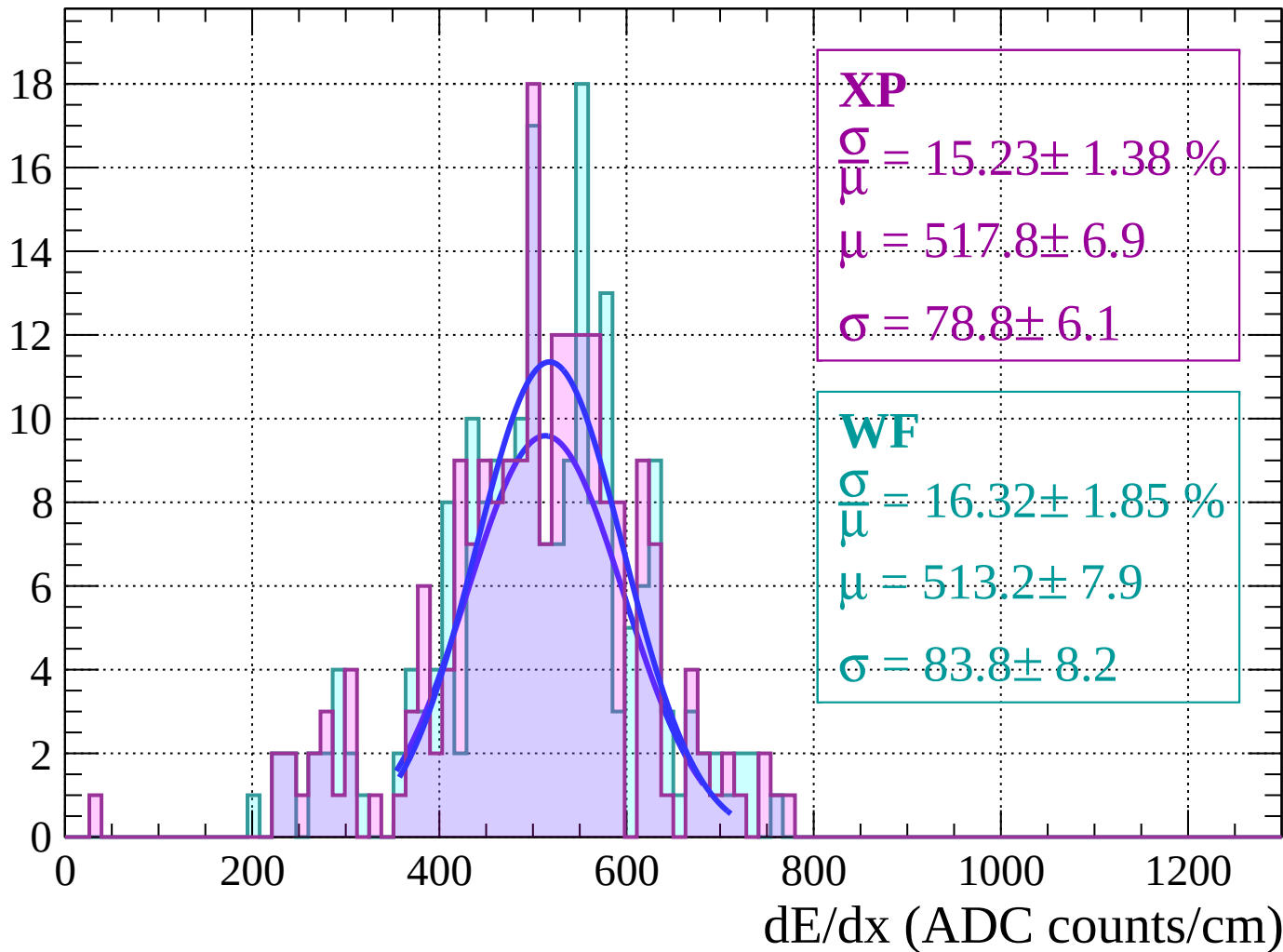
Energy loss | $56 < \theta < 58$

Count



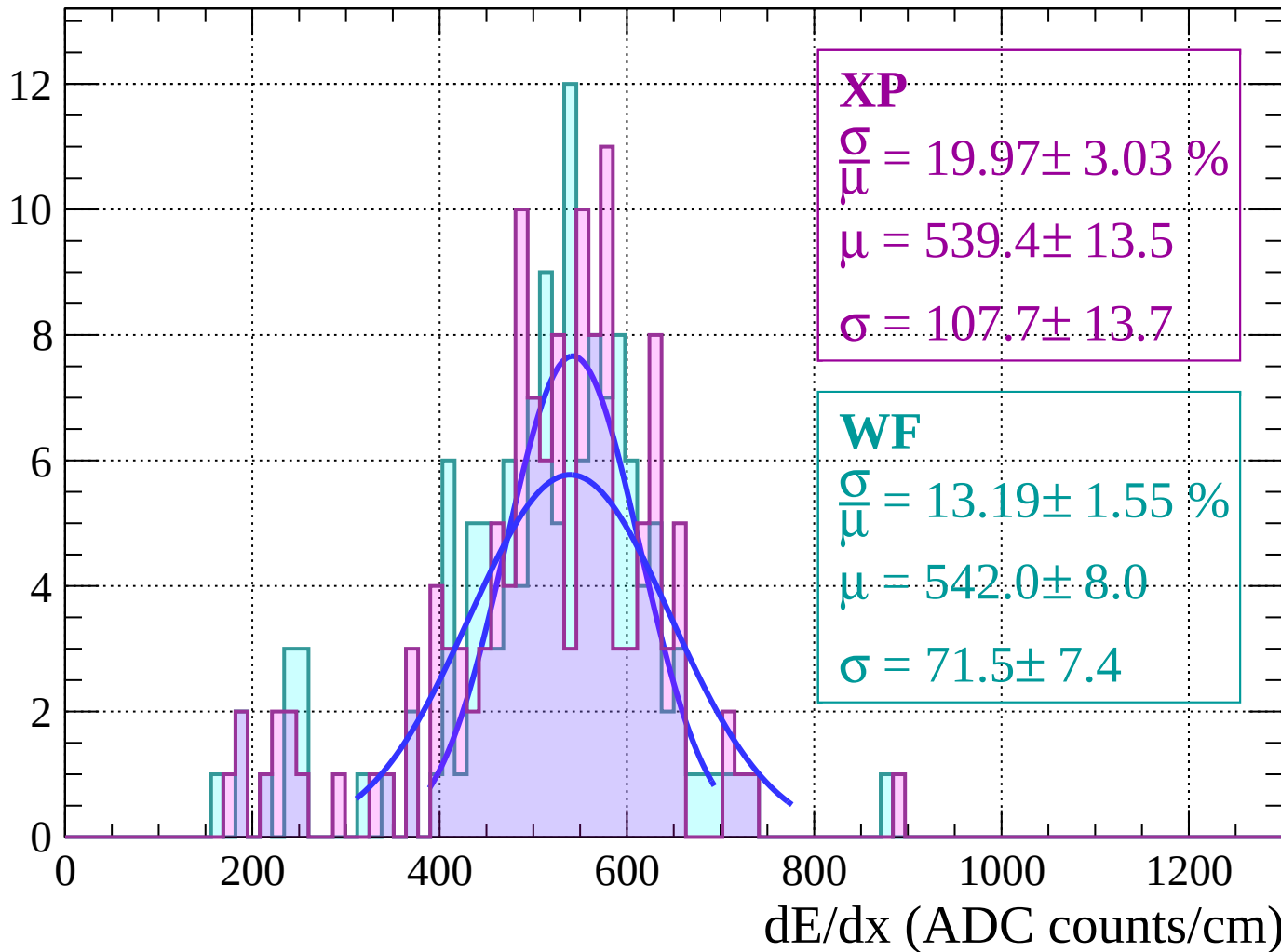
Energy loss | $58 < \theta < 60$

Count



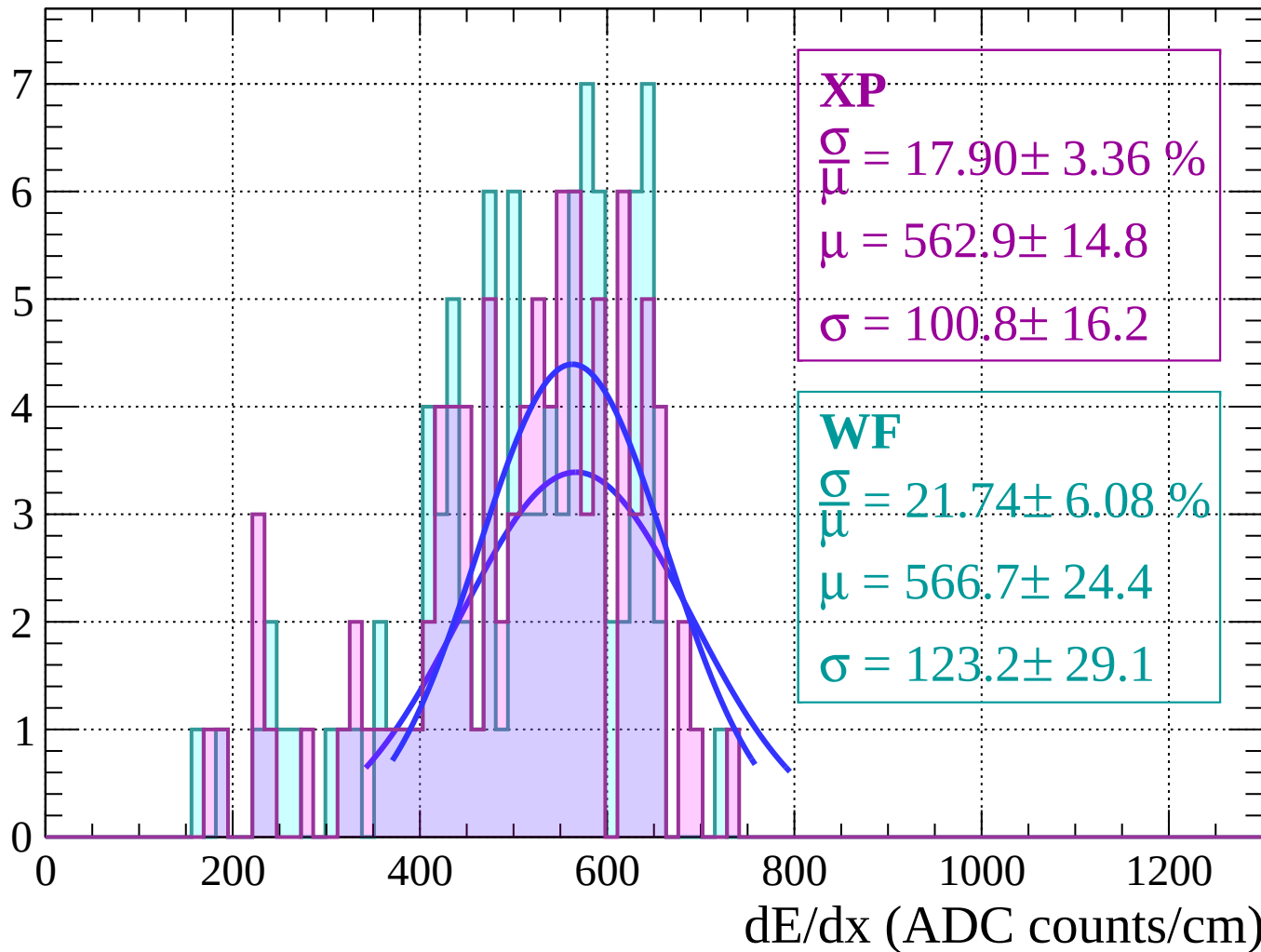
Energy loss | $60 < \theta < 62$

Count



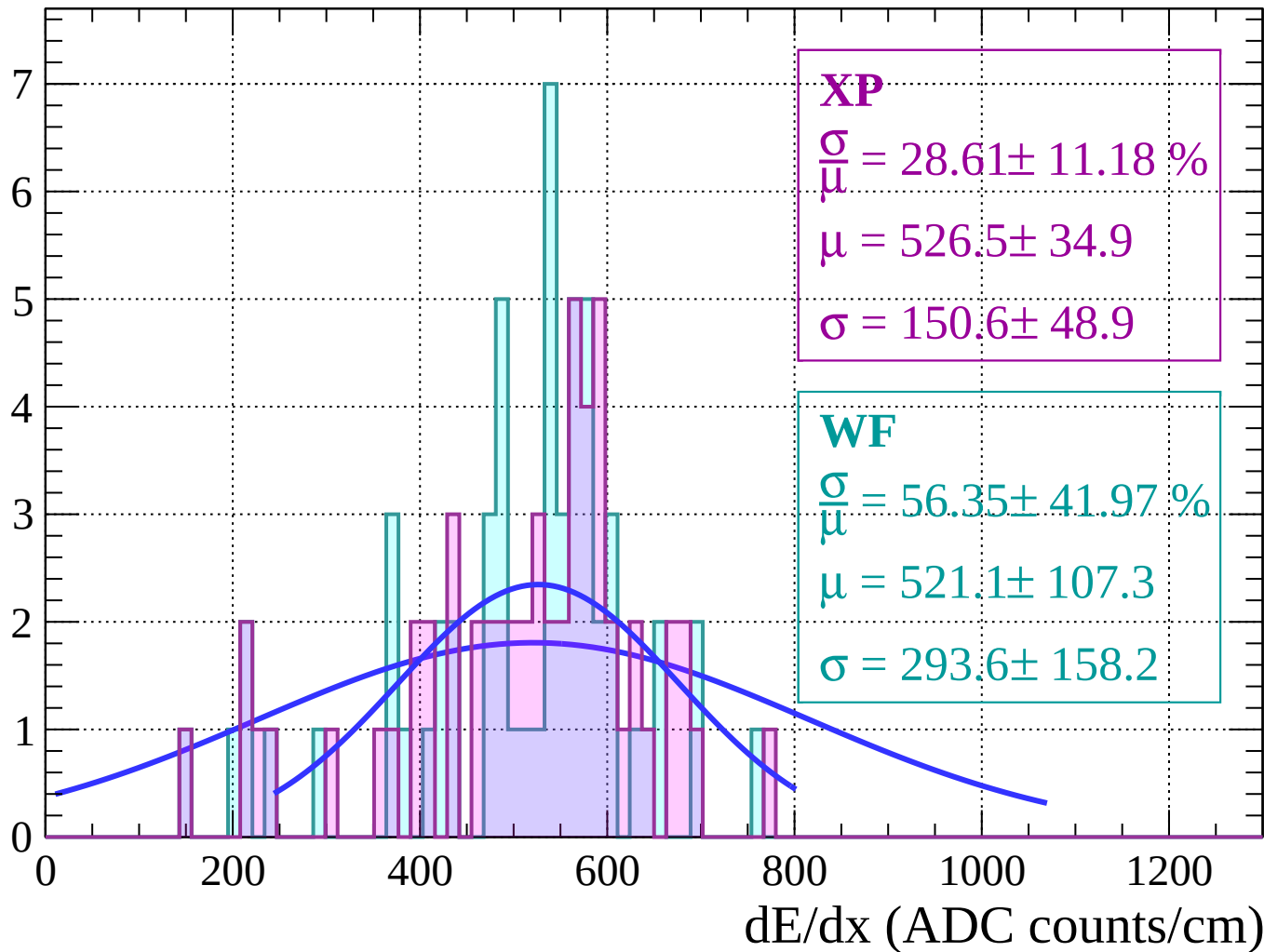
Energy loss | $62 < \theta < 64$

Count



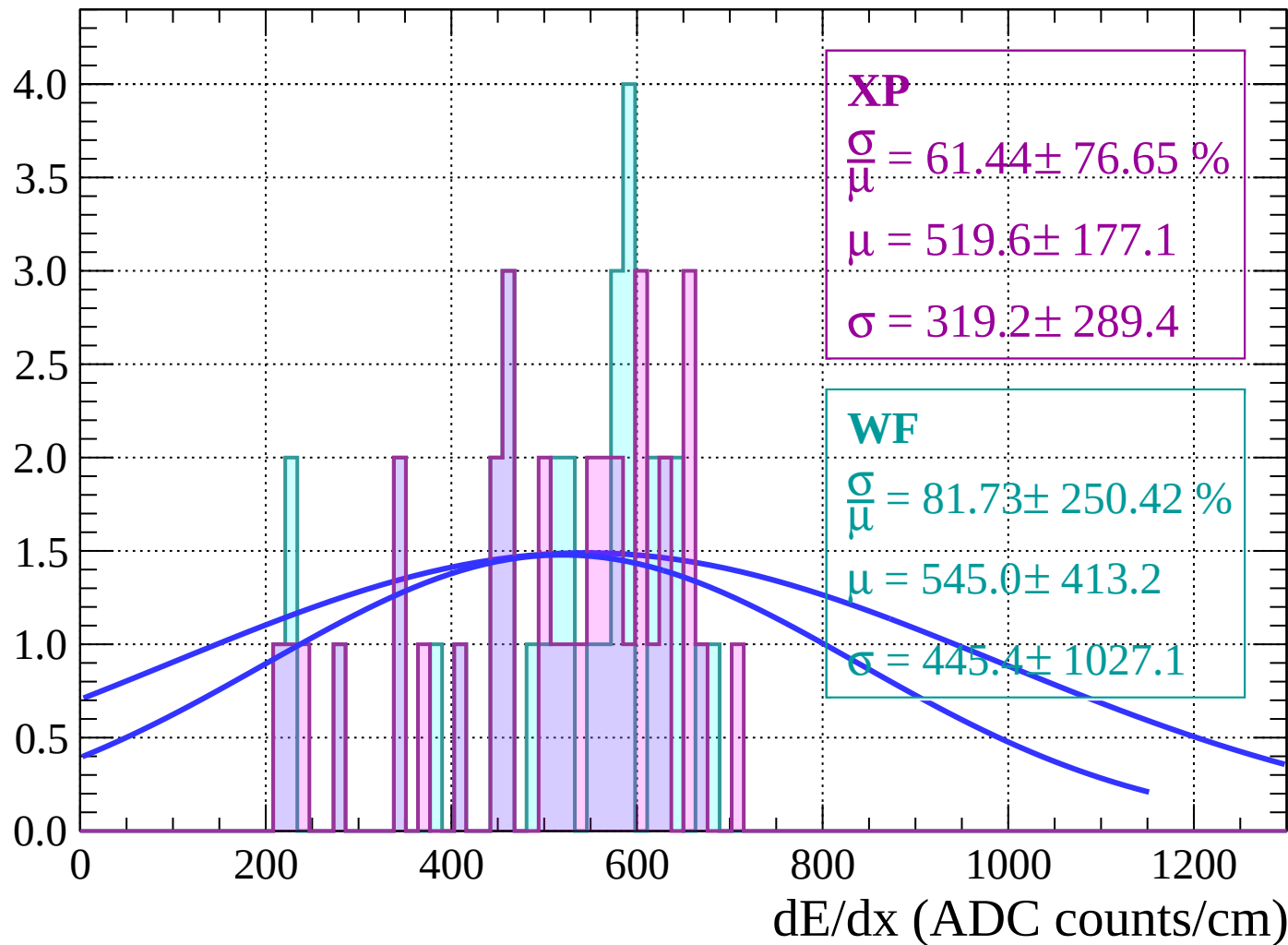
Energy loss | $64 < \theta < 66$

Count



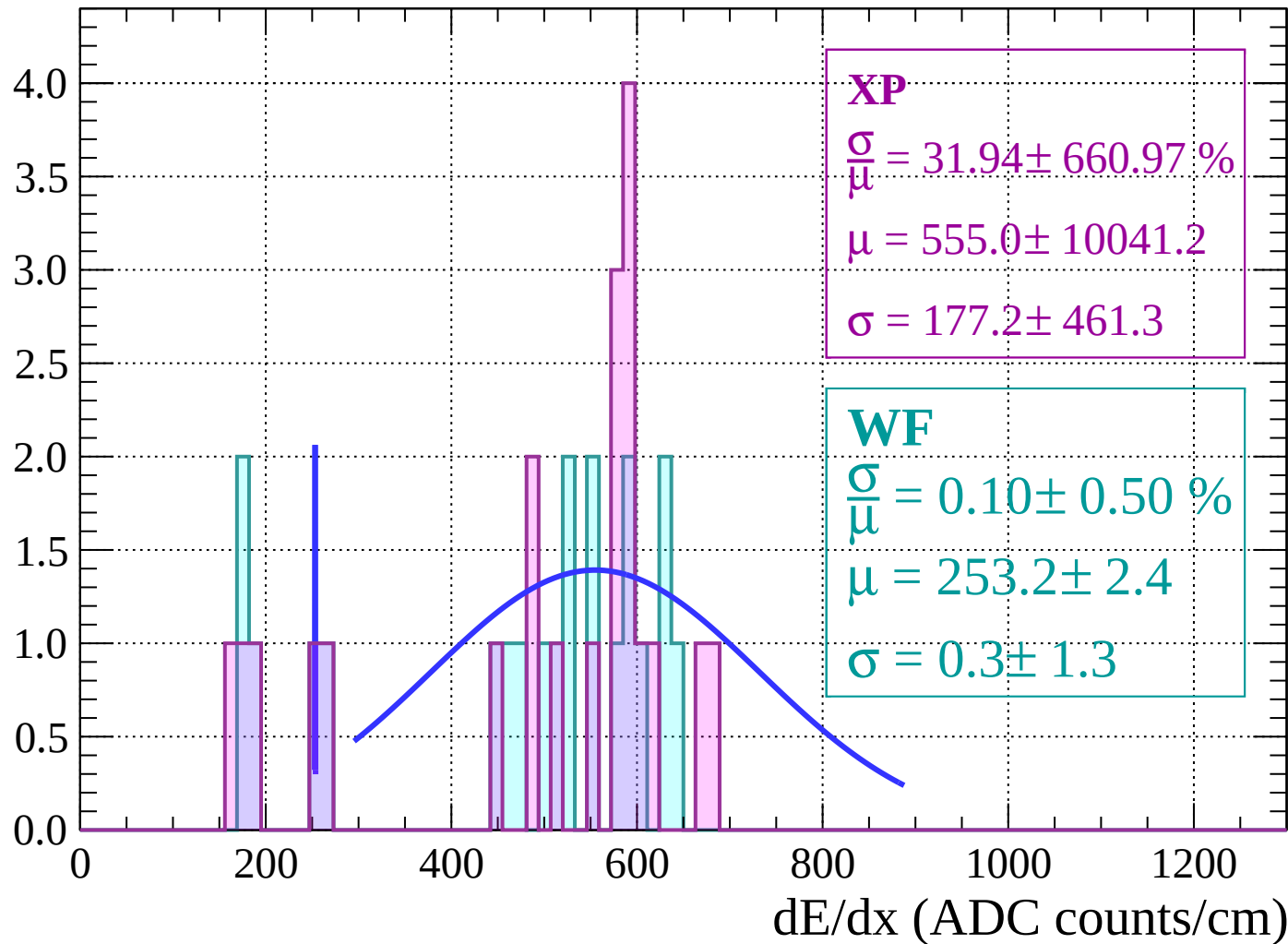
Energy loss | $66 < \theta < 68$

Count



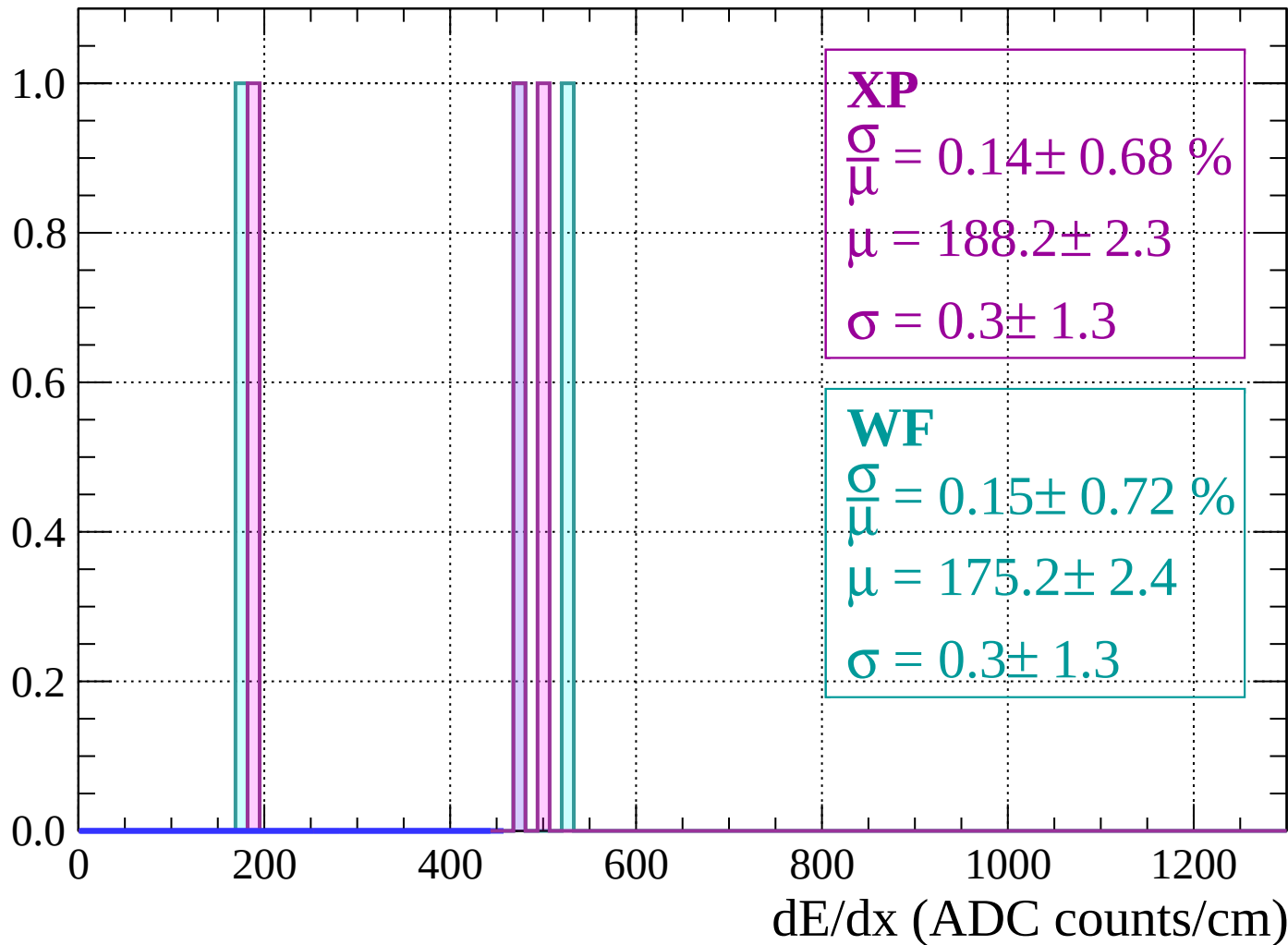
Energy loss | $68 < \theta < 70$

Count



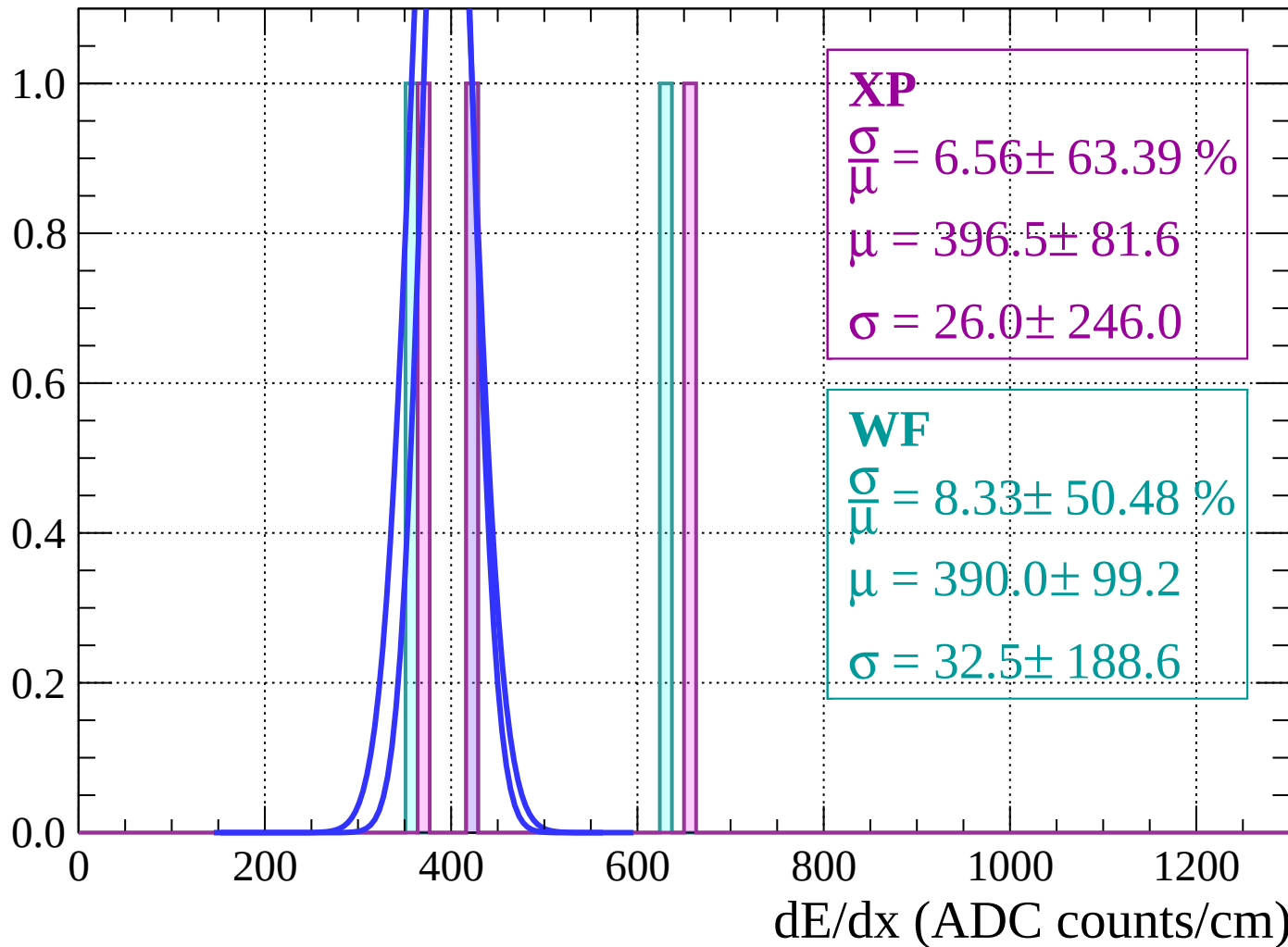
Energy loss | $70 < \theta < 72$

Count



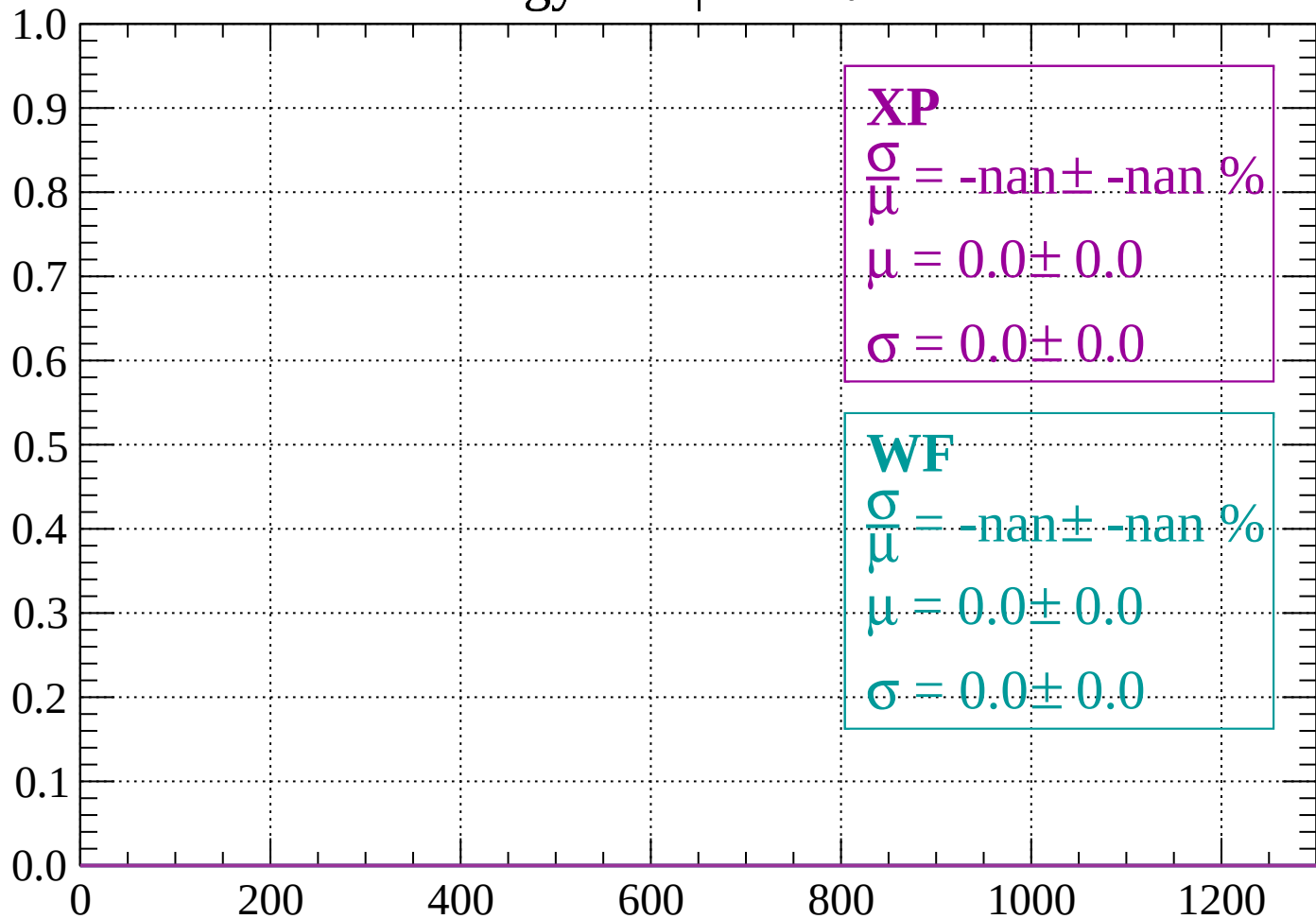
Energy loss | $72 < \theta < 74$

Count



Energy loss | $74 < \theta < 76$

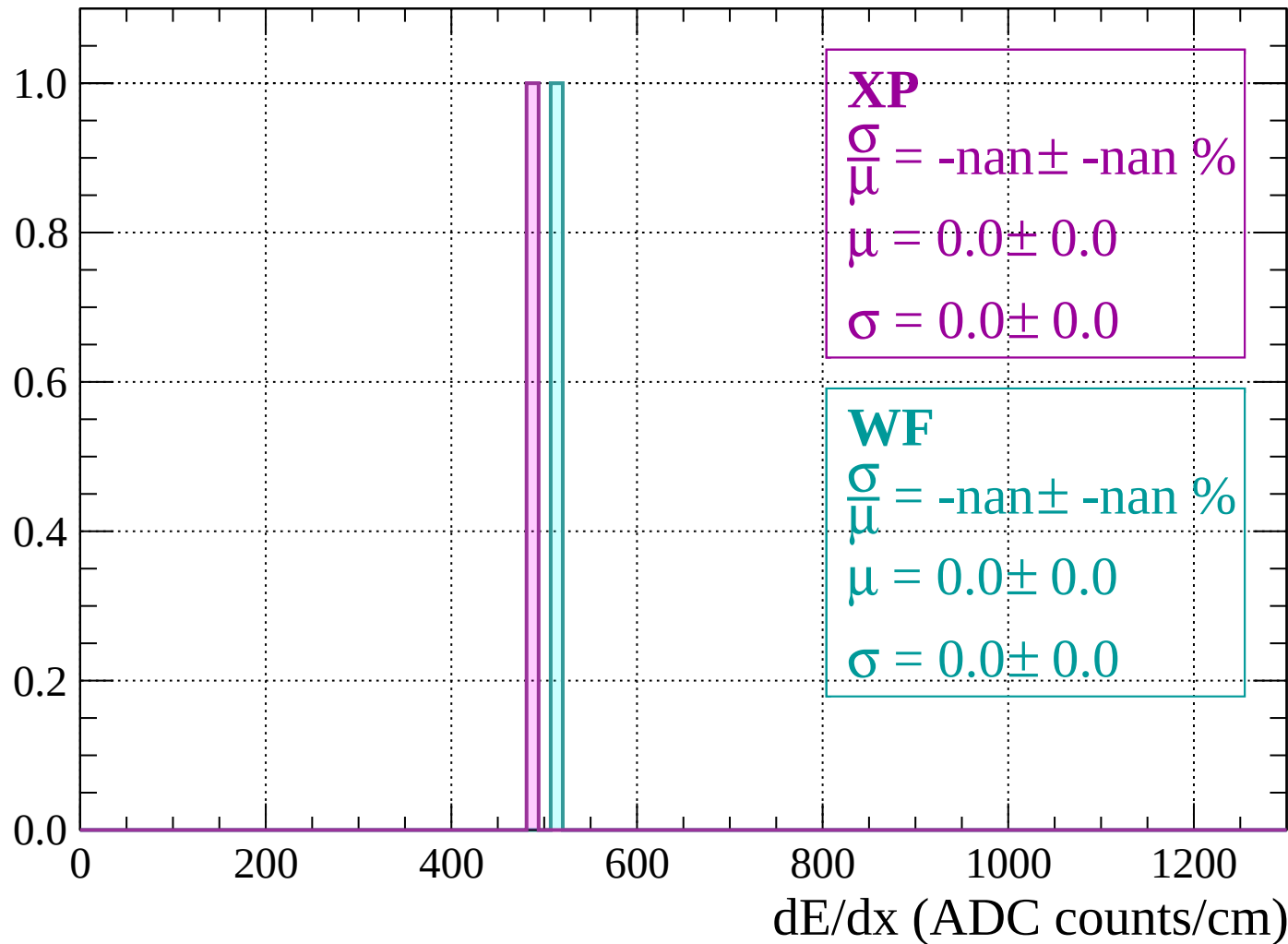
Count



dE/dx (ADC counts/cm)

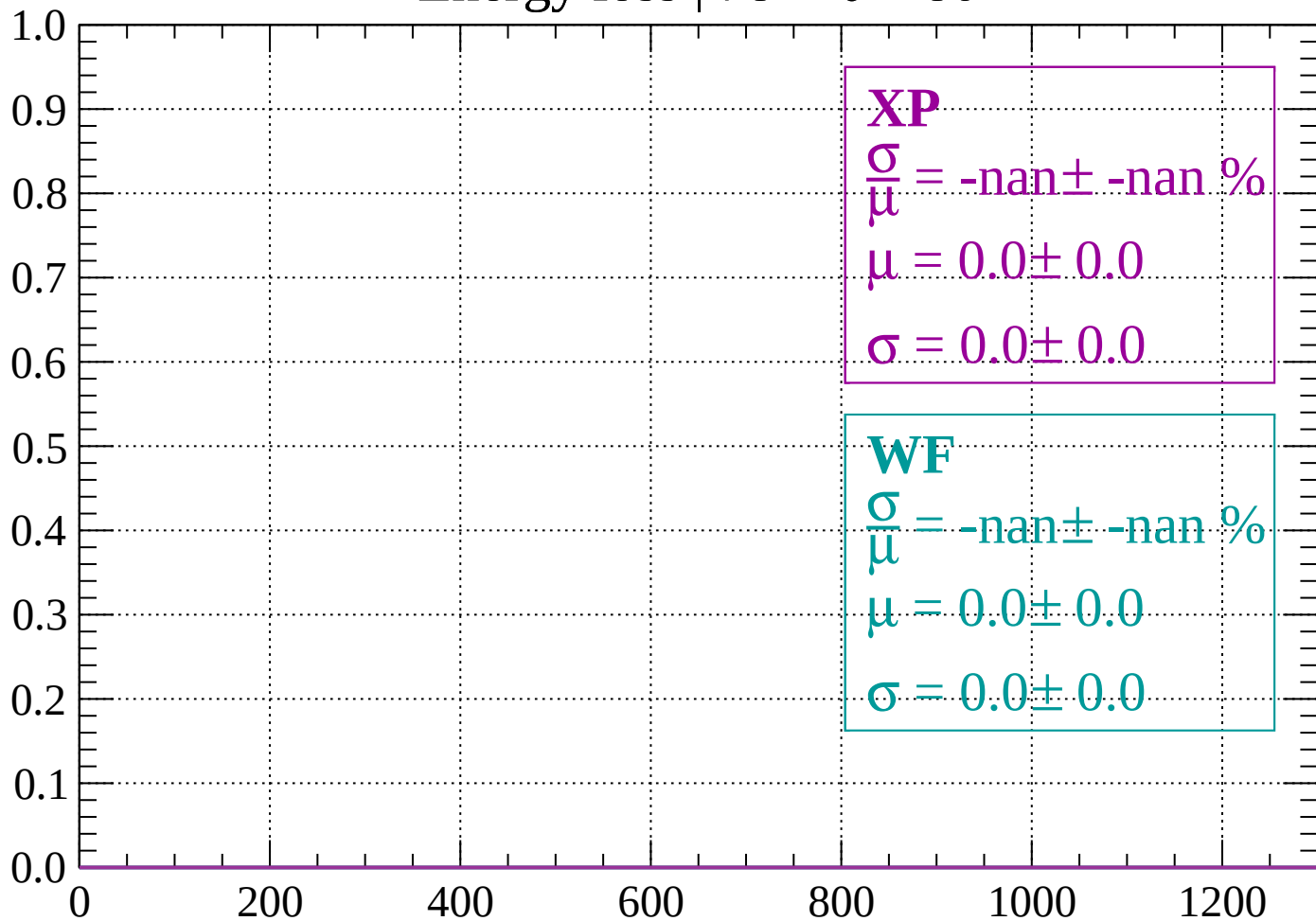
Energy loss | $76 < \theta < 78$

Count



Energy loss | $78 < \theta < 80$

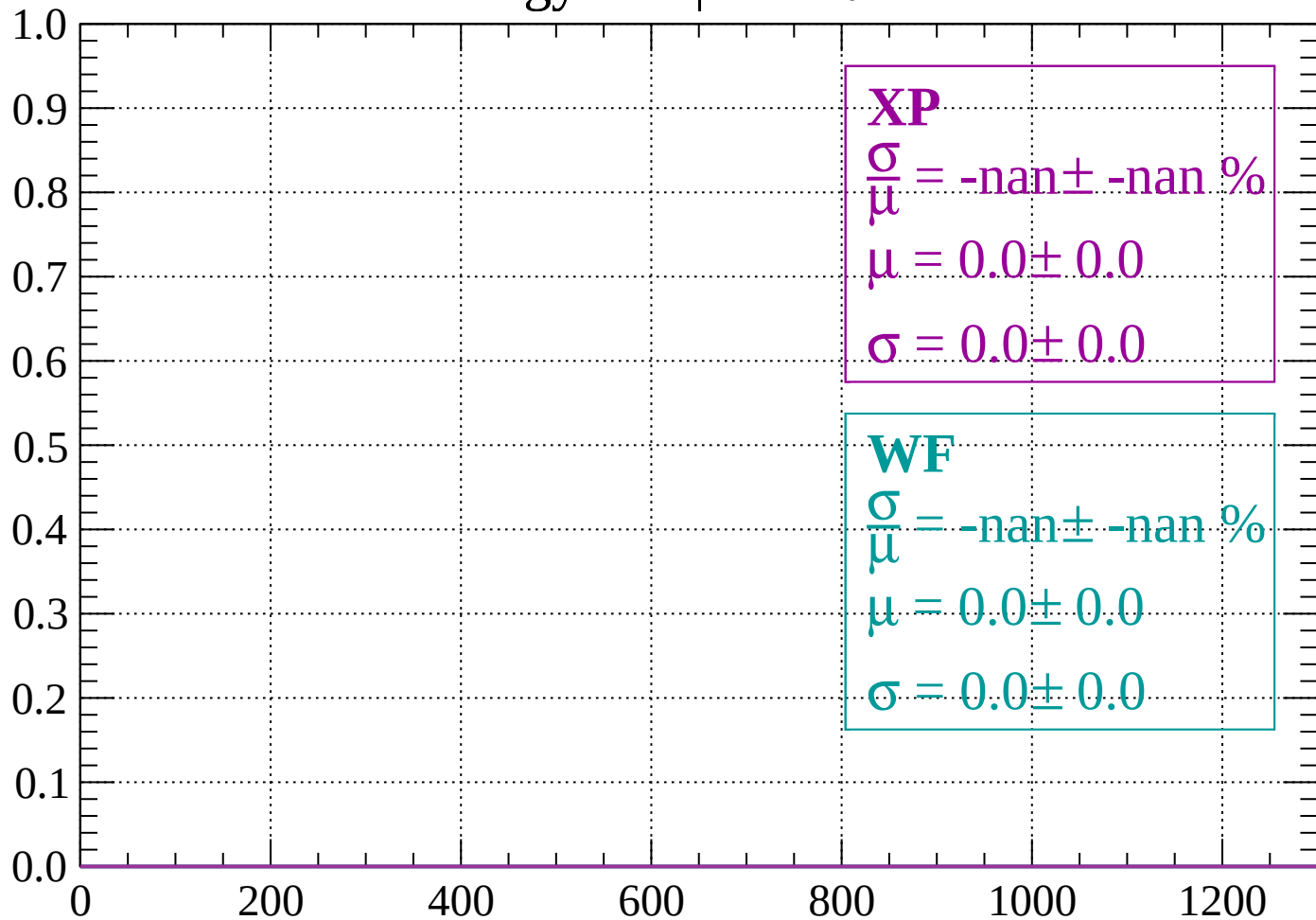
Count



dE/dx (ADC counts/cm)

Energy loss | $80 < \theta < 82$

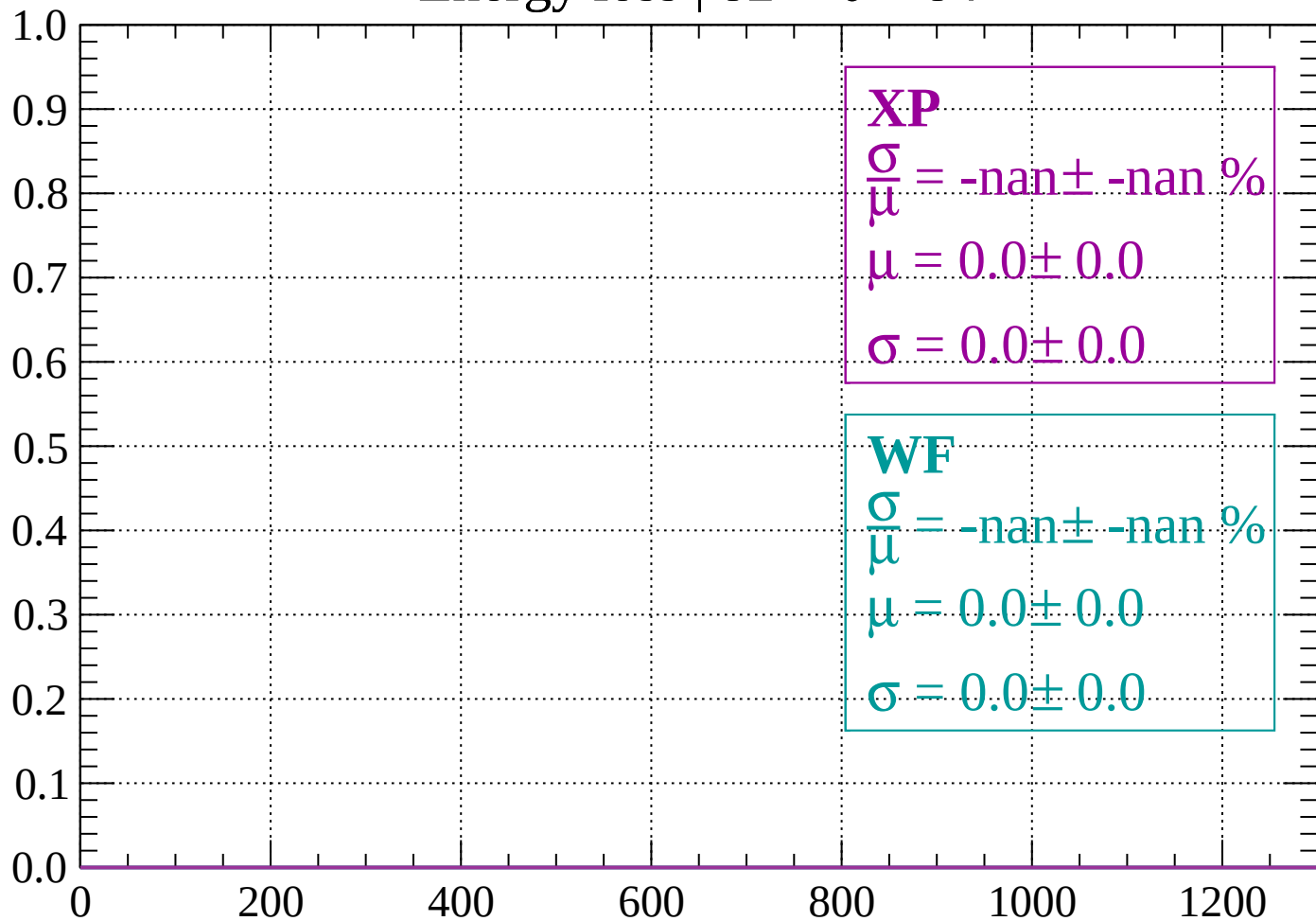
Count



dE/dx (ADC counts/cm)

Energy loss | $82 < \theta < 84$

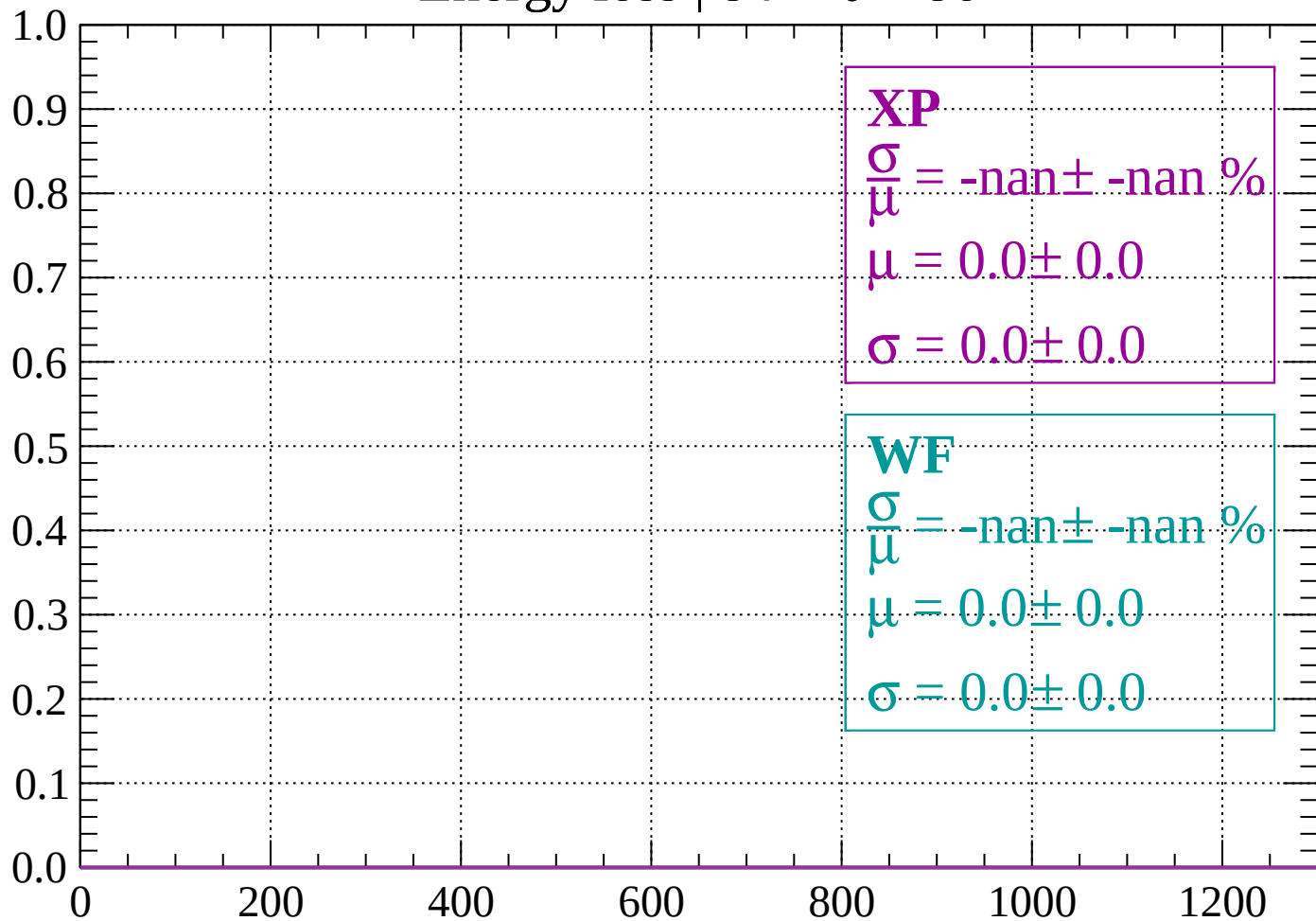
Count



dE/dx (ADC counts/cm)

Energy loss | $84 < \theta < 86$

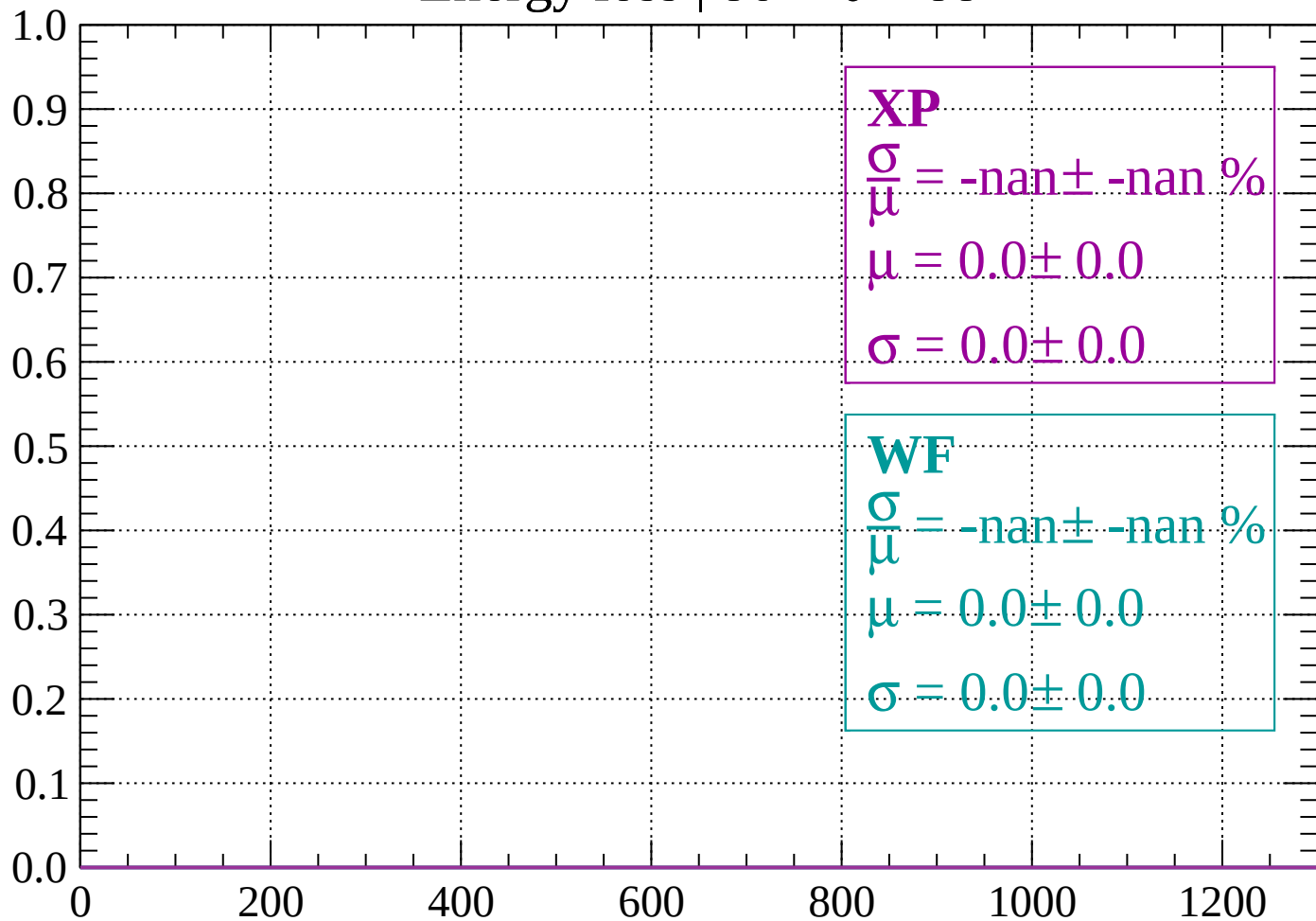
Count



dE/dx (ADC counts/cm)

Energy loss | $86 < \theta < 88$

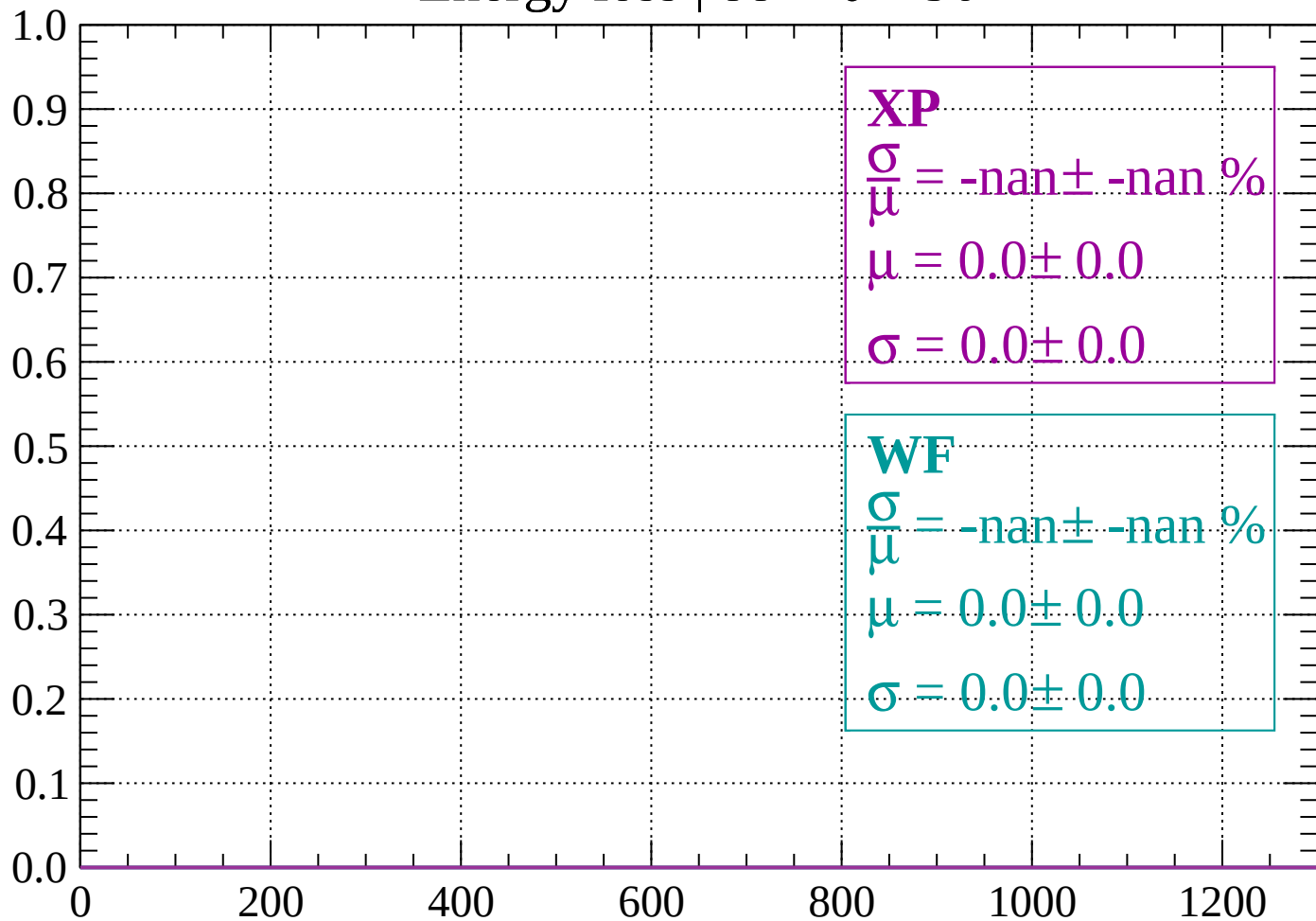
Count



dE/dx (ADC counts/cm)

Energy loss | $88 < \theta < 90$

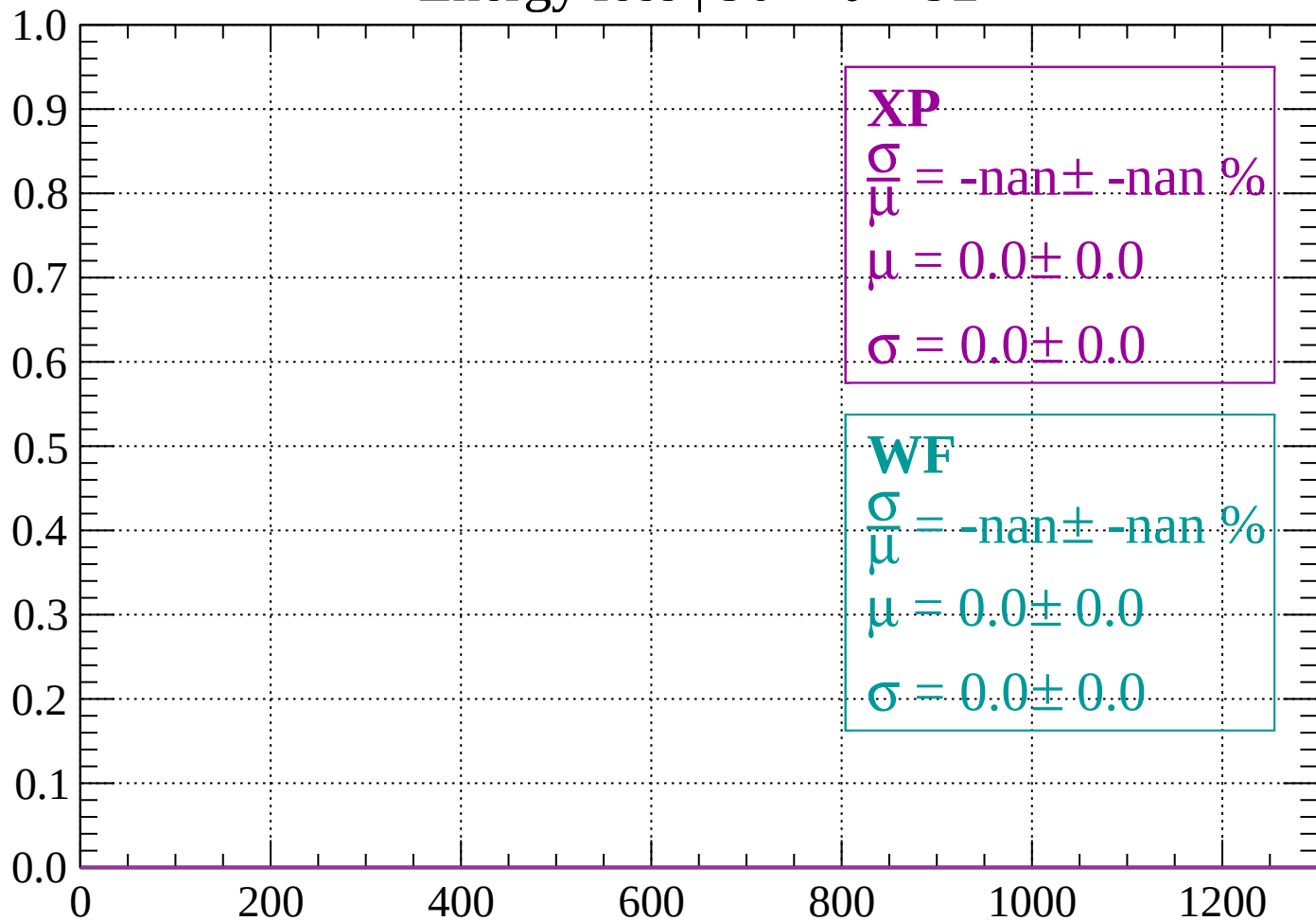
Count



dE/dx (ADC counts/cm)

Energy loss | $90 < \theta < 92$

Count



dE/dx (ADC counts/cm)

